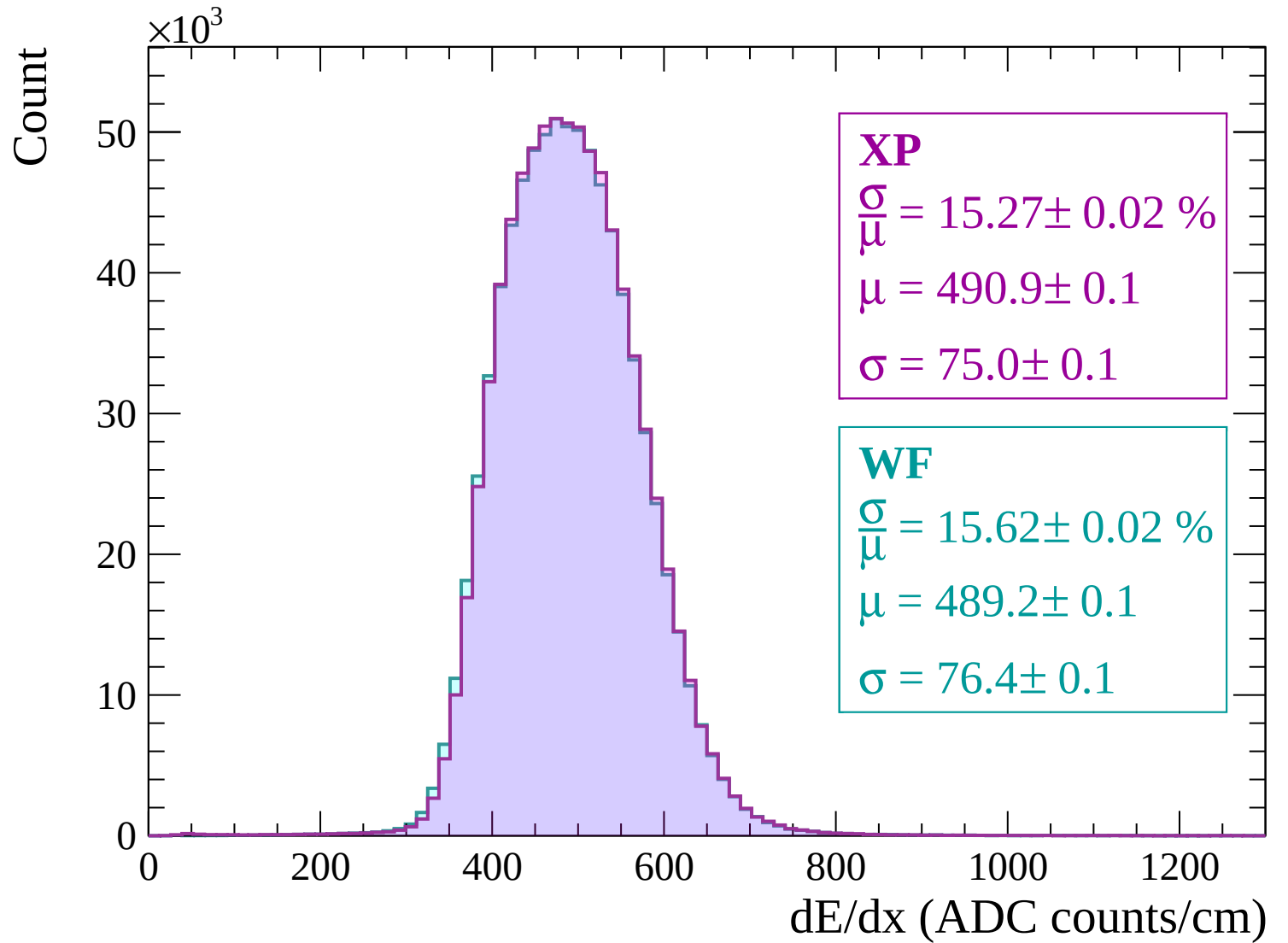
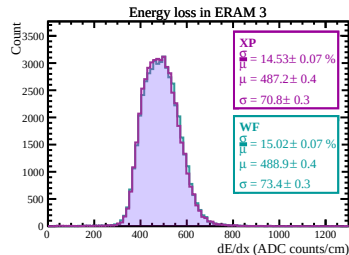
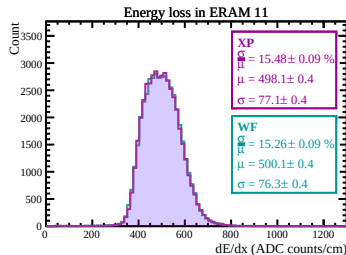
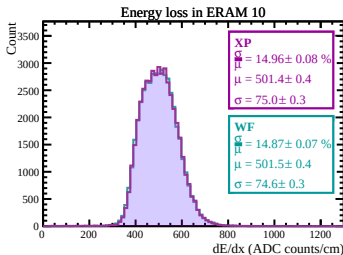
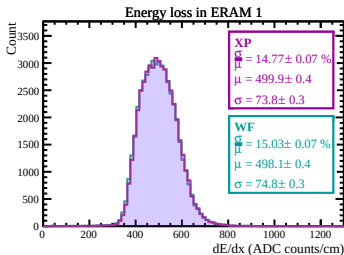
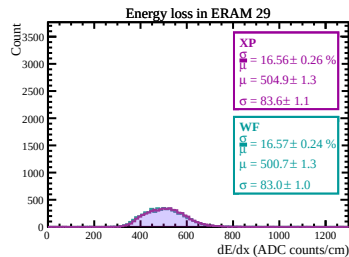
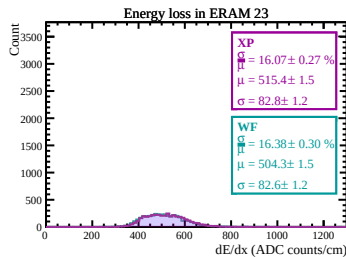
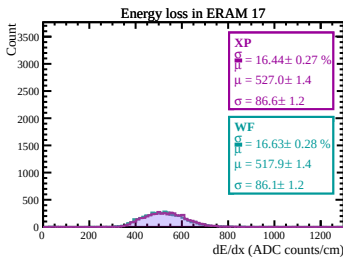
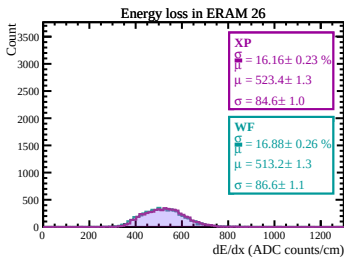
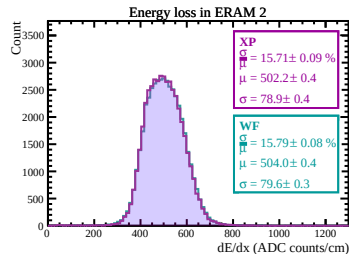
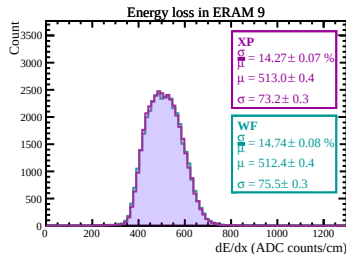
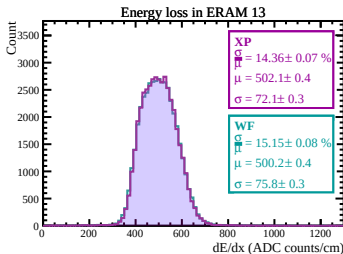
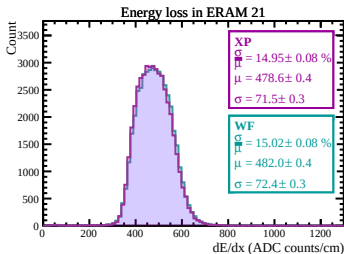
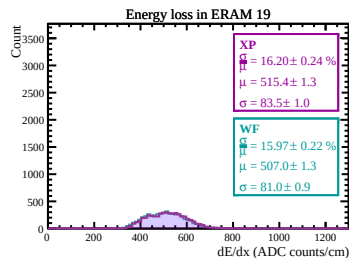
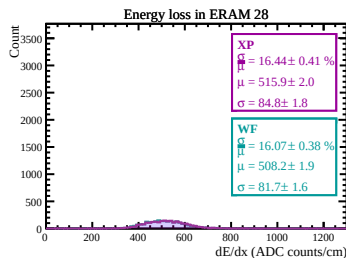
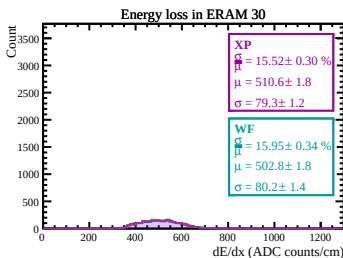
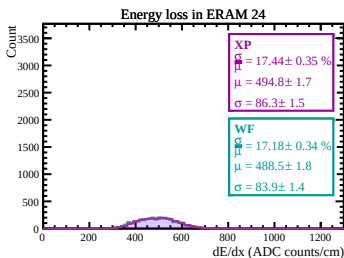
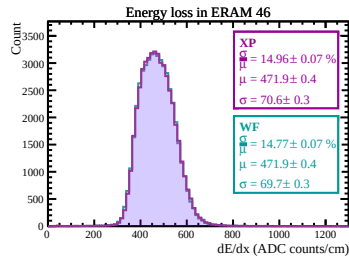
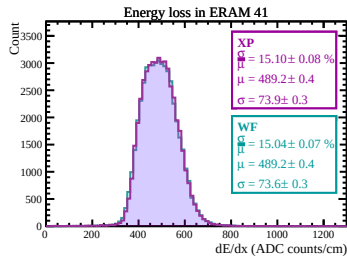
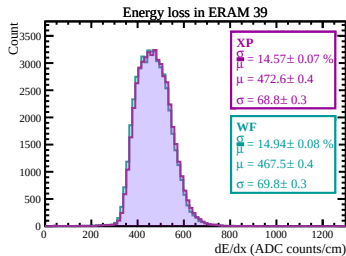
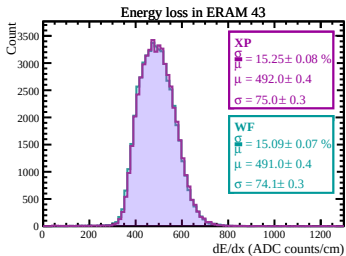
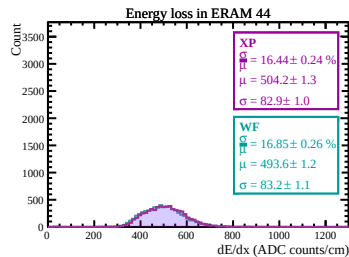
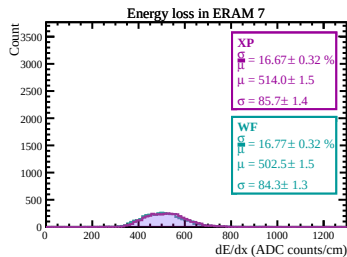
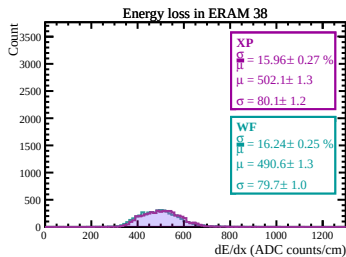
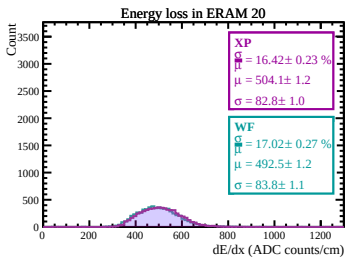
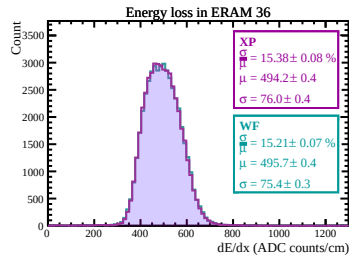
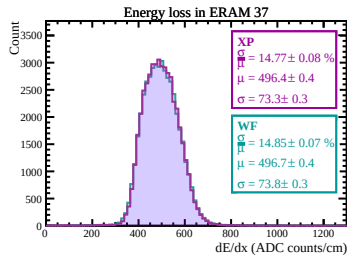
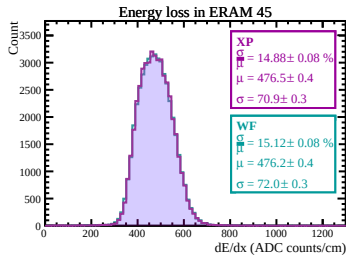
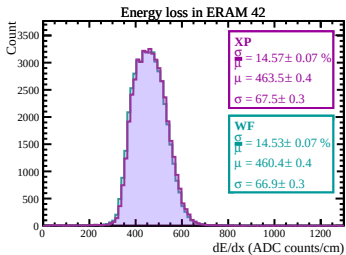
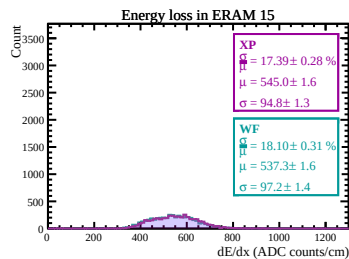
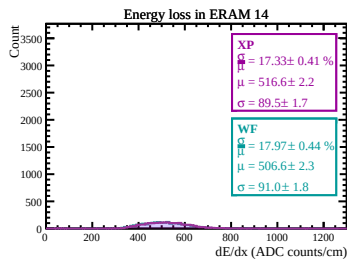
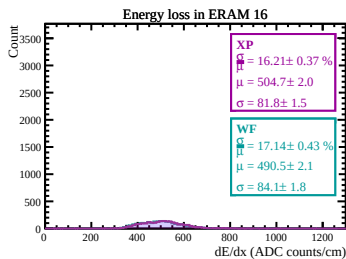
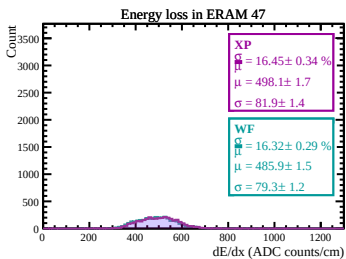
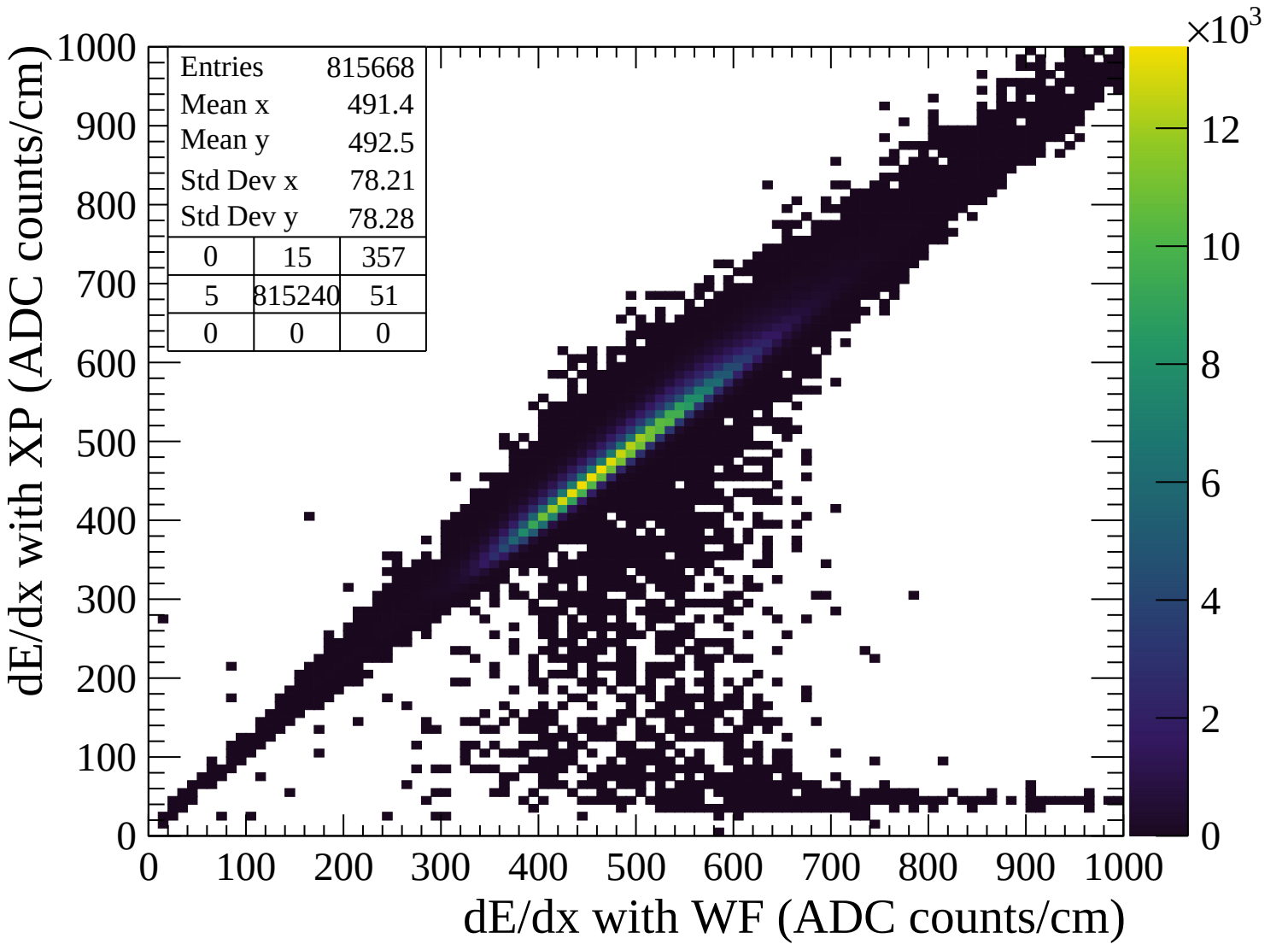
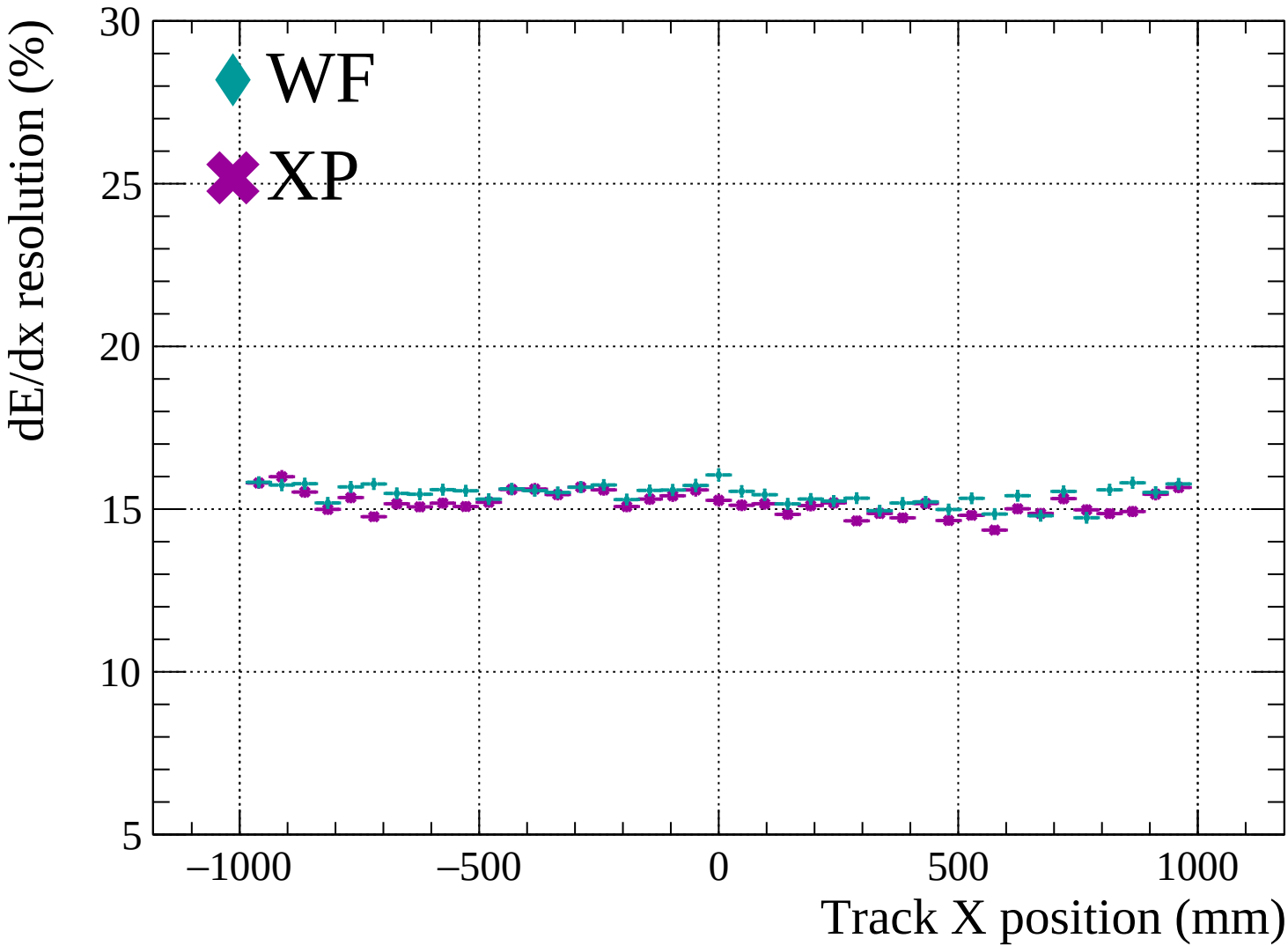


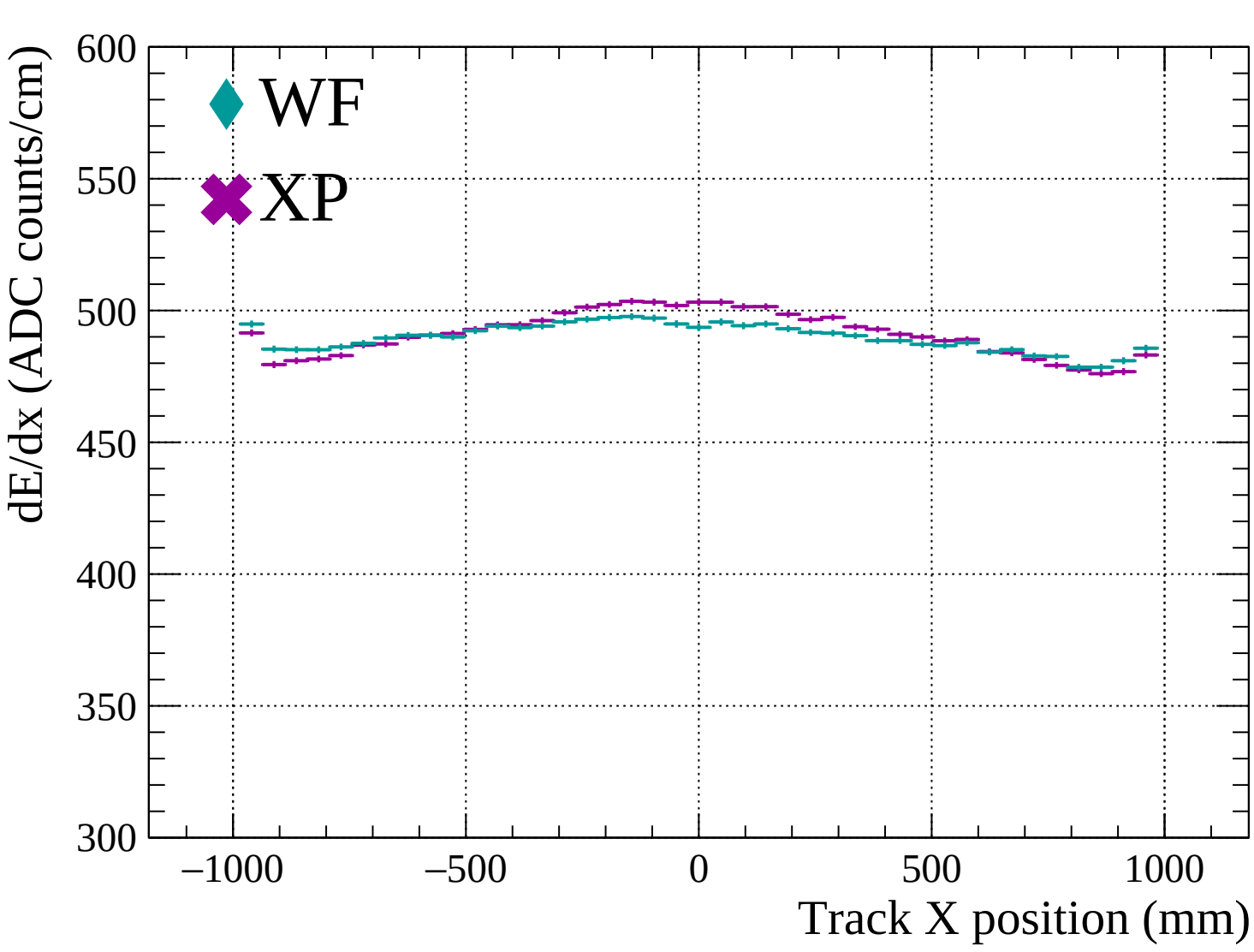


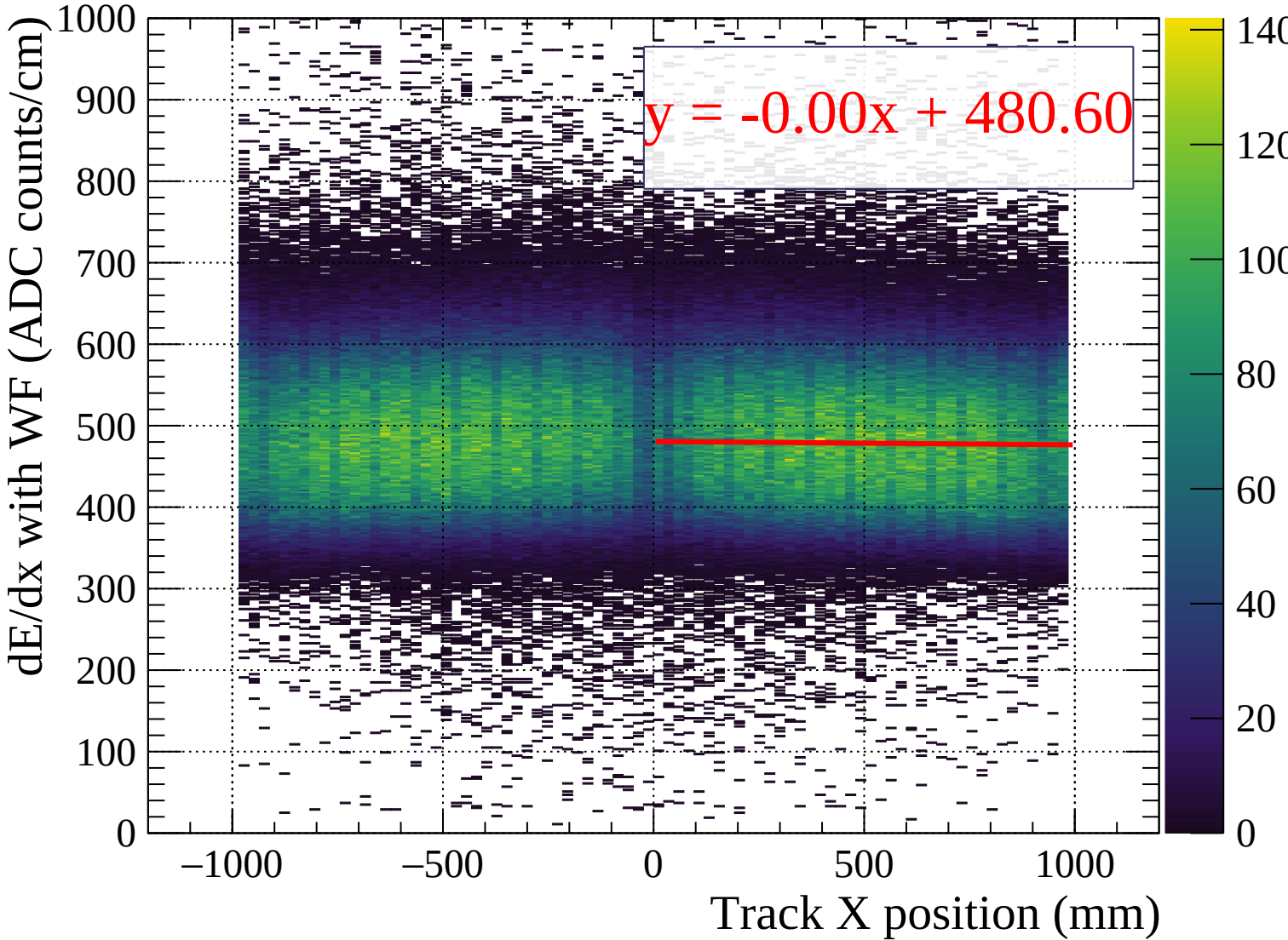


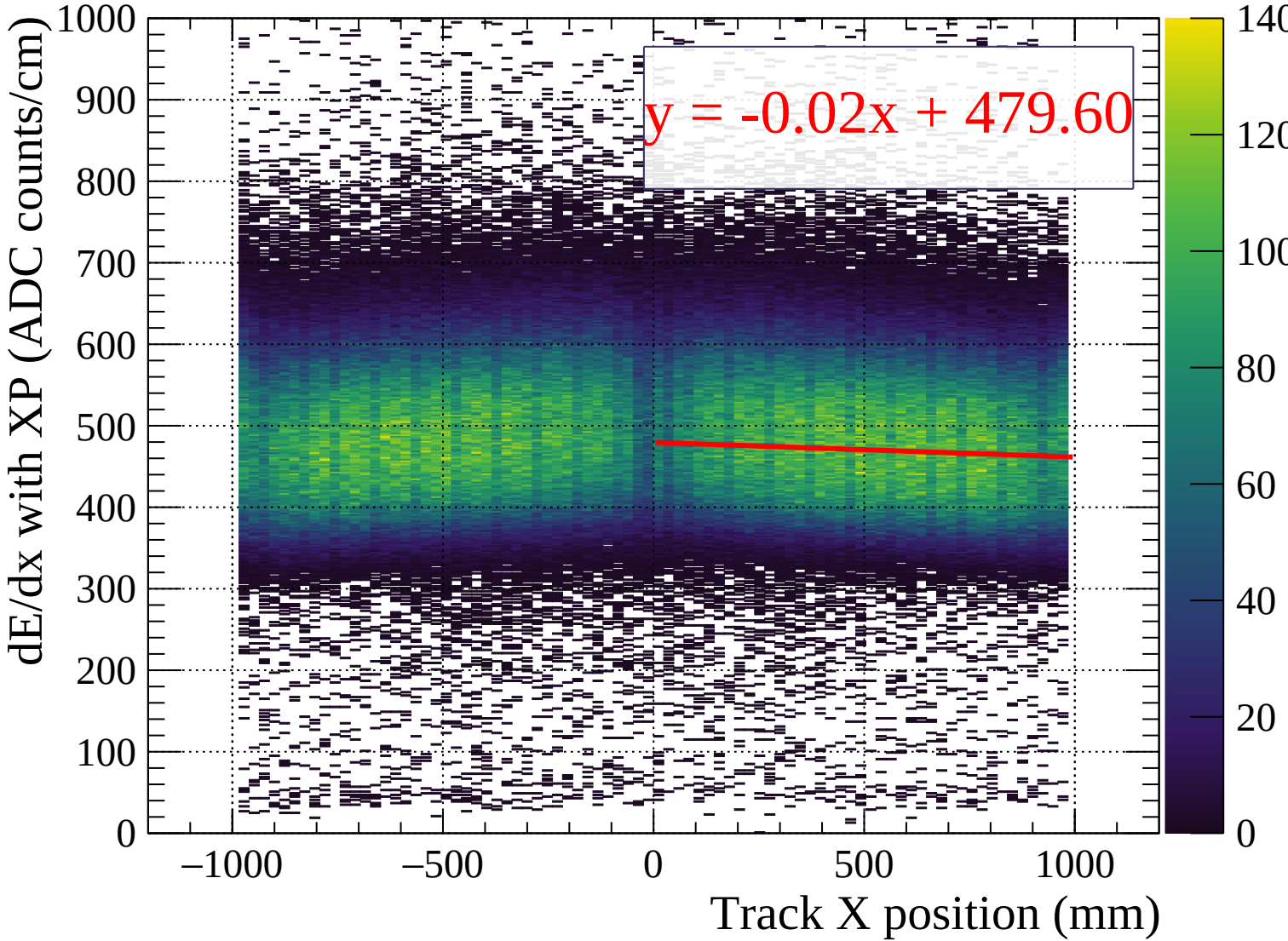


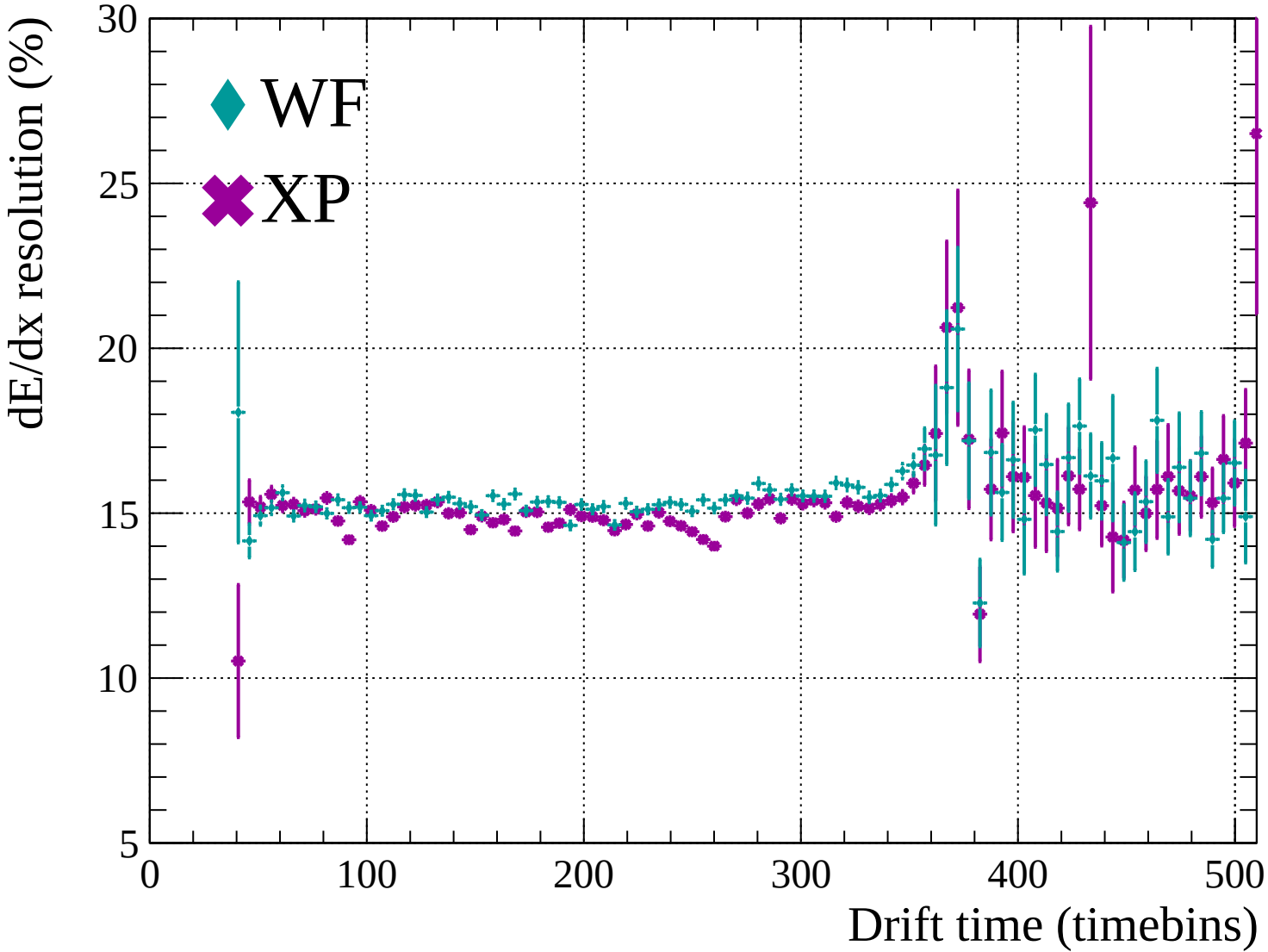


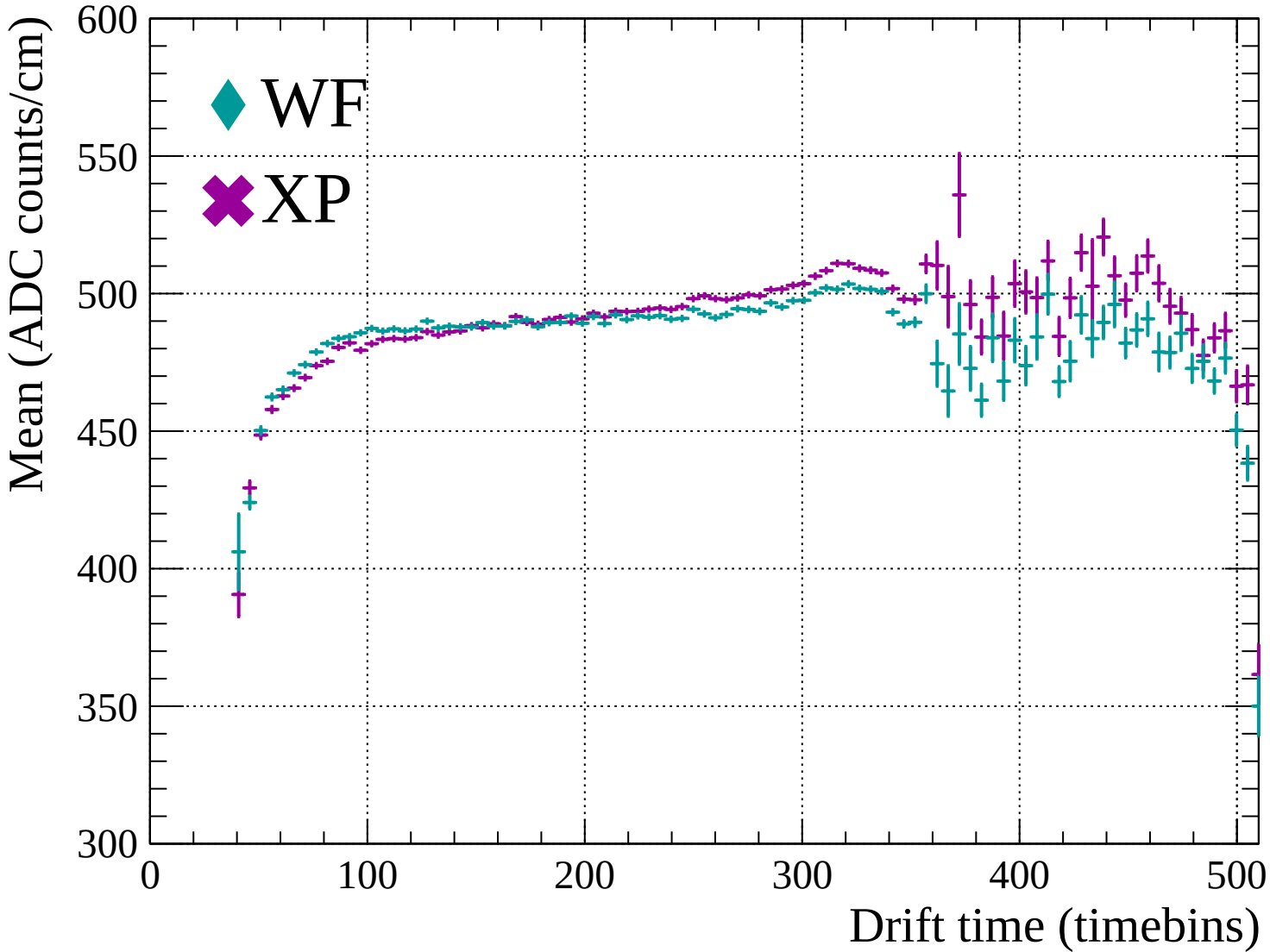


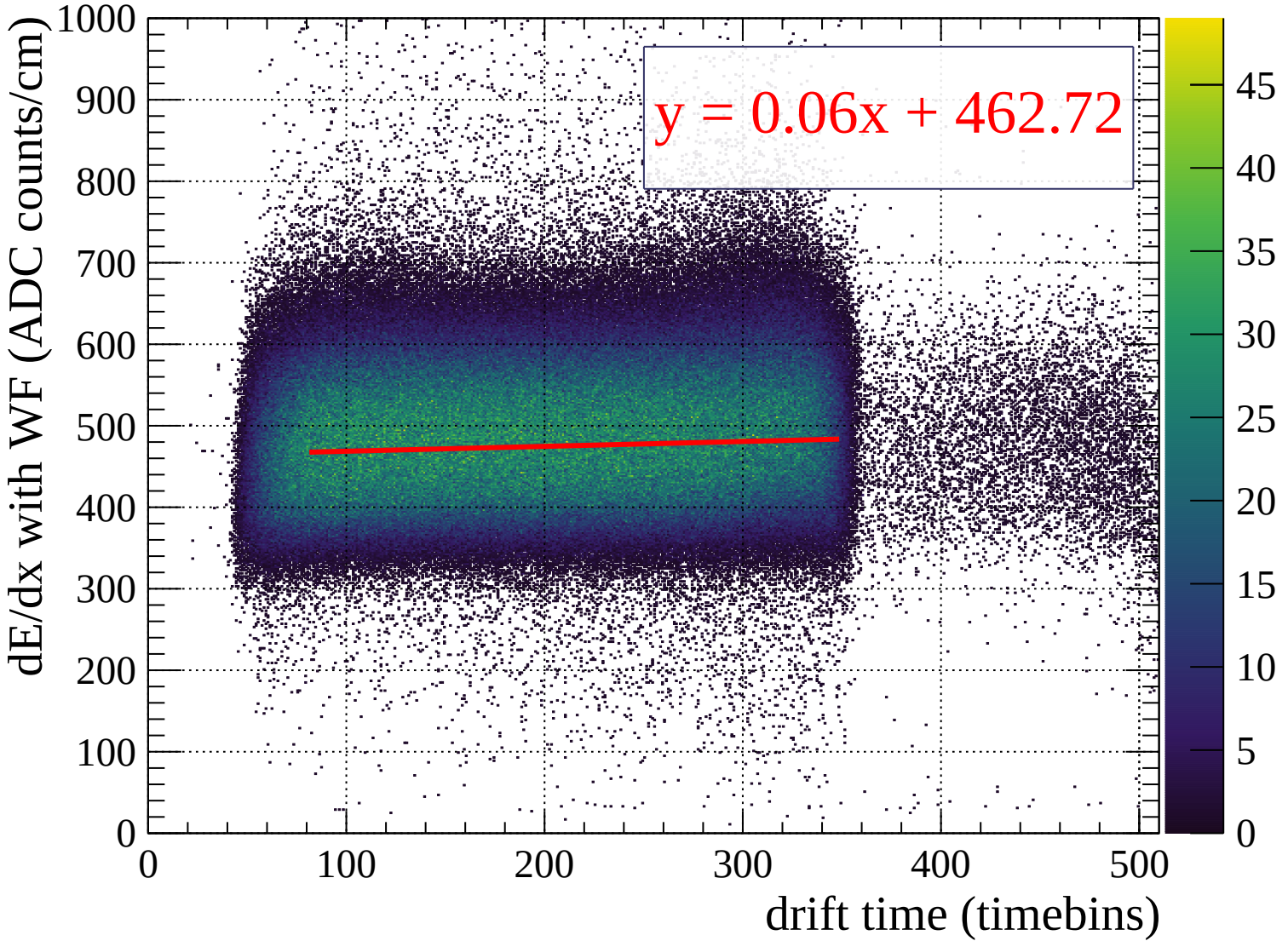


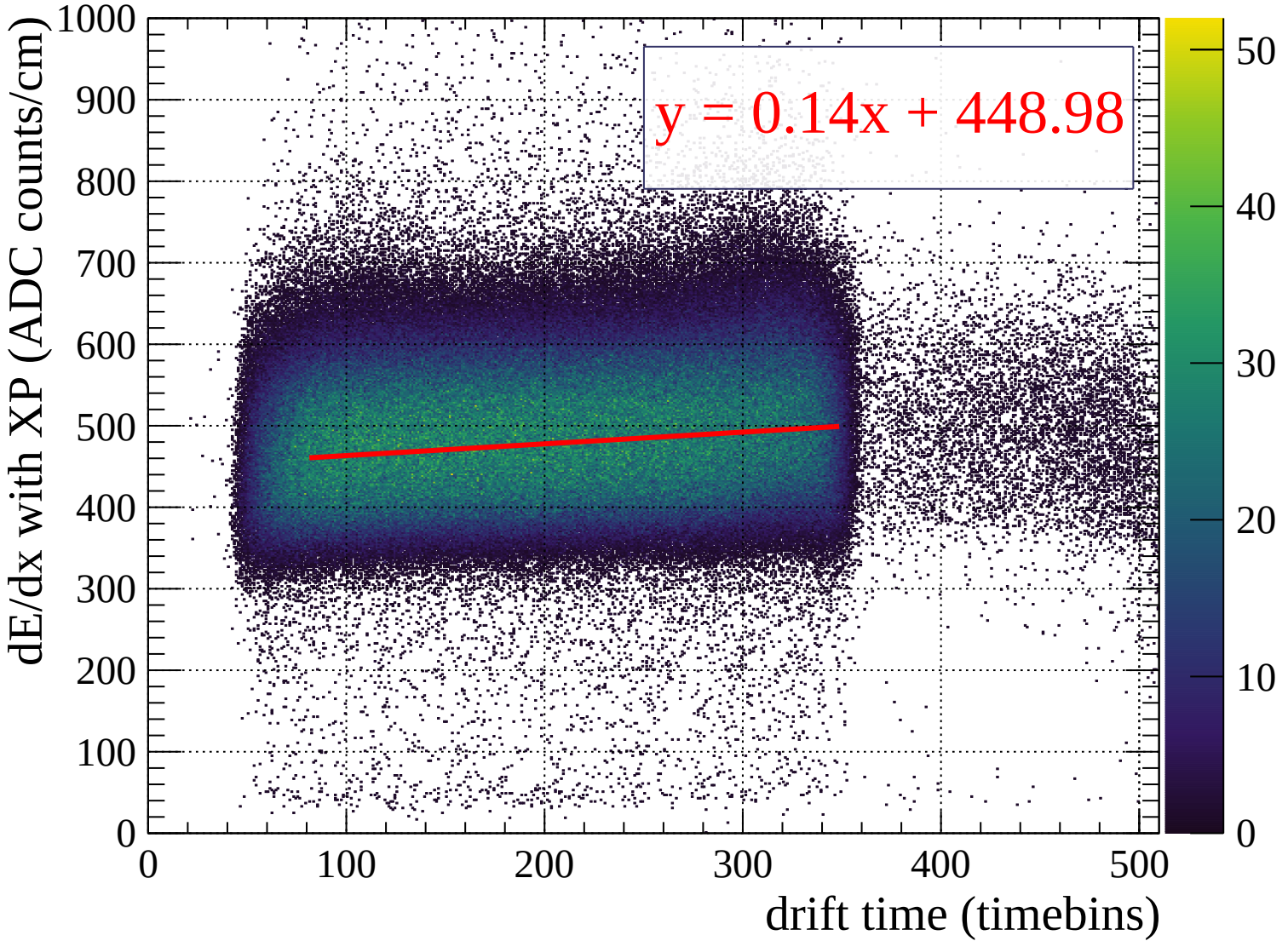


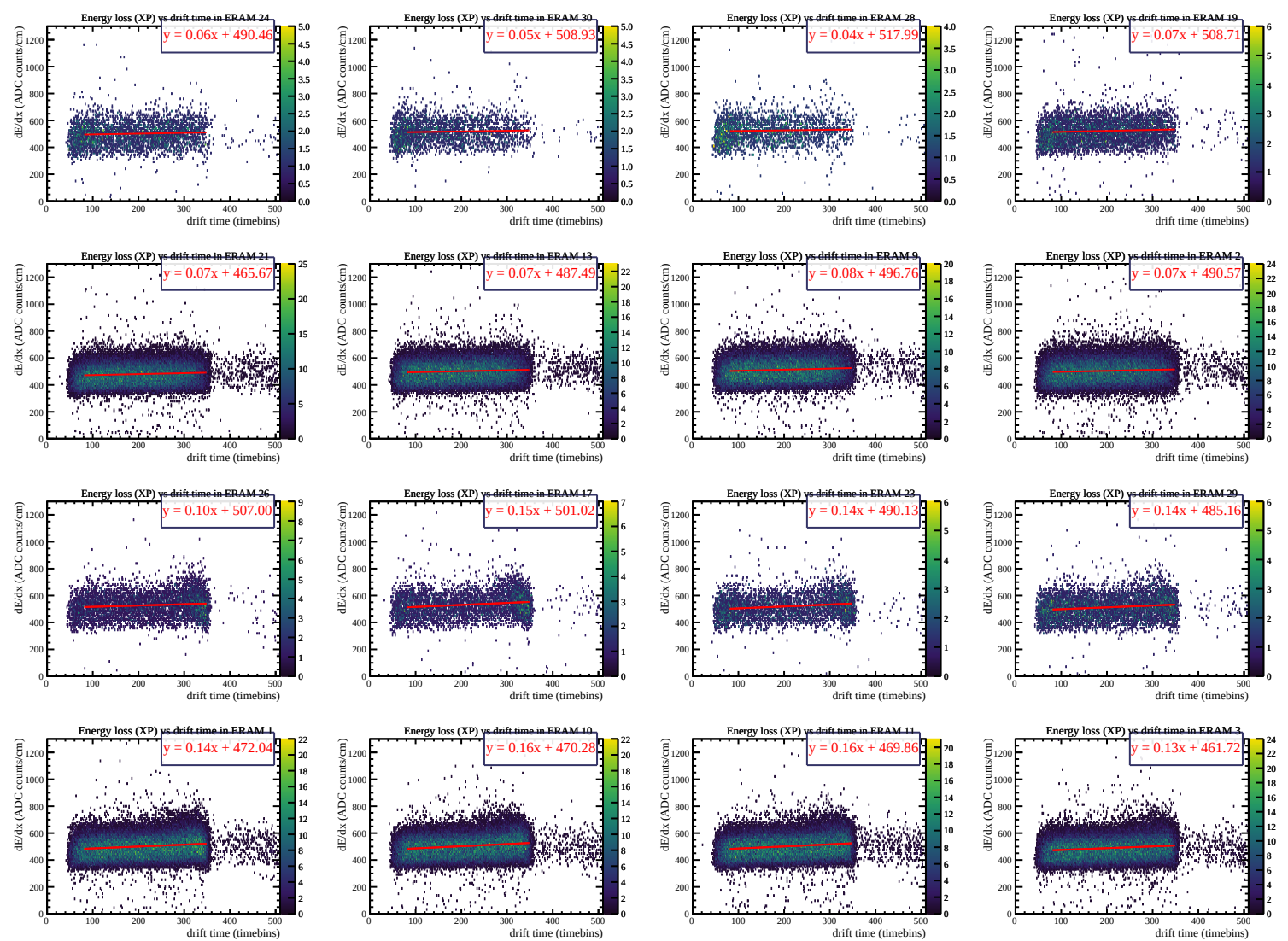


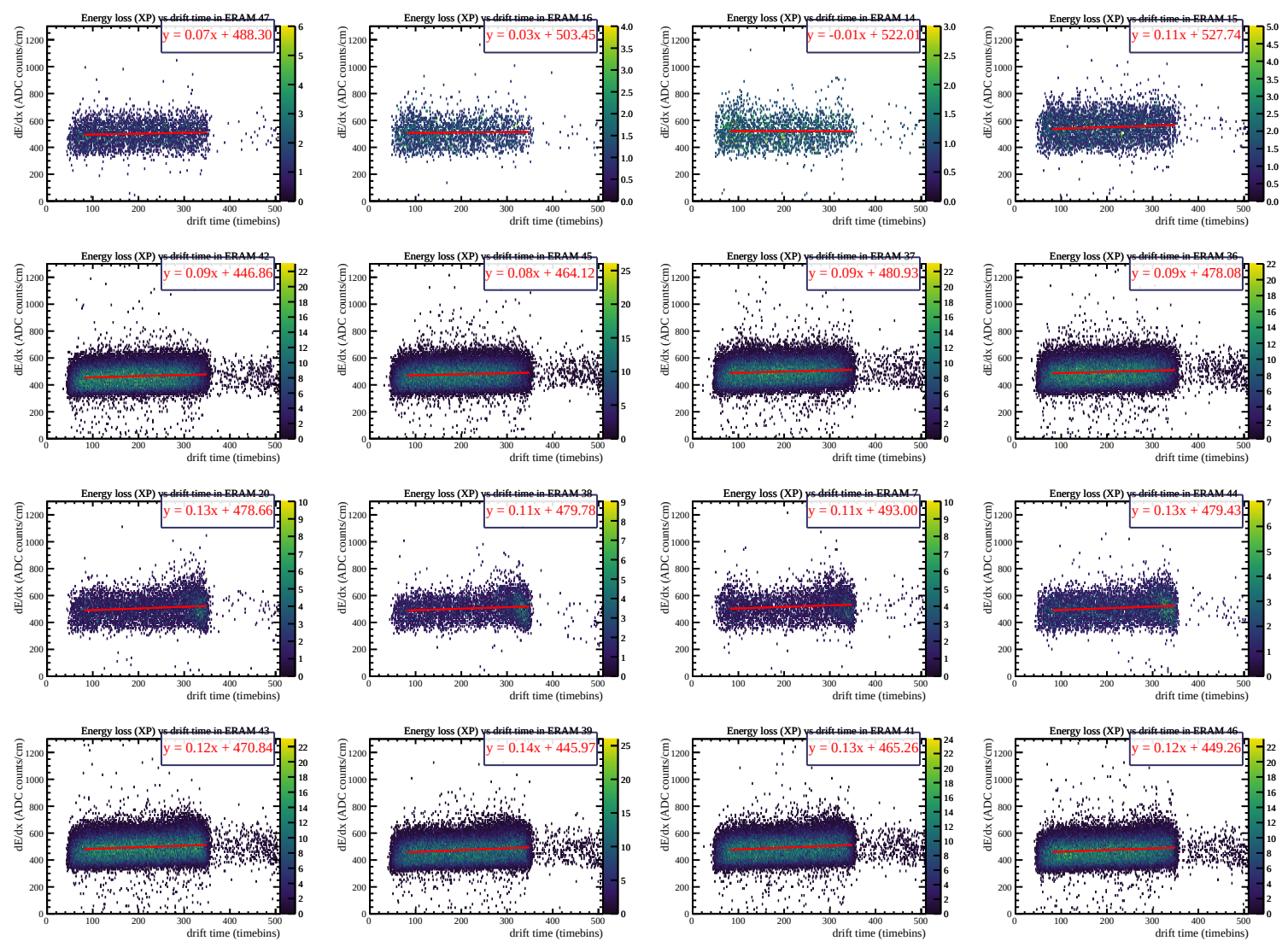


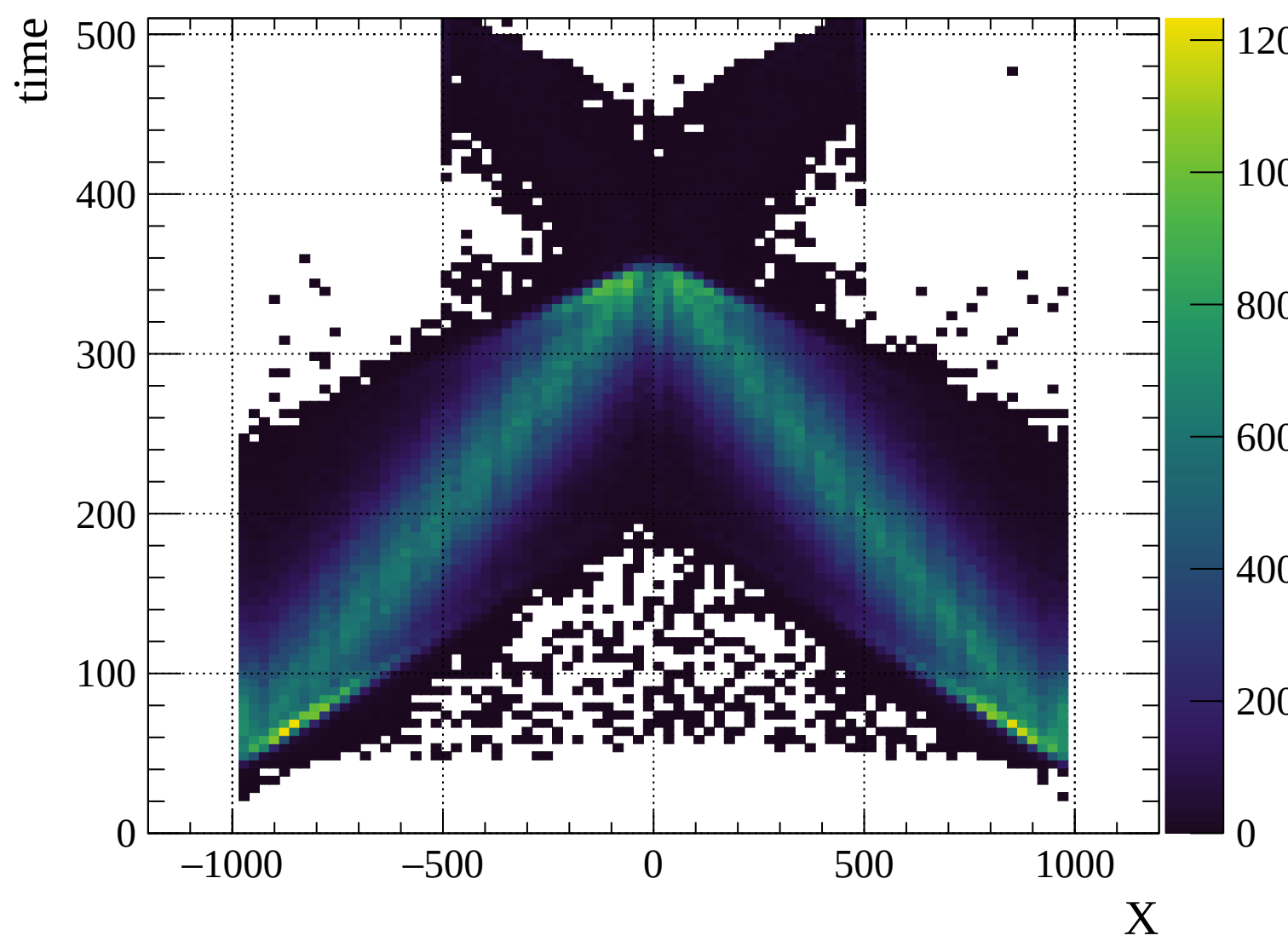


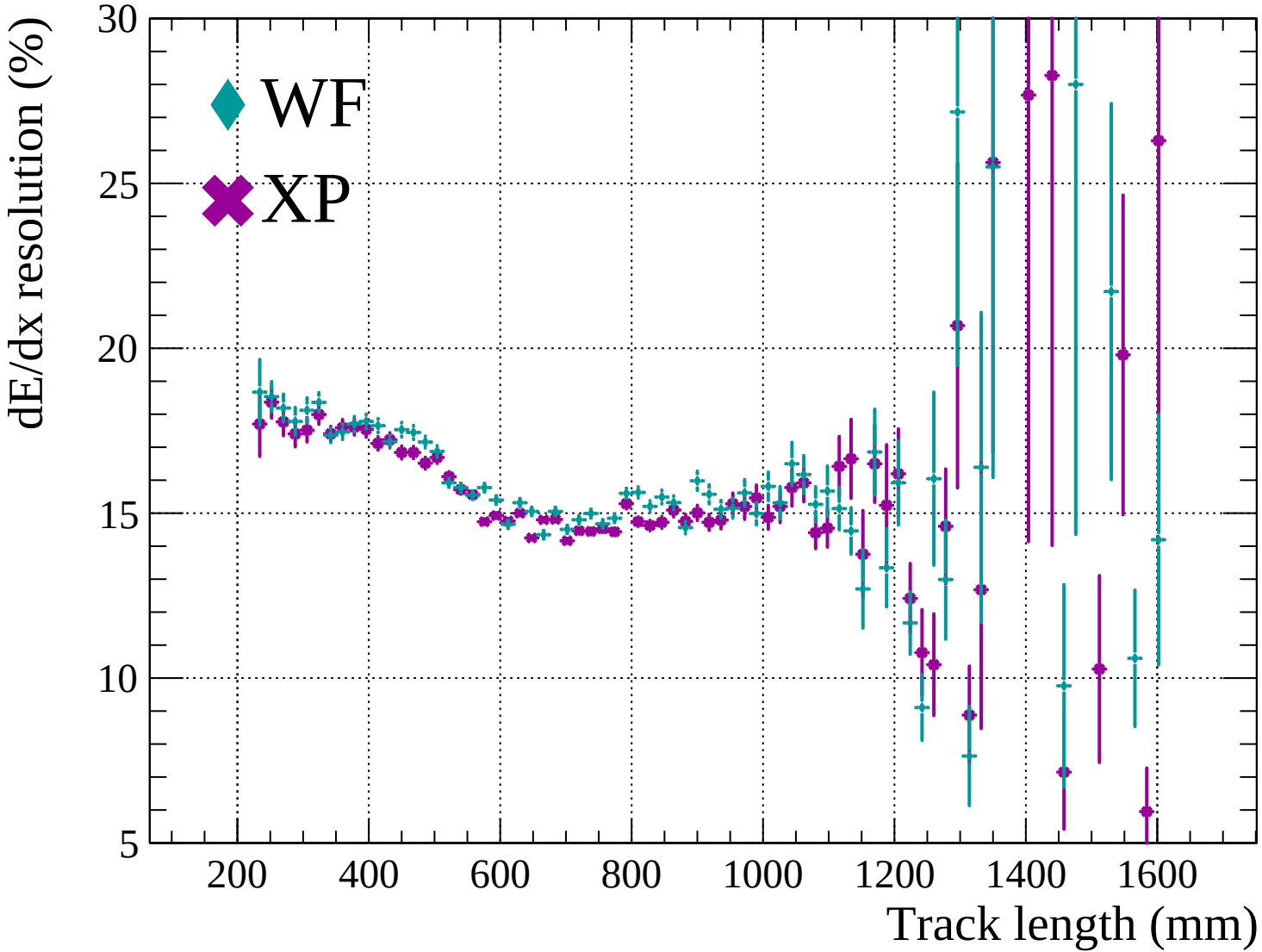


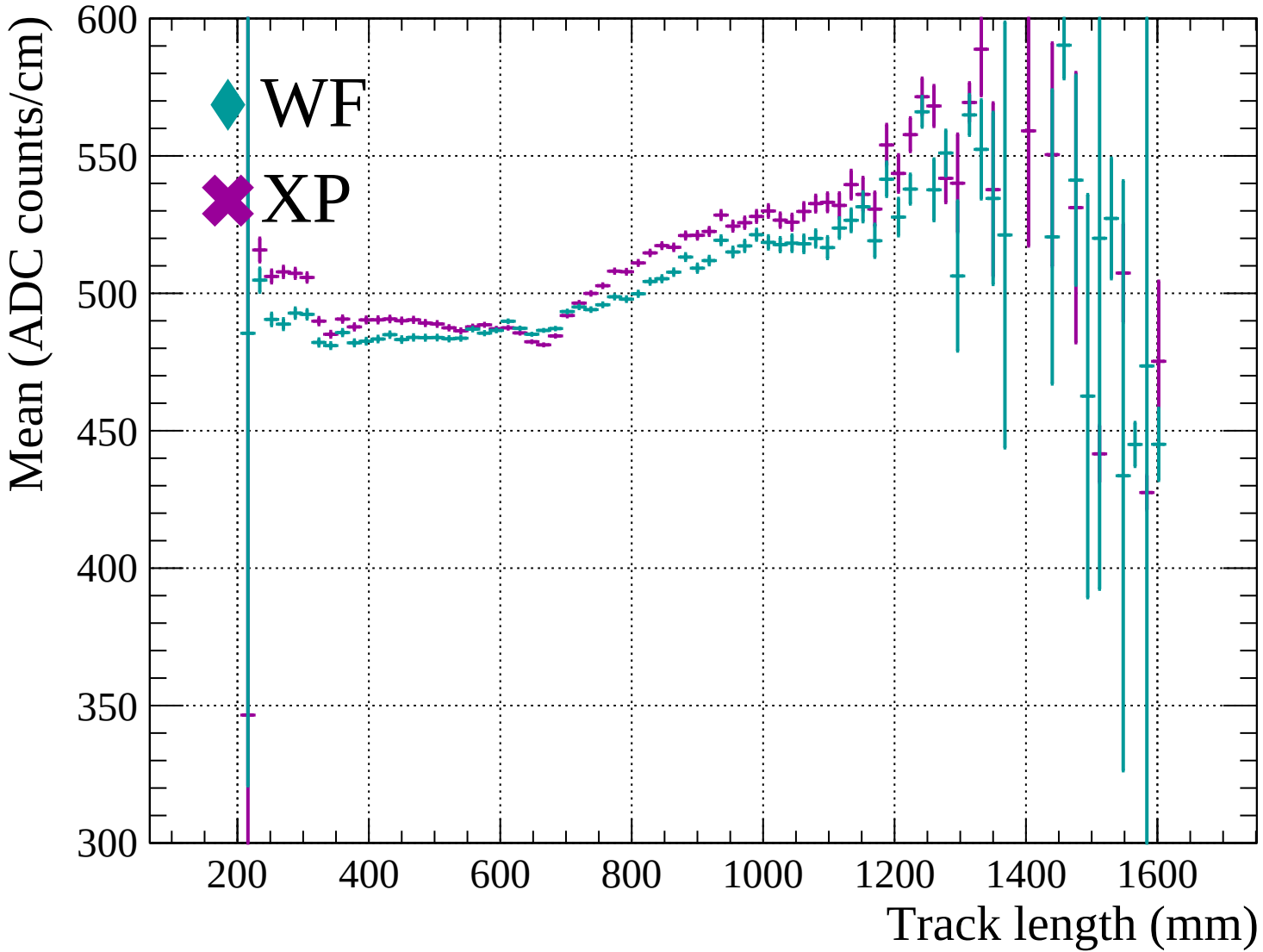


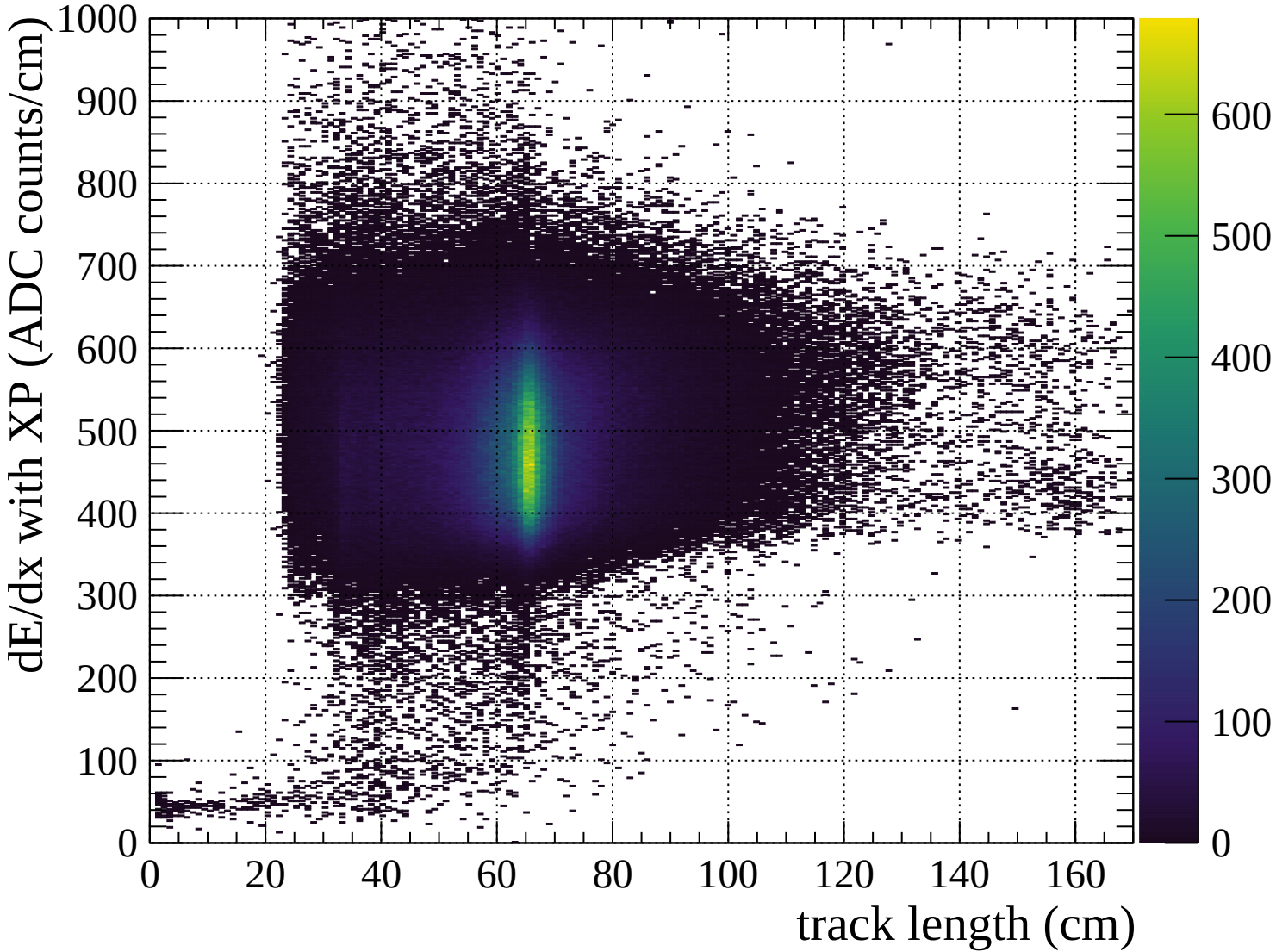


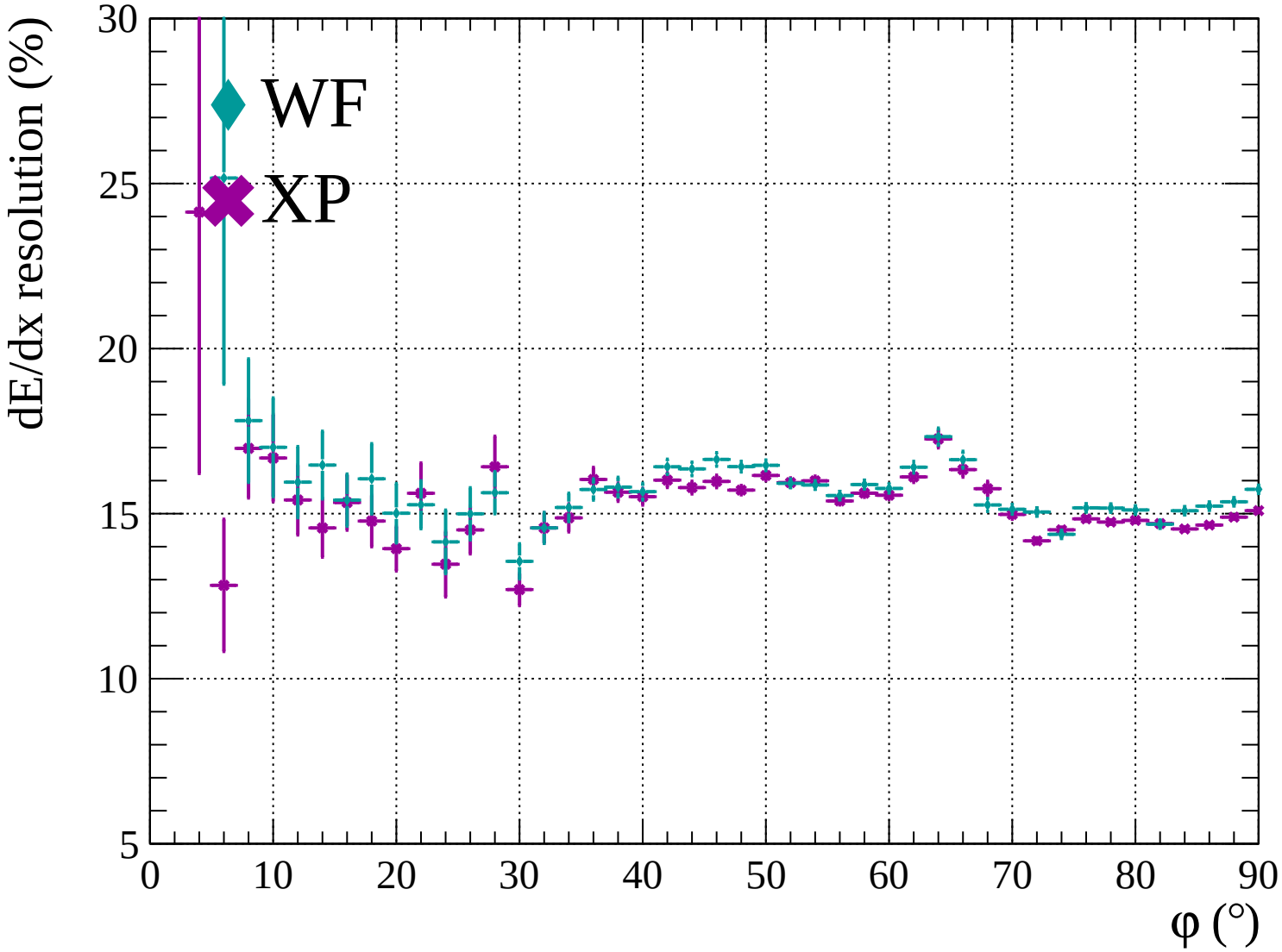


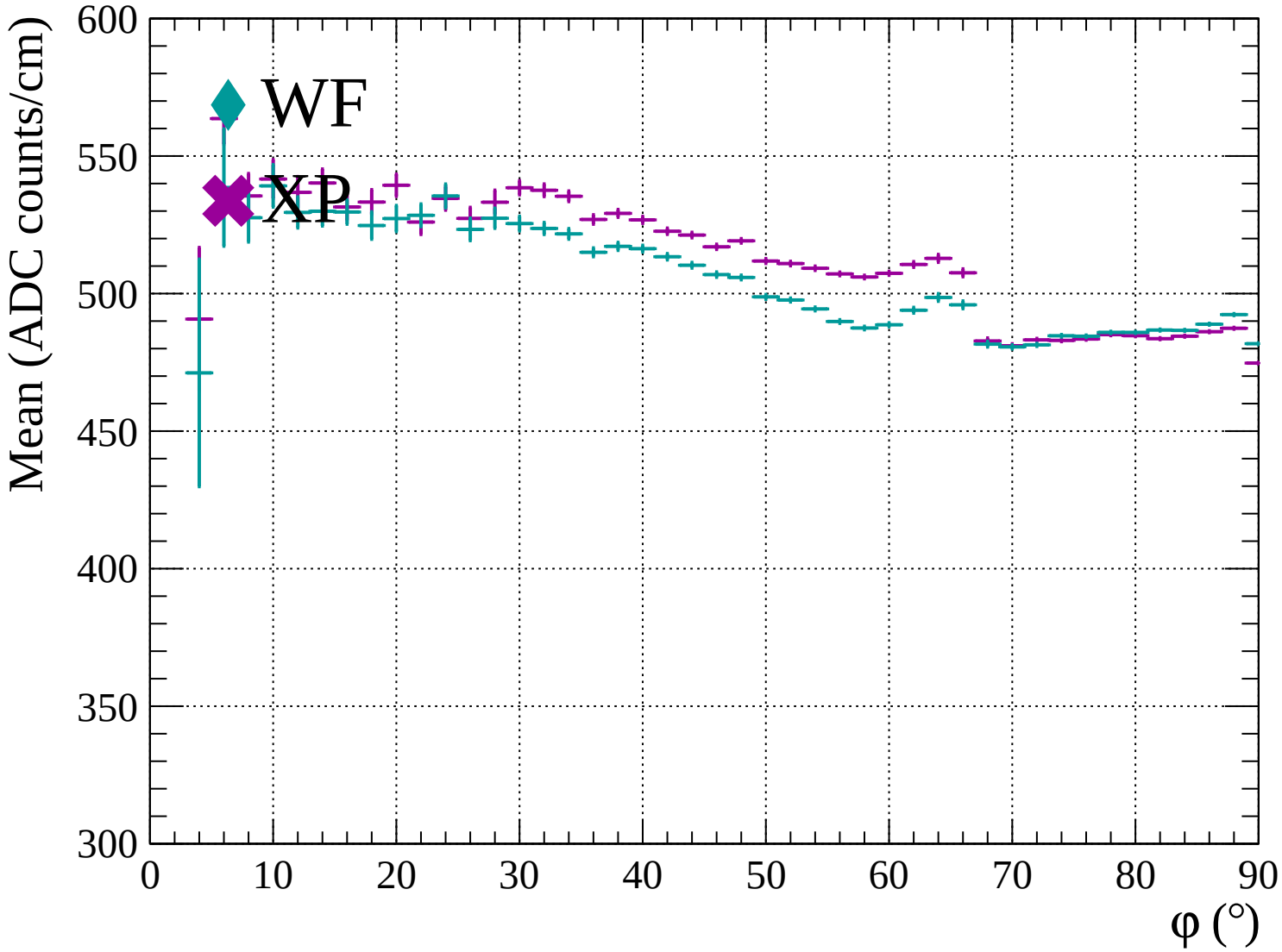


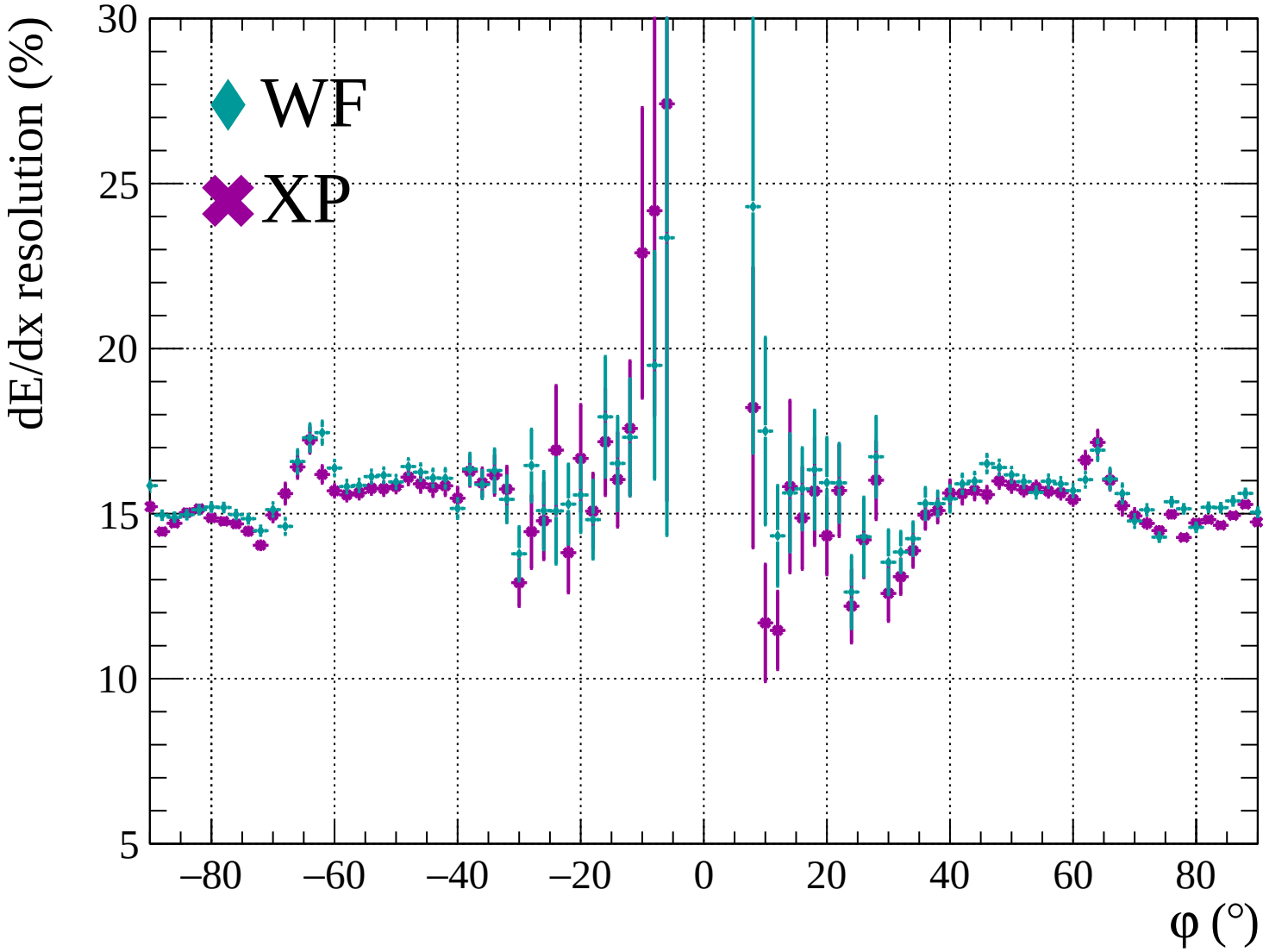


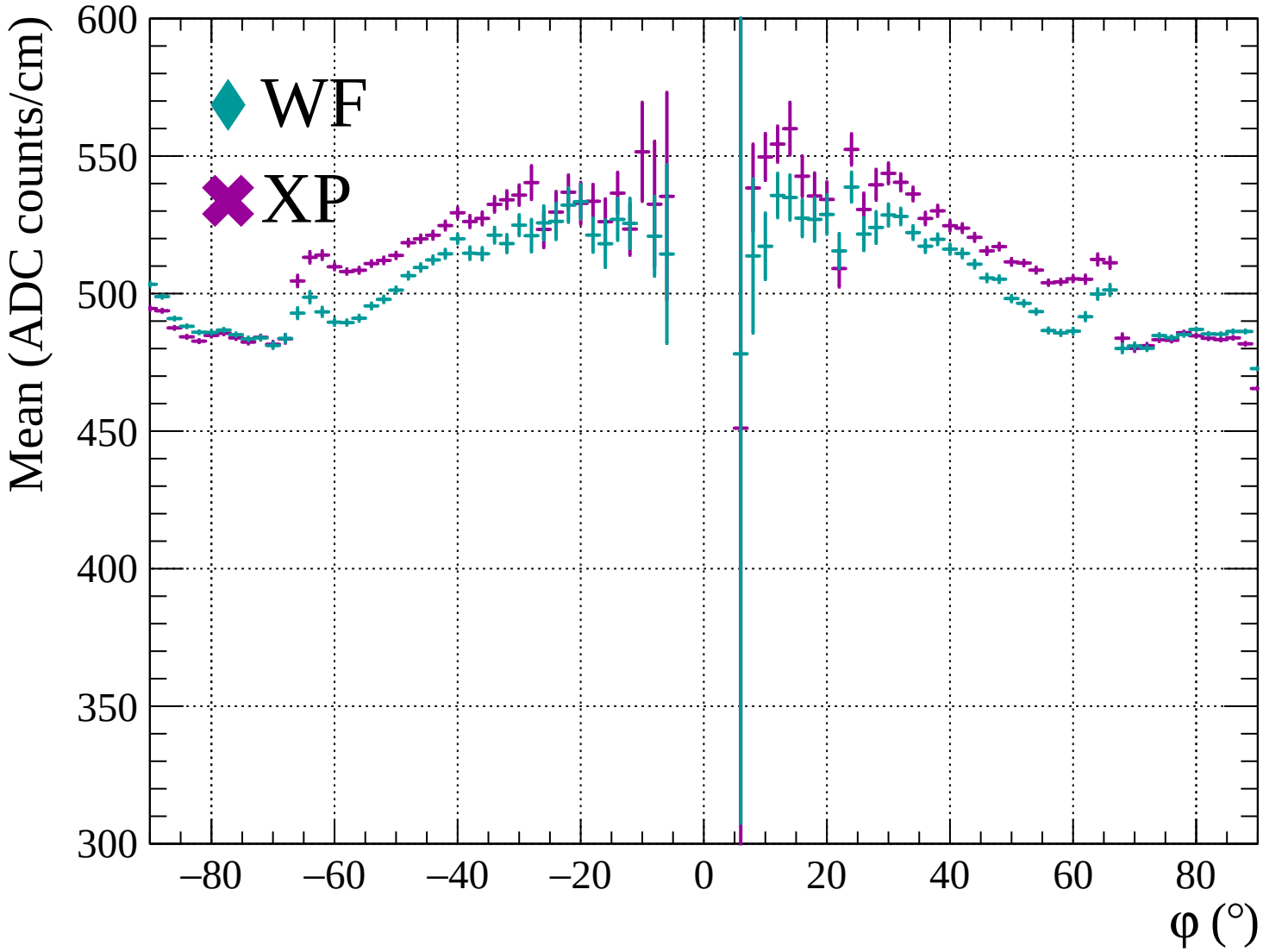


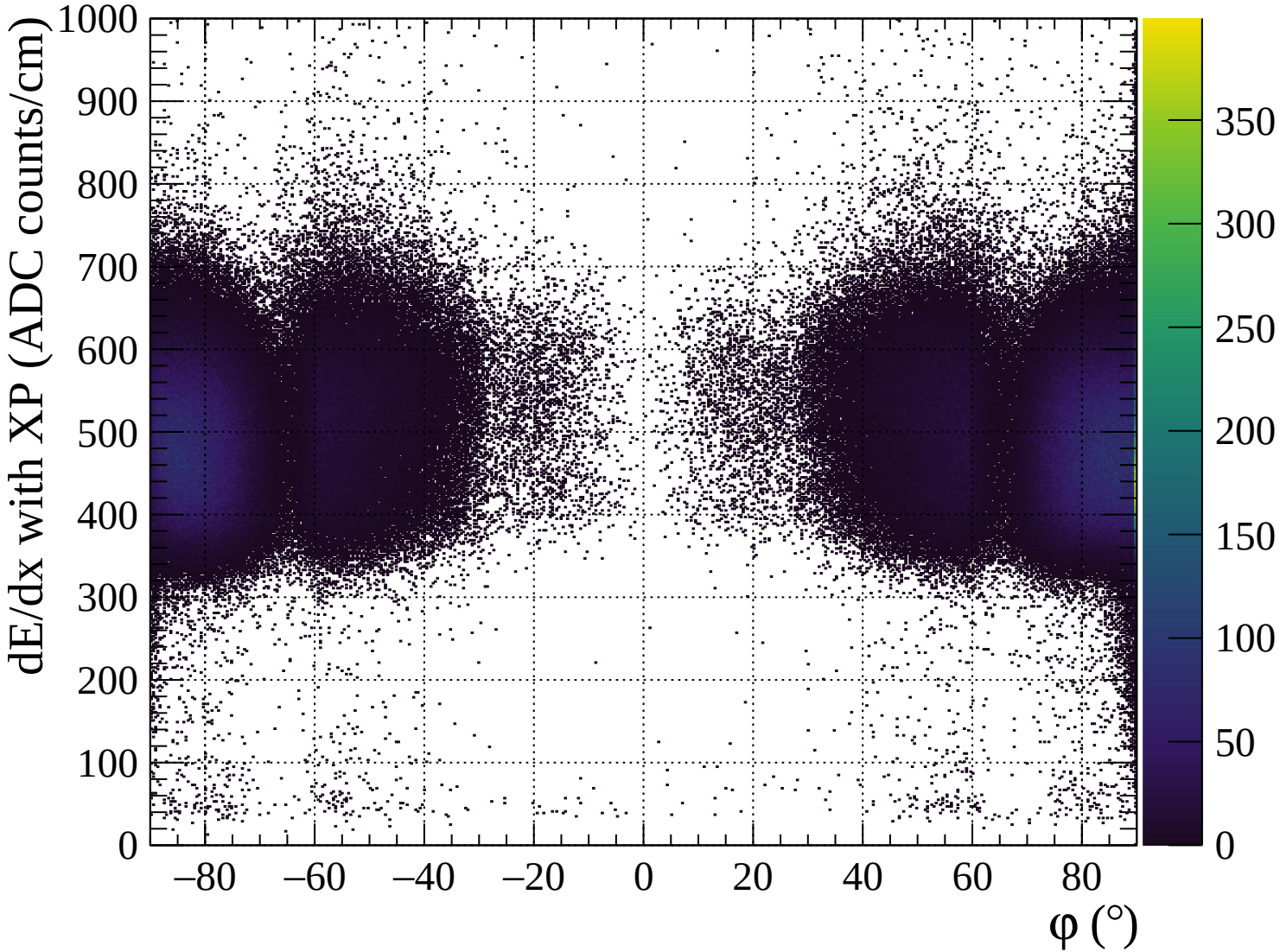


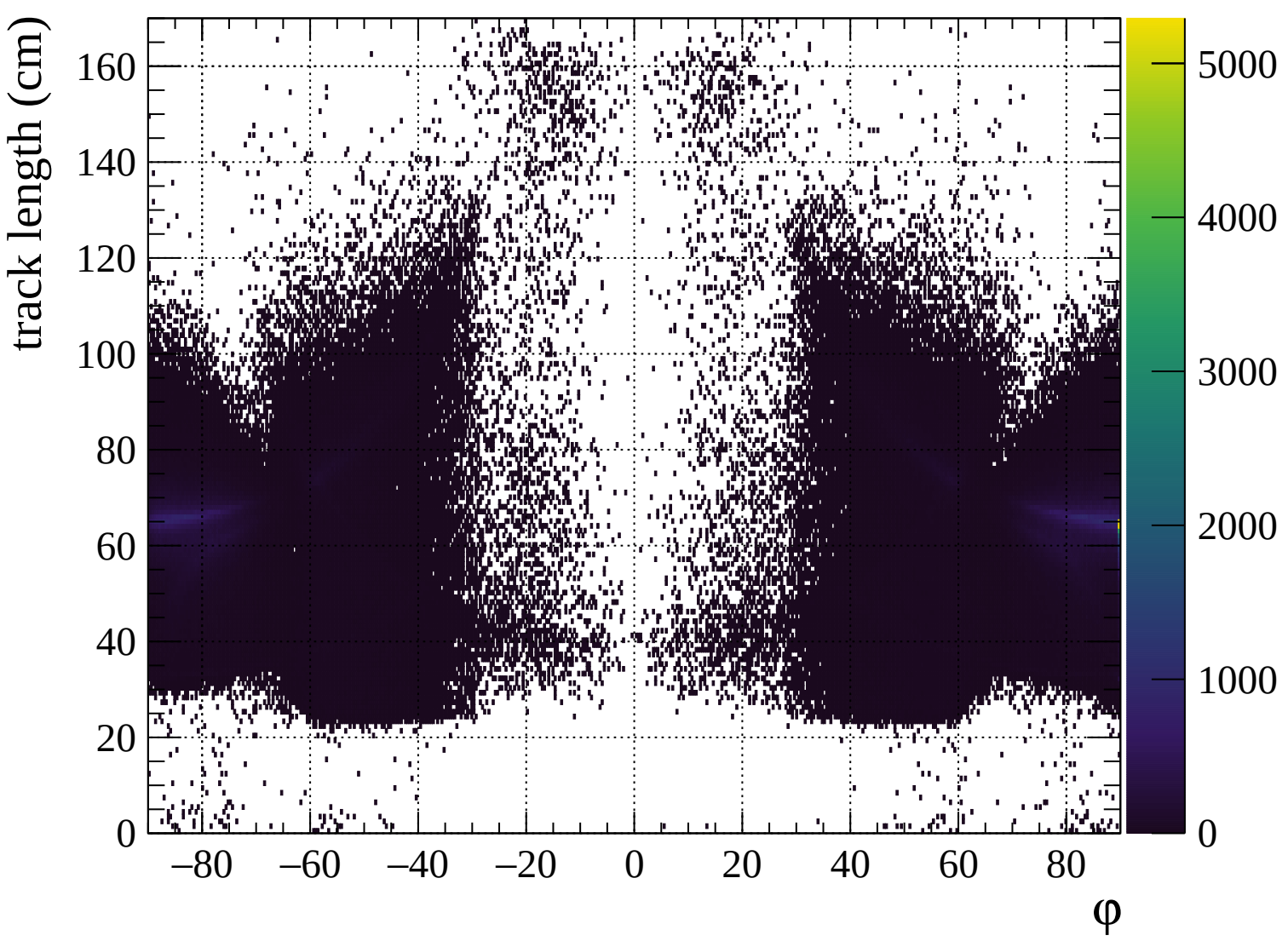


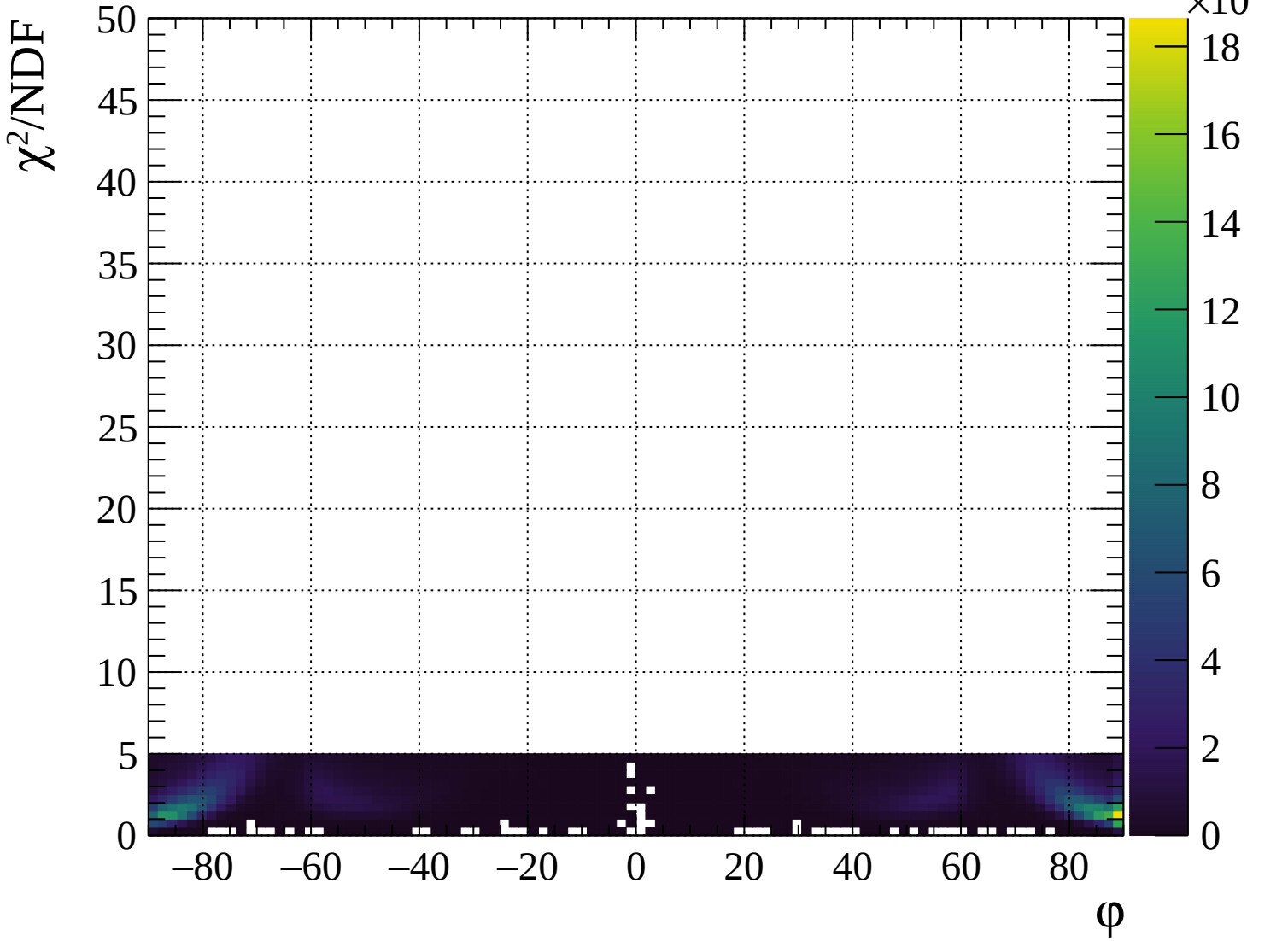


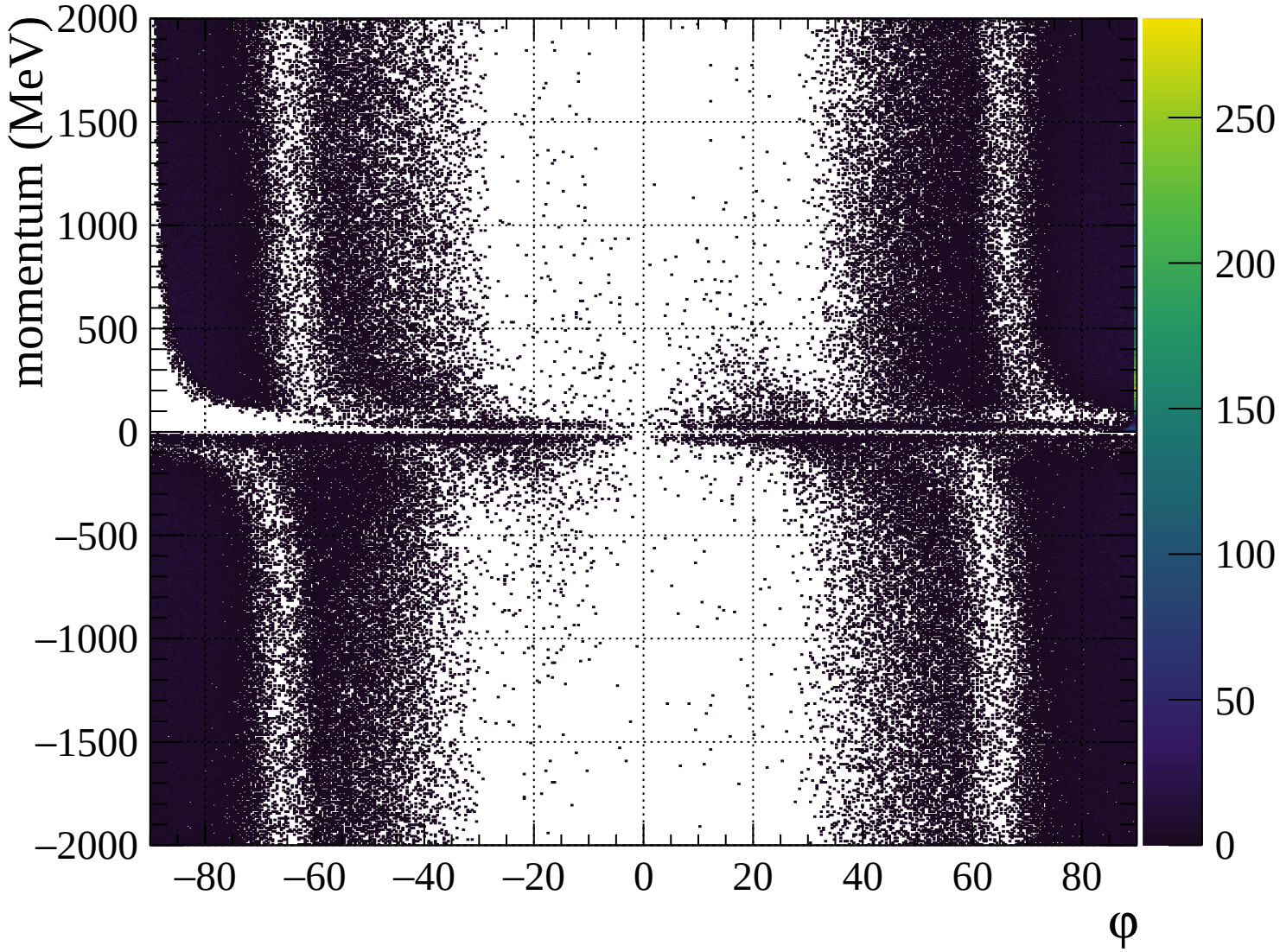


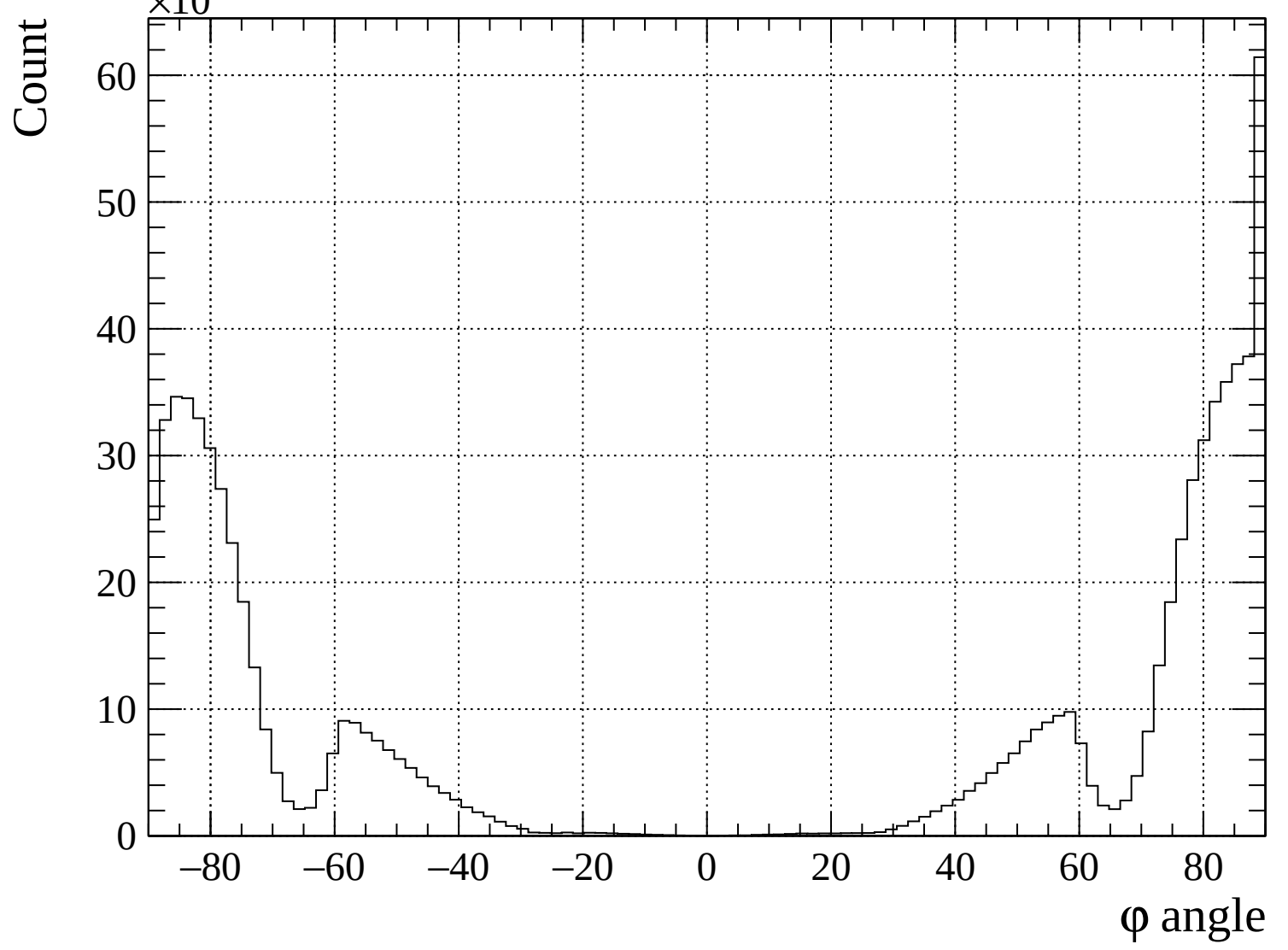


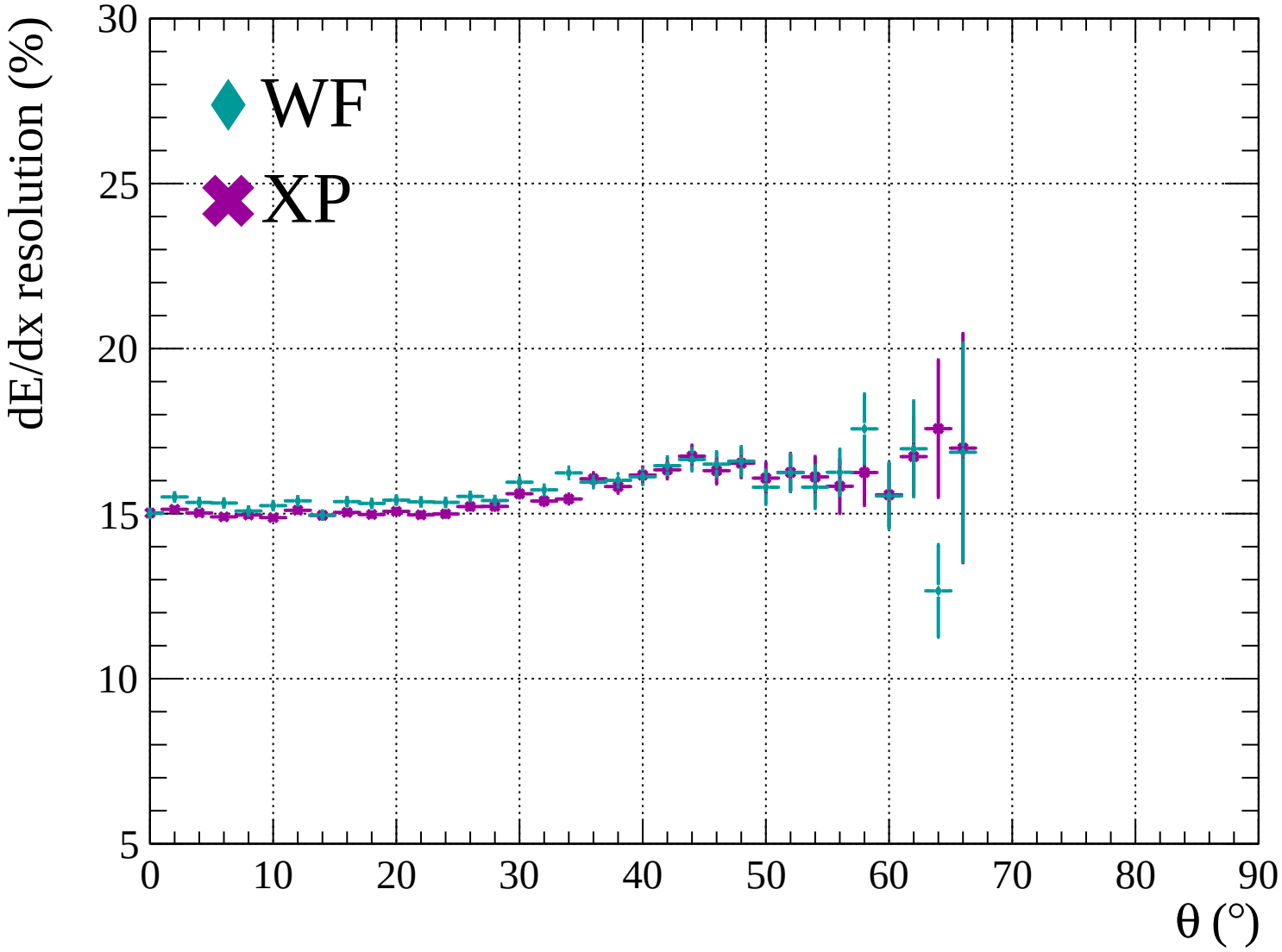


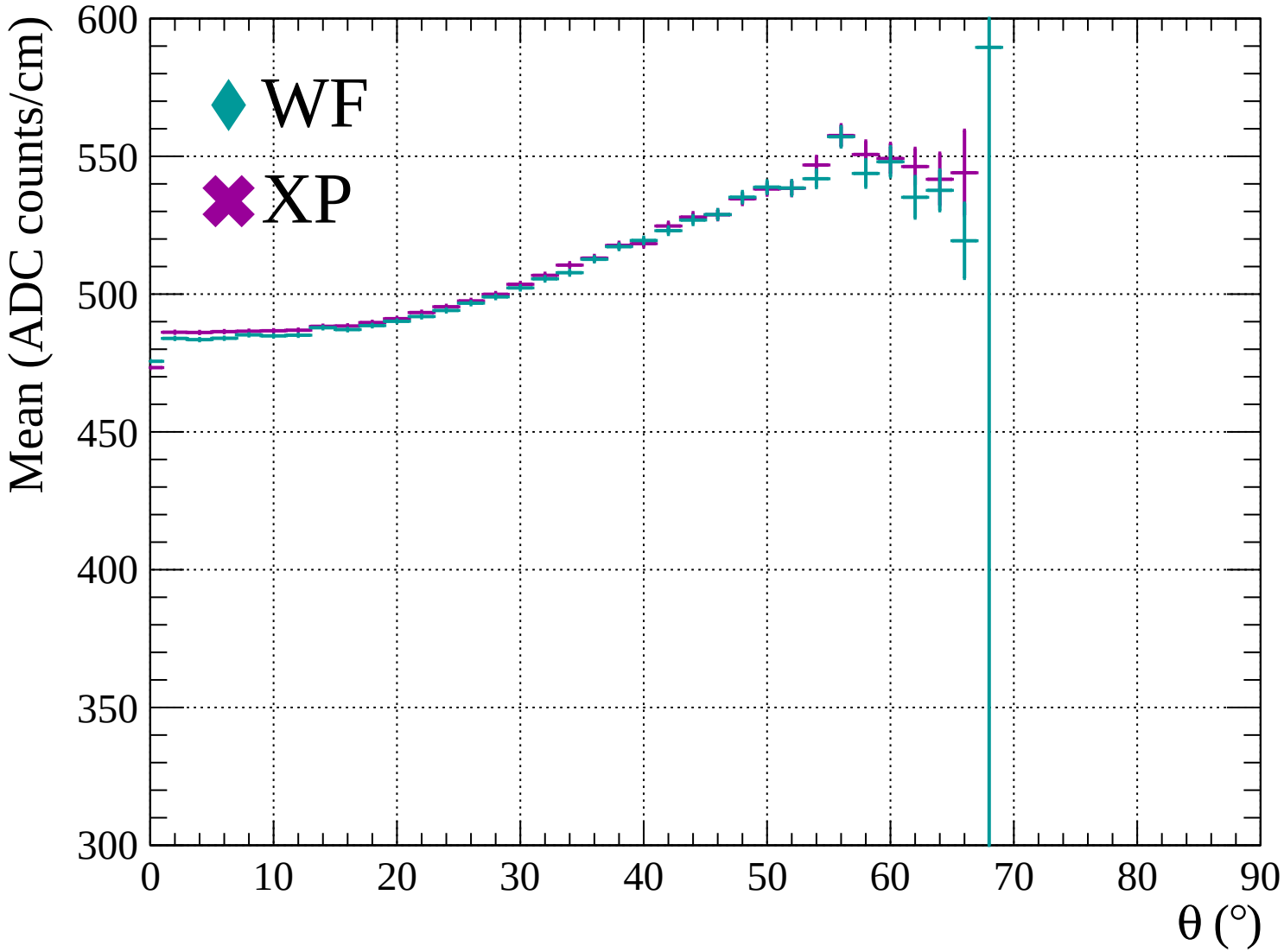


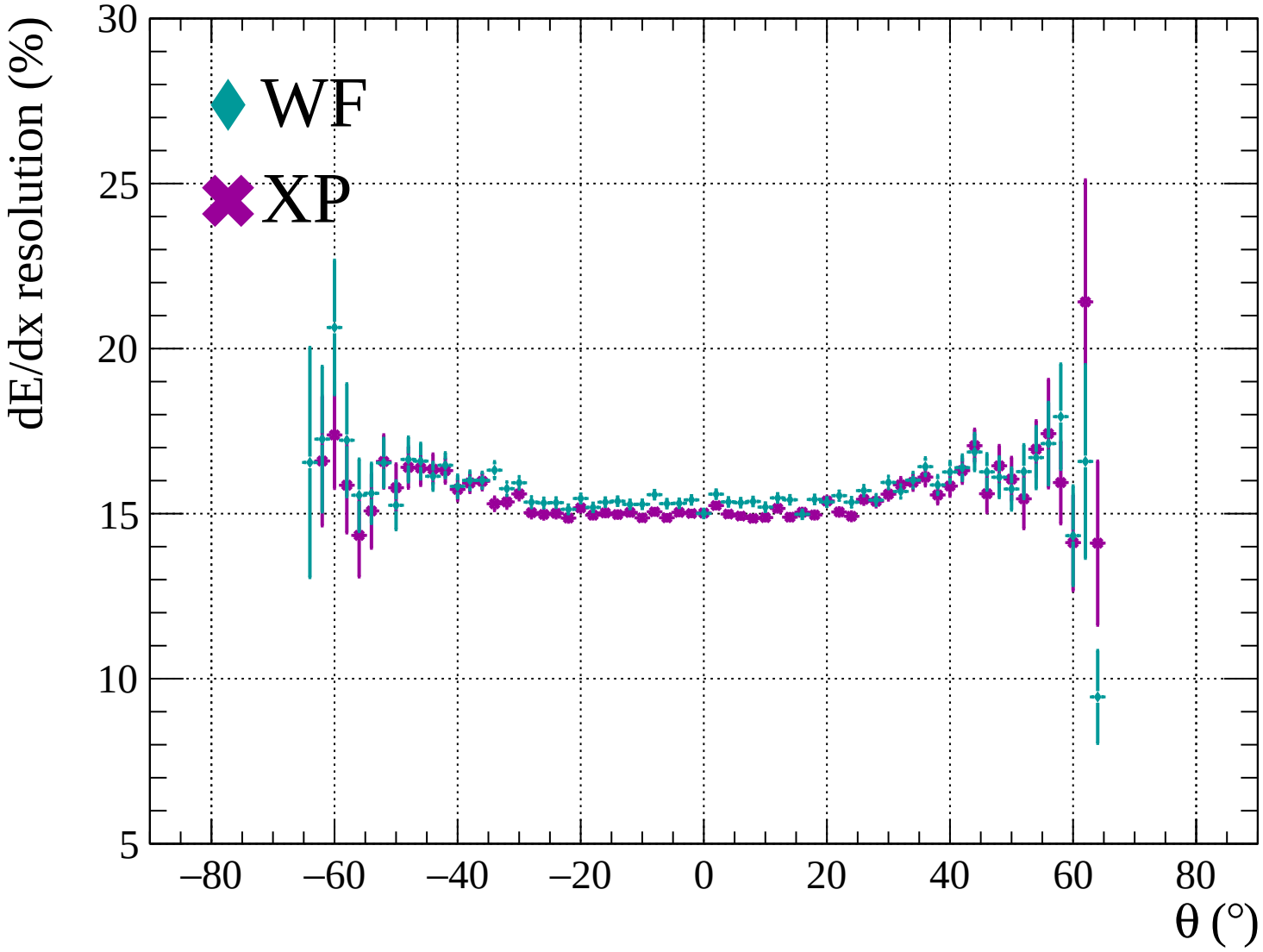


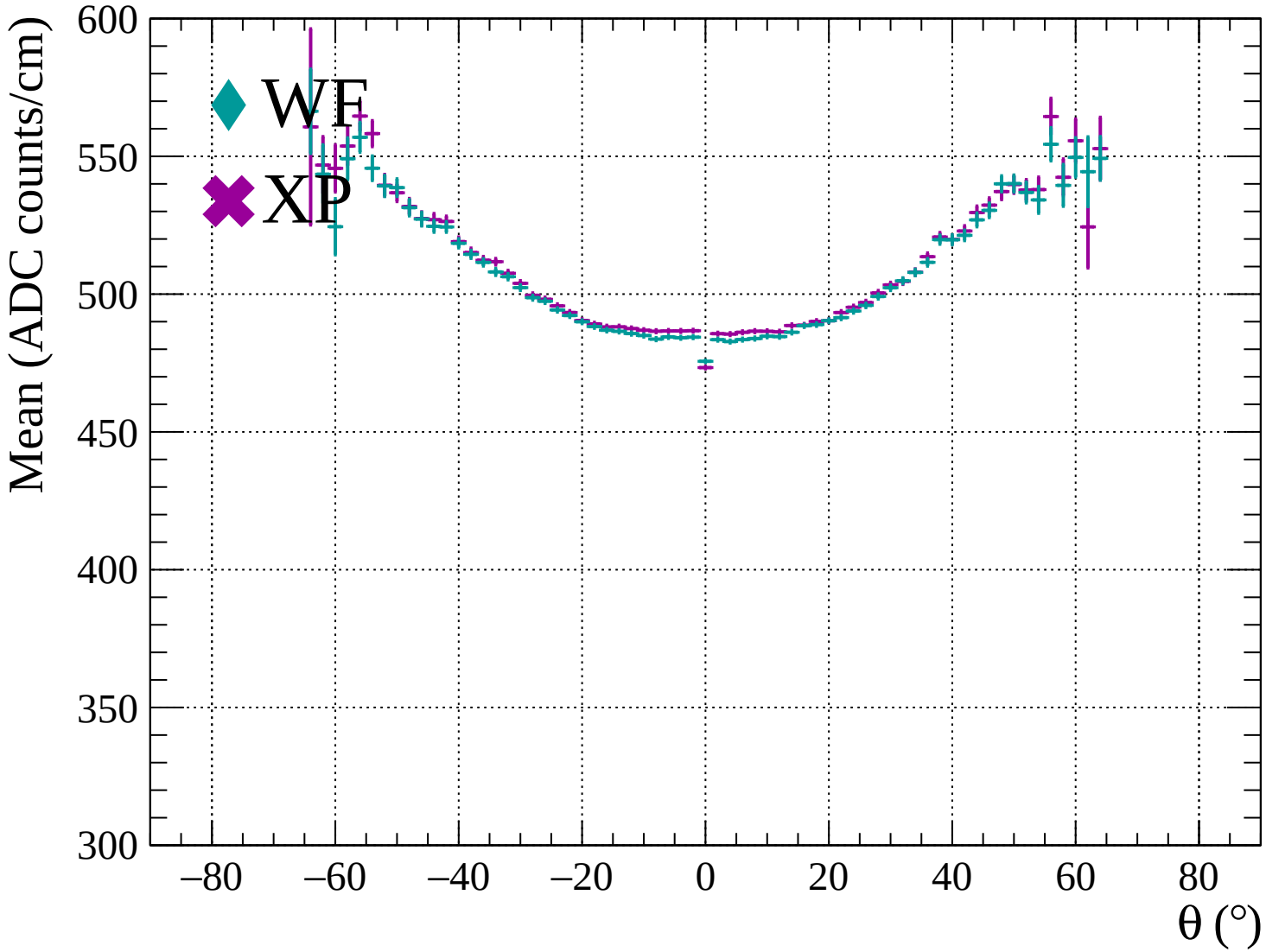


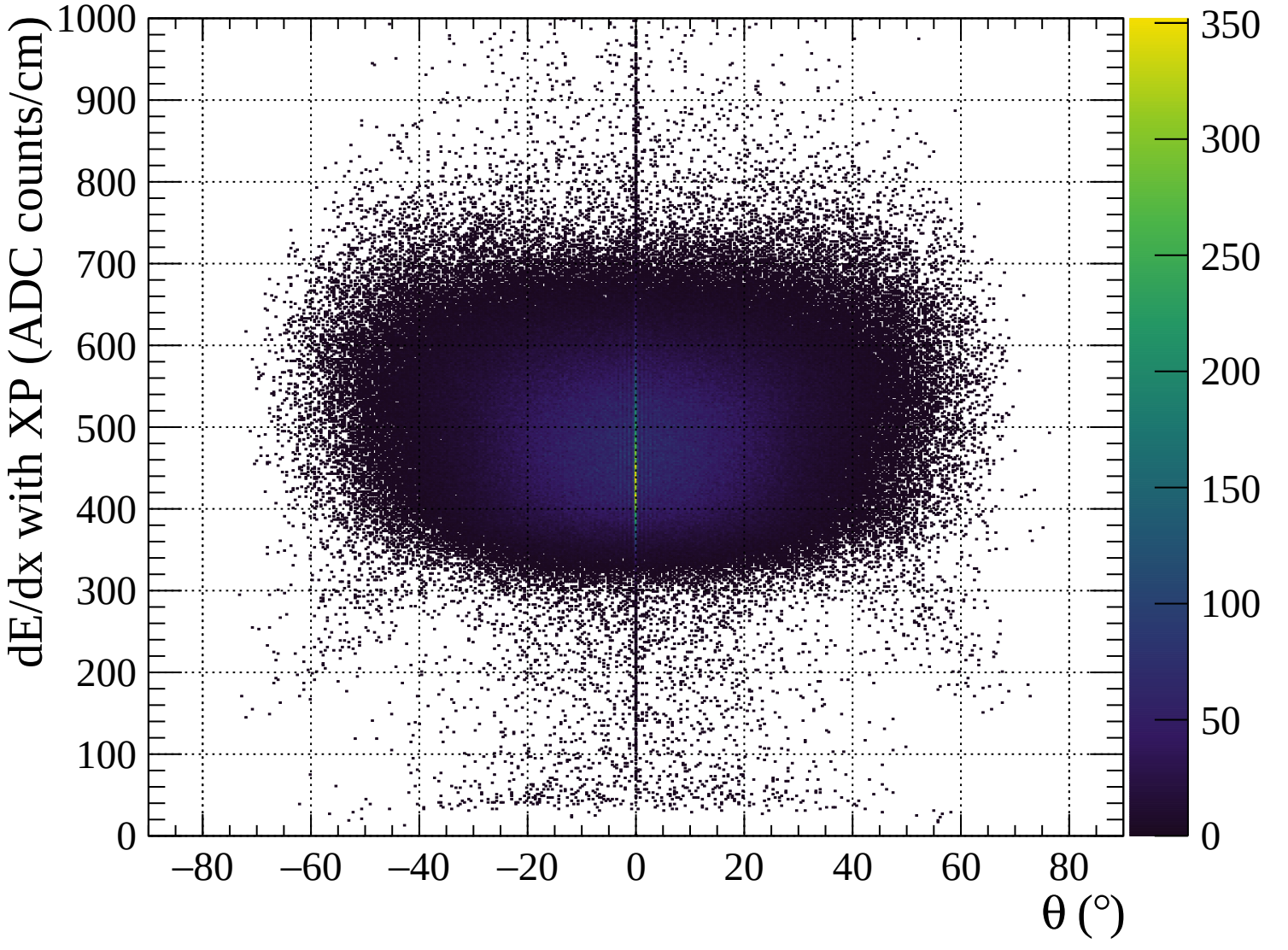


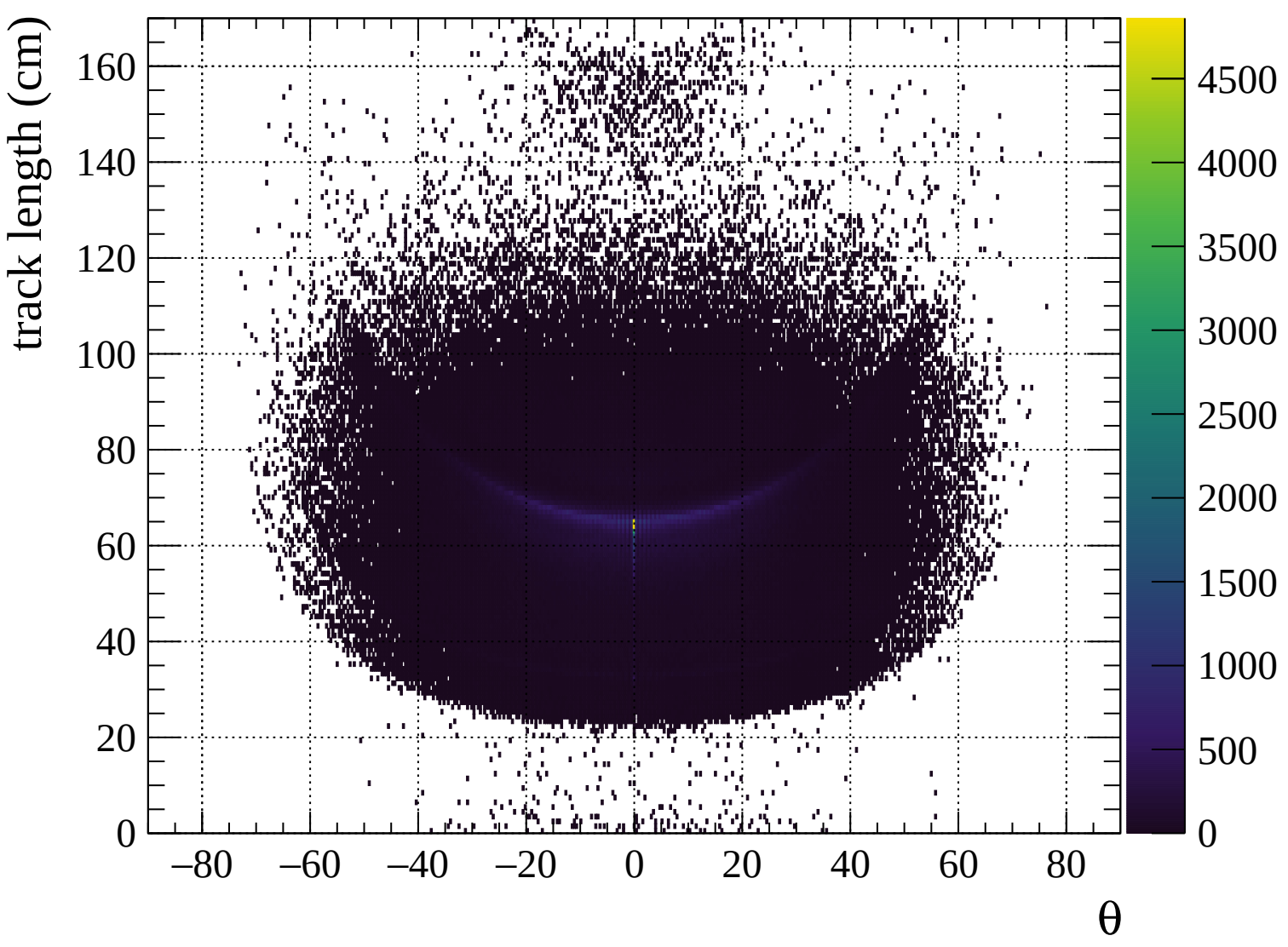


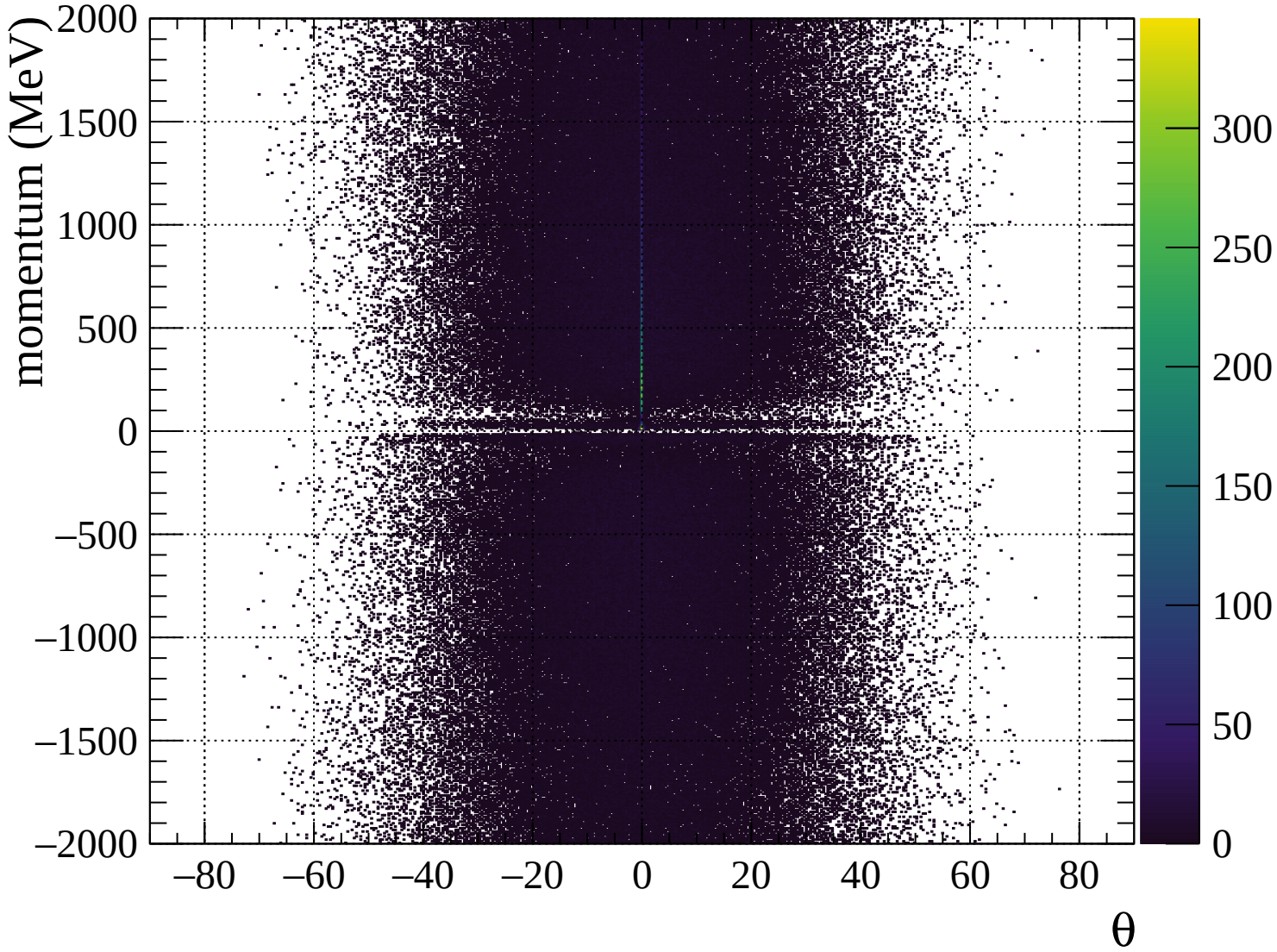


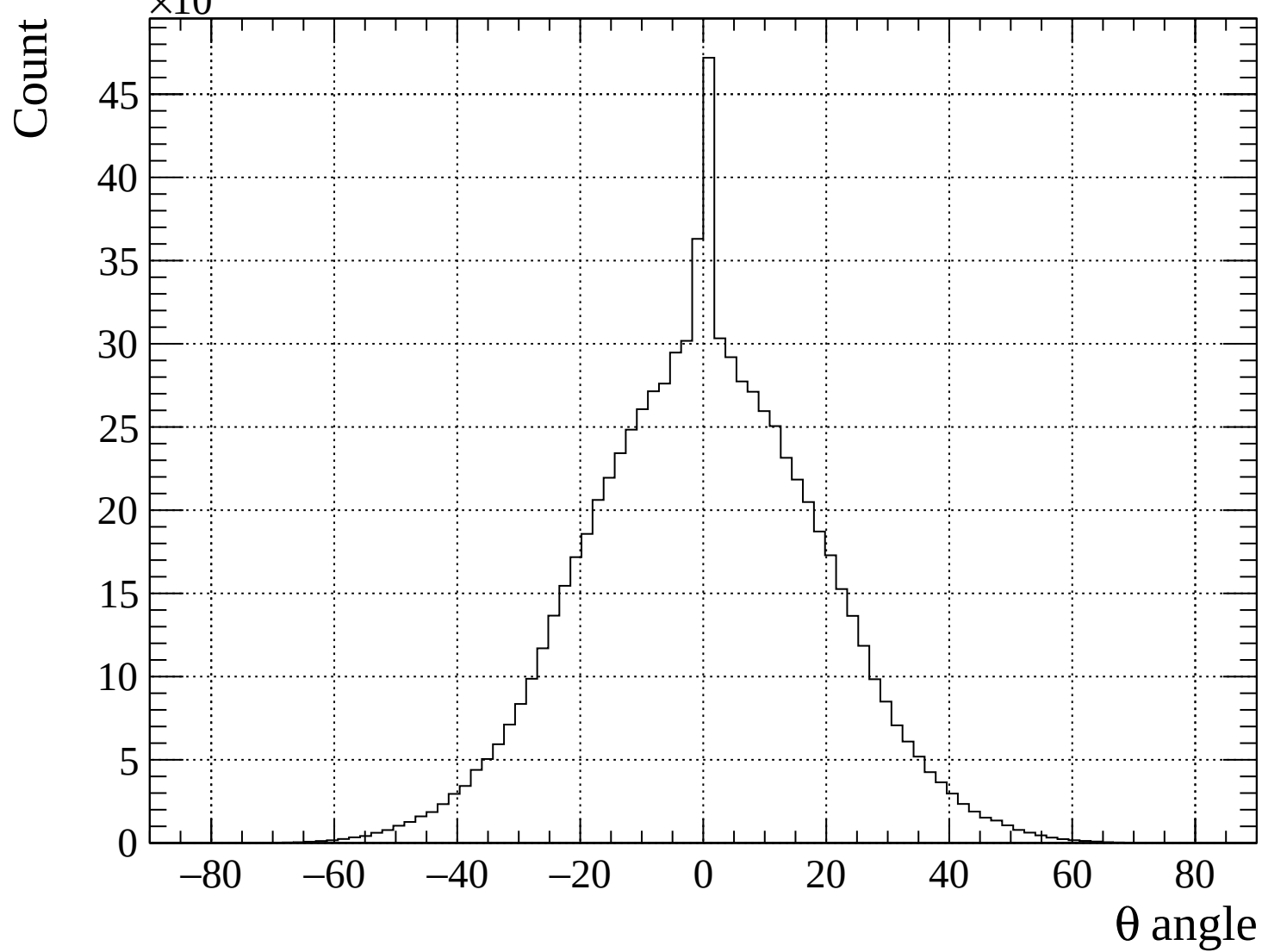


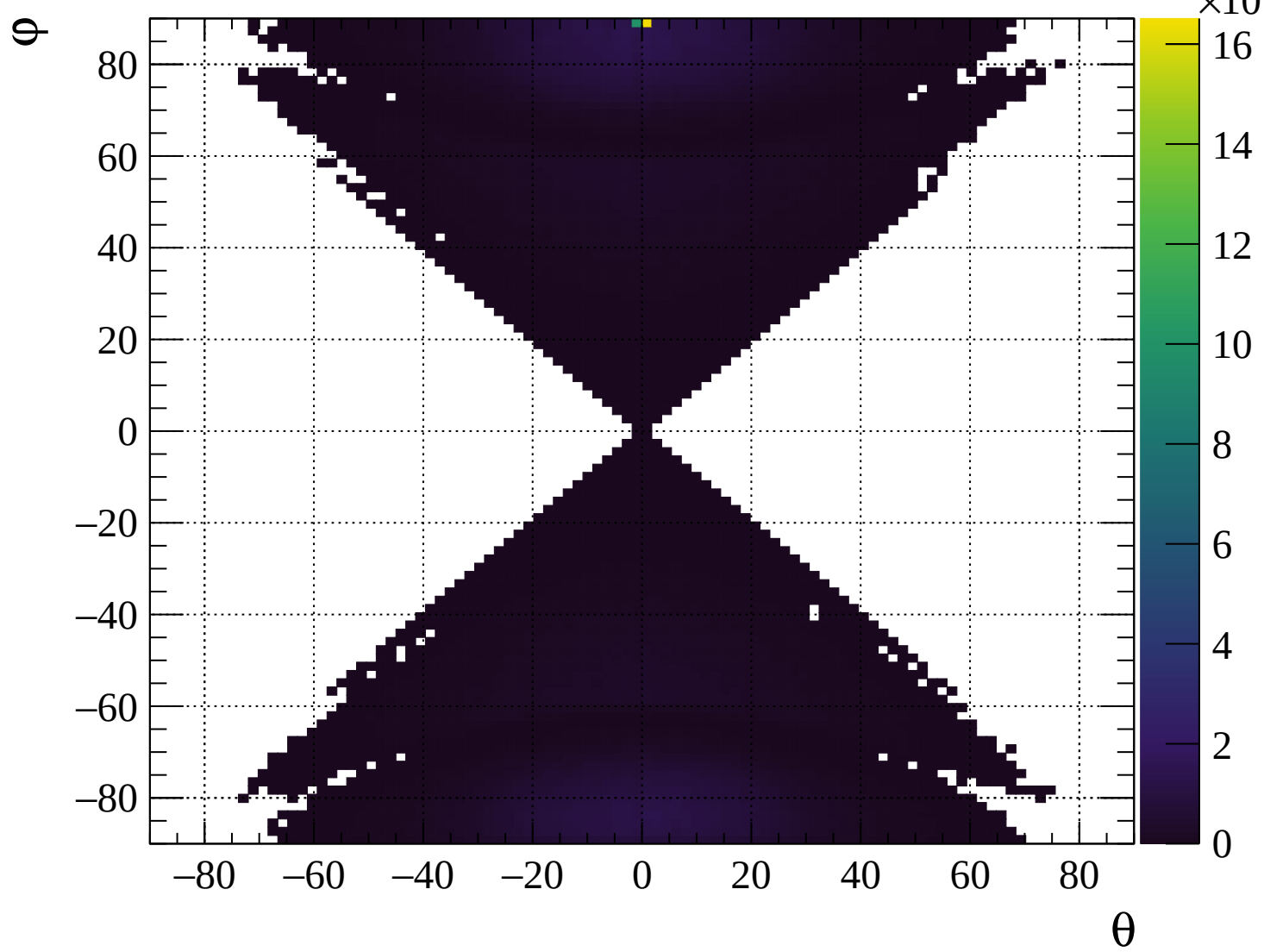


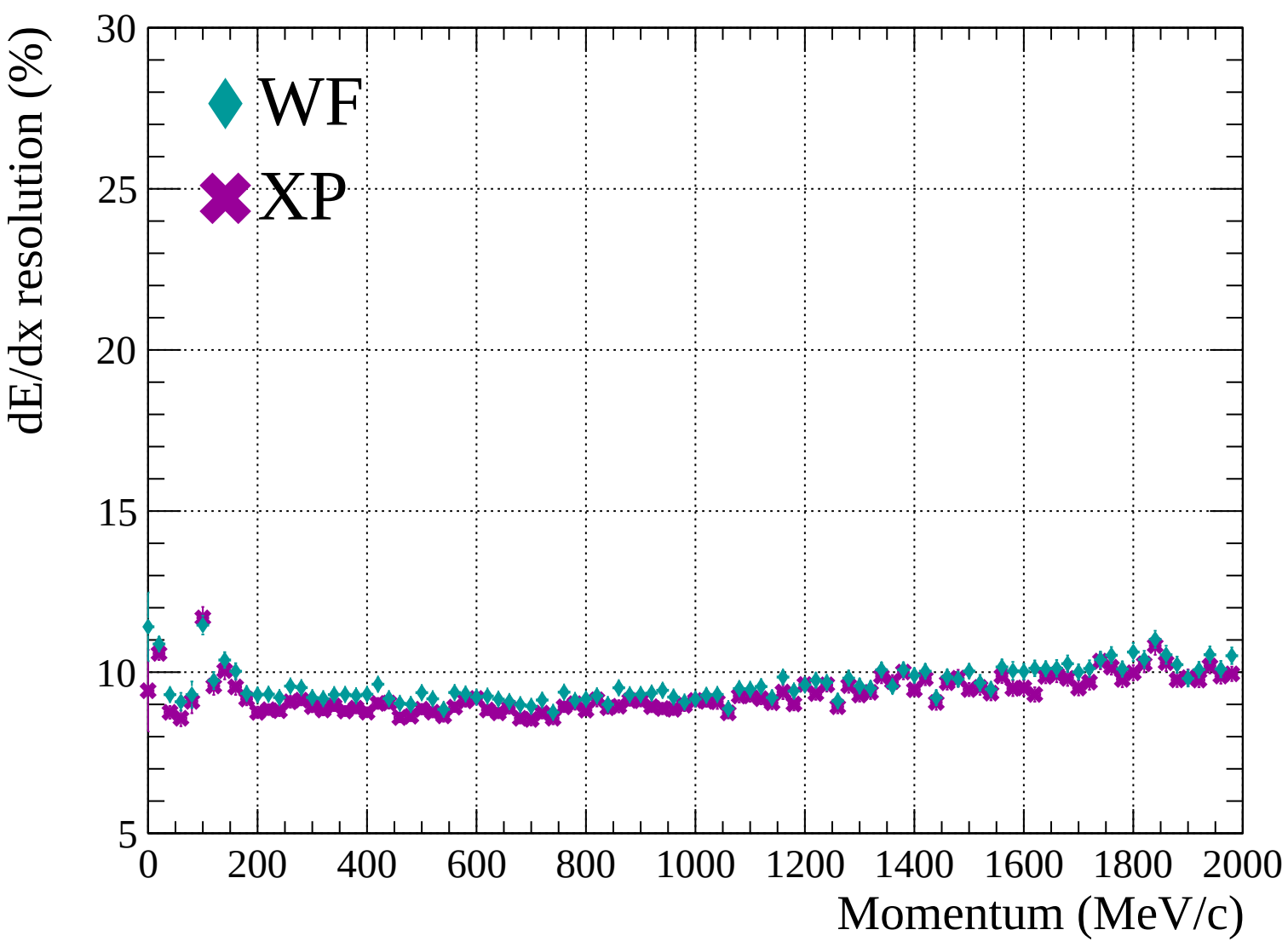


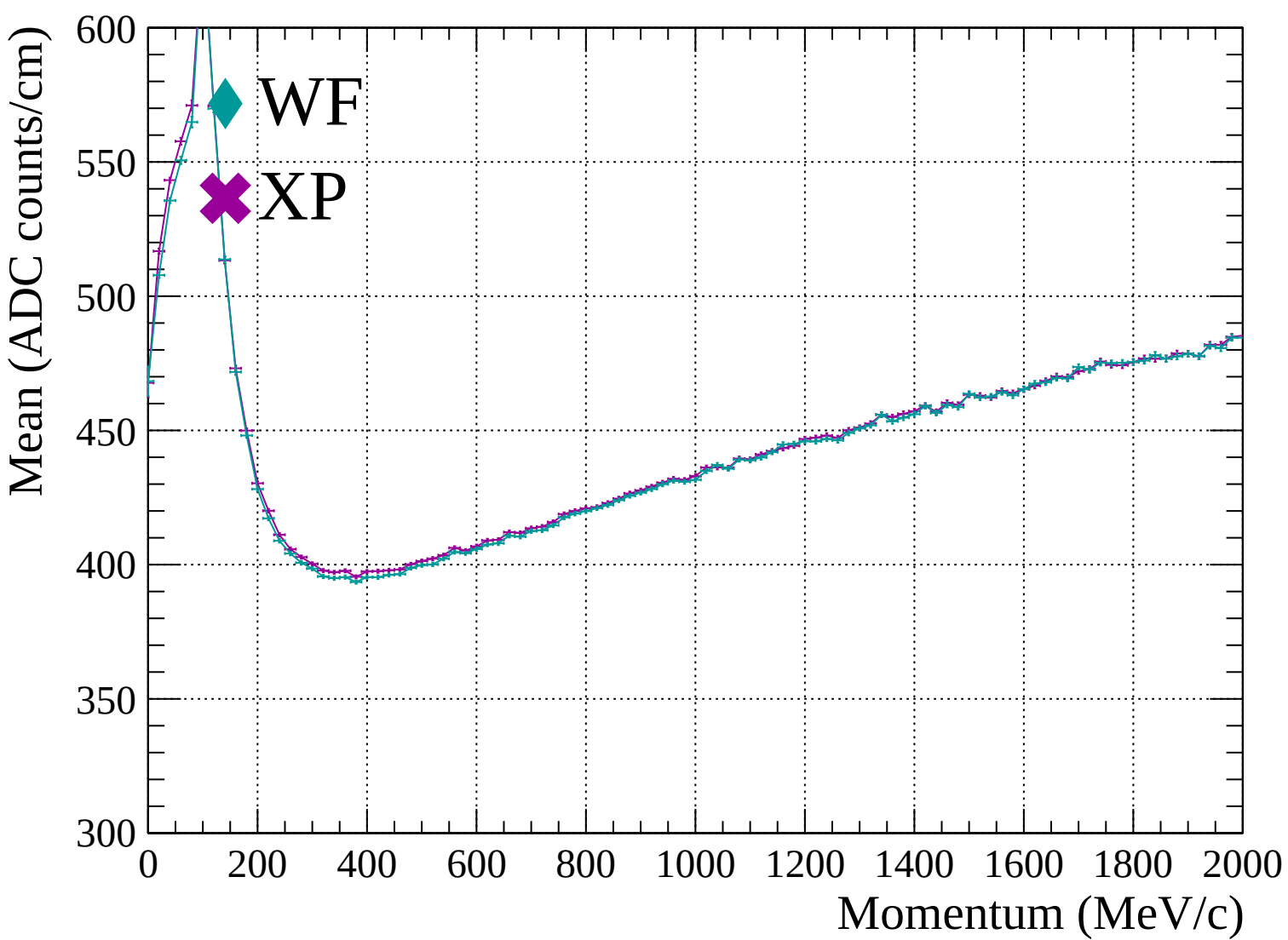


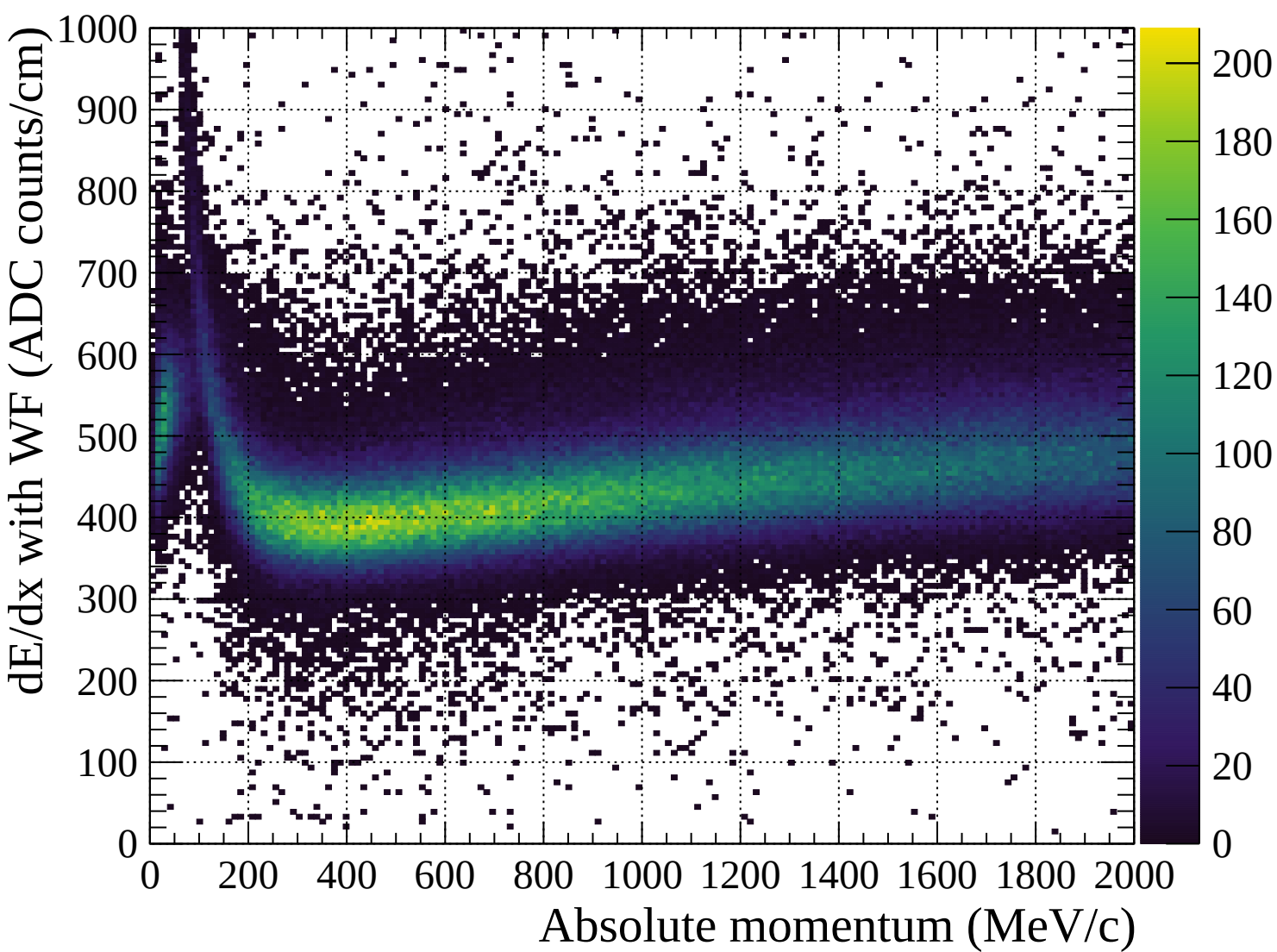


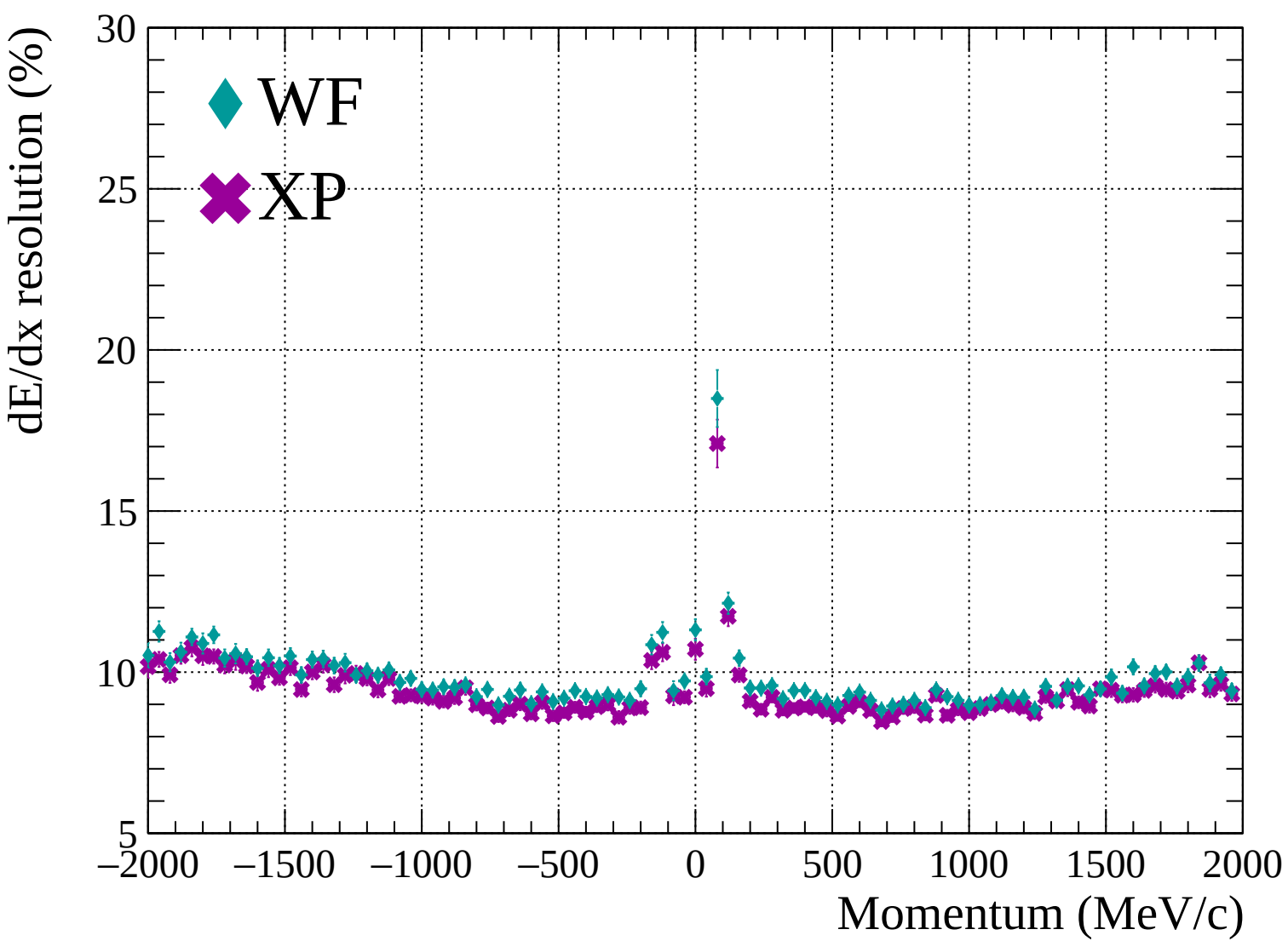


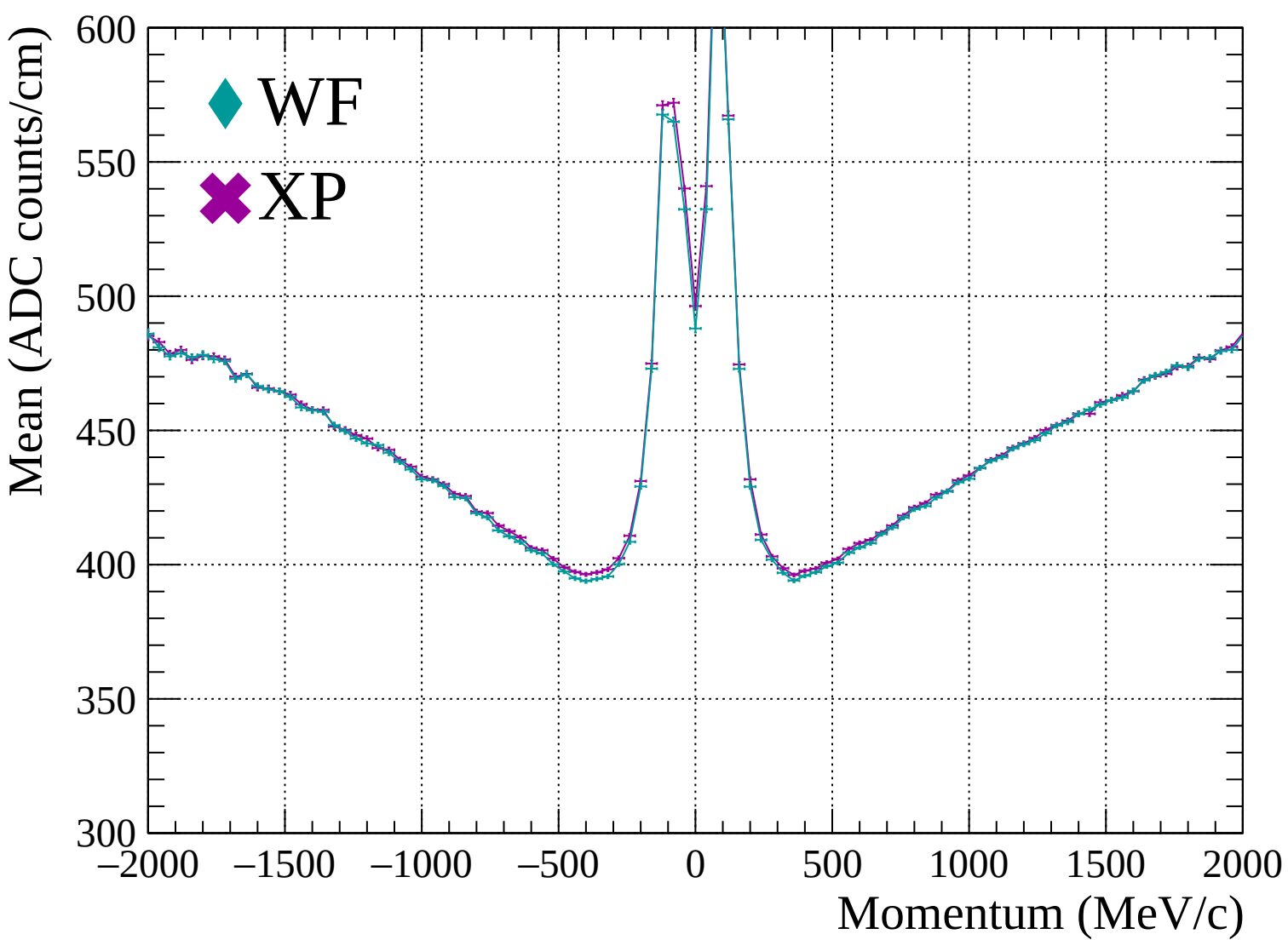


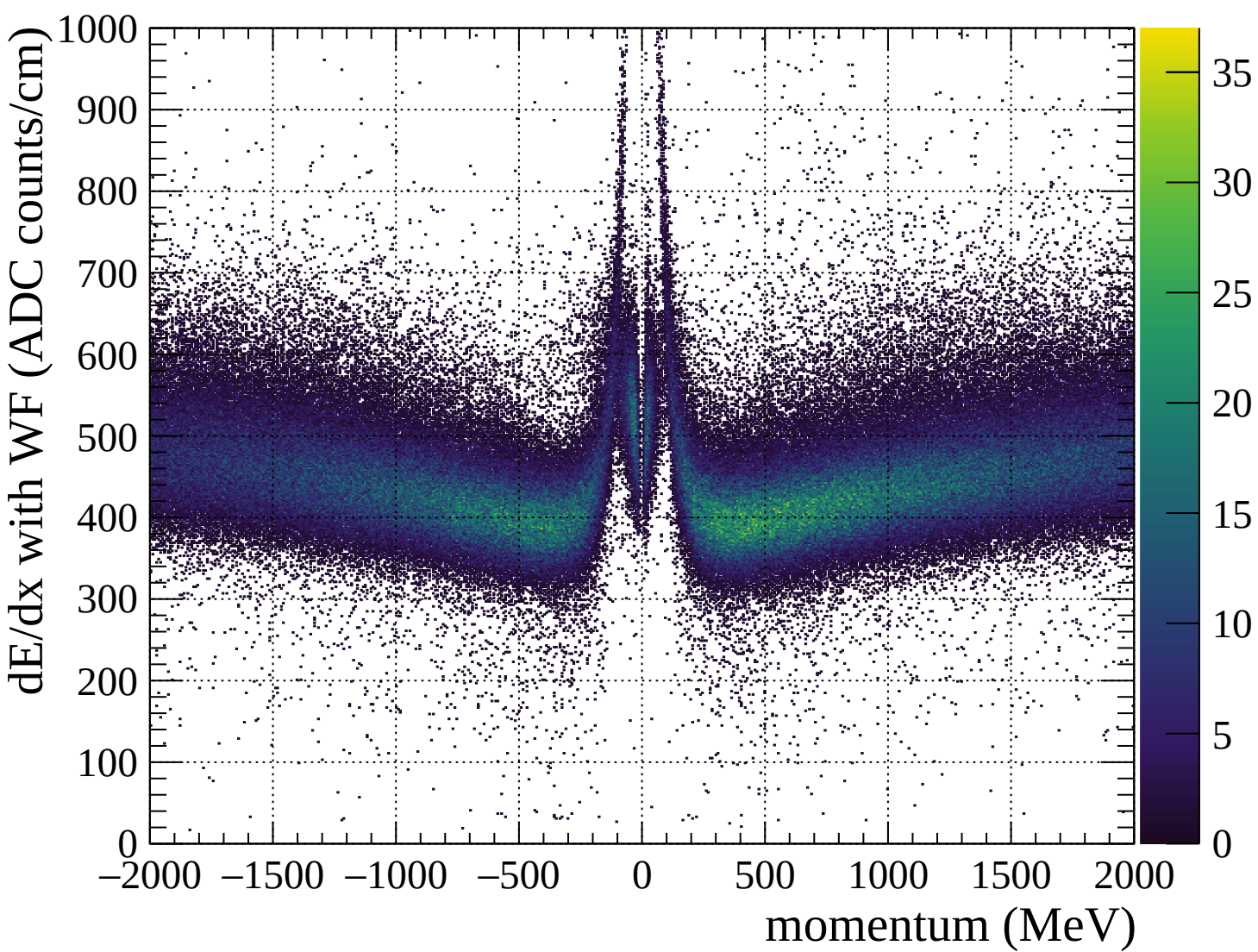


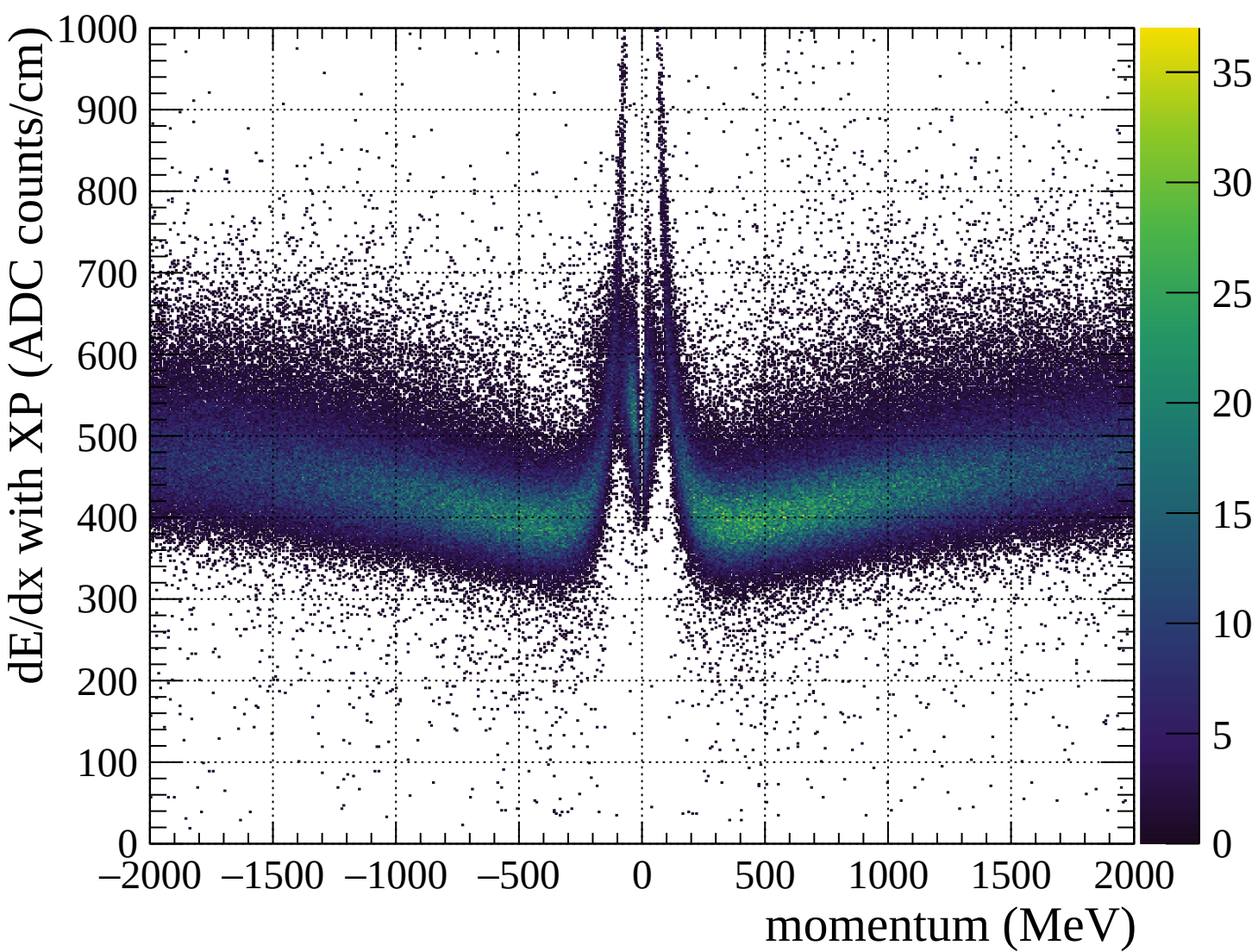


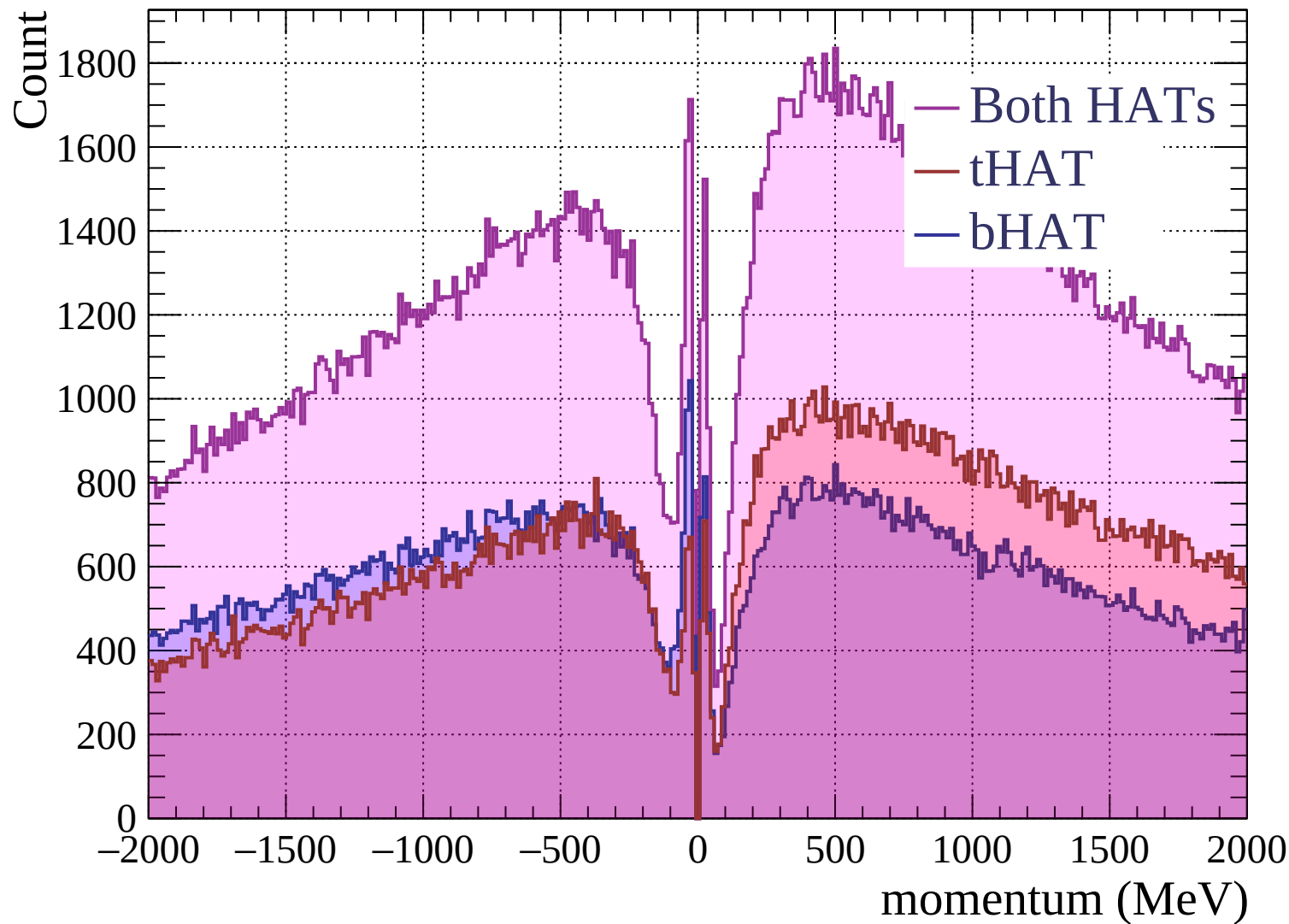


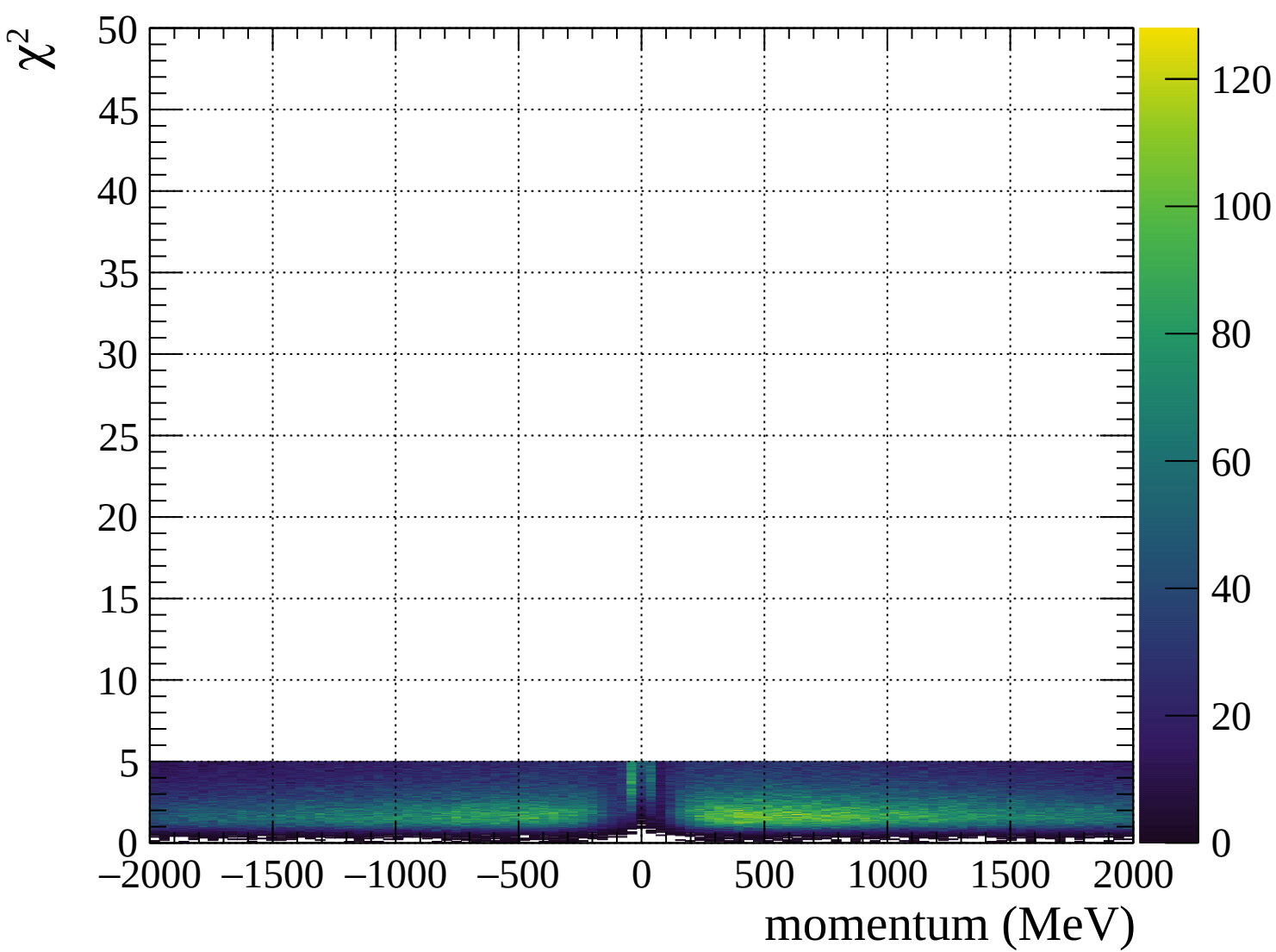


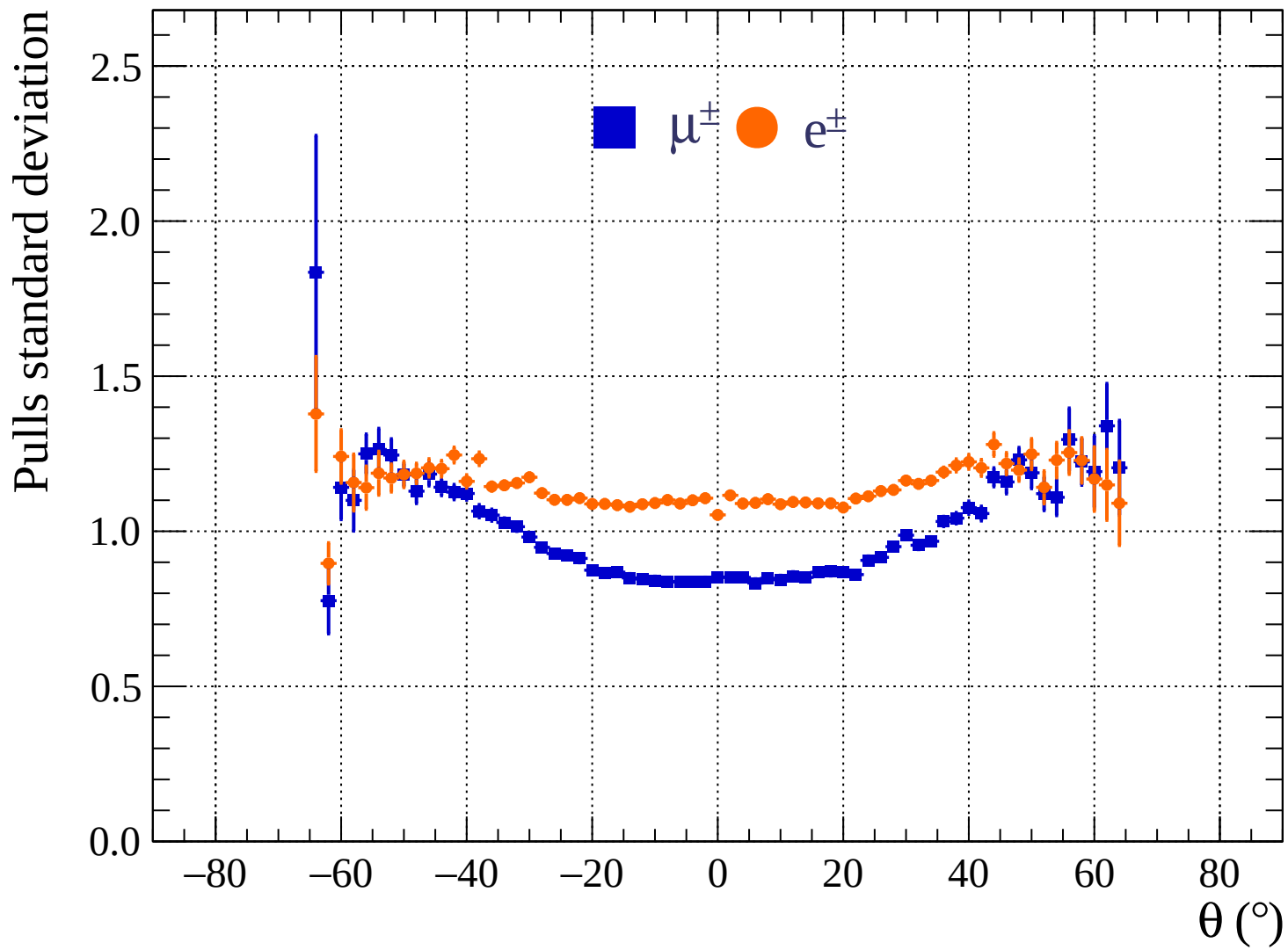


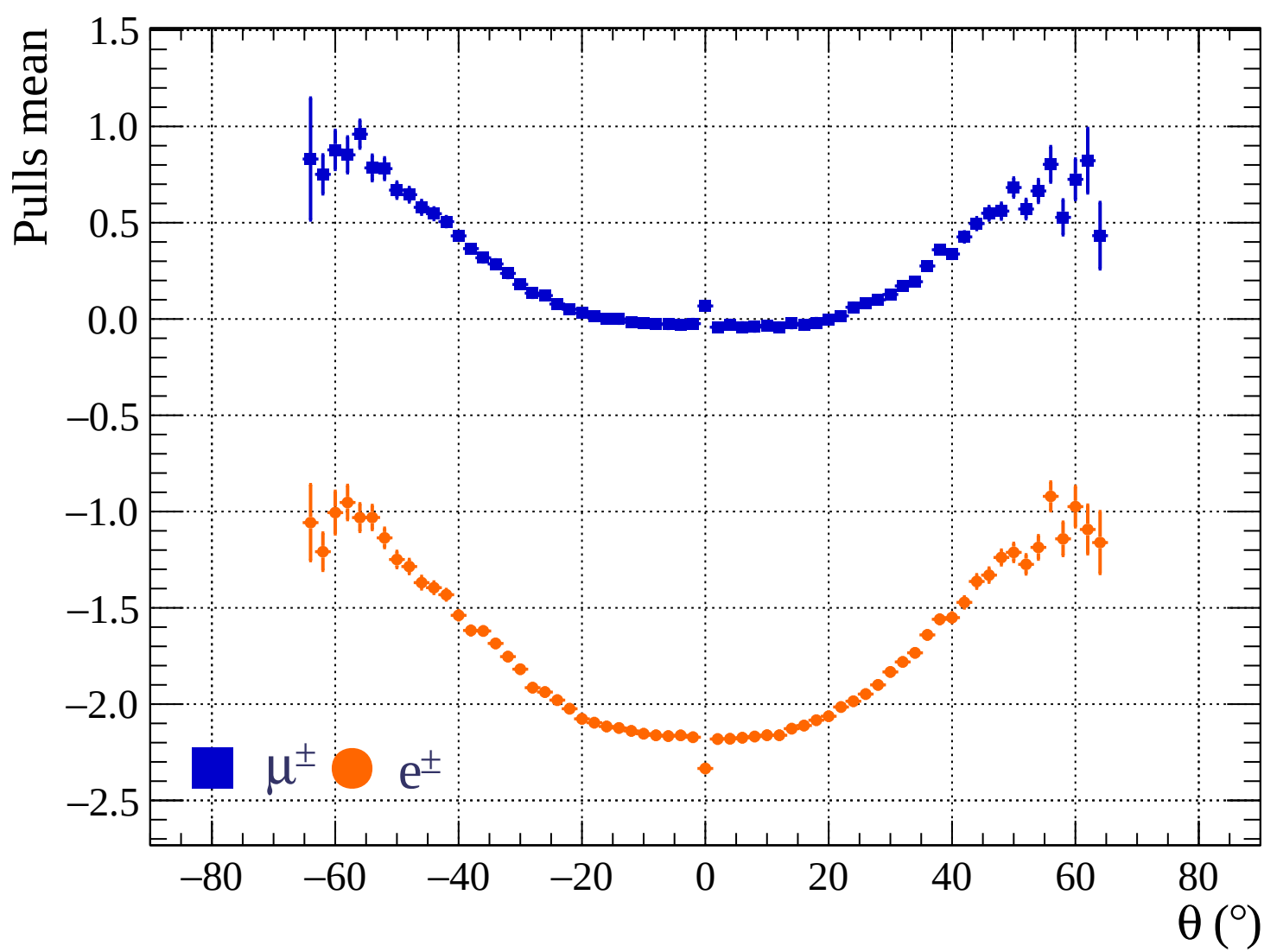




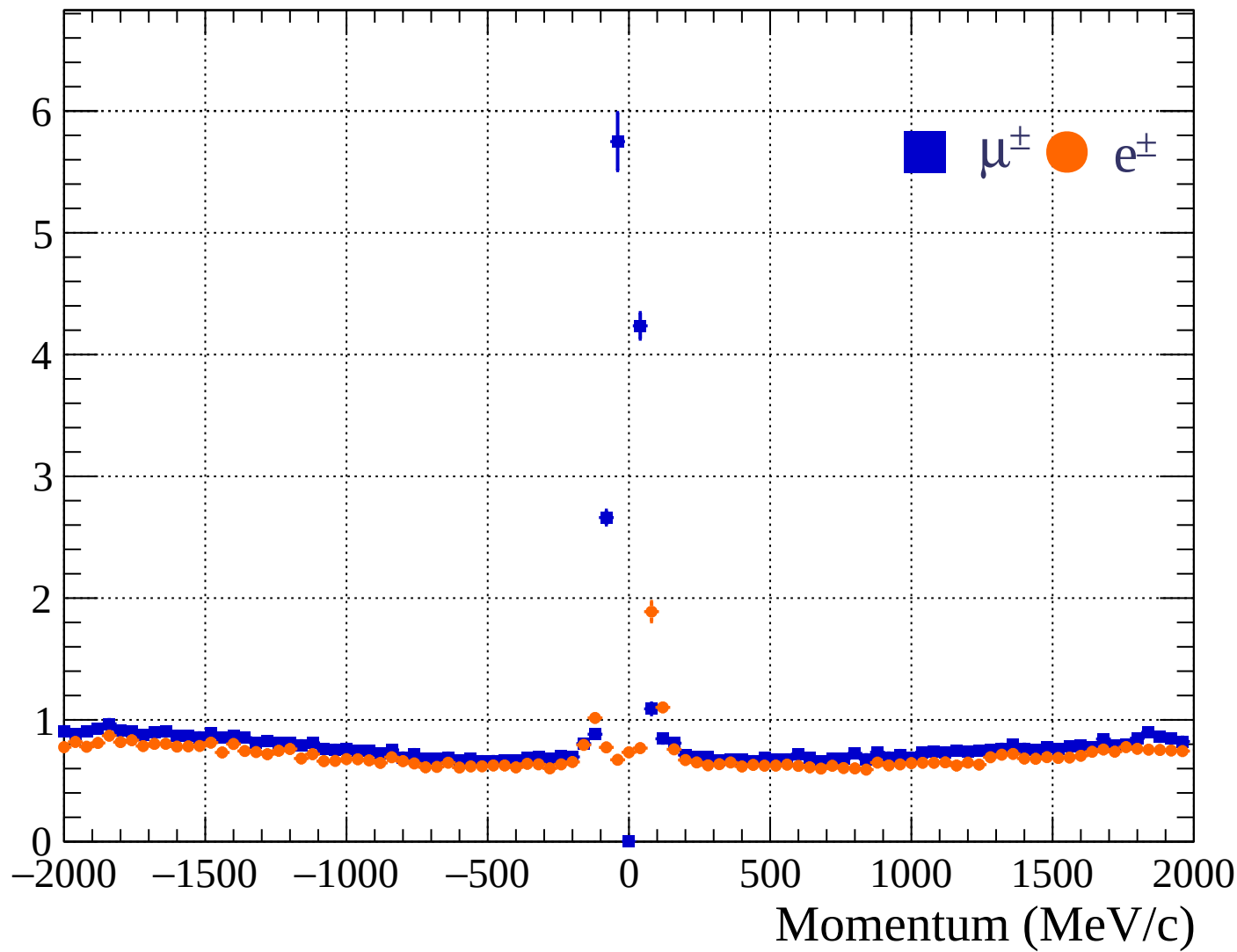




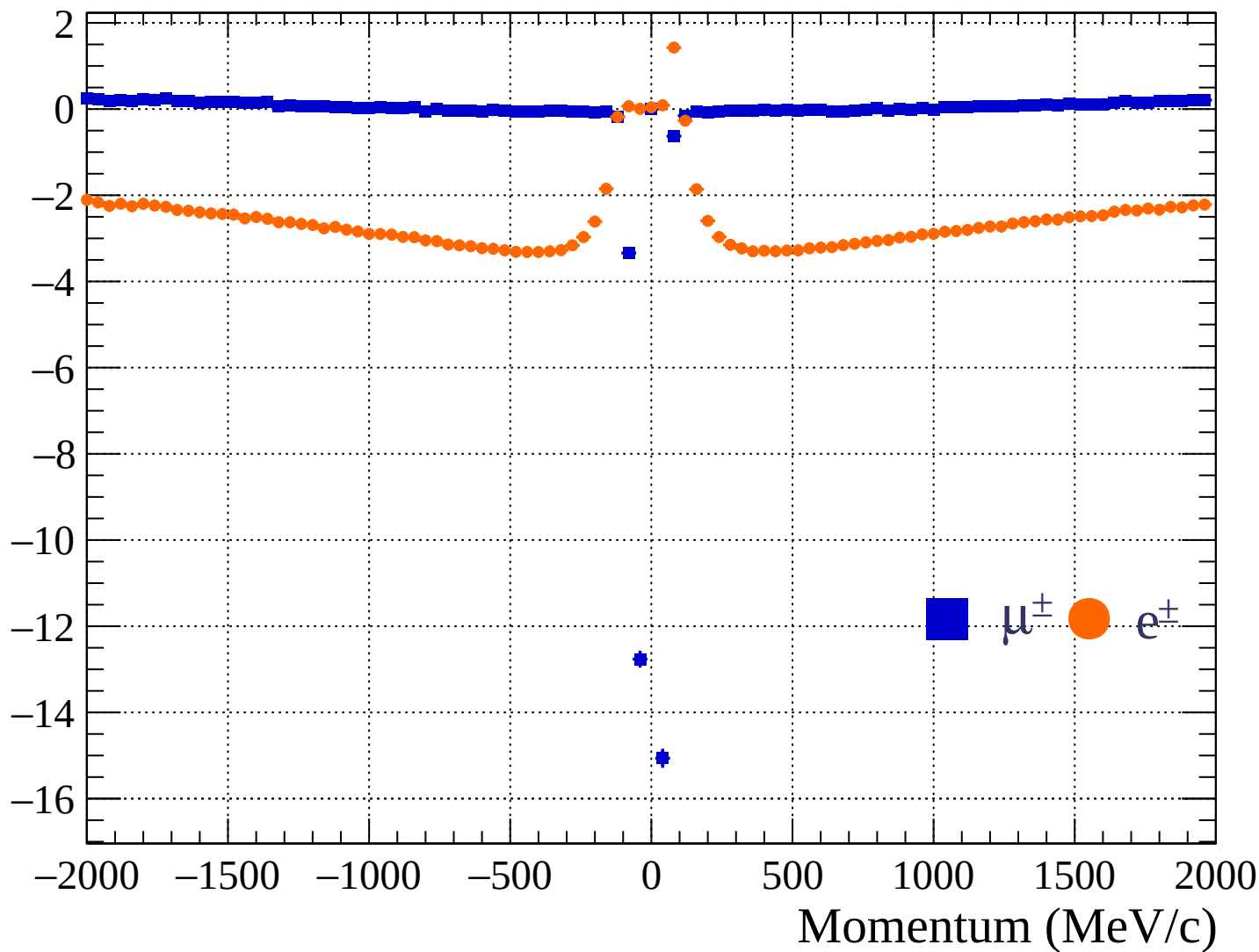


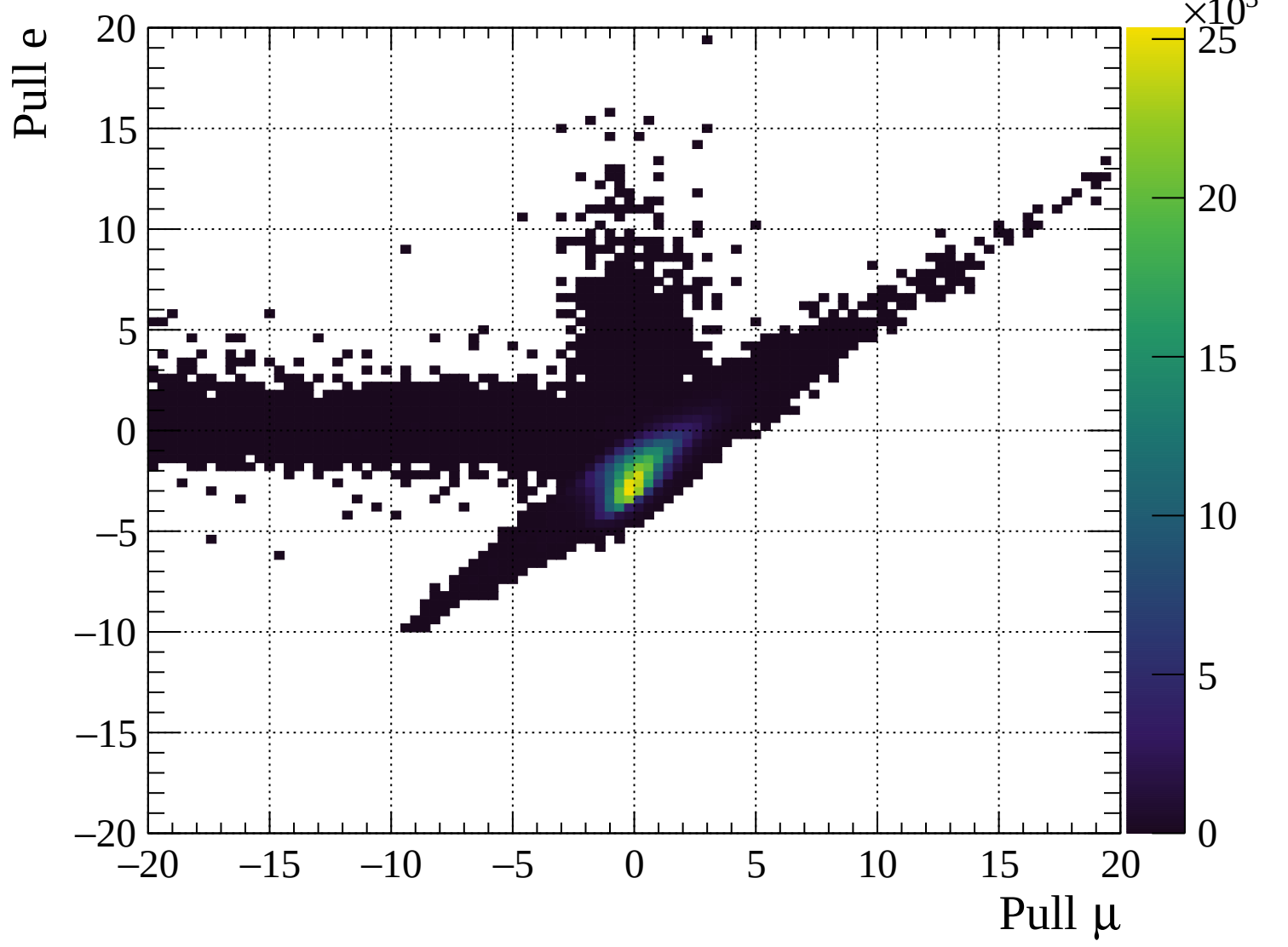


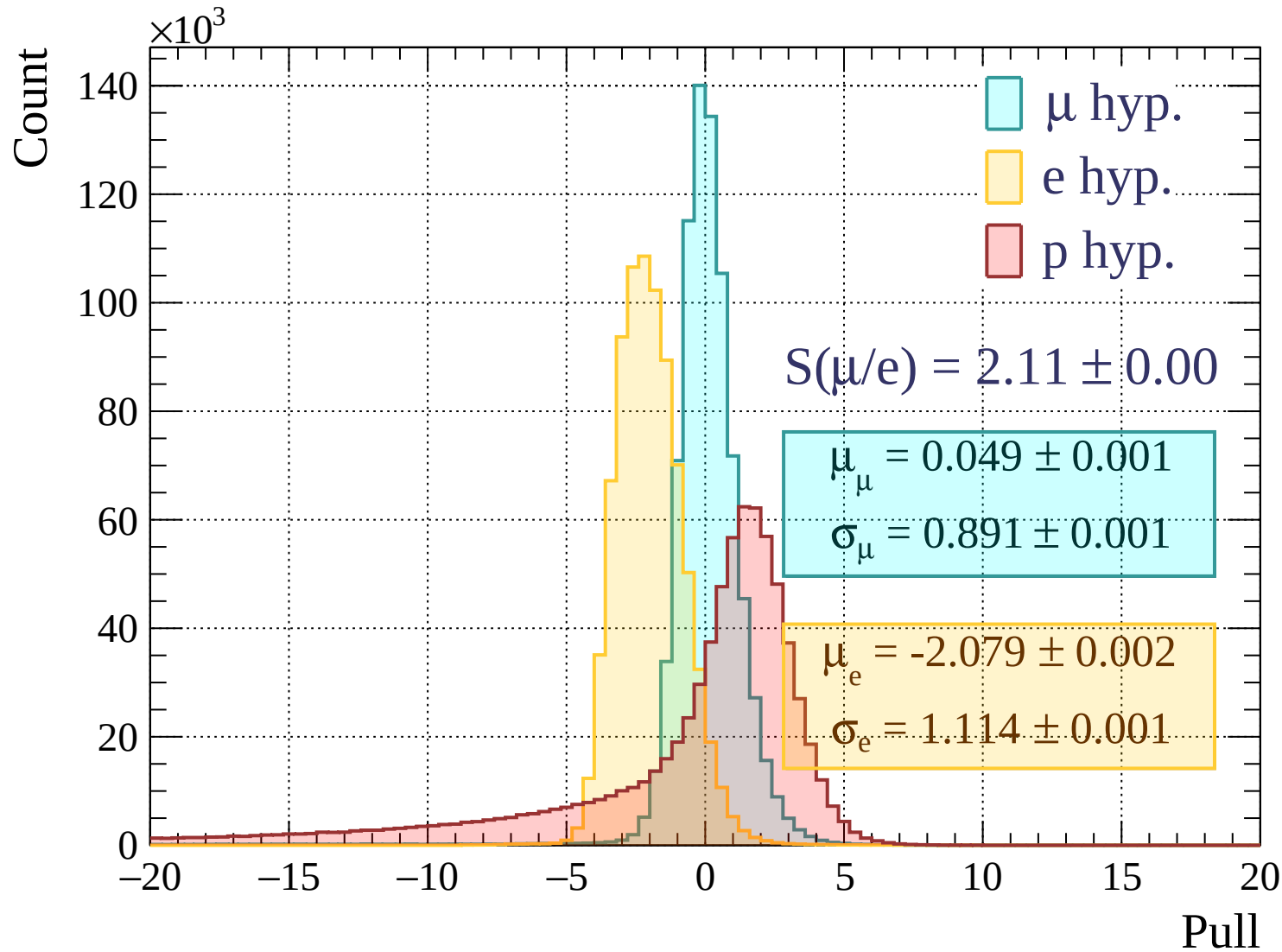
Pulls standard deviation

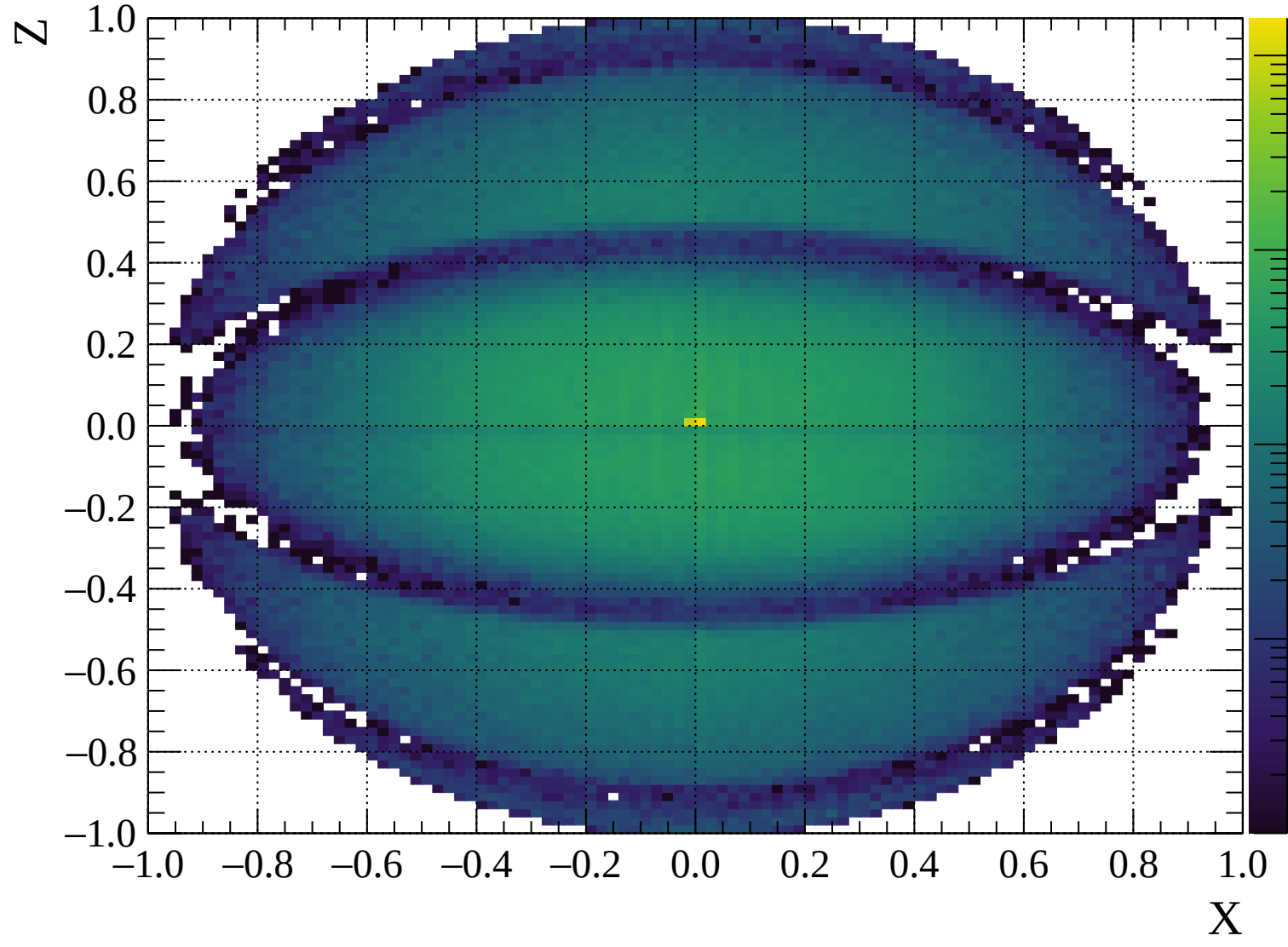


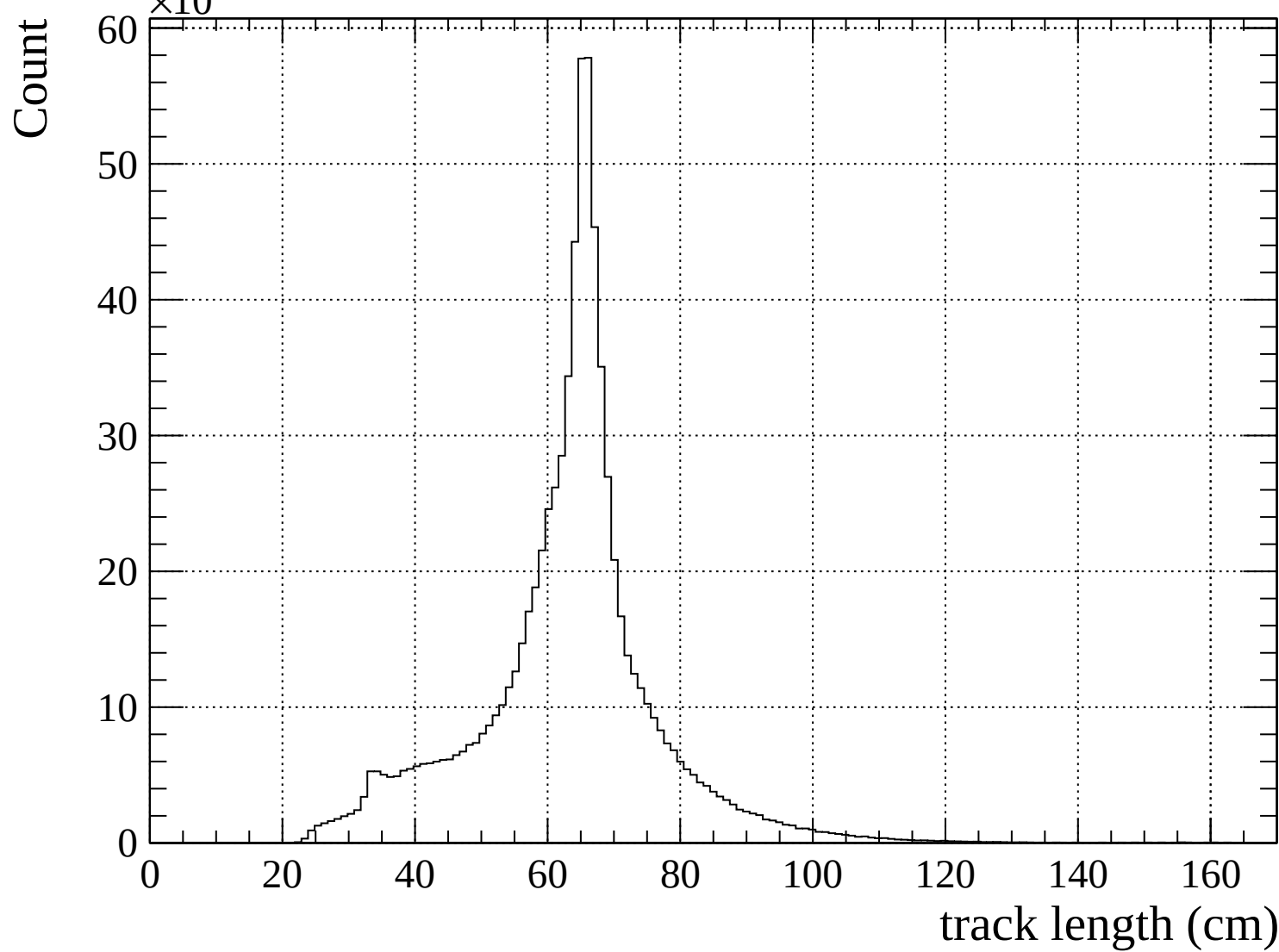
Pulls mean

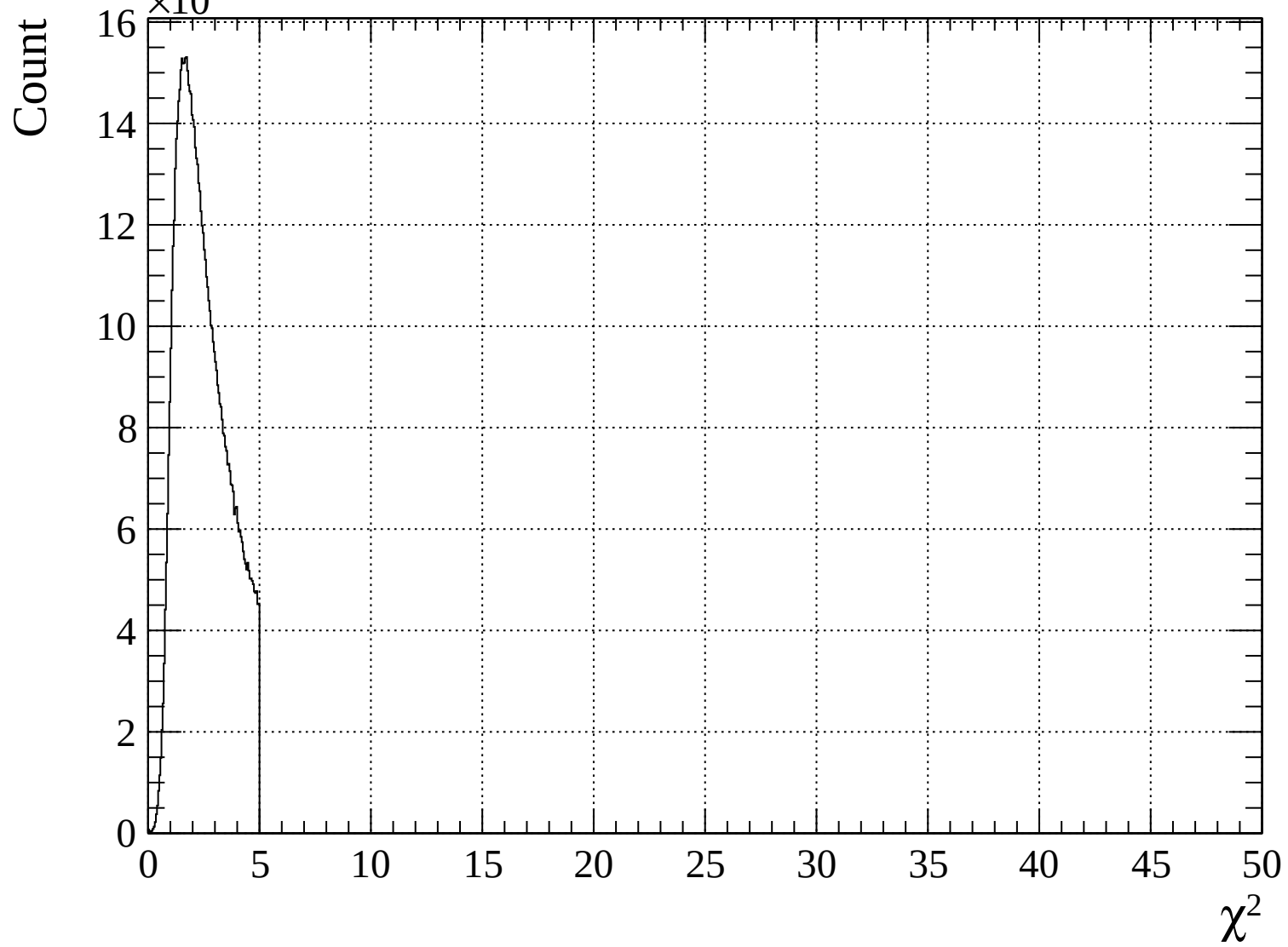


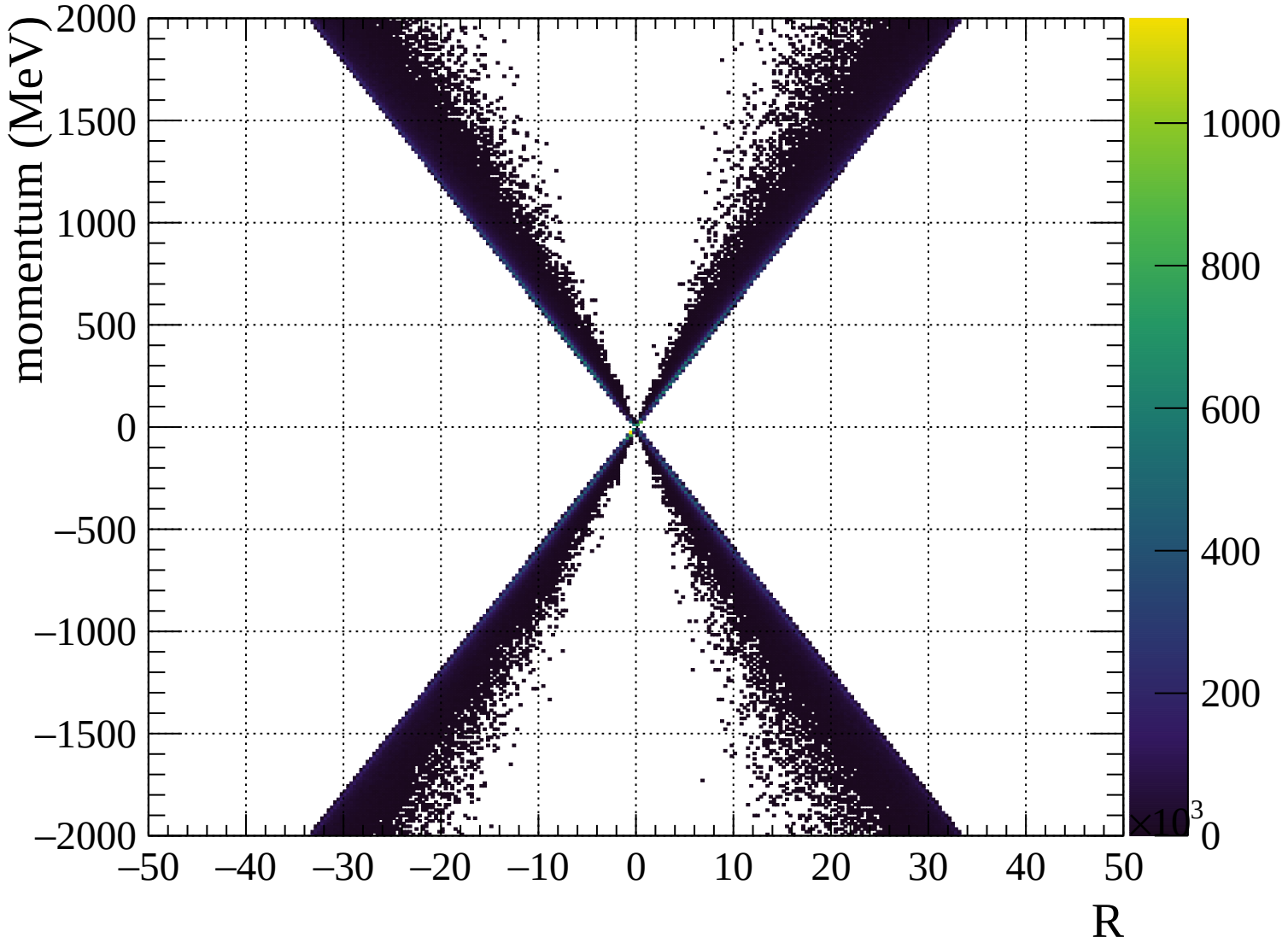


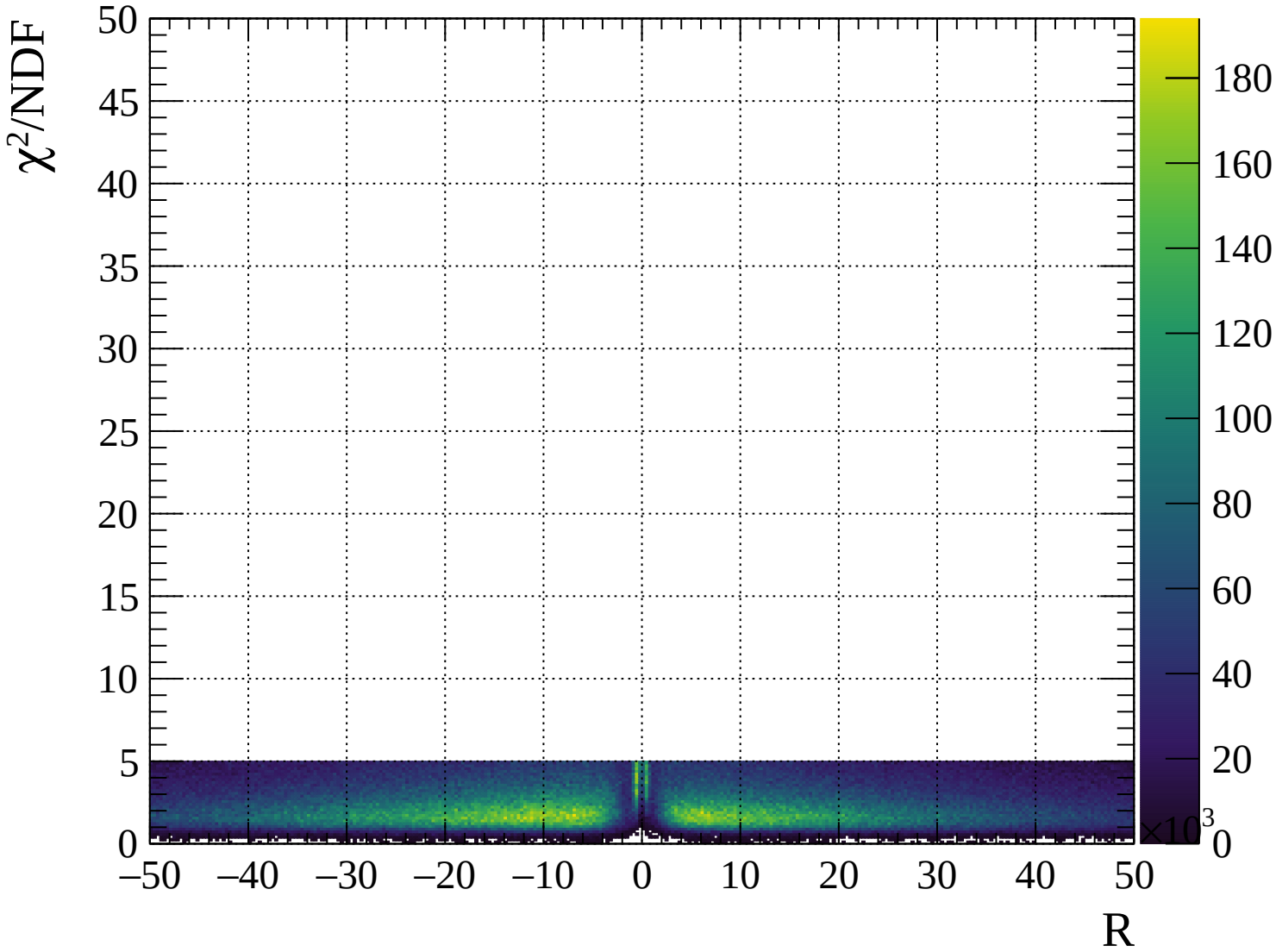






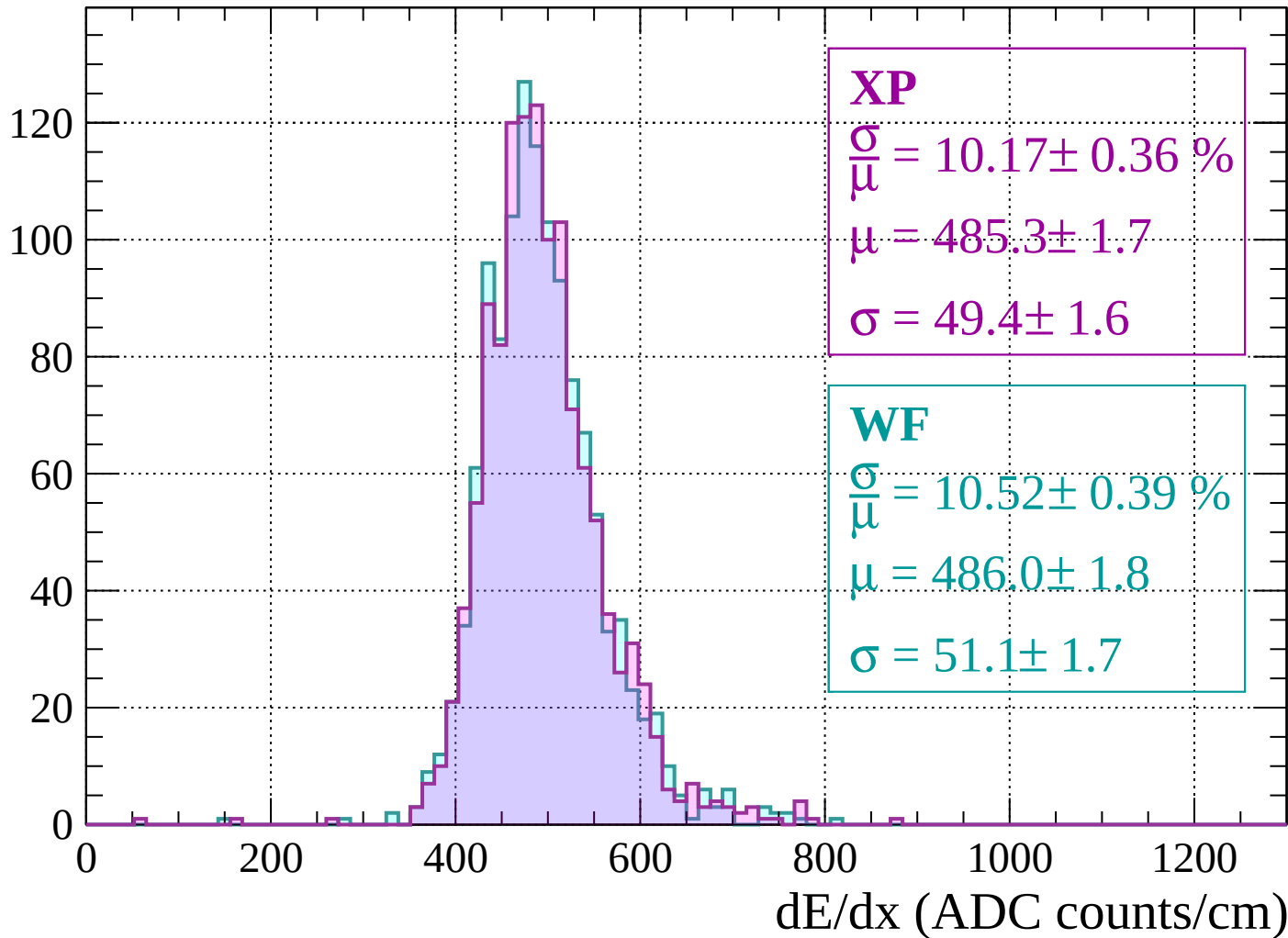




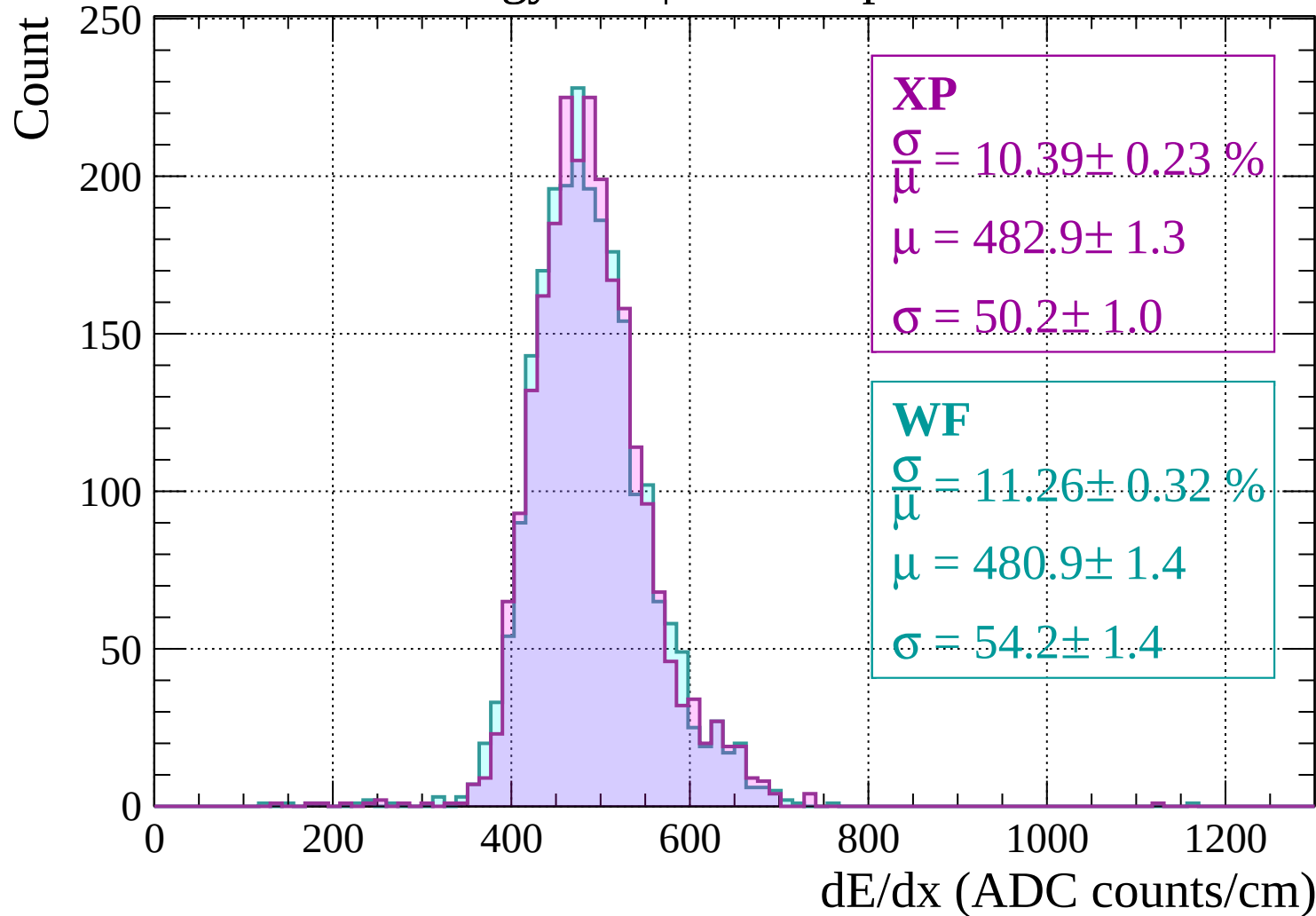


Energy loss | -2000 < p < -1960

Count

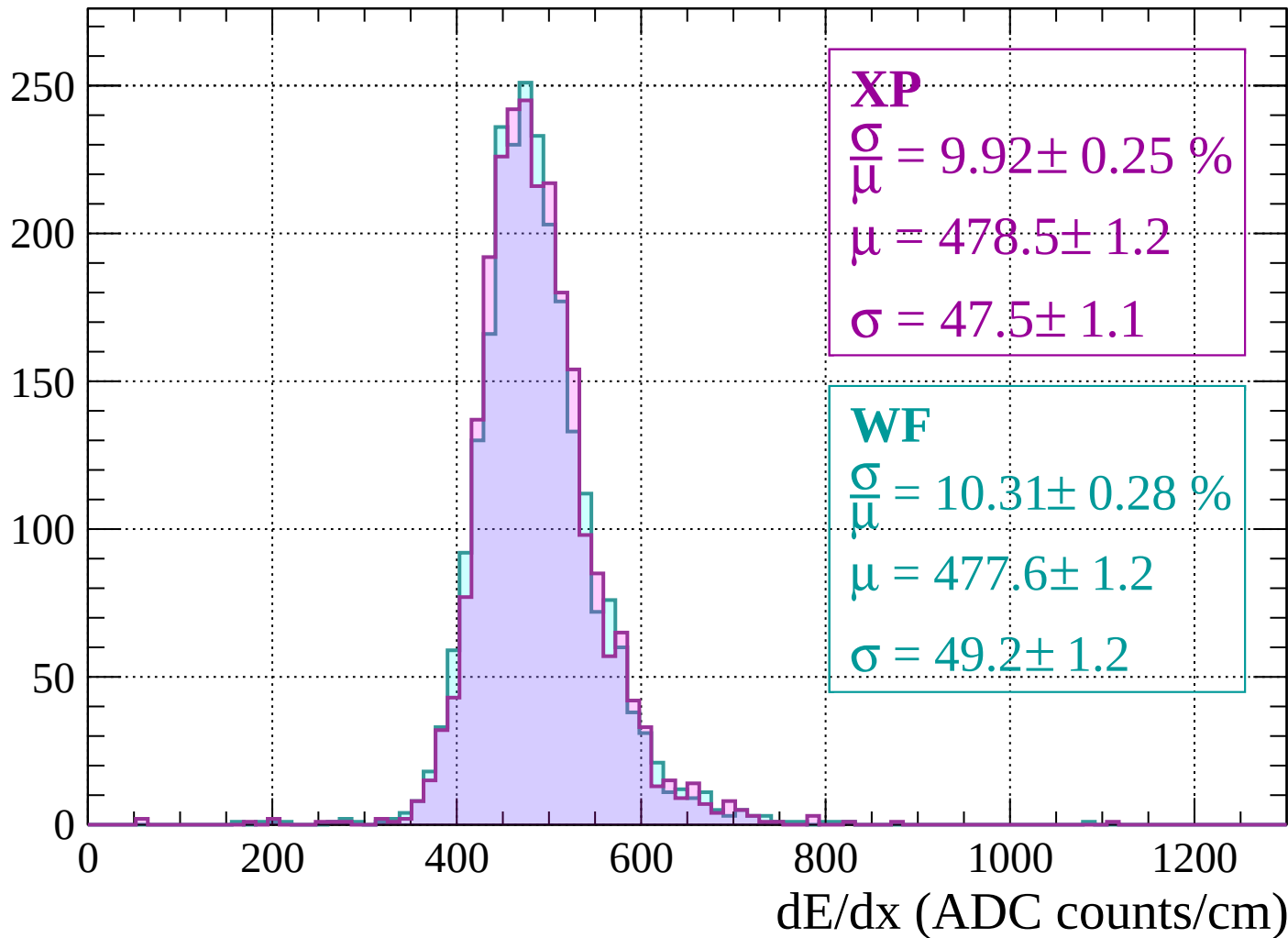


Energy loss | -1960 < p < -1920



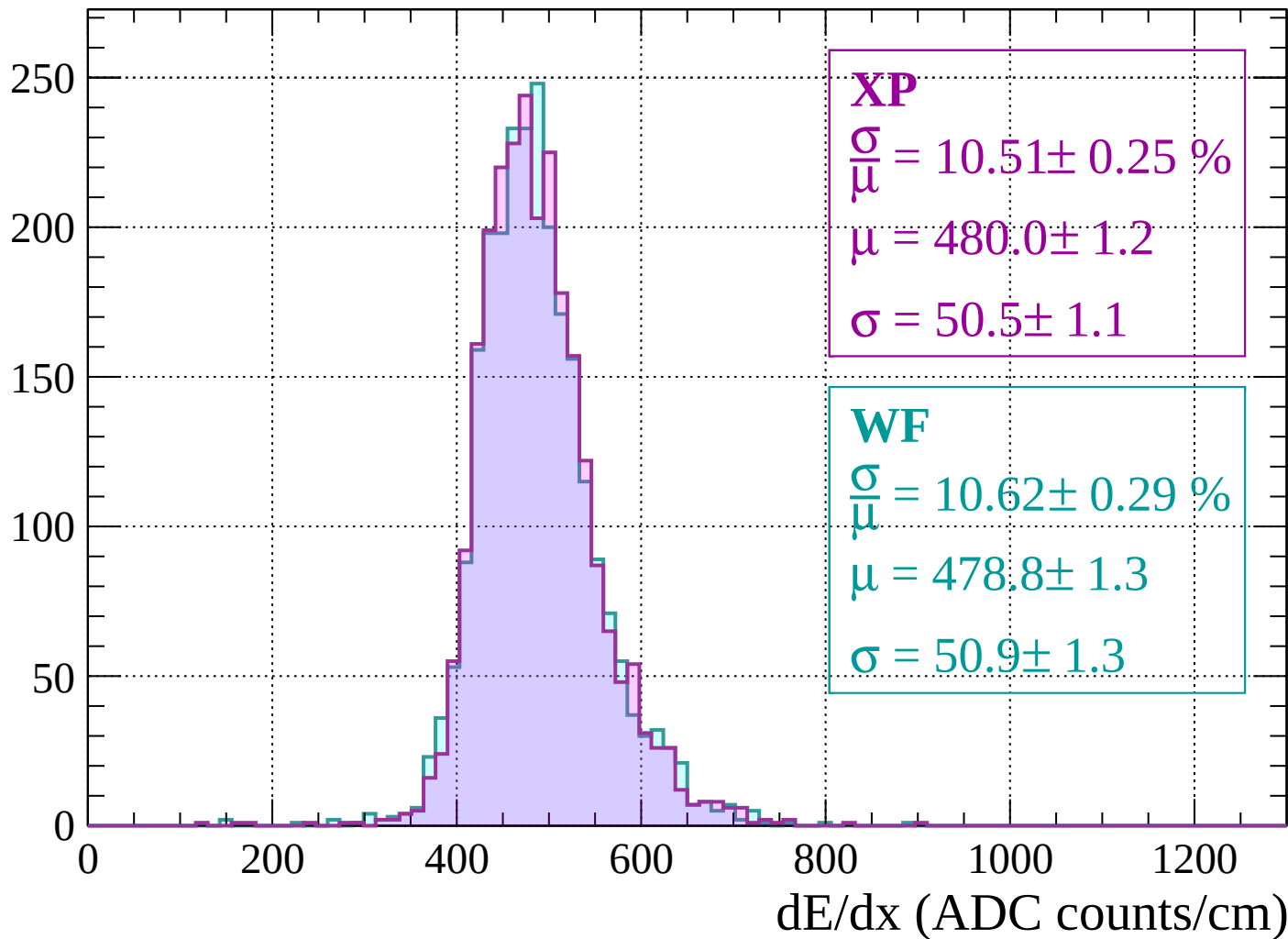
Energy loss | $-1920 < p < -1880$

Count



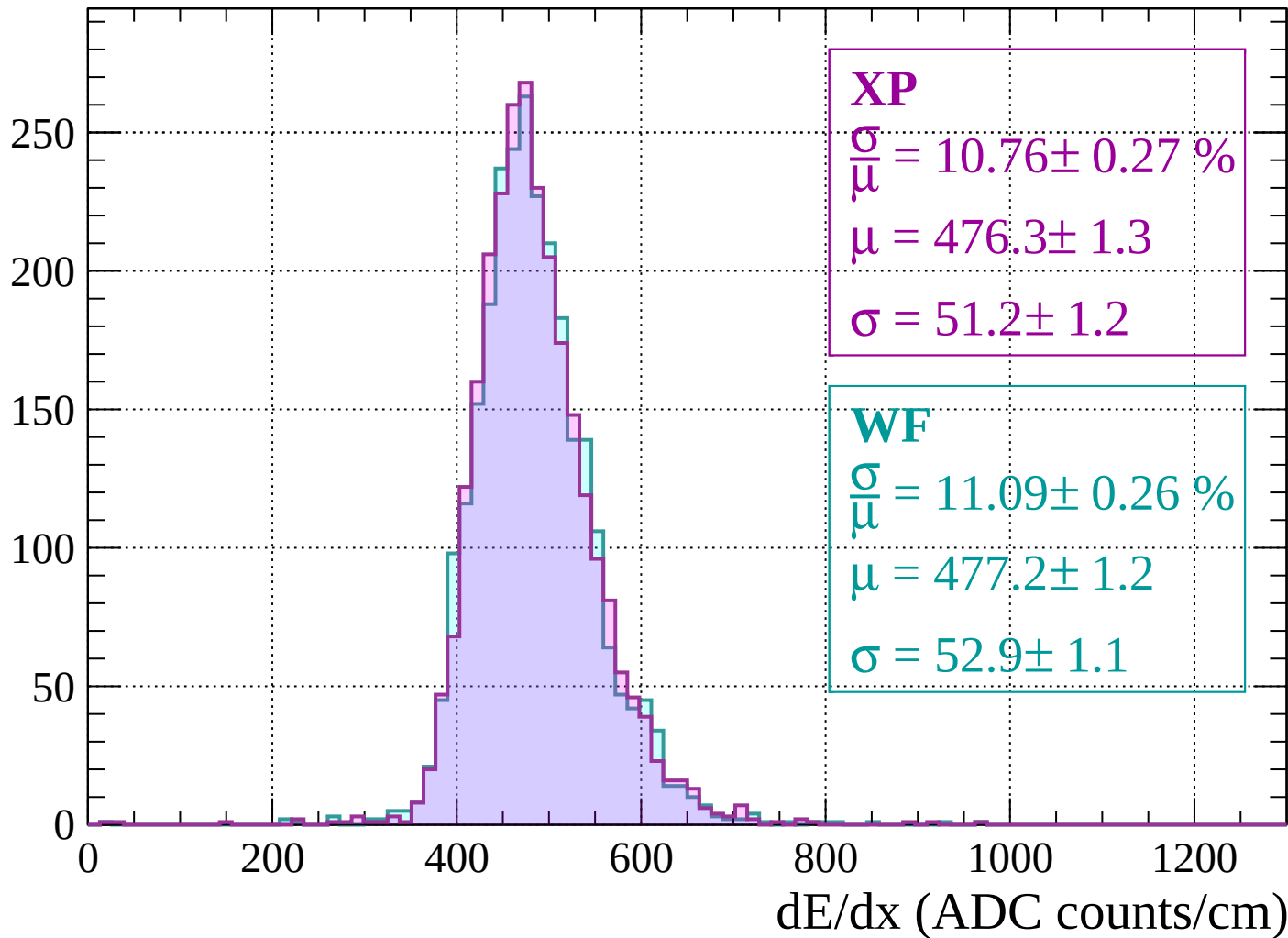
Energy loss | $-1880 < p < -1840$

Count



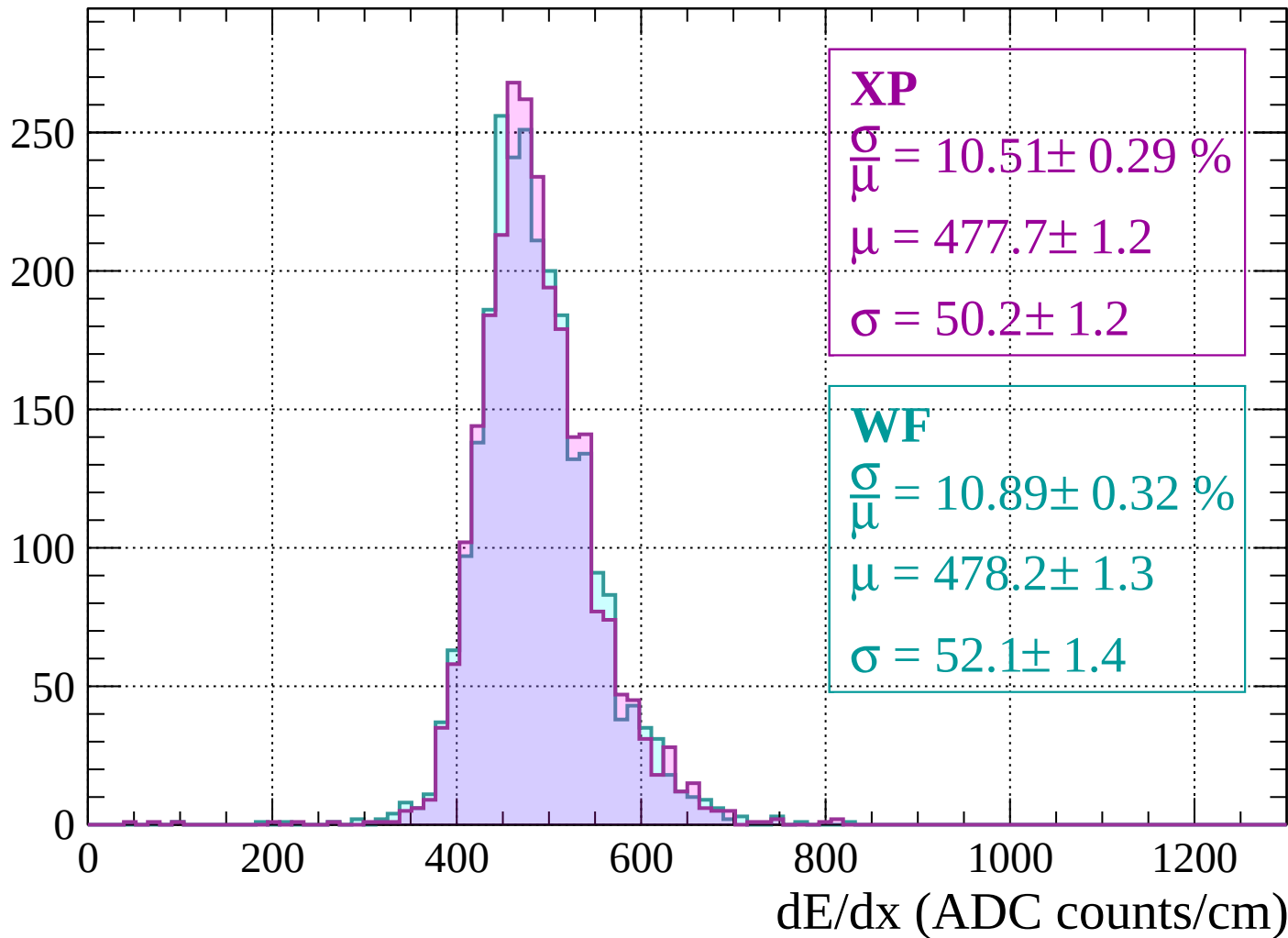
Energy loss | $-1840 < p < -1800$

Count



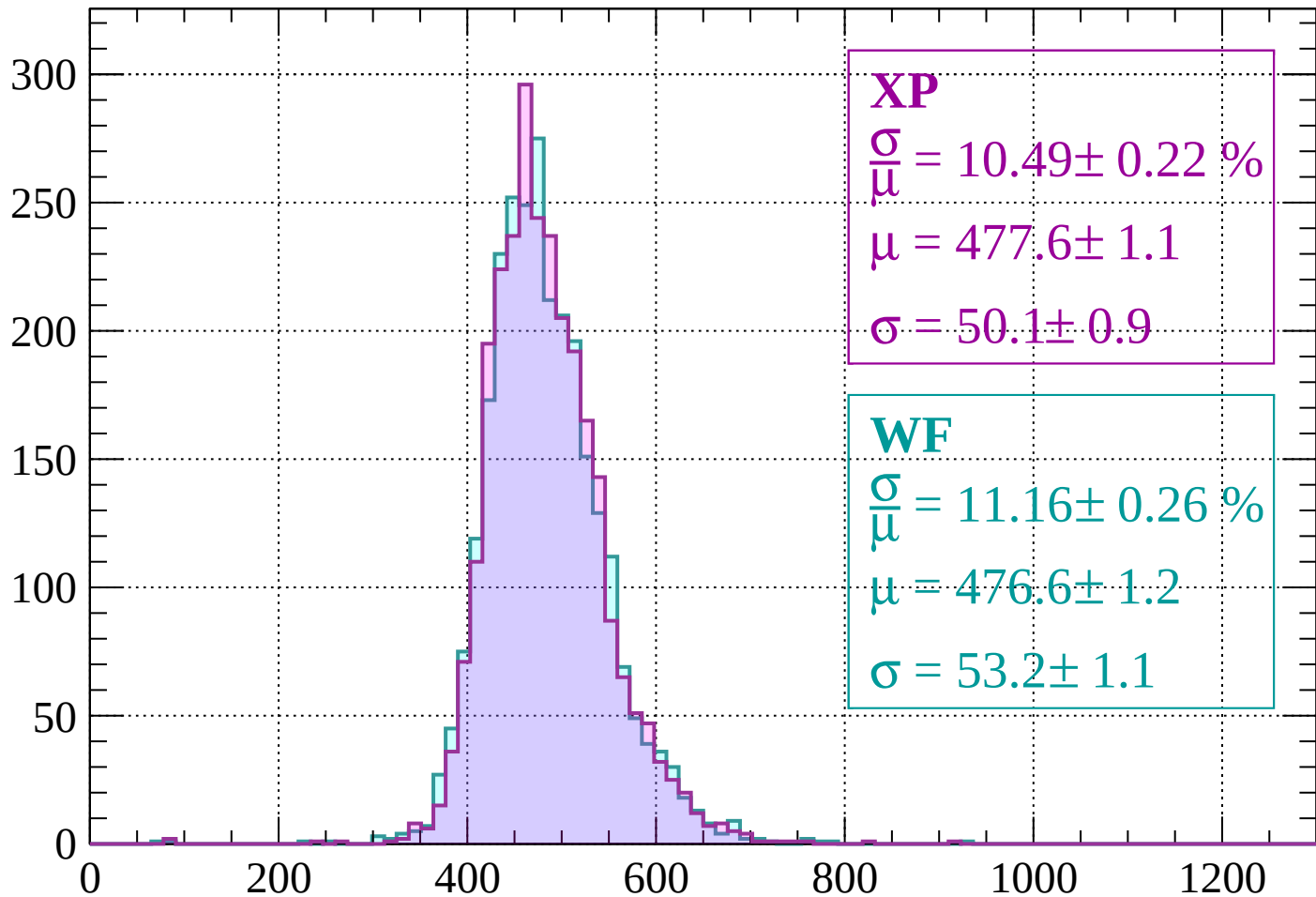
Energy loss | $-1800 < p < -1760$

Count



Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.22 \%$$

$$\mu = 477.6 \pm 1.1$$

$$\sigma = 50.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.16 \pm 0.26 \%$$

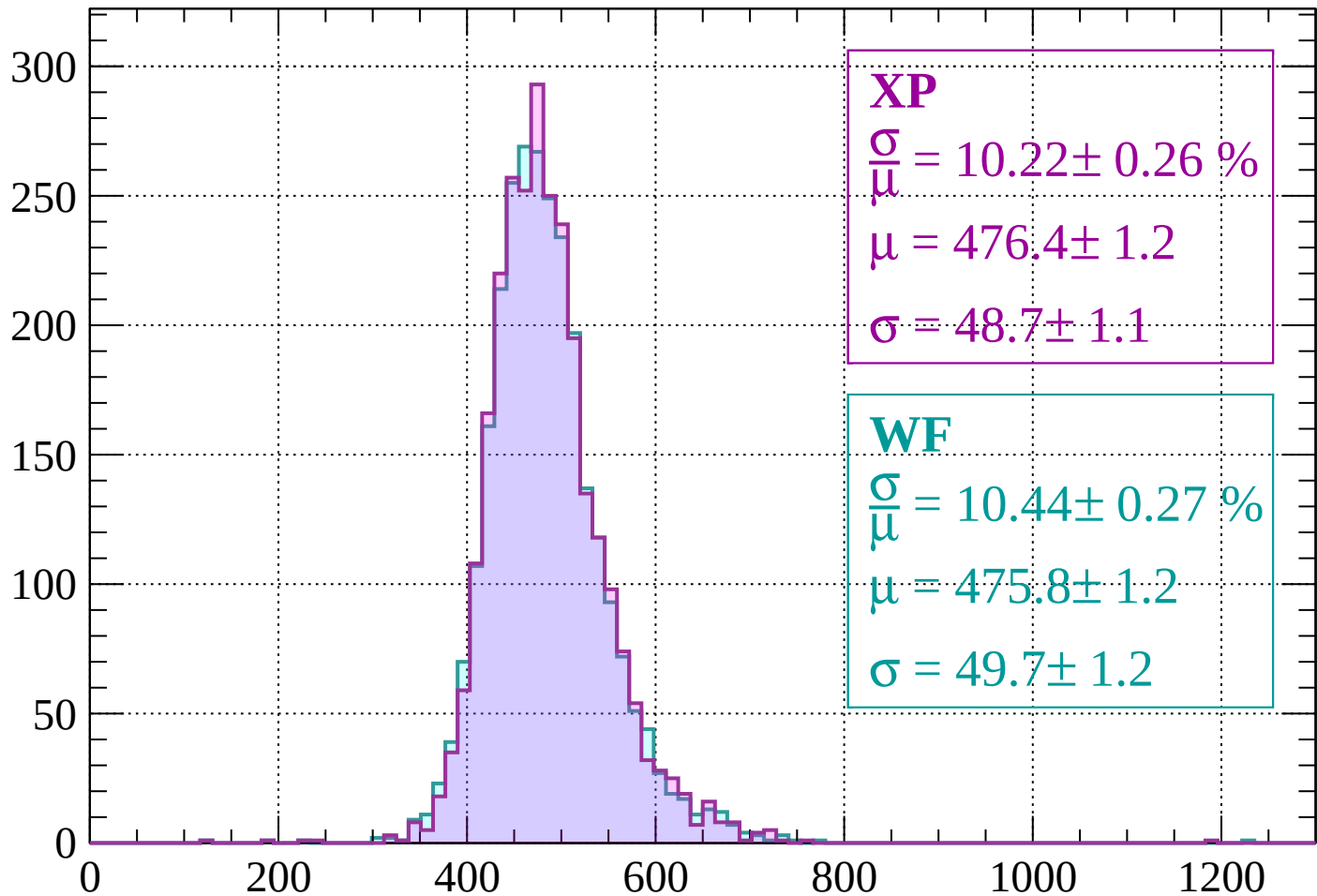
$$\mu = 476.6 \pm 1.2$$

$$\sigma = 53.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.22 \pm 0.26 \%$$

$$\mu = 476.4 \pm 1.2$$

$$\sigma = 48.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.44 \pm 0.27 \%$$

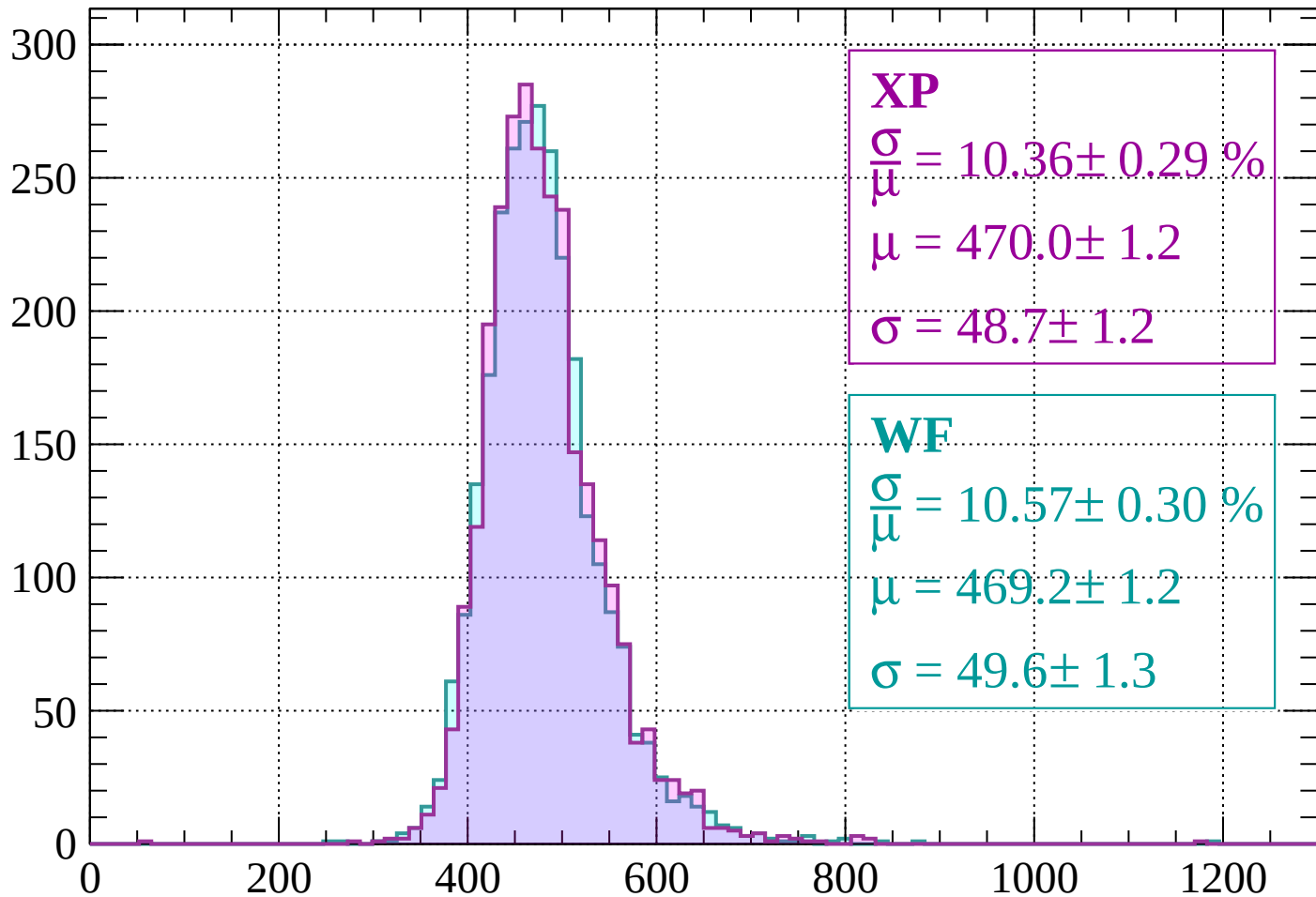
$$\mu = 475.8 \pm 1.2$$

$$\sigma = 49.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.29 \%$$

$$\mu = 470.0 \pm 1.2$$

$$\sigma = 48.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.57 \pm 0.30 \%$$

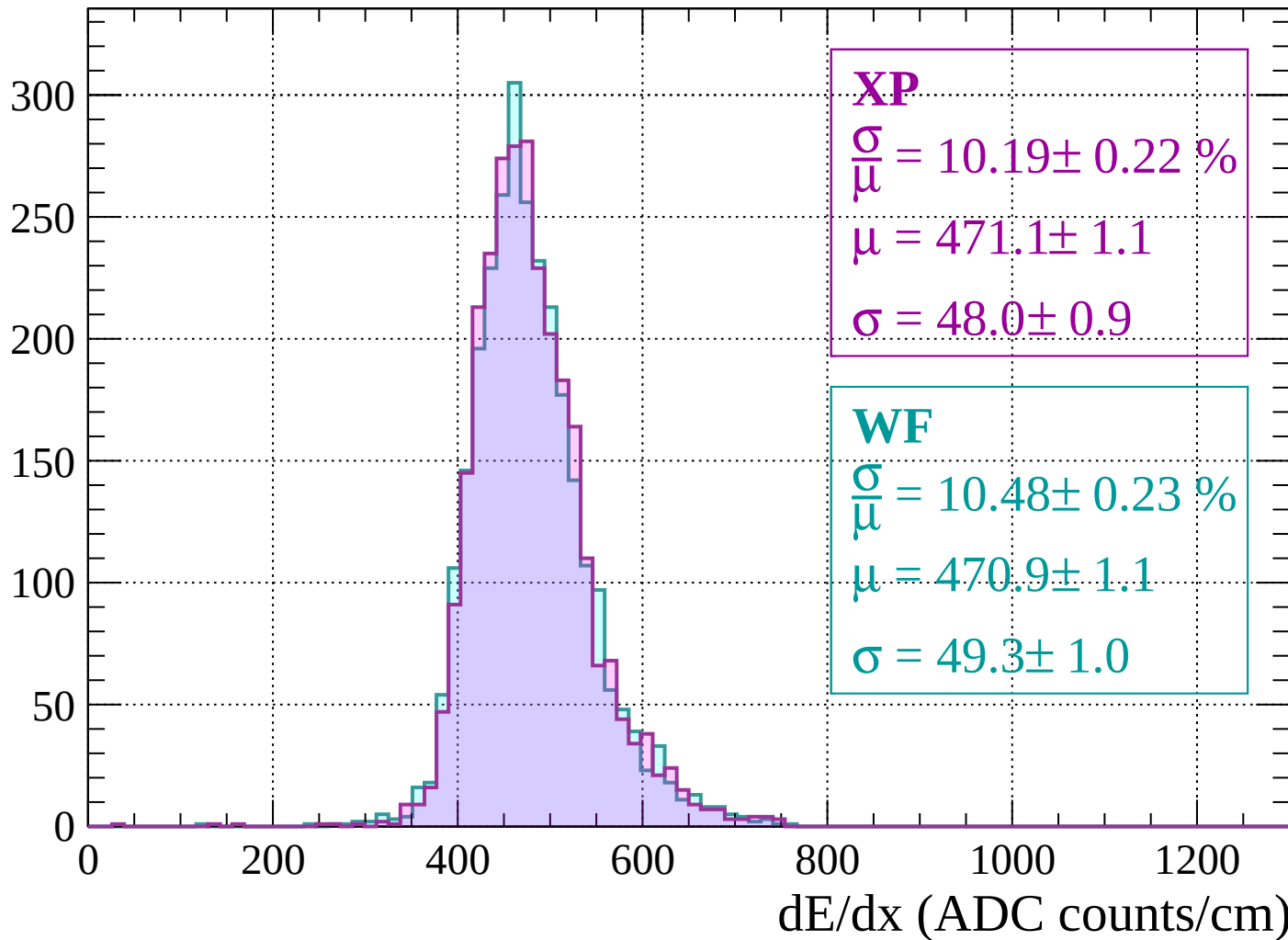
$$\mu = 469.2 \pm 1.2$$

$$\sigma = 49.6 \pm 1.3$$

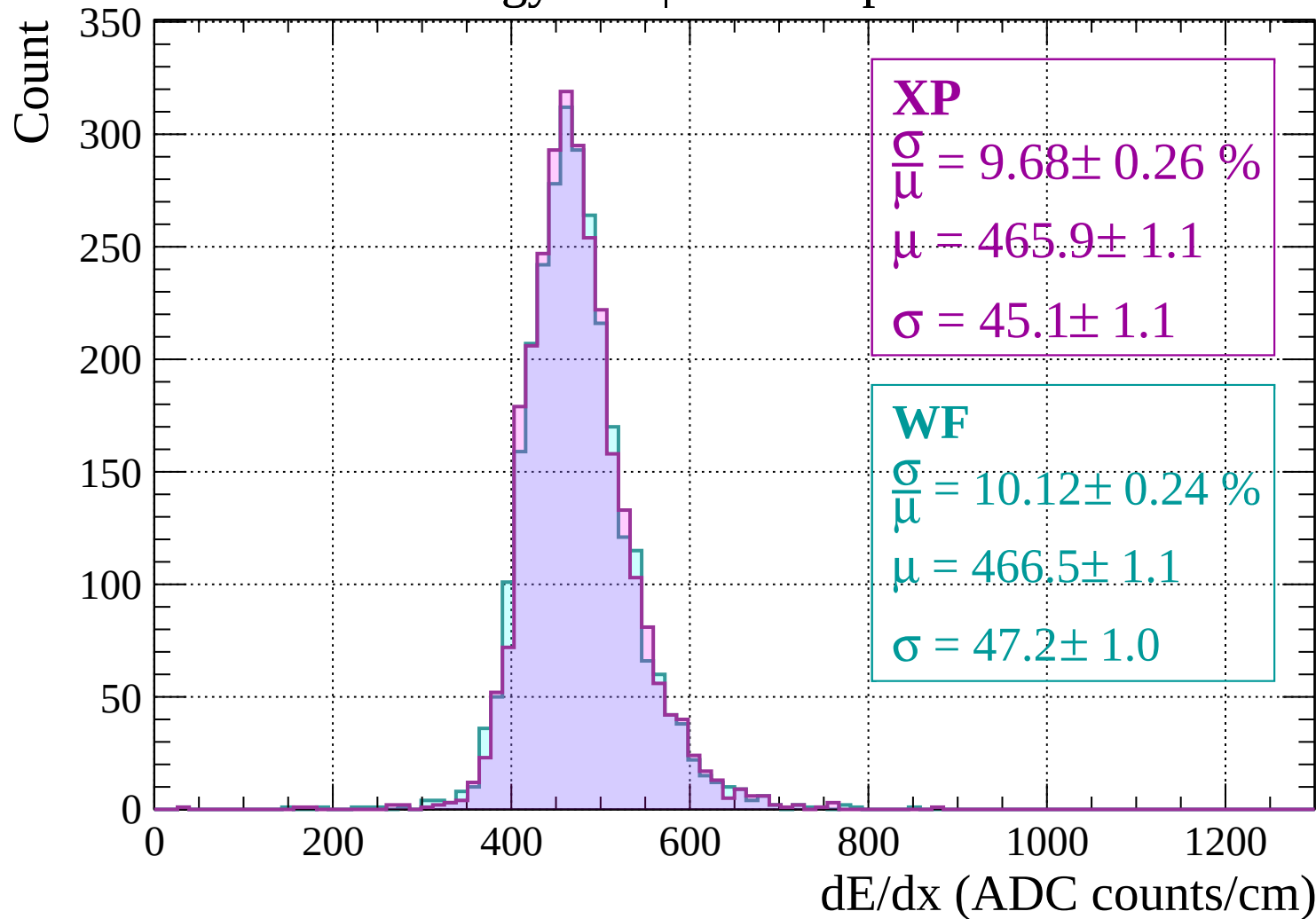
dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

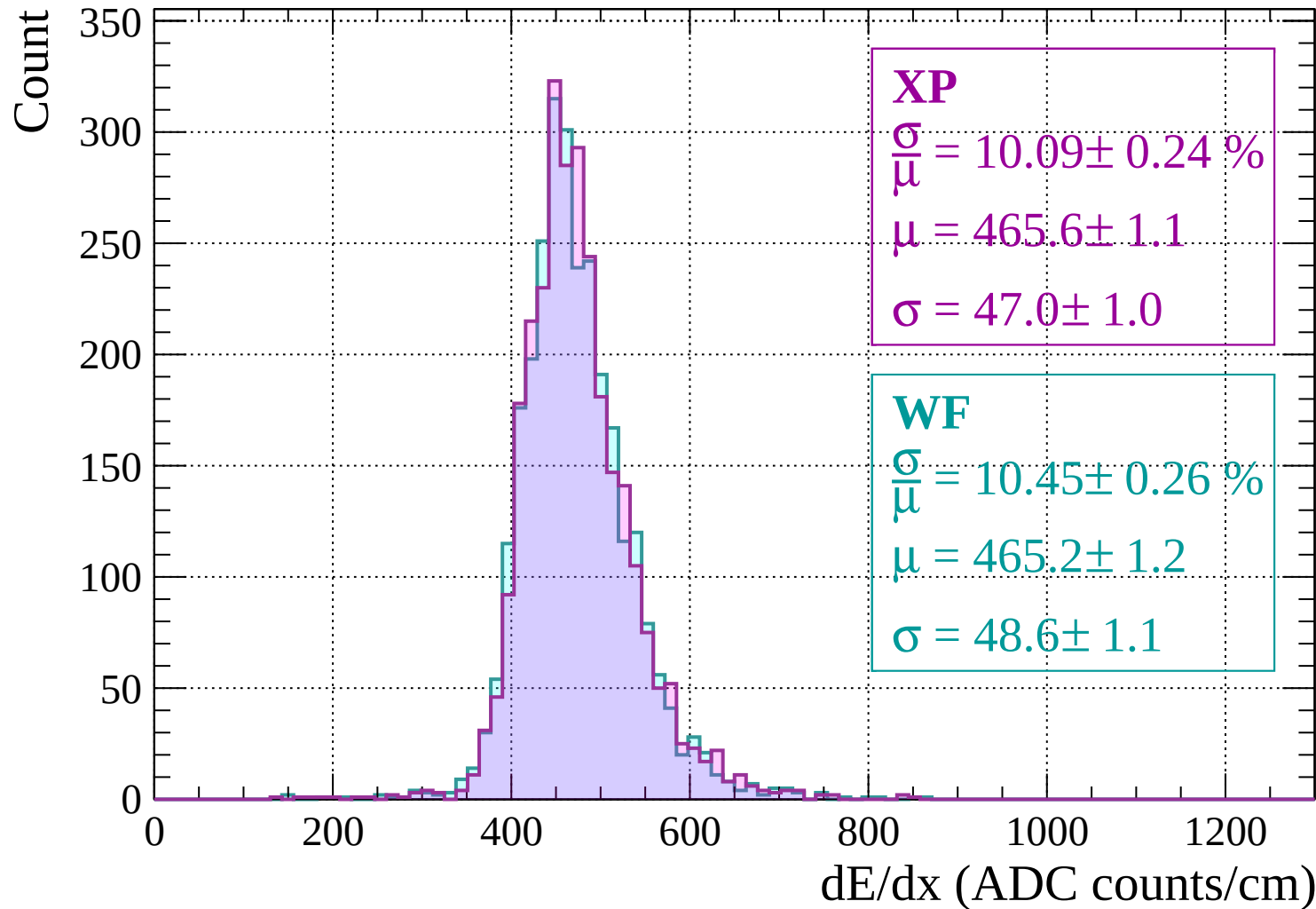
Count



Energy loss | $-1600 < p < -1560$

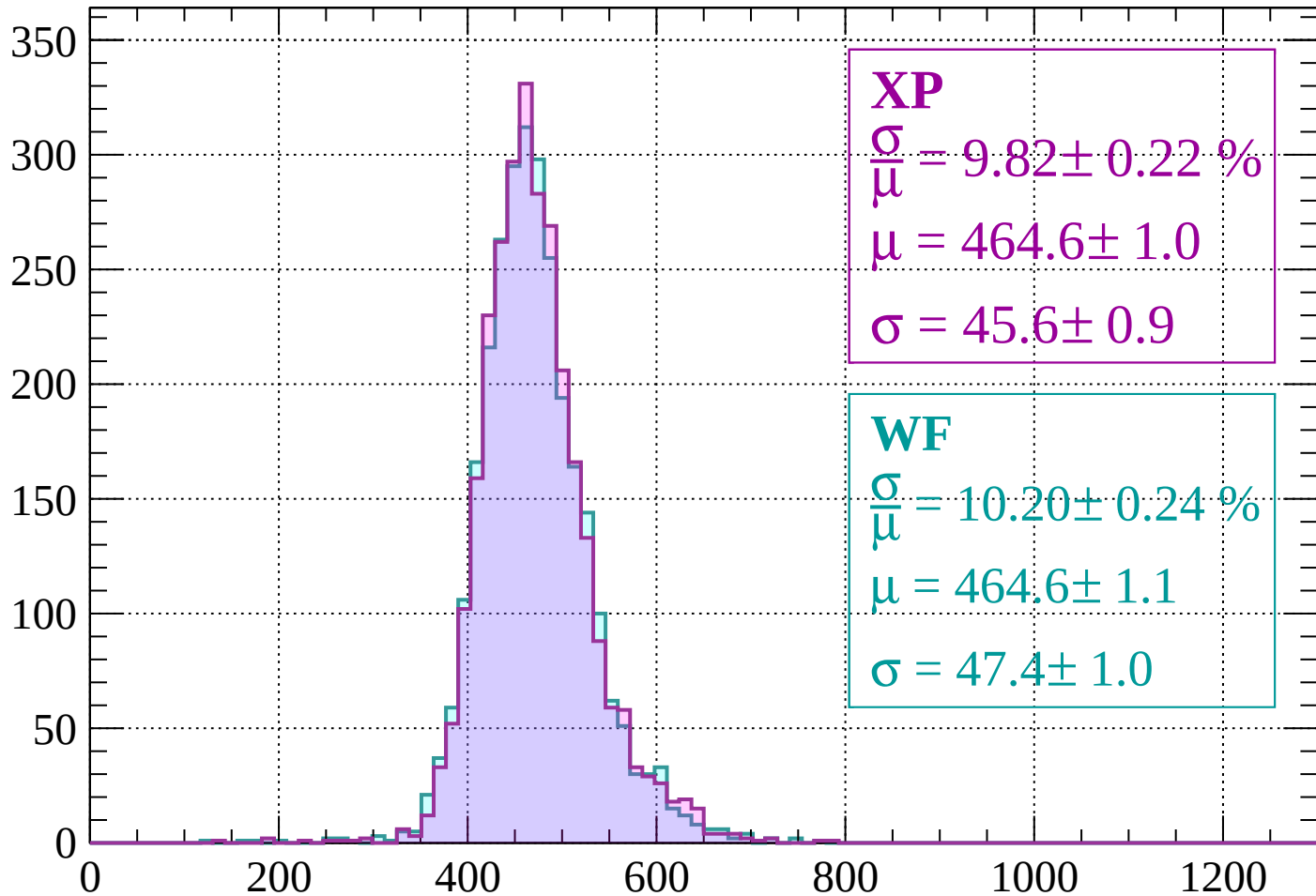


Energy loss | $-1560 < p < -1520$



Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.22 \%$$

$$\mu = 464.6 \pm 1.0$$

$$\sigma = 45.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.24 \%$$

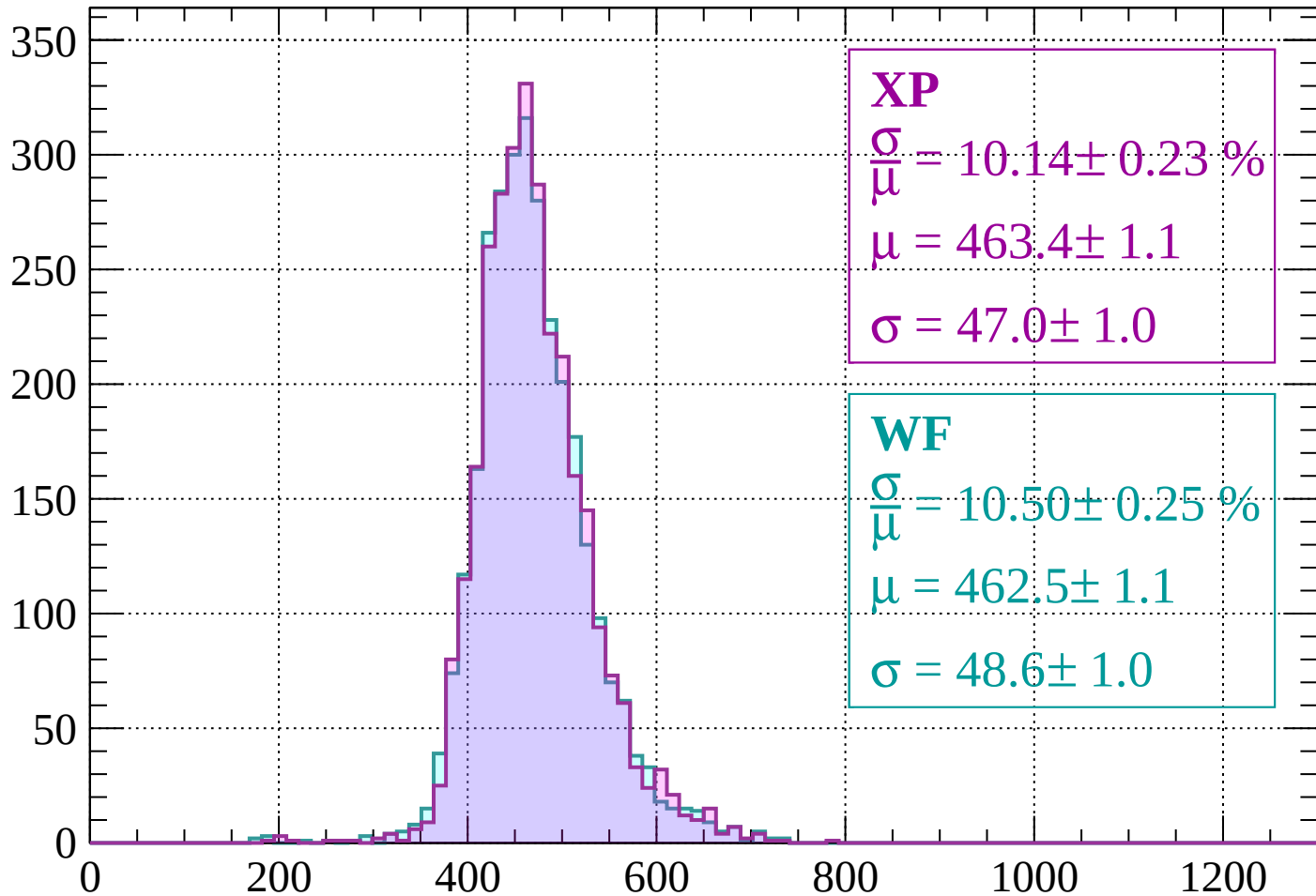
$$\mu = 464.6 \pm 1.1$$

$$\sigma = 47.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.23 \%$$

$$\mu = 463.4 \pm 1.1$$

$$\sigma = 47.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.50 \pm 0.25 \%$$

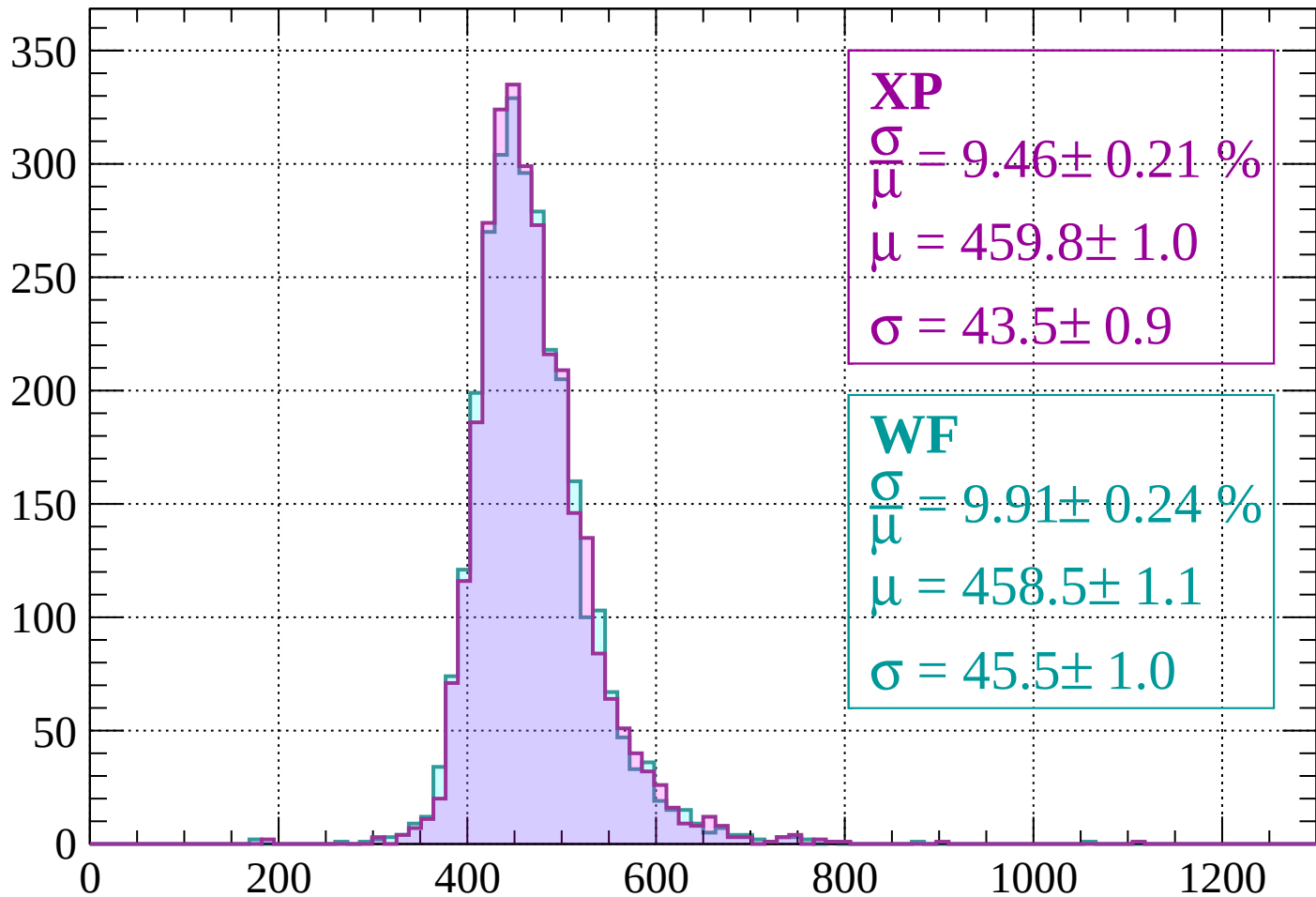
$$\mu = 462.5 \pm 1.1$$

$$\sigma = 48.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.21 \%$$

$$\mu = 459.8 \pm 1.0$$

$$\sigma = 43.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.24 \%$$

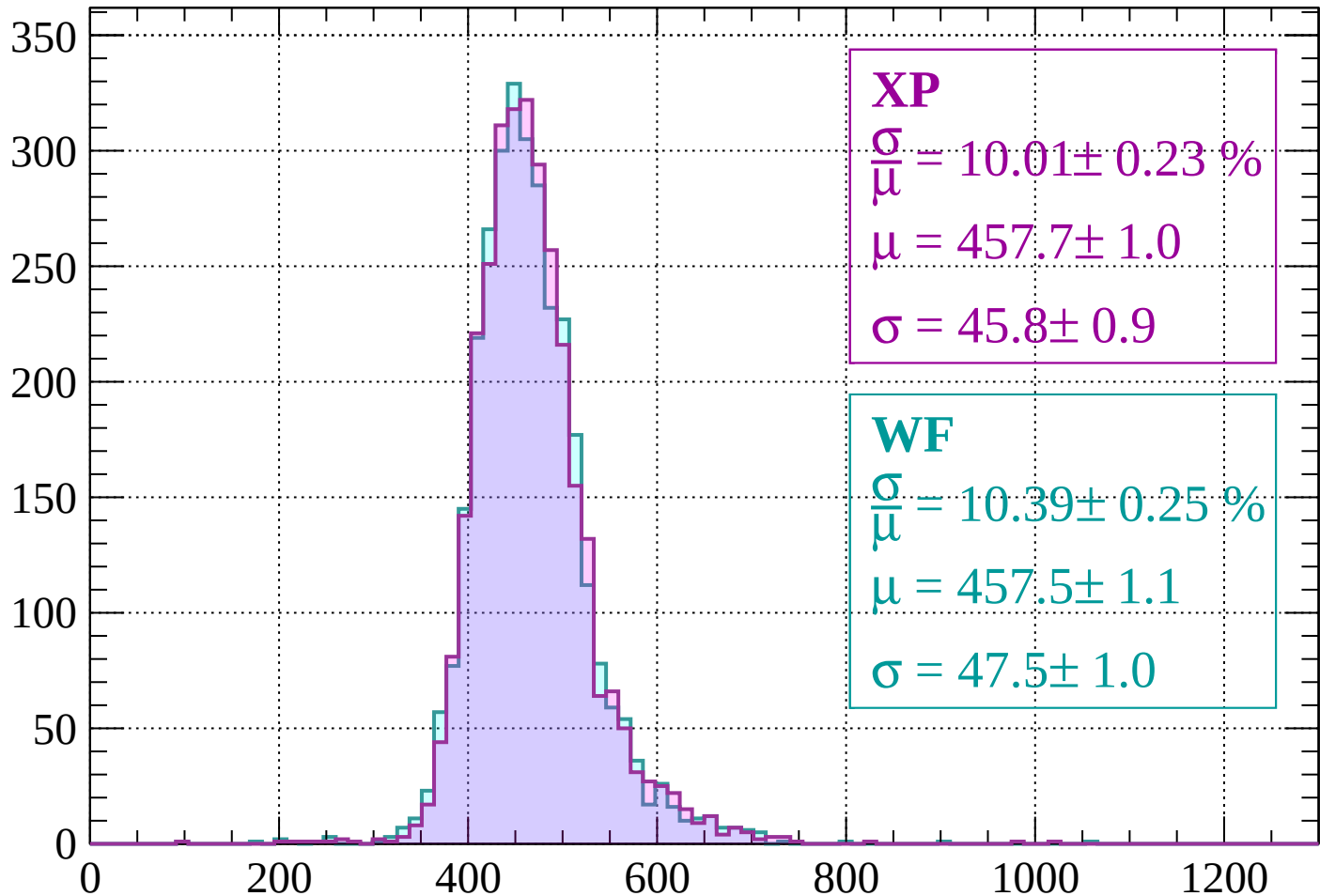
$$\mu = 458.5 \pm 1.1$$

$$\sigma = 45.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.01 \pm 0.23 \%$$

$$\mu = 457.7 \pm 1.0$$

$$\sigma = 45.8 \pm 0.9$$

WF

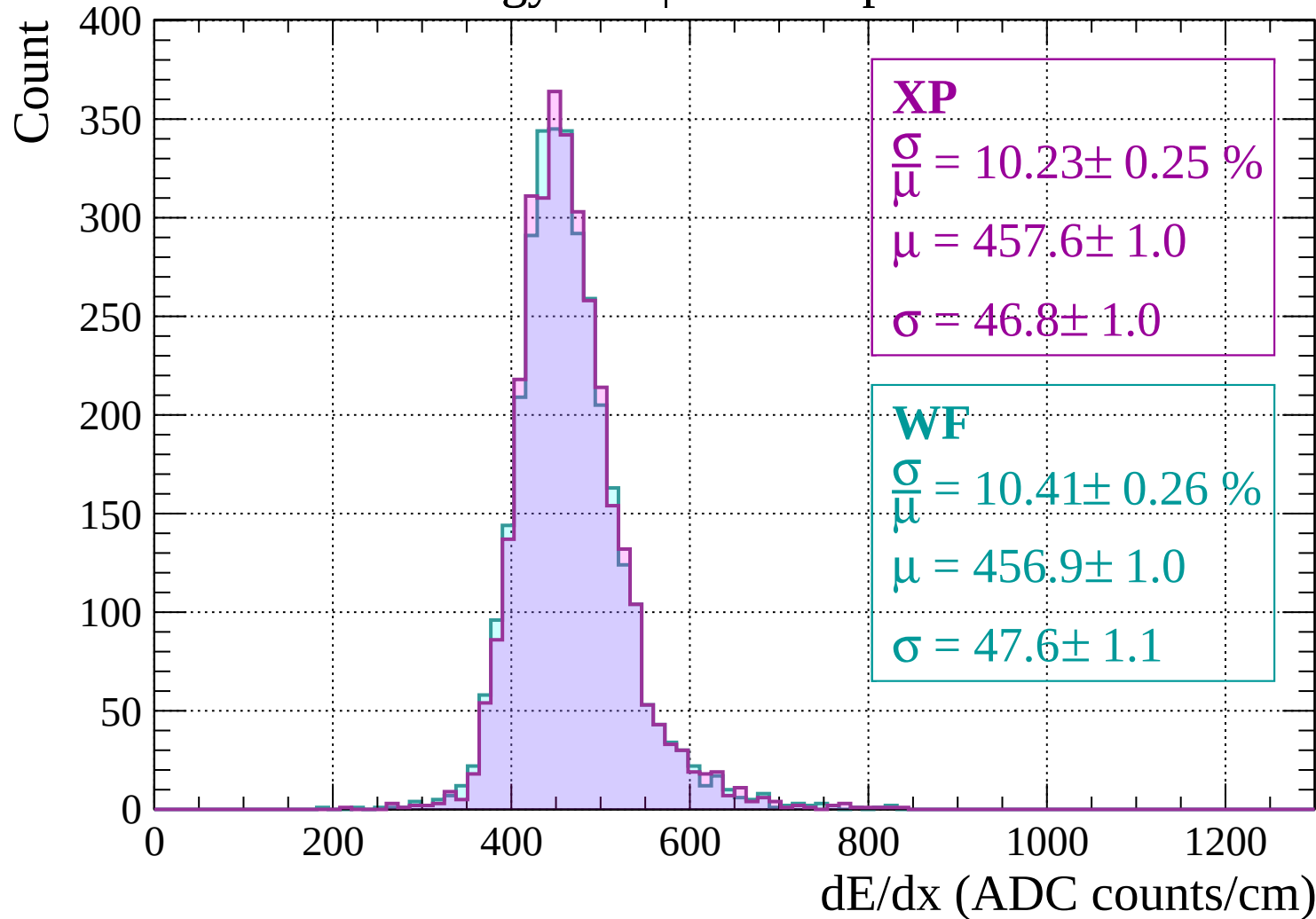
$$\frac{\sigma}{\bar{\mu}} = 10.39 \pm 0.25 \%$$

$$\mu = 457.5 \pm 1.1$$

$$\sigma = 47.5 \pm 1.0$$

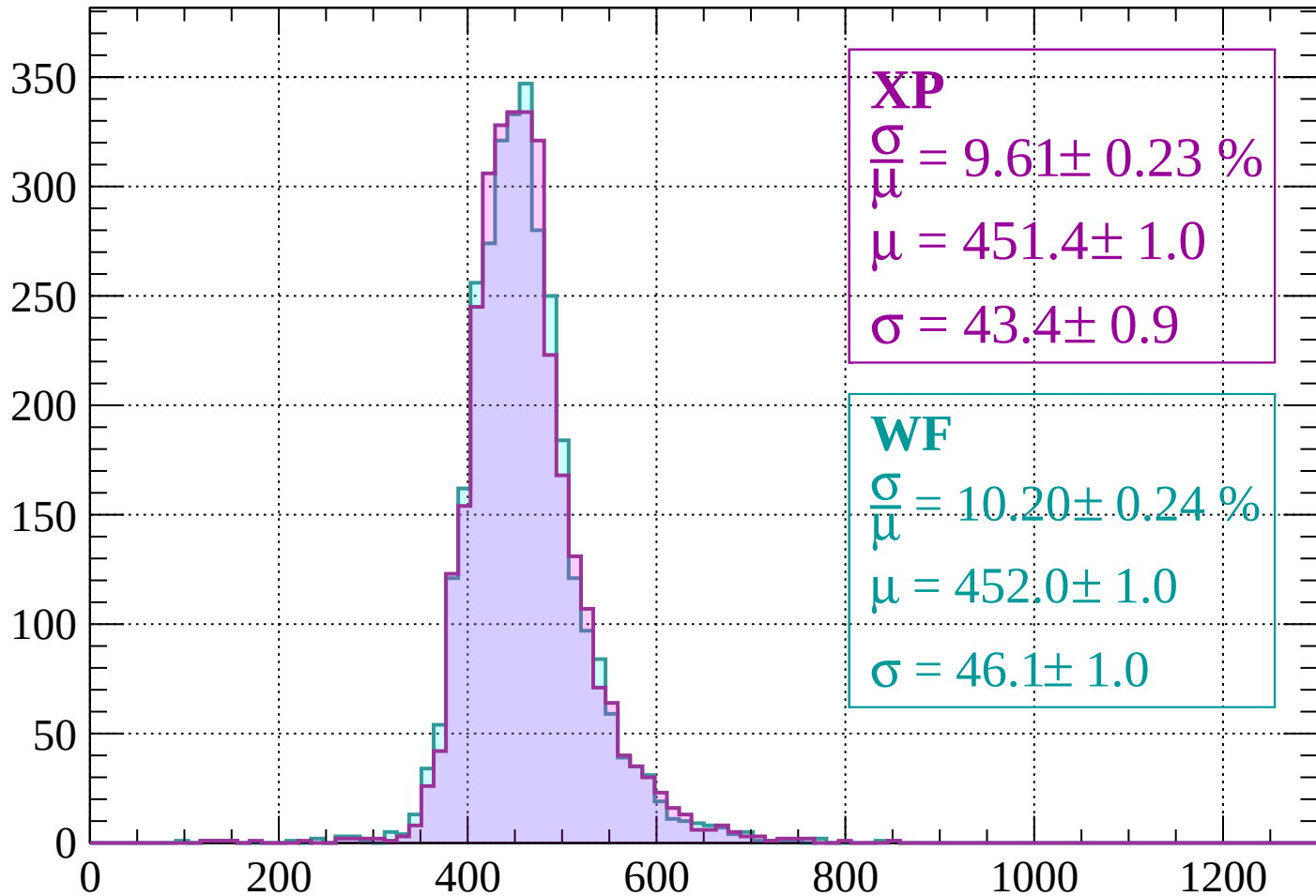
dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.23 \%$$

$$\mu = 451.4 \pm 1.0$$

$$\sigma = 43.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.24 \%$$

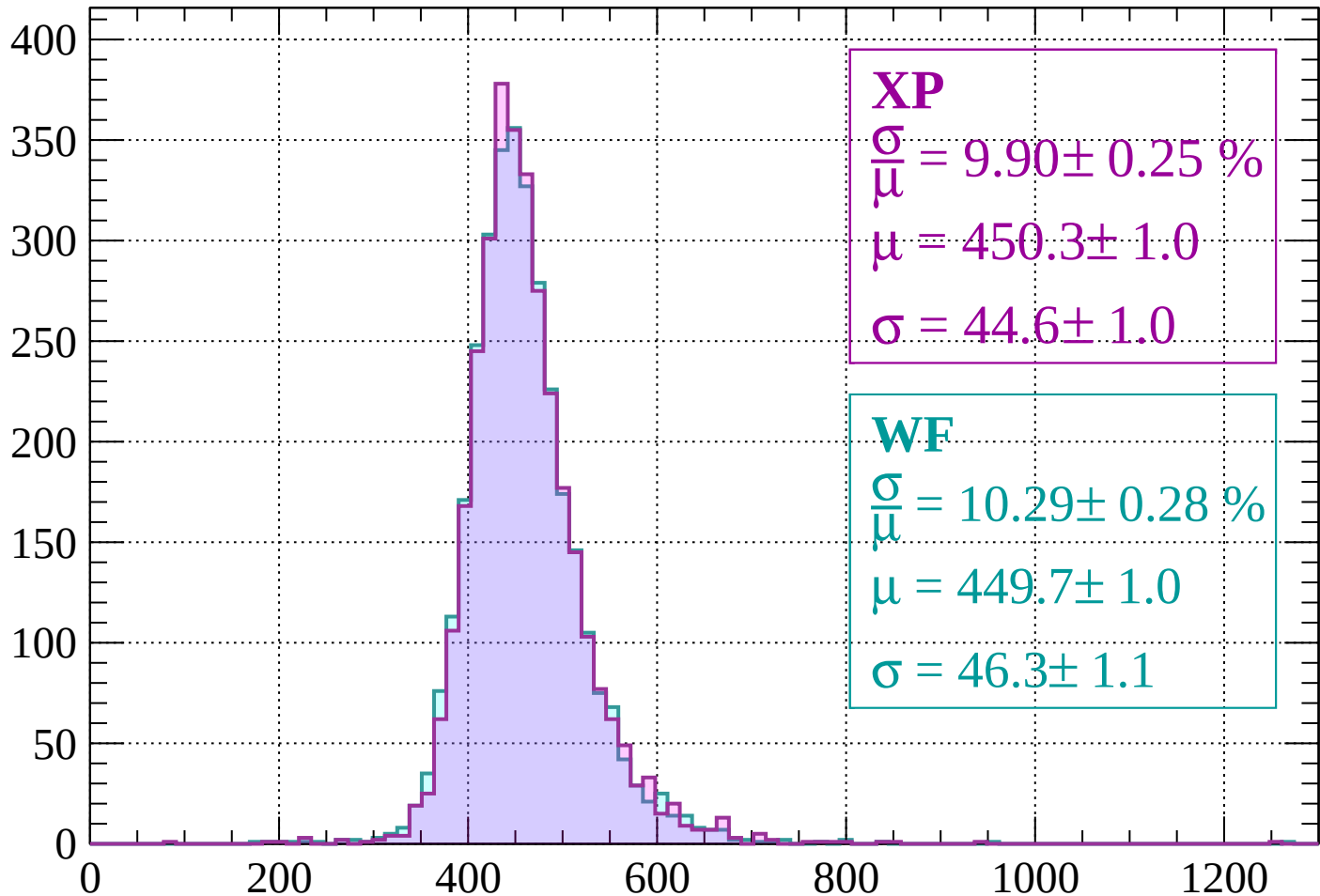
$$\mu = 452.0 \pm 1.0$$

$$\sigma = 46.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.90 \pm 0.25 \%$$

$$\mu = 450.3 \pm 1.0$$

$$\sigma = 44.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 0.28 \%$$

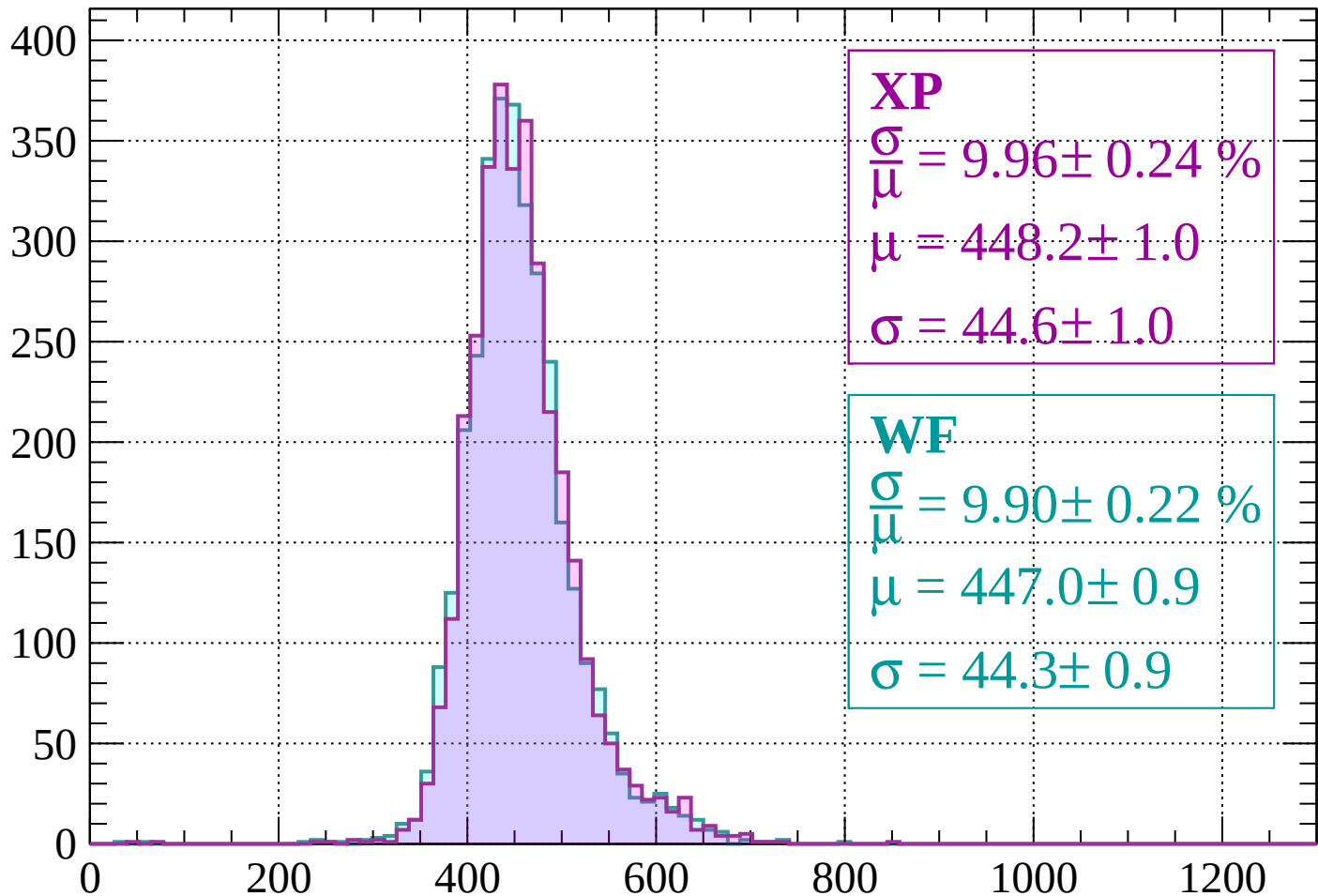
$$\mu = 449.7 \pm 1.0$$

$$\sigma = 46.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.24 \%$$

$$\mu = 448.2 \pm 1.0$$

$$\sigma = 44.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.90 \pm 0.22 \%$$

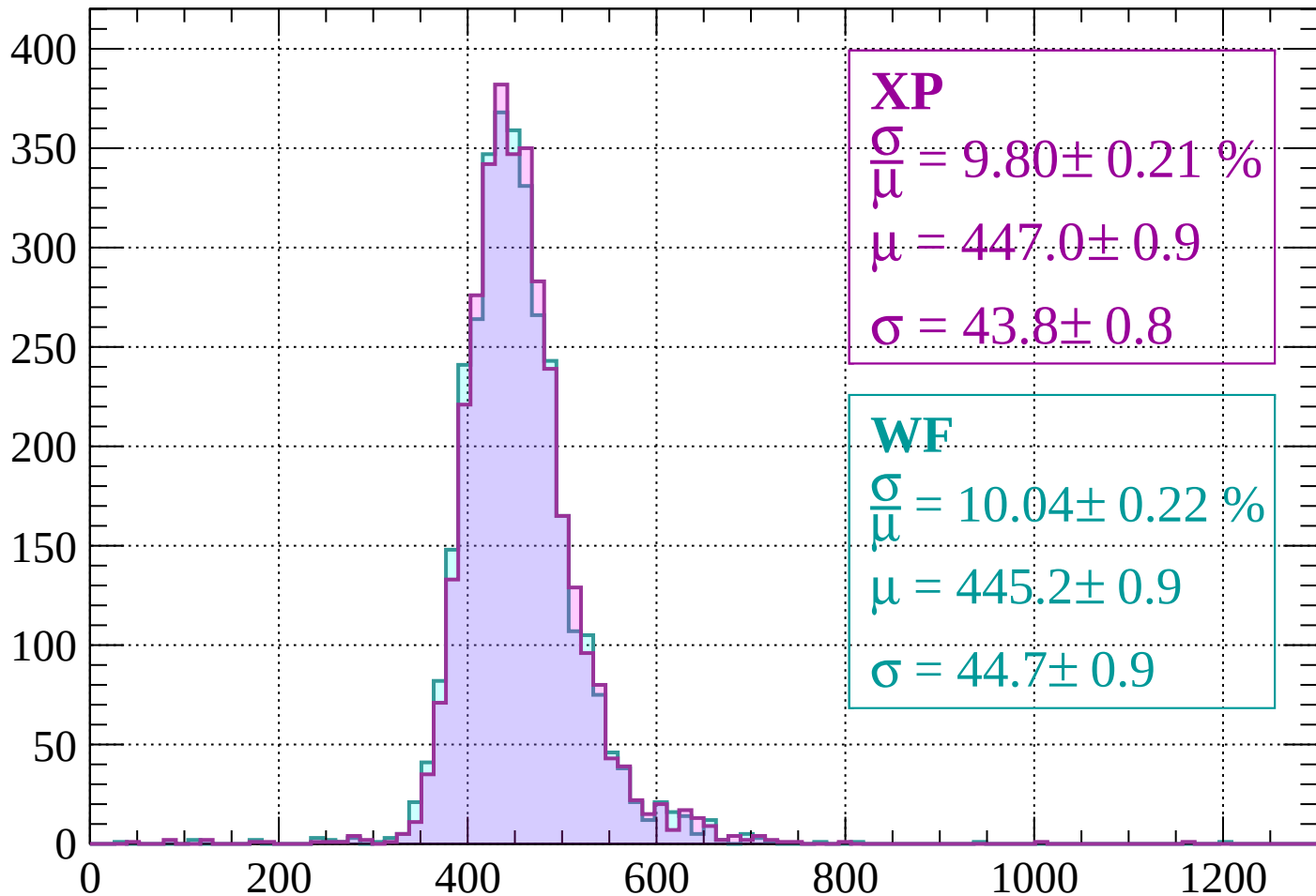
$$\mu = 447.0 \pm 0.9$$

$$\sigma = 44.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.21 \%$$

$$\mu = 447.0 \pm 0.9$$

$$\sigma = 43.8 \pm 0.8$$

WF

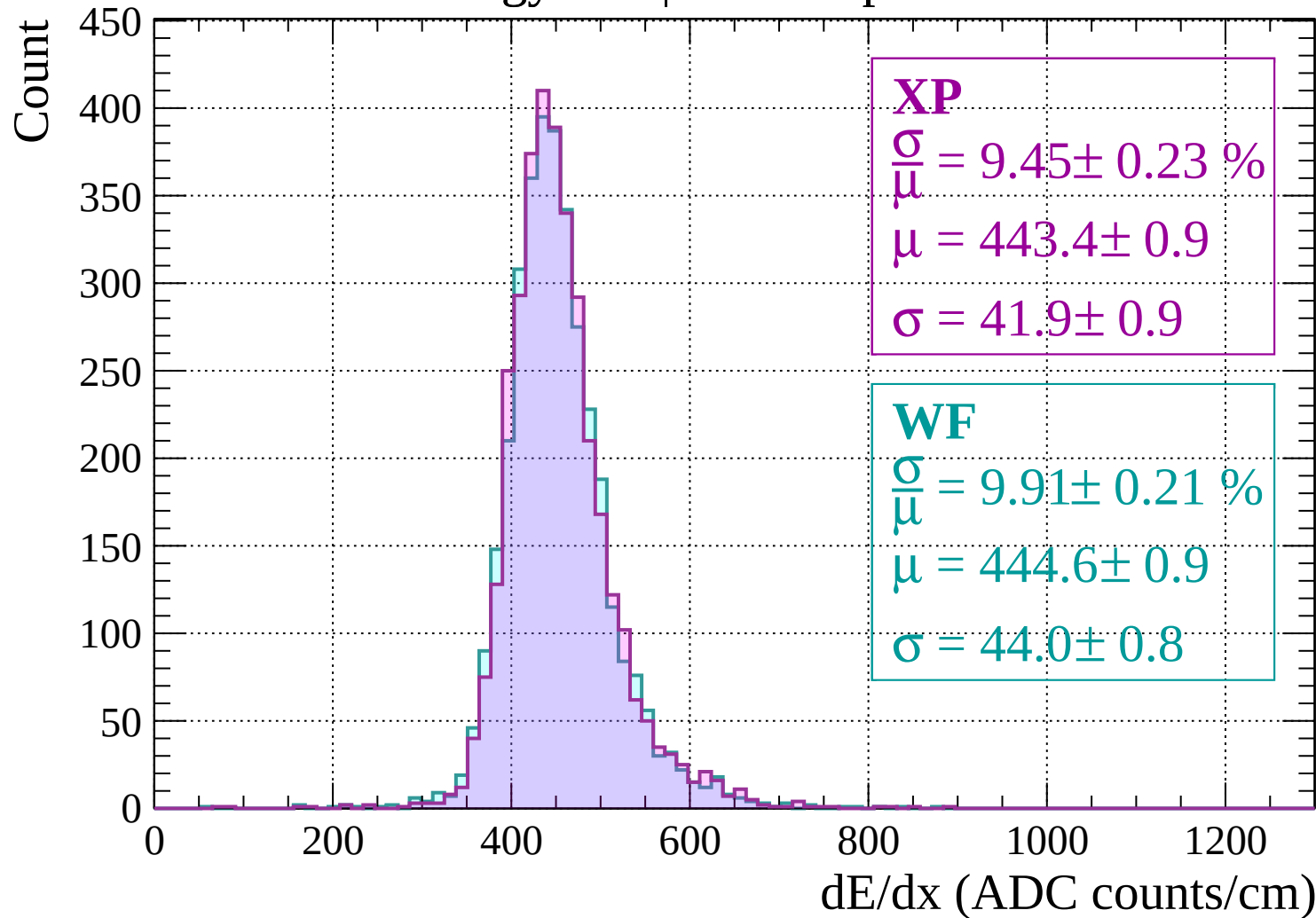
$$\frac{\sigma}{\mu} = 10.04 \pm 0.22 \%$$

$$\mu = 445.2 \pm 0.9$$

$$\sigma = 44.7 \pm 0.9$$

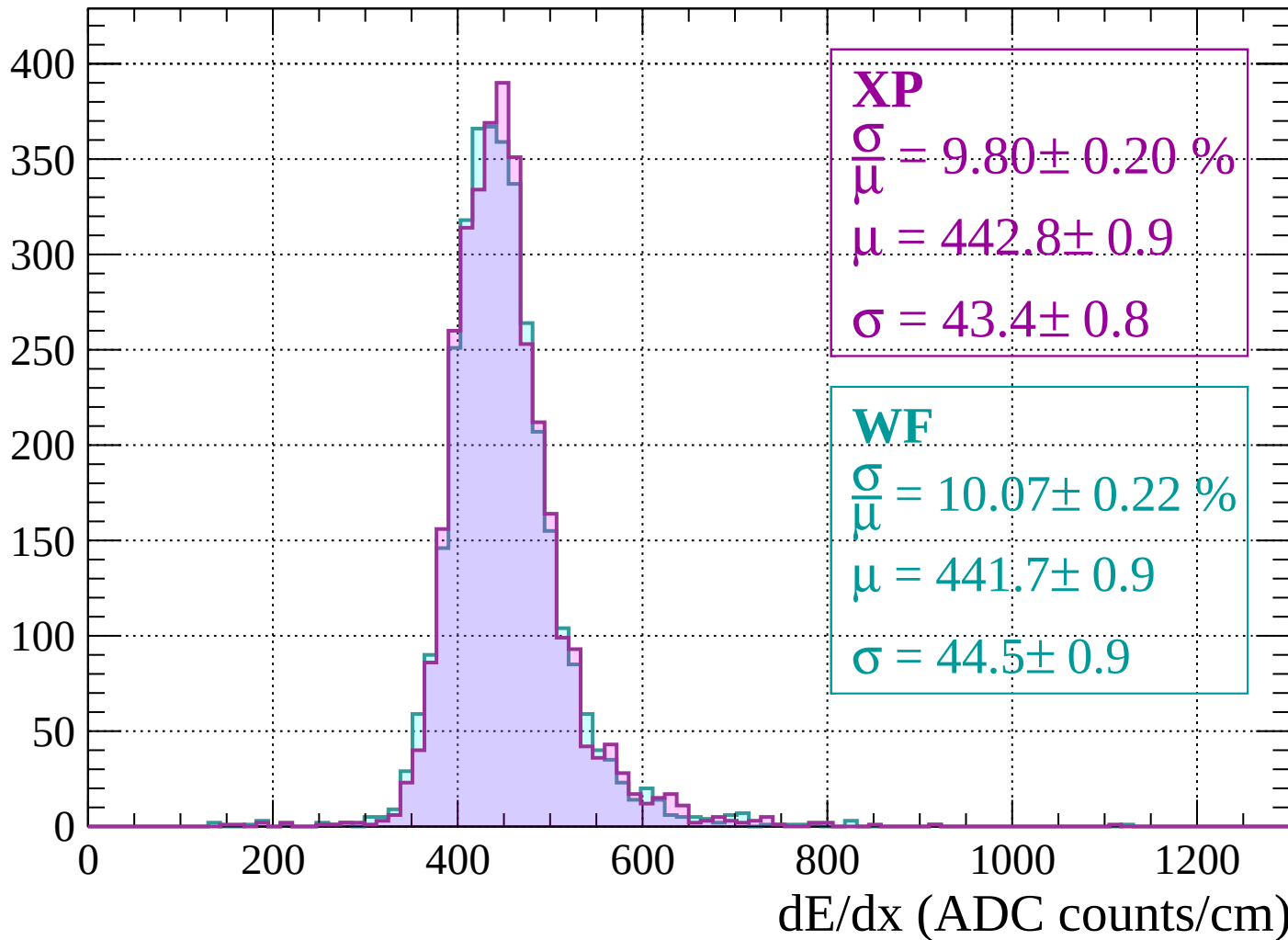
dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$



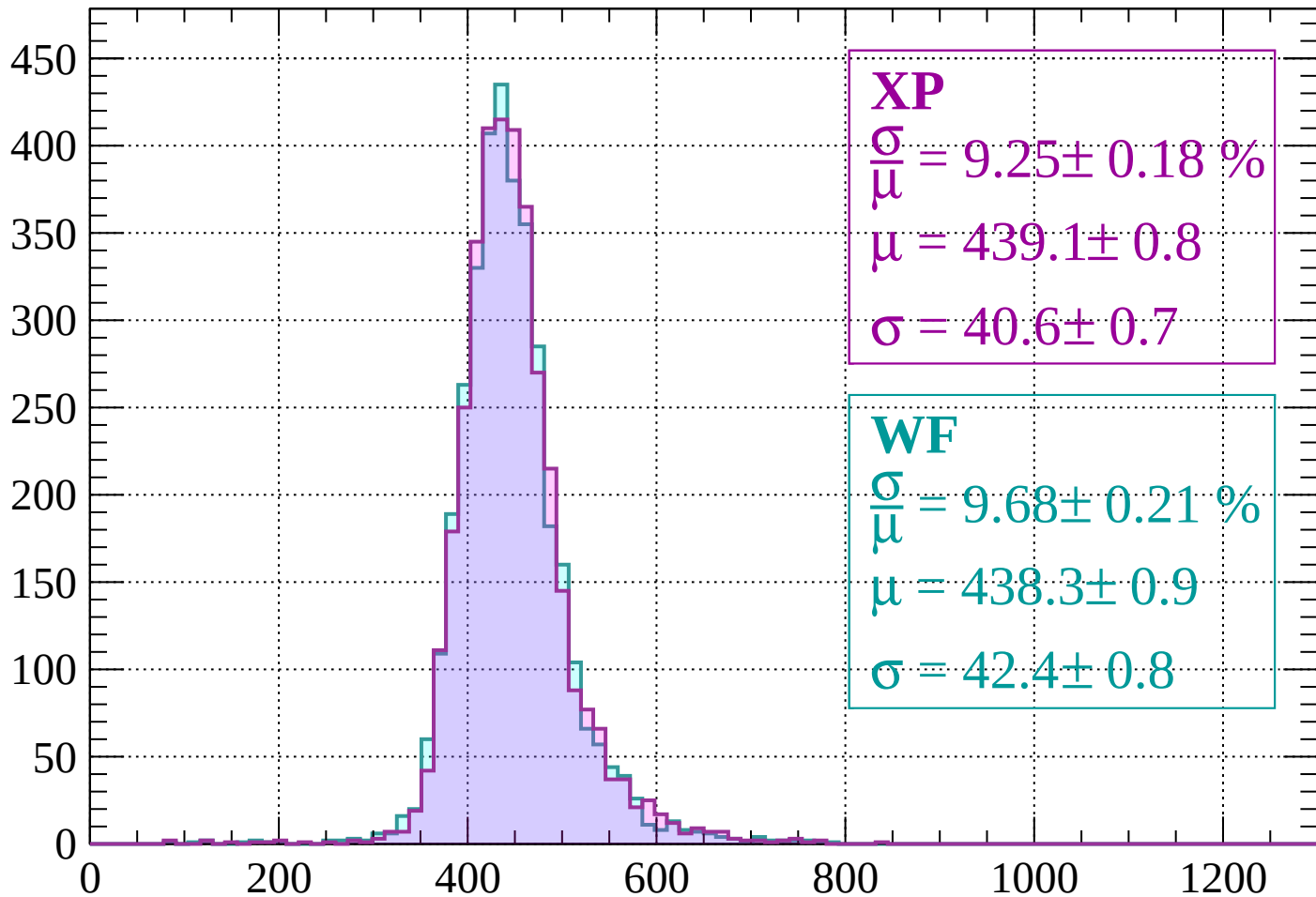
Energy loss | $-1120 < p < -1080$

Count



Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.18 \%$$

$$\mu = 439.1 \pm 0.8$$

$$\sigma = 40.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.21 \%$$

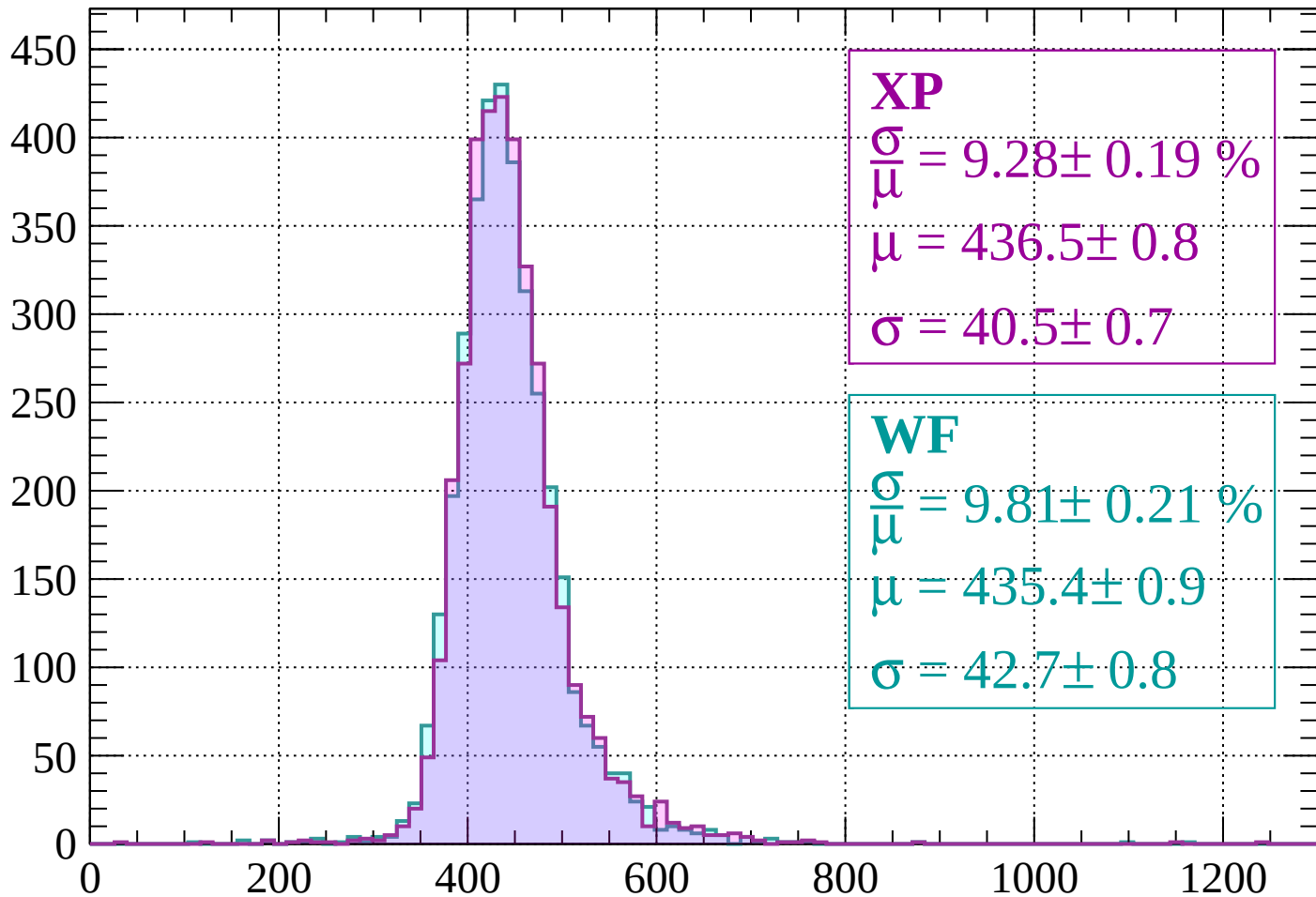
$$\mu = 438.3 \pm 0.9$$

$$\sigma = 42.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 0.19 \%$$

$$\mu = 436.5 \pm 0.8$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.21 \%$$

$$\mu = 435.4 \pm 0.9$$

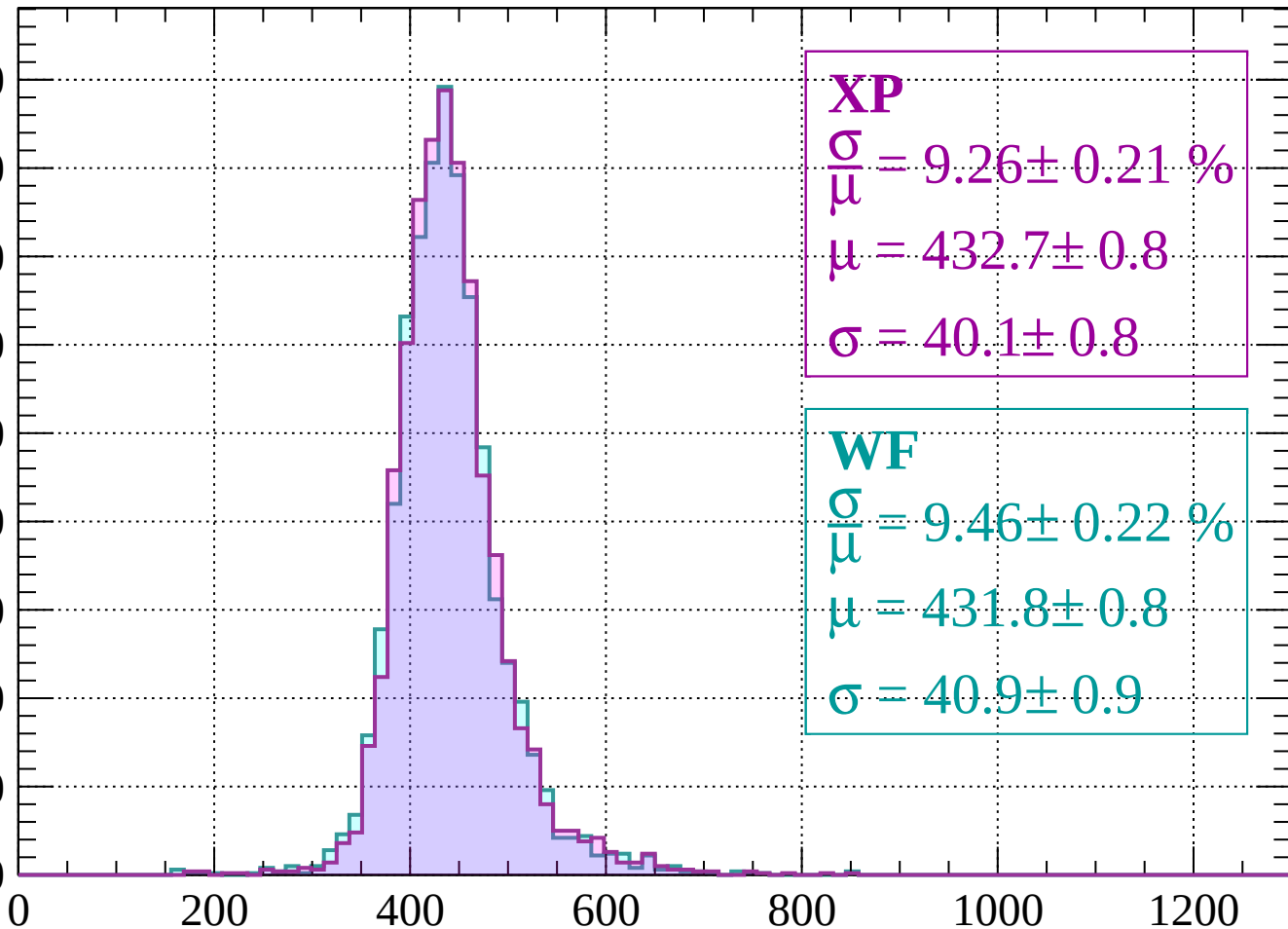
$$\sigma = 42.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.21 \%$$

$$\mu = 432.7 \pm 0.8$$

$$\sigma = 40.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

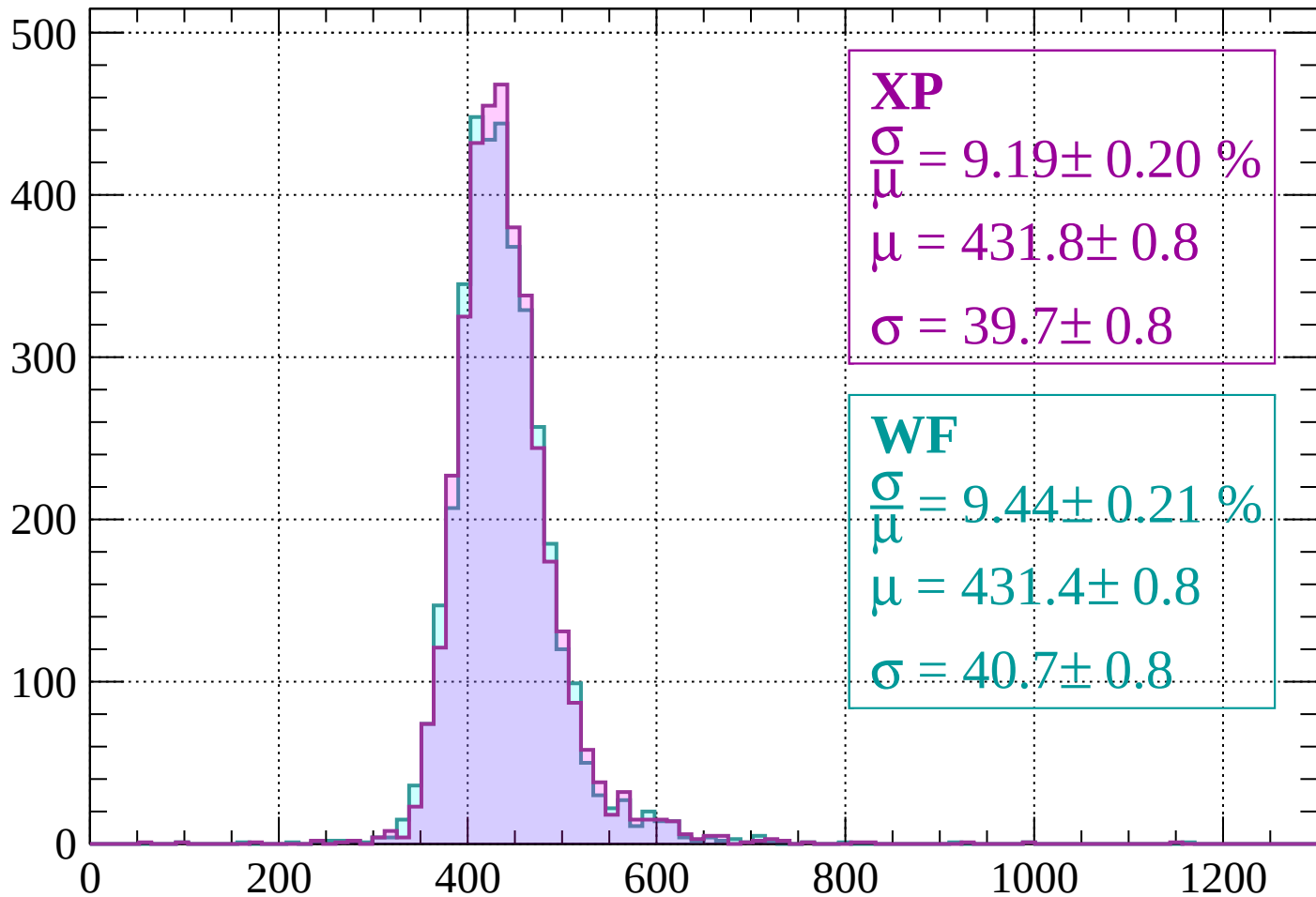
$$\mu = 431.8 \pm 0.8$$

$$\sigma = 40.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.20 \%$$

$$\mu = 431.8 \pm 0.8$$

$$\sigma = 39.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.21 \%$$

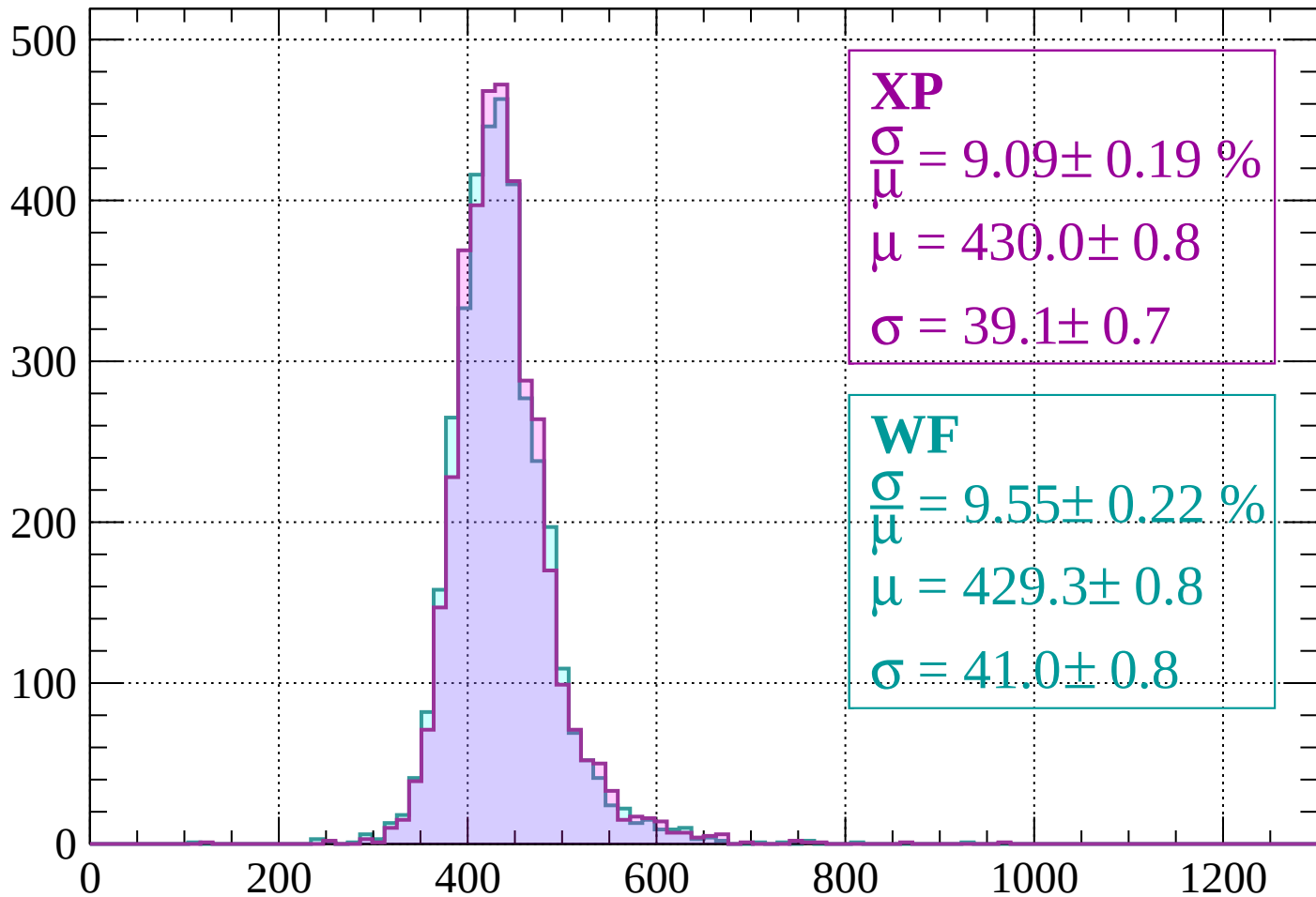
$$\mu = 431.4 \pm 0.8$$

$$\sigma = 40.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.09 \pm 0.19 \%$$

$$\mu = 430.0 \pm 0.8$$

$$\sigma = 39.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.22 \%$$

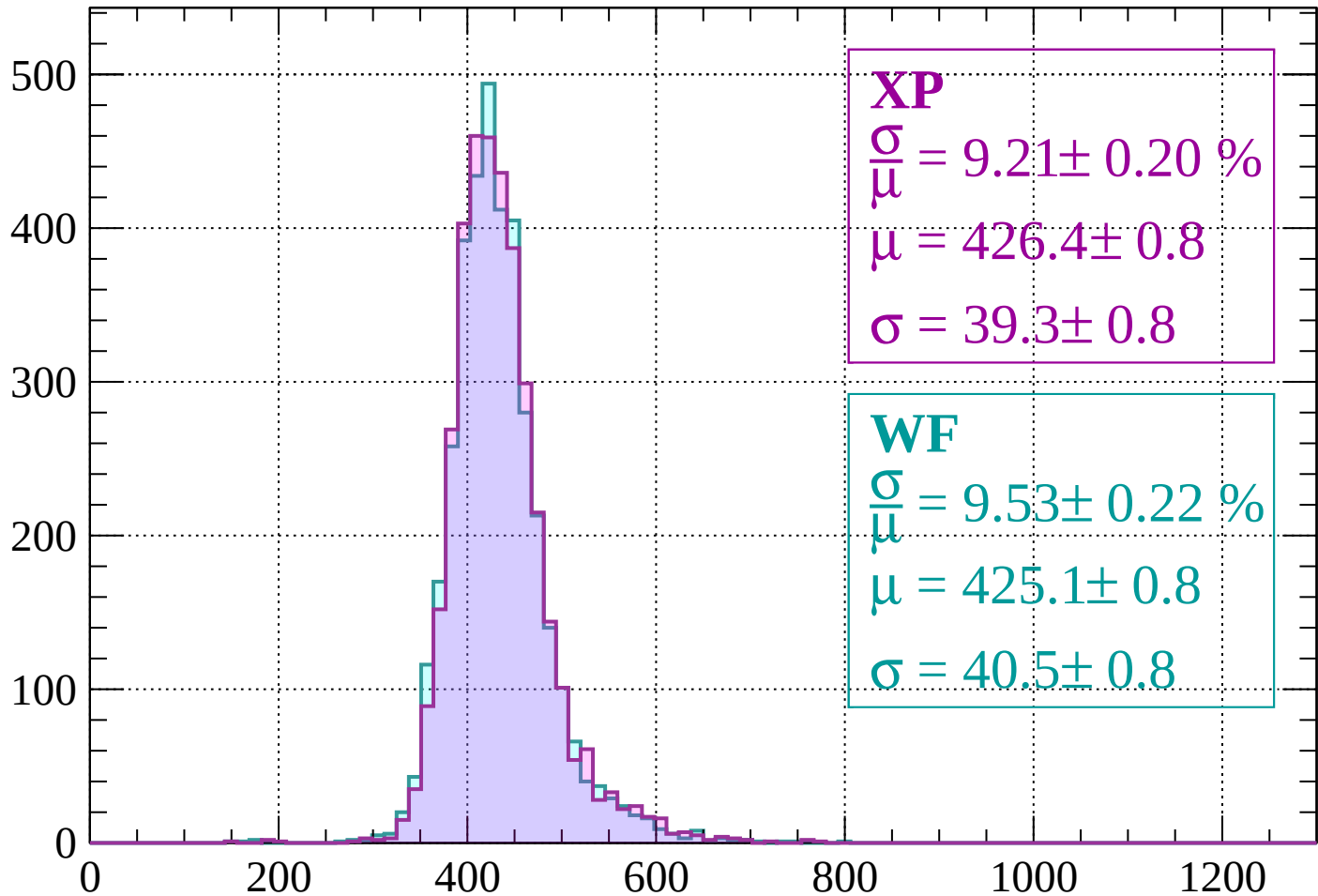
$$\mu = 429.3 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.20 \%$$

$$\mu = 426.4 \pm 0.8$$

$$\sigma = 39.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.53 \pm 0.22 \%$$

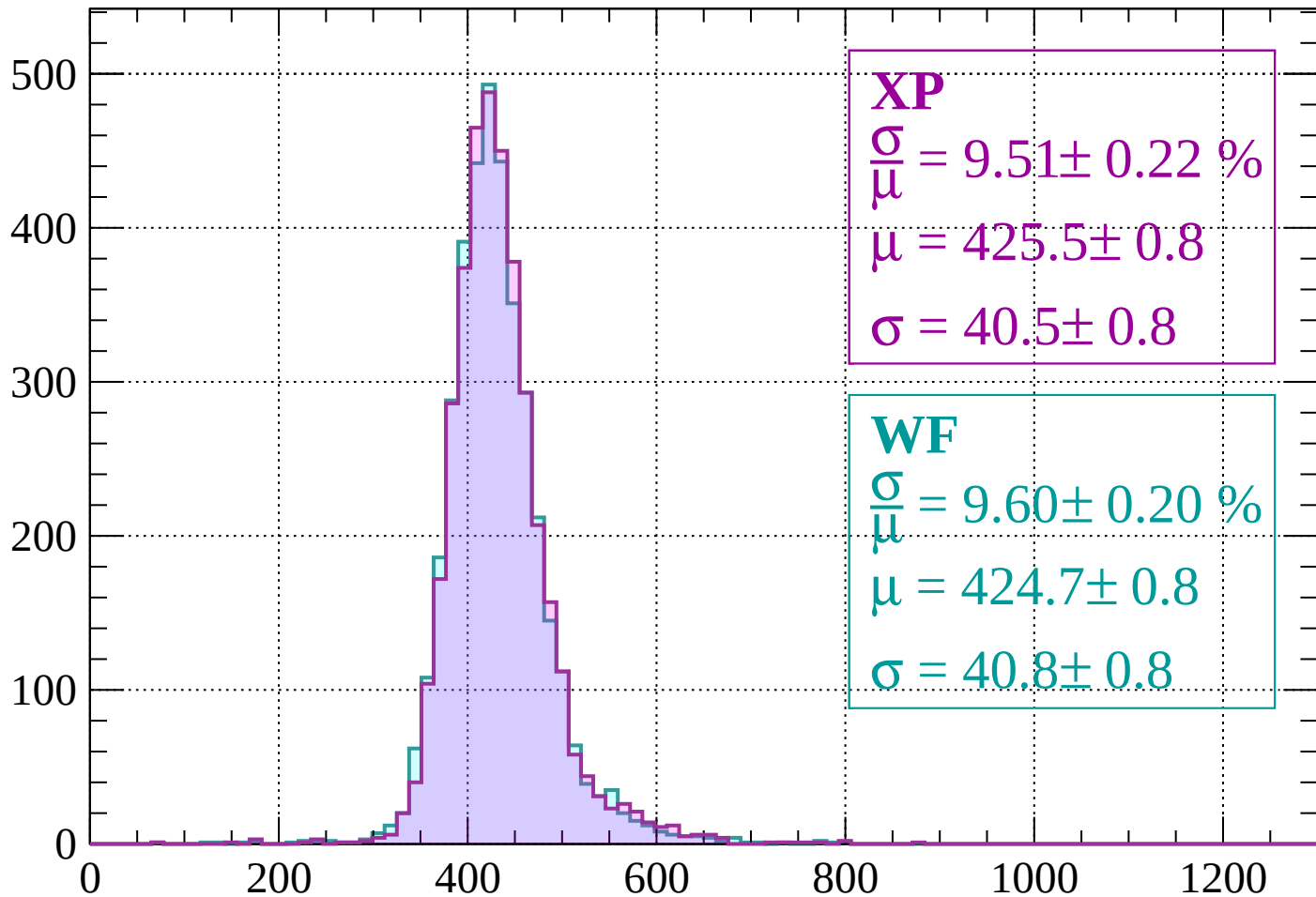
$$\mu = 425.1 \pm 0.8$$

$$\sigma = 40.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 9.51 \pm 0.22 \%$$

$$\mu = 425.5 \pm 0.8$$

$$\sigma = 40.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.20 \%$$

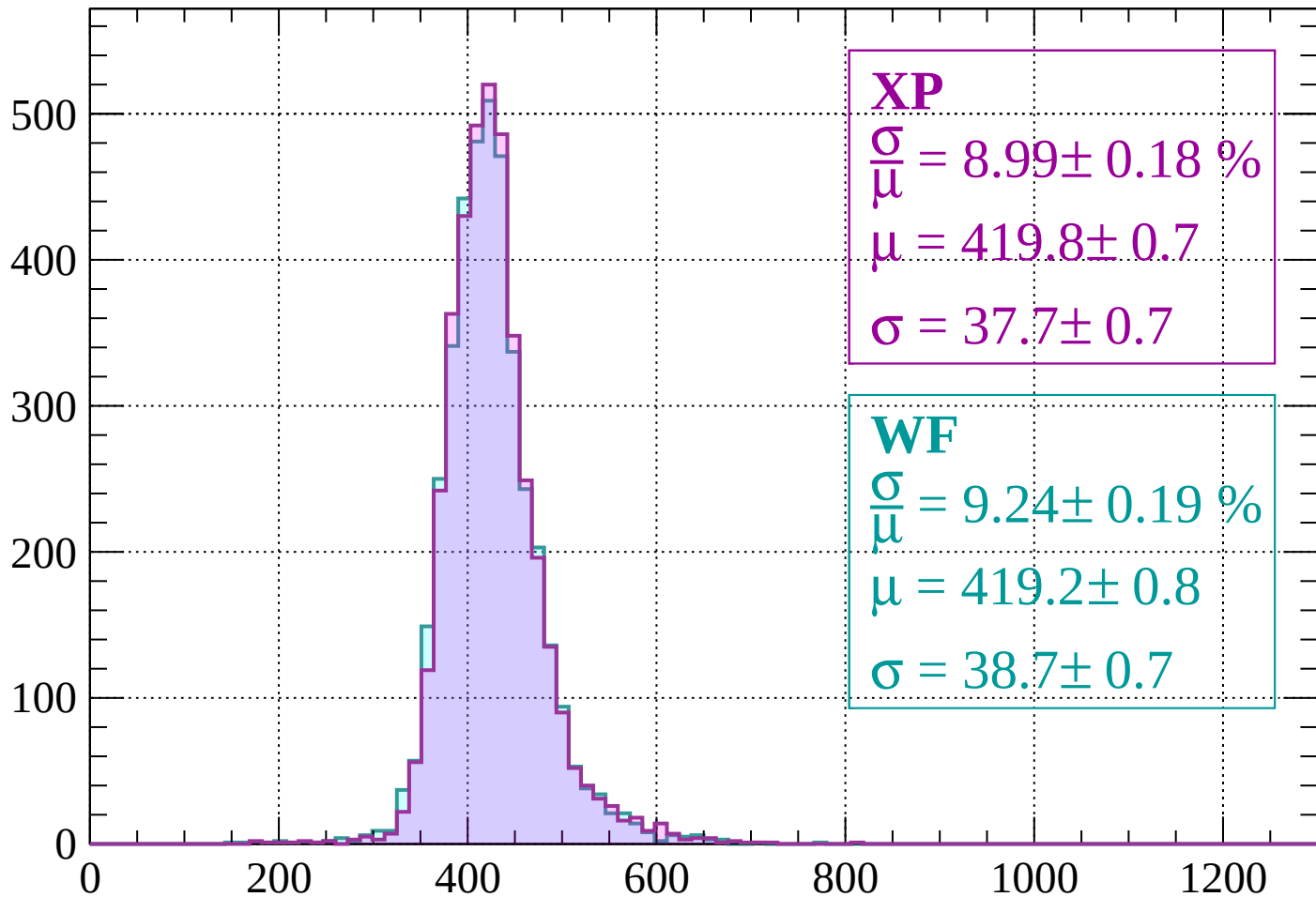
$$\mu = 424.7 \pm 0.8$$

$$\sigma = 40.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.18 \%$$

$$\mu = 419.8 \pm 0.7$$

$$\sigma = 37.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.19 \%$$

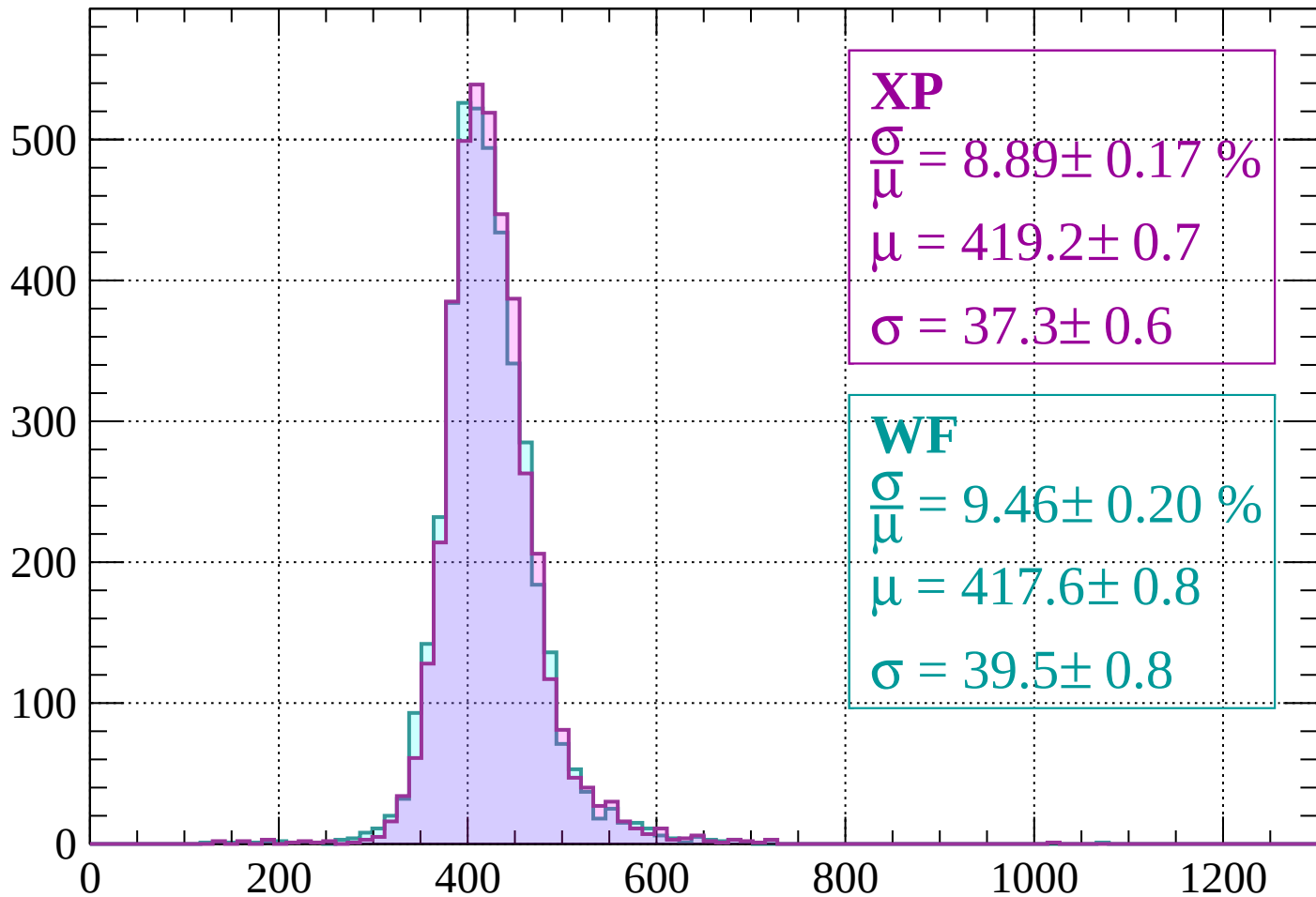
$$\mu = 419.2 \pm 0.8$$

$$\sigma = 38.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 419.2 \pm 0.7$$

$$\sigma = 37.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

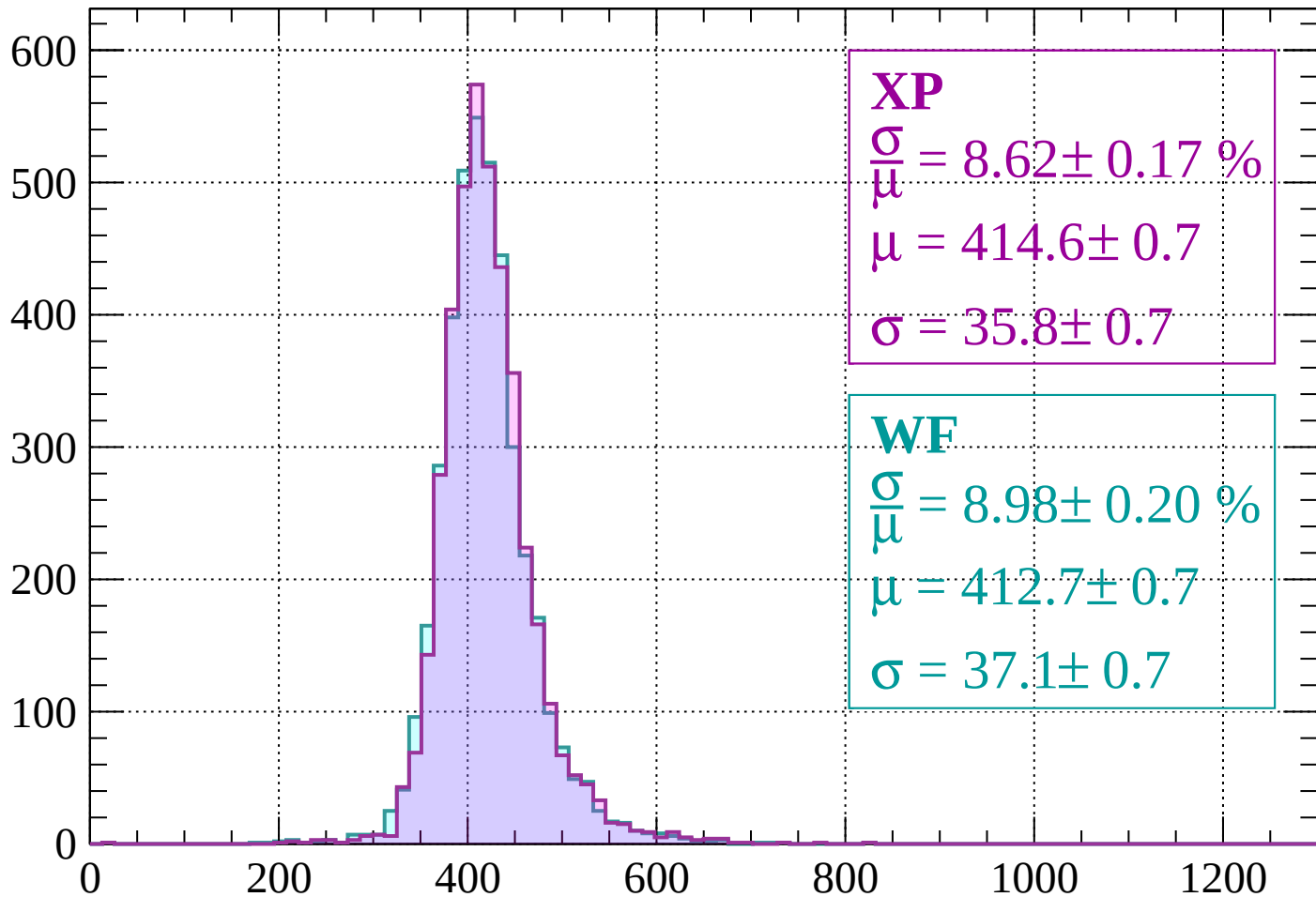
$$\mu = 417.6 \pm 0.8$$

$$\sigma = 39.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.17 \%$$

$$\mu = 414.6 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

WF

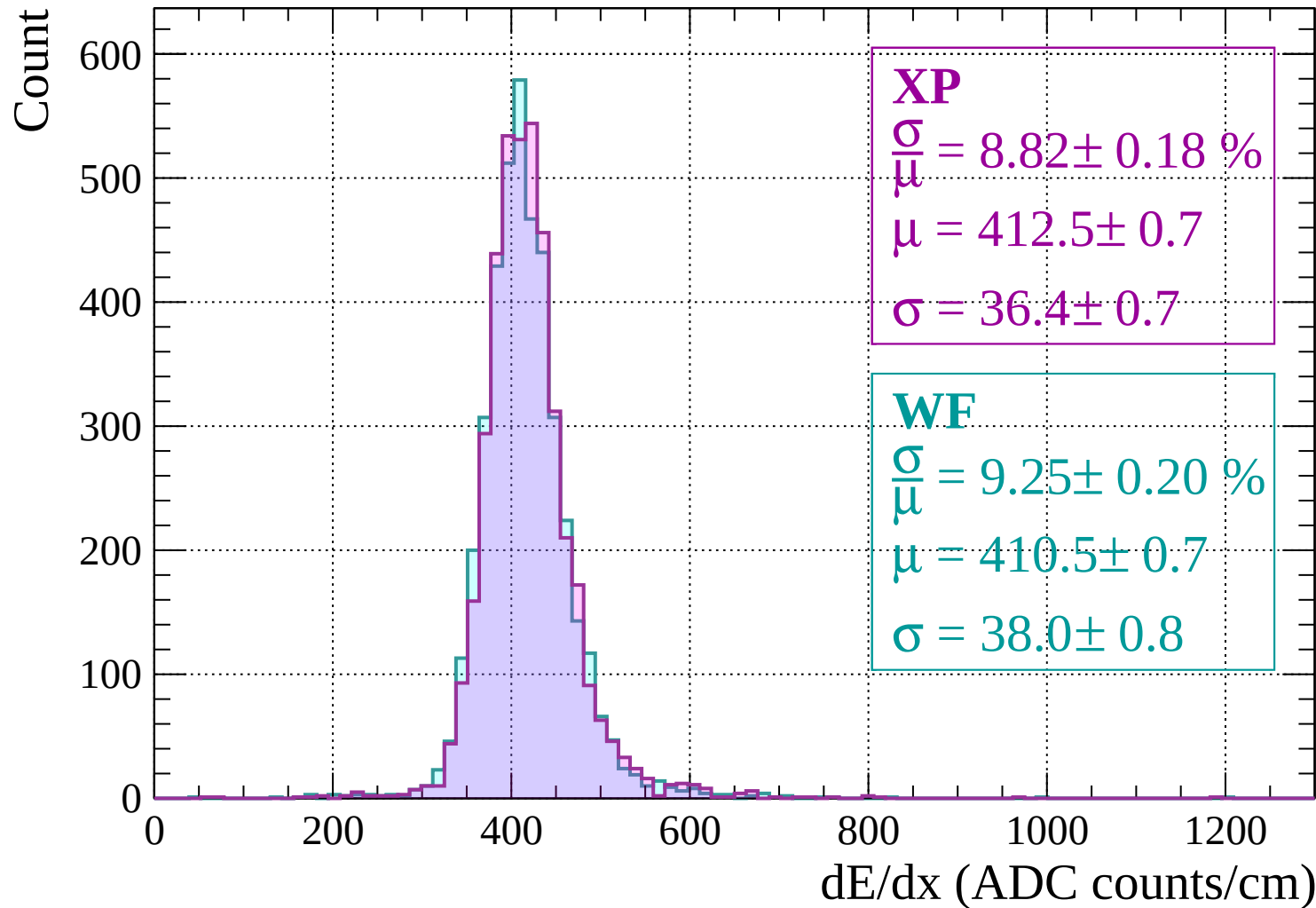
$$\frac{\sigma}{\mu} = 8.98 \pm 0.20 \%$$

$$\mu = 412.7 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

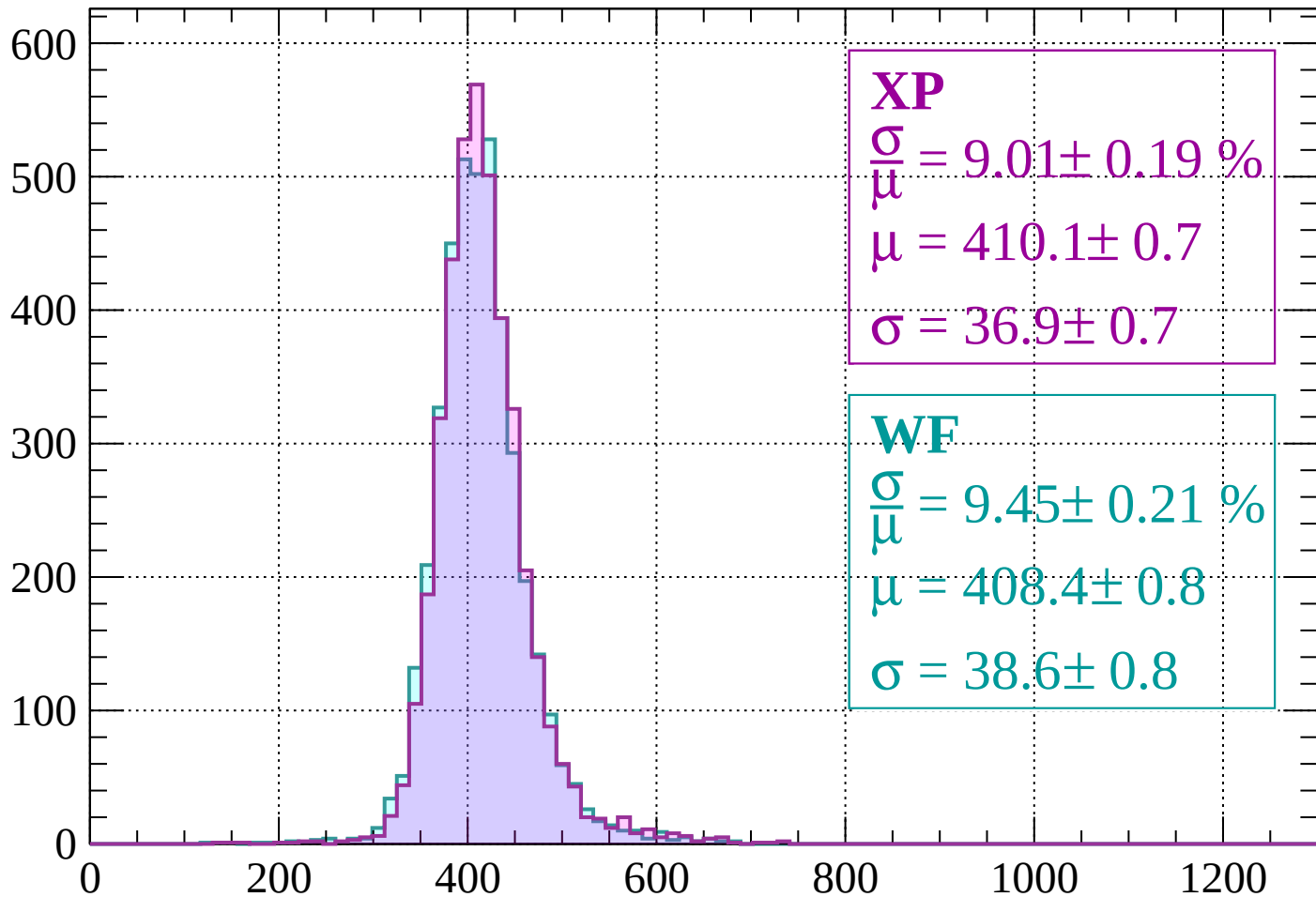
dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.19 \%$$

$$\mu = 410.1 \pm 0.7$$

$$\sigma = 36.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.45 \pm 0.21 \%$$

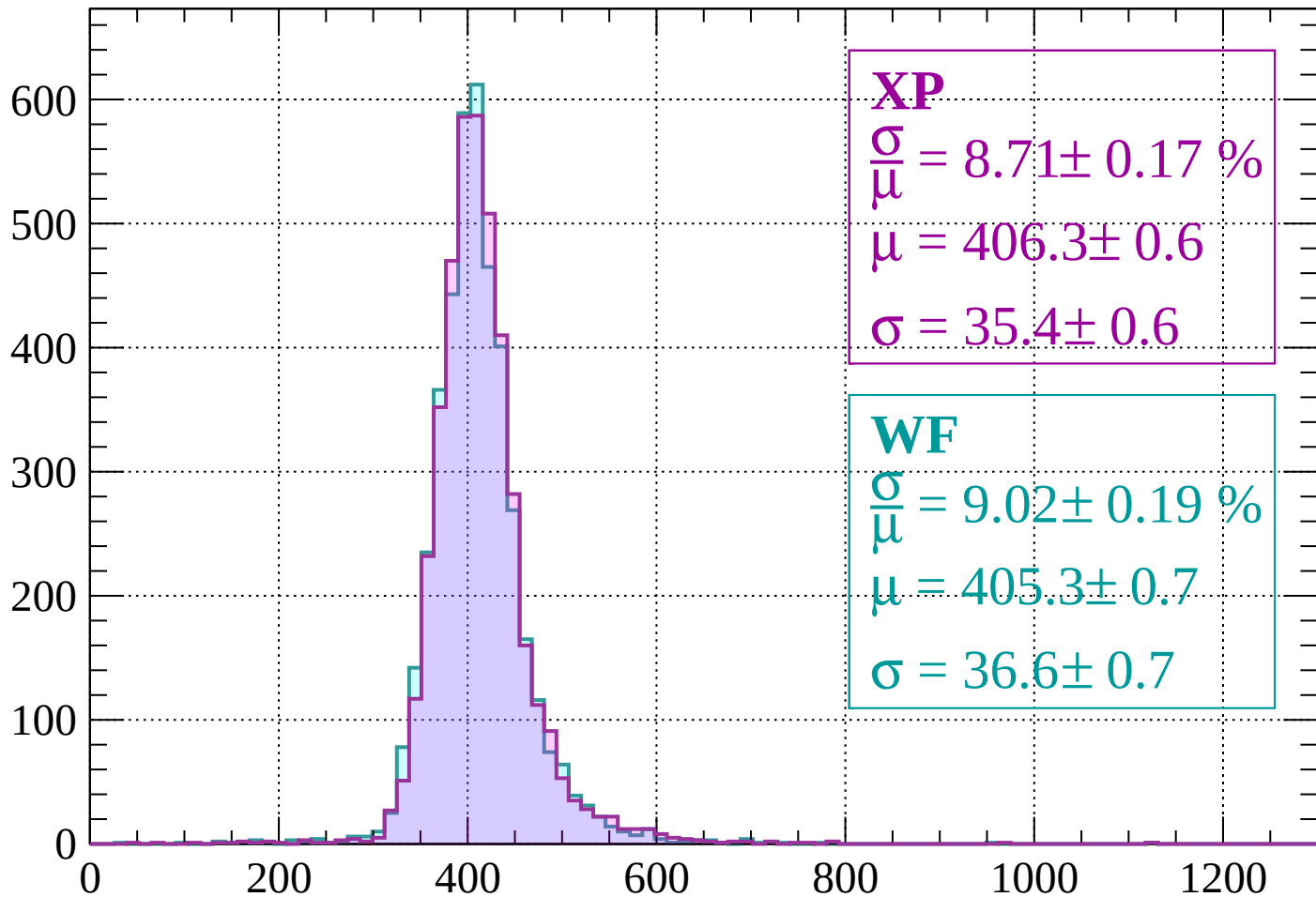
$$\mu = 408.4 \pm 0.8$$

$$\sigma = 38.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.71 \pm 0.17 \%$$

$$\mu = 406.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.02 \pm 0.19 \%$$

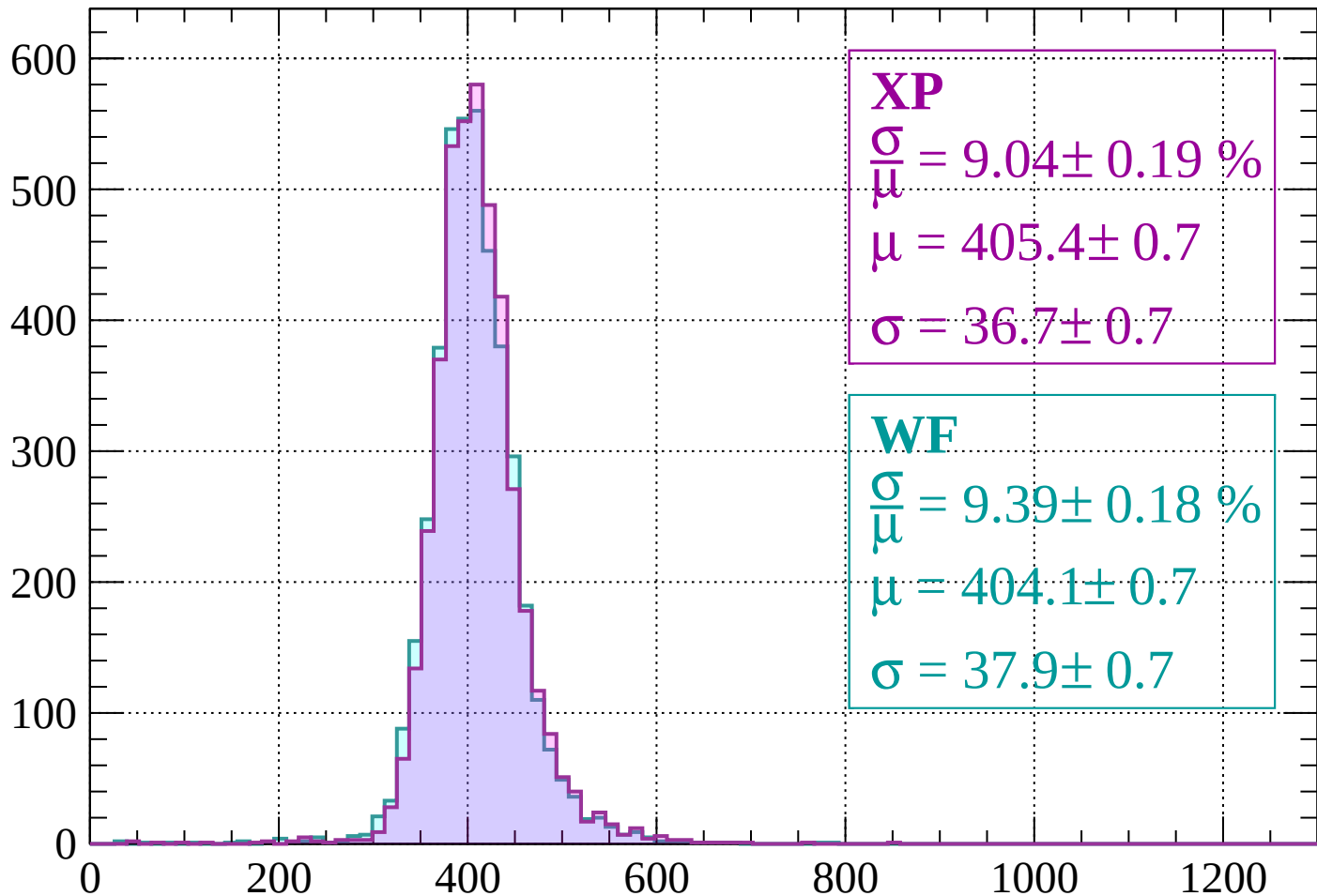
$$\mu = 405.3 \pm 0.7$$

$$\sigma = 36.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$$

$$\mu = 405.4 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.39 \pm 0.18 \%$$

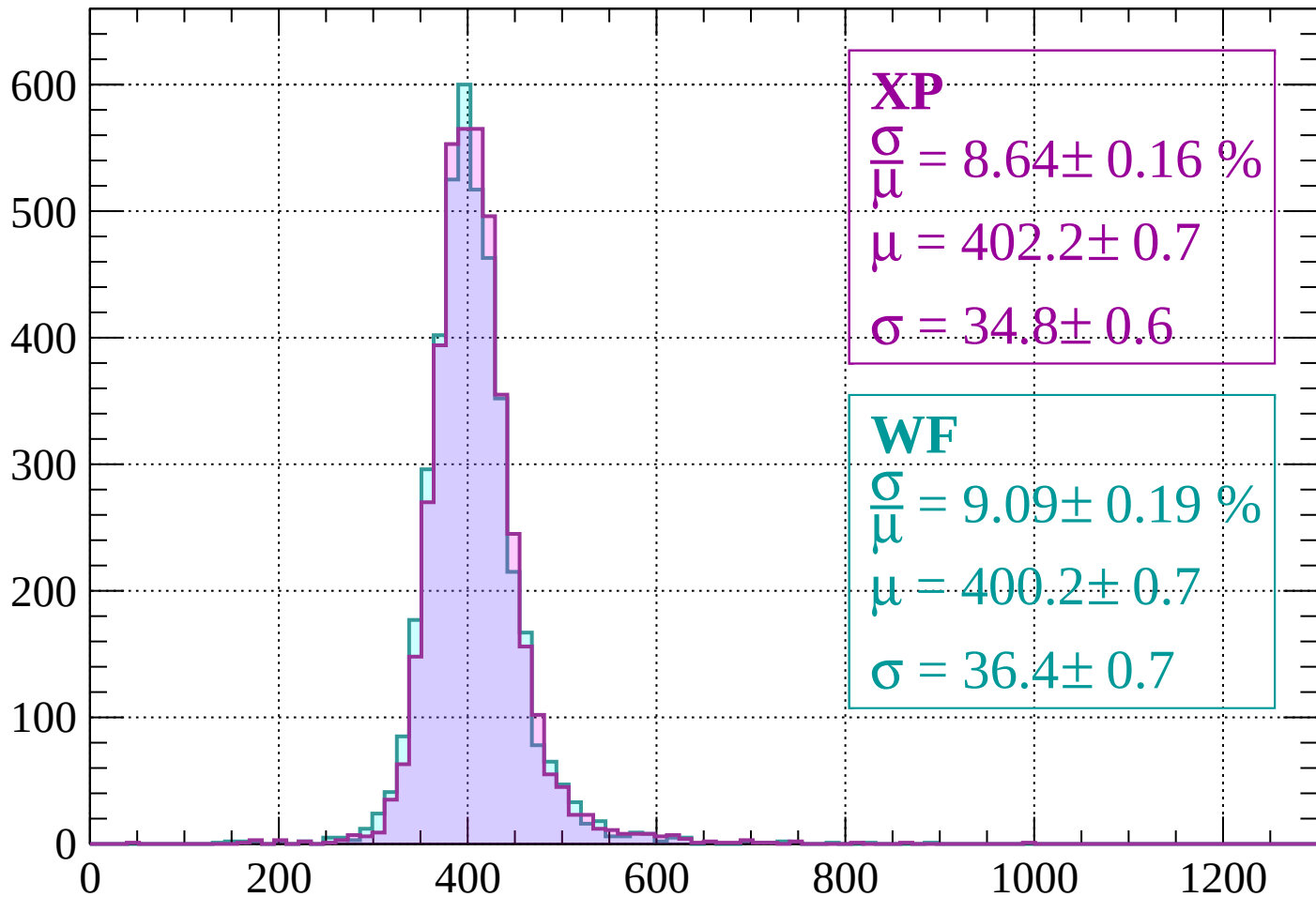
$$\mu = 404.1 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.64 \pm 0.16 \%$$

$$\mu = 402.2 \pm 0.7$$

$$\sigma = 34.8 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.09 \pm 0.19 \%$$

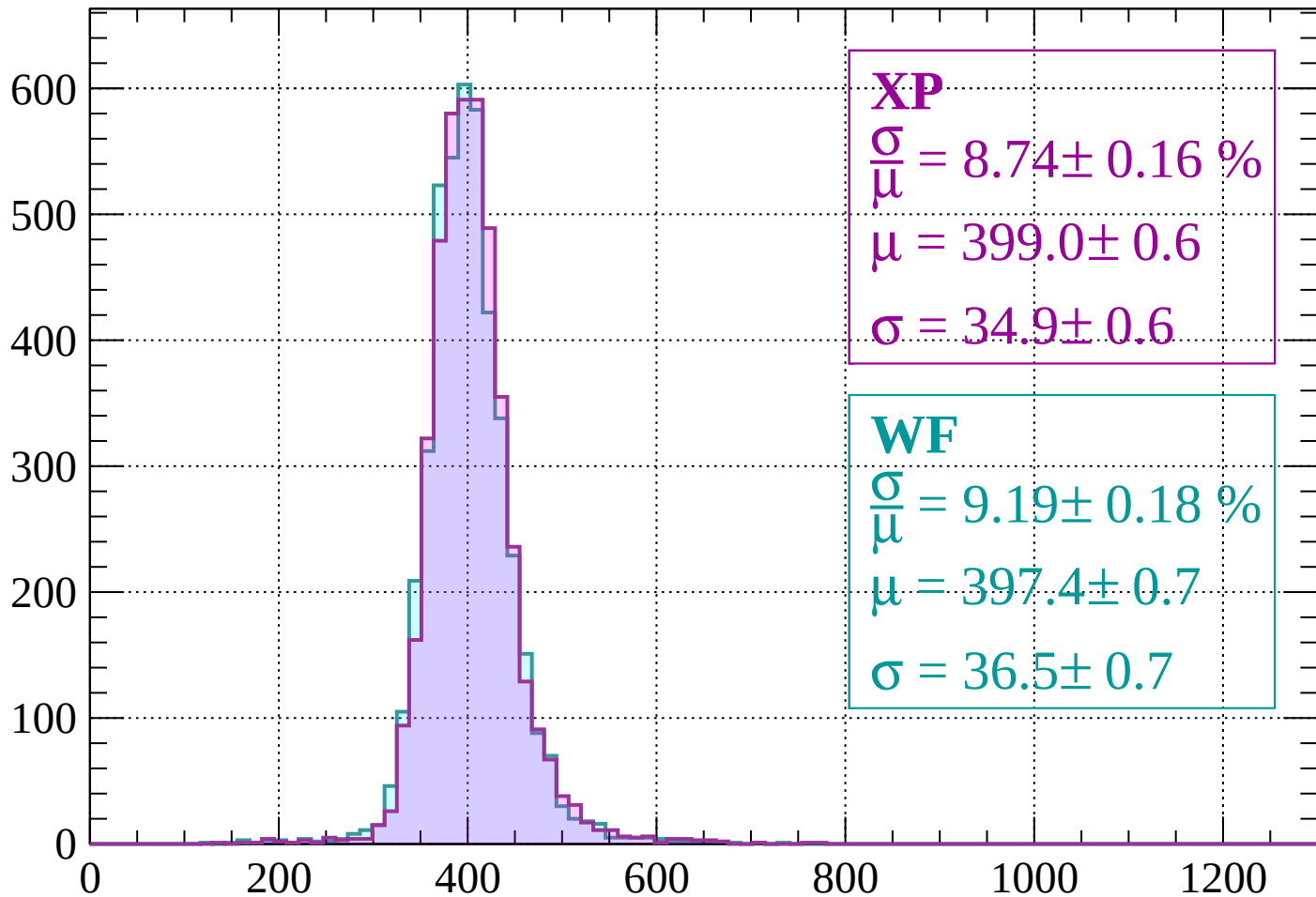
$$\mu = 400.2 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.16 \%$$

$$\mu = 399.0 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.18 \%$$

$$\mu = 397.4 \pm 0.7$$

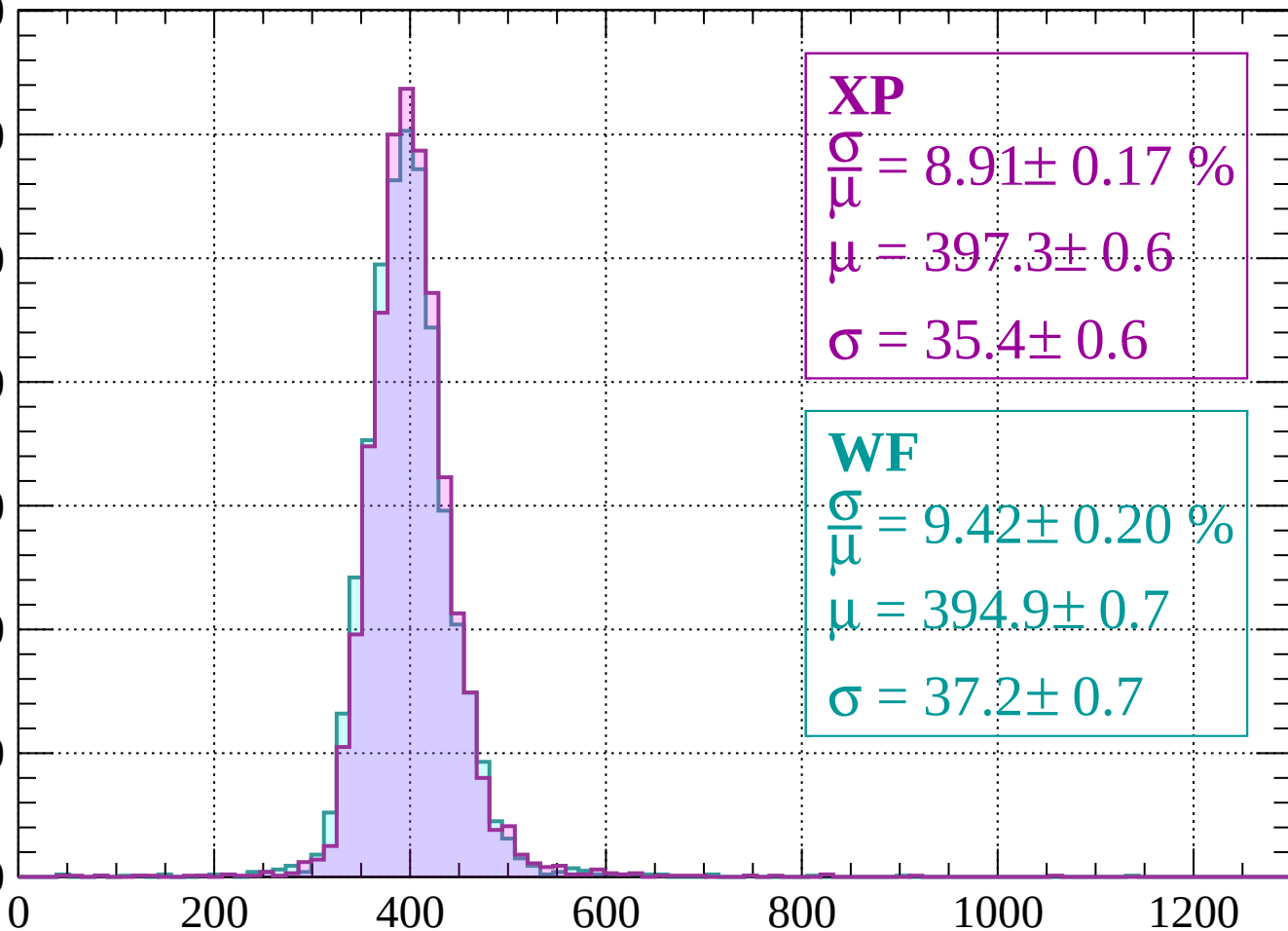
$$\sigma = 36.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.17 \%$$

$$\mu = 397.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.20 \%$$

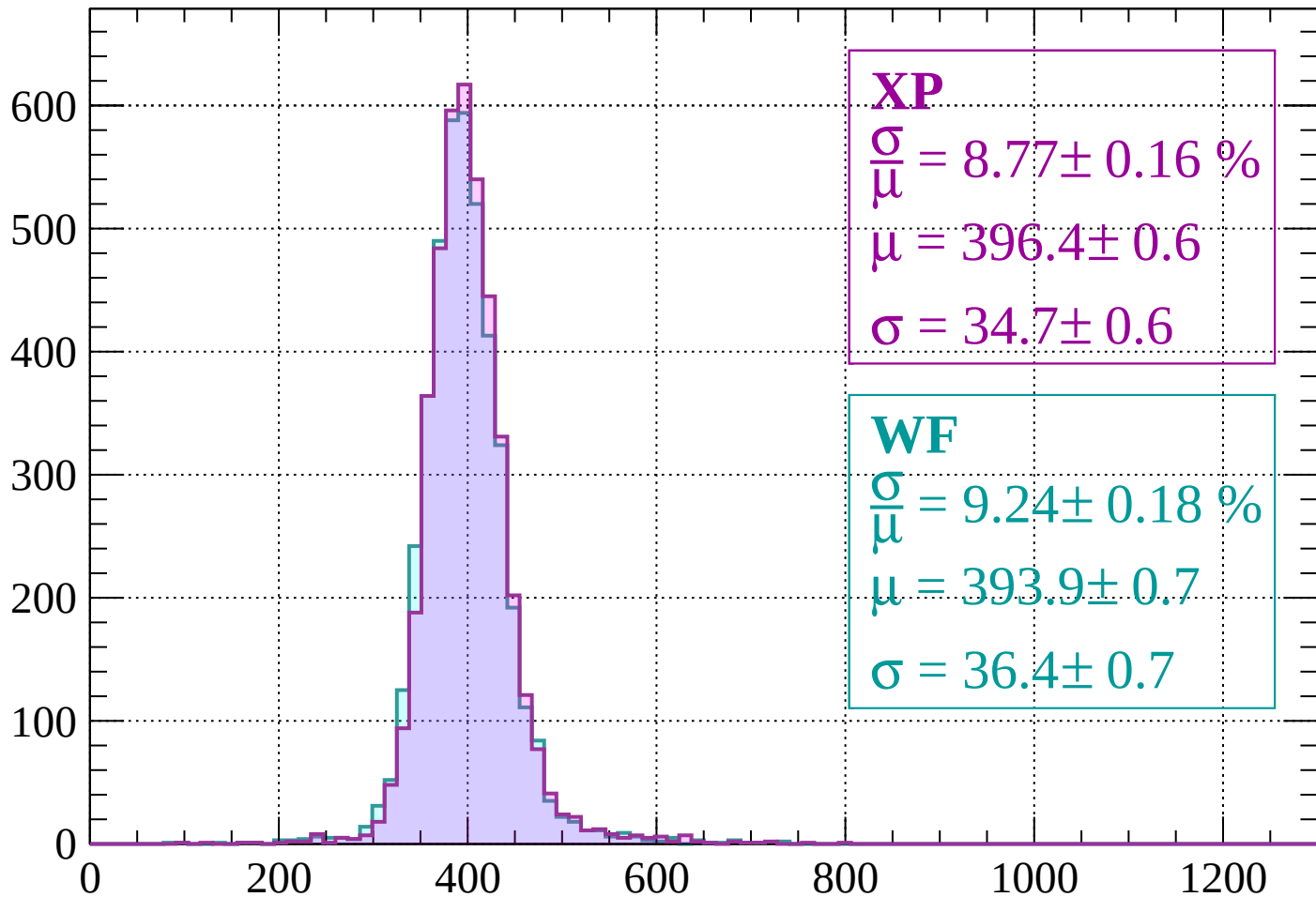
$$\mu = 394.9 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.16 \%$$

$$\mu = 396.4 \pm 0.6$$

$$\sigma = 34.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.18 \%$$

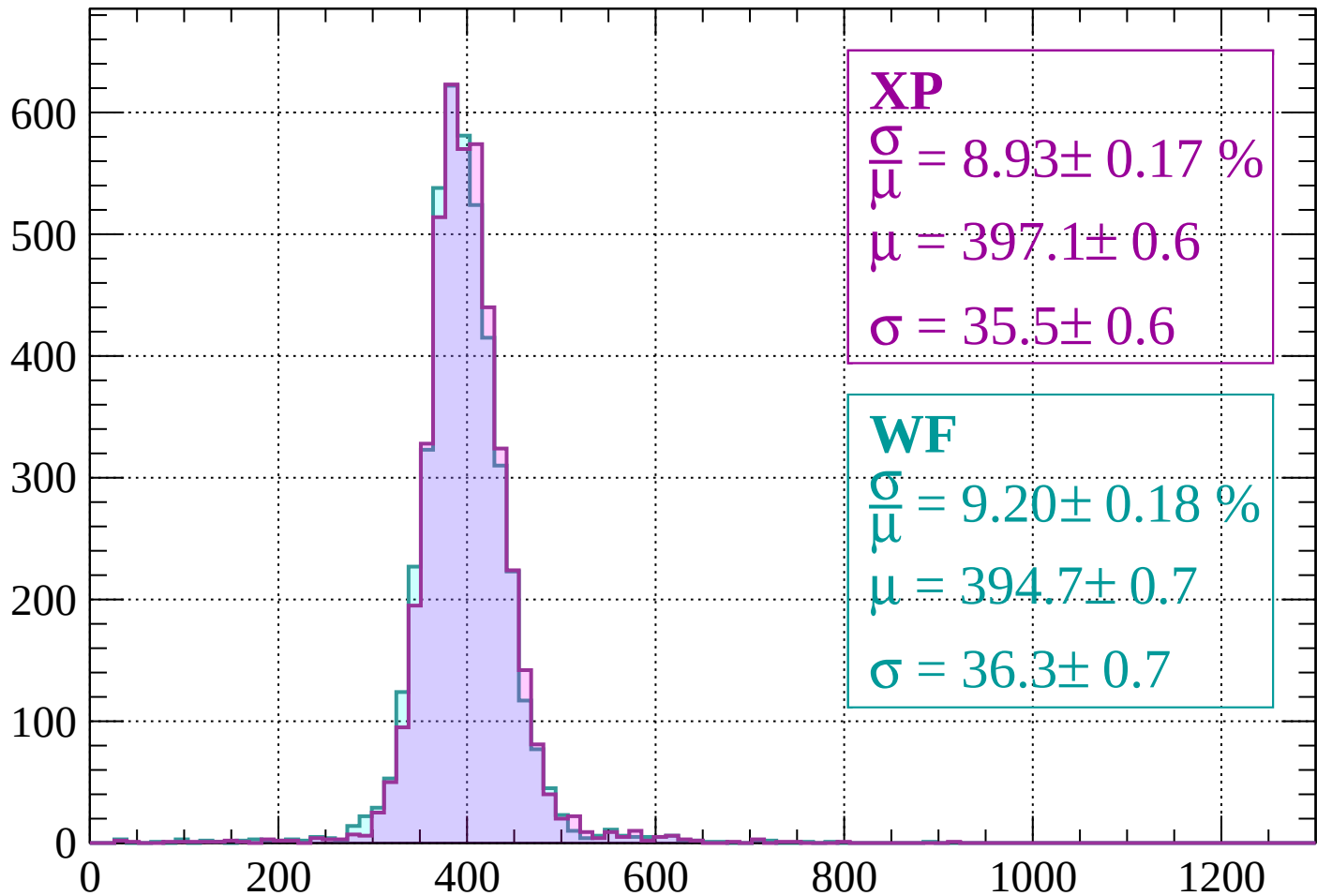
$$\mu = 393.9 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.93 \pm 0.17 \%$$

$$\mu = 397.1 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.18 \%$$

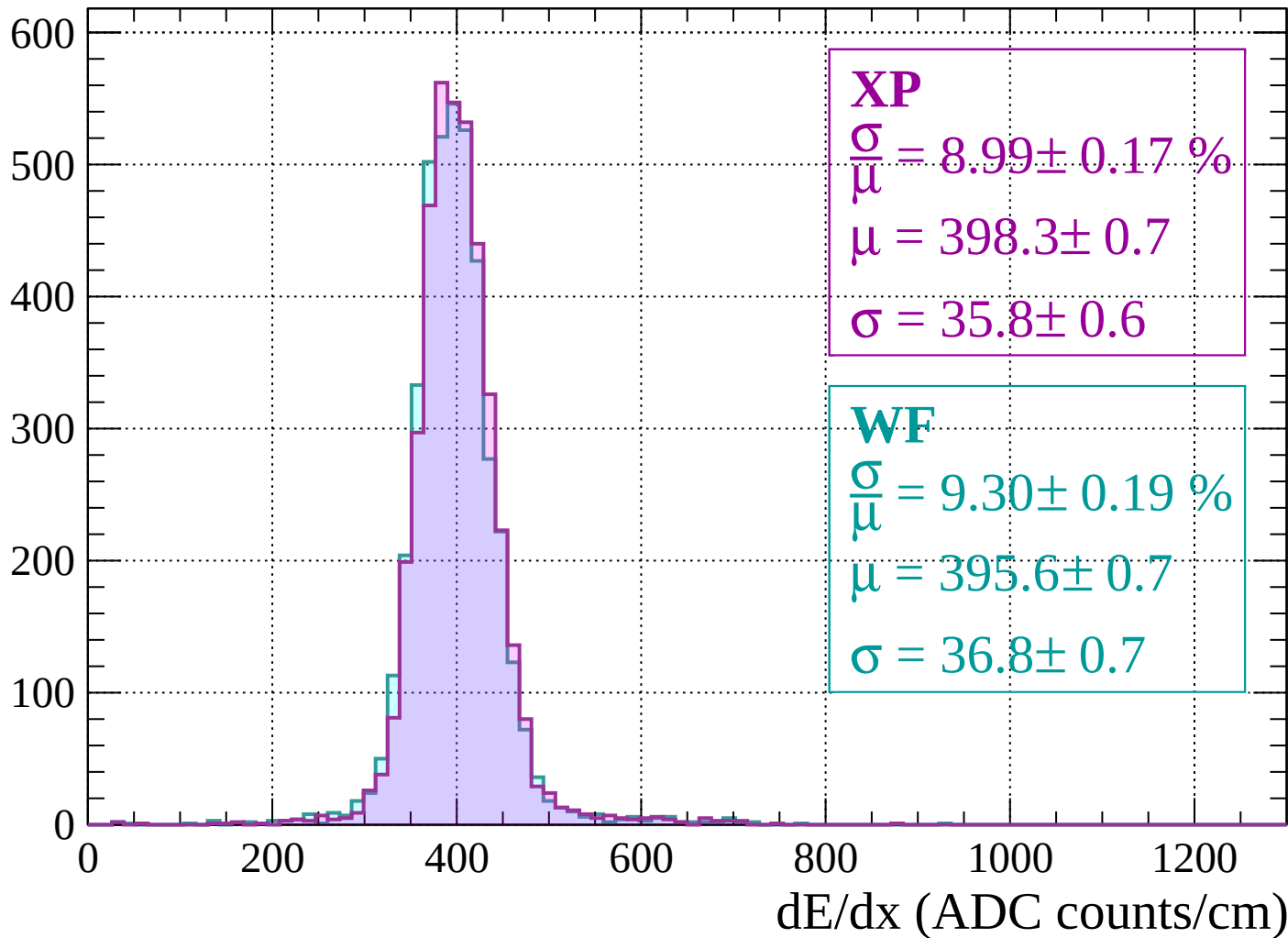
$$\mu = 394.7 \pm 0.7$$

$$\sigma = 36.3 \pm 0.7$$

dE/dx (ADC counts/cm)

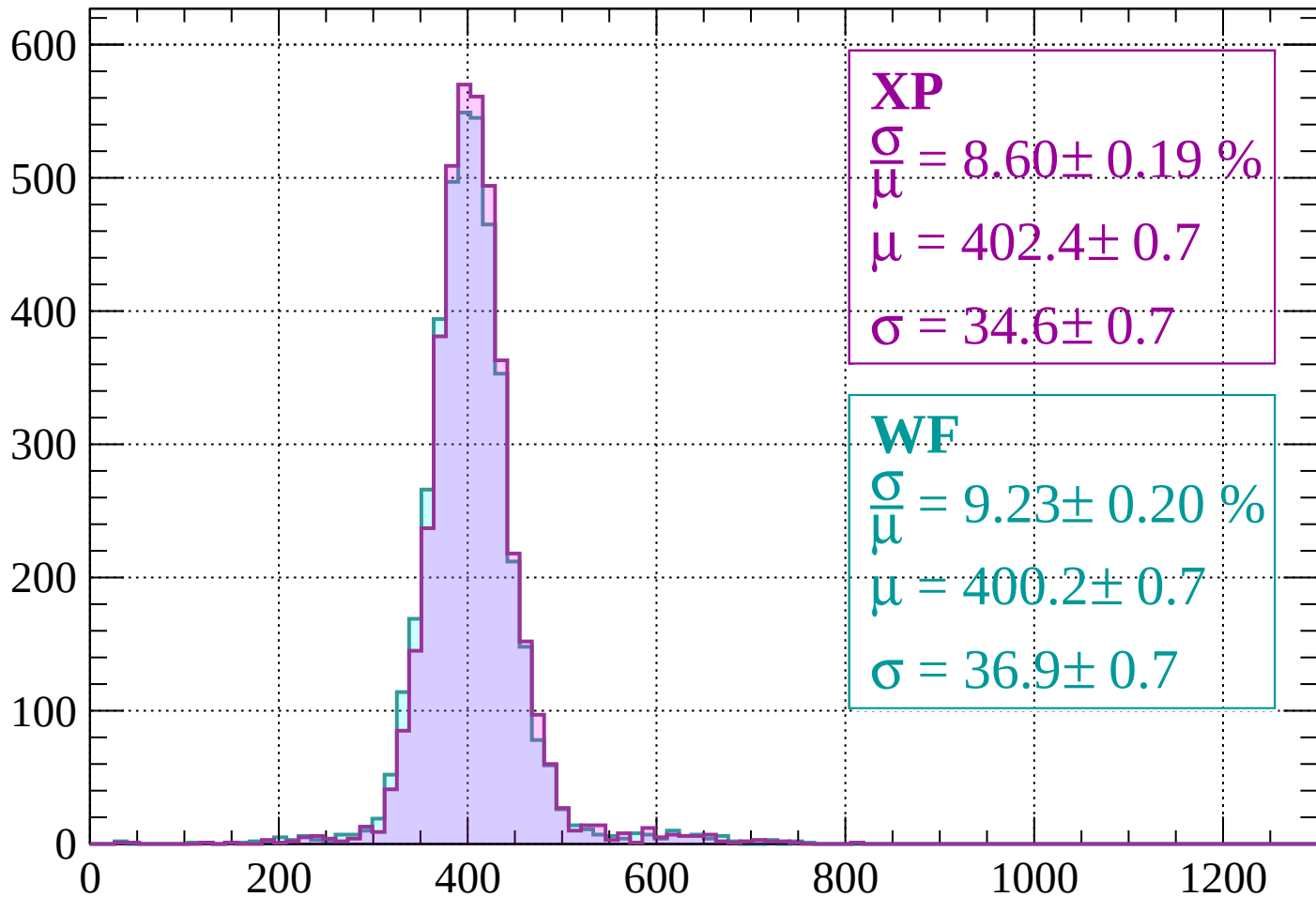
Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.60 \pm 0.19 \%$$

$$\mu = 402.4 \pm 0.7$$

$$\sigma = 34.6 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.23 \pm 0.20 \%$$

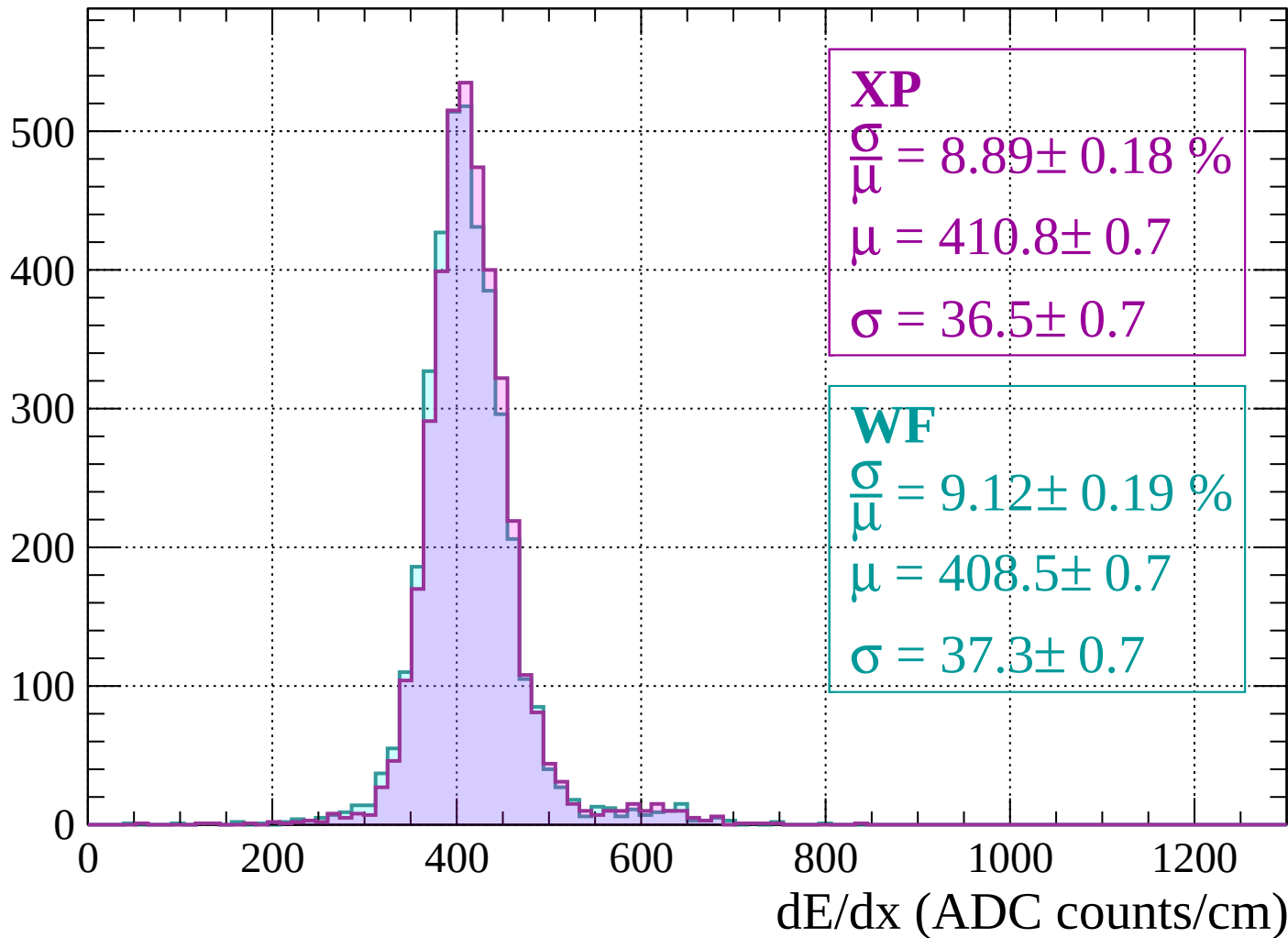
$$\mu = 400.2 \pm 0.7$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

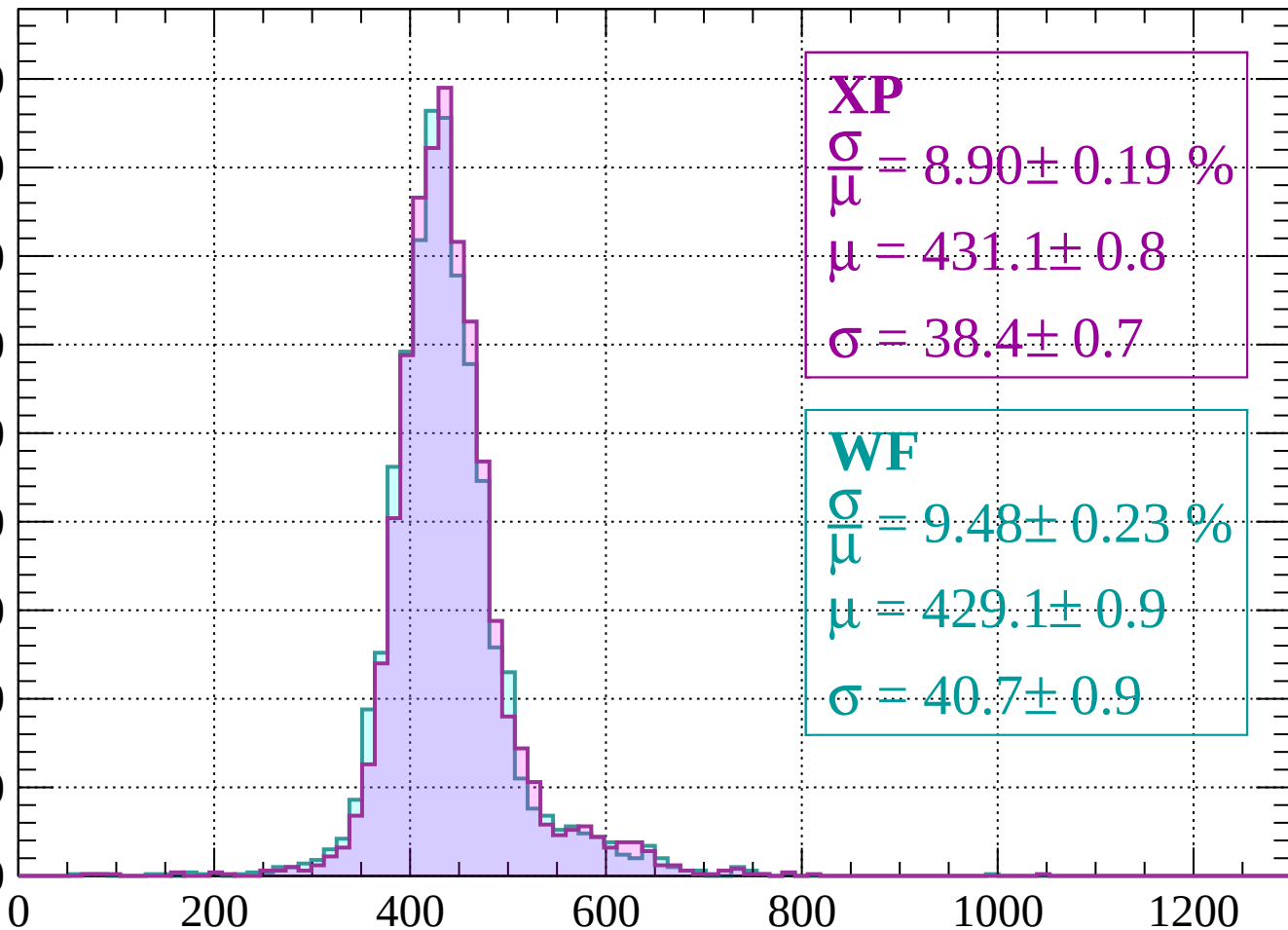
Count



Energy loss | $-200 < p < -160$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.19 \%$$

$$\mu = 431.1 \pm 0.8$$

$$\sigma = 38.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.23 \%$$

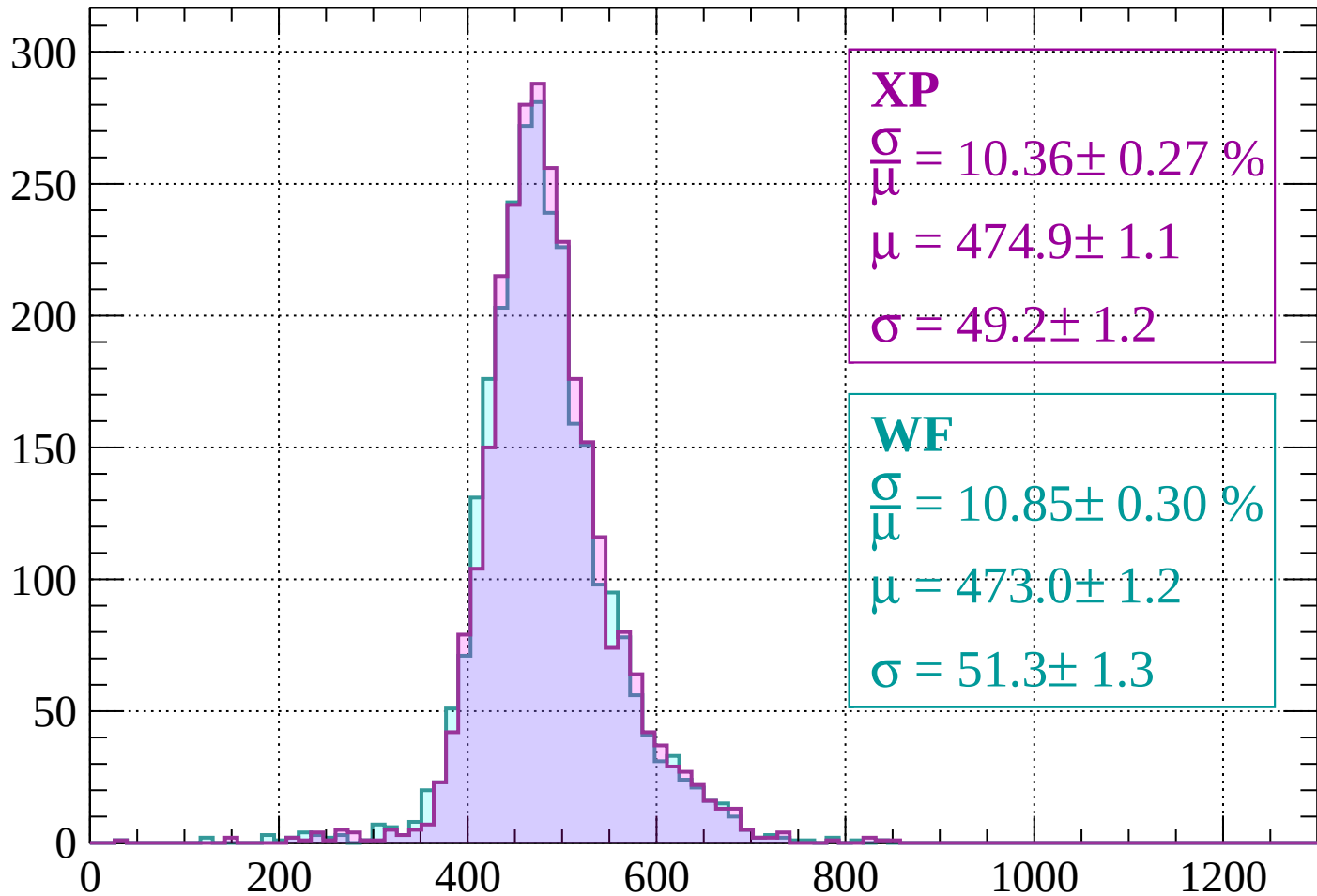
$$\mu = 429.1 \pm 0.9$$

$$\sigma = 40.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -160 < p < -120

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.27 \%$$

$$\mu = 474.9 \pm 1.1$$

$$\sigma = 49.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.85 \pm 0.30 \%$$

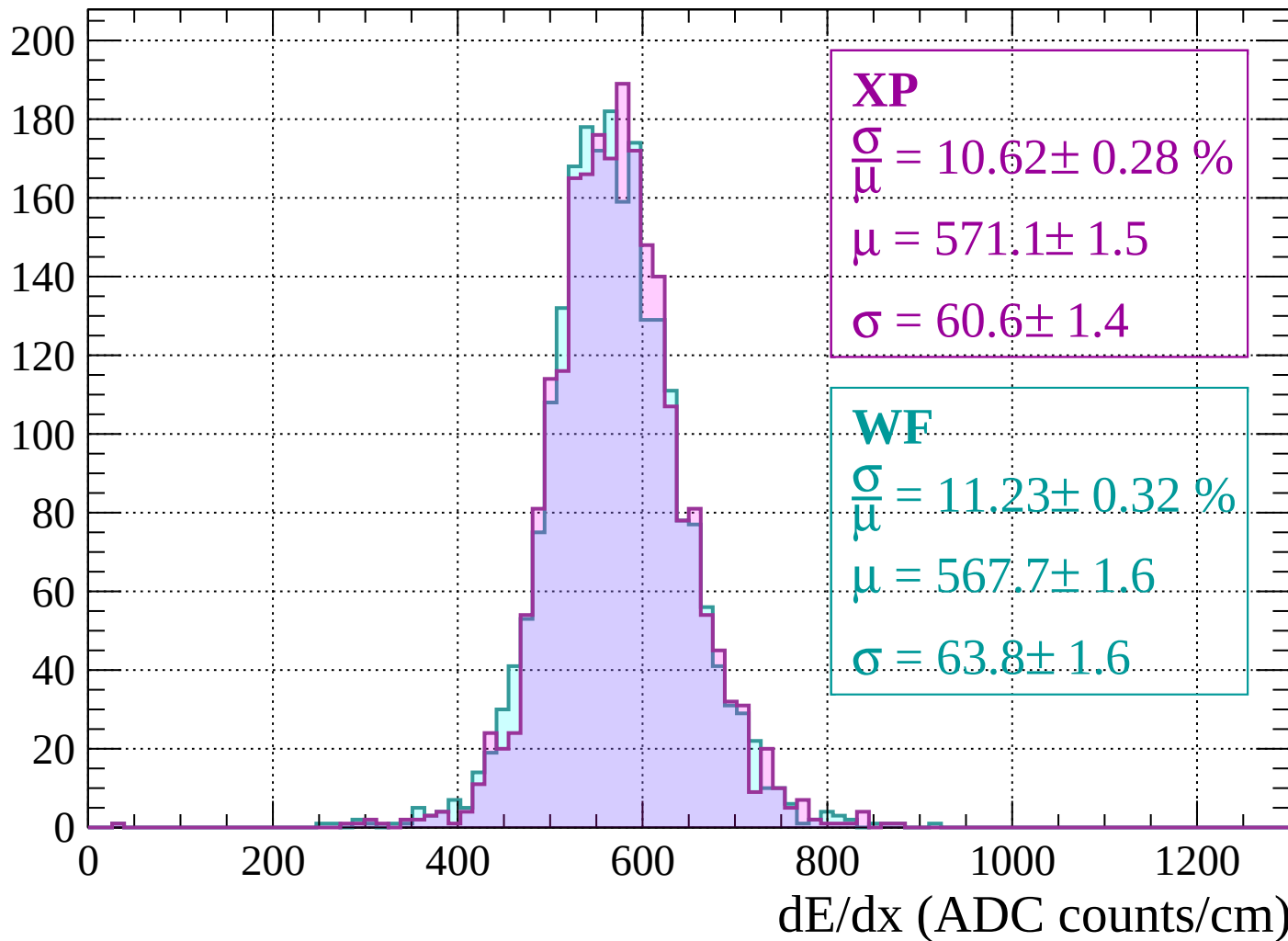
$$\mu = 473.0 \pm 1.2$$

$$\sigma = 51.3 \pm 1.3$$

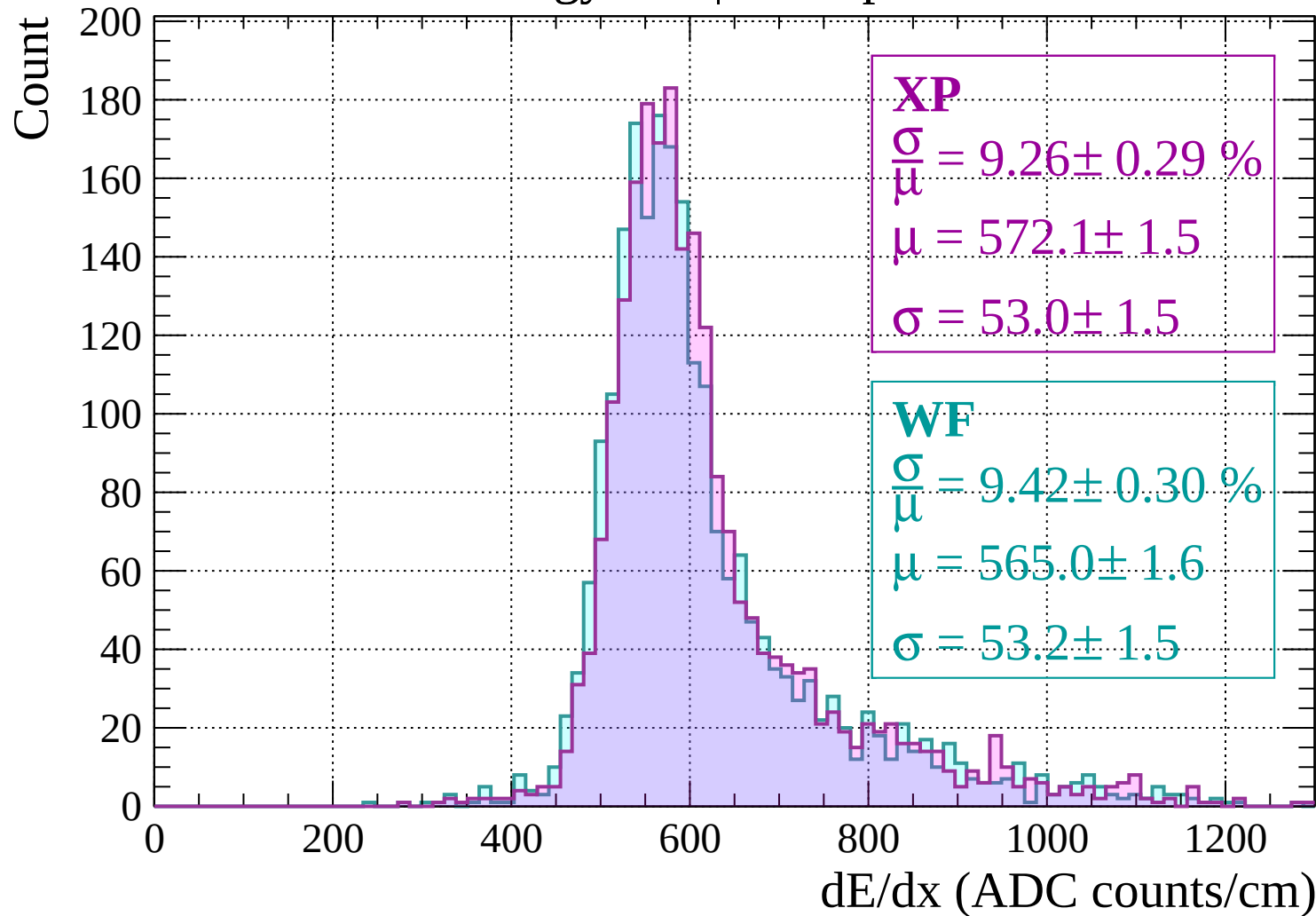
dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count

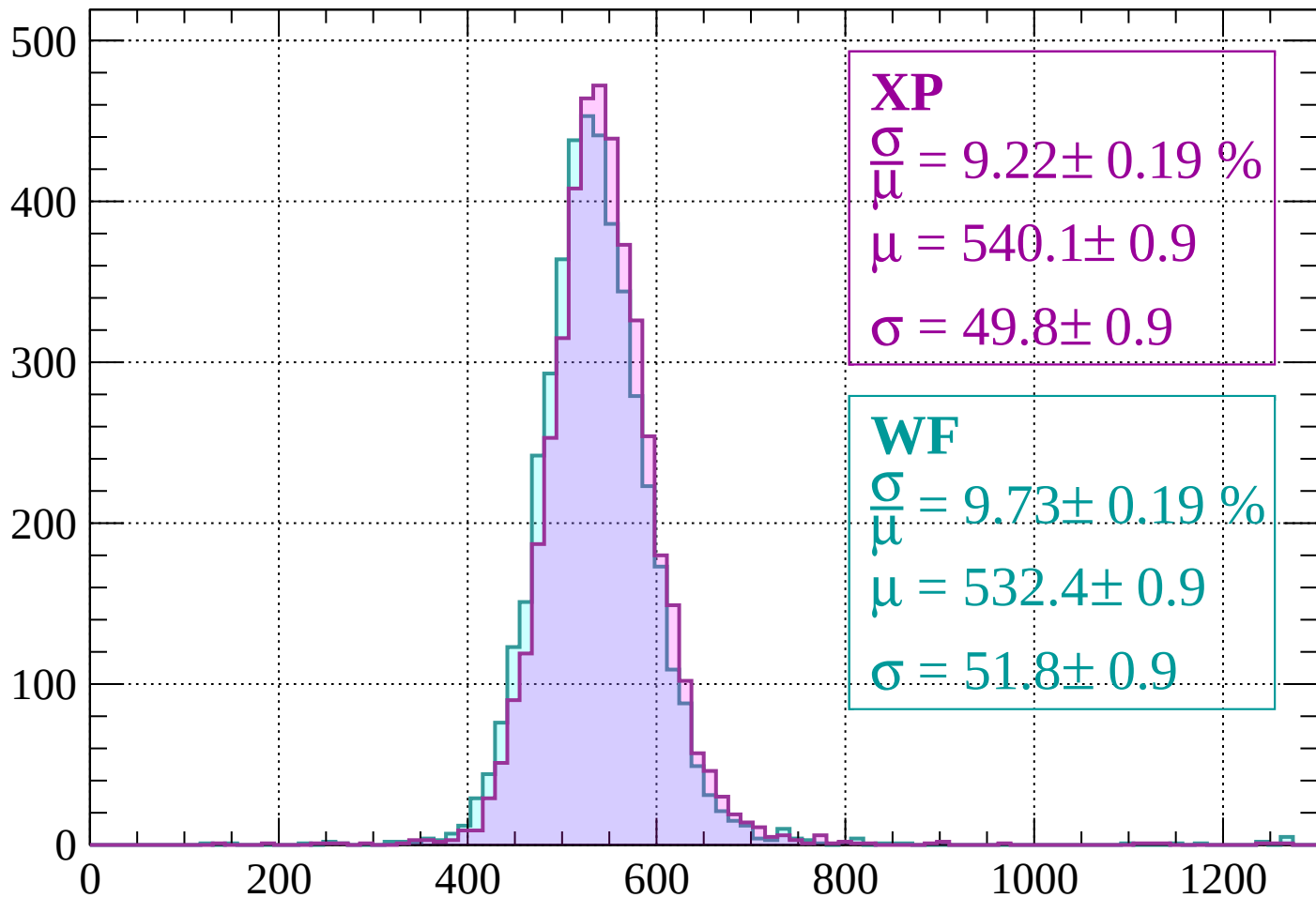


Energy loss | $-80 < p < -40$



Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

$$\mu = 540.1 \pm 0.9$$

$$\sigma = 49.8 \pm 0.9$$

WF

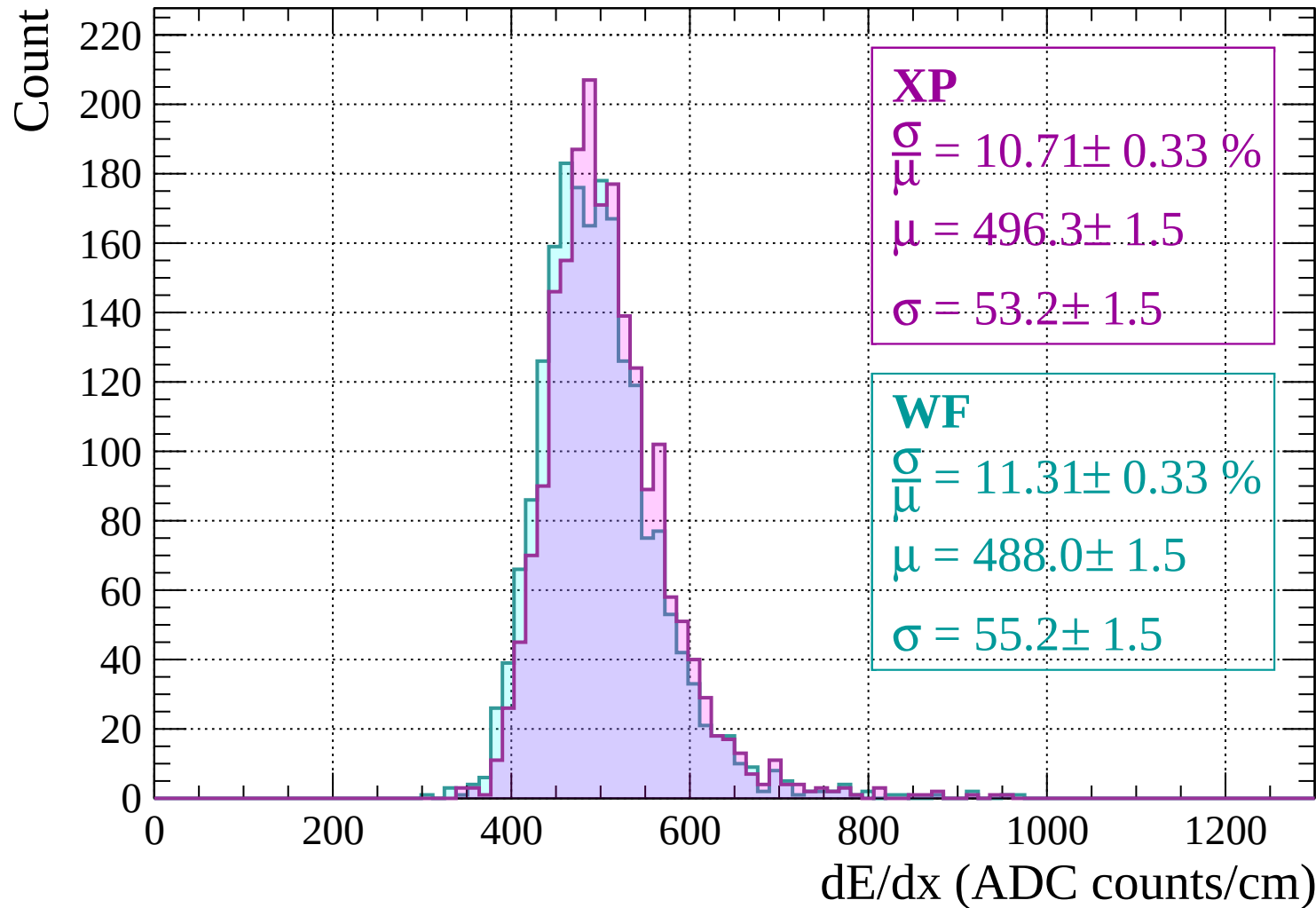
$$\frac{\sigma}{\mu} = 9.73 \pm 0.19 \%$$

$$\mu = 532.4 \pm 0.9$$

$$\sigma = 51.8 \pm 0.9$$

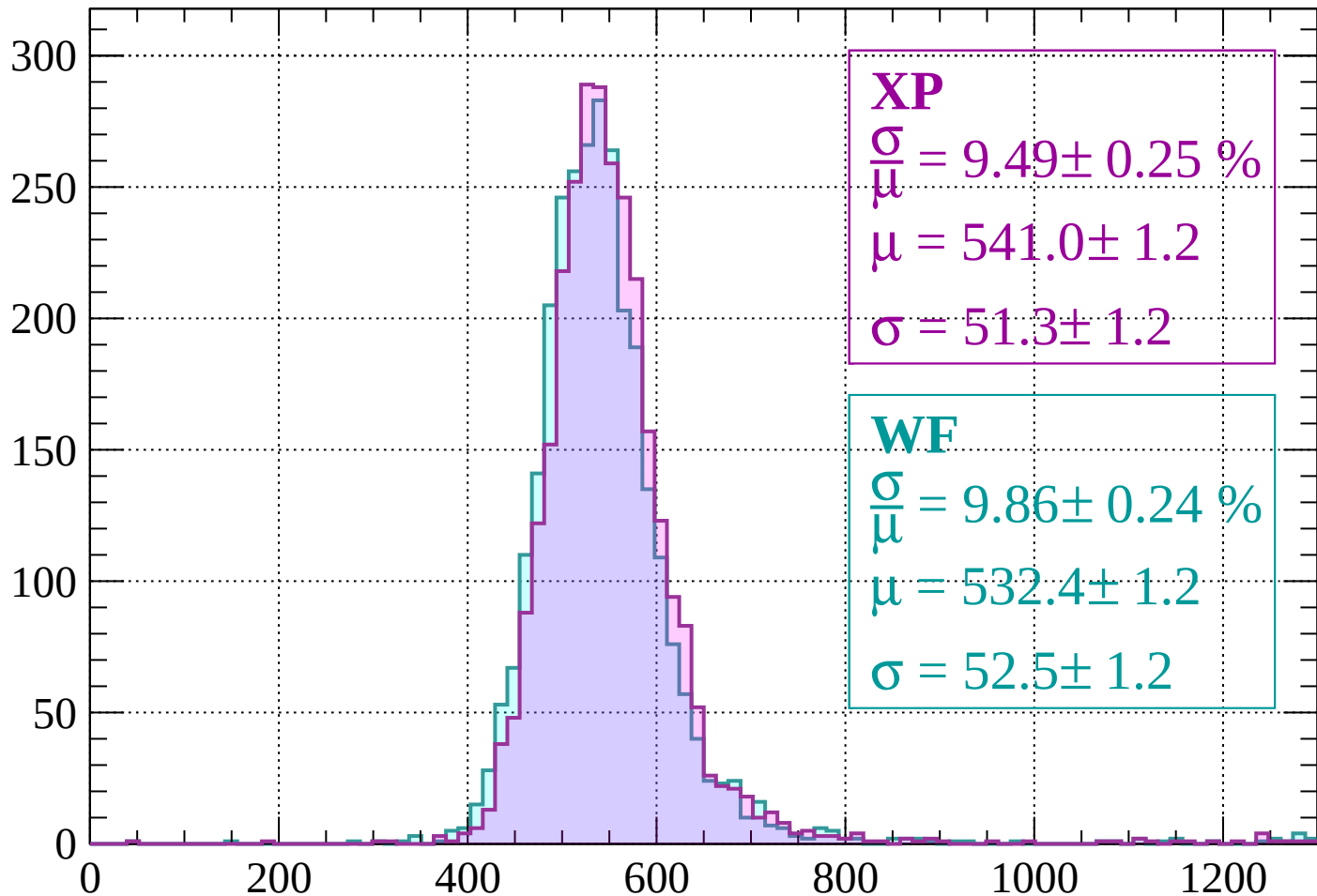
dE/dx (ADC counts/cm)

Energy loss | $0 < p < 40$



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.25 \%$$

$$\mu = 541.0 \pm 1.2$$

$$\sigma = 51.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.24 \%$$

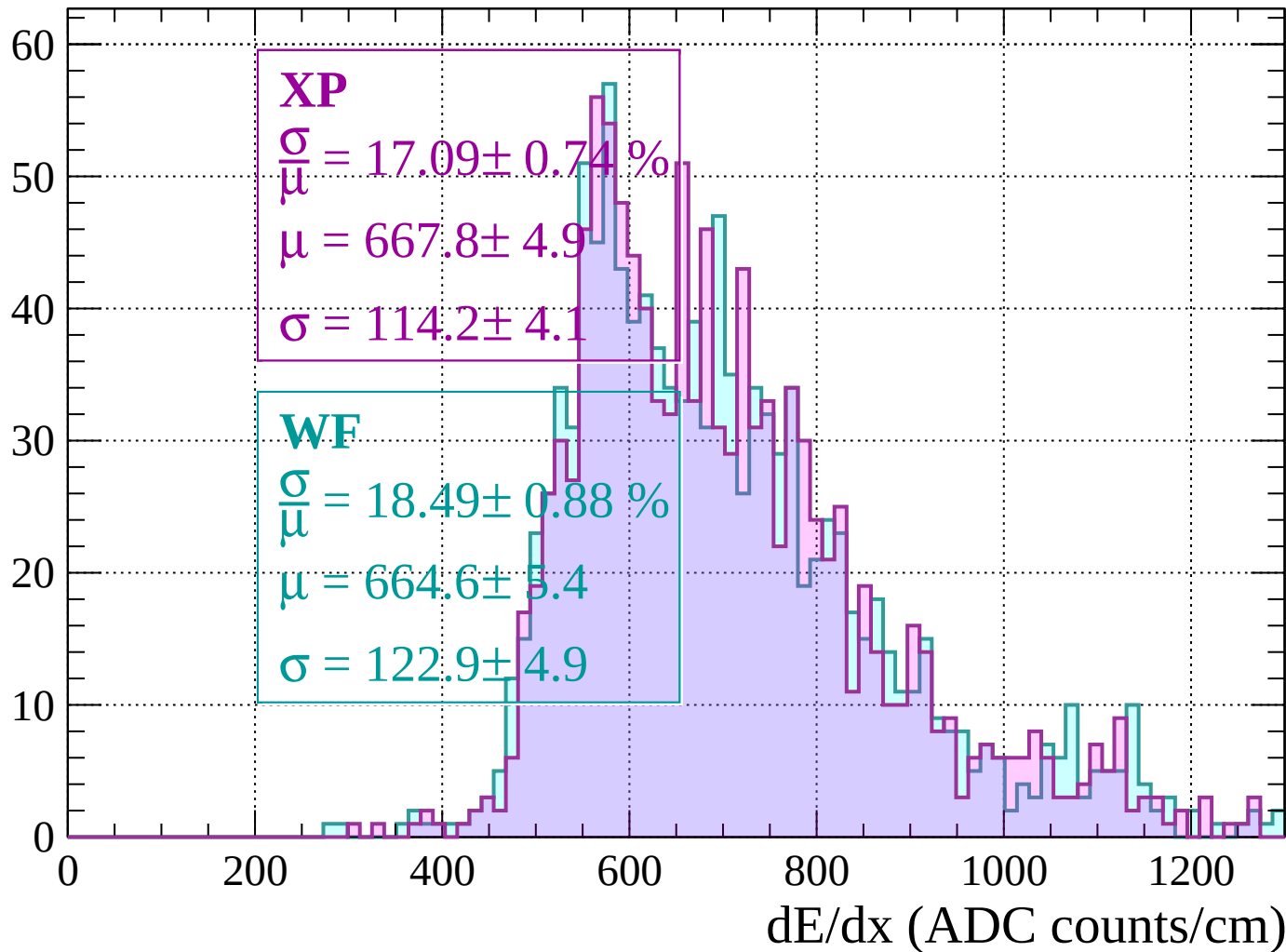
$$\mu = 532.4 \pm 1.2$$

$$\sigma = 52.5 \pm 1.2$$

dE/dx (ADC counts/cm)

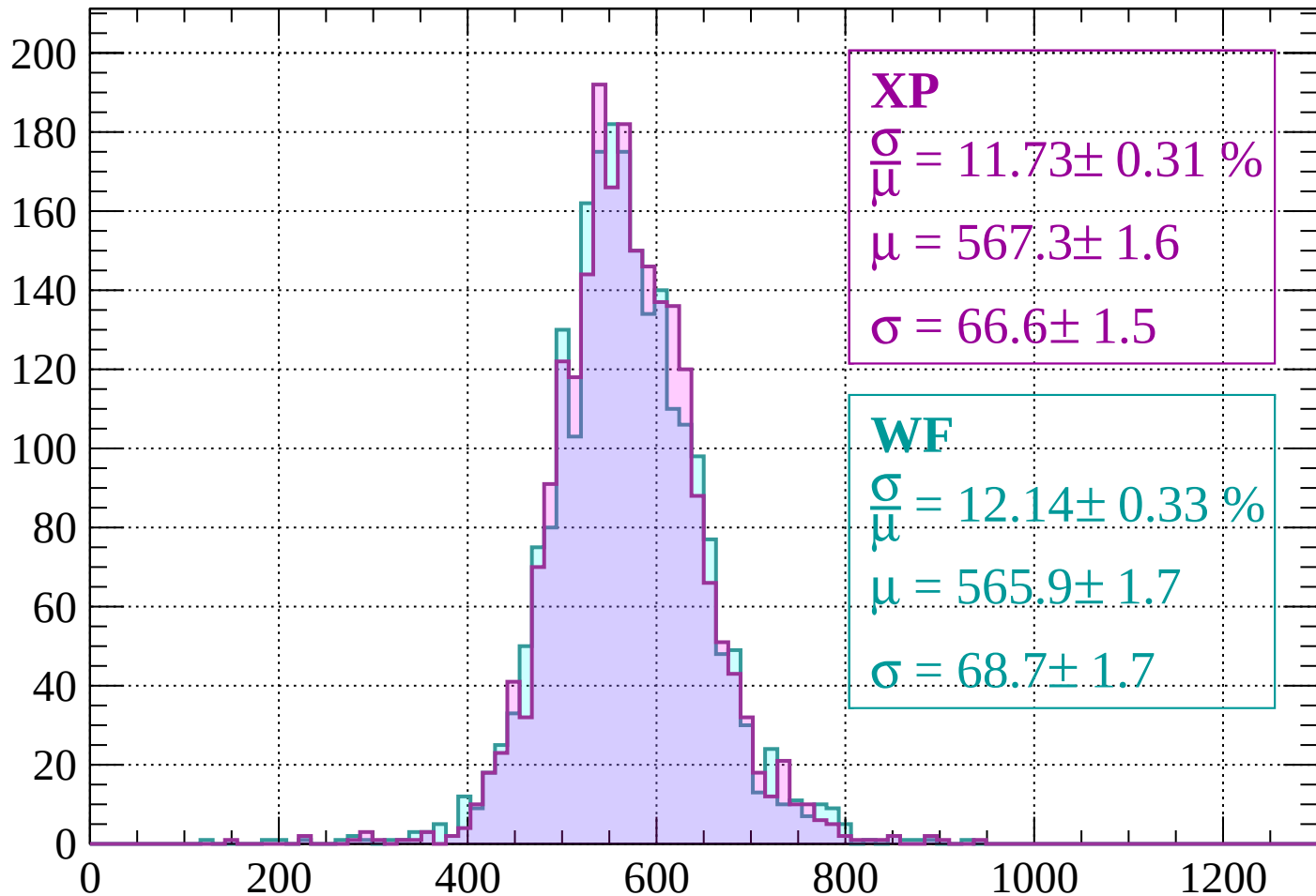
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 11.73 \pm 0.31 \%$$

$$\mu = 567.3 \pm 1.6$$

$$\sigma = 66.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 12.14 \pm 0.33 \%$$

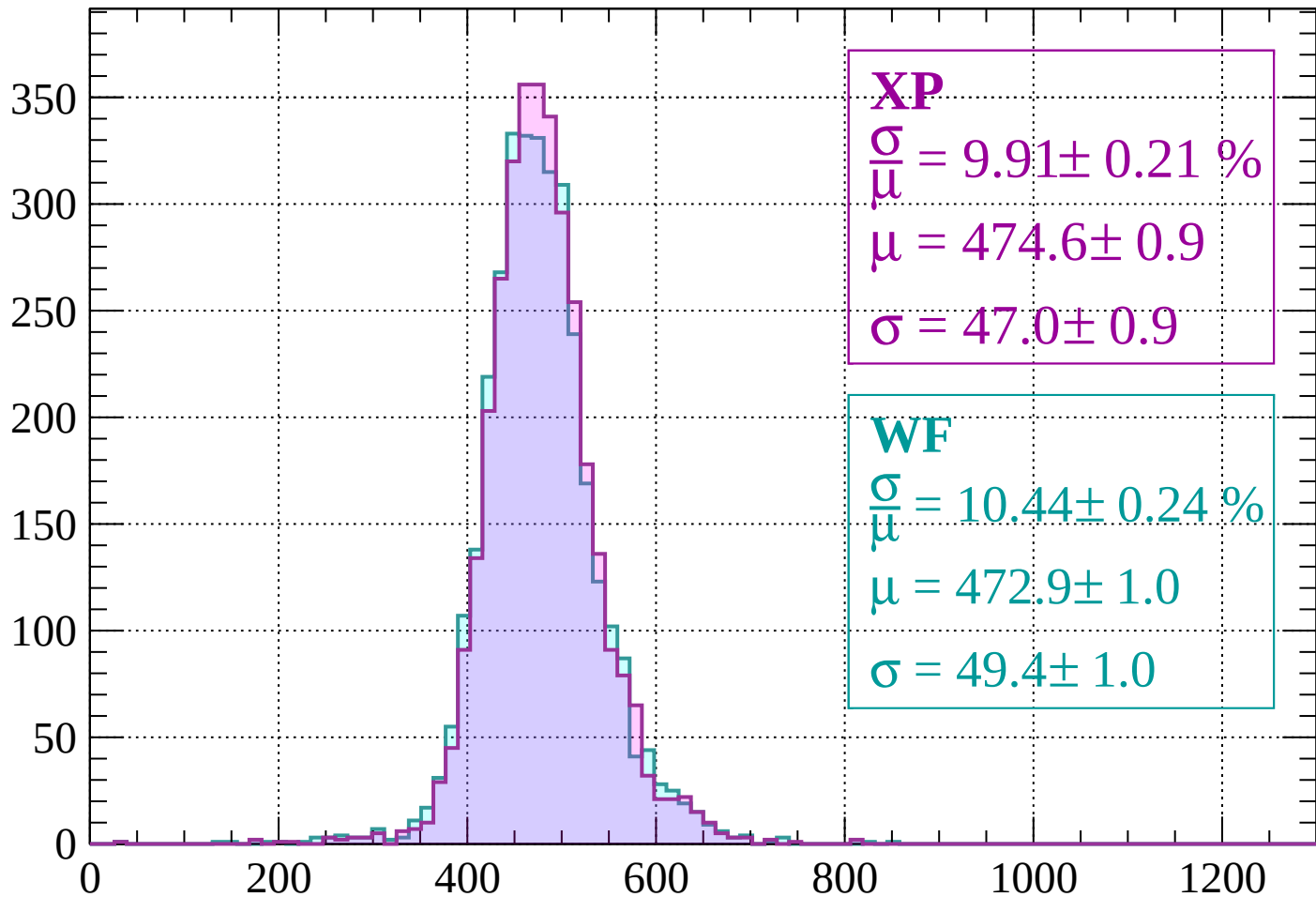
$$\mu = 565.9 \pm 1.7$$

$$\sigma = 68.7 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 9.91 \pm 0.21 \%$$

$$\mu = 474.6 \pm 0.9$$

$$\sigma = 47.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.44 \pm 0.24 \%$$

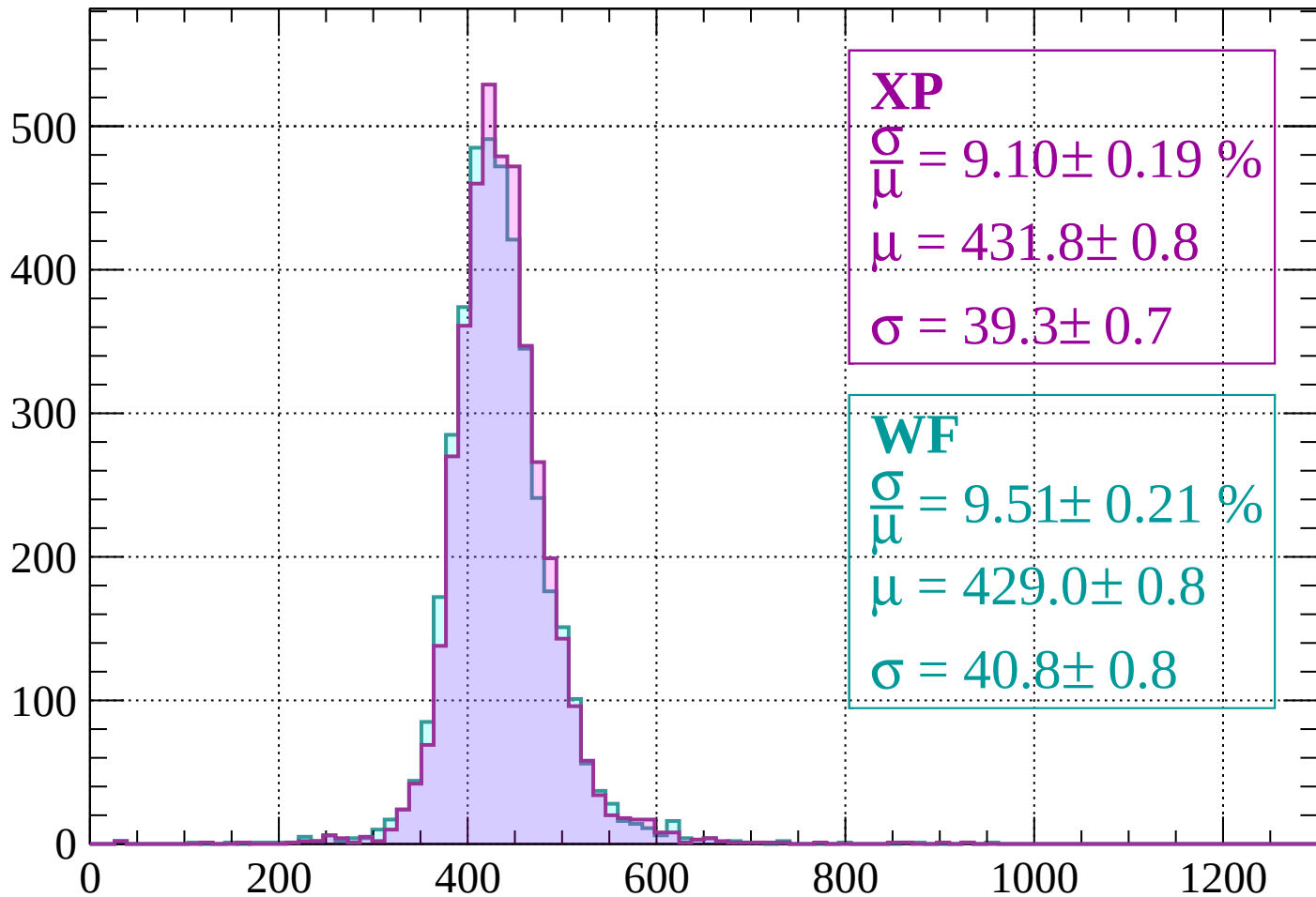
$$\mu = 472.9 \pm 1.0$$

$$\sigma = 49.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.19 \%$$

$$\mu = 431.8 \pm 0.8$$

$$\sigma = 39.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.21 \%$$

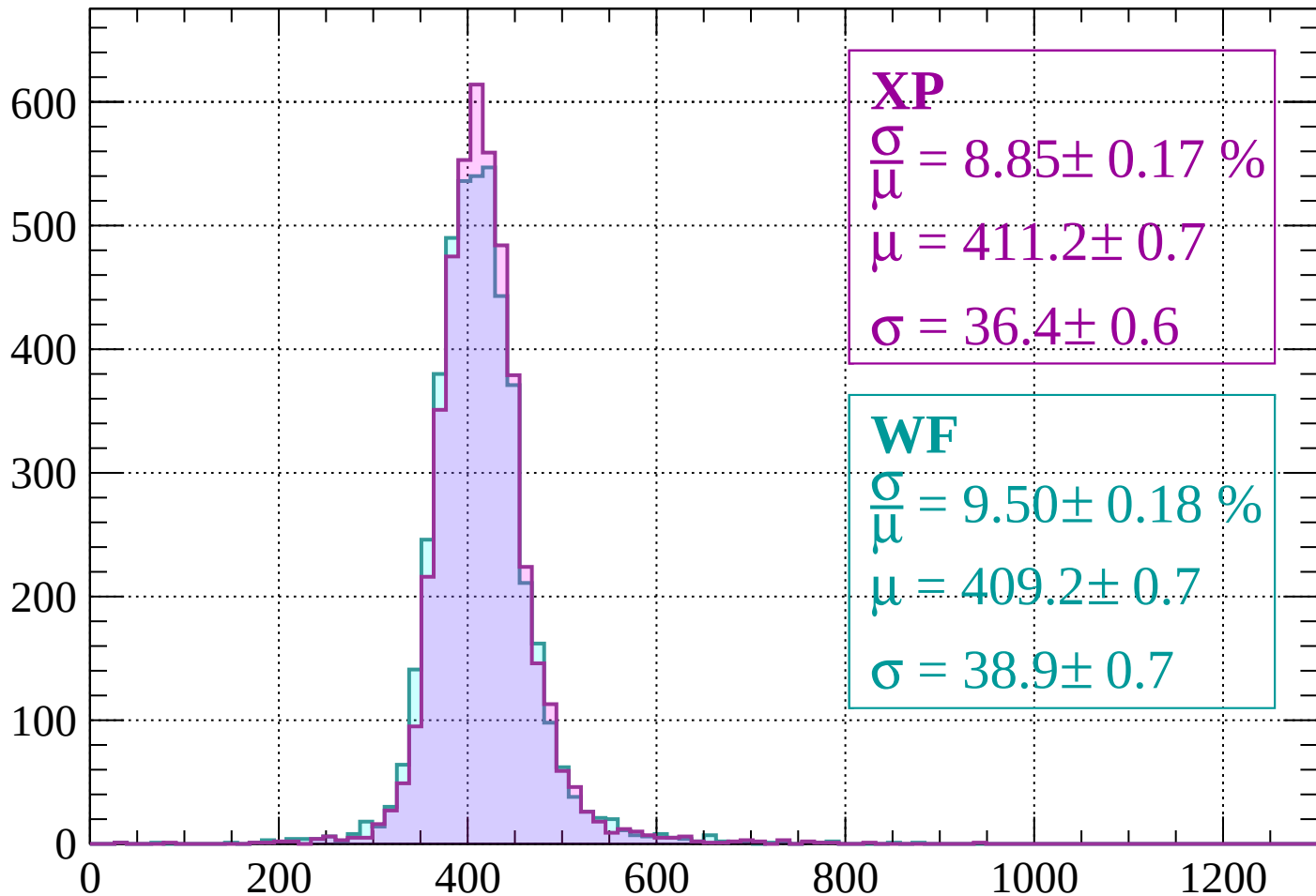
$$\mu = 429.0 \pm 0.8$$

$$\sigma = 40.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

$$\mu = 411.2 \pm 0.7$$

$$\sigma = 36.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.18 \%$$

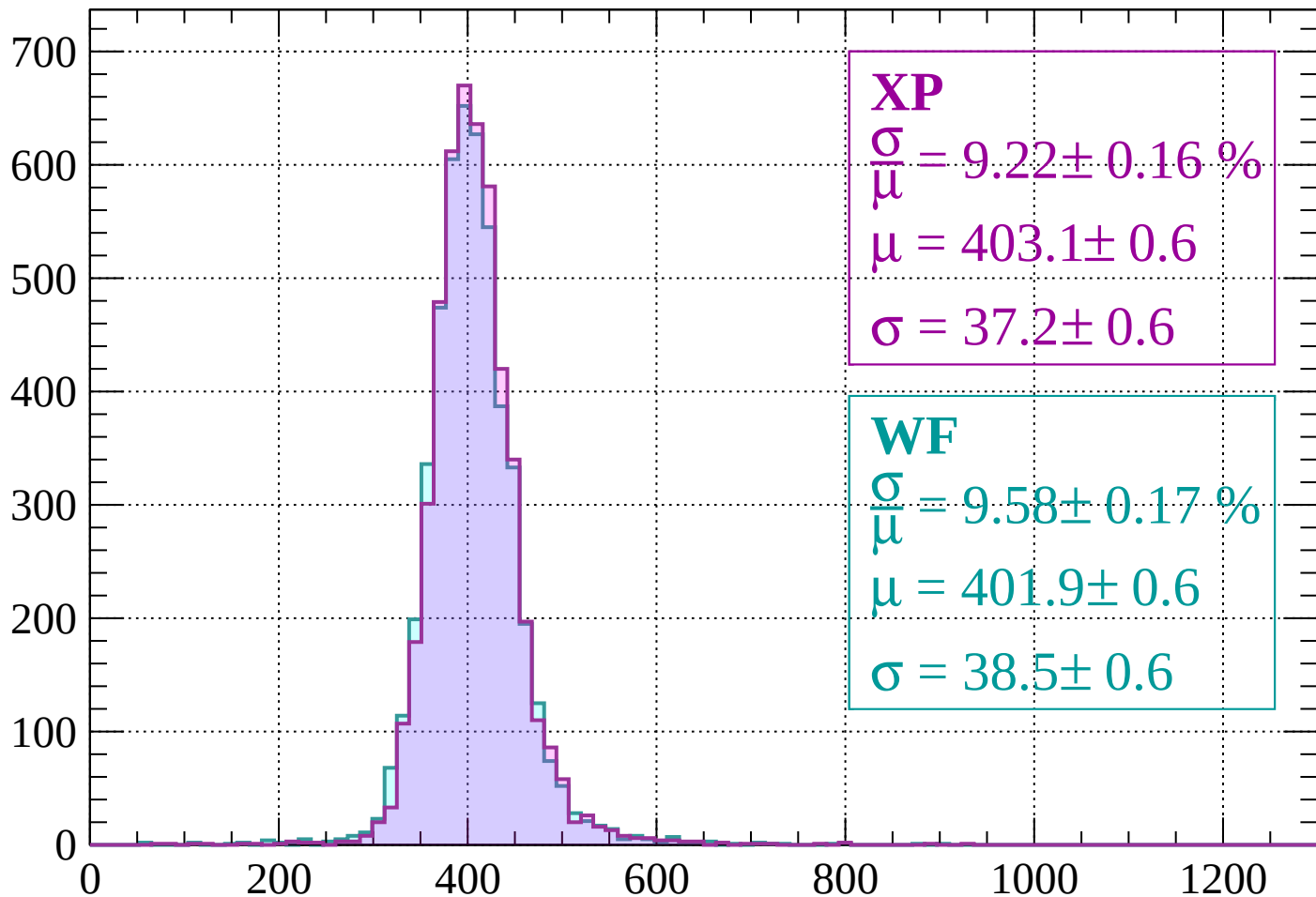
$$\mu = 409.2 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.16 \%$$

$$\mu = 403.1 \pm 0.6$$

$$\sigma = 37.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.58 \pm 0.17 \%$$

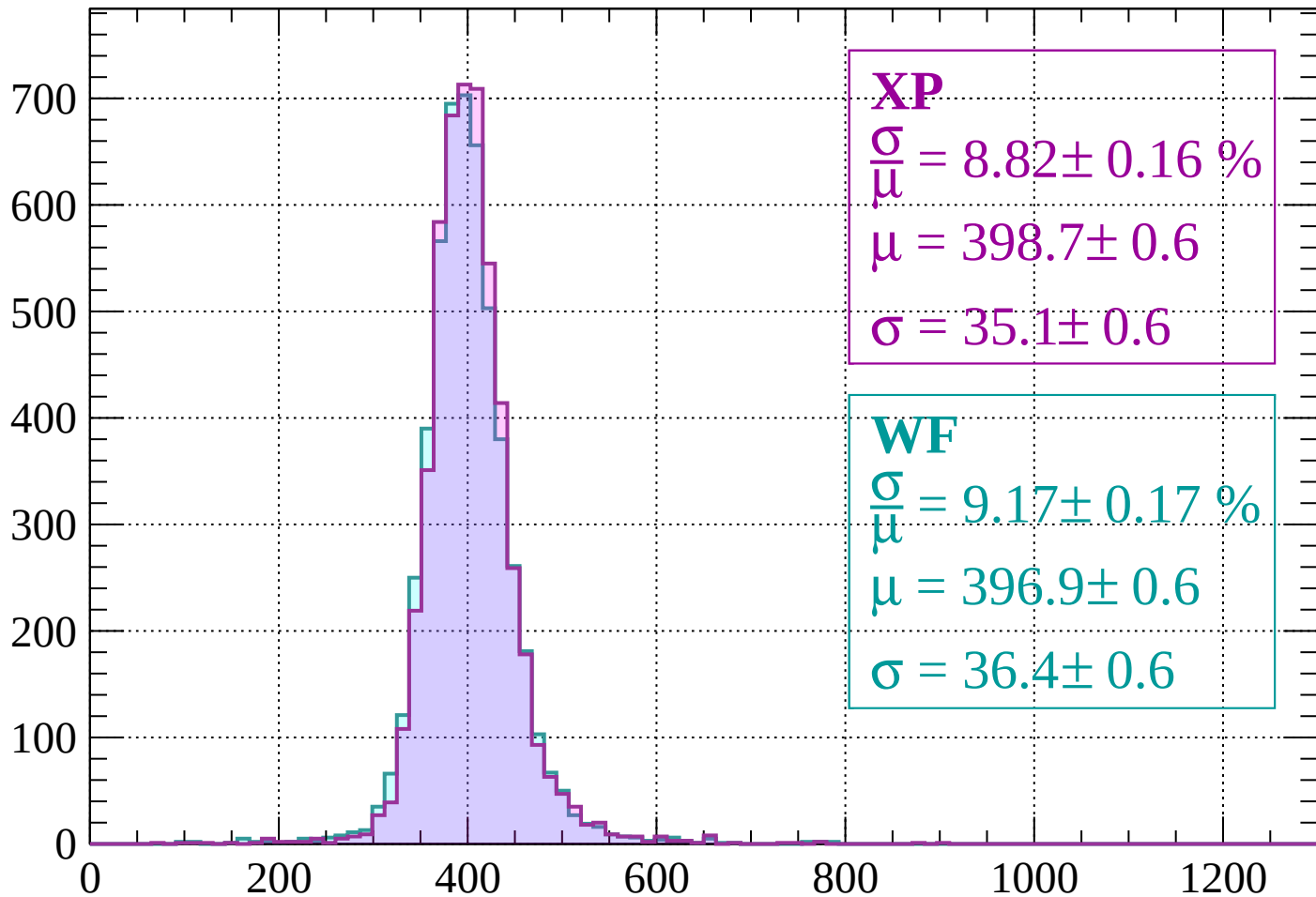
$$\mu = 401.9 \pm 0.6$$

$$\sigma = 38.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.16 \%$$

$$\mu = 398.7 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.17 \pm 0.17 \%$$

$$\mu = 396.9 \pm 0.6$$

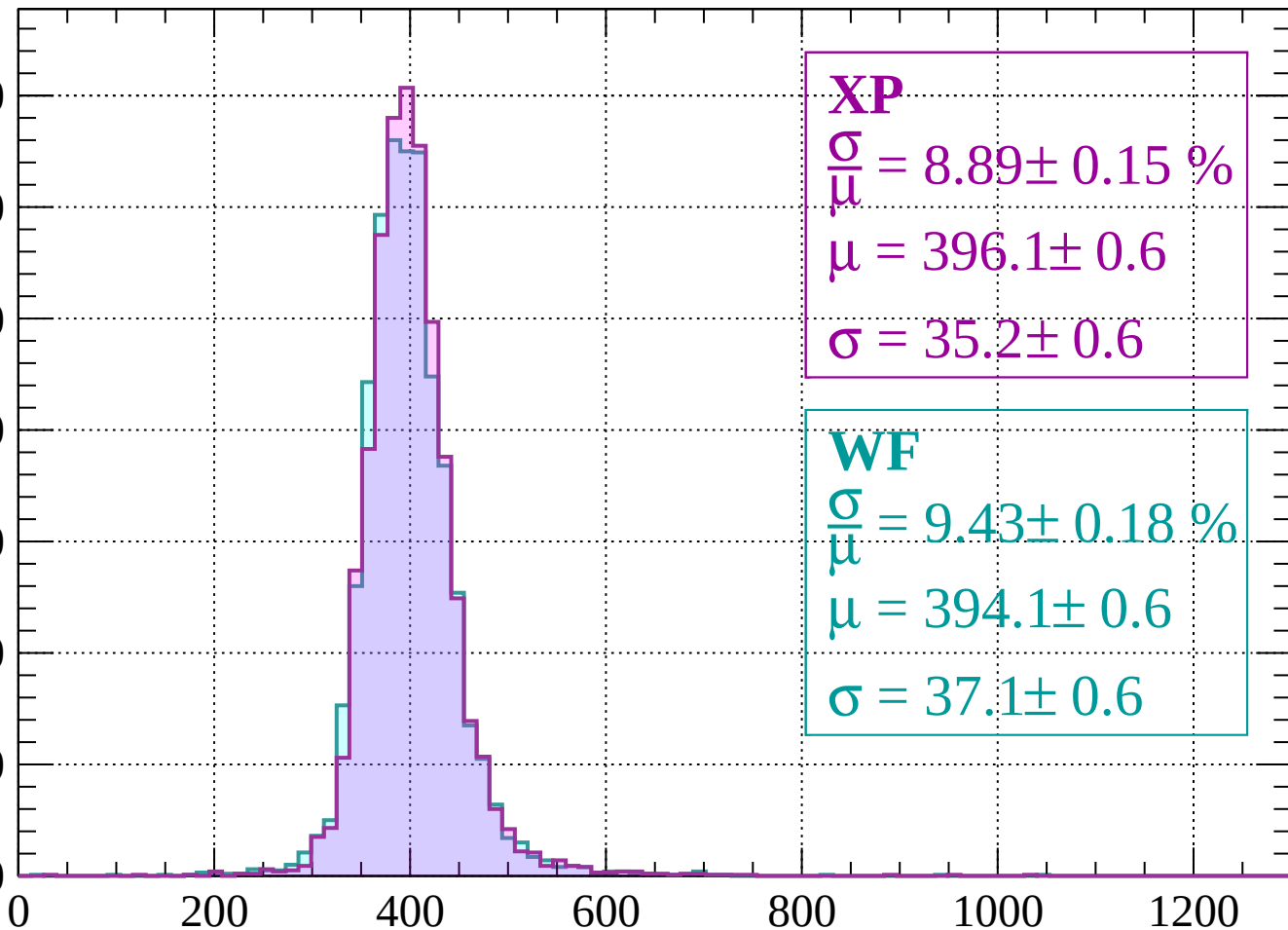
$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.15 \%$$

$$\mu = 396.1 \pm 0.6$$

$$\sigma = 35.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

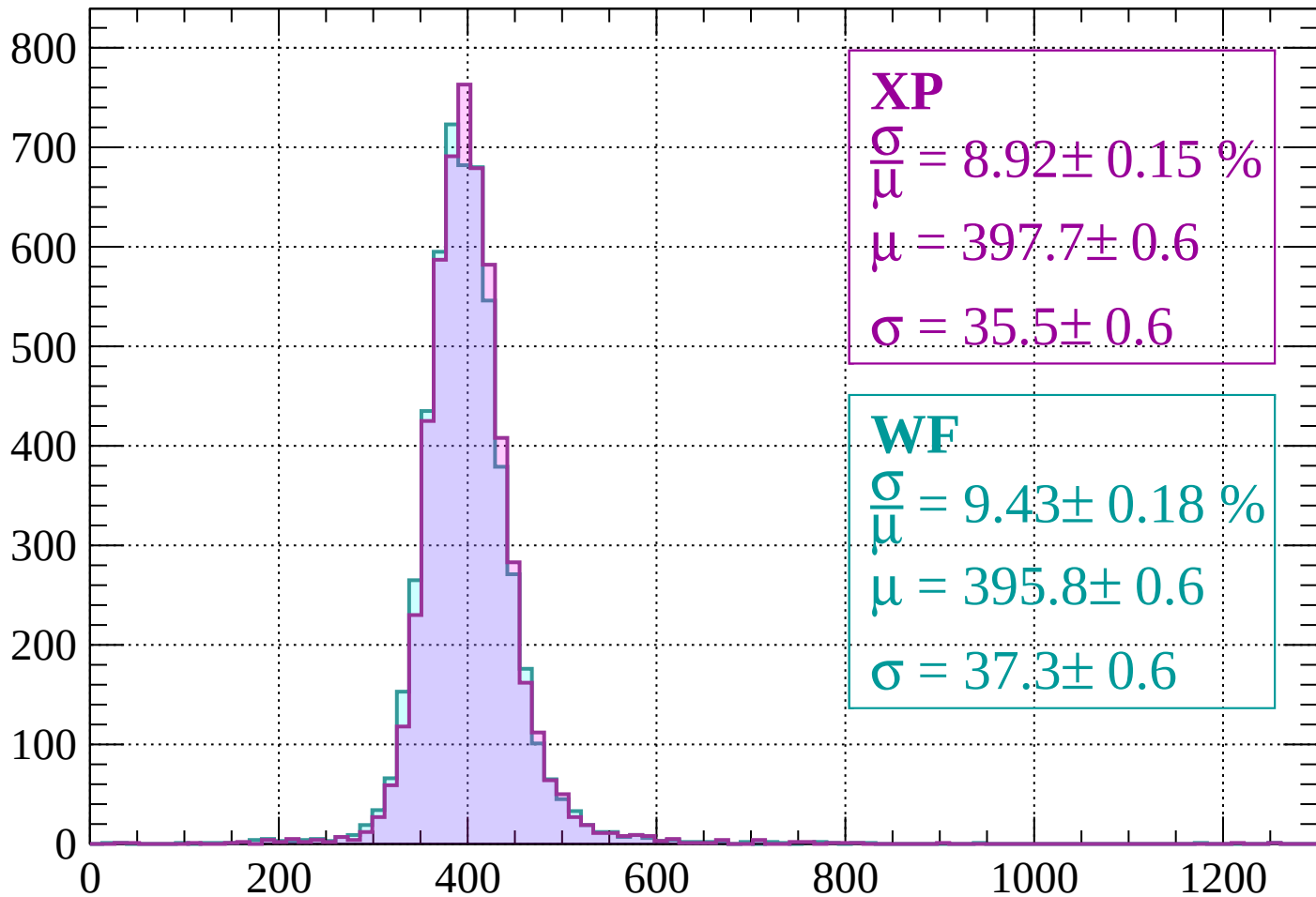
$$\mu = 394.1 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.15 \%$$

$$\mu = 397.7 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

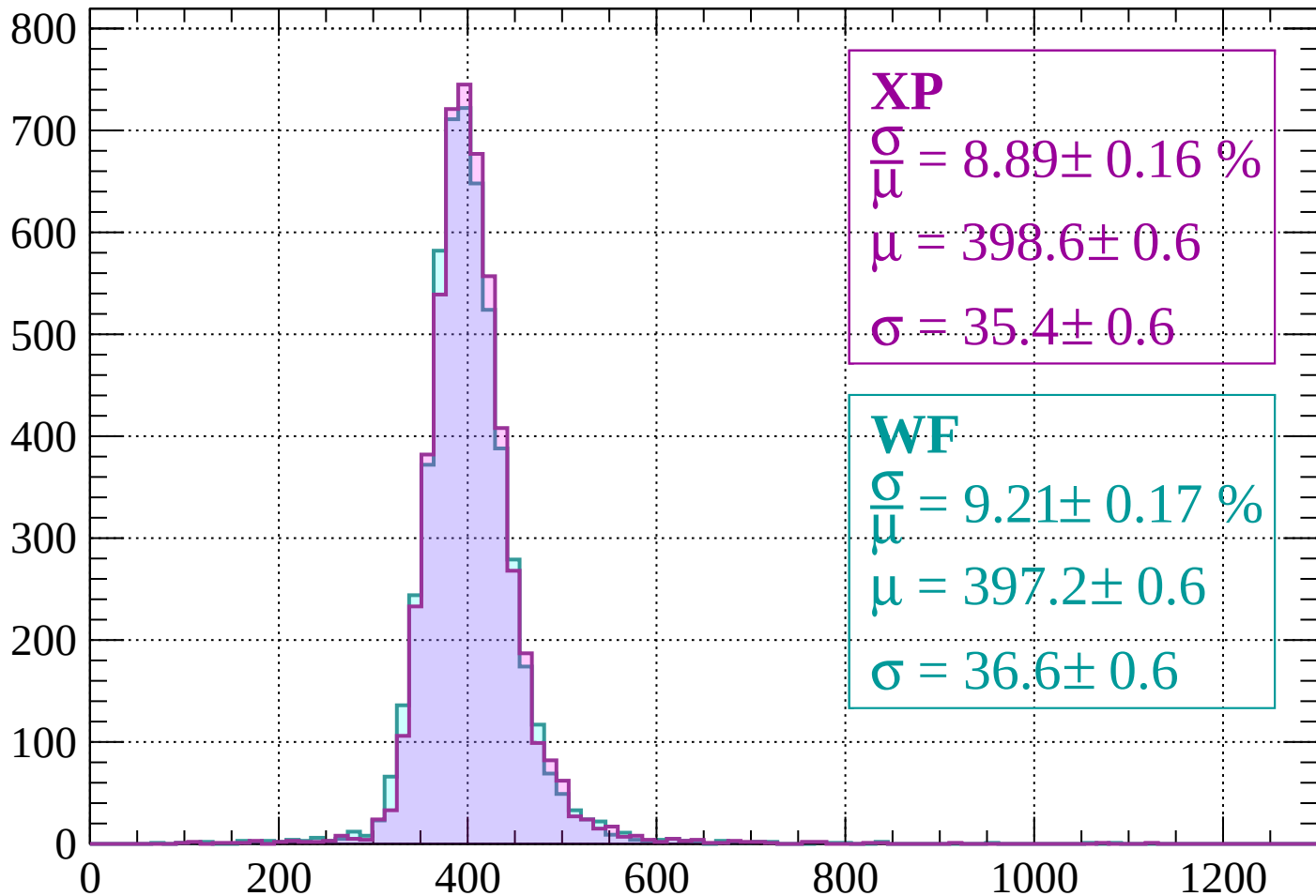
$$\mu = 395.8 \pm 0.6$$

$$\sigma = 37.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.17 \%$$

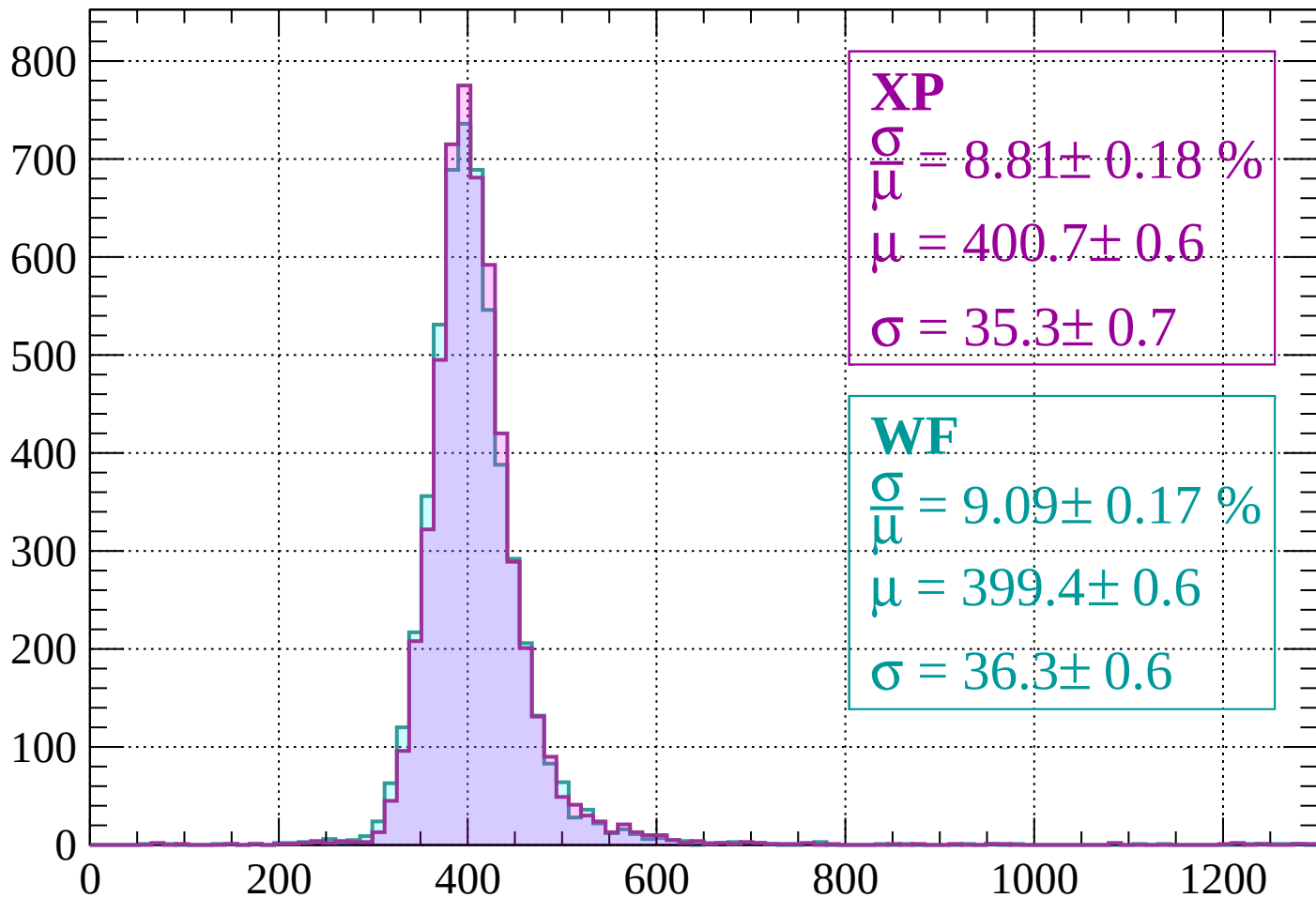
$$\mu = 397.2 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.18 \%$$

$$\mu = 400.7 \pm 0.6$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.17 \%$$

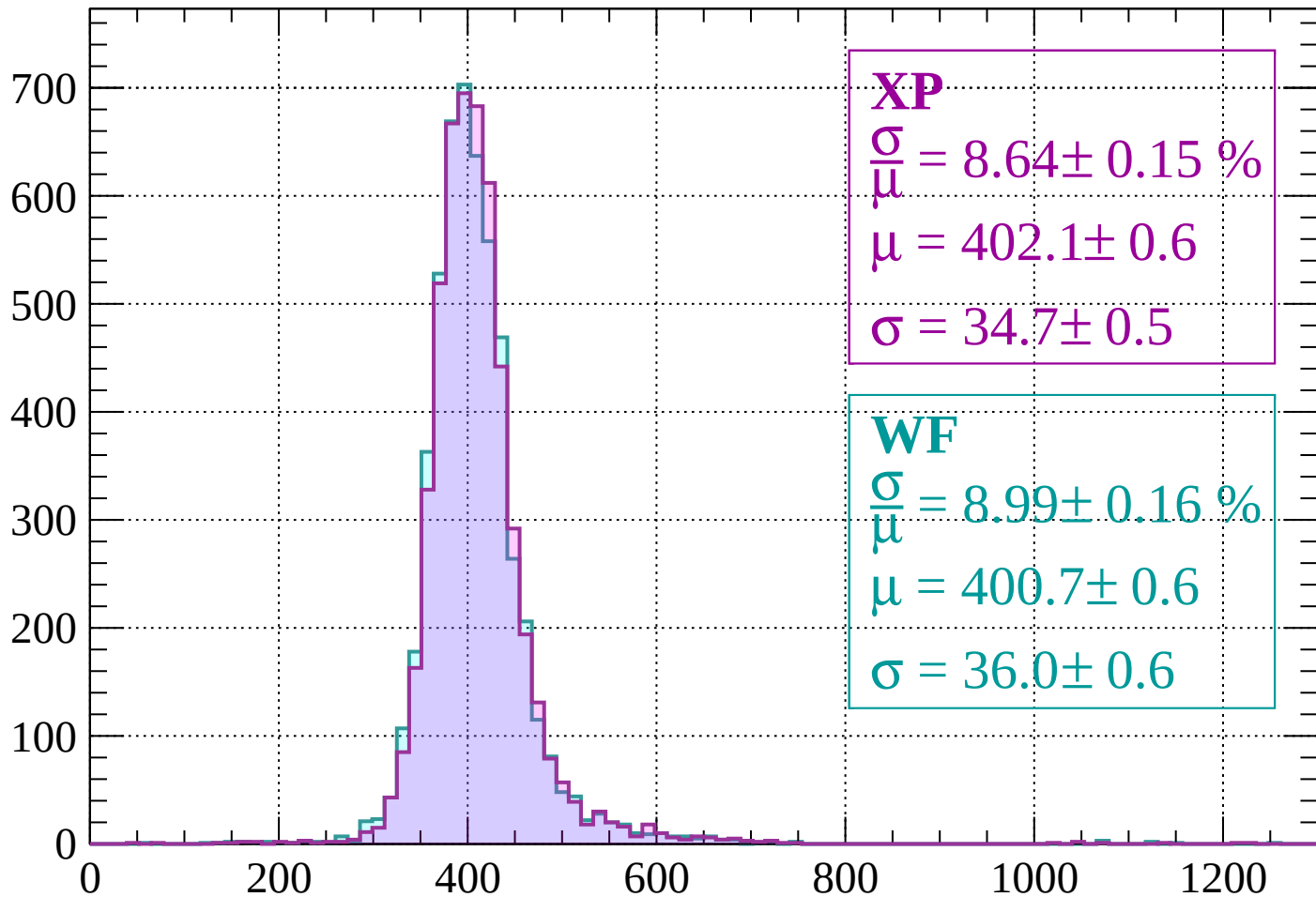
$$\mu = 399.4 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.15 \%$$

$$\mu = 402.1 \pm 0.6$$

$$\sigma = 34.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

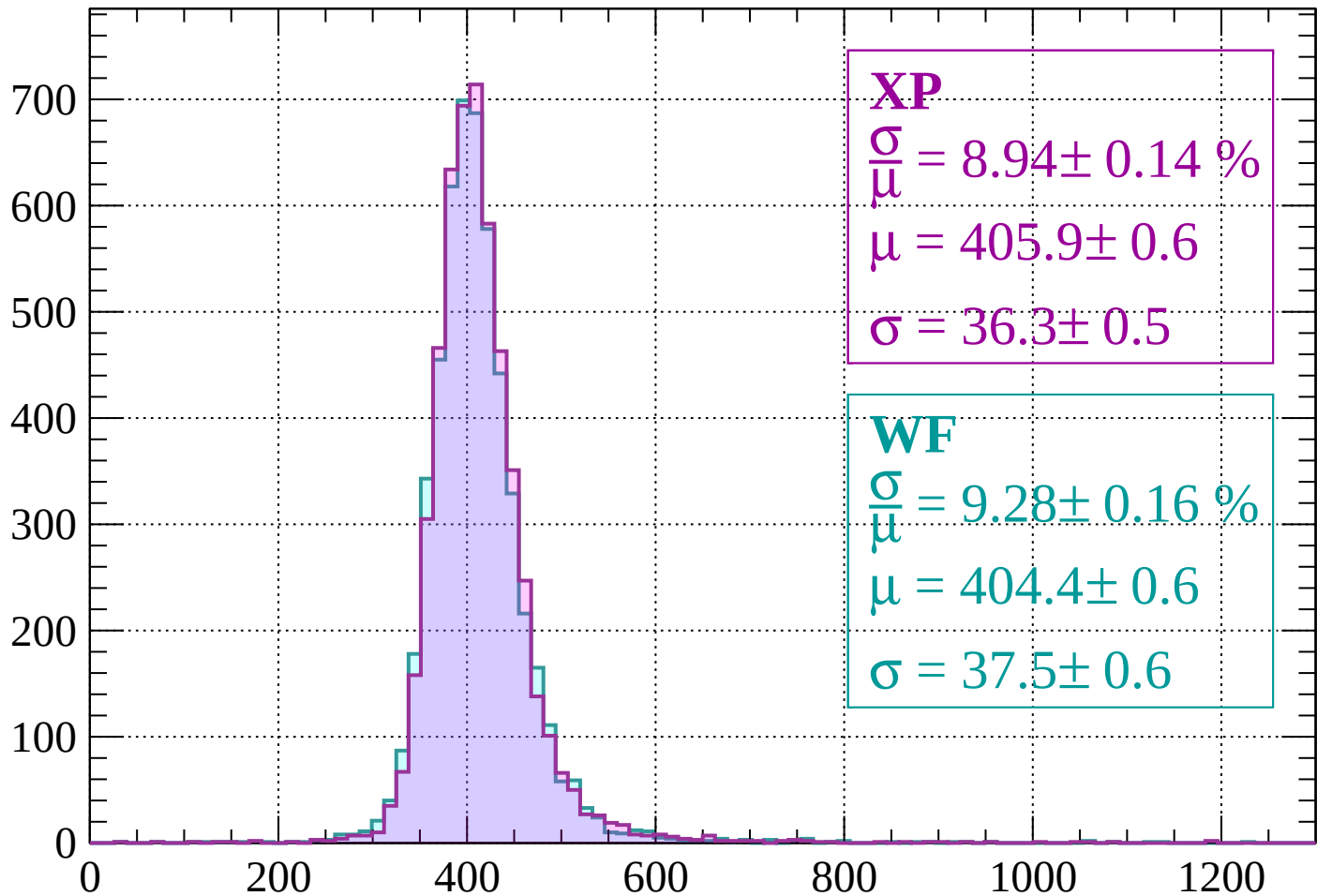
$$\mu = 400.7 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 0.14 \%$$

$$\mu = 405.9 \pm 0.6$$

$$\sigma = 36.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.16 \%$$

$$\mu = 404.4 \pm 0.6$$

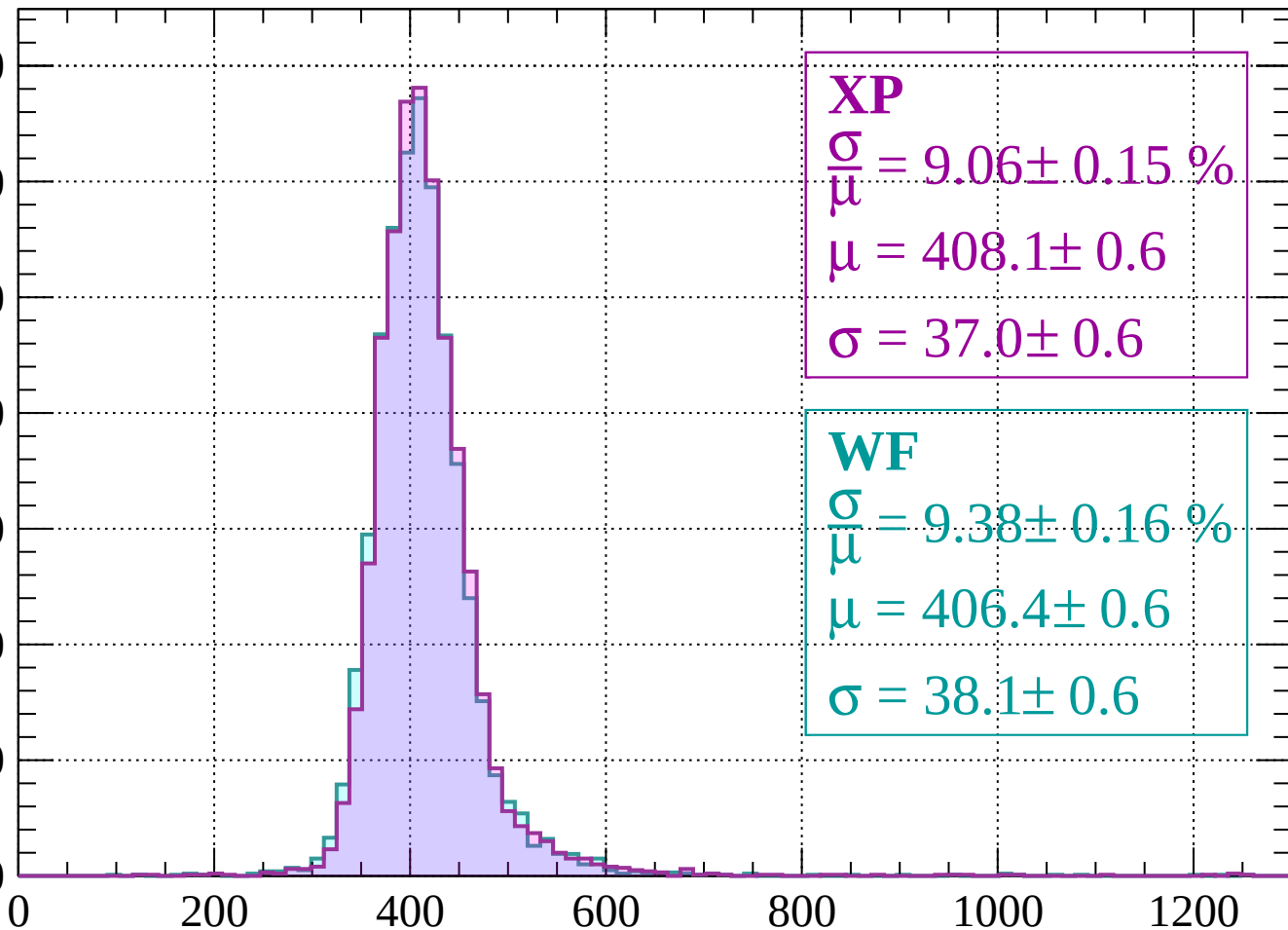
$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.15 \%$$

$$\mu = 408.1 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.38 \pm 0.16 \%$$

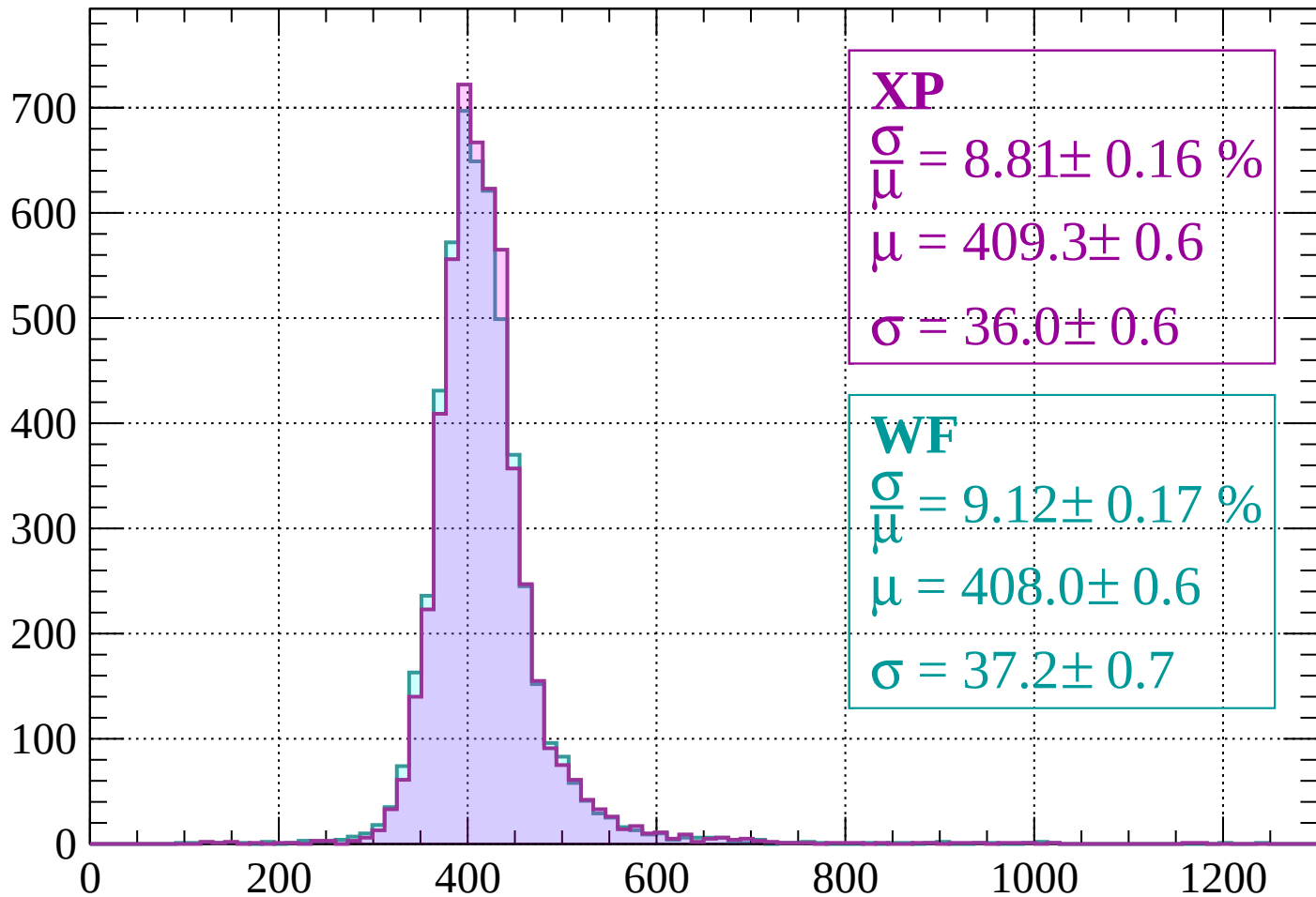
$$\mu = 406.4 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.81 \pm 0.16 \%$$

$$\mu = 409.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.12 \pm 0.17 \%$$

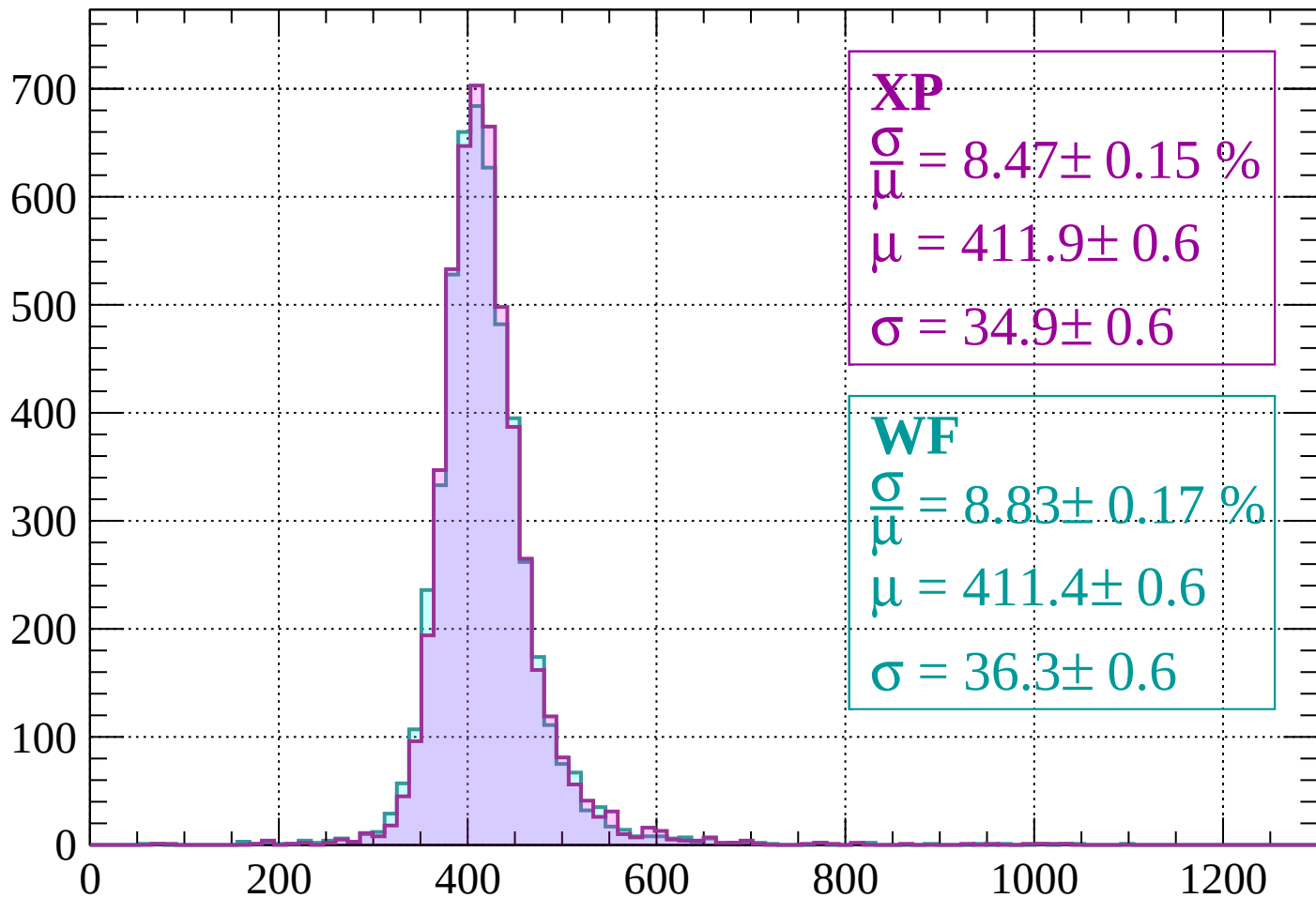
$$\mu = 408.0 \pm 0.6$$

$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.15 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.83 \pm 0.17 \%$$

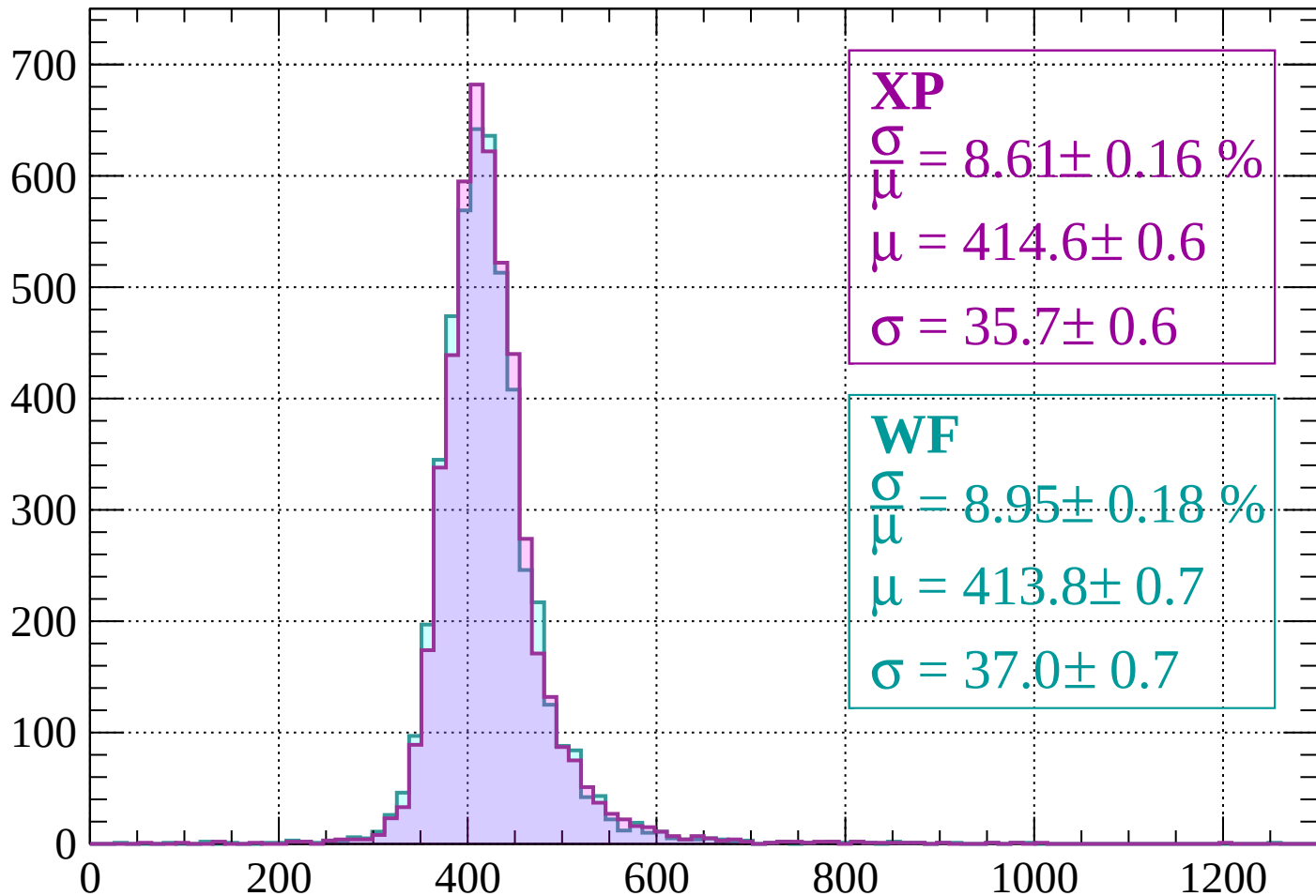
$$\mu = 411.4 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.16 \%$$

$$\mu = 414.6 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.18 \%$$

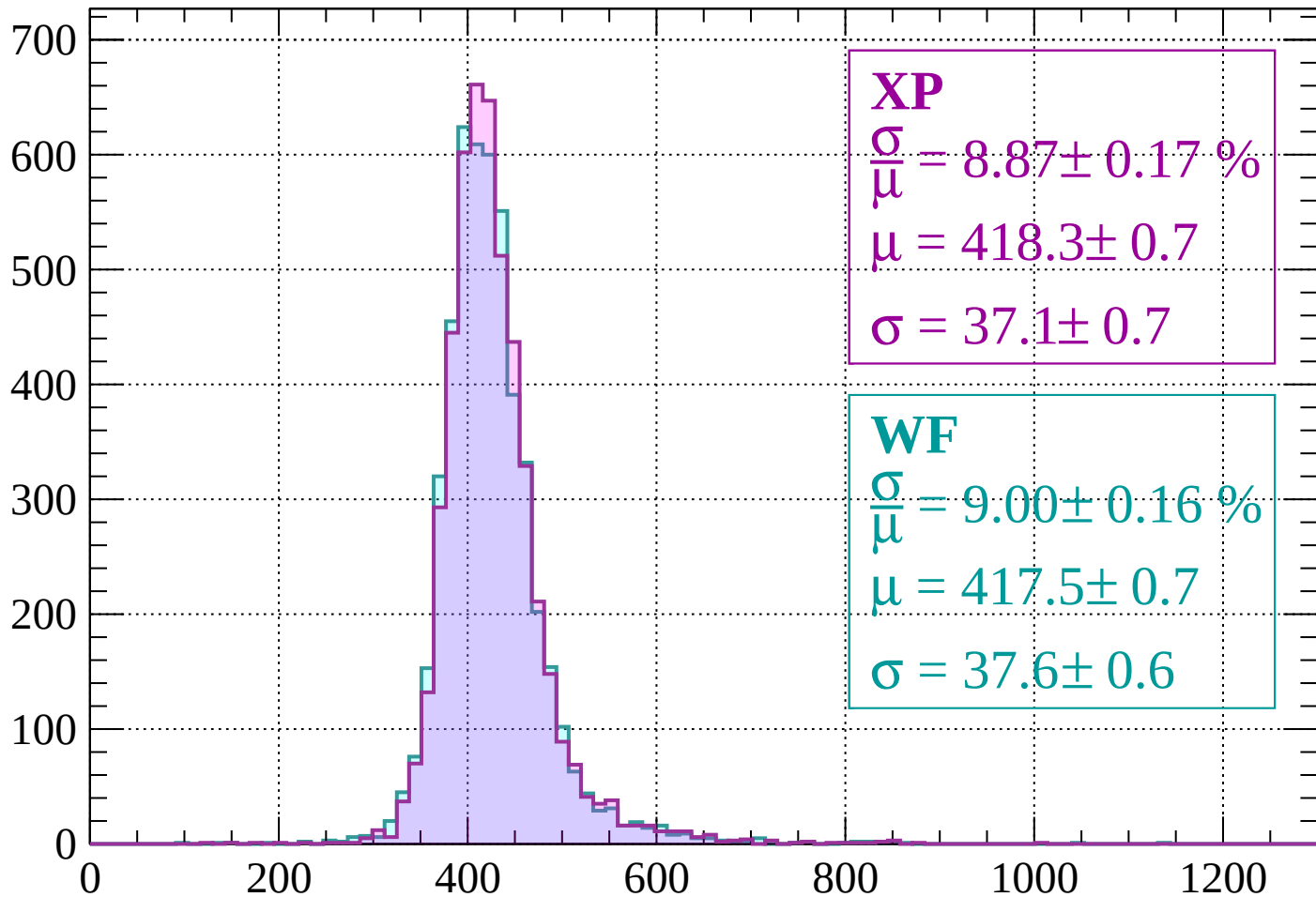
$$\mu = 413.8 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.87 \pm 0.17 \%$$

$$\mu = 418.3 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.00 \pm 0.16 \%$$

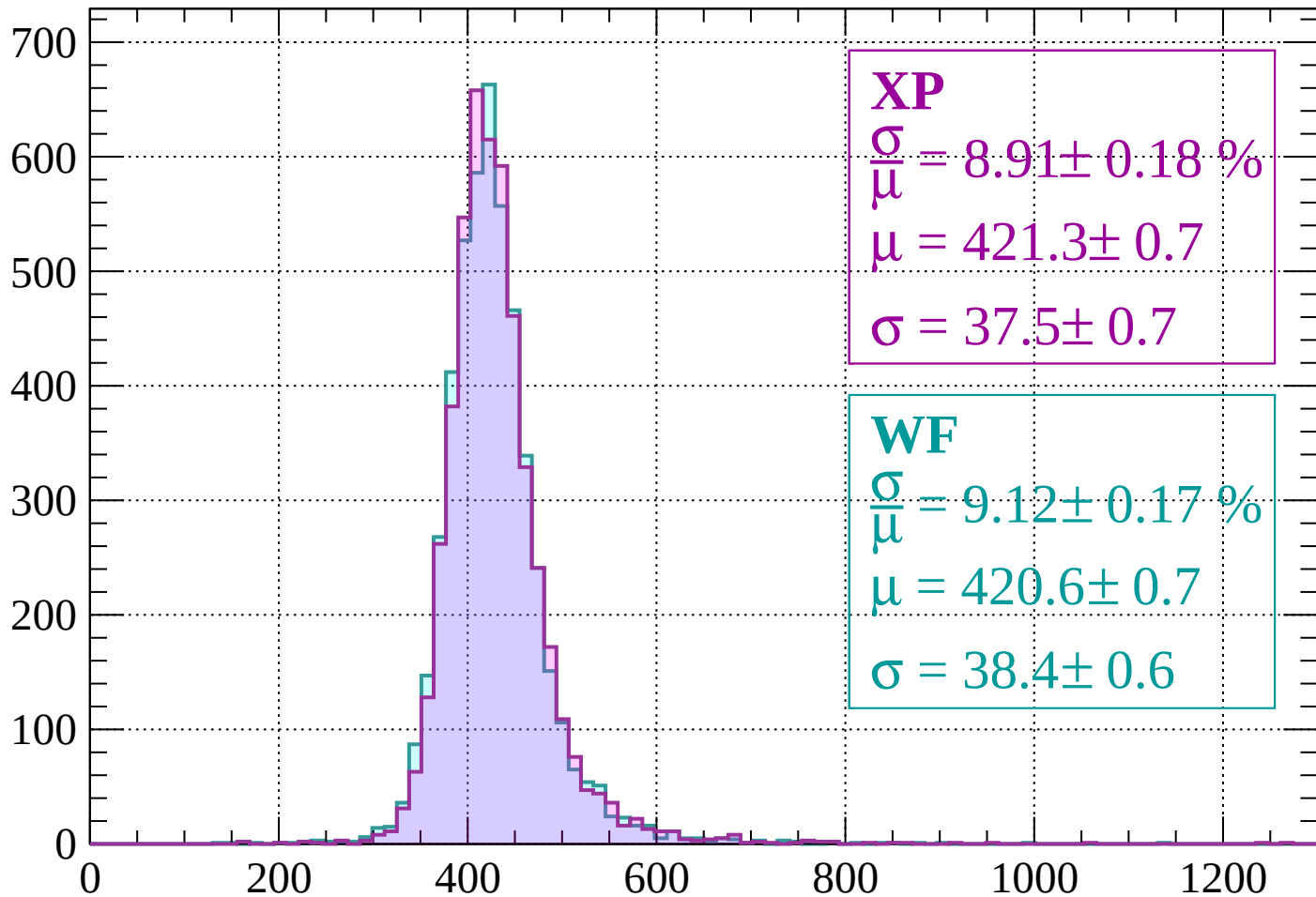
$$\mu = 417.5 \pm 0.7$$

$$\sigma = 37.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.91 \pm 0.18 \%$$

$$\mu = 421.3 \pm 0.7$$

$$\sigma = 37.5 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.12 \pm 0.17 \%$$

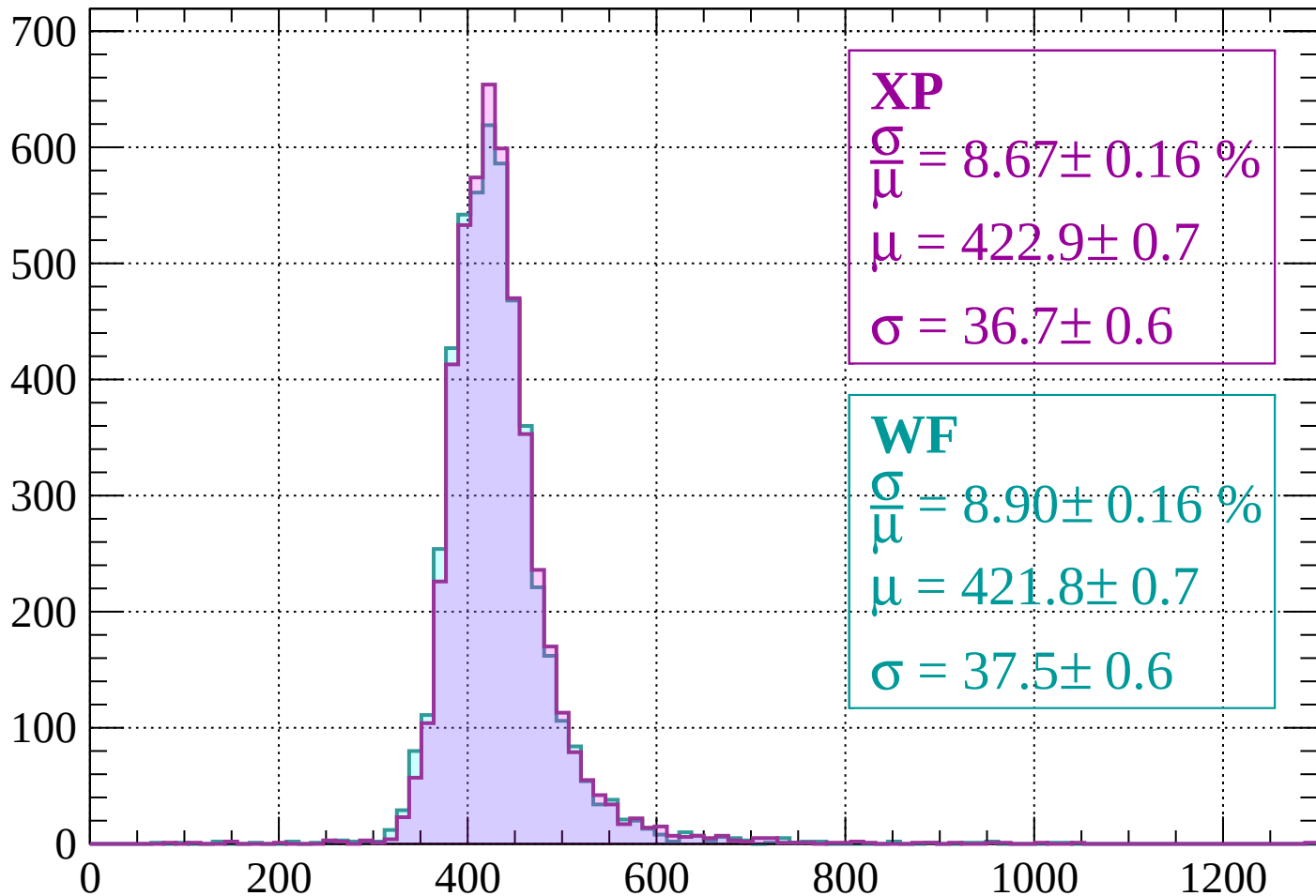
$$\mu = 420.6 \pm 0.7$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.67 \pm 0.16 \%$$

$$\mu = 422.9 \pm 0.7$$

$$\sigma = 36.7 \pm 0.6$$

WF

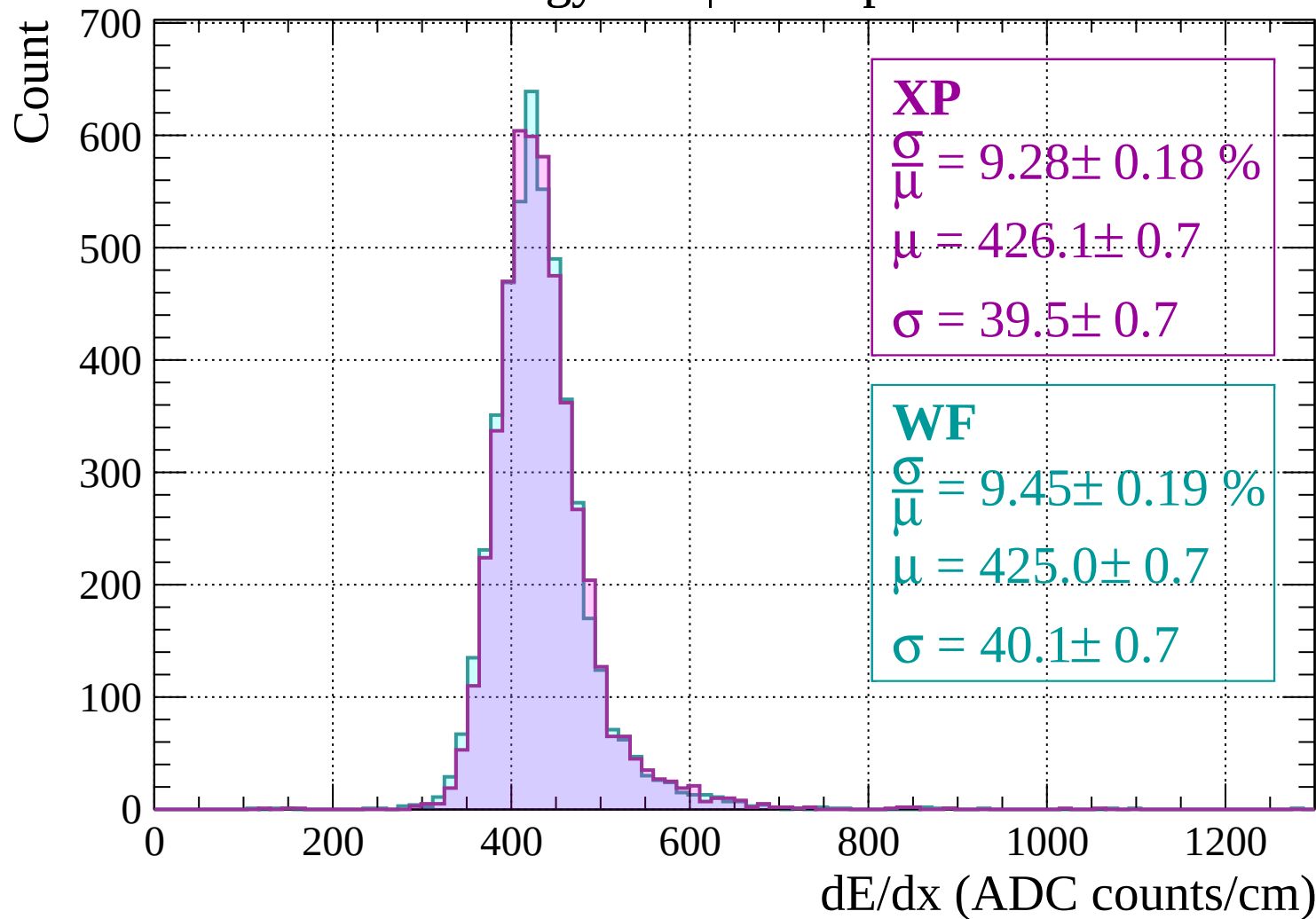
$$\frac{\sigma}{\bar{\mu}} = 8.90 \pm 0.16 \%$$

$$\mu = 421.8 \pm 0.7$$

$$\sigma = 37.5 \pm 0.6$$

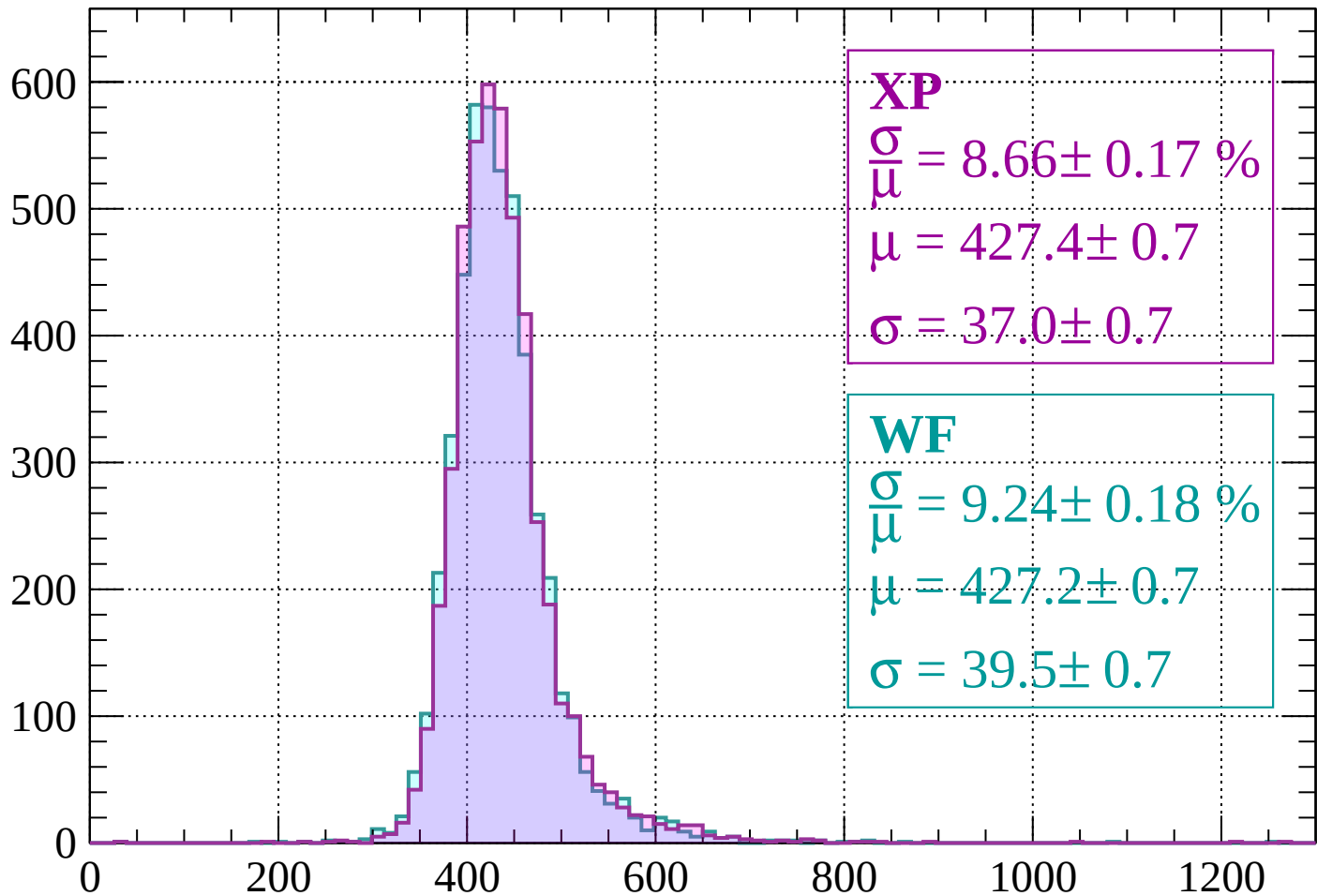
dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$



Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.17 \%$$

$$\mu = 427.4 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.18 \%$$

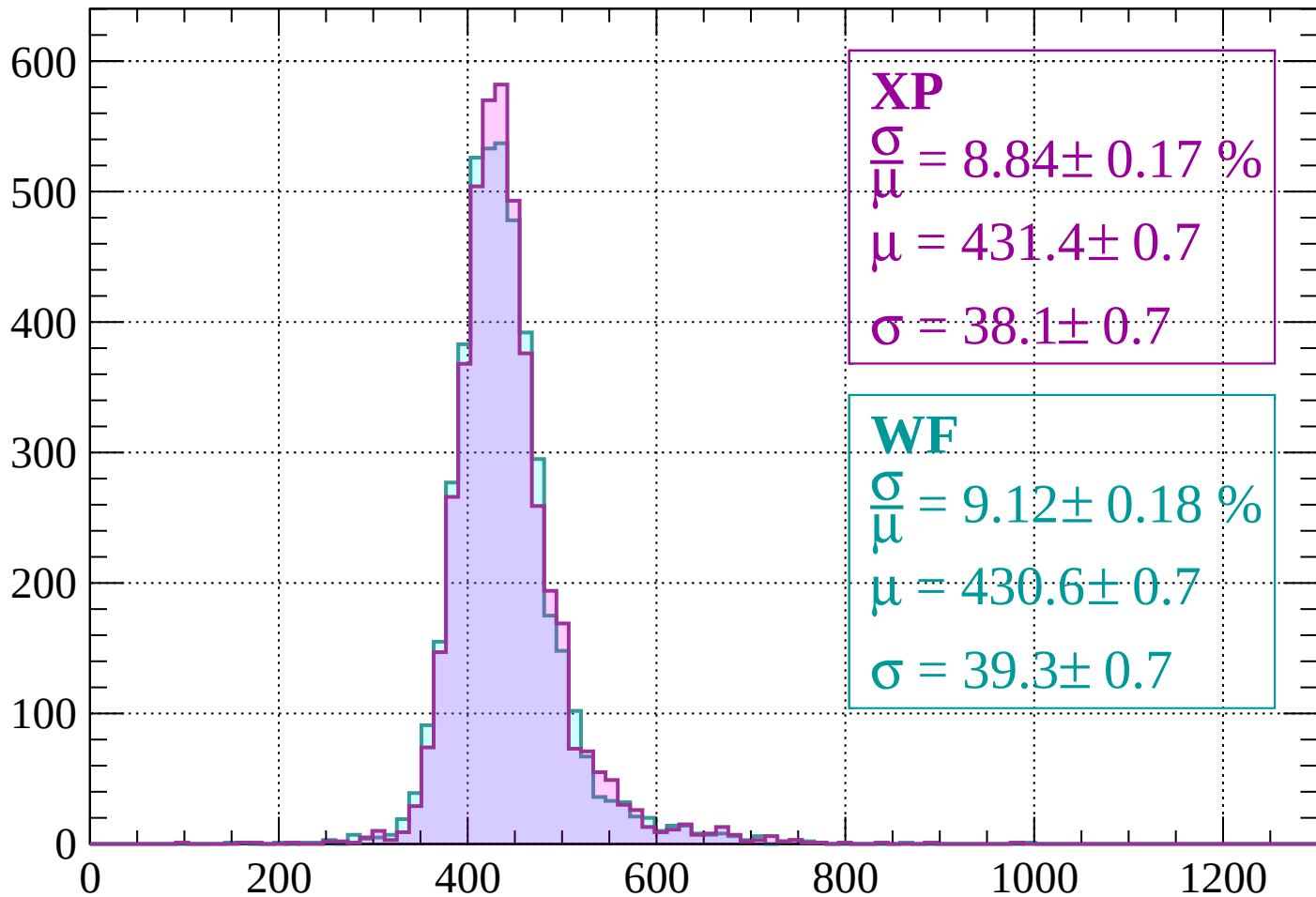
$$\mu = 427.2 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$$

$$\mu = 431.4 \pm 0.7$$

$$\sigma = 38.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.18 \%$$

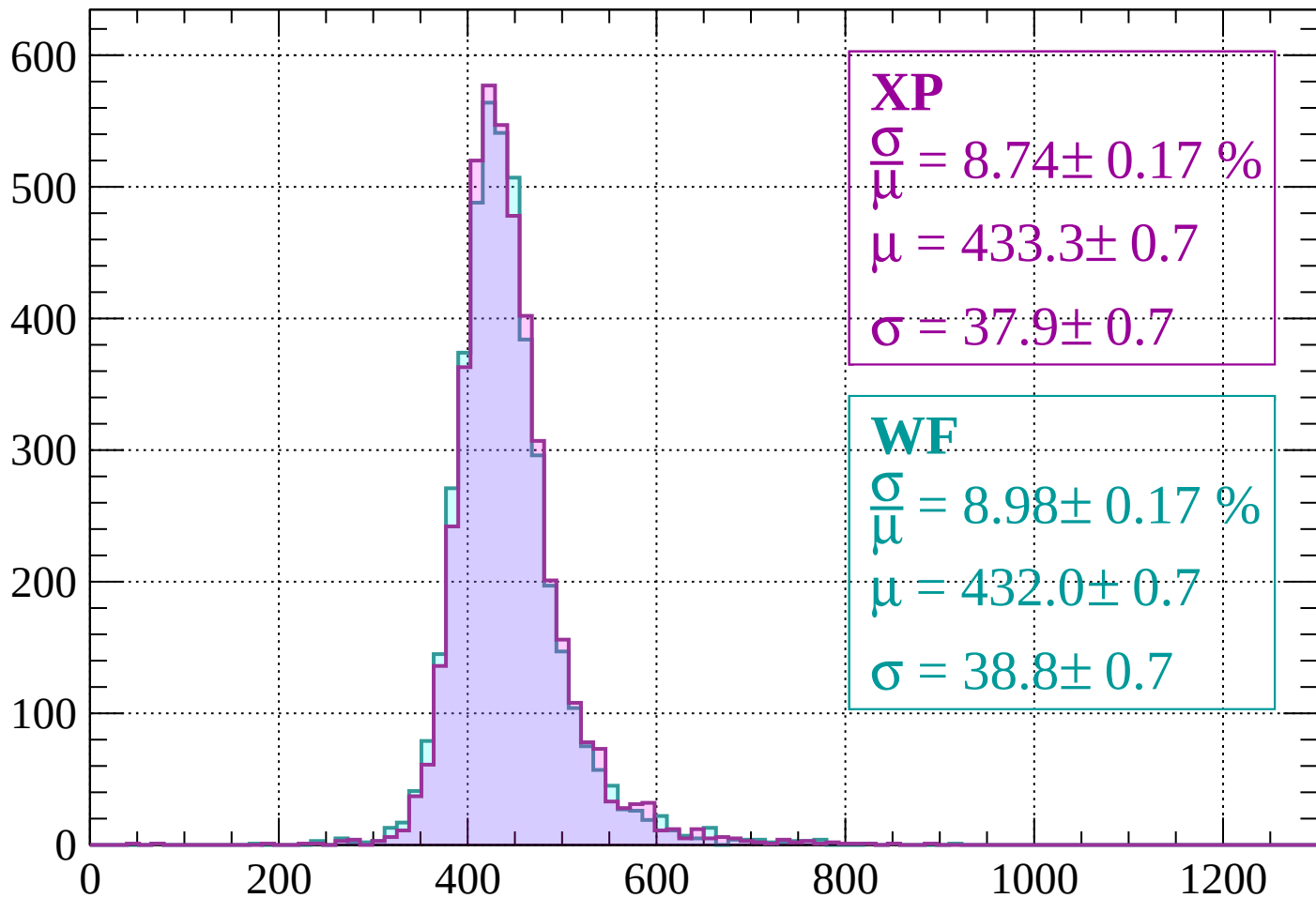
$$\mu = 430.6 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.17 \%$$

$$\mu = 433.3 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.17 \%$$

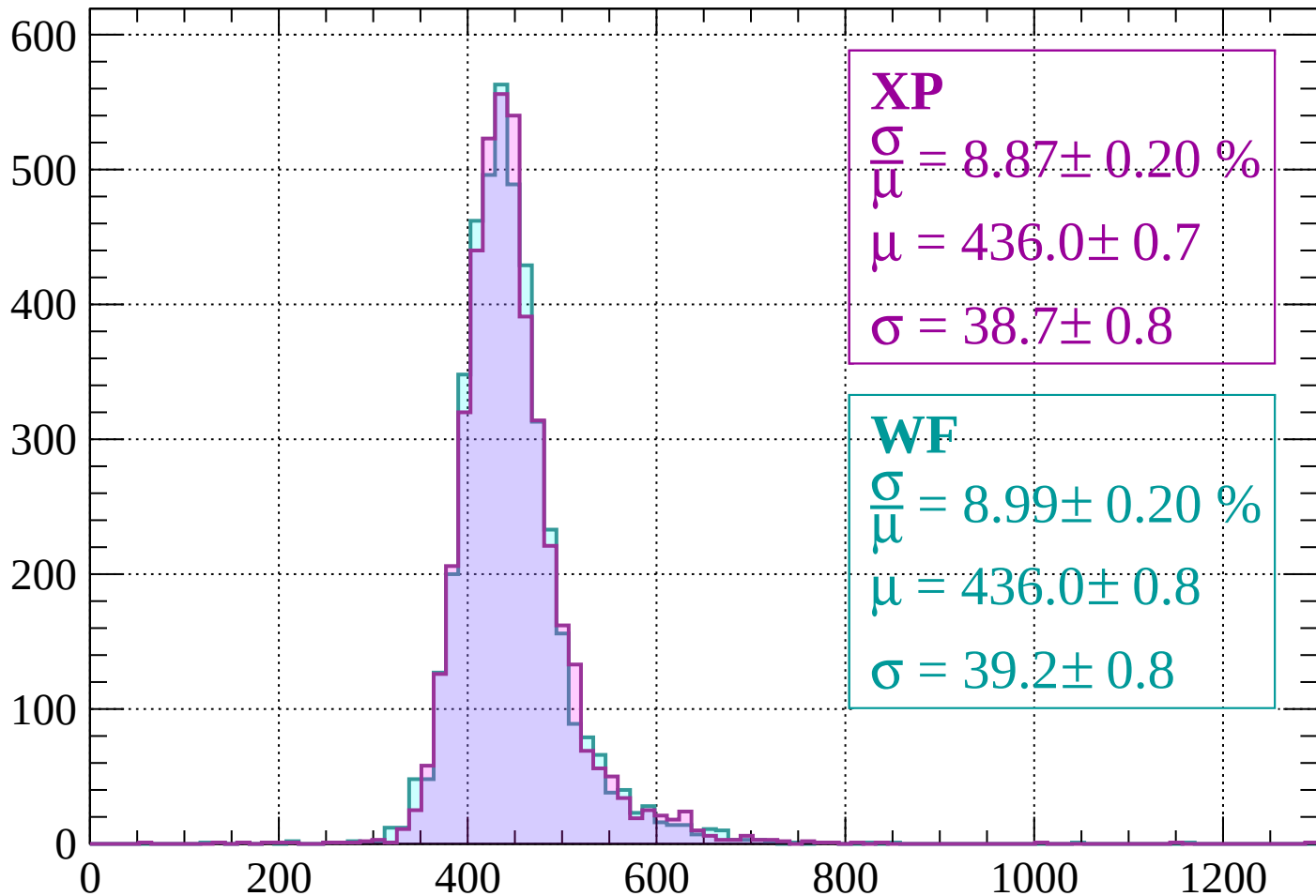
$$\mu = 432.0 \pm 0.7$$

$$\sigma = 38.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.87 \pm 0.20 \%$$

$$\mu = 436.0 \pm 0.7$$

$$\sigma = 38.7 \pm 0.8$$

WF

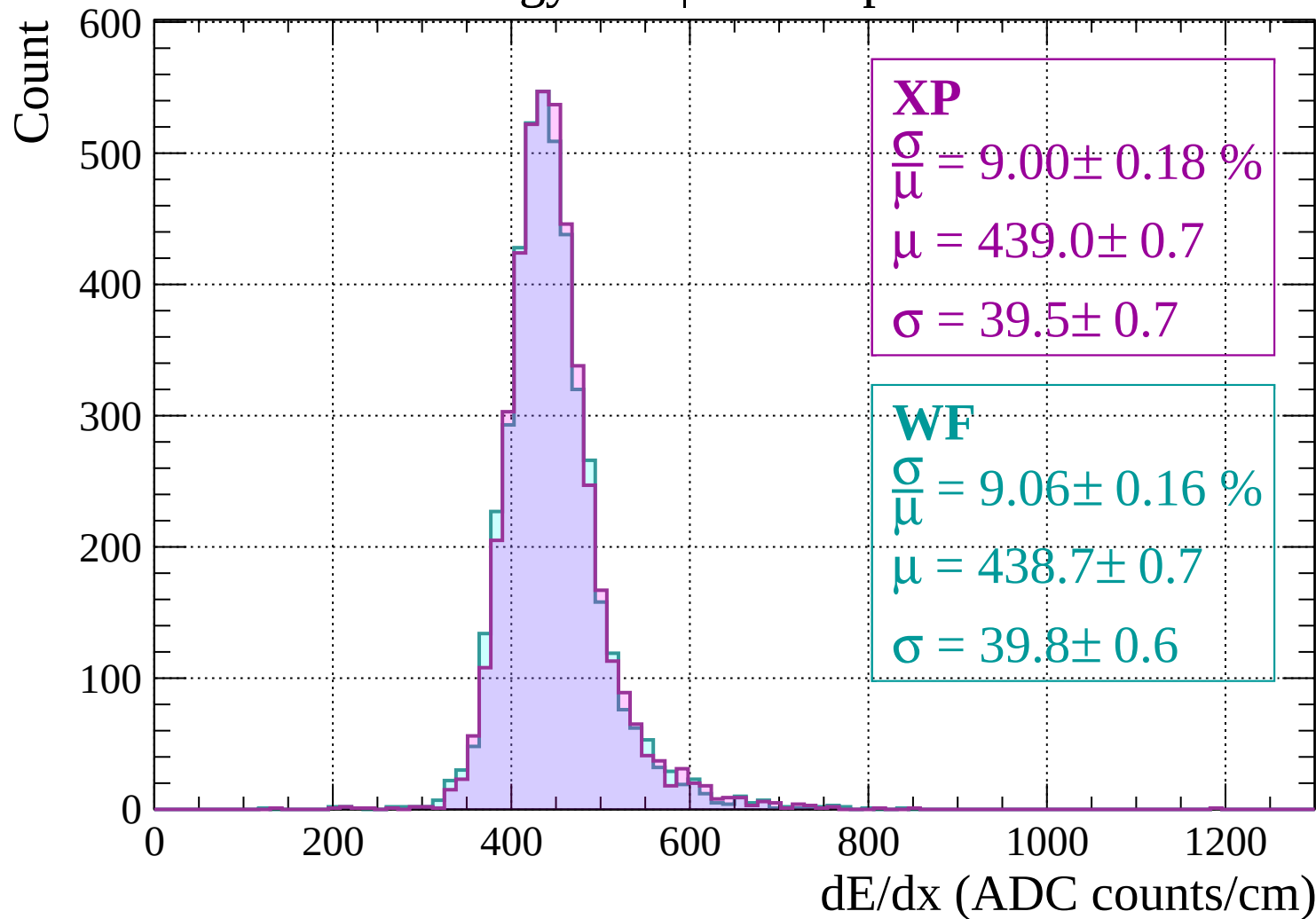
$$\frac{\sigma}{\bar{\mu}} = 8.99 \pm 0.20 \%$$

$$\mu = 436.0 \pm 0.8$$

$$\sigma = 39.2 \pm 0.8$$

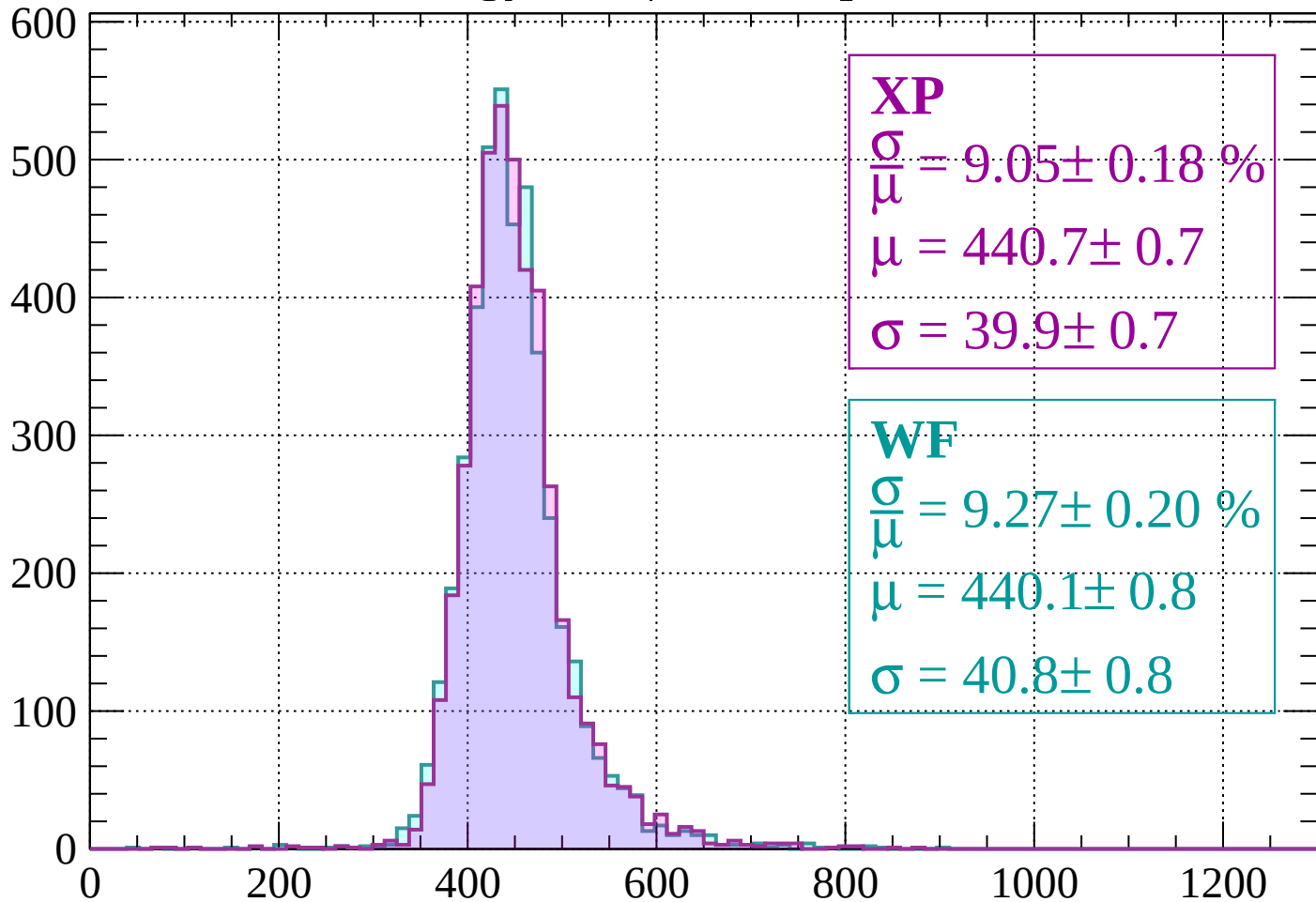
dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$



Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 9.05 \pm 0.18 \%$$

$$\mu = 440.7 \pm 0.7$$

$$\sigma = 39.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.27 \pm 0.20 \%$$

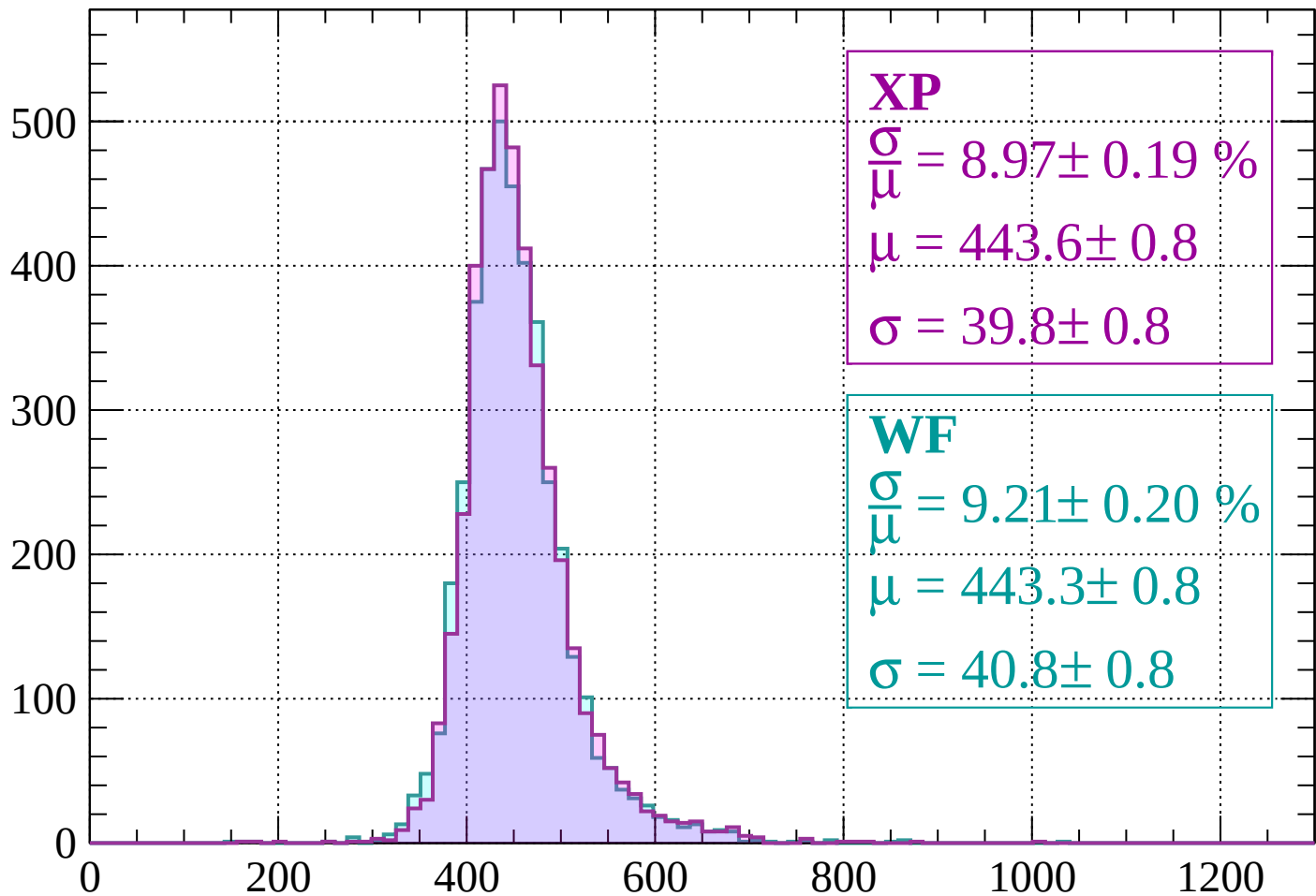
$$\mu = 440.1 \pm 0.8$$

$$\sigma = 40.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.97 \pm 0.19 \%$$

$$\mu = 443.6 \pm 0.8$$

$$\sigma = 39.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.20 \%$$

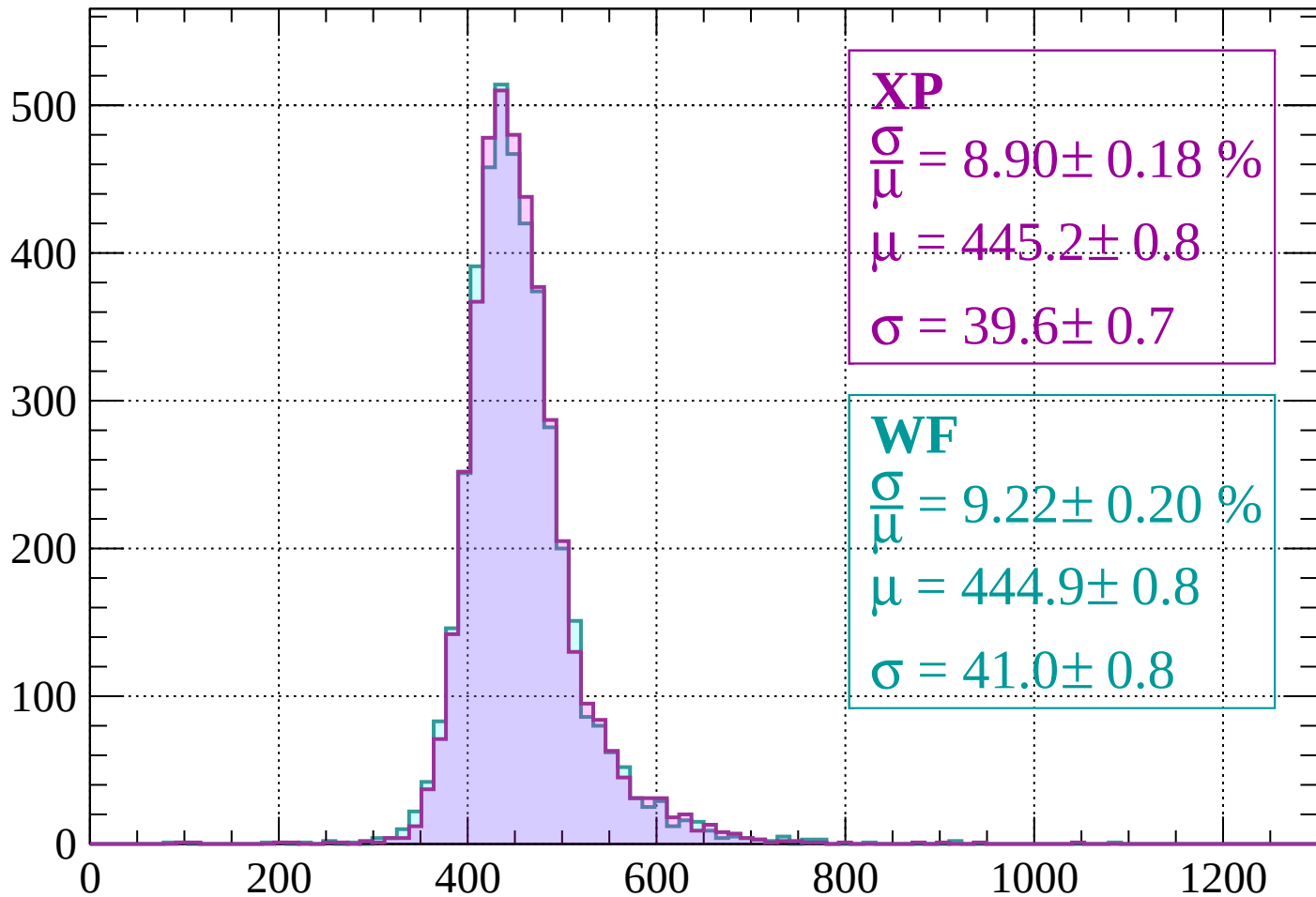
$$\mu = 443.3 \pm 0.8$$

$$\sigma = 40.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.90 \pm 0.18 \%$$

$$\mu = 445.2 \pm 0.8$$

$$\sigma = 39.6 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.22 \pm 0.20 \%$$

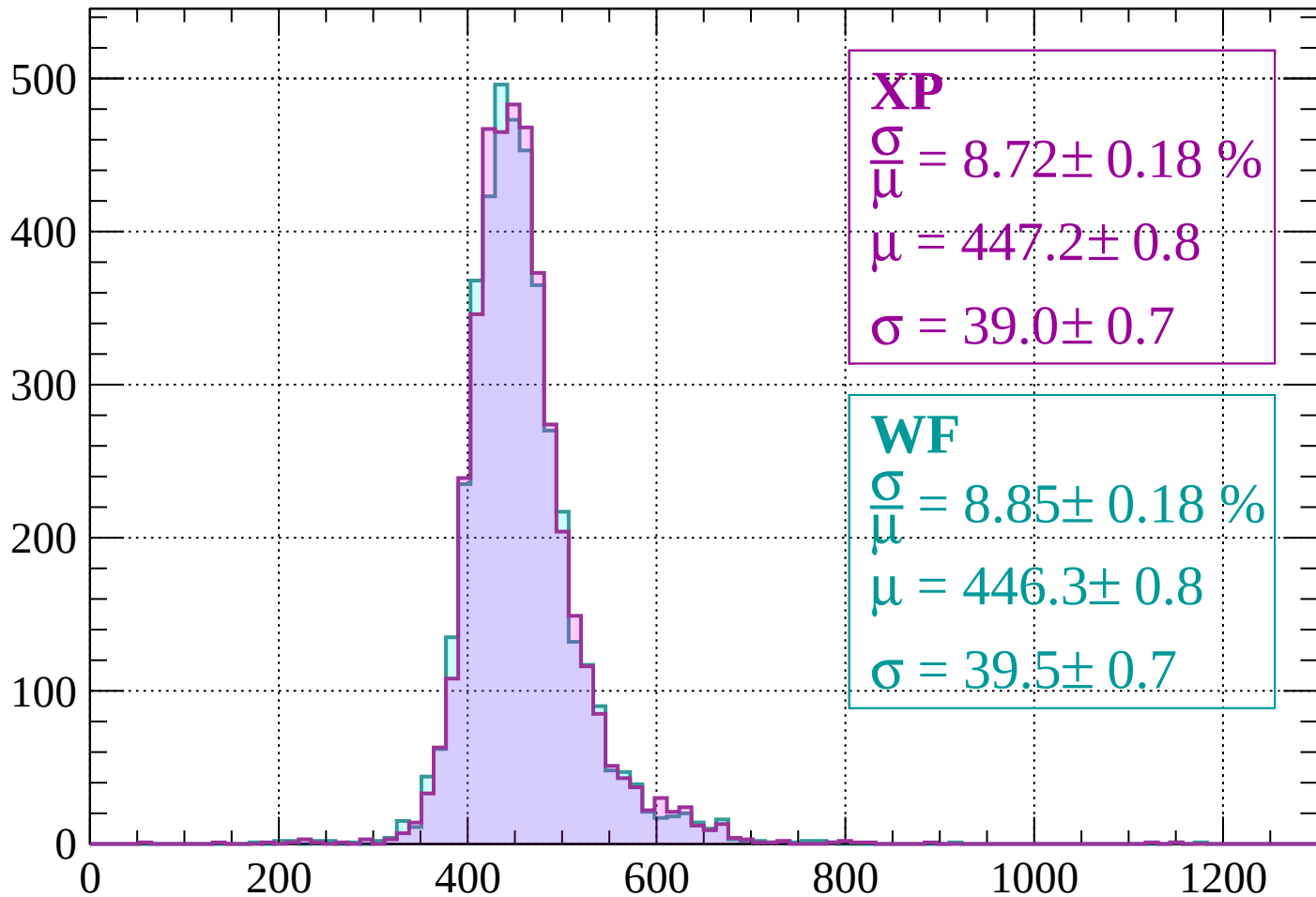
$$\mu = 444.9 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.18 \%$$

$$\mu = 447.2 \pm 0.8$$

$$\sigma = 39.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.85 \pm 0.18 \%$$

$$\mu = 446.3 \pm 0.8$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.19 \%$$

$$\mu = 450.2 \pm 0.8$$

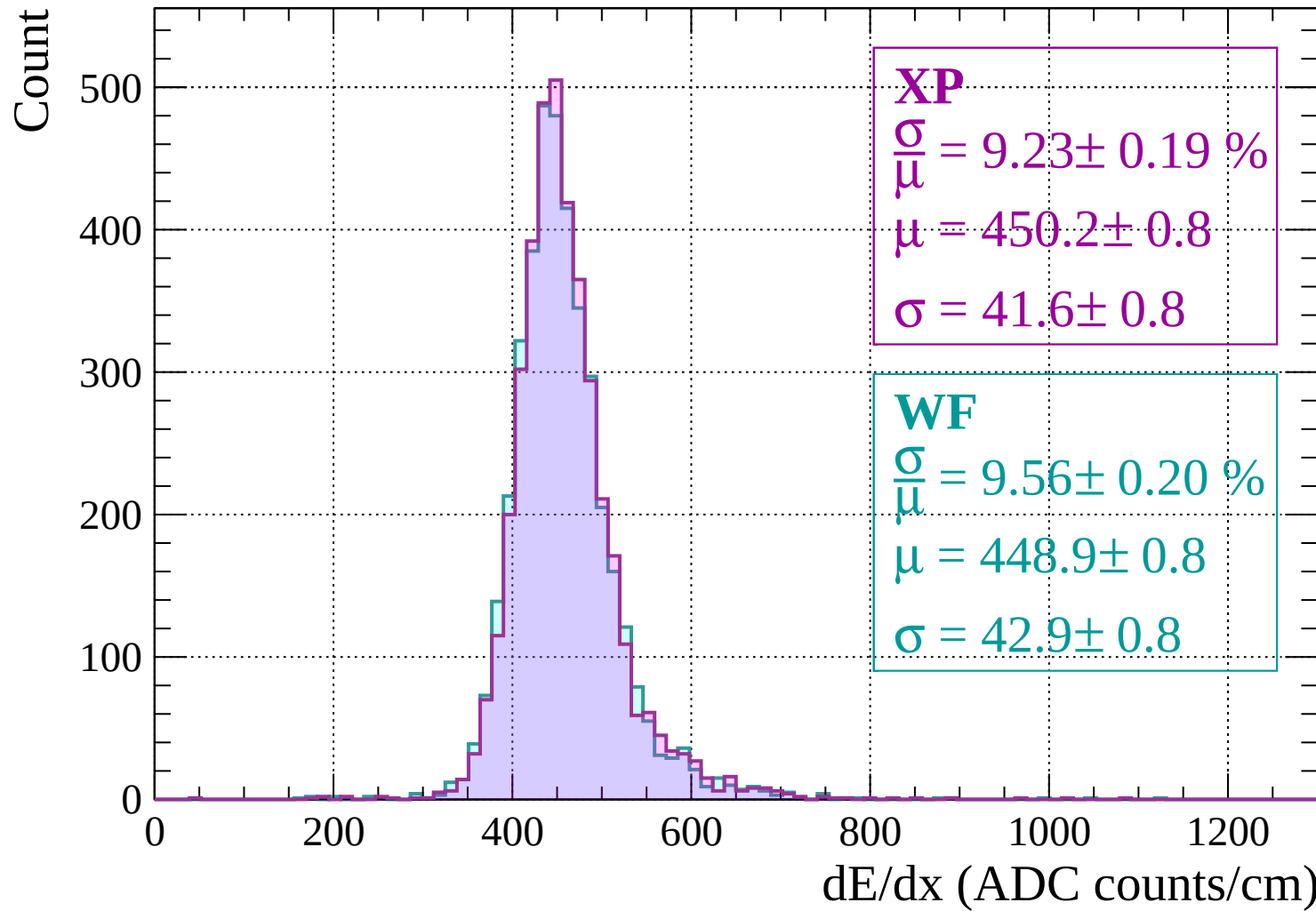
$$\sigma = 41.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.20 \%$$

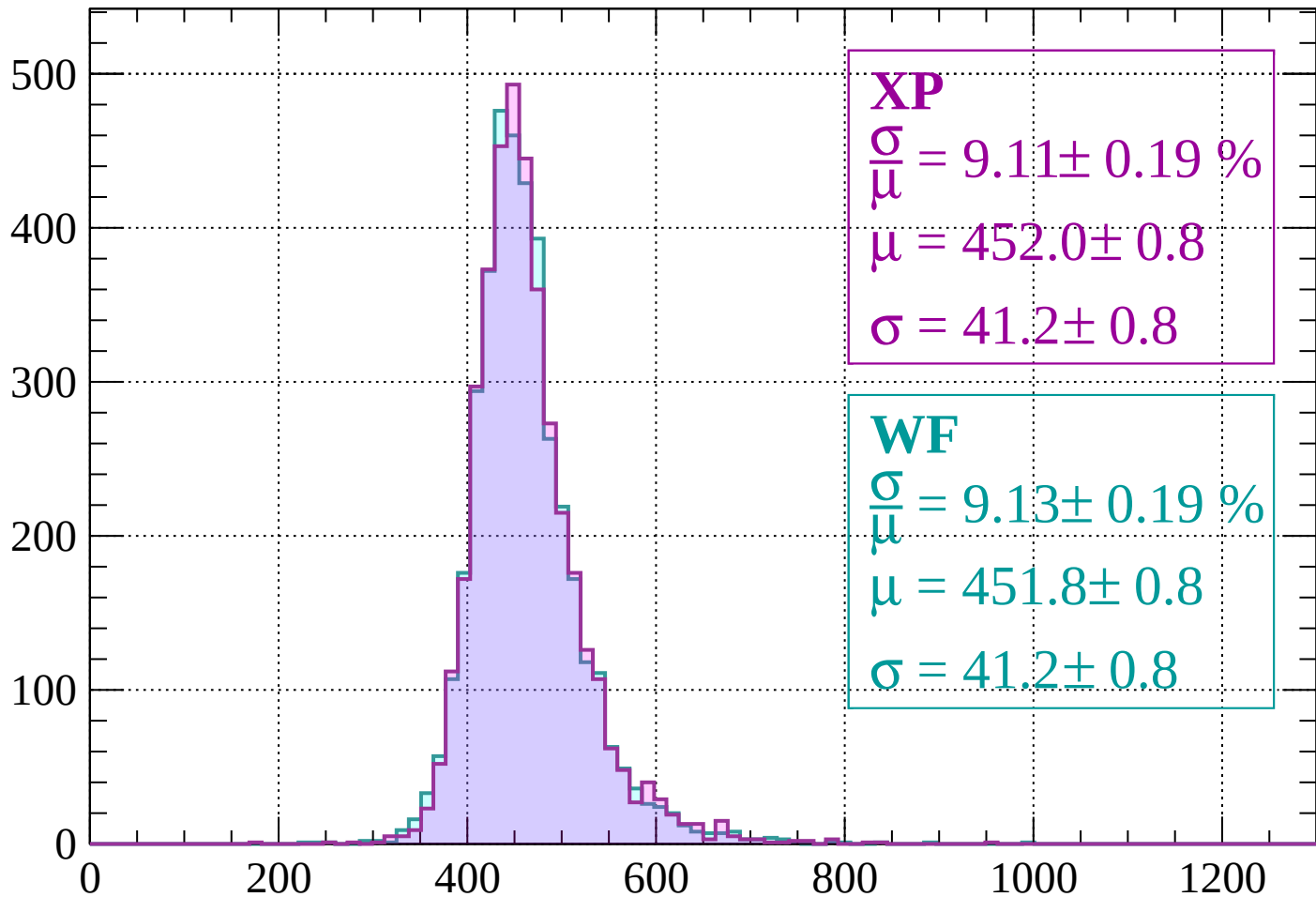
$$\mu = 448.9 \pm 0.8$$

$$\sigma = 42.9 \pm 0.8$$



Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.19 \%$$

$$\mu = 452.0 \pm 0.8$$

$$\sigma = 41.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.13 \pm 0.19 \%$$

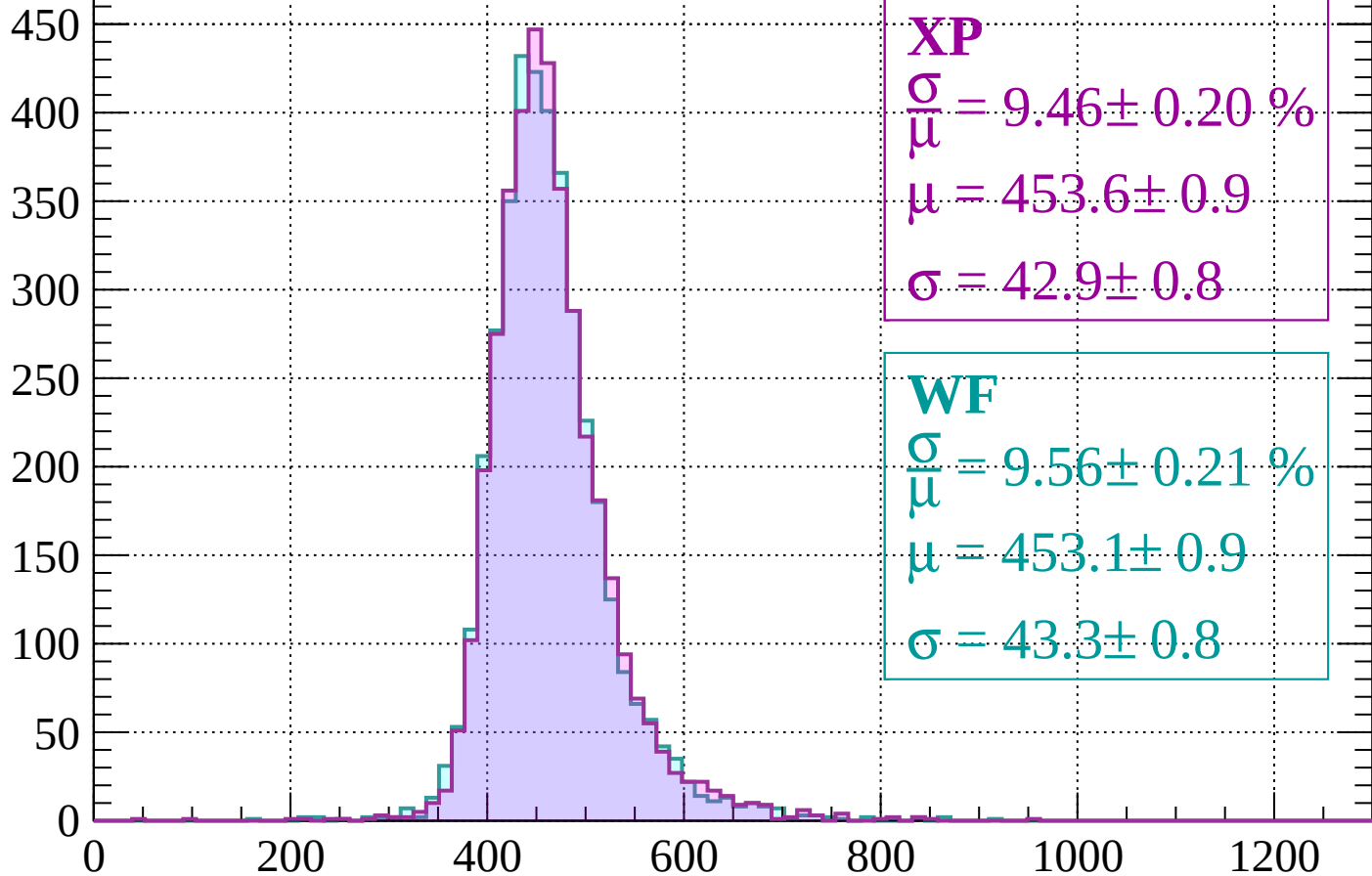
$$\mu = 451.8 \pm 0.8$$

$$\sigma = 41.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

$$\mu = 453.6 \pm 0.9$$

$$\sigma = 42.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.21 \%$$

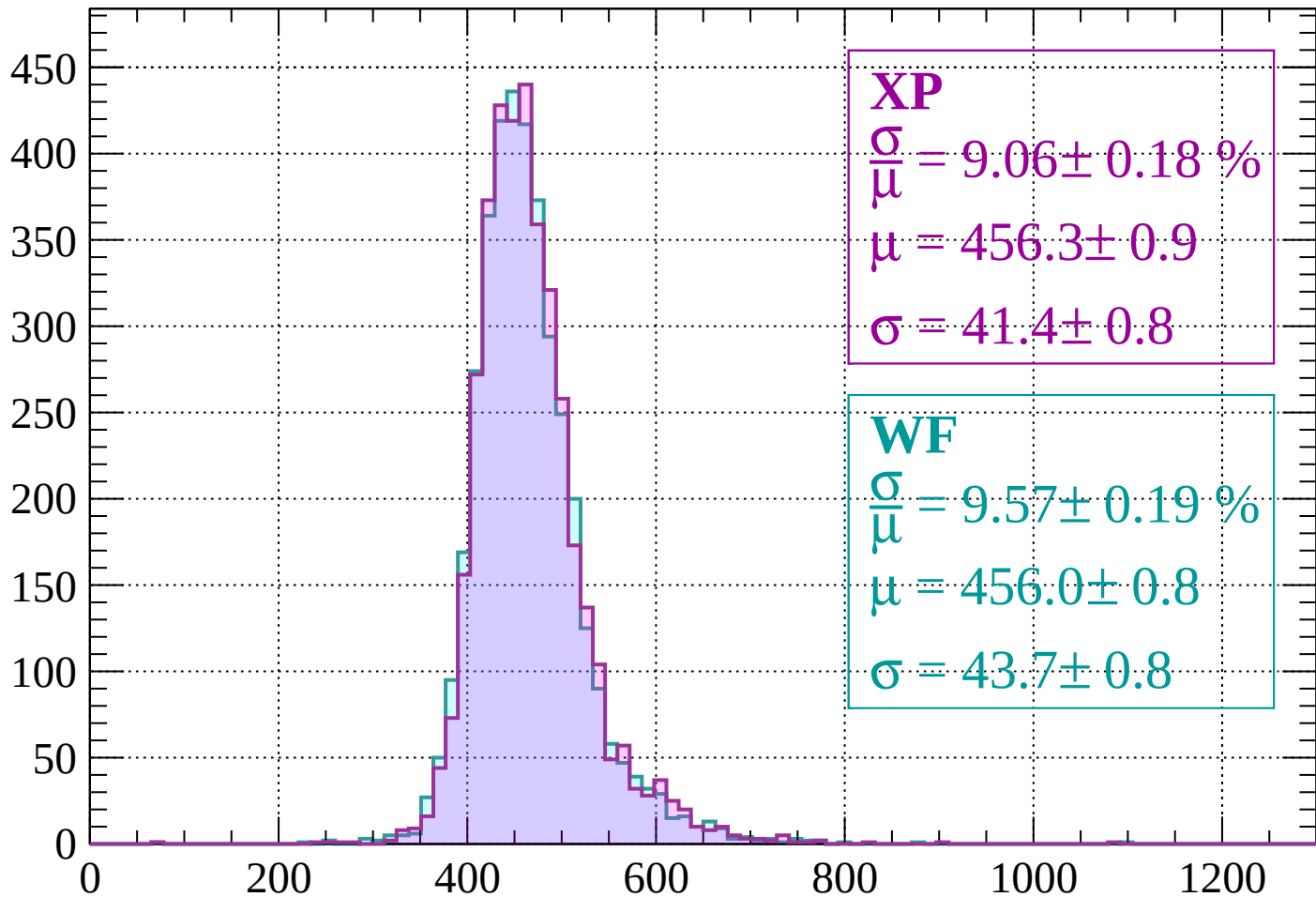
$$\mu = 453.1 \pm 0.9$$

$$\sigma = 43.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.06 \pm 0.18 \%$$

$$\mu = 456.3 \pm 0.9$$

$$\sigma = 41.4 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.57 \pm 0.19 \%$$

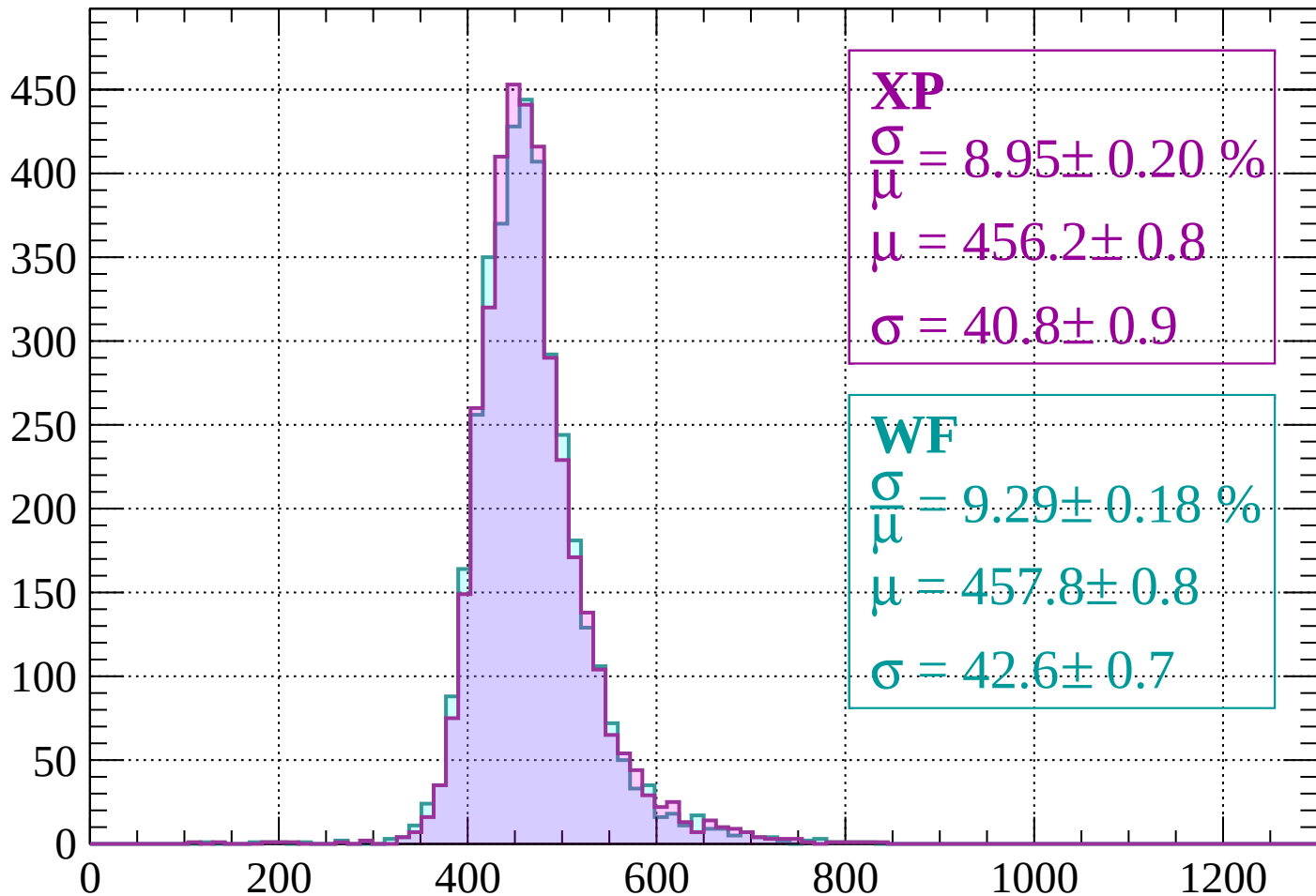
$$\mu = 456.0 \pm 0.8$$

$$\sigma = 43.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.20 \%$$

$$\mu = 456.2 \pm 0.8$$

$$\sigma = 40.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.29 \pm 0.18 \%$$

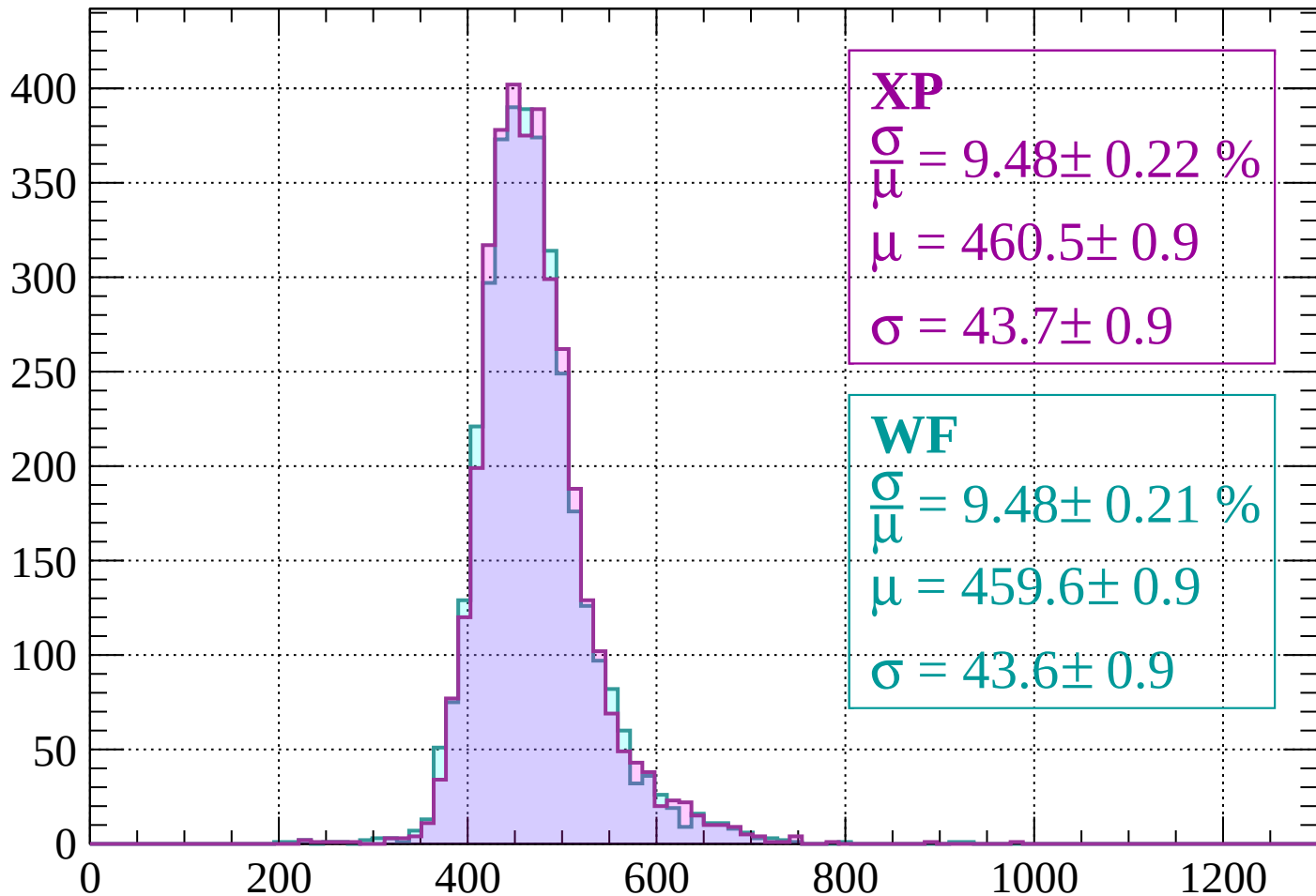
$$\mu = 457.8 \pm 0.8$$

$$\sigma = 42.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.22 \%$$

$$\mu = 460.5 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.21 \%$$

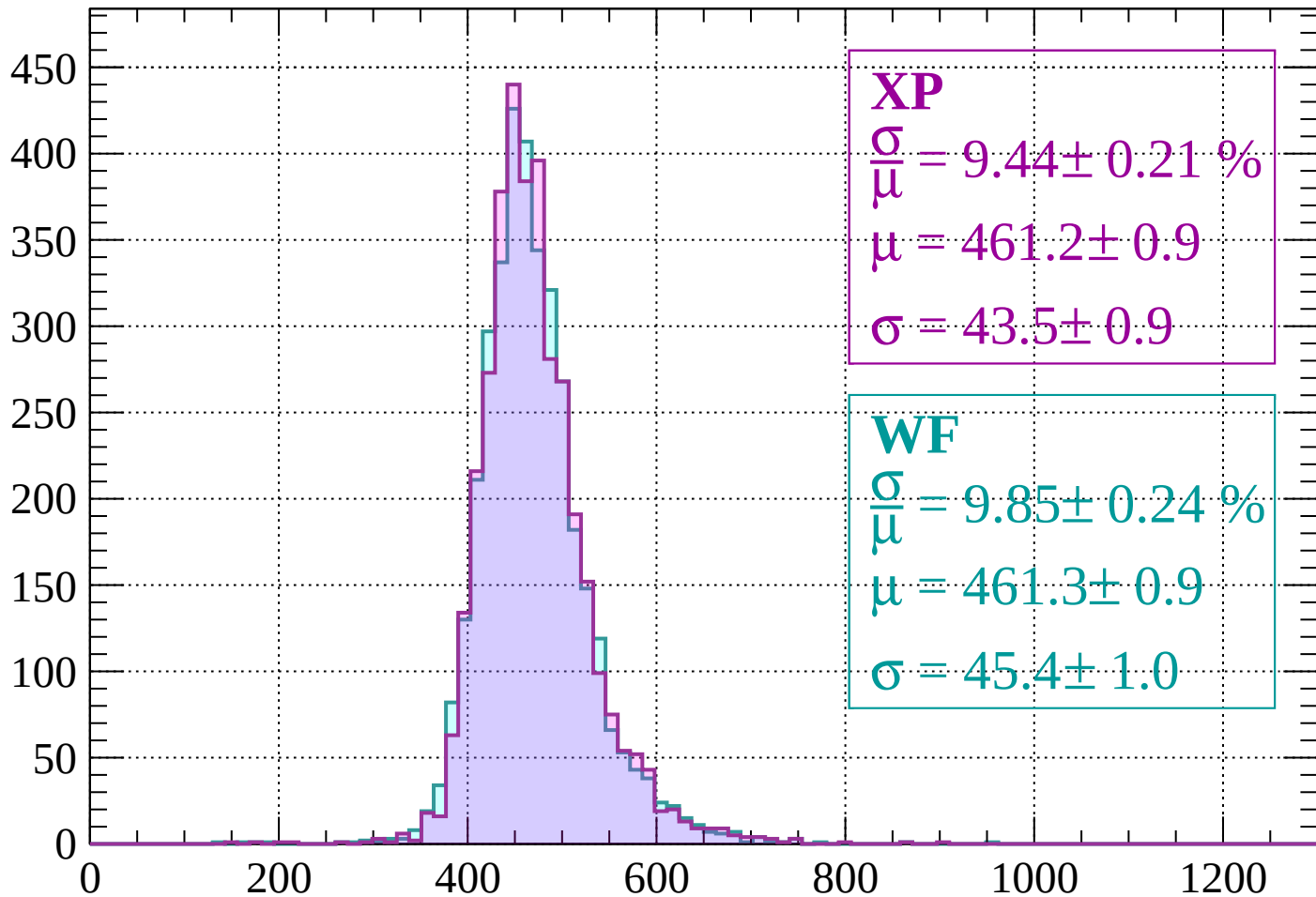
$$\mu = 459.6 \pm 0.9$$

$$\sigma = 43.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.44 \pm 0.21 \%$$

$$\mu = 461.2 \pm 0.9$$

$$\sigma = 43.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.24 \%$$

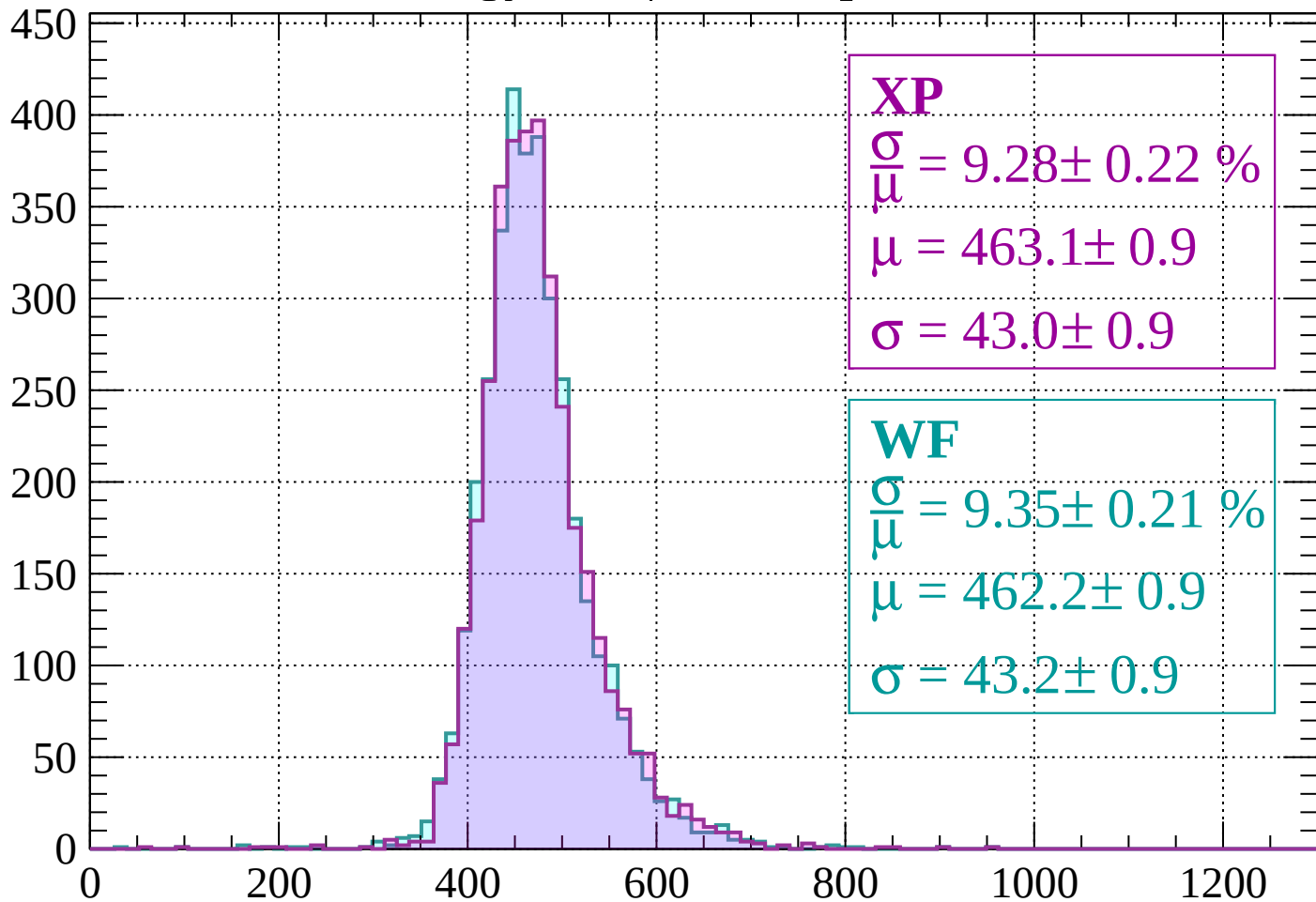
$$\mu = 461.3 \pm 0.9$$

$$\sigma = 45.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 0.22 \%$$

$$\mu = 463.1 \pm 0.9$$

$$\sigma = 43.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.21 \%$$

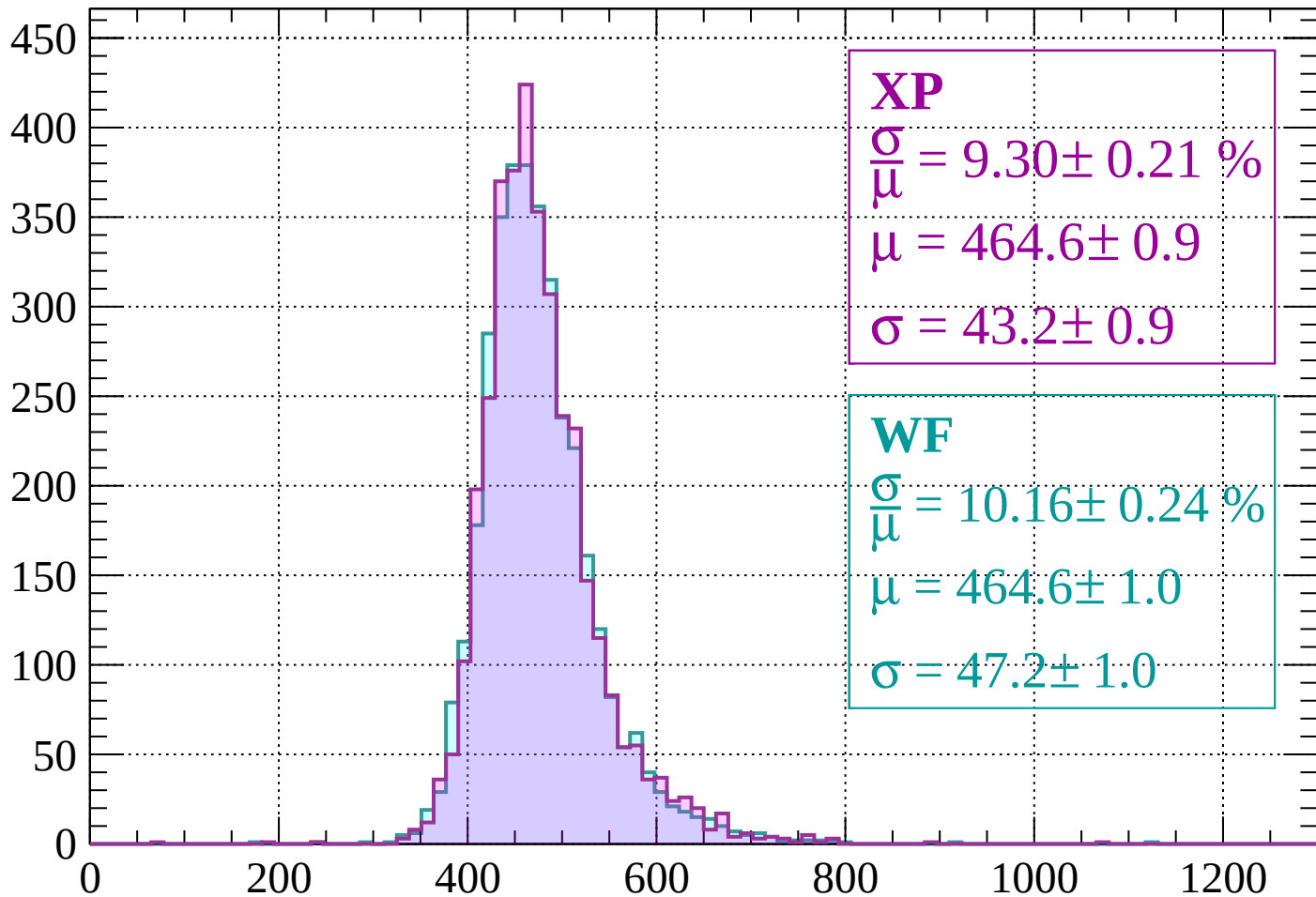
$$\mu = 462.2 \pm 0.9$$

$$\sigma = 43.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.21 \%$$

$$\mu = 464.6 \pm 0.9$$

$$\sigma = 43.2 \pm 0.9$$

WF

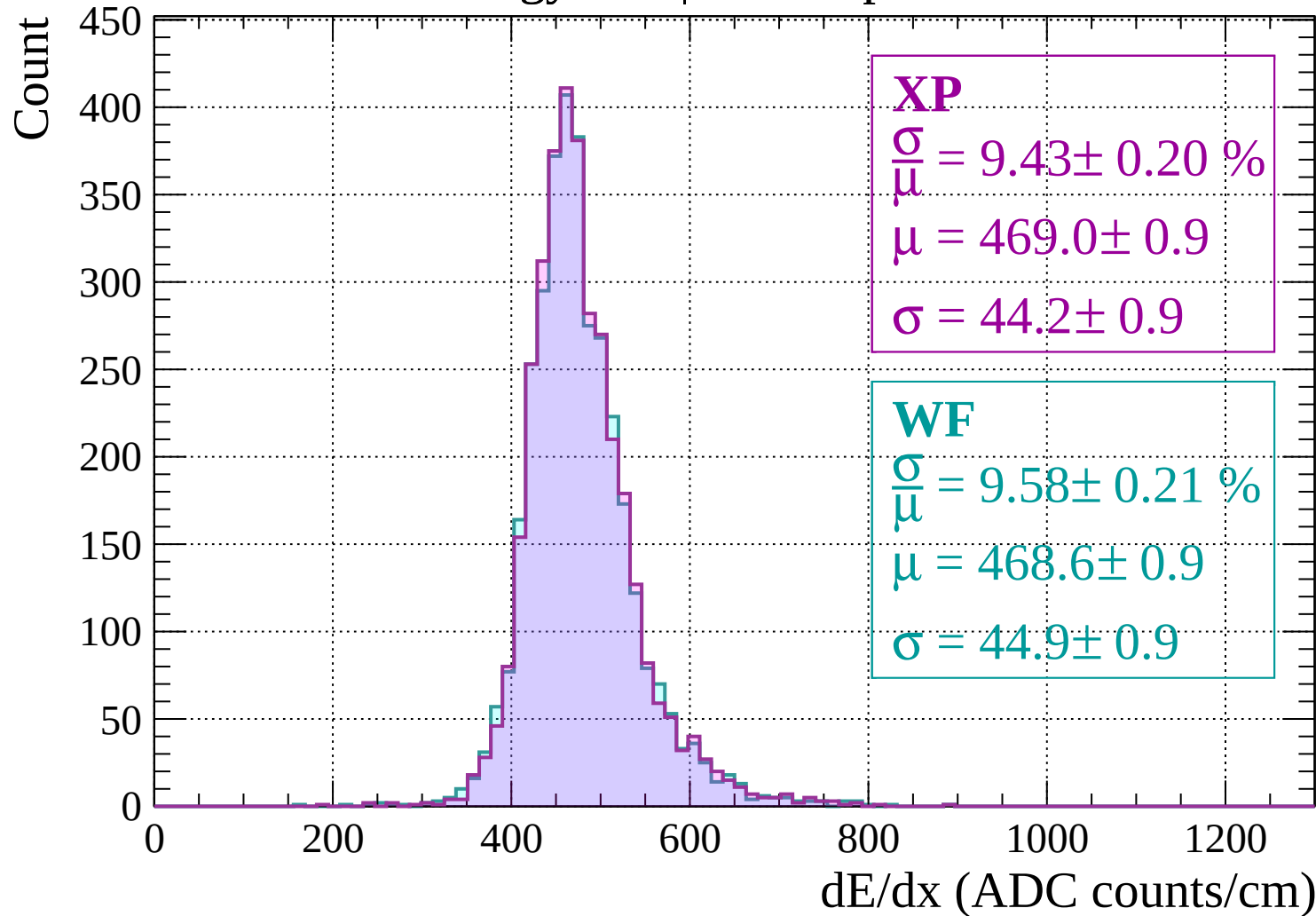
$$\frac{\sigma}{\mu} = 10.16 \pm 0.24 \%$$

$$\mu = 464.6 \pm 1.0$$

$$\sigma = 47.2 \pm 1.0$$

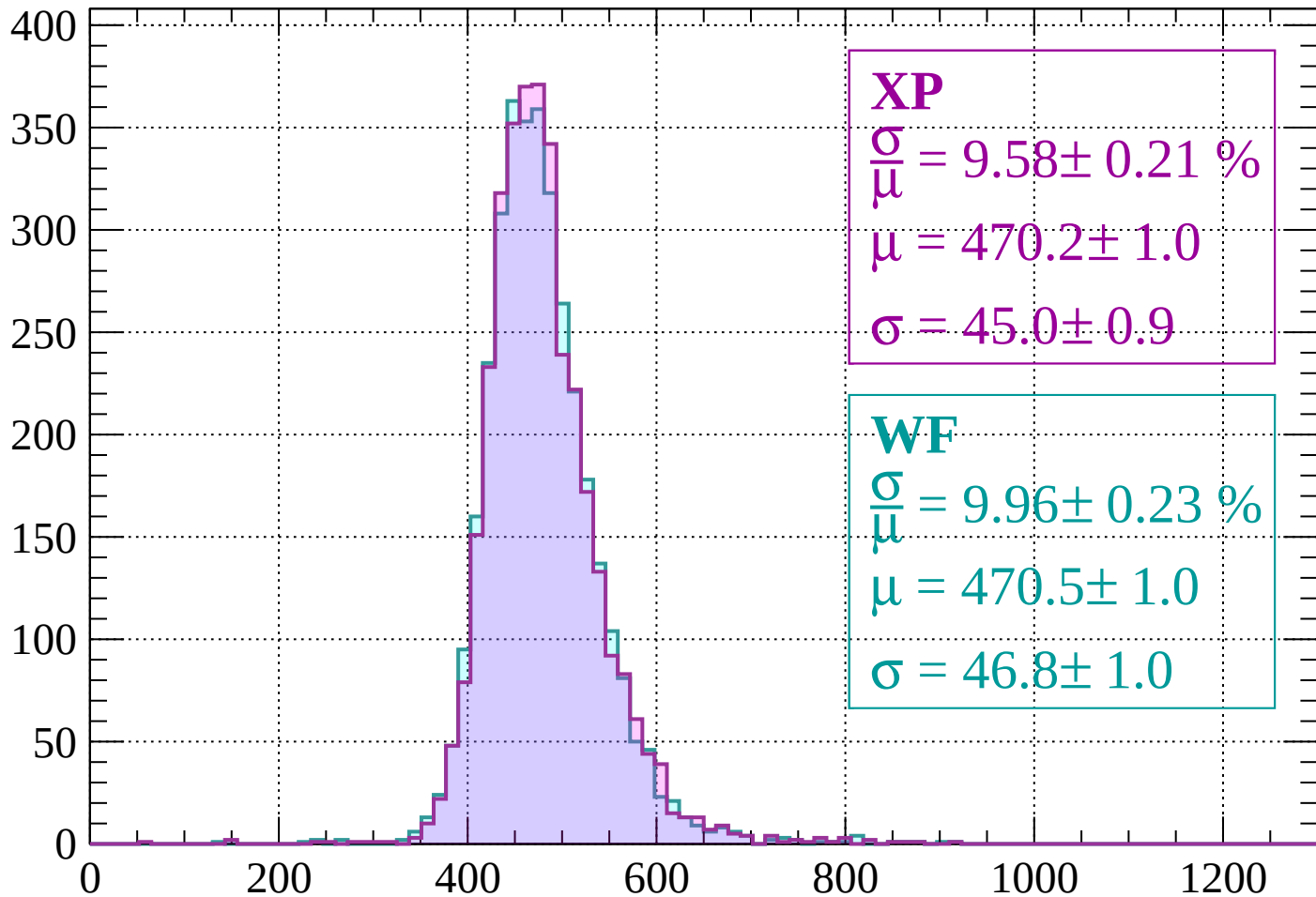
dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$



Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 9.58 \pm 0.21 \%$$

$$\mu = 470.2 \pm 1.0$$

$$\sigma = 45.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.96 \pm 0.23 \%$$

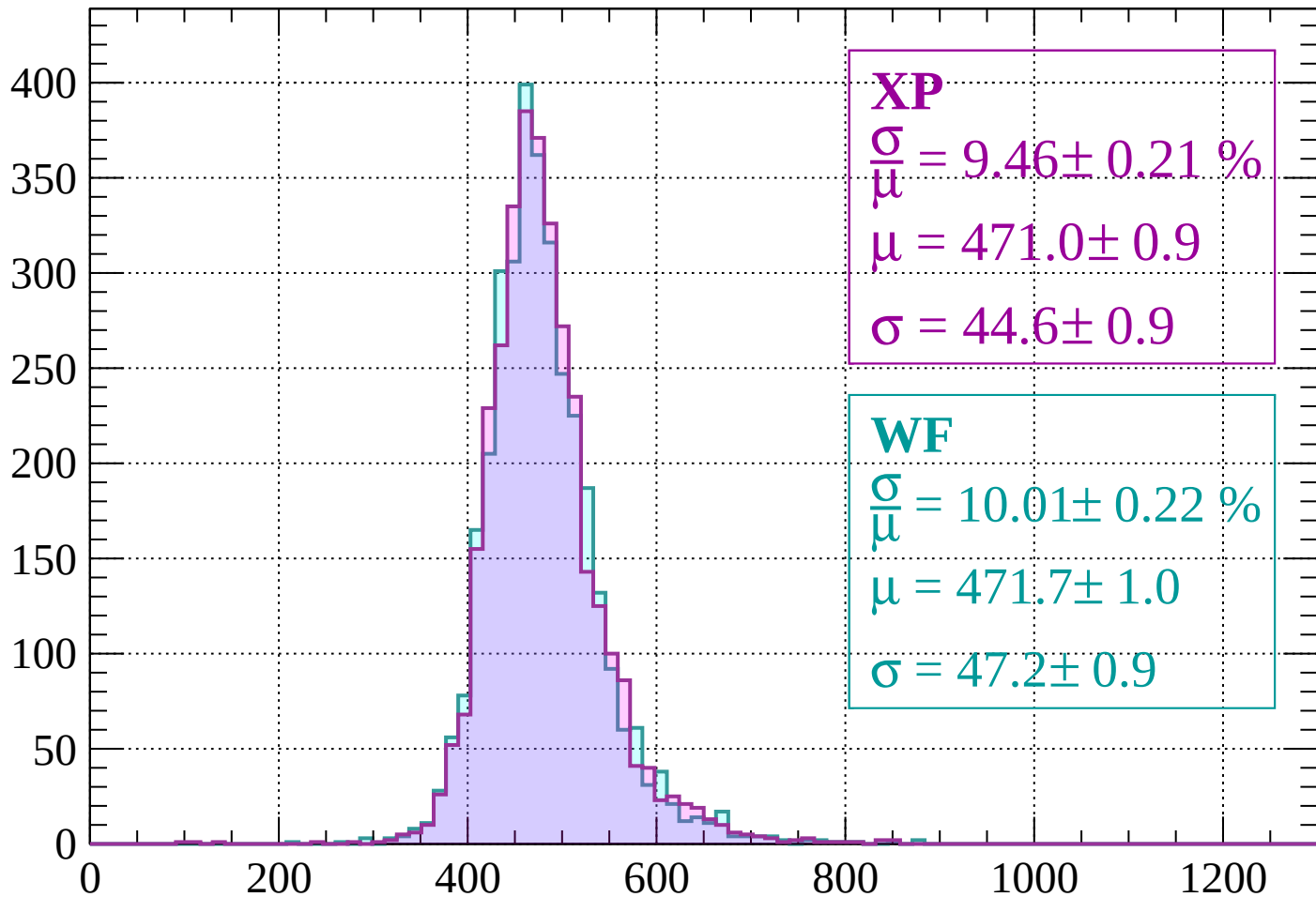
$$\mu = 470.5 \pm 1.0$$

$$\sigma = 46.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.21 \%$$

$$\mu = 471.0 \pm 0.9$$

$$\sigma = 44.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.01 \pm 0.22 \%$$

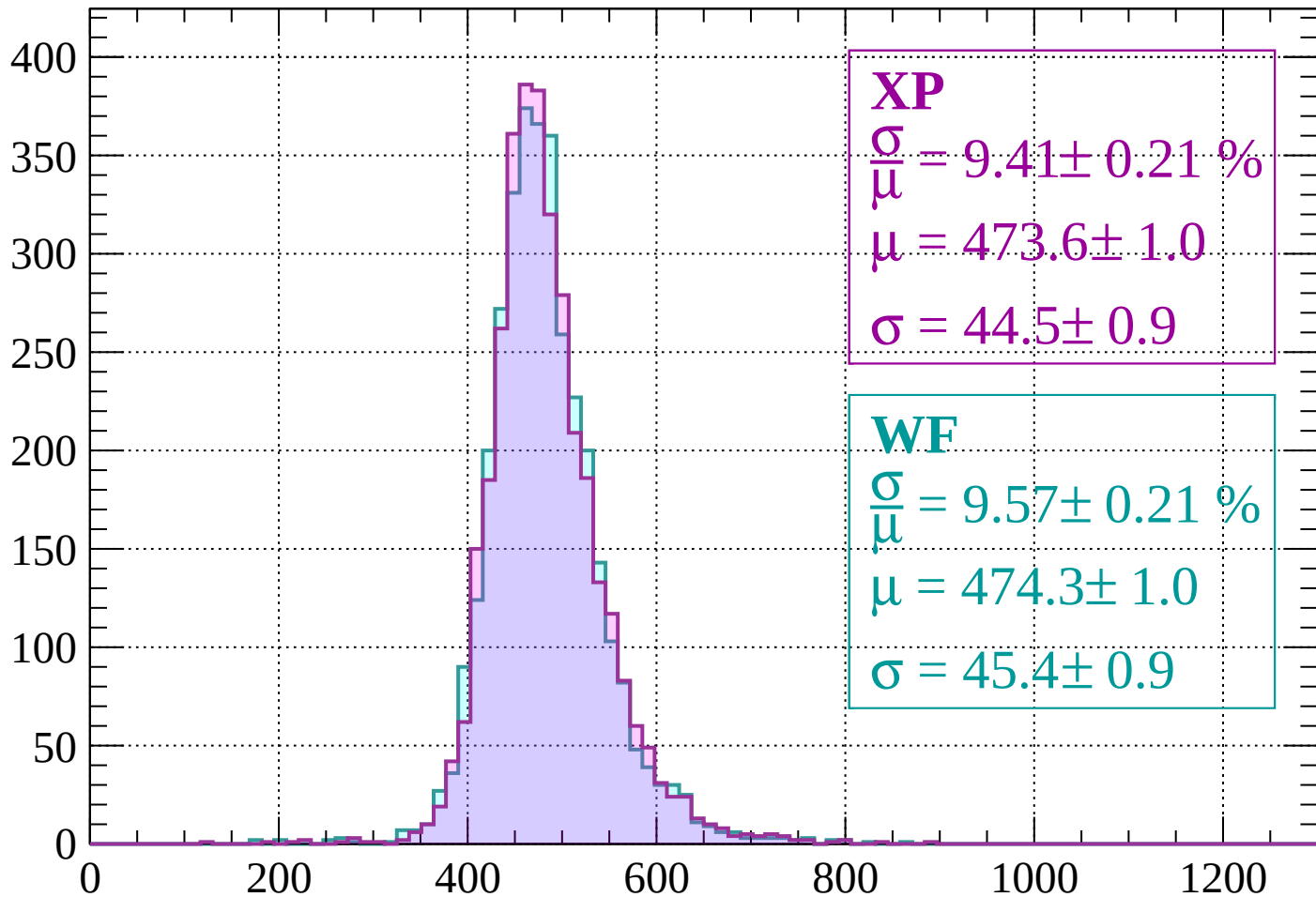
$$\mu = 471.7 \pm 1.0$$

$$\sigma = 47.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.21 \%$$

$$\mu = 473.6 \pm 1.0$$

$$\sigma = 44.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.57 \pm 0.21 \%$$

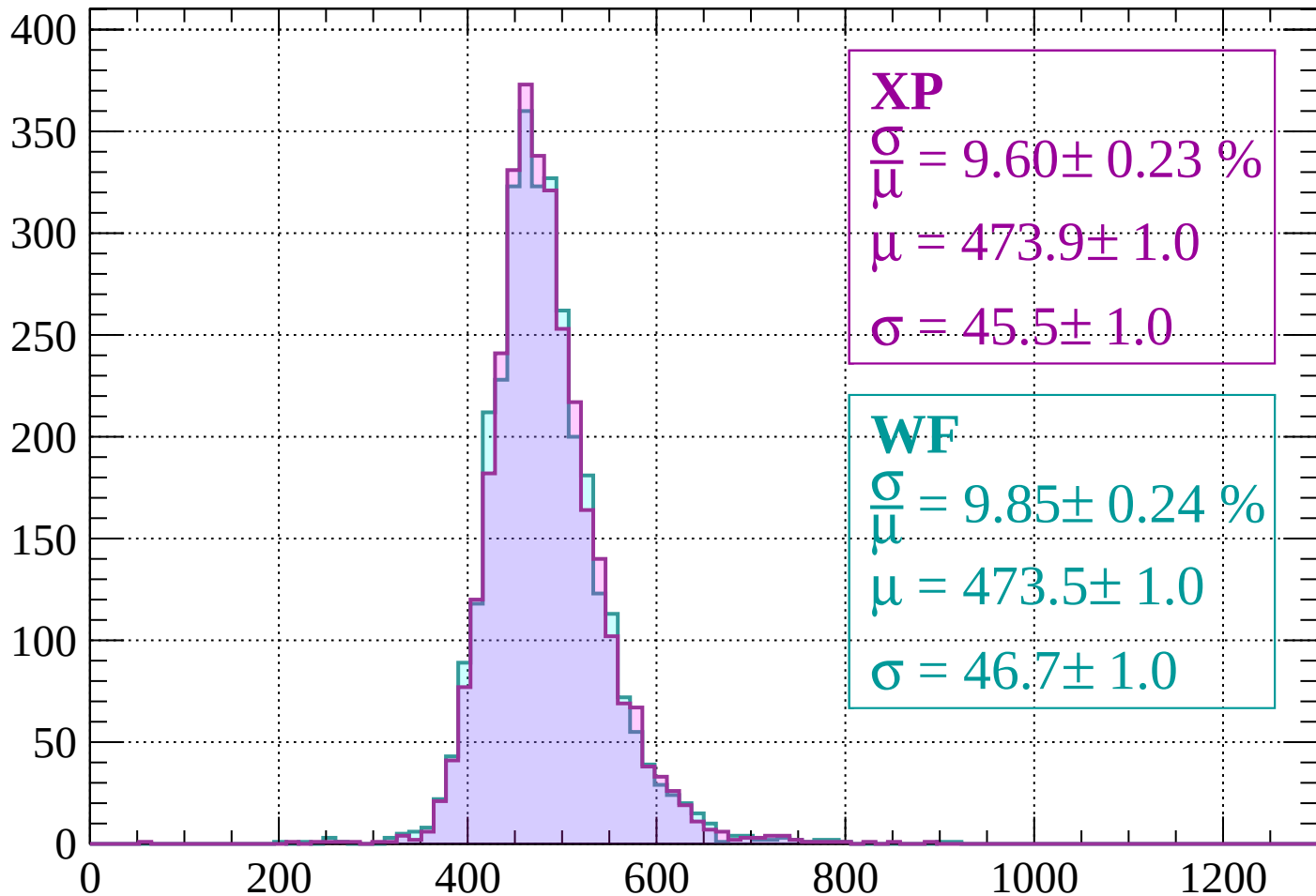
$$\mu = 474.3 \pm 1.0$$

$$\sigma = 45.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.60 \pm 0.23 \%$$

$$\mu = 473.9 \pm 1.0$$

$$\sigma = 45.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.24 \%$$

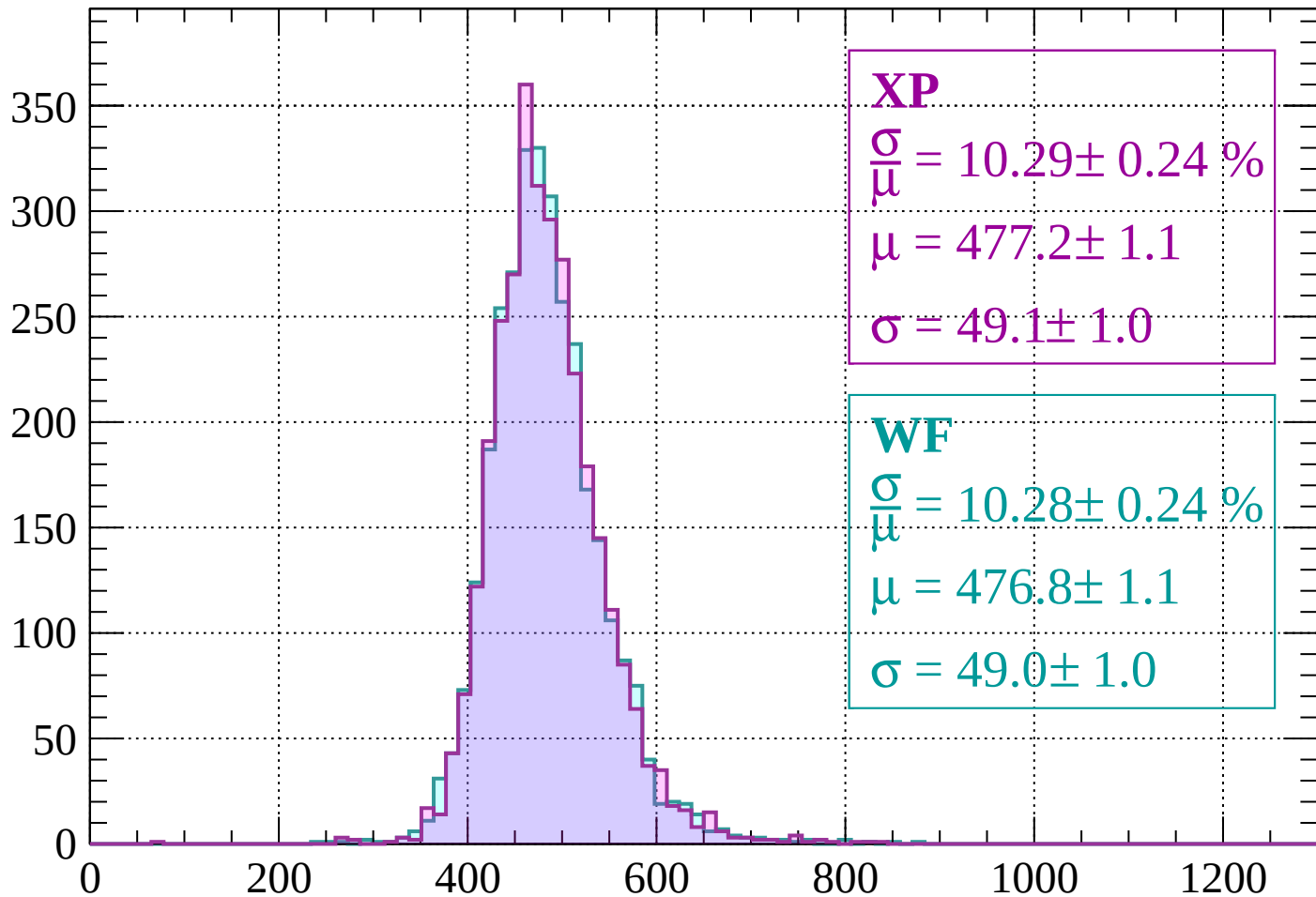
$$\mu = 473.5 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 10.29 \pm 0.24 \%$$

$$\mu = 477.2 \pm 1.1$$

$$\sigma = 49.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.28 \pm 0.24 \%$$

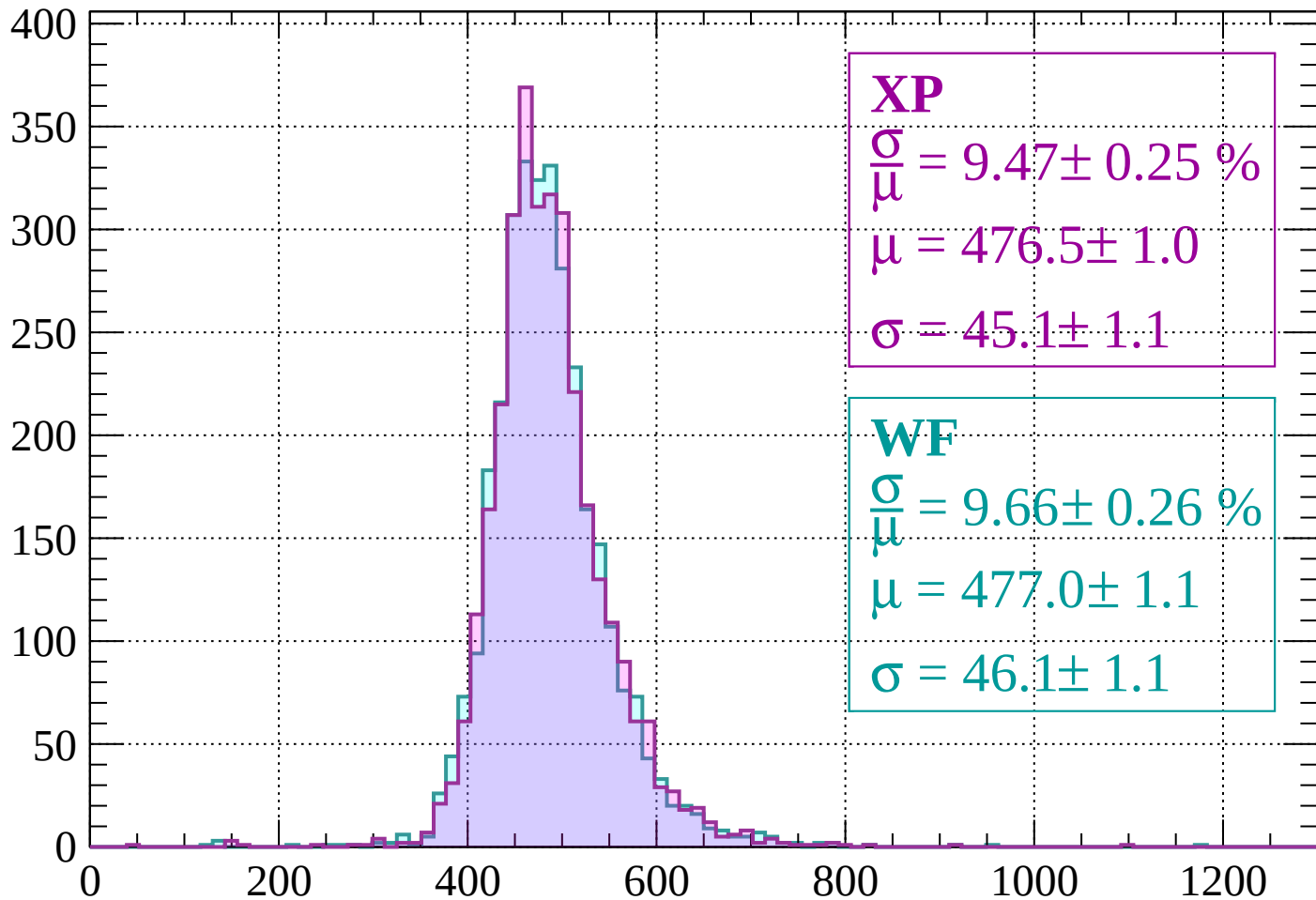
$$\mu = 476.8 \pm 1.1$$

$$\sigma = 49.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.47 \pm 0.25 \%$$

$$\mu = 476.5 \pm 1.0$$

$$\sigma = 45.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.26 \%$$

$$\mu = 477.0 \pm 1.1$$

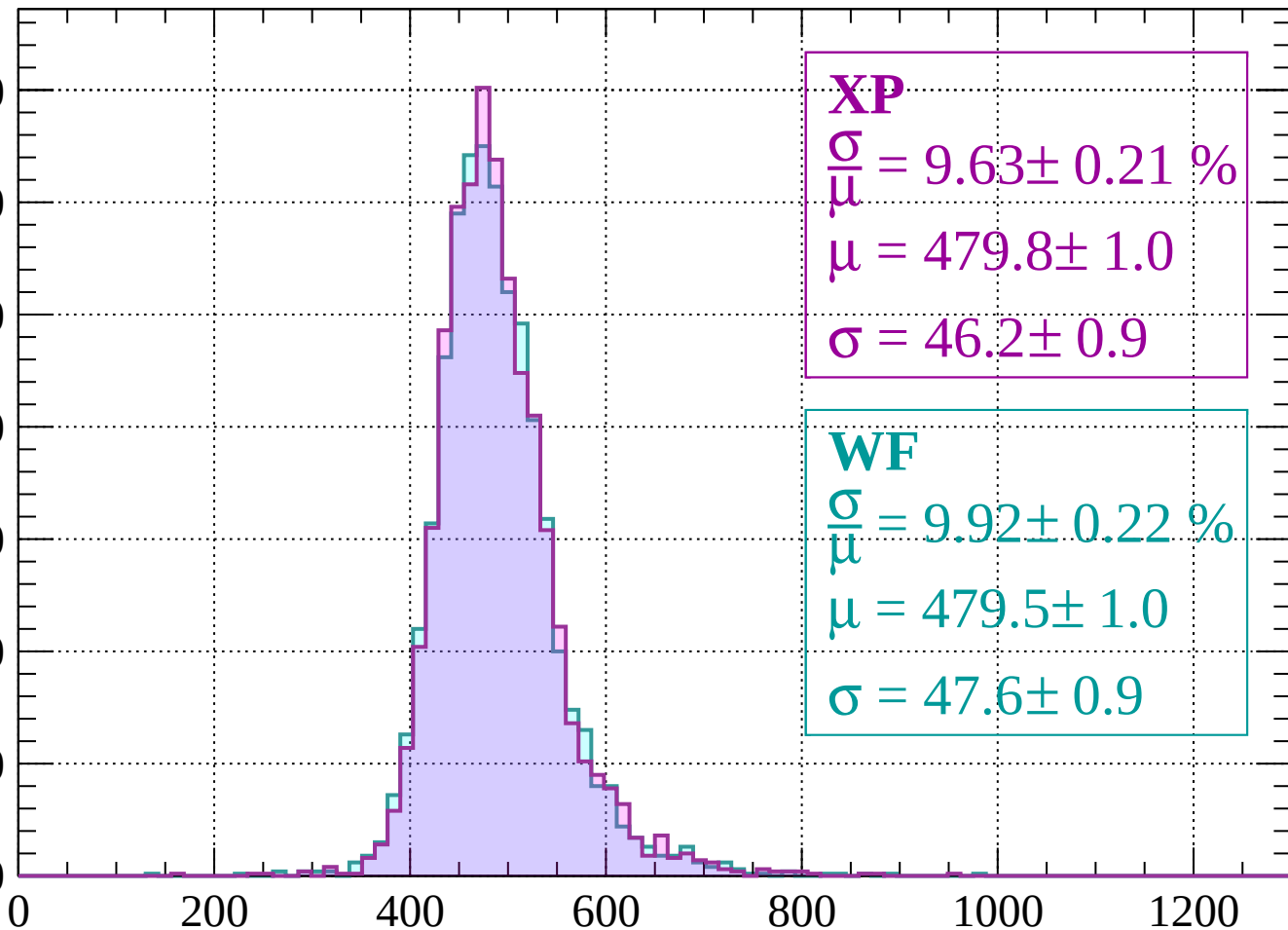
$$\sigma = 46.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.63 \pm 0.21 \%$$

$$\mu = 479.8 \pm 1.0$$

$$\sigma = 46.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.92 \pm 0.22 \%$$

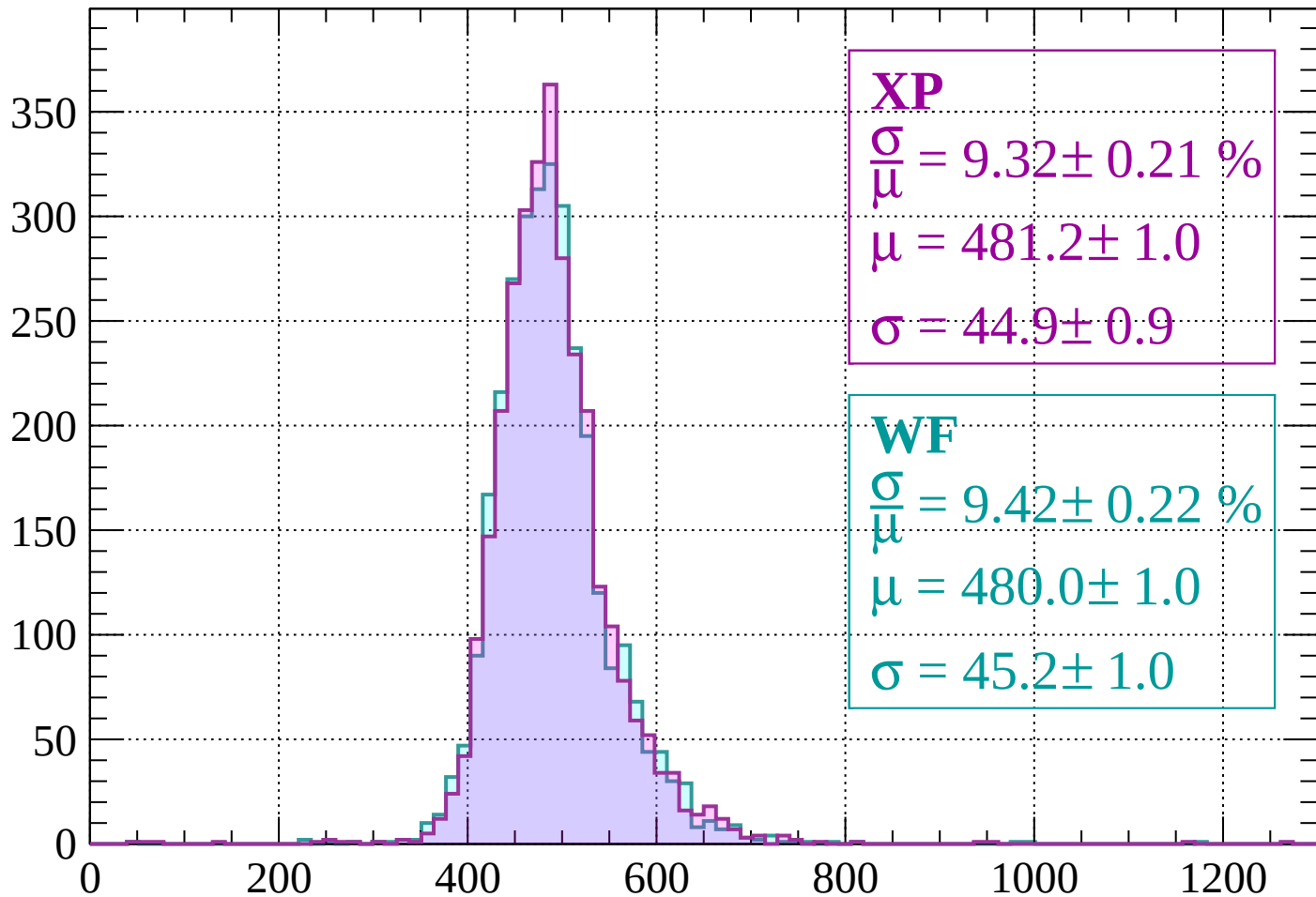
$$\mu = 479.5 \pm 1.0$$

$$\sigma = 47.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1960 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 9.32 \pm 0.21 \%$$

$$\mu = 481.2 \pm 1.0$$

$$\sigma = 44.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.22 \%$$

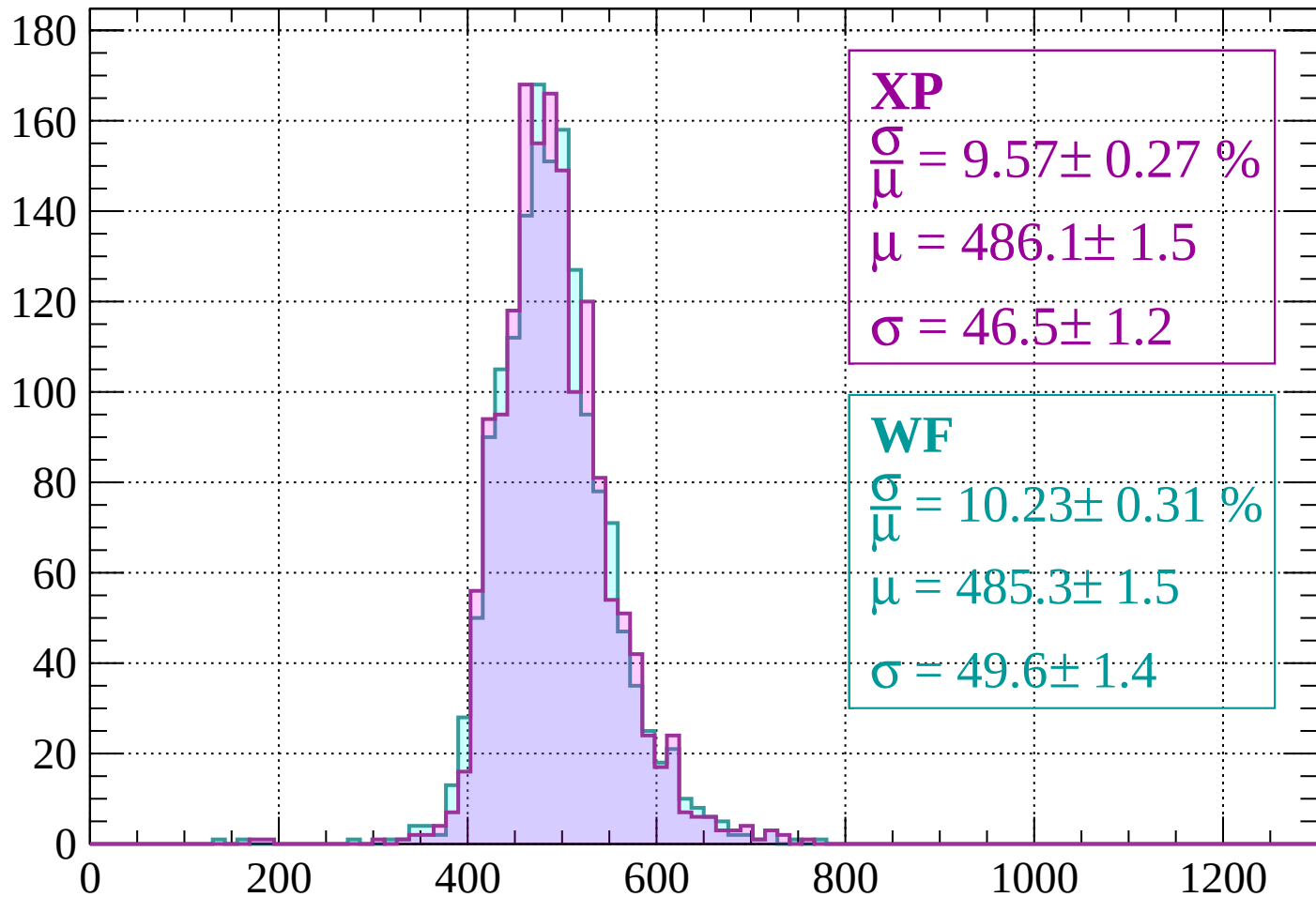
$$\mu = 480.0 \pm 1.0$$

$$\sigma = 45.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$

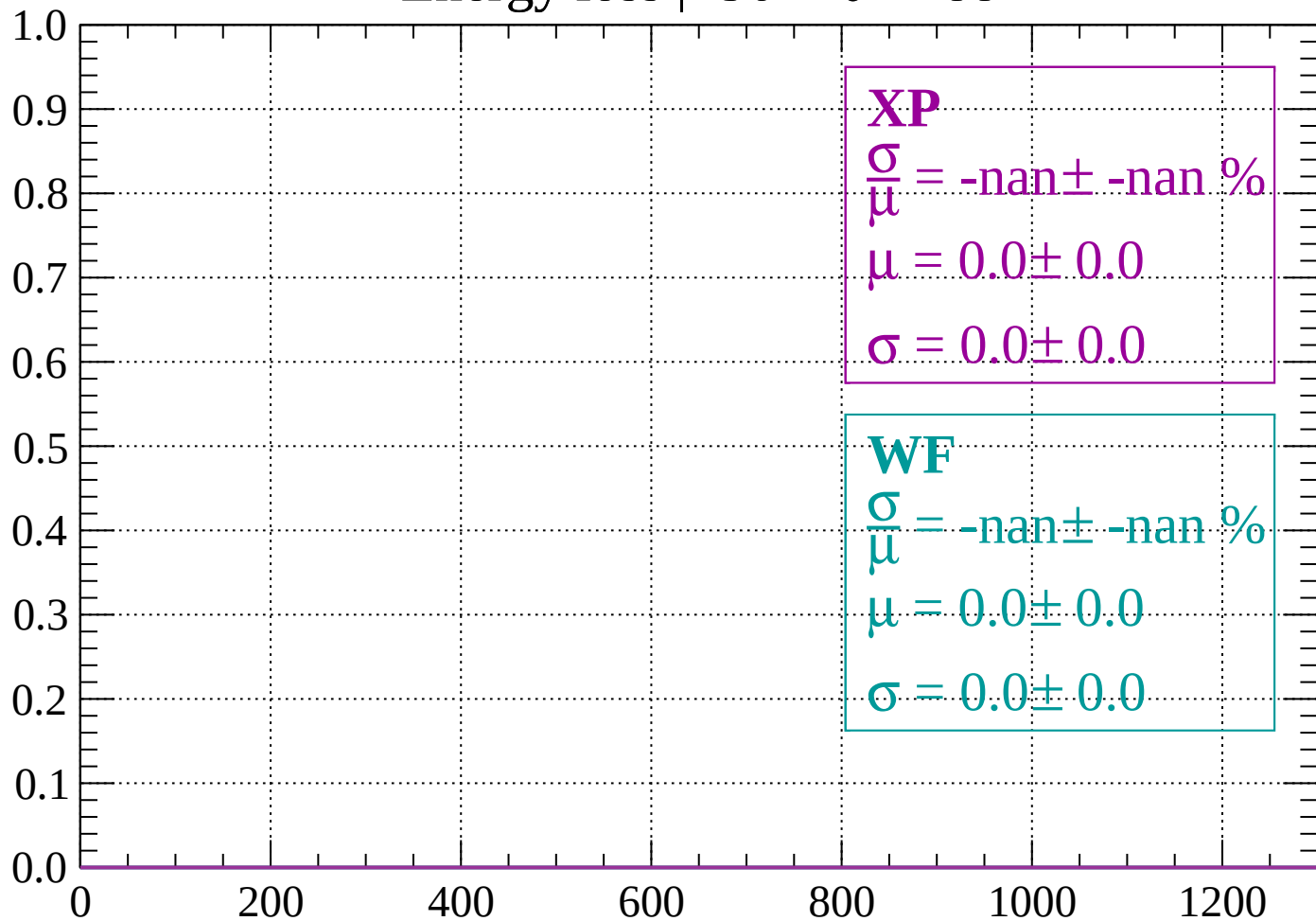
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

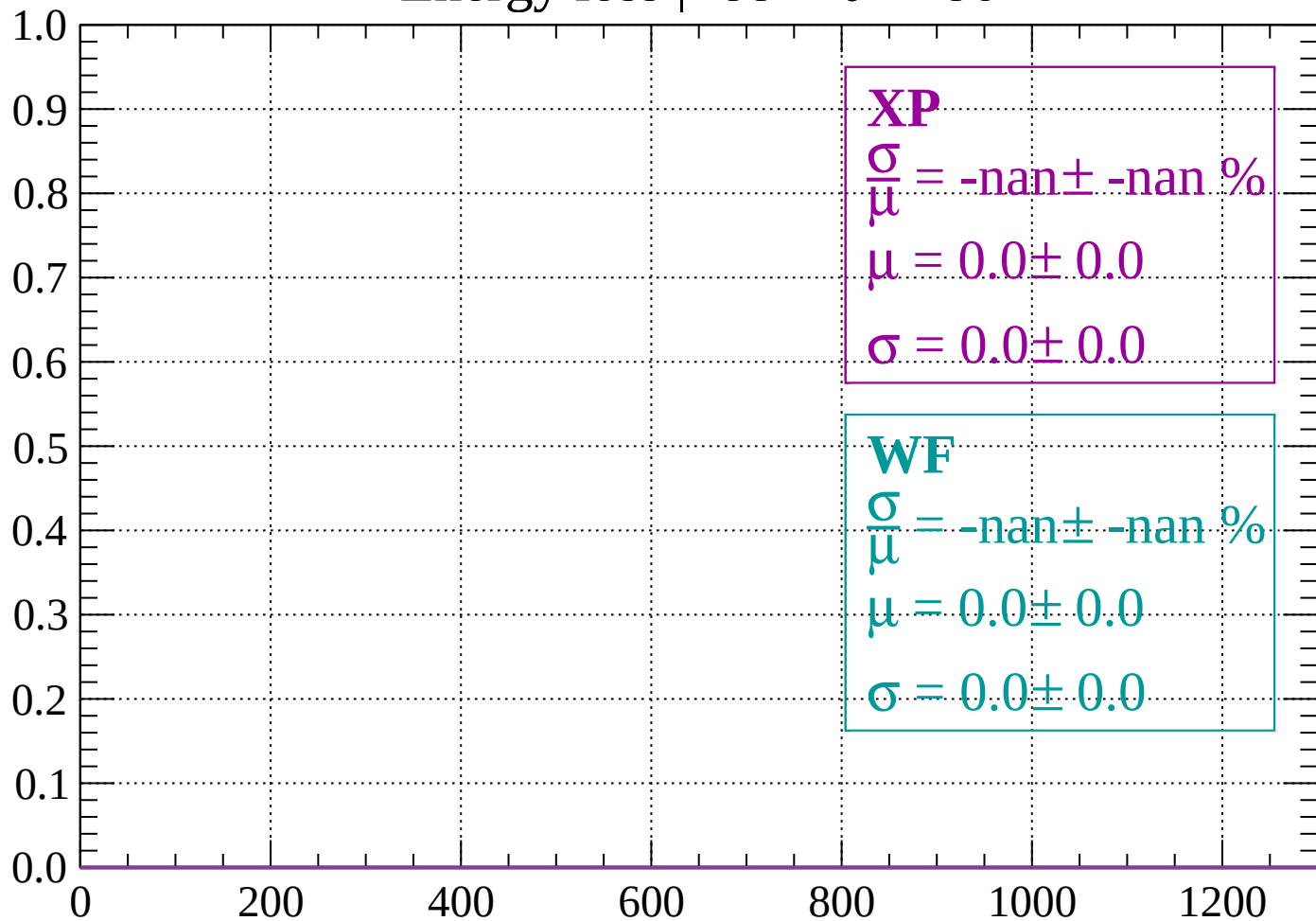
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

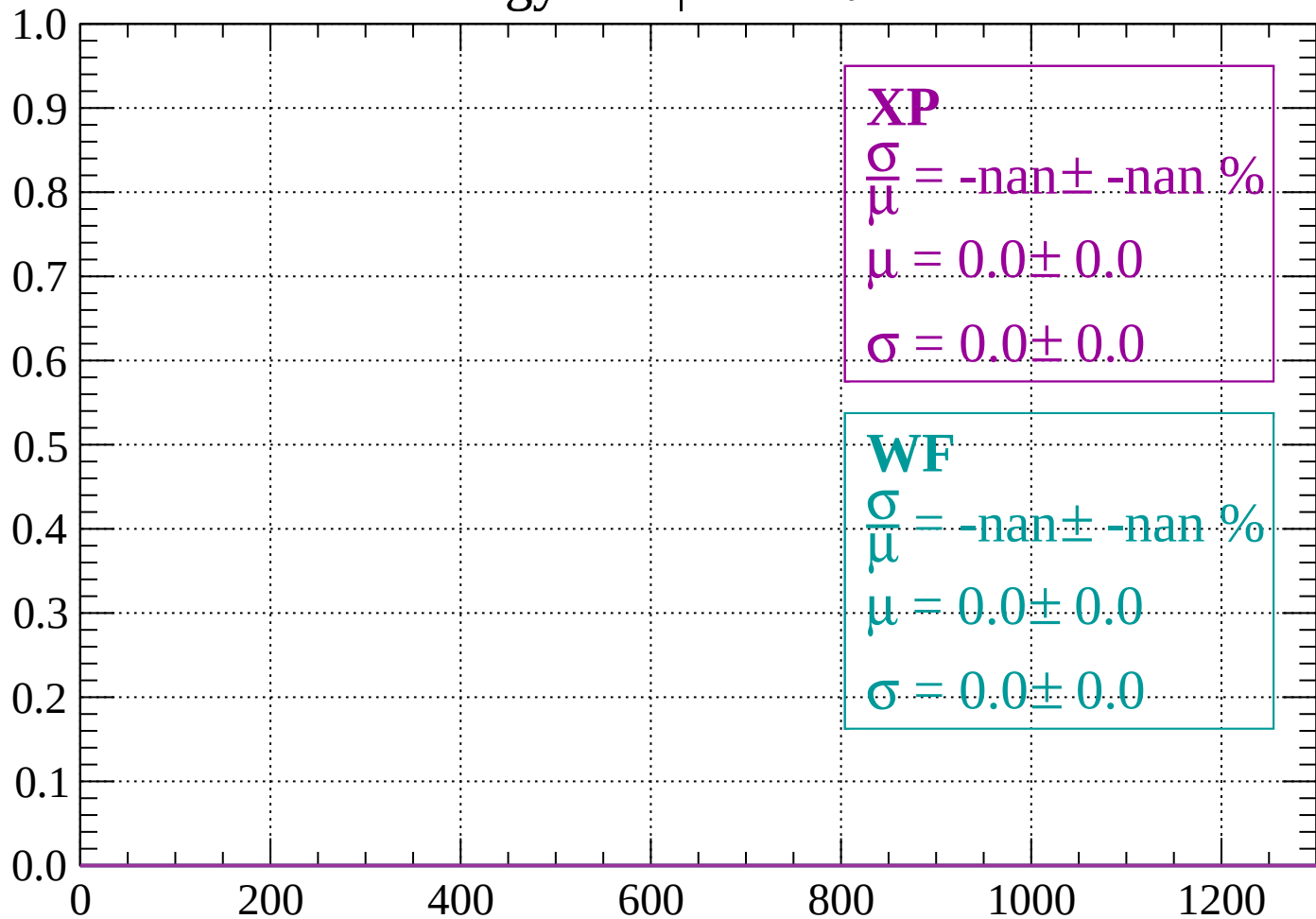
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

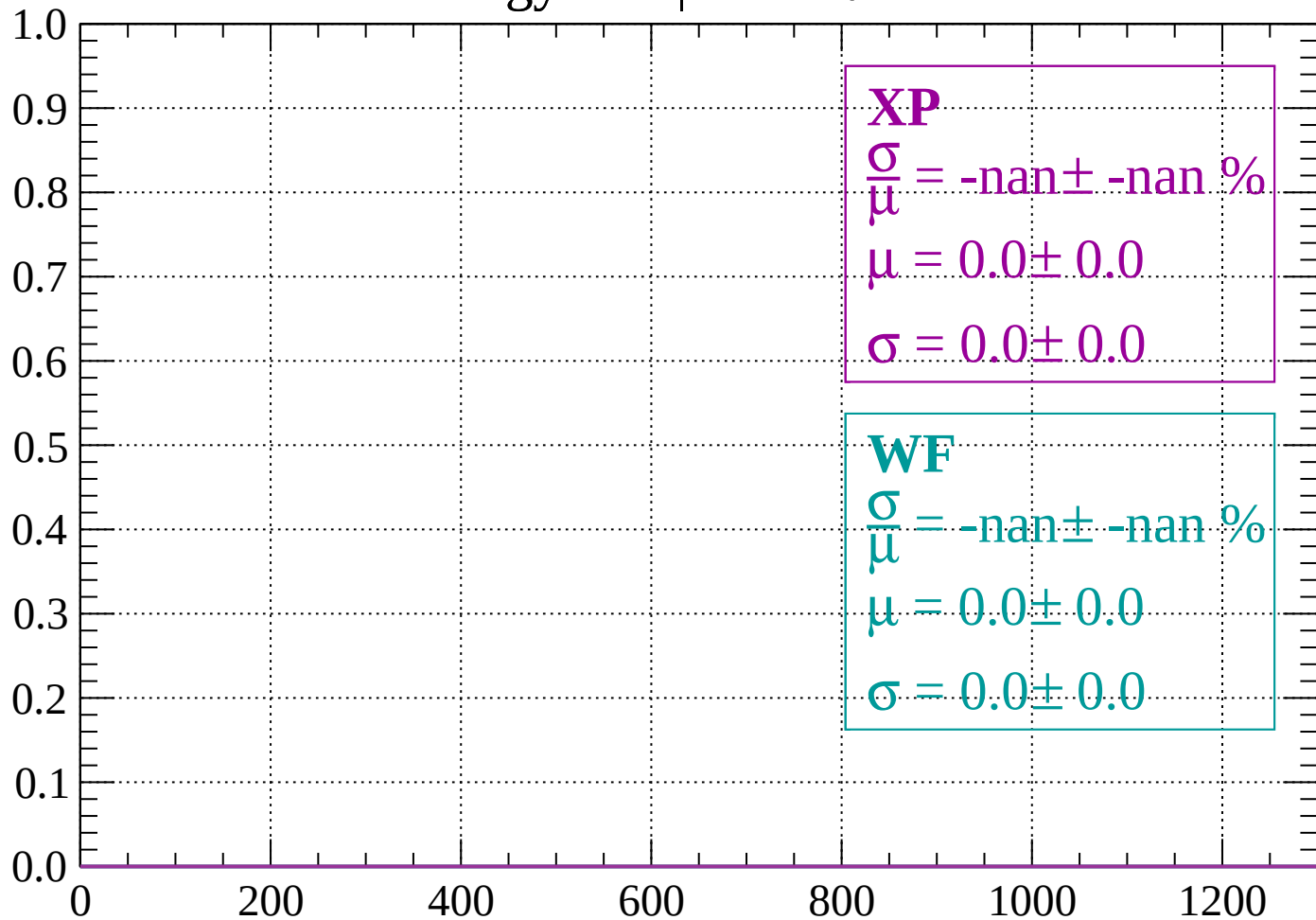
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

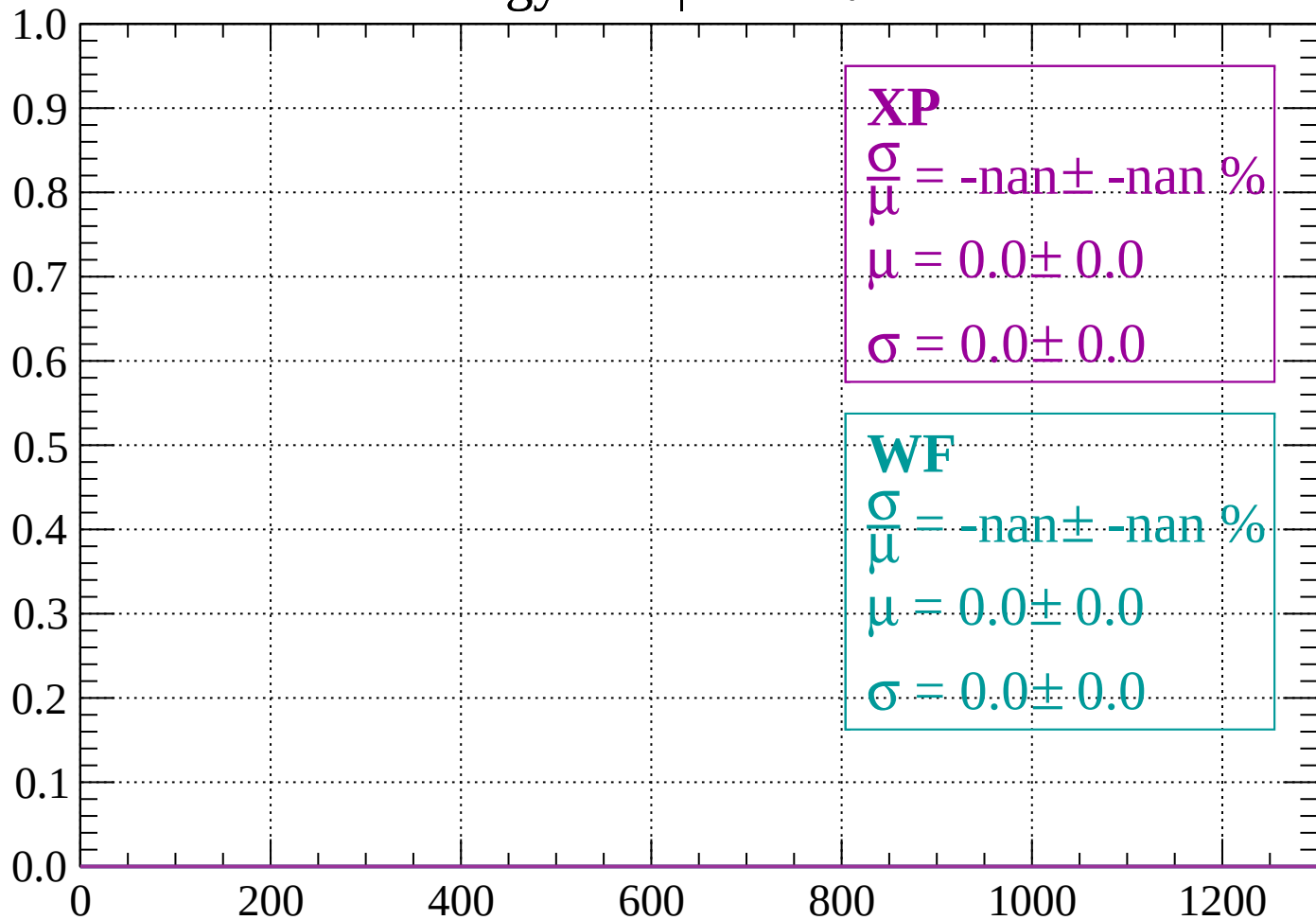
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

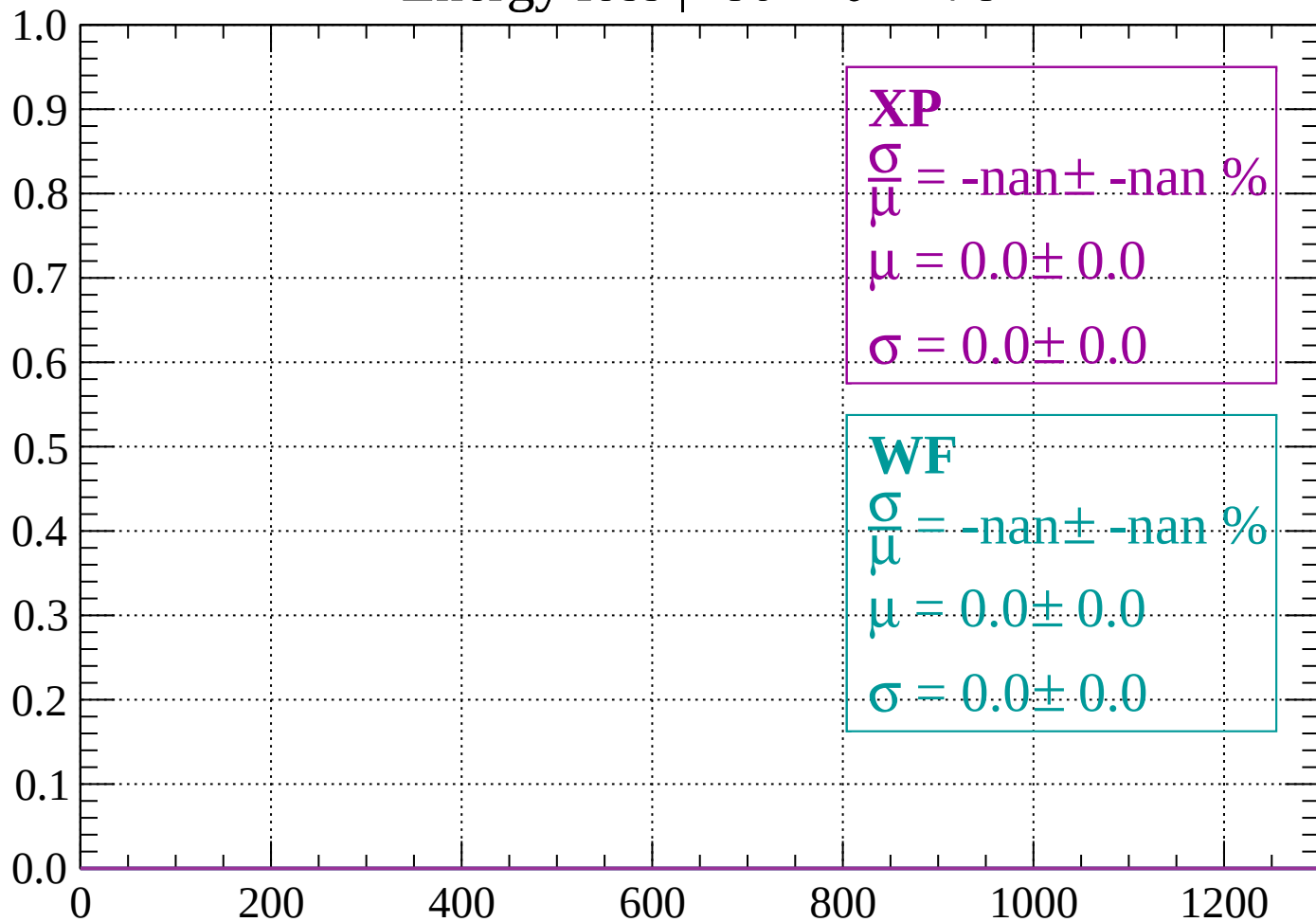
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

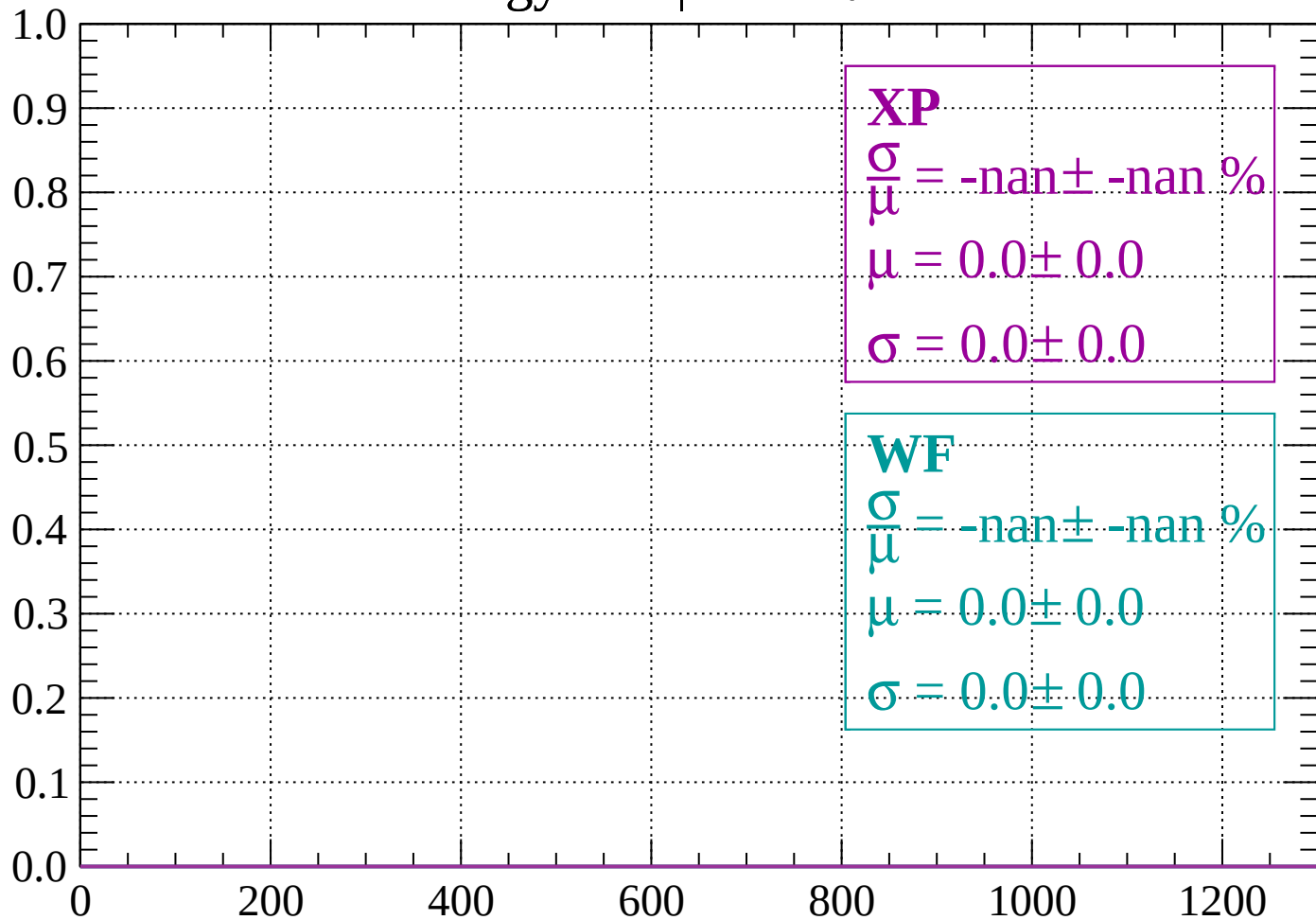
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

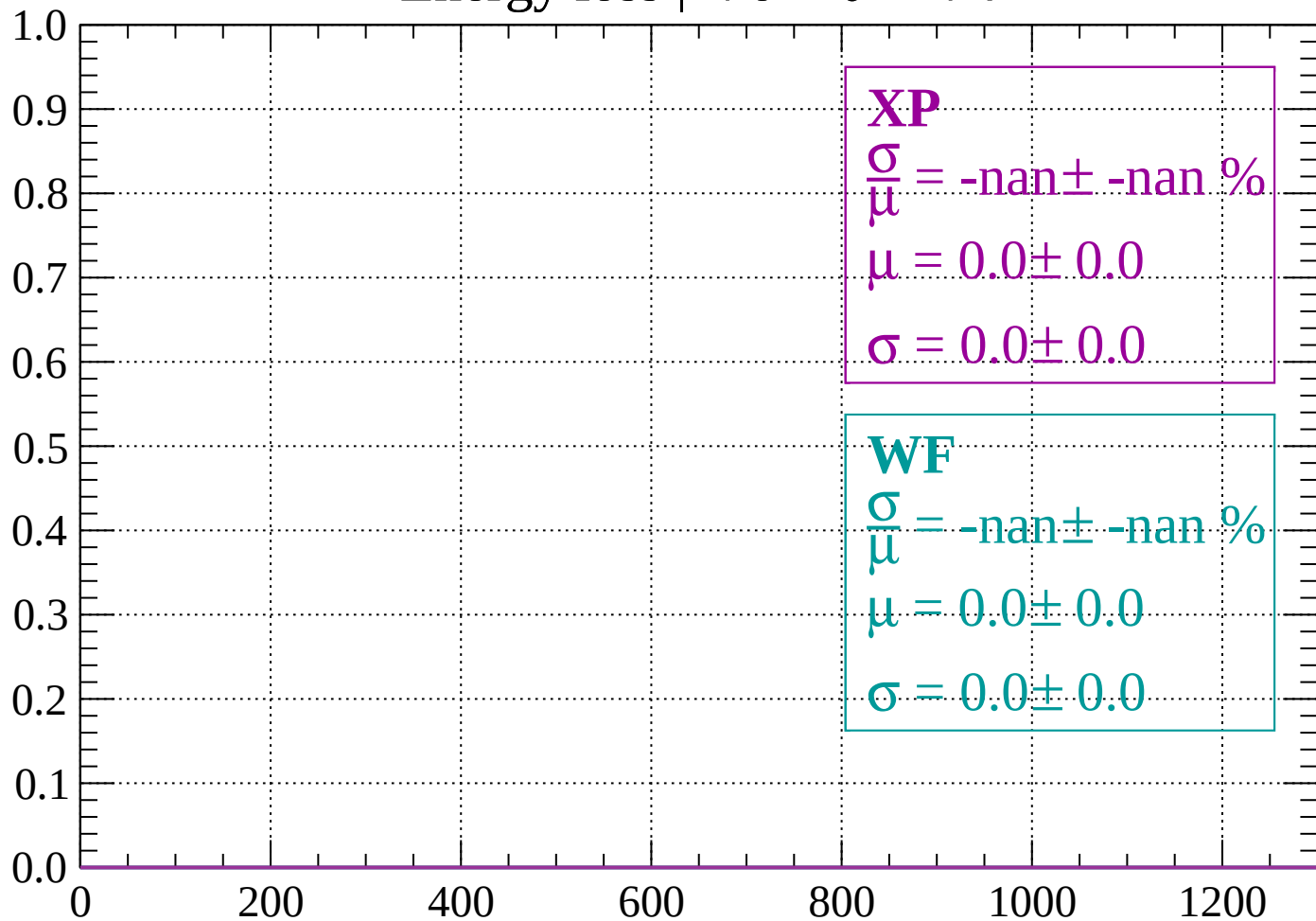
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

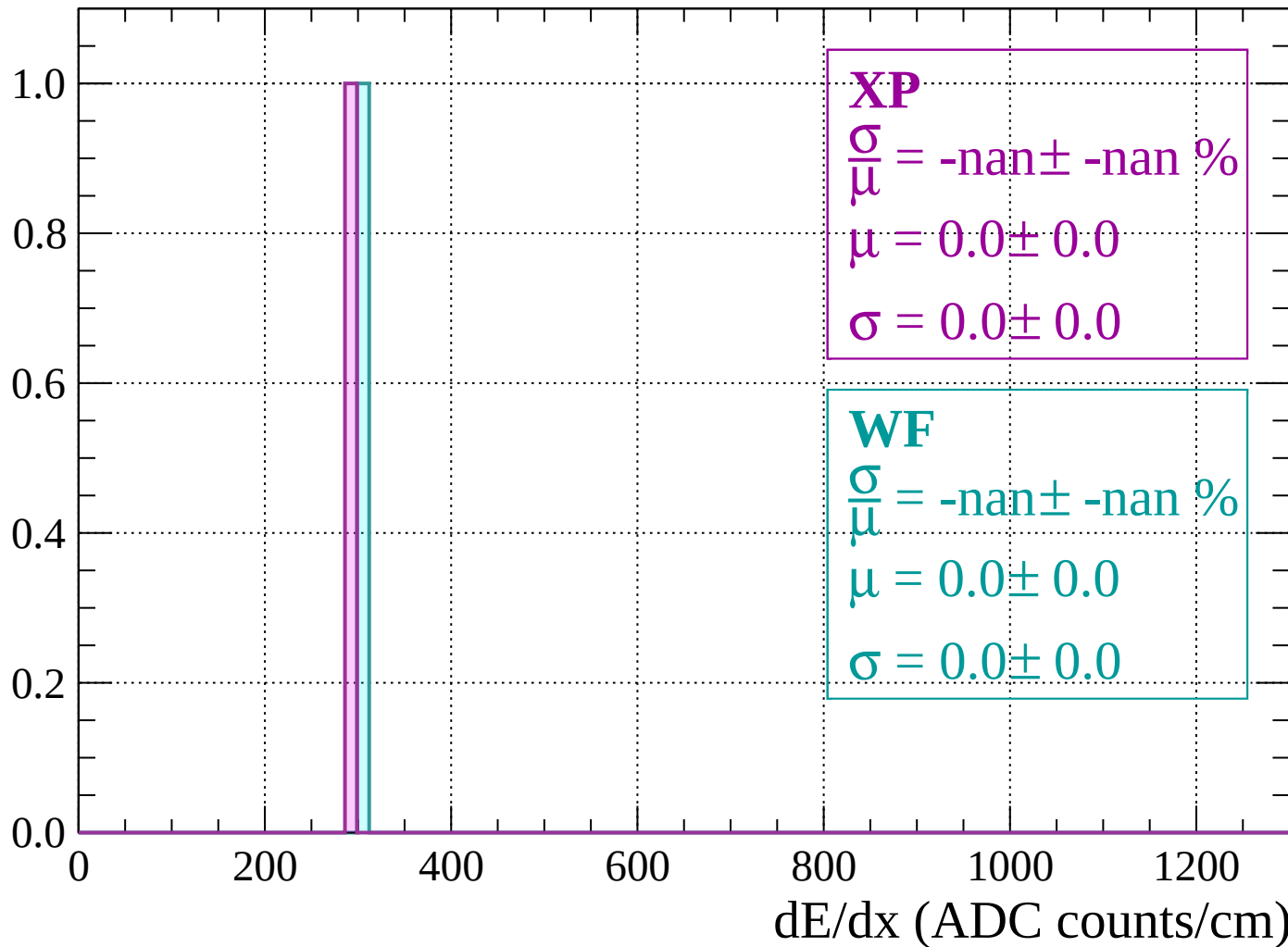
Count



dE/dx (ADC counts/cm)

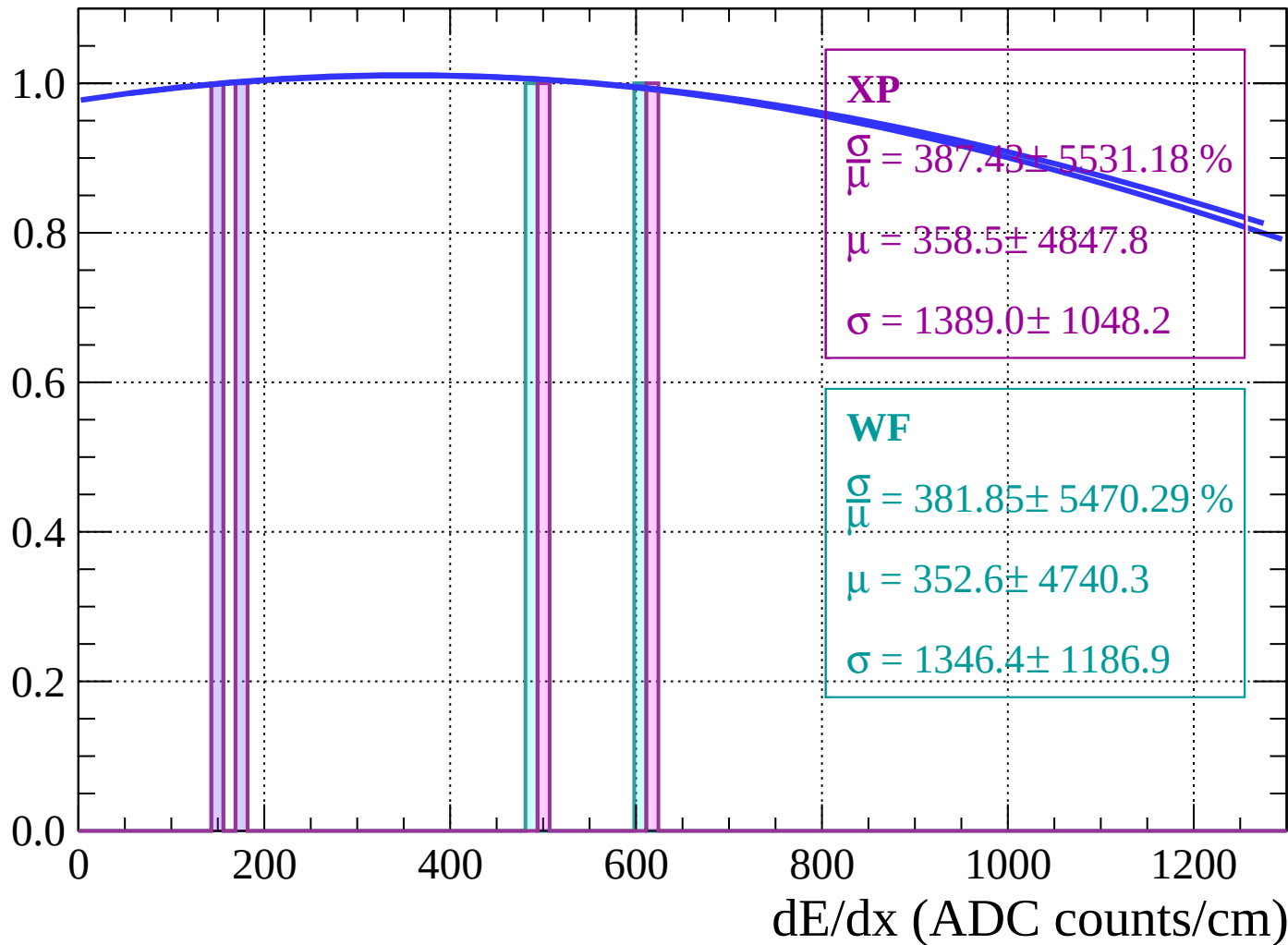
Energy loss | $-74 < \theta < -72$

Count



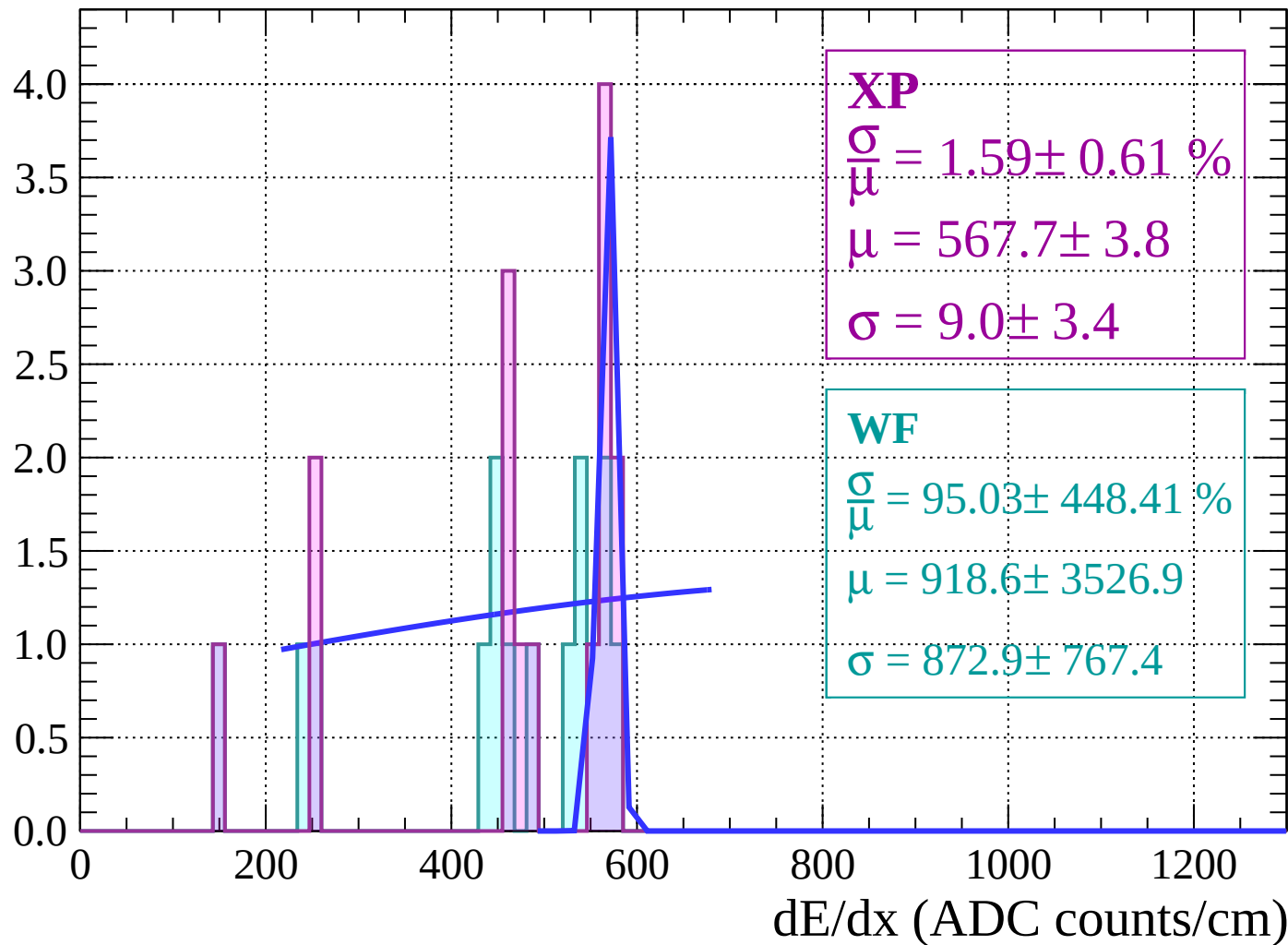
Energy loss $|-72 < \theta < -70$

Count



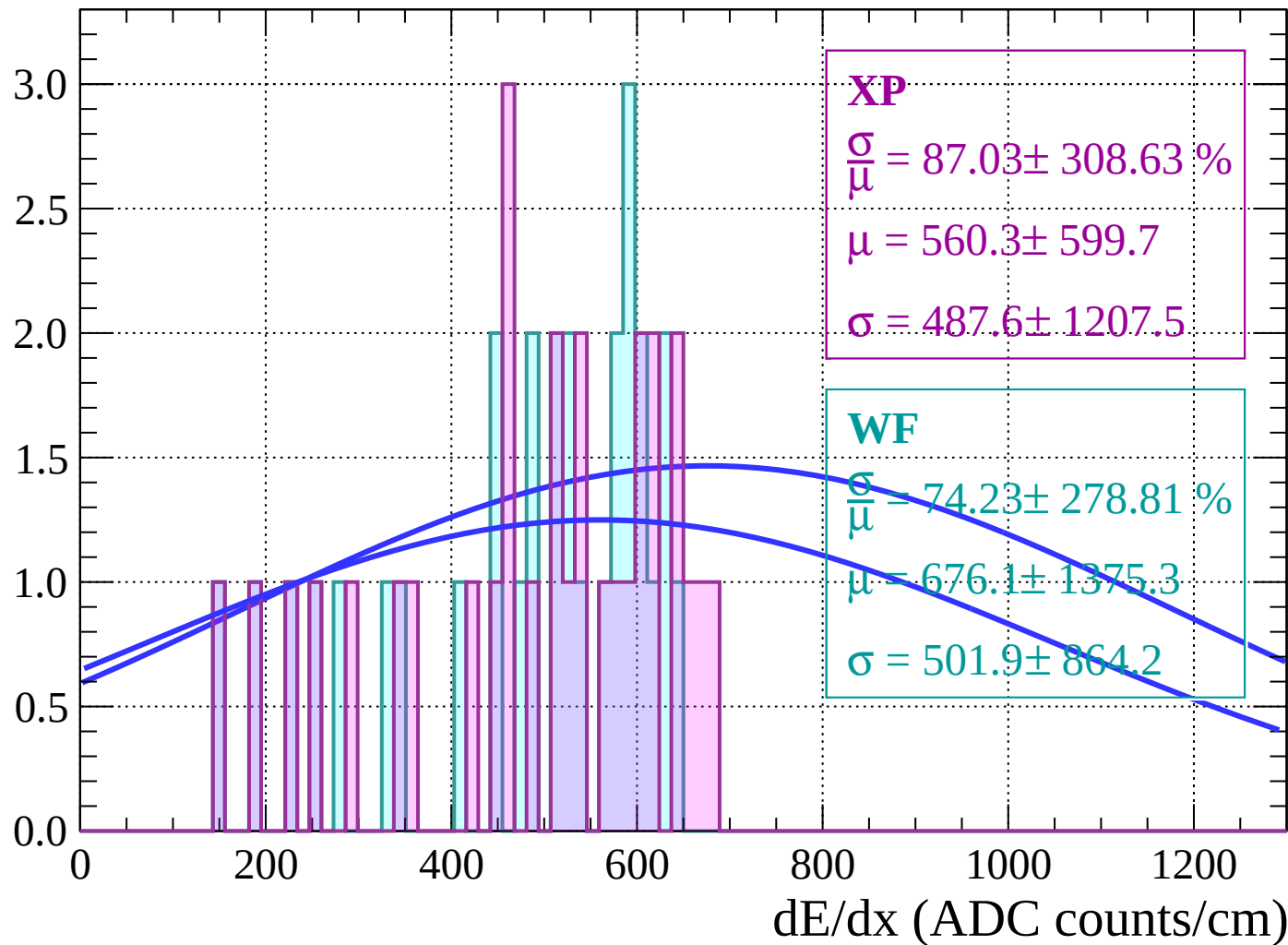
Energy loss | $-70 < \theta < -68$

Count



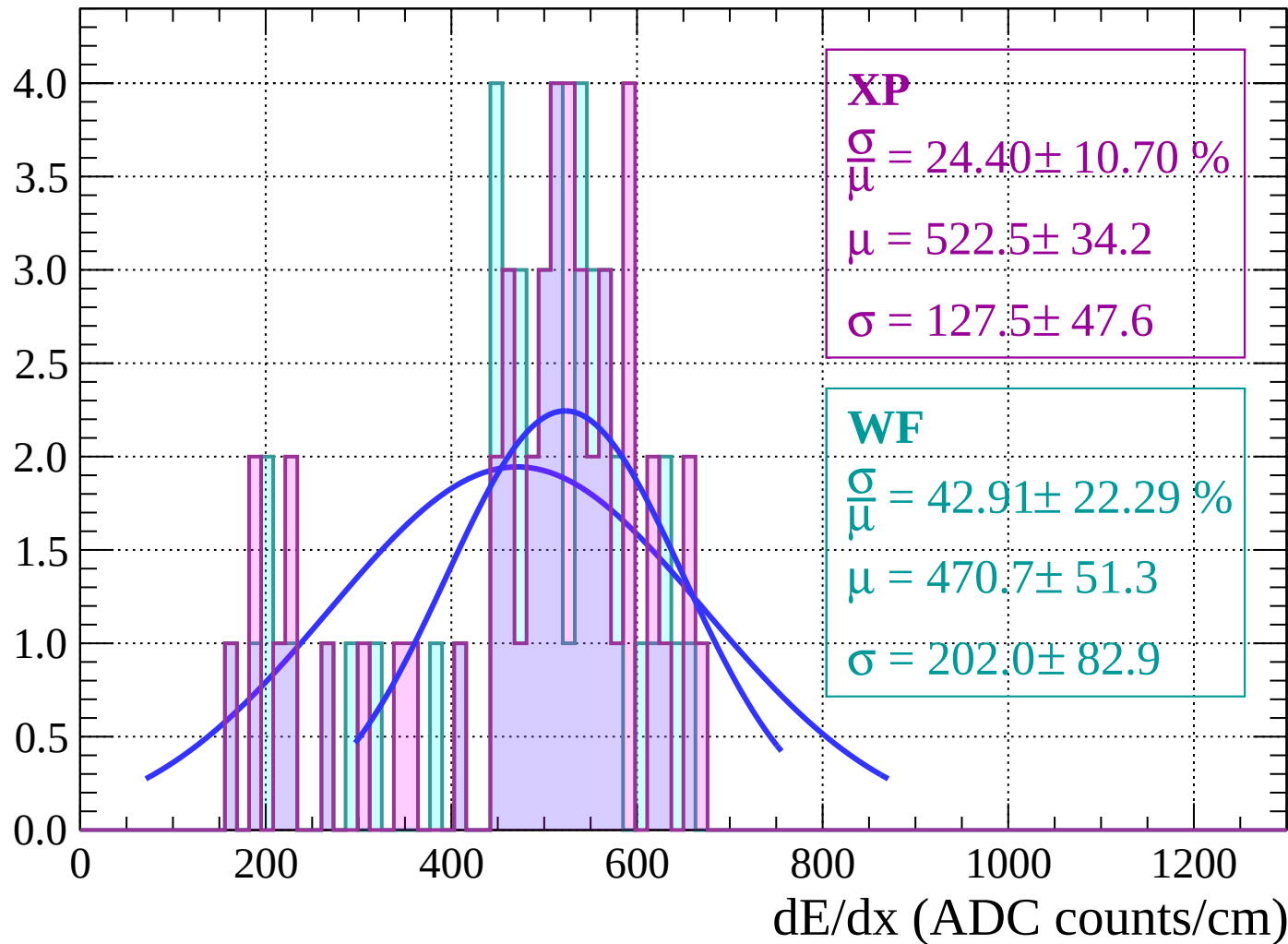
Energy loss | $-68 < \theta < -66$

Count



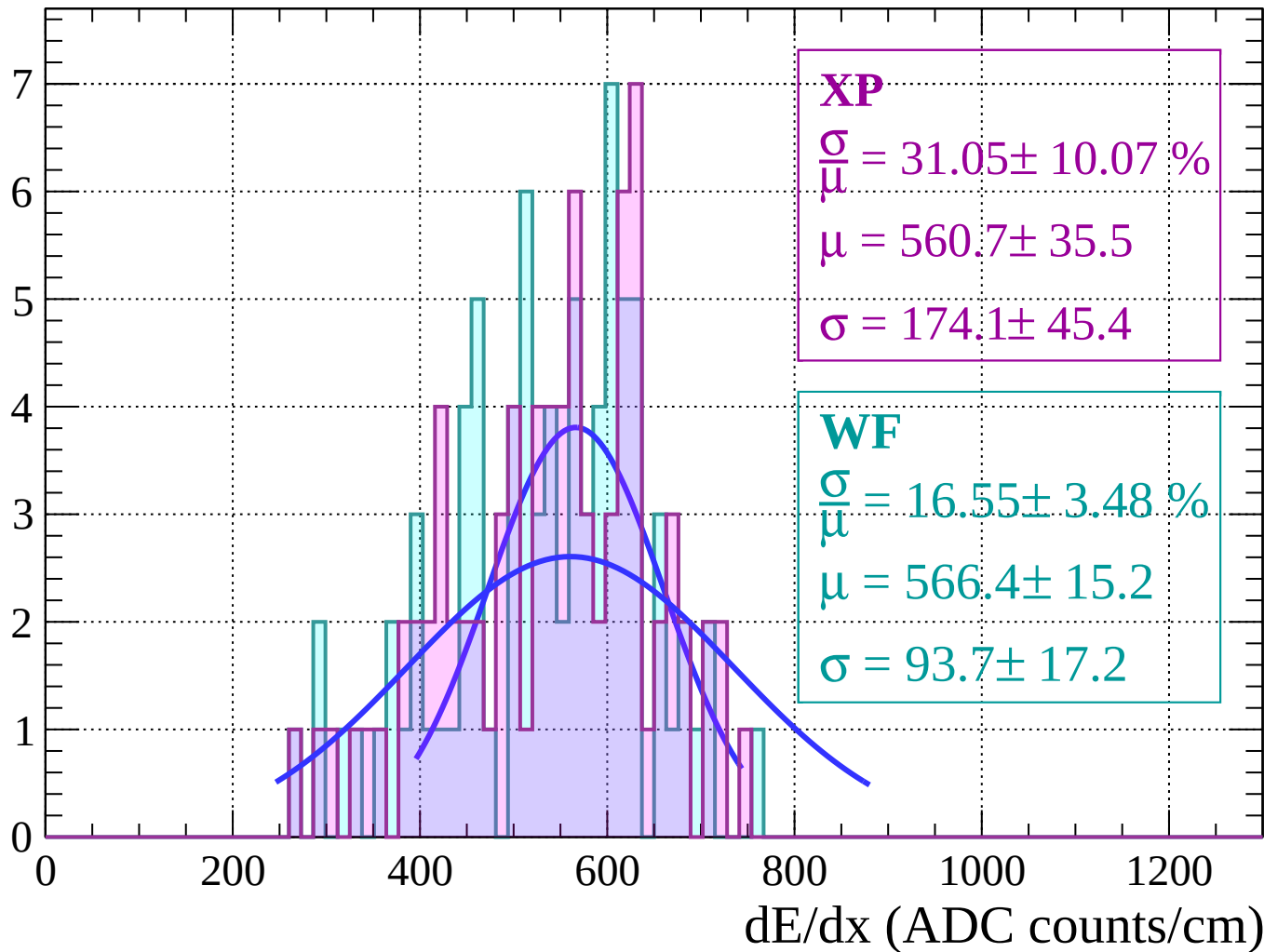
Energy loss | $-66 < \theta < -64$

Count



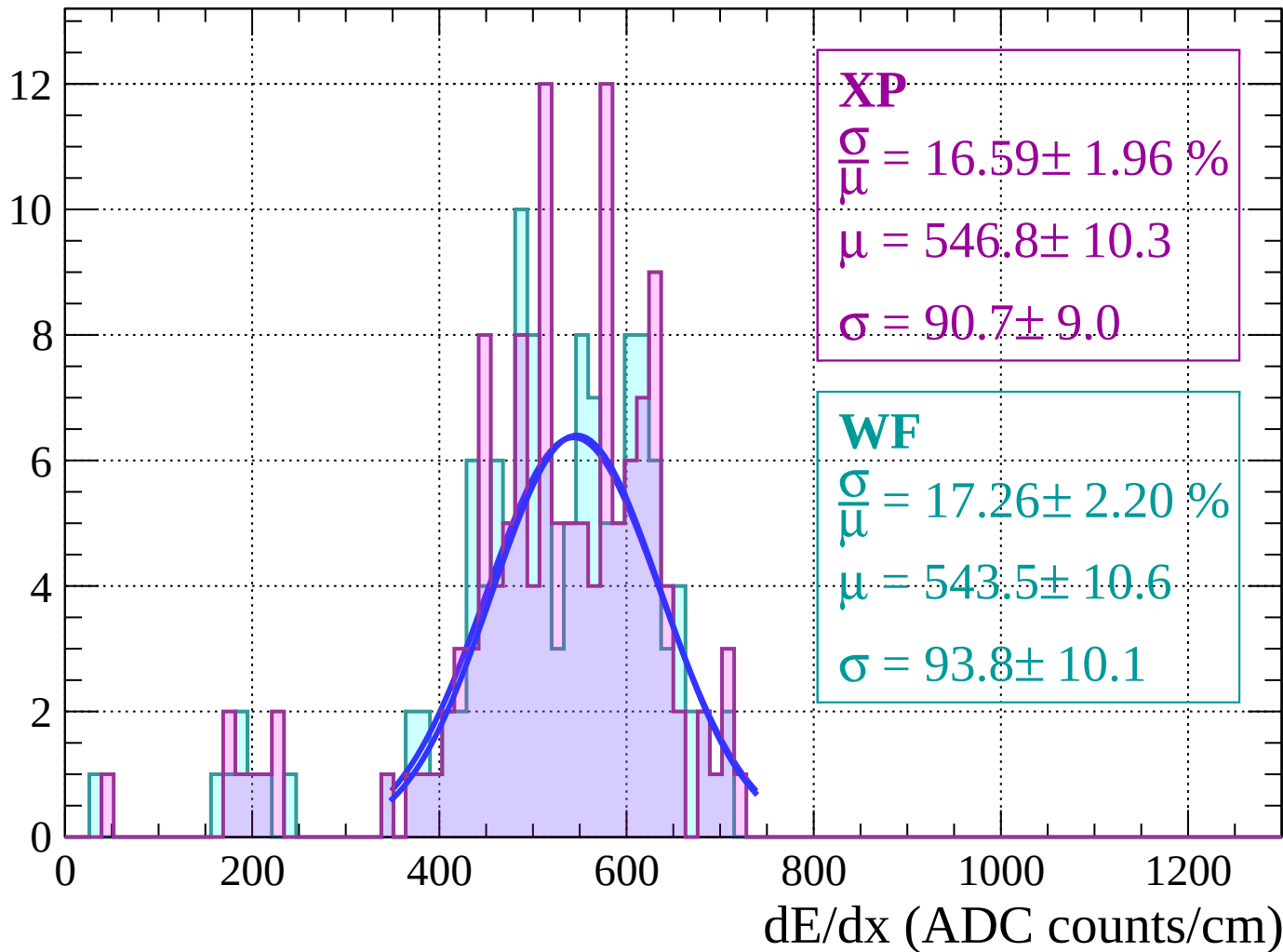
Energy loss $|-64 < \theta < -62$

Count



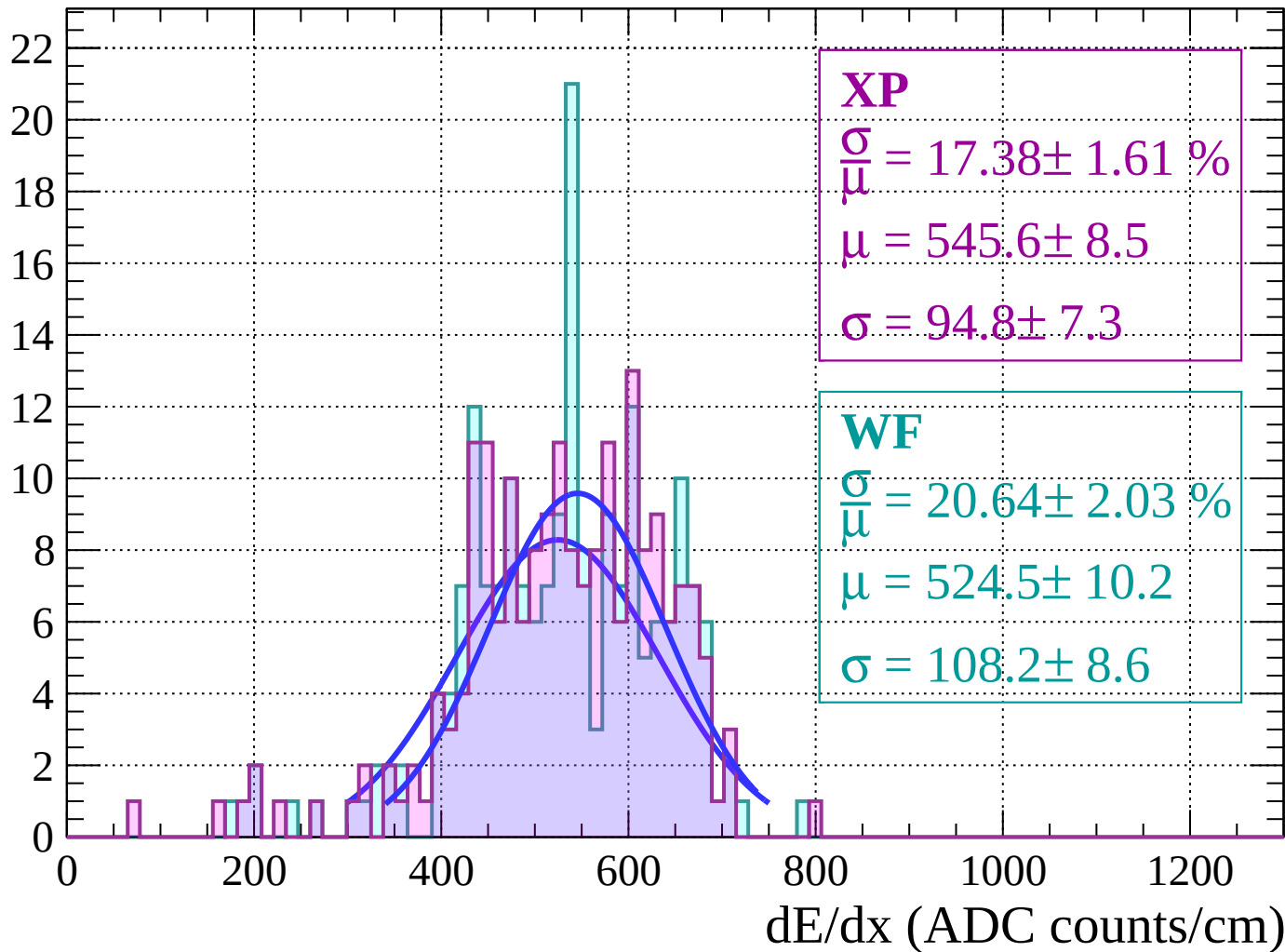
Energy loss | $-62 < \theta < -60$

Count



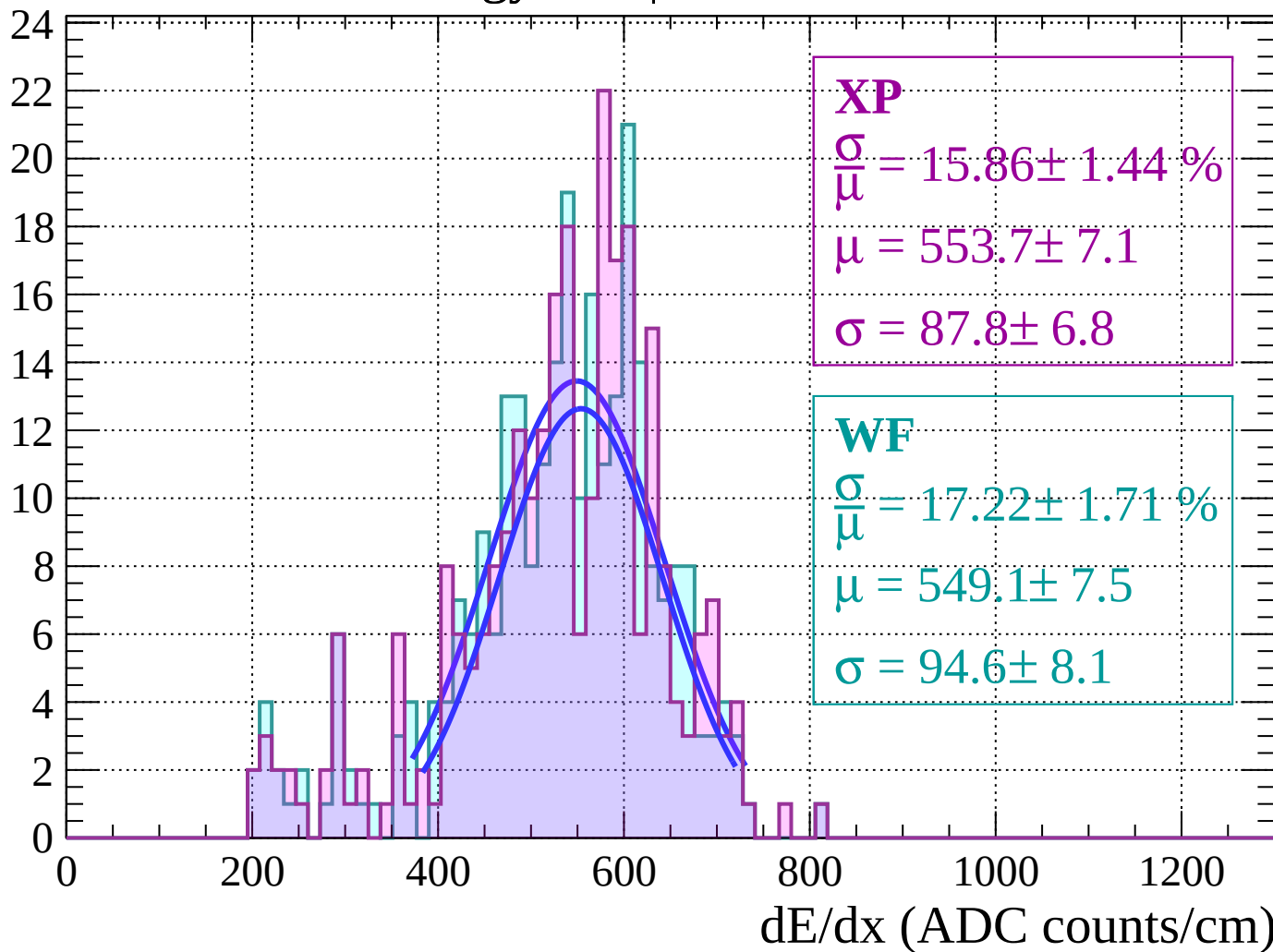
Energy loss | $-60 < \theta < -58$

Count



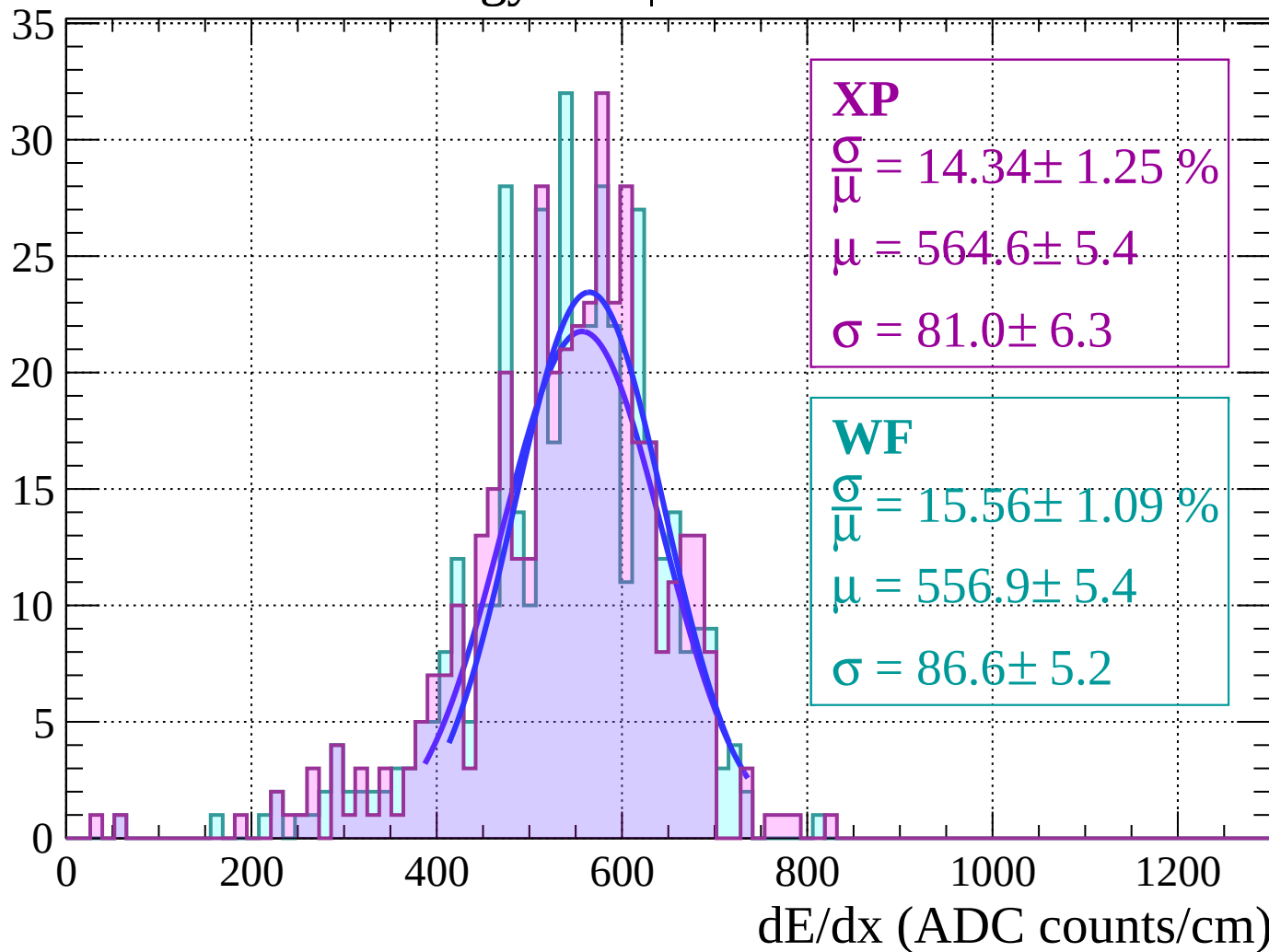
Energy loss $|-58 < \theta < -56$

Count



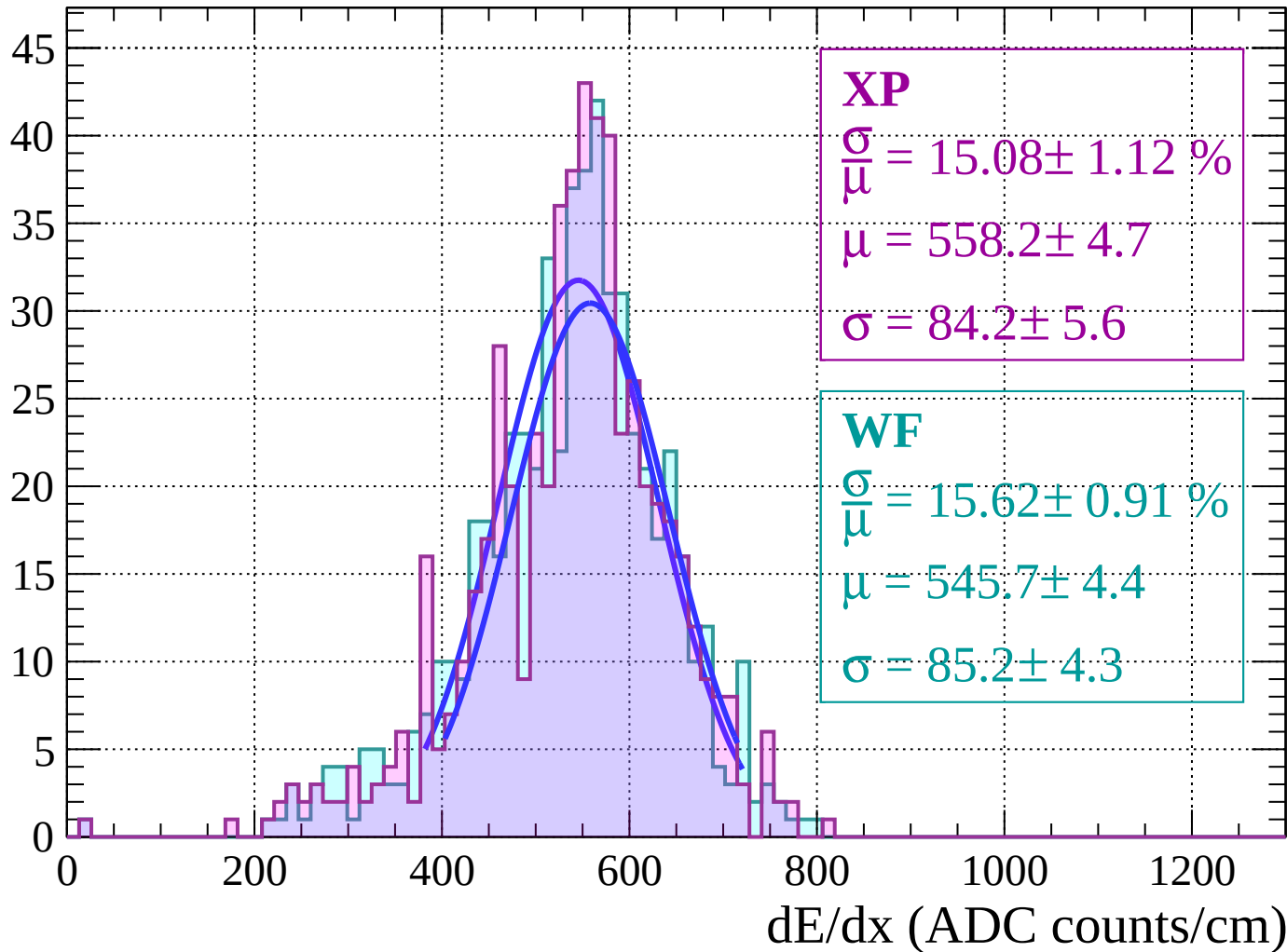
Energy loss $|-56 < \theta < -54$

Count



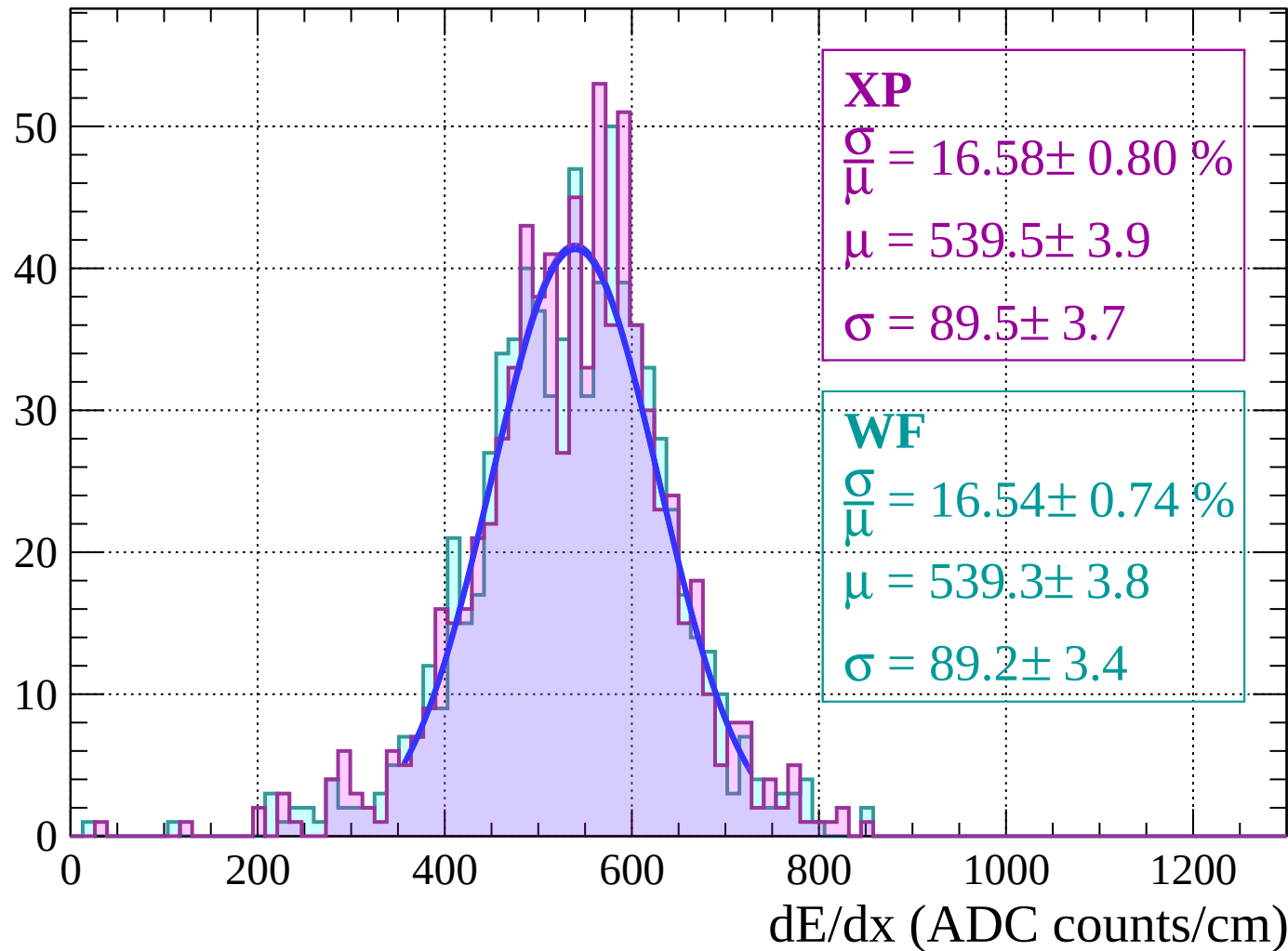
Energy loss $|-54 < \theta < -52$

Count



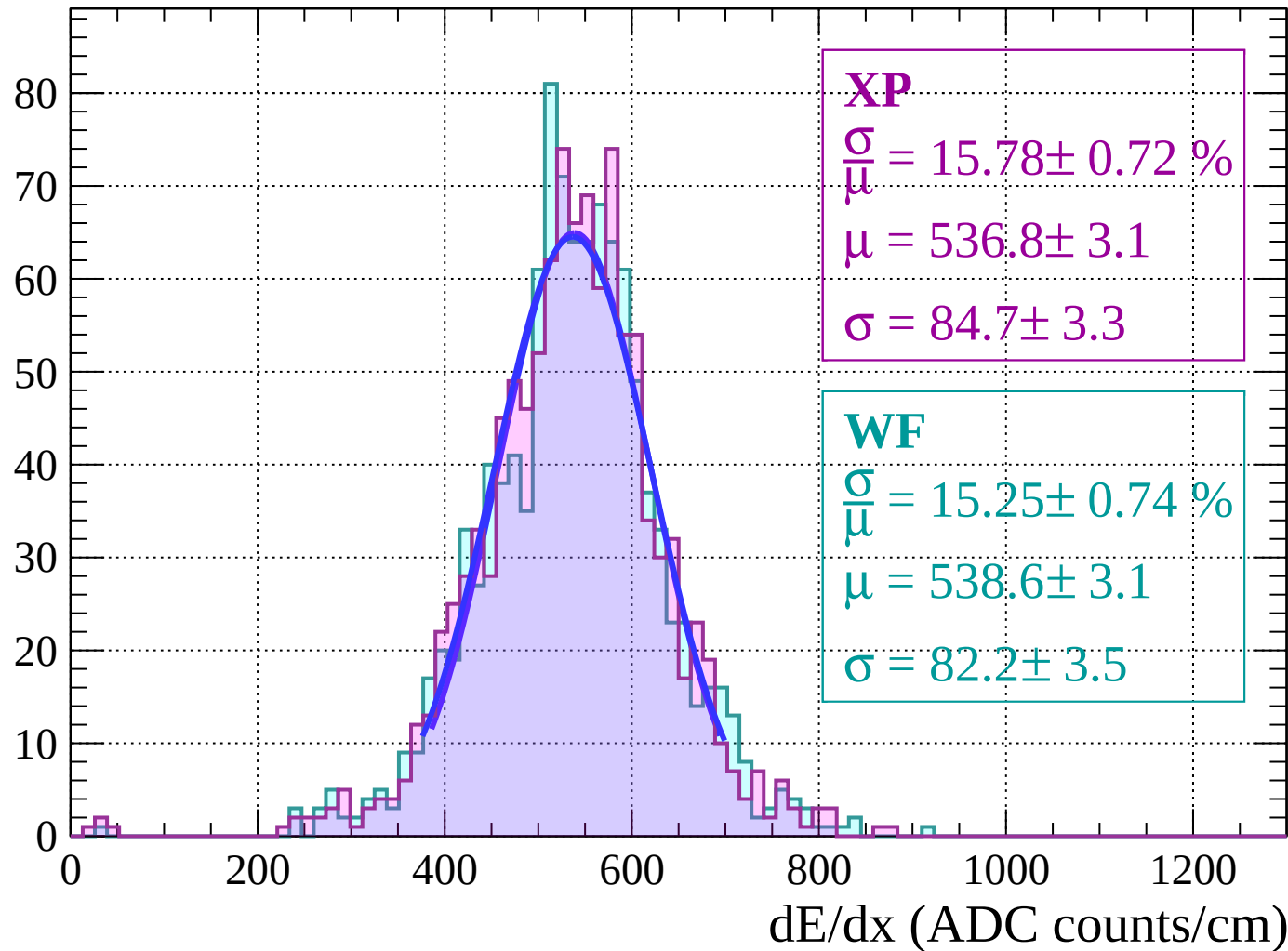
Energy loss $|-52 < \theta < -50$

Count



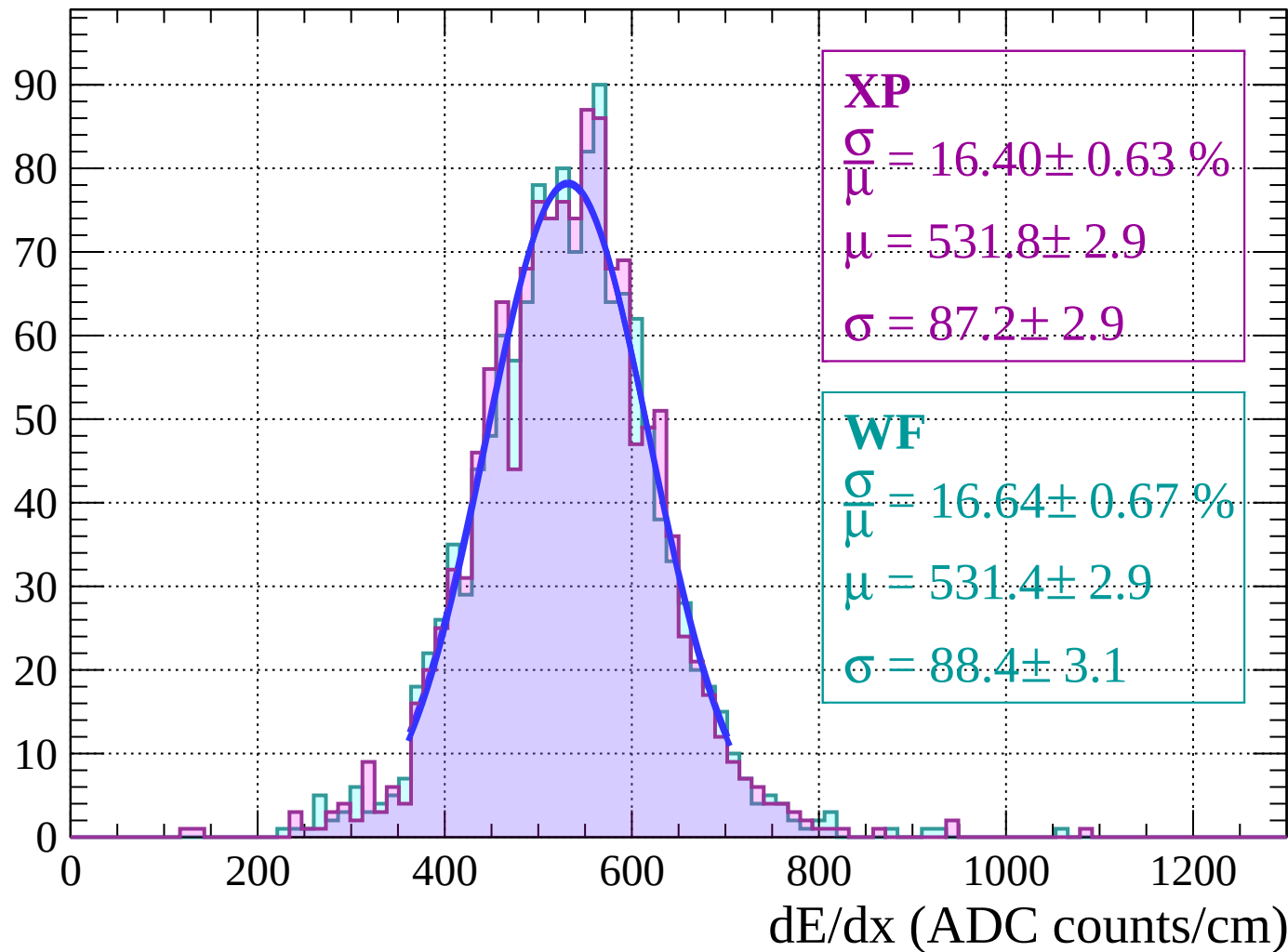
Energy loss | $-50 < \theta < -48$

Count



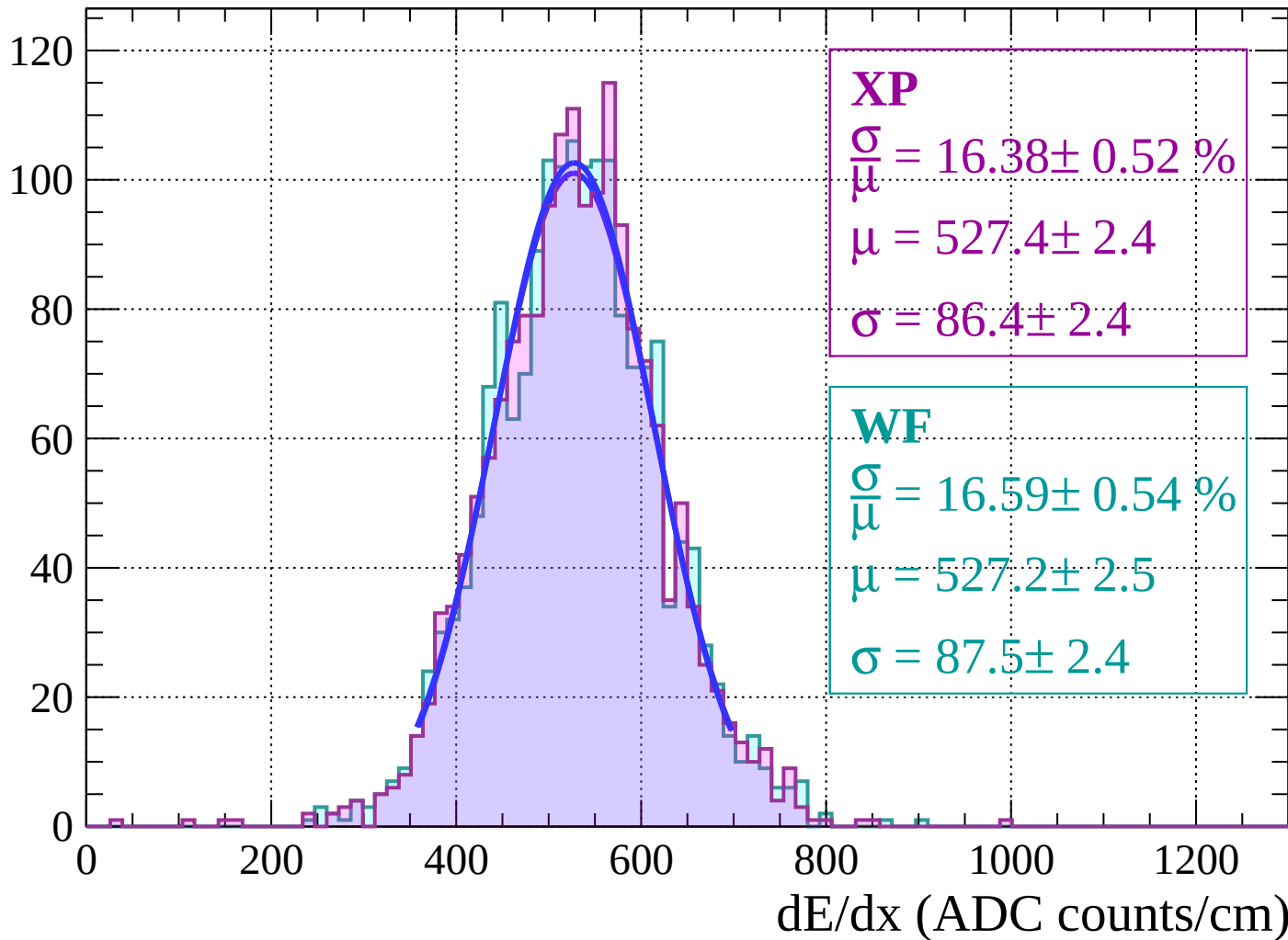
Energy loss $|-48 < \theta < -46$

Count



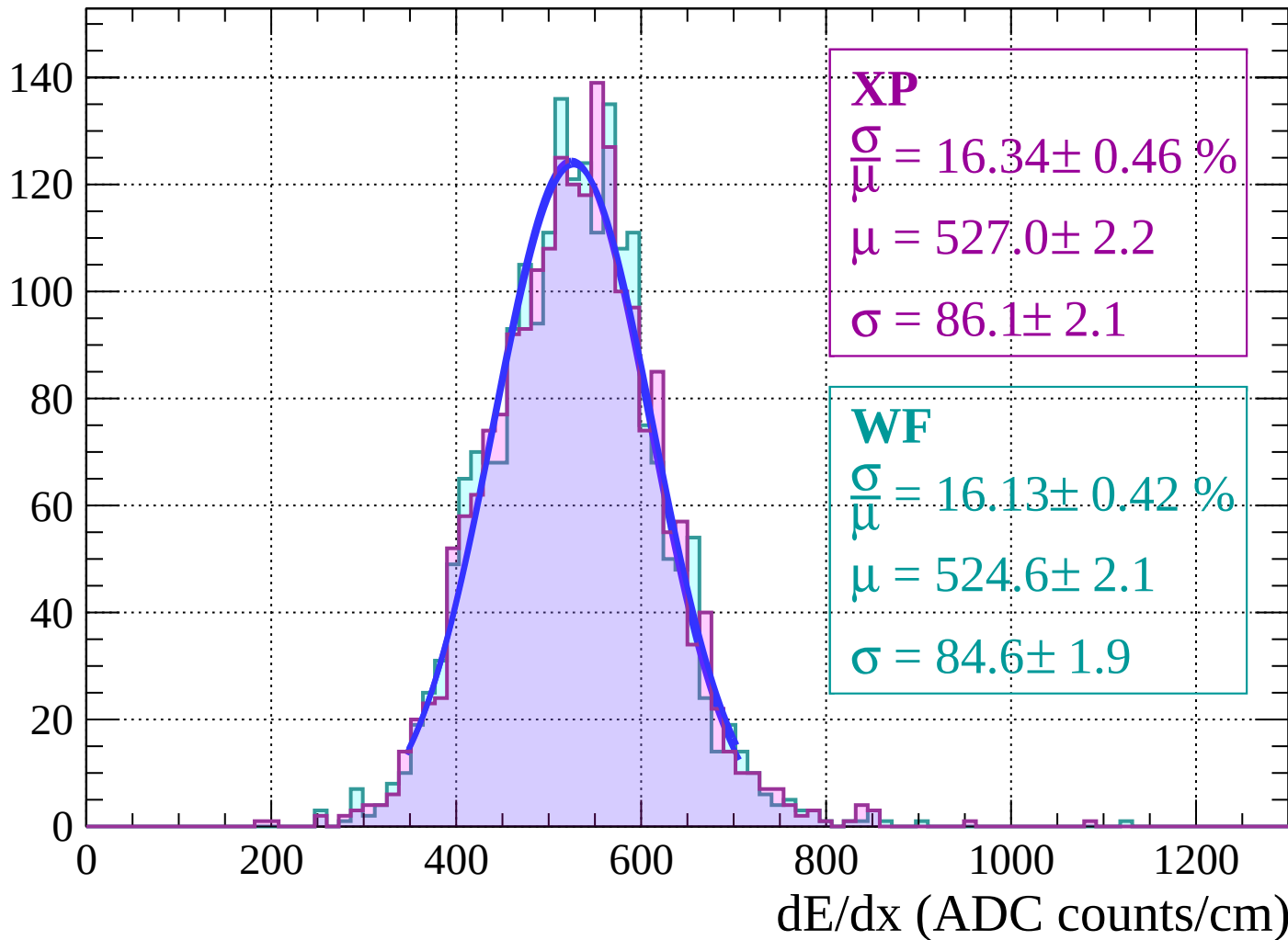
Energy loss | $-46 < \theta < -44$

Count



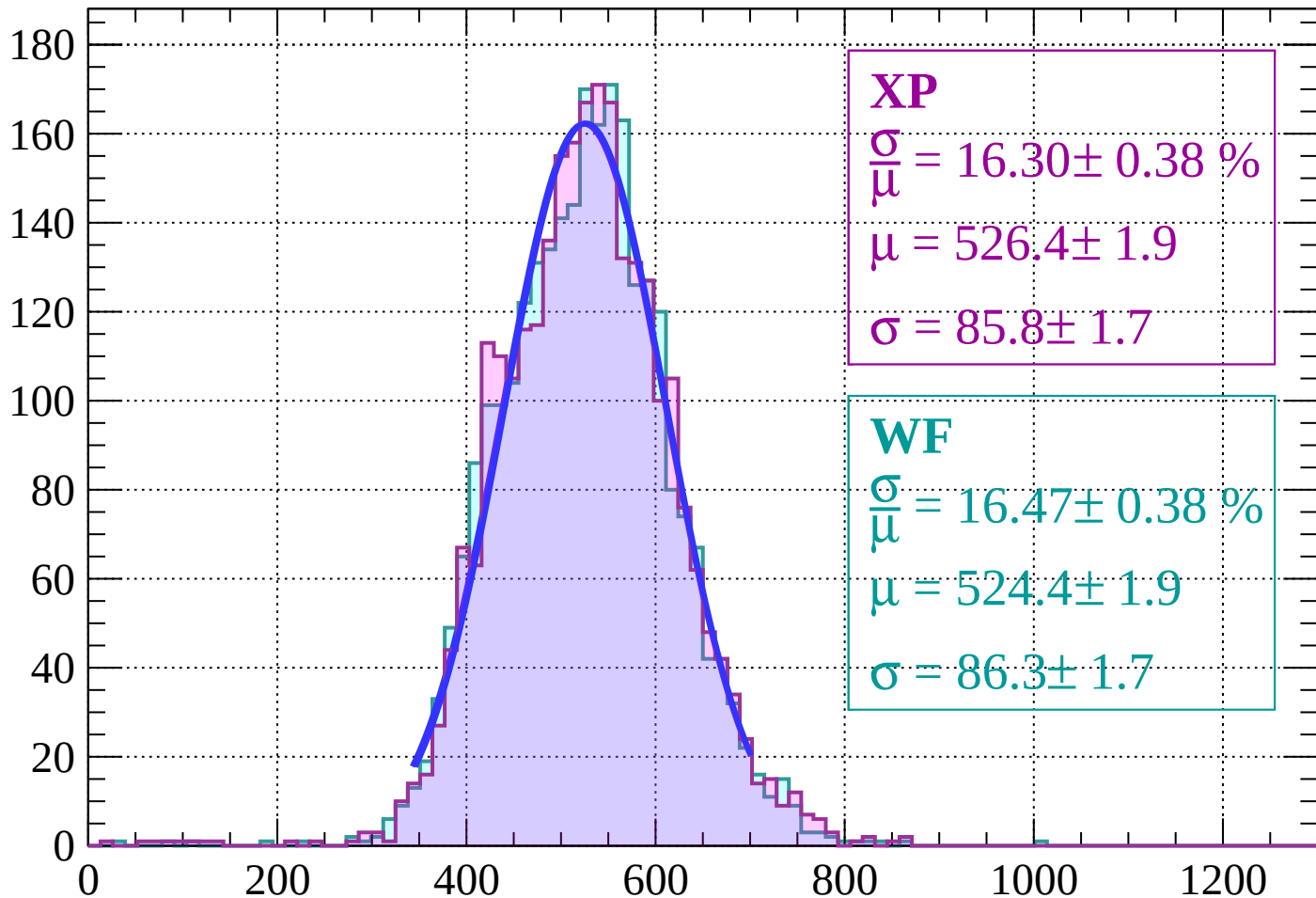
Energy loss $|-44 < \theta < -42$

Count



Energy loss | $-42 < \theta < -40$

Count



XP

$$\frac{\sigma}{\mu} = 16.30 \pm 0.38 \%$$

$$\mu = 526.4 \pm 1.9$$

$$\sigma = 85.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 16.47 \pm 0.38 \%$$

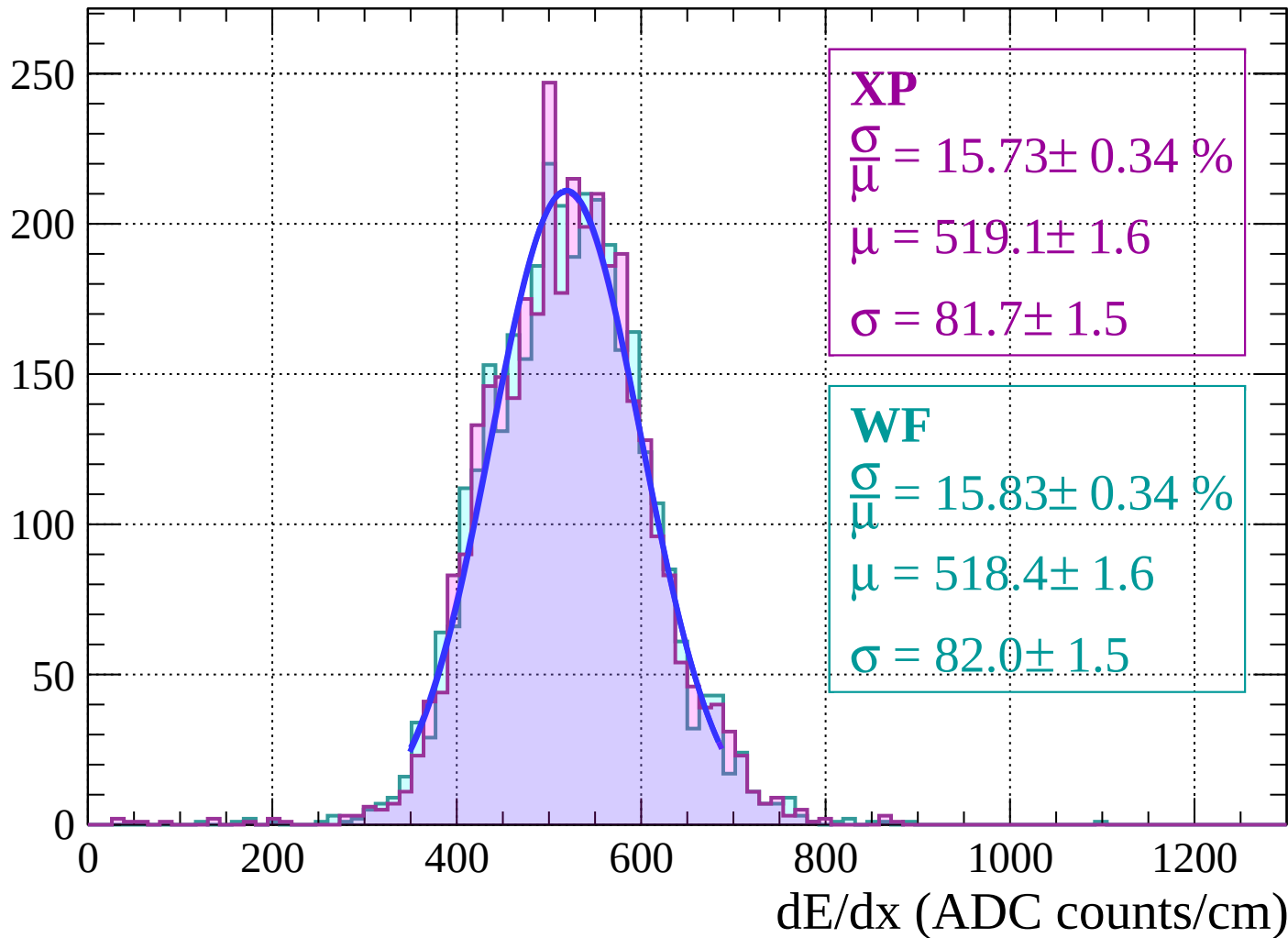
$$\mu = 524.4 \pm 1.9$$

$$\sigma = 86.3 \pm 1.7$$

dE/dx (ADC counts/cm)

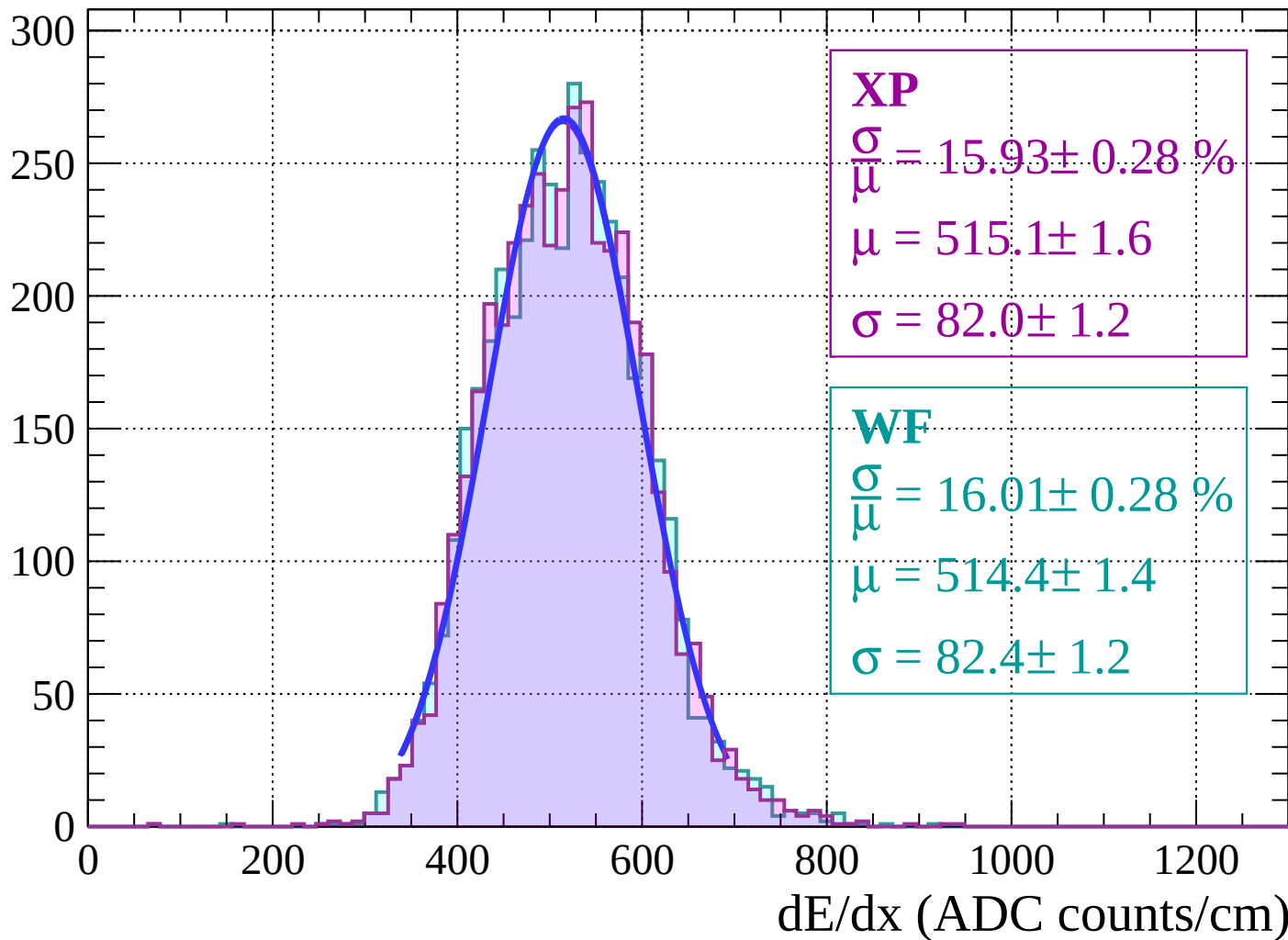
Energy loss | $-40 < \theta < -38$

Count



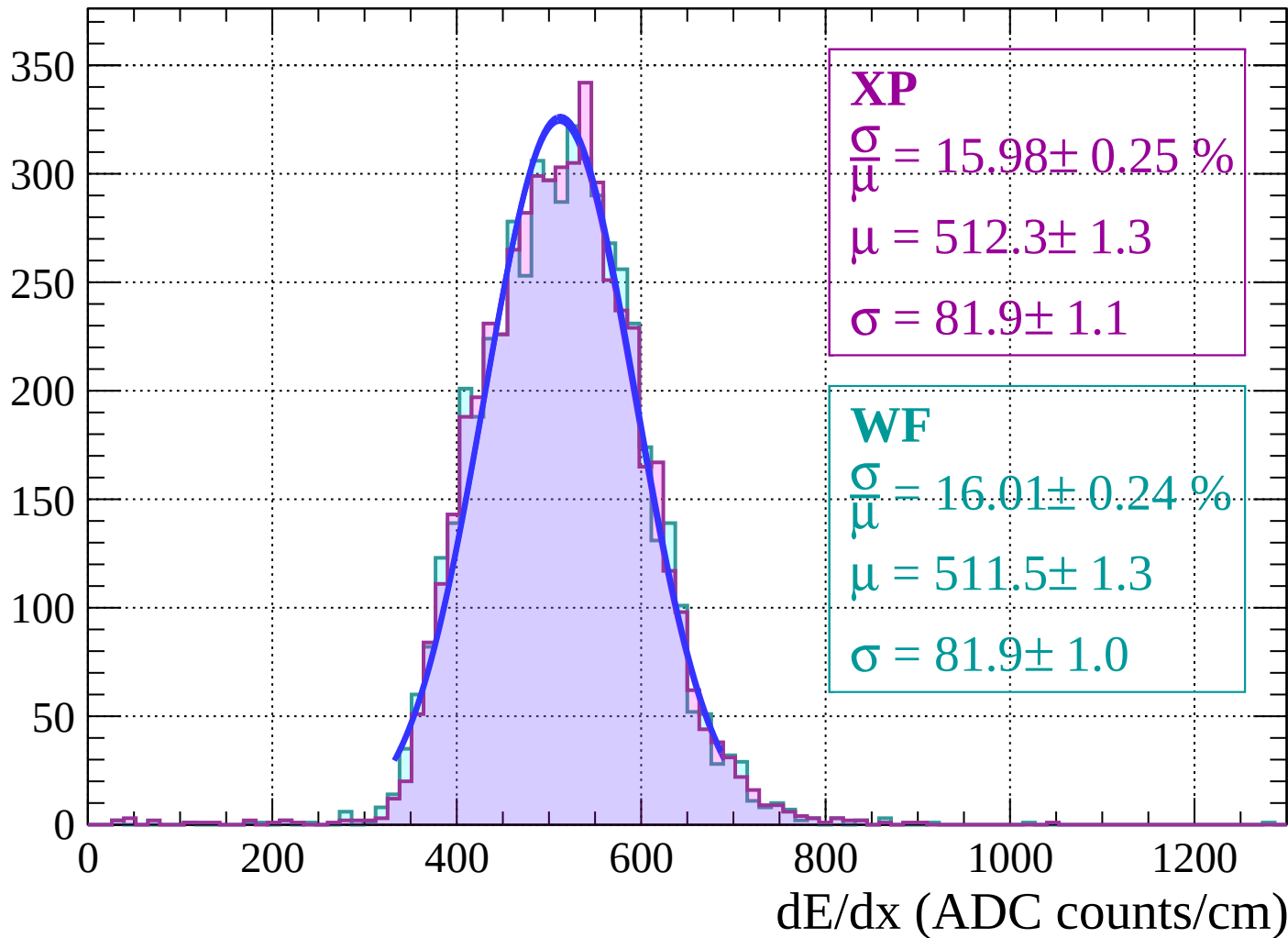
Energy loss | $-38 < \theta < -36$

Count

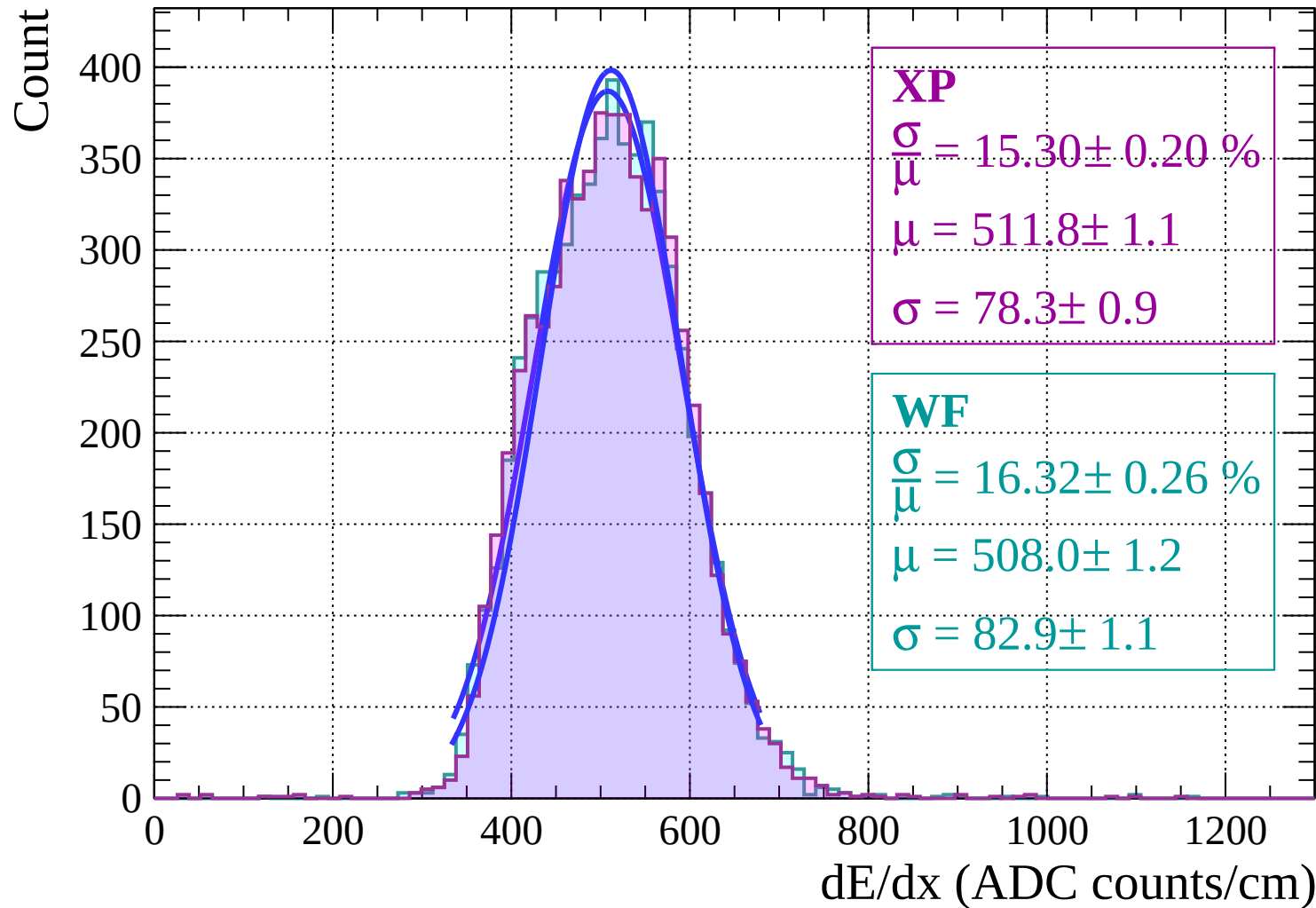


Energy loss | $-36 < \theta < -34$

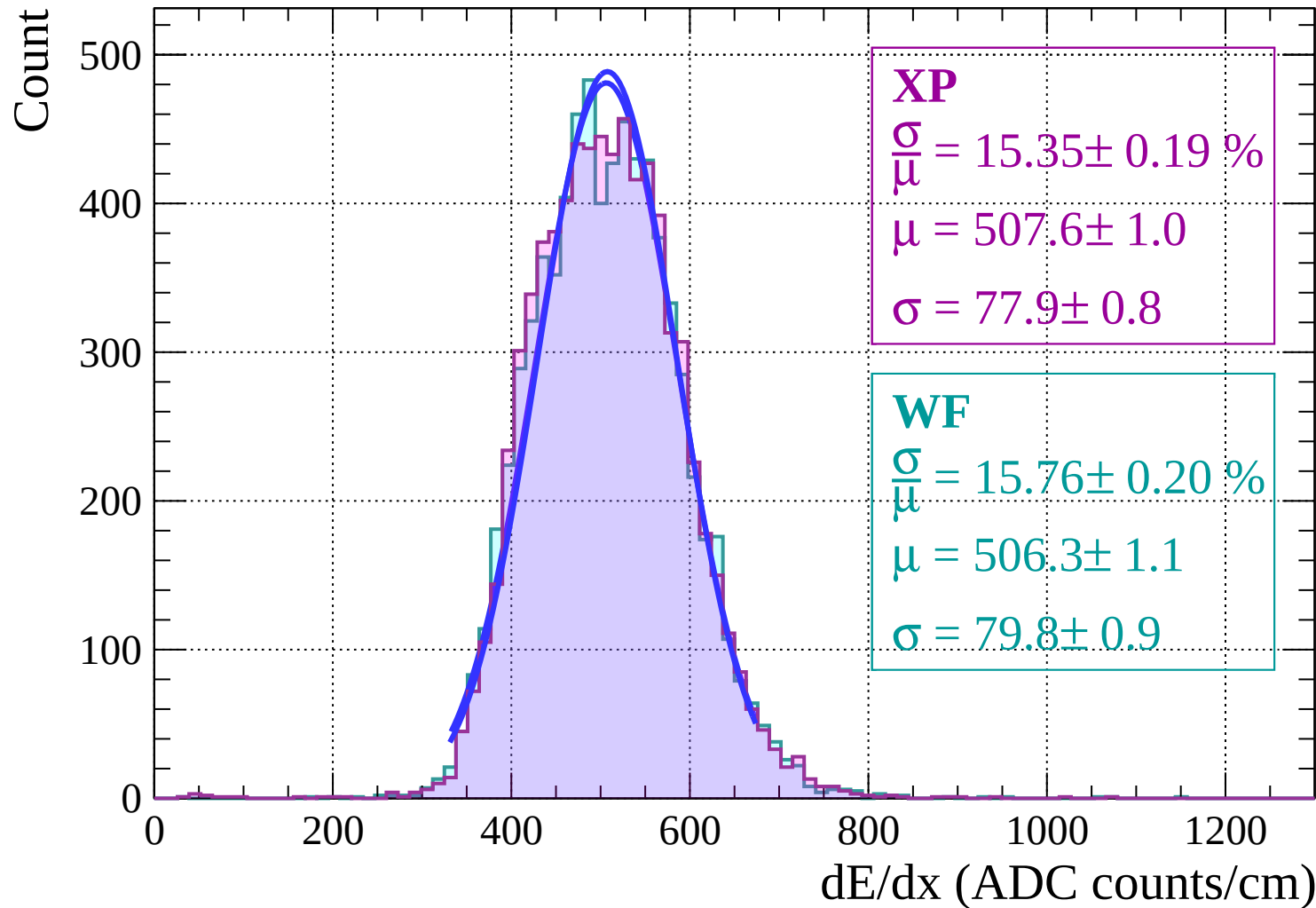
Count



Energy loss $|-34 < \theta < -32$

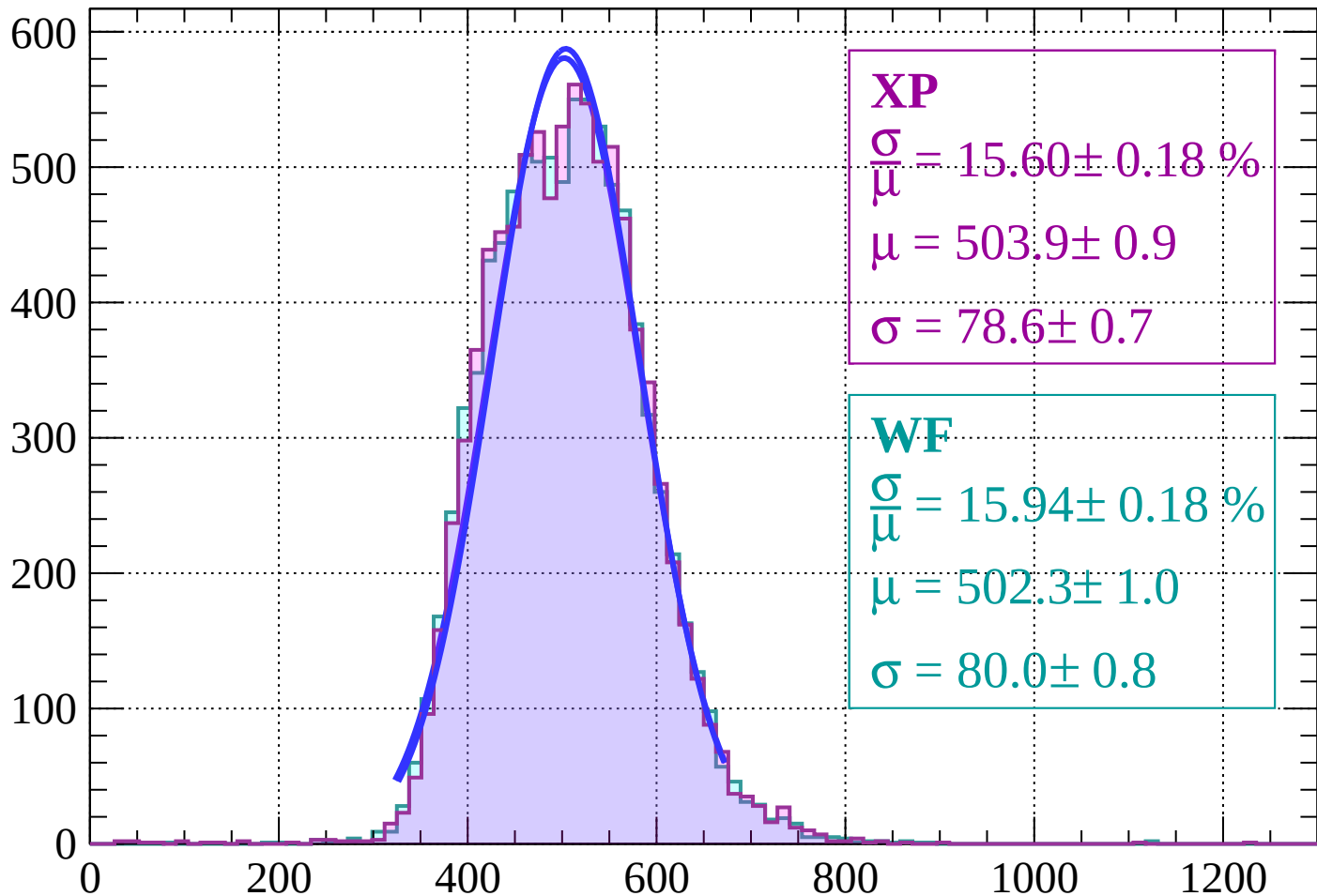


Energy loss | $-32 < \theta < -30$



Energy loss | $-30 < \theta < -28$

Count



XP

$$\frac{\sigma}{\mu} = 15.60 \pm 0.18 \%$$

$$\mu = 503.9 \pm 0.9$$

$$\sigma = 78.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 15.94 \pm 0.18 \%$$

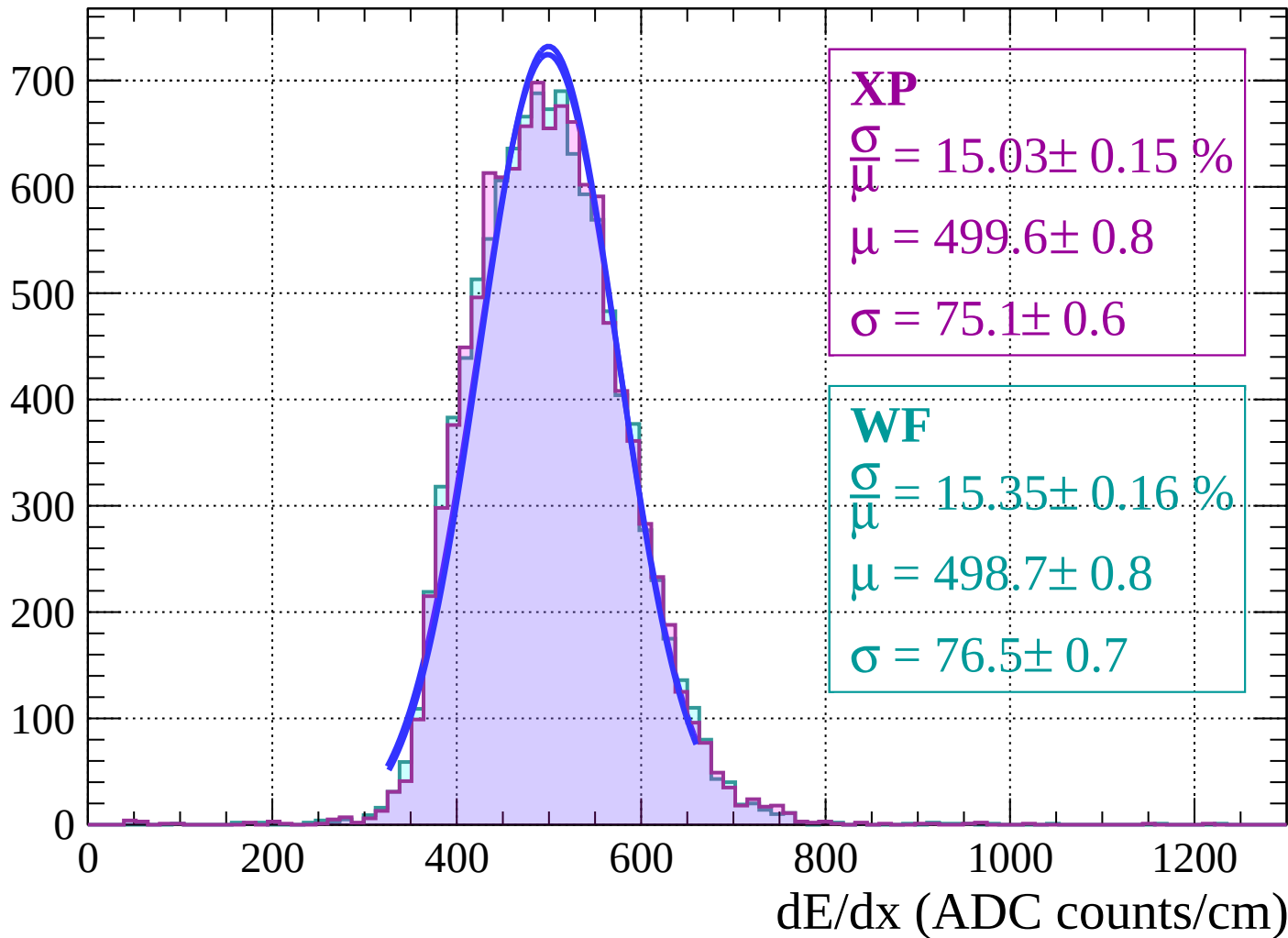
$$\mu = 502.3 \pm 1.0$$

$$\sigma = 80.0 \pm 0.8$$

dE/dx (ADC counts/cm)

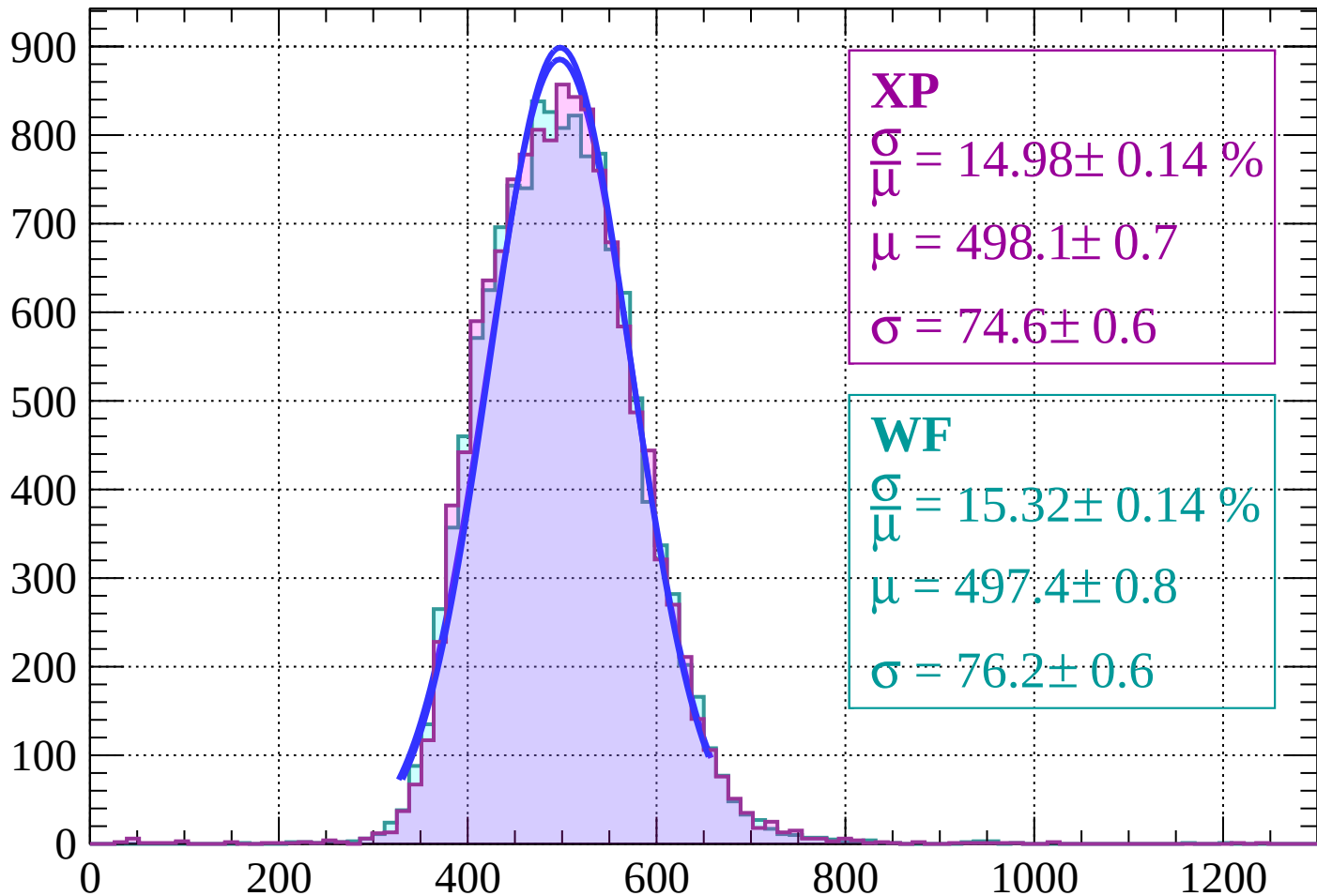
Energy loss $|-28 < \theta < -26$

Count



Energy loss | $-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 14.98 \pm 0.14 \%$$

$$\mu = 498.1 \pm 0.7$$

$$\sigma = 74.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.32 \pm 0.14 \%$$

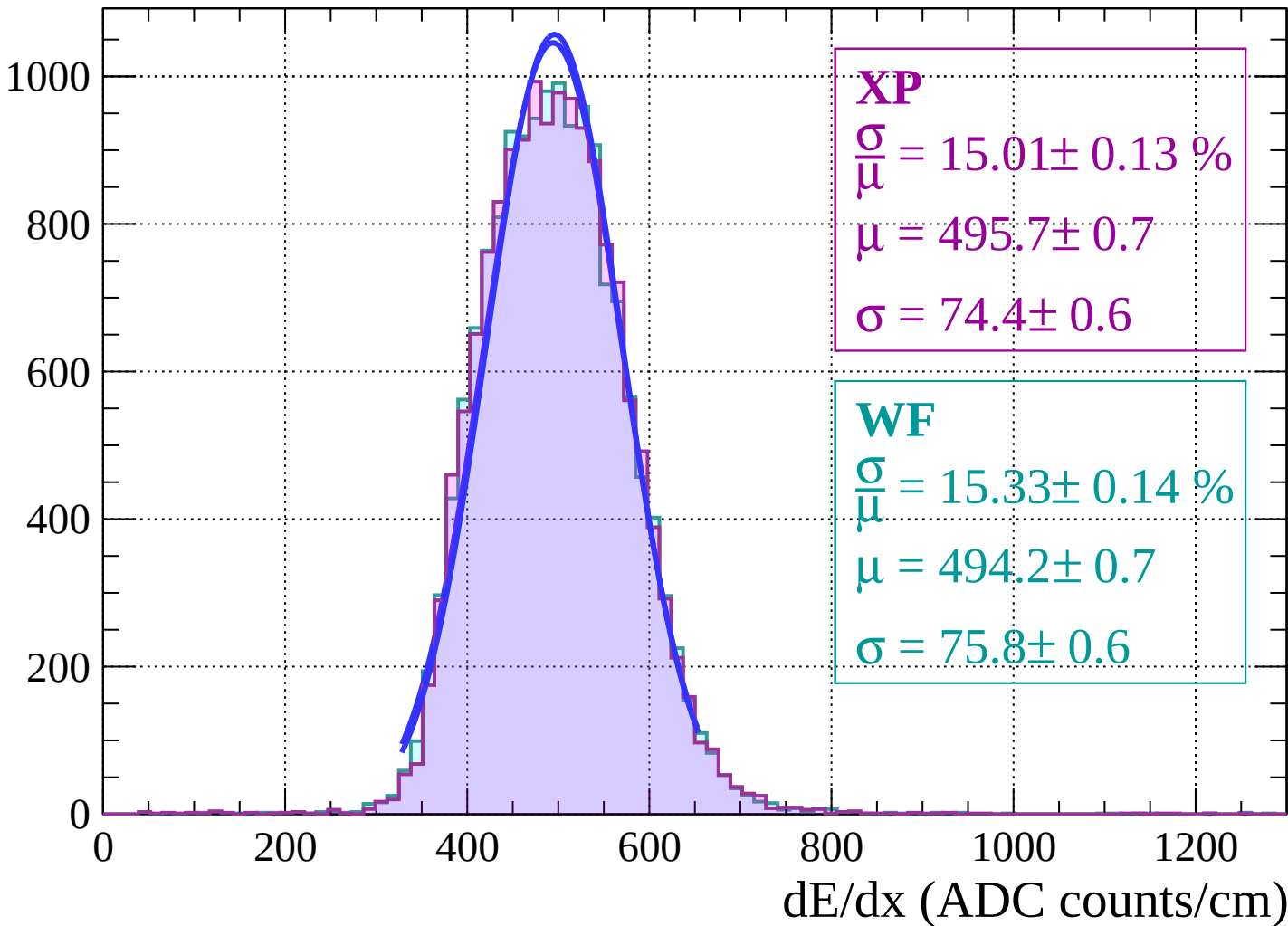
$$\mu = 497.4 \pm 0.8$$

$$\sigma = 76.2 \pm 0.6$$

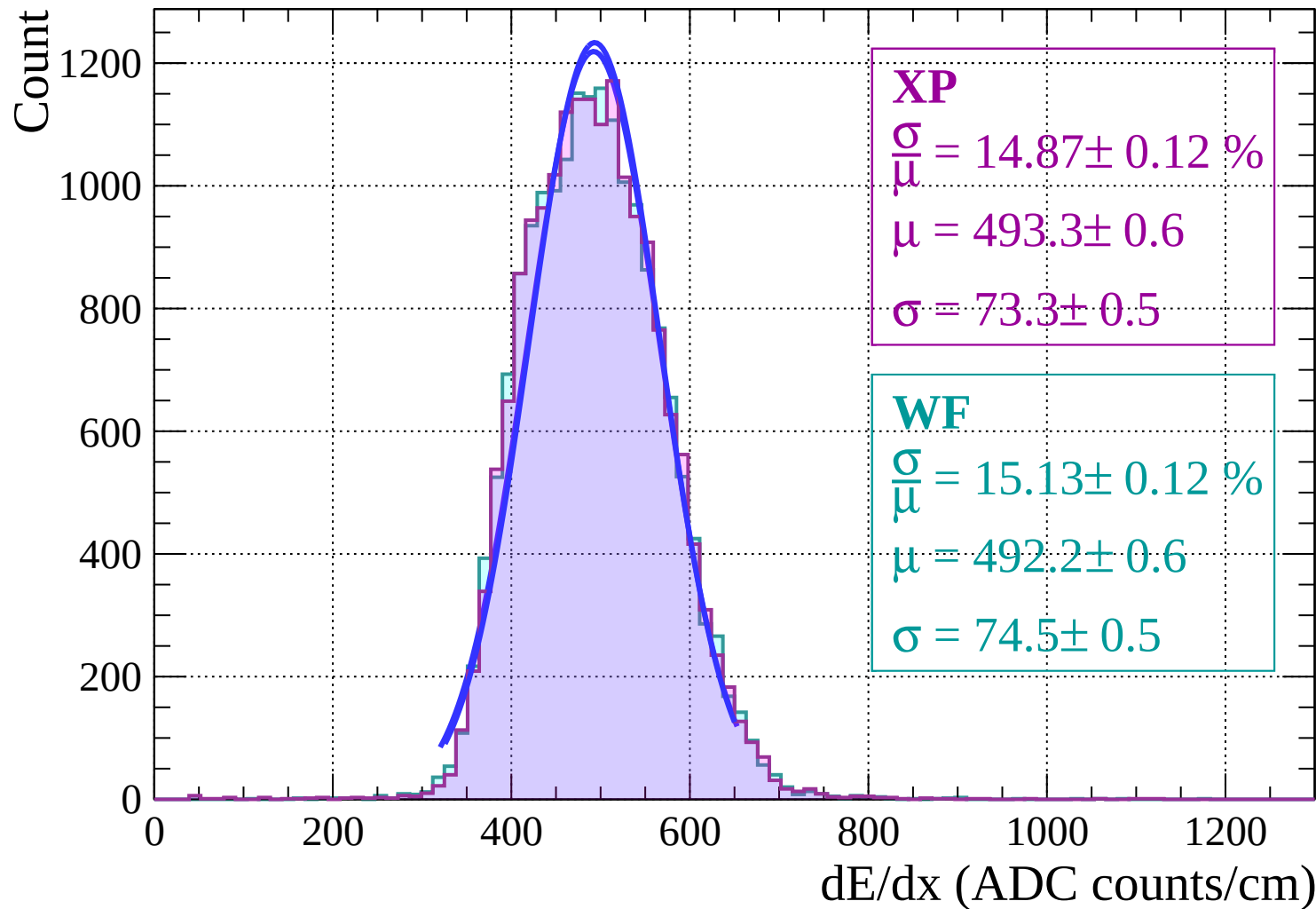
dE/dx (ADC counts/cm)

Energy loss $|-24 < \theta < -22$

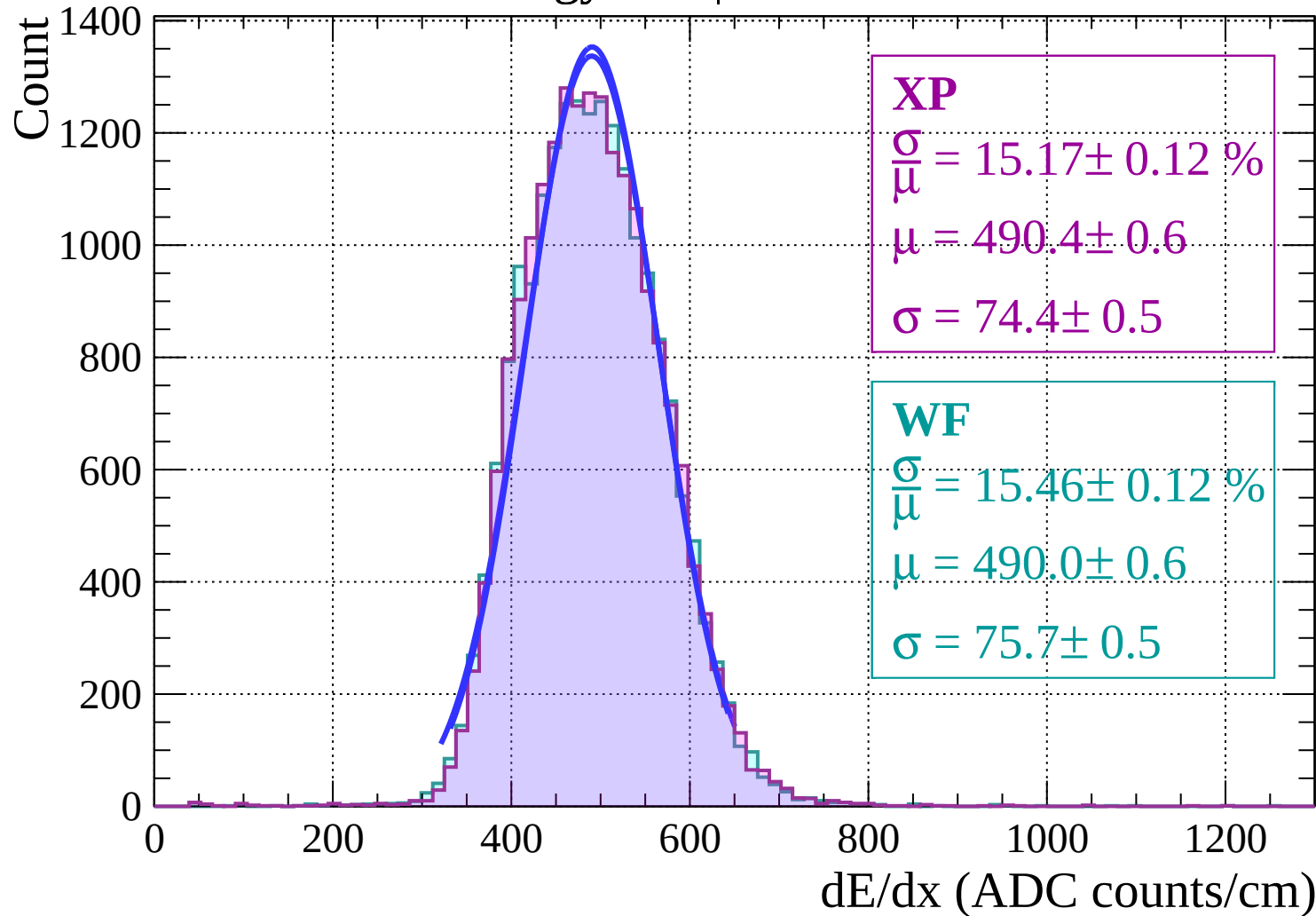
Count



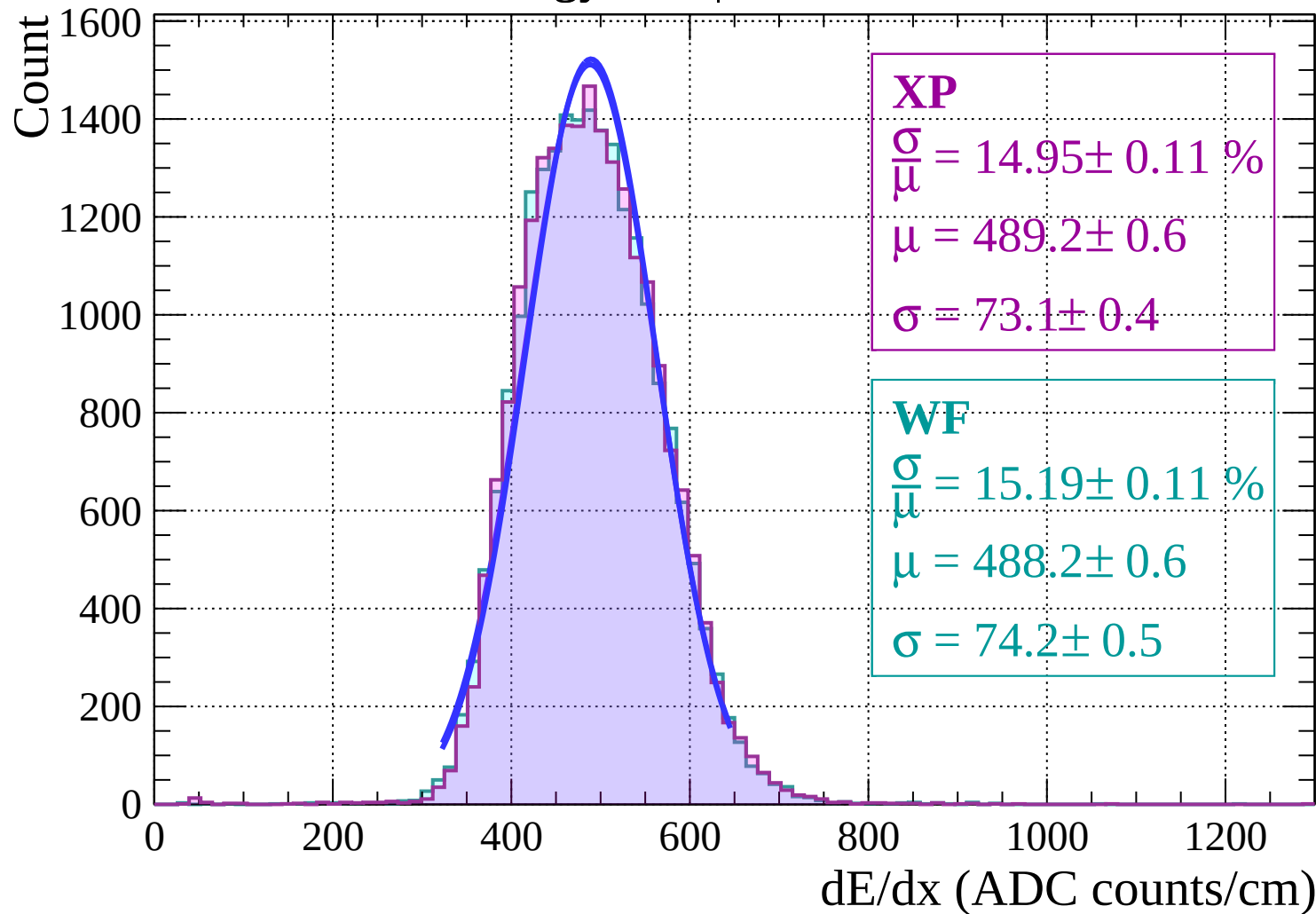
Energy loss | $-22 < \theta < -20$



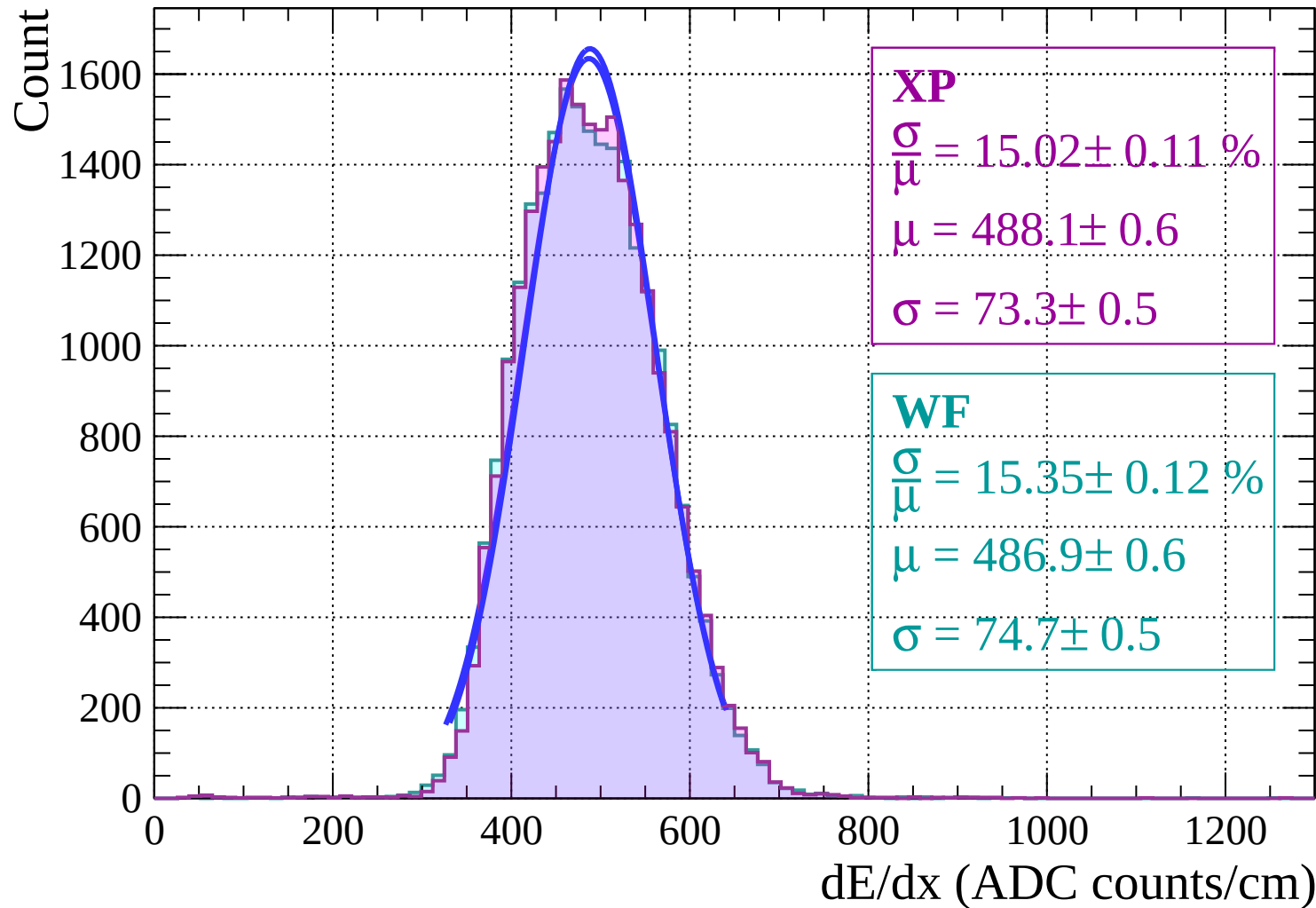
Energy loss | $-20 < \theta < -18$



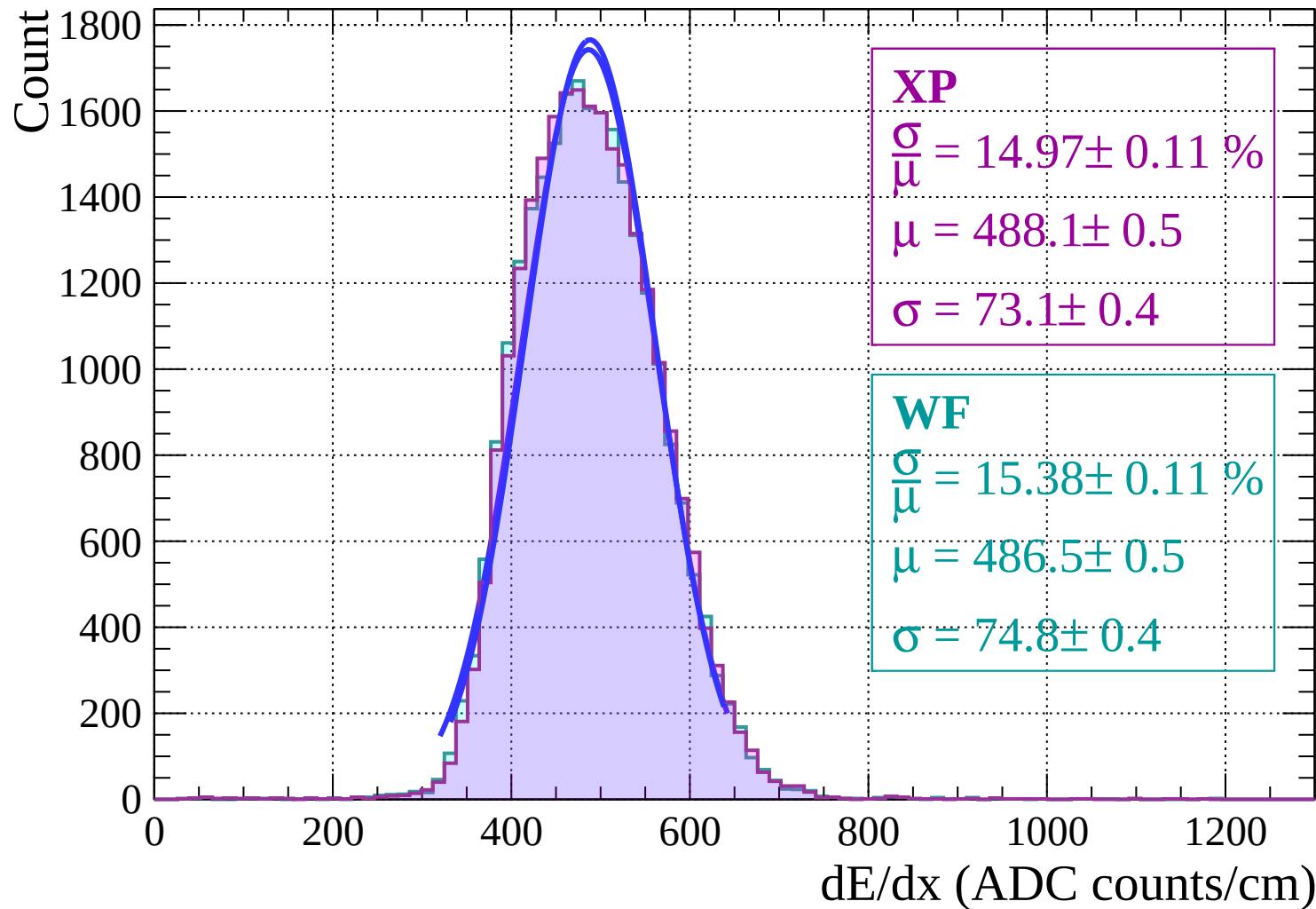
Energy loss | $-18 < \theta < -16$



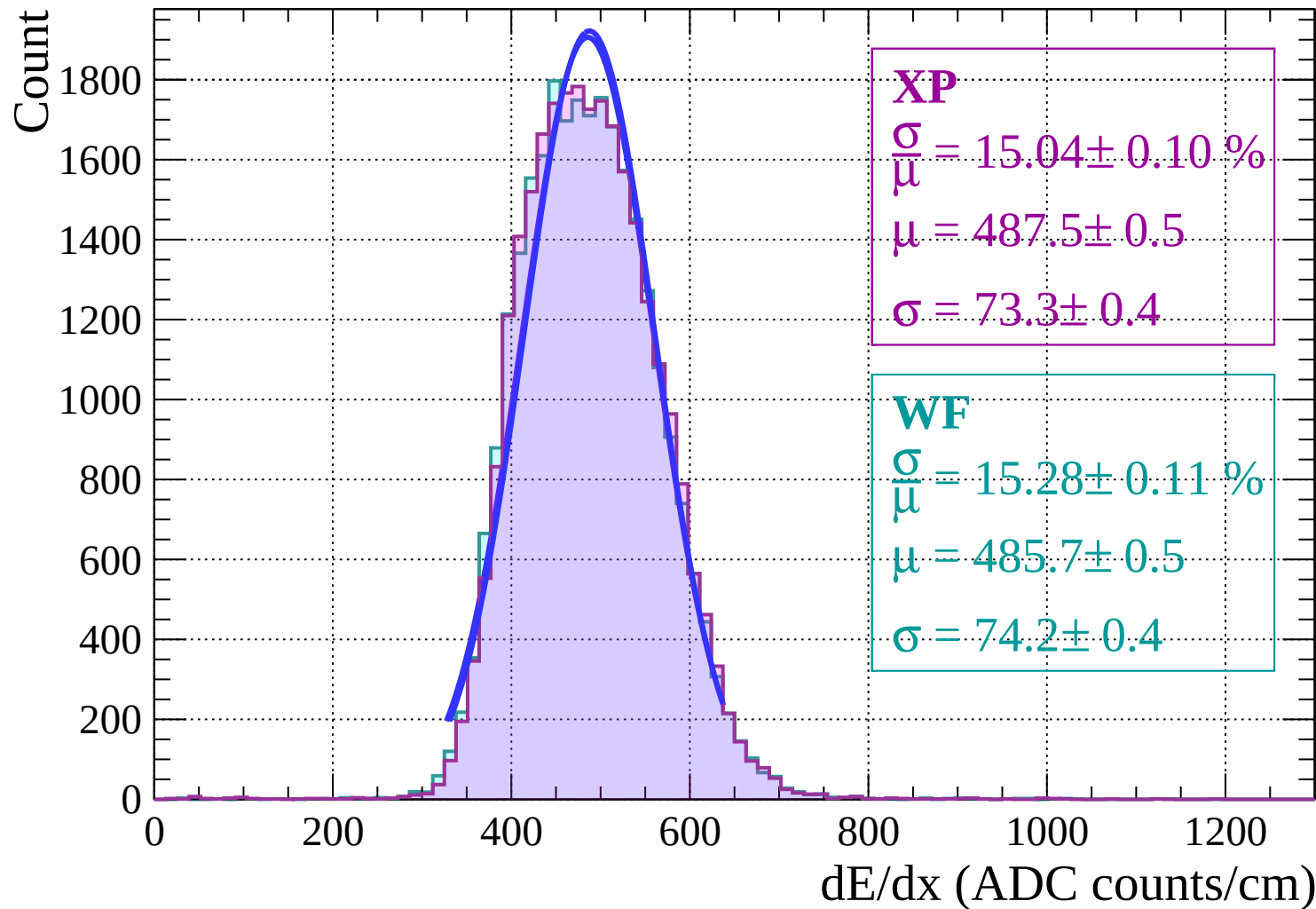
Energy loss | $-16 < \theta < -14$



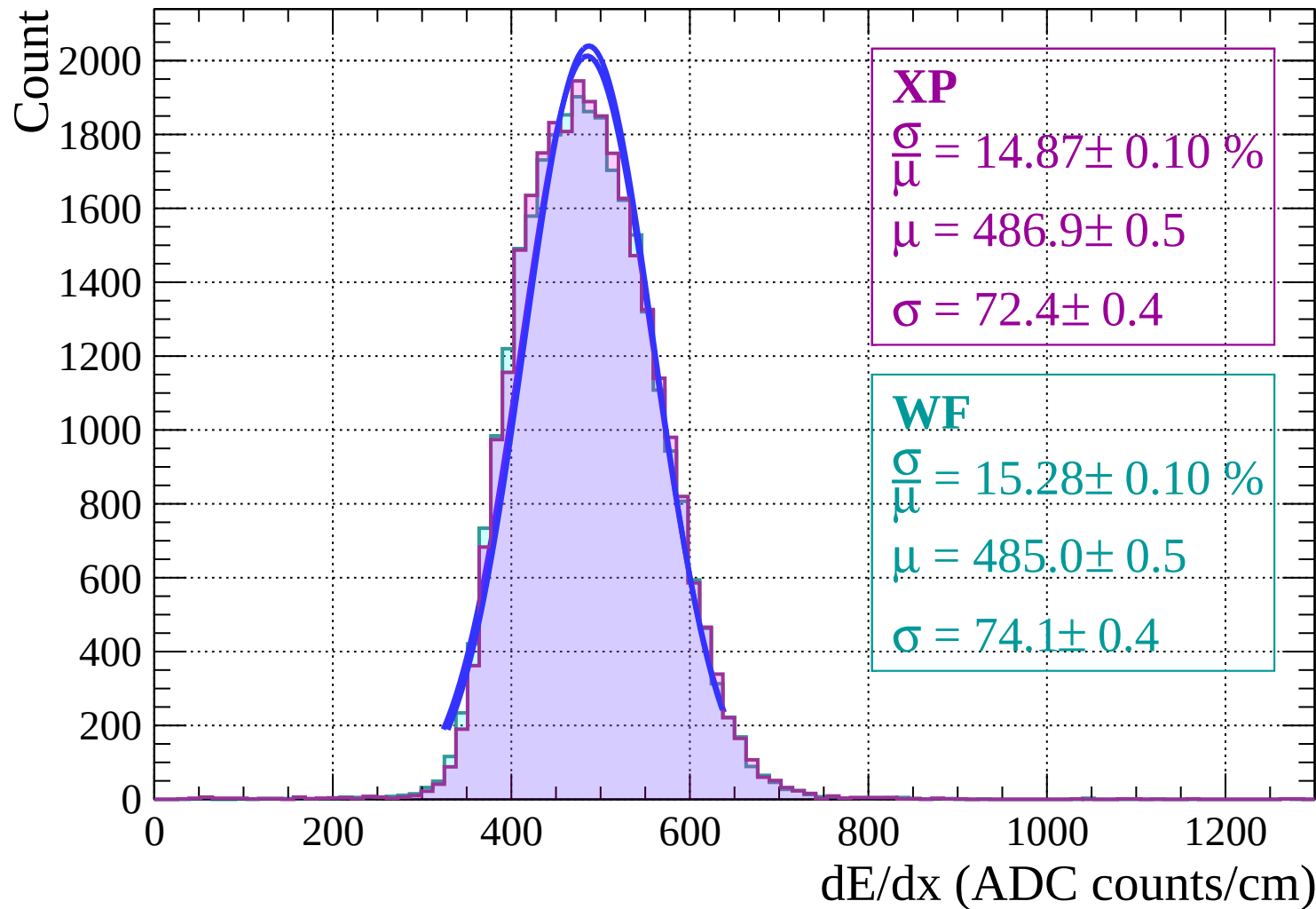
Energy loss | $-14 < \theta < -12$



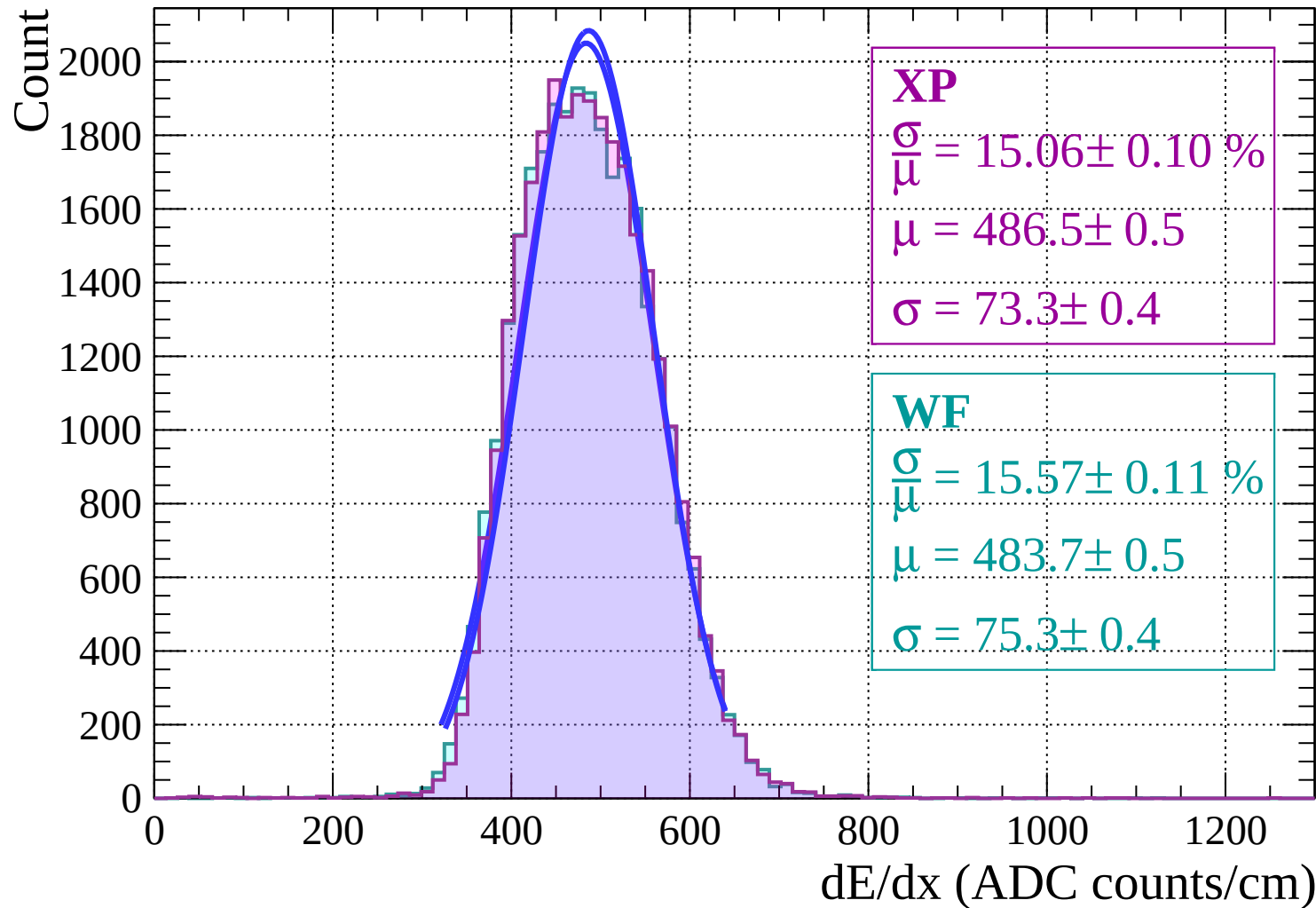
Energy loss | $-12 < \theta < -10$



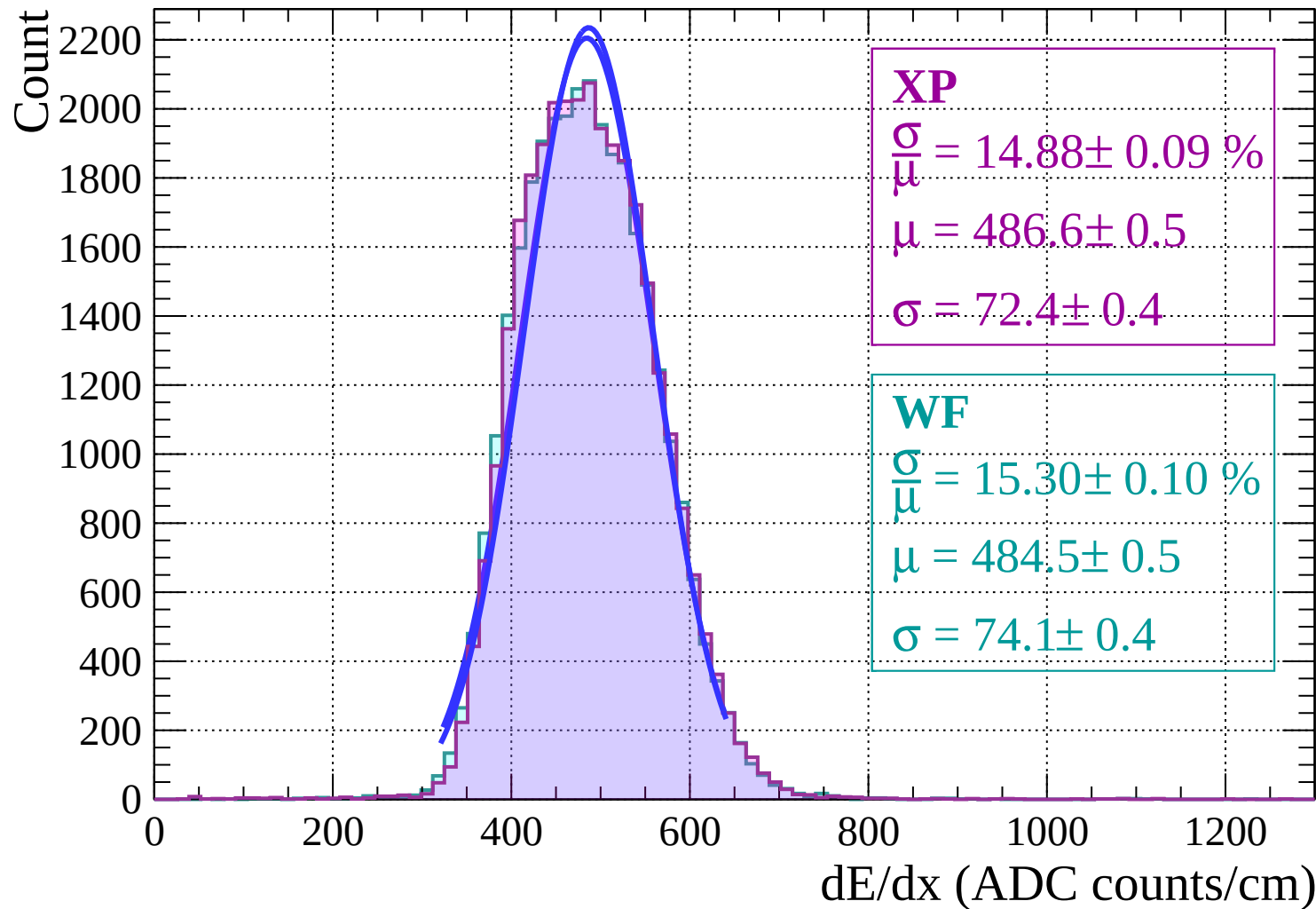
Energy loss | $-10 < \theta < -8$



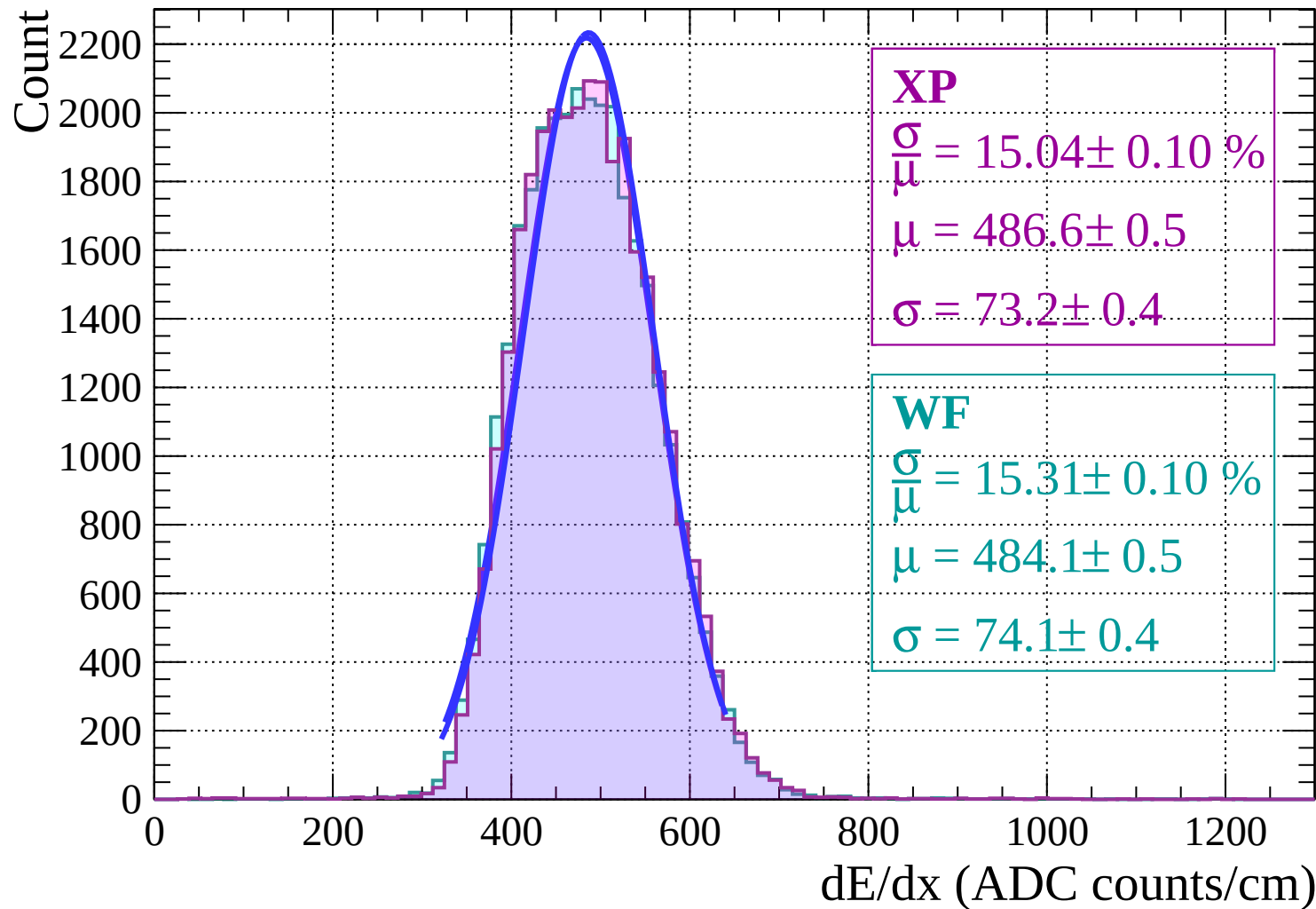
Energy loss | $-8 < \theta < -6$



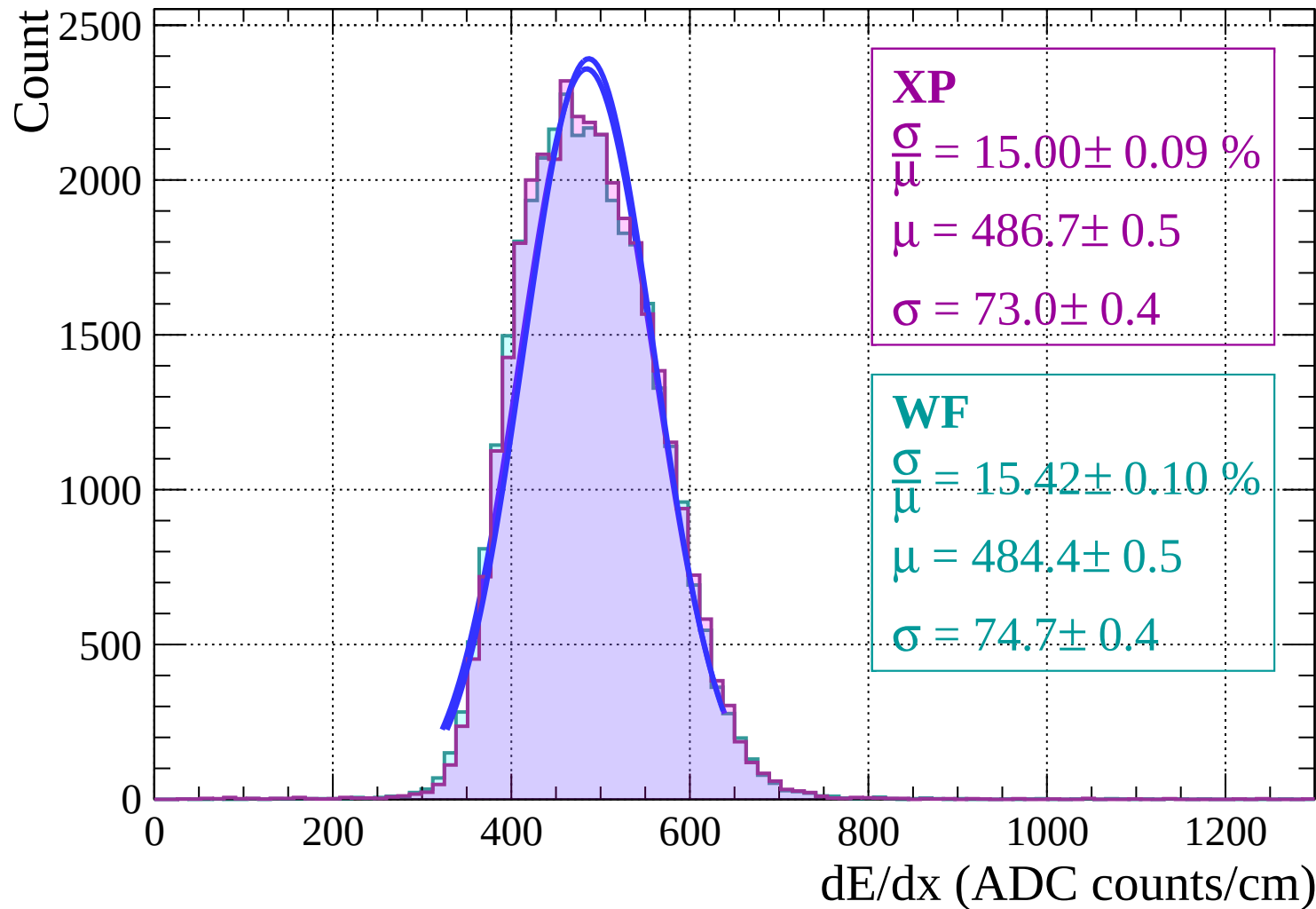
Energy loss | $-6 < \theta < -4$



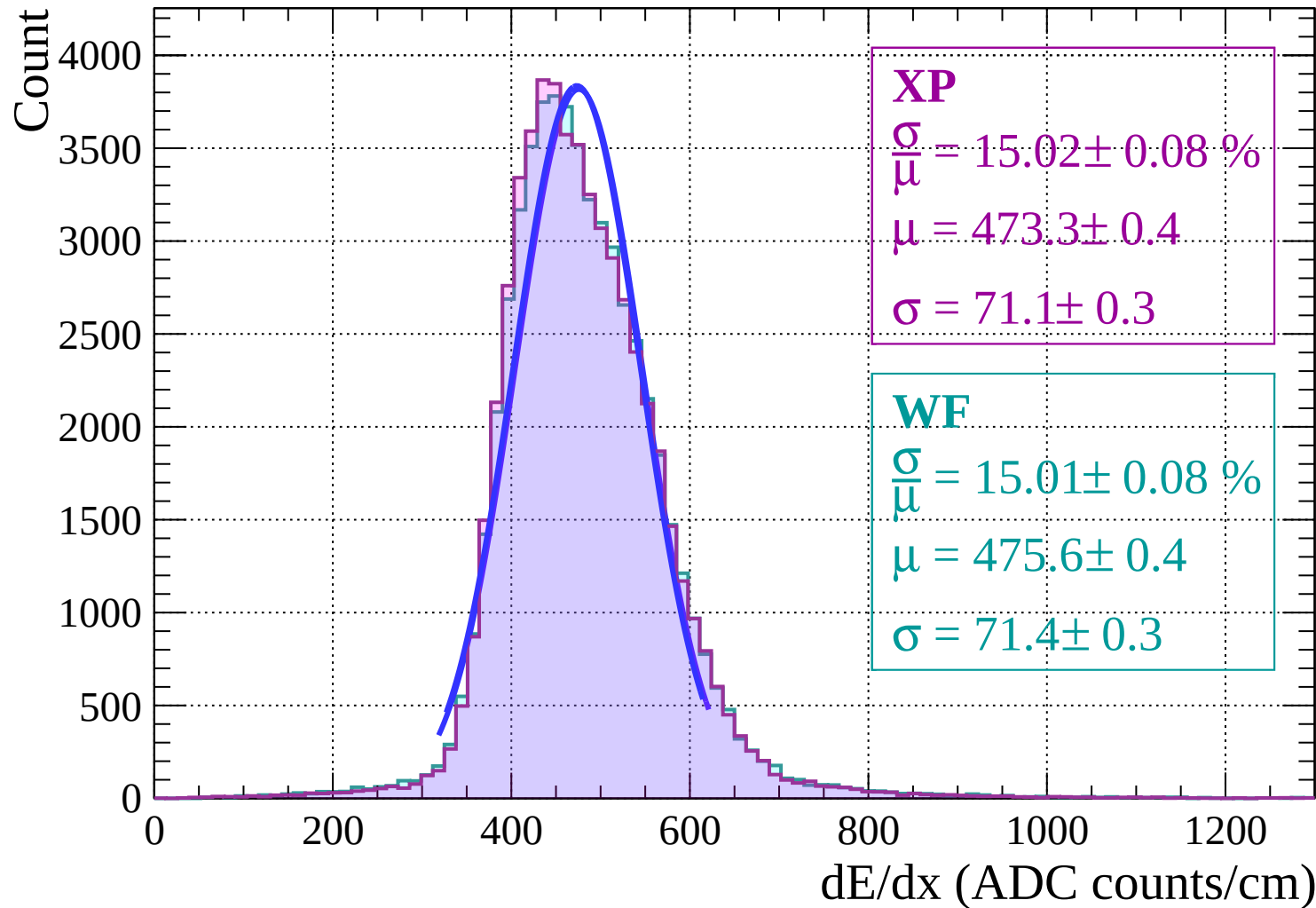
Energy loss | $-4 < \theta < -2$



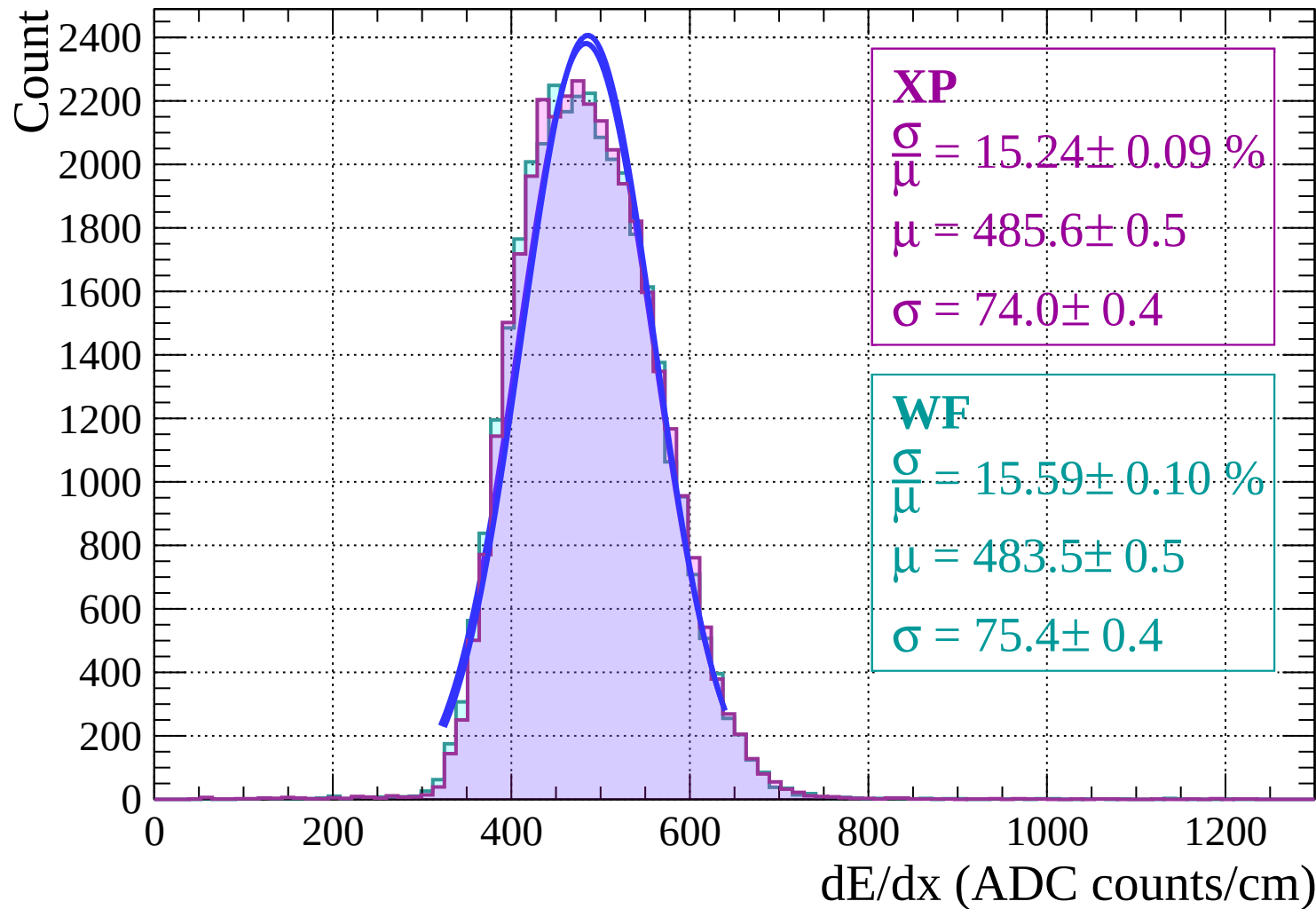
Energy loss | $-2 < \theta < 0$



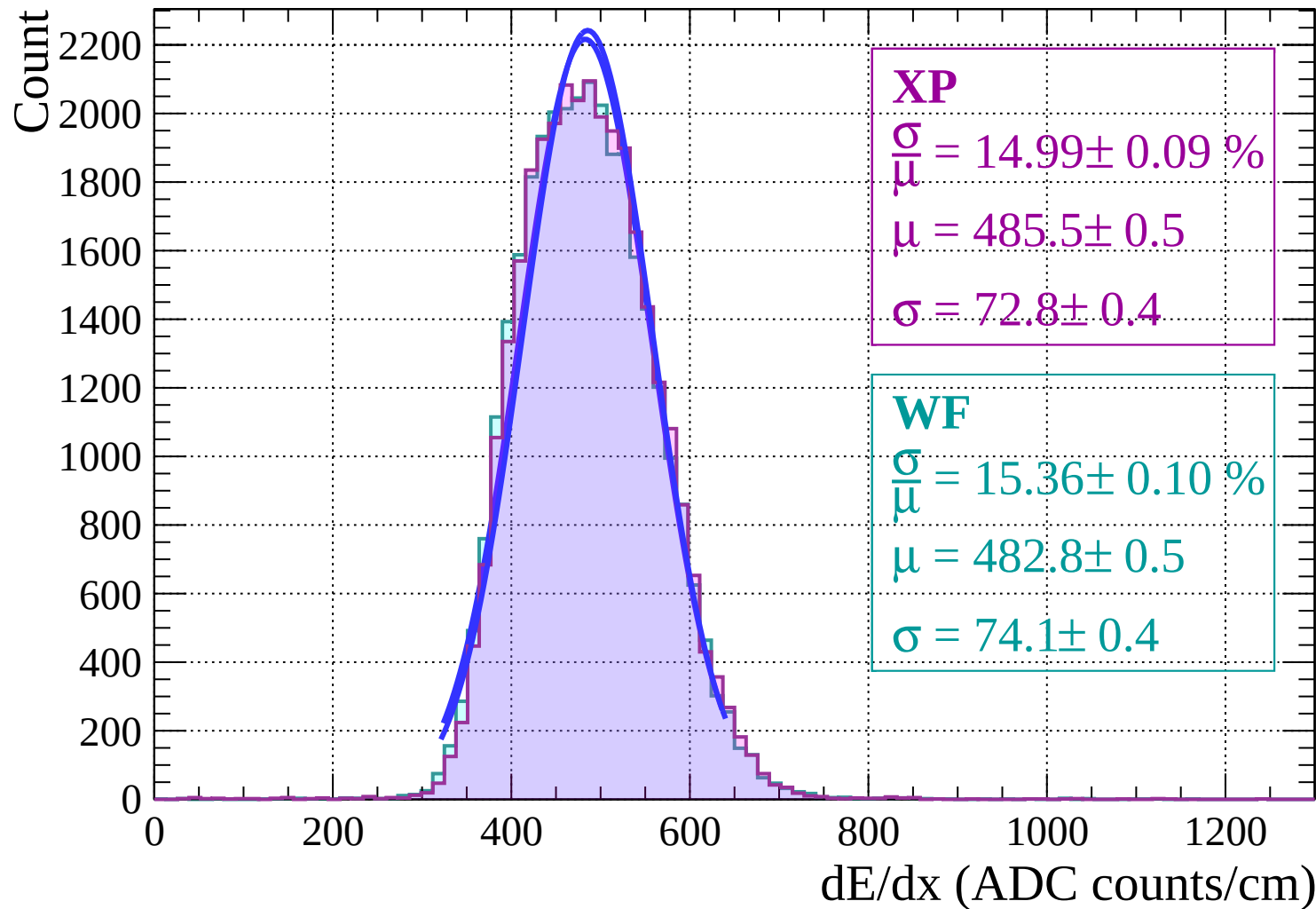
Energy loss | $0 < \theta < 2$



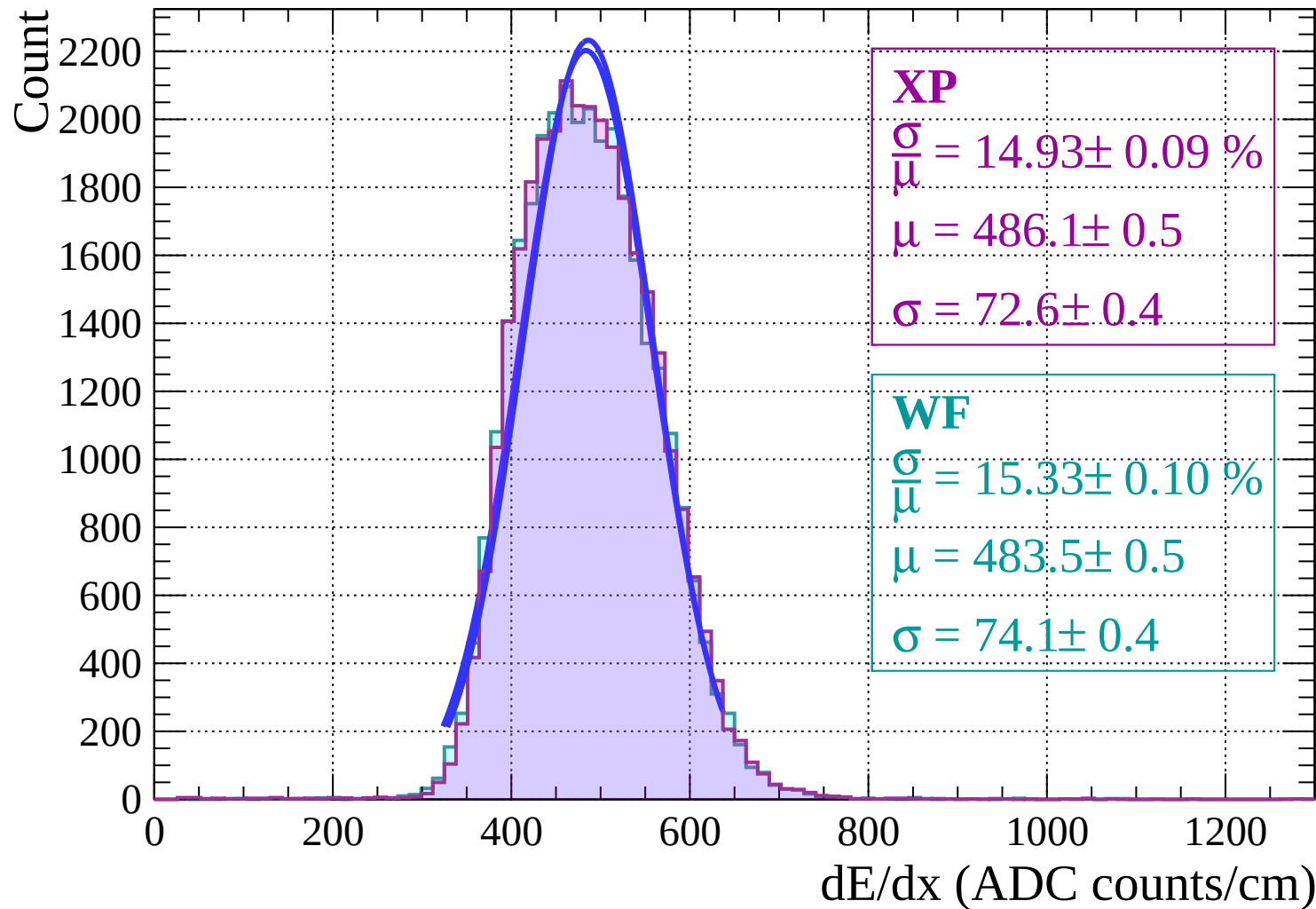
Energy loss | $2 < \theta < 4$



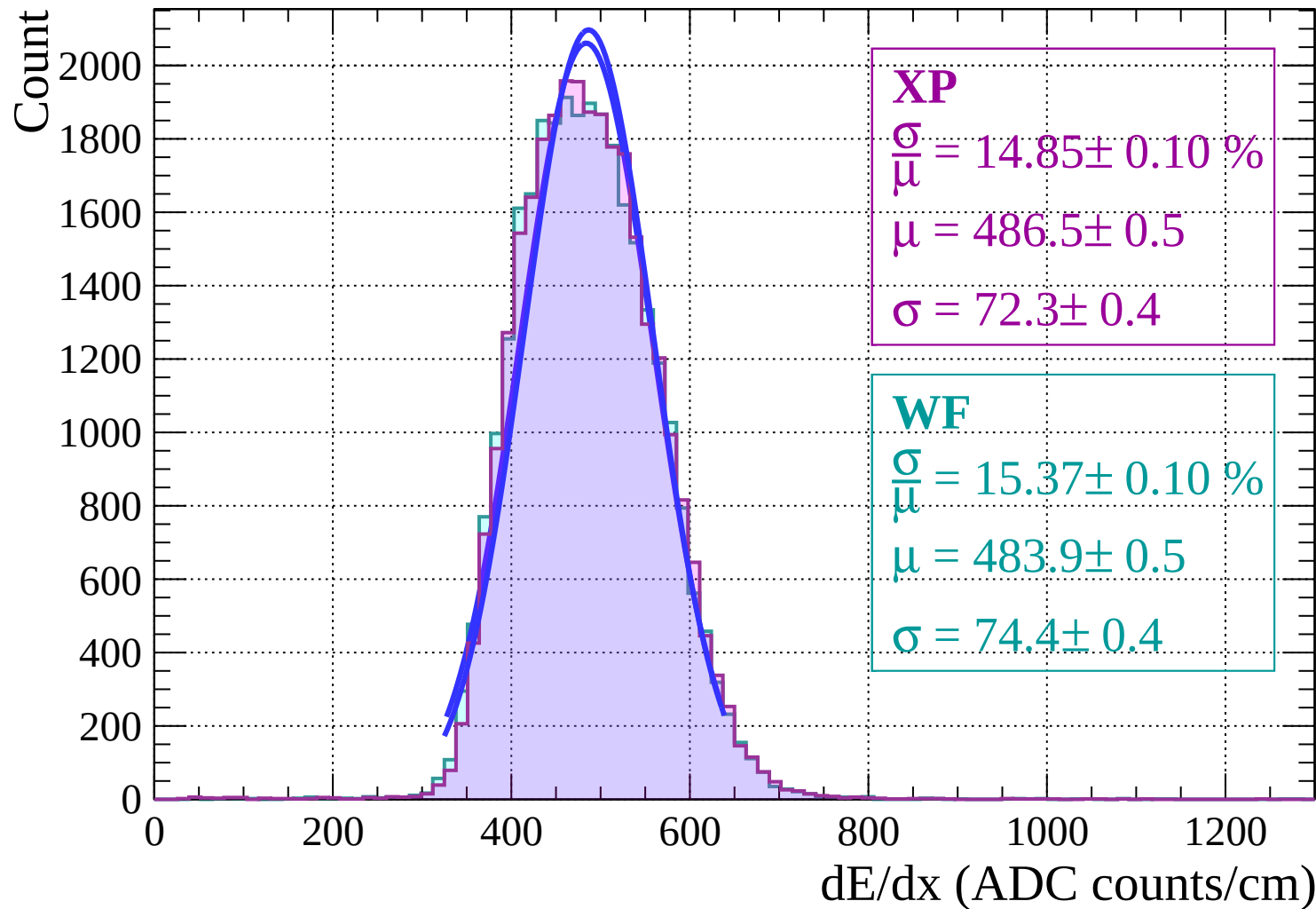
Energy loss | $4 < \theta < 6$



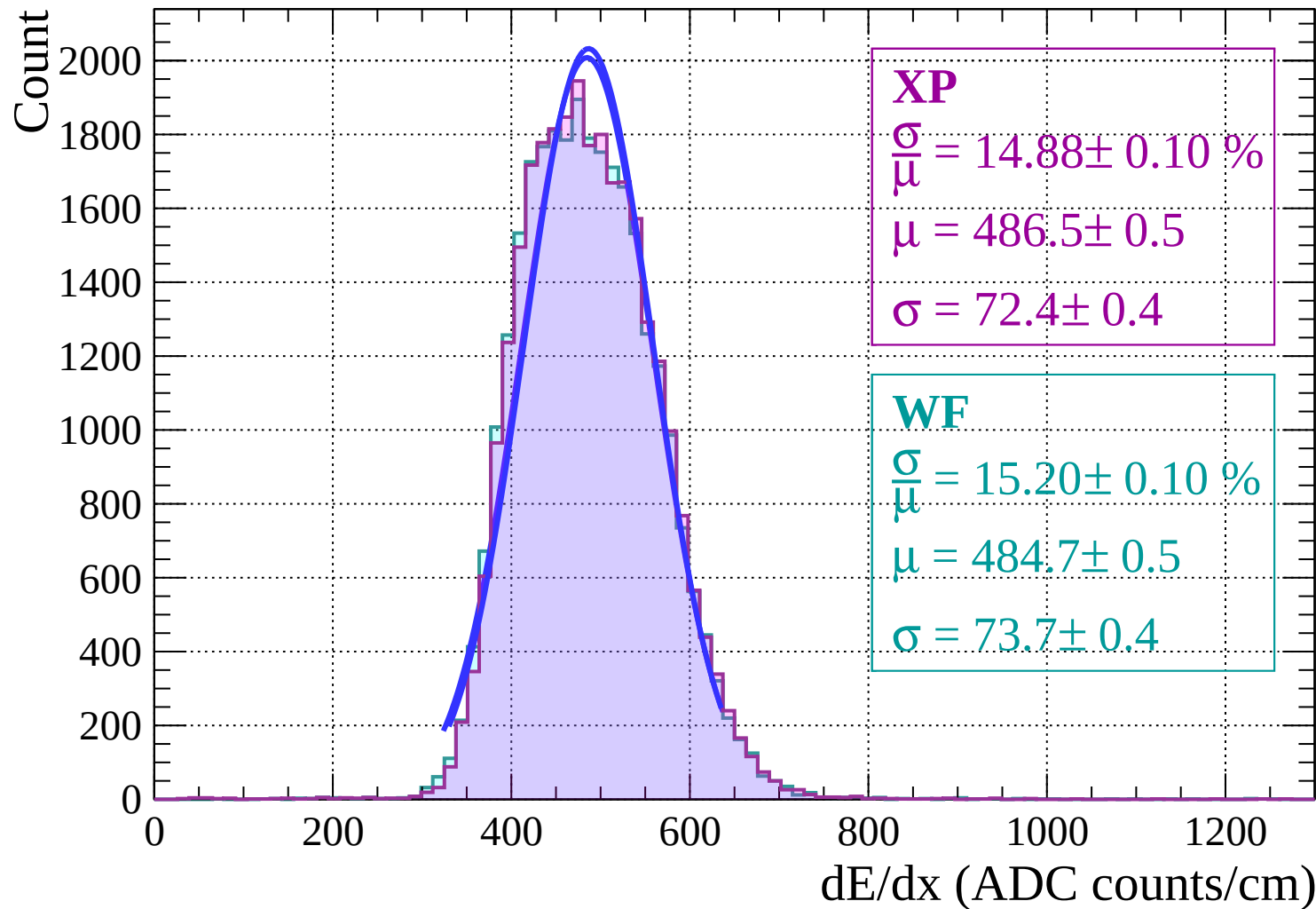
Energy loss | $6 < \theta < 8$



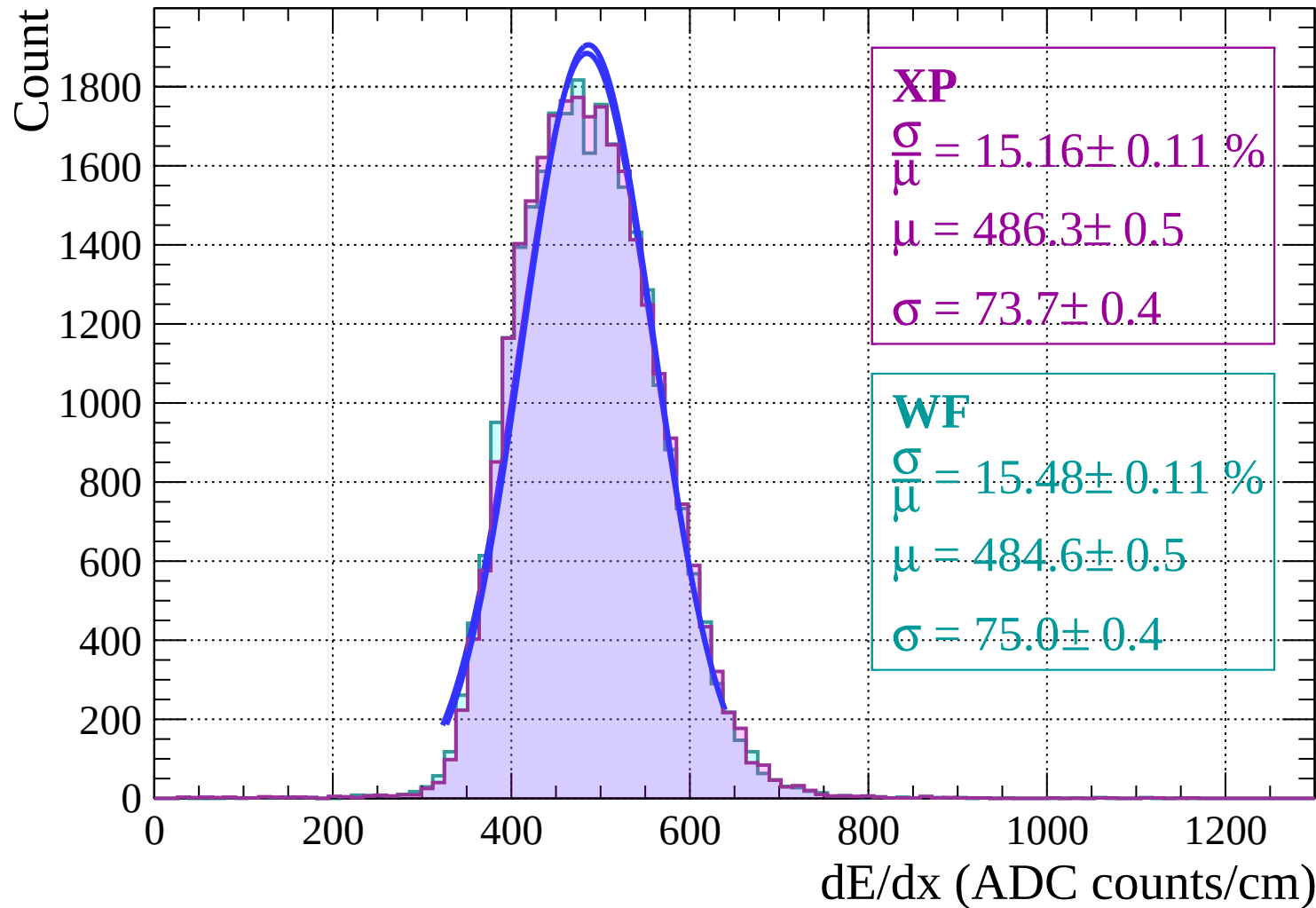
Energy loss | $8 < \theta < 10$



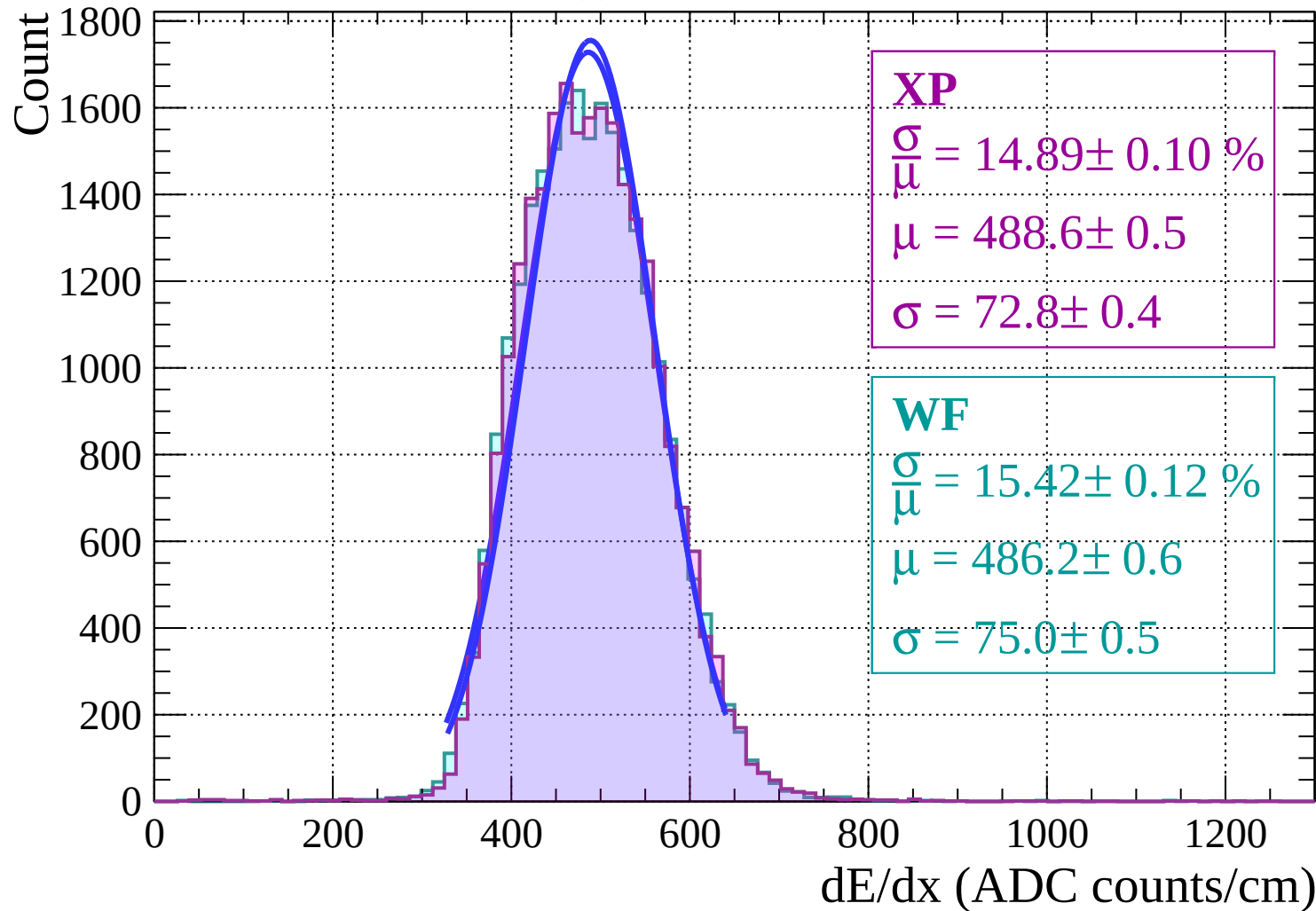
Energy loss | $10 < \theta < 12$



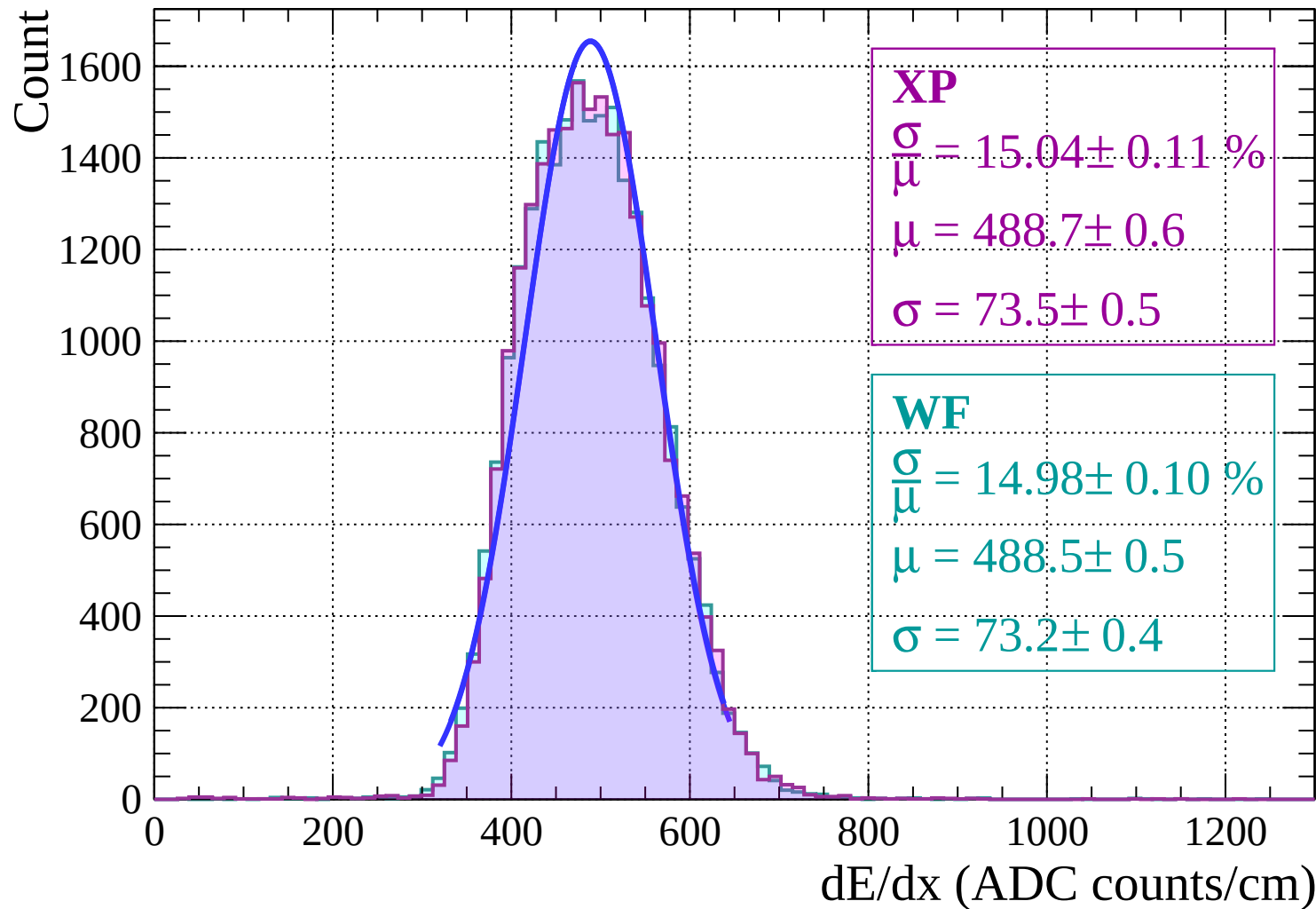
Energy loss | $12 < \theta < 14$



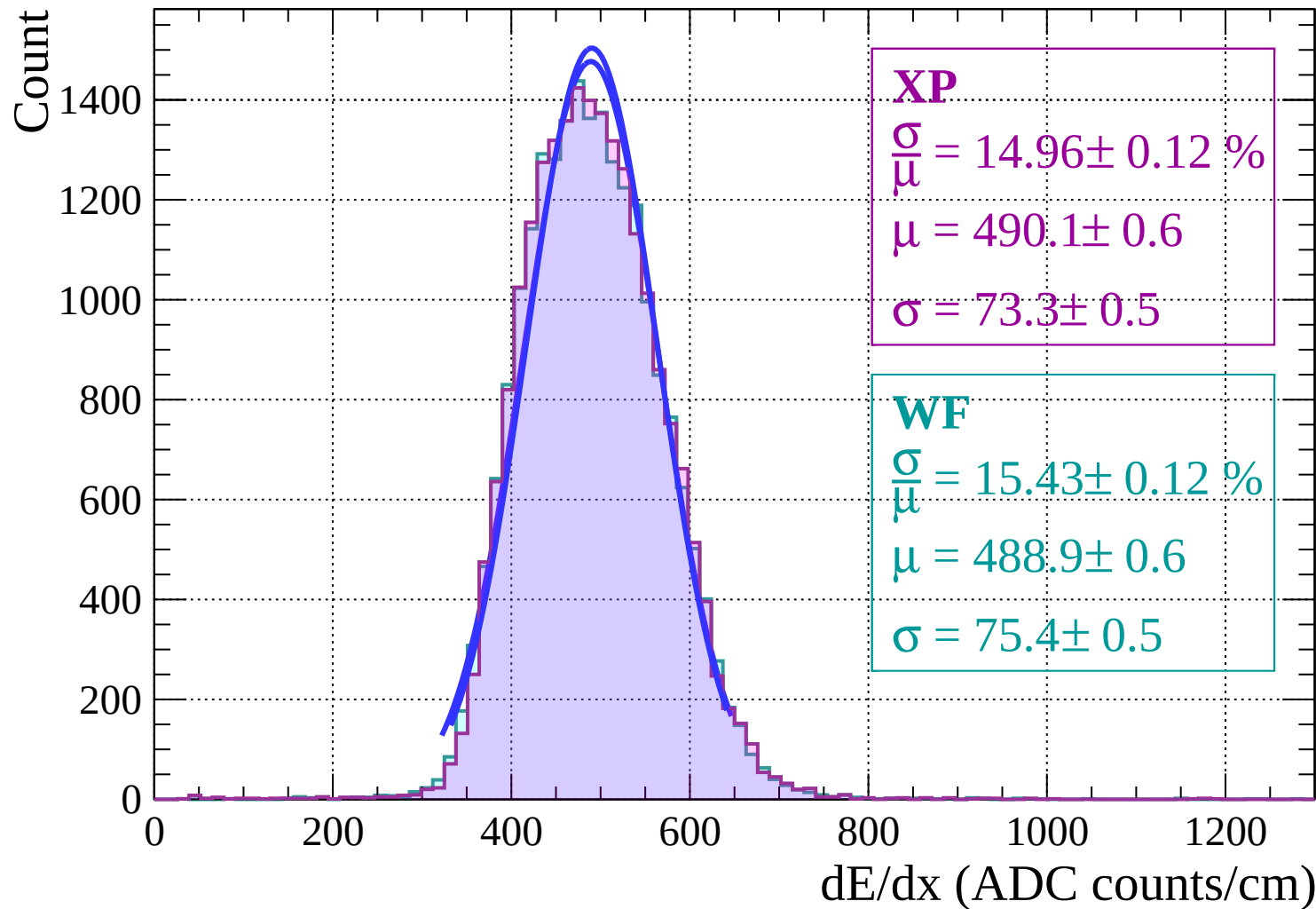
Energy loss | $14 < \theta < 16$



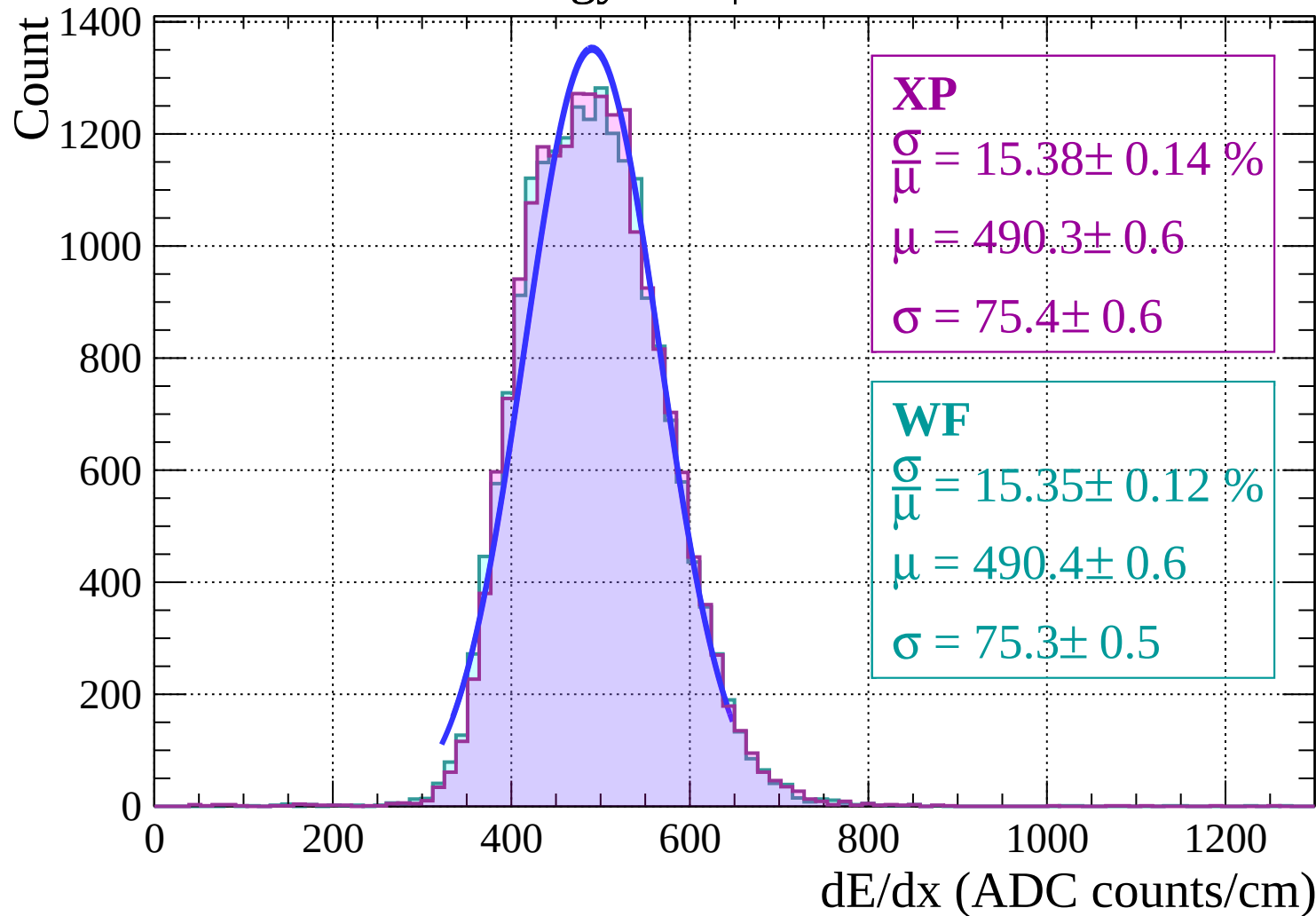
Energy loss | $16 < \theta < 18$



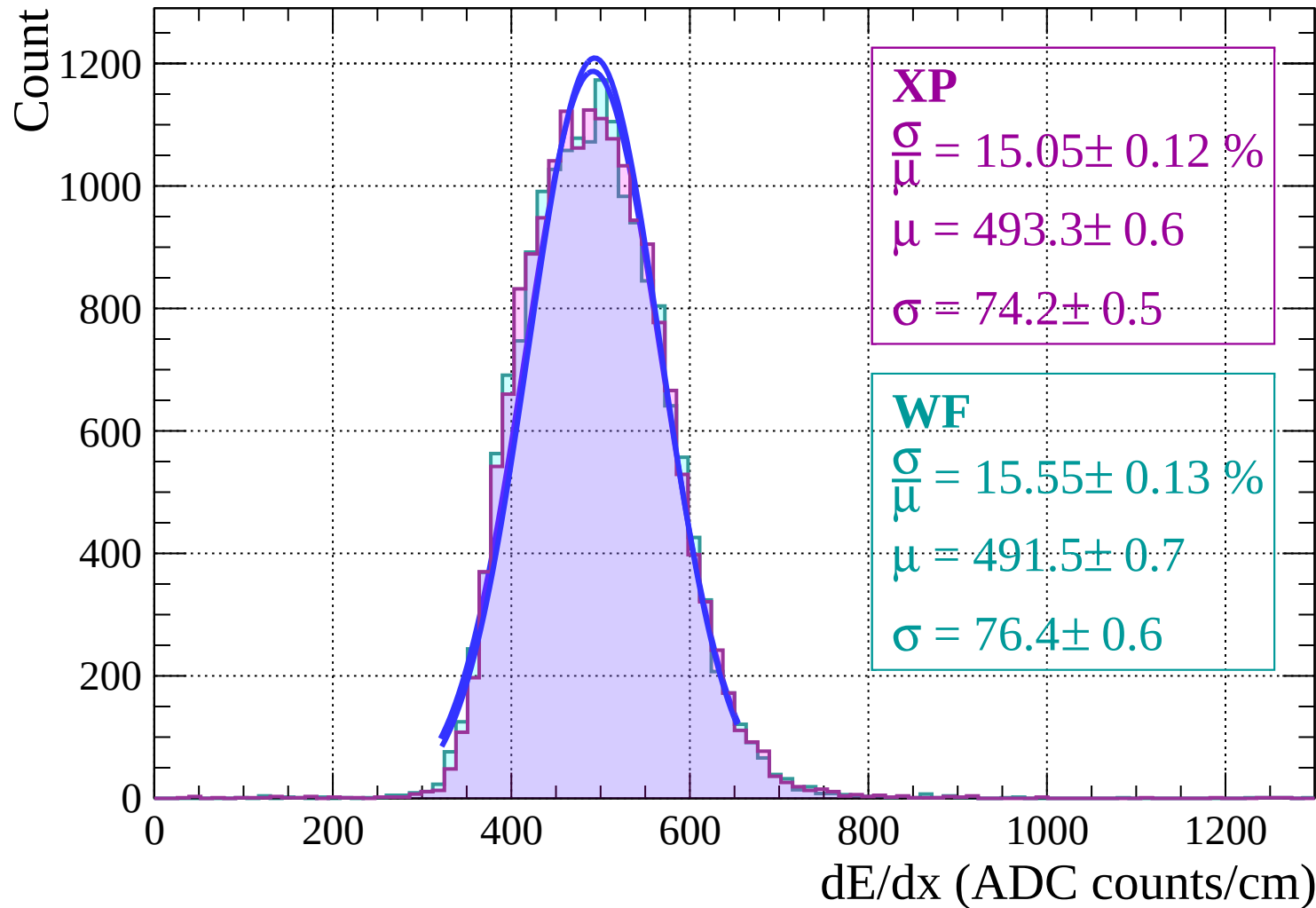
Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$



Energy loss | $24 < \theta < 26$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 14.92 \pm 0.13 \%$$

$$\mu = 495.2 \pm 0.7$$

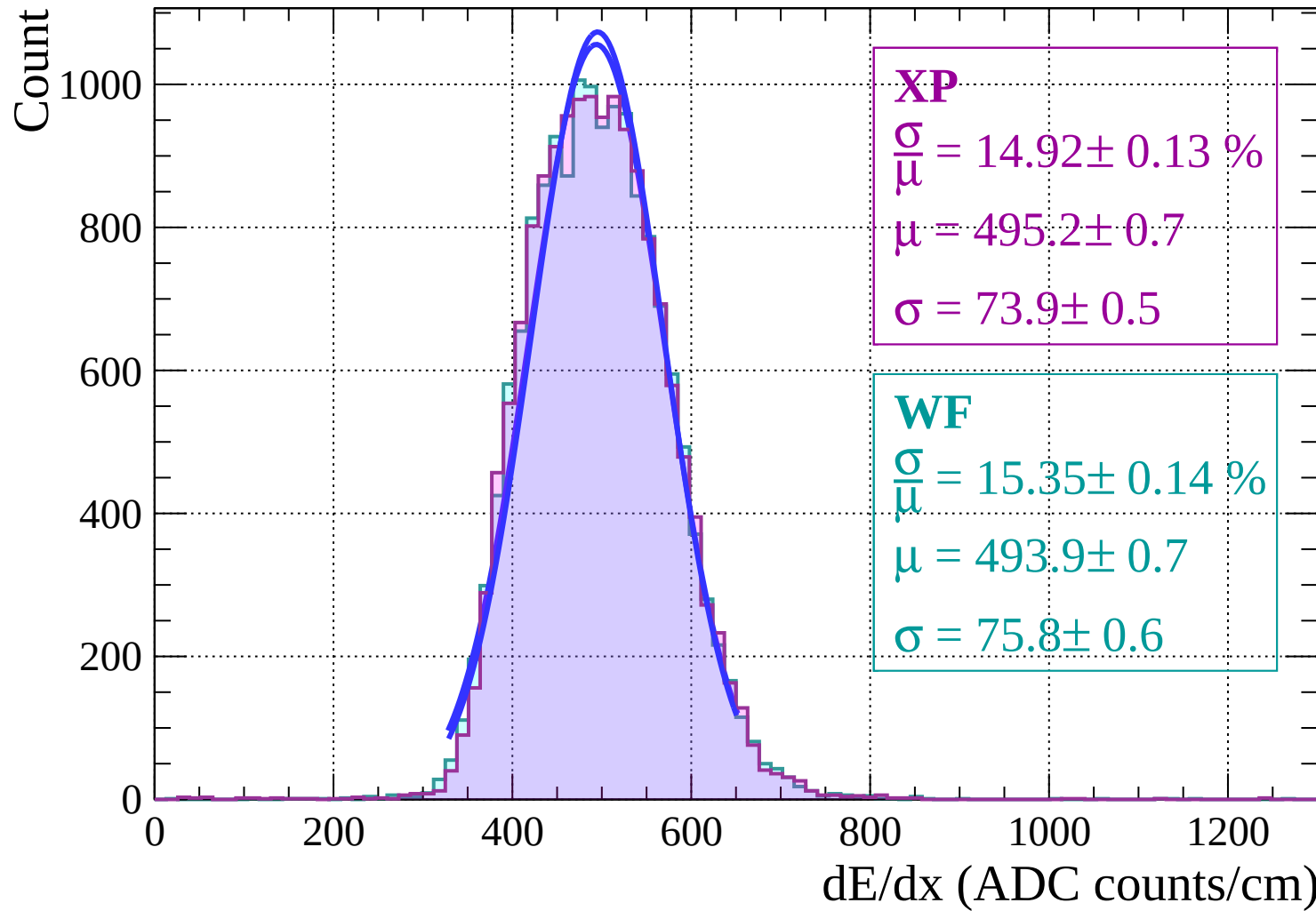
$$\sigma = 73.9 \pm 0.5$$

WF

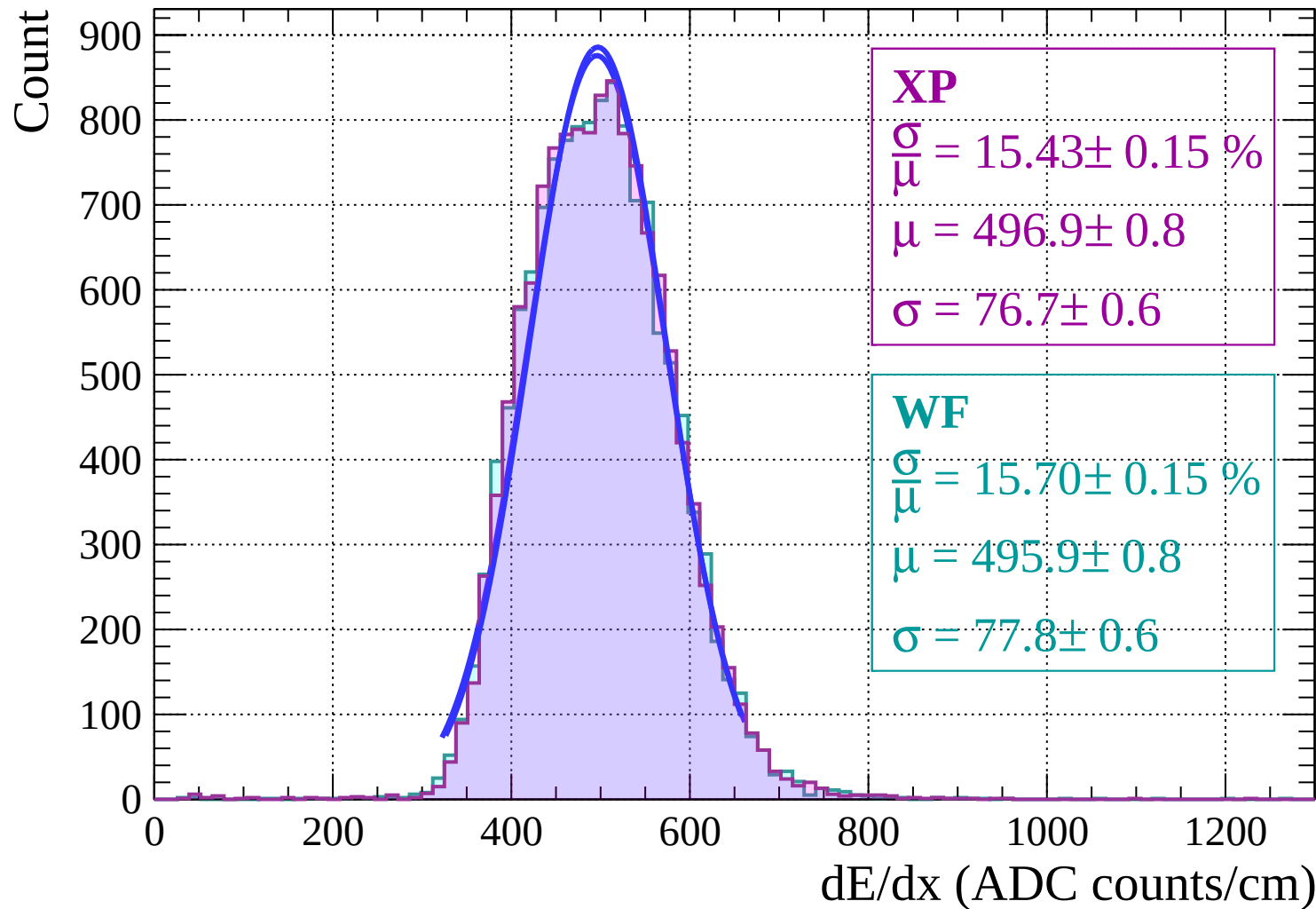
$$\frac{\sigma}{\mu} = 15.35 \pm 0.14 \%$$

$$\mu = 493.9 \pm 0.7$$

$$\sigma = 75.8 \pm 0.6$$



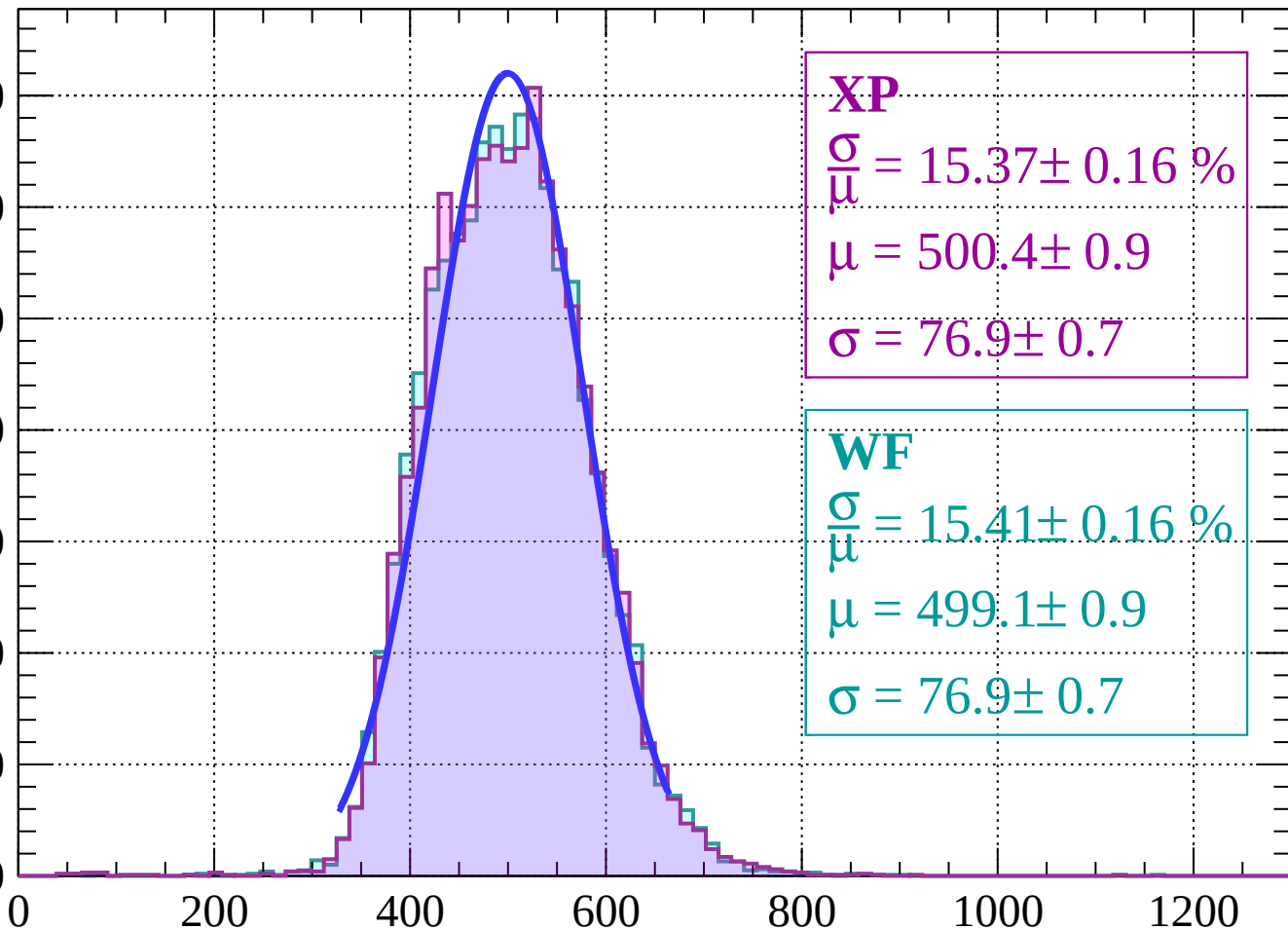
Energy loss | $26 < \theta < 28$



Energy loss | $28 < \theta < 30$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 15.37 \pm 0.16 \%$$

$$\mu = 500.4 \pm 0.9$$

$$\sigma = 76.9 \pm 0.7$$

WF

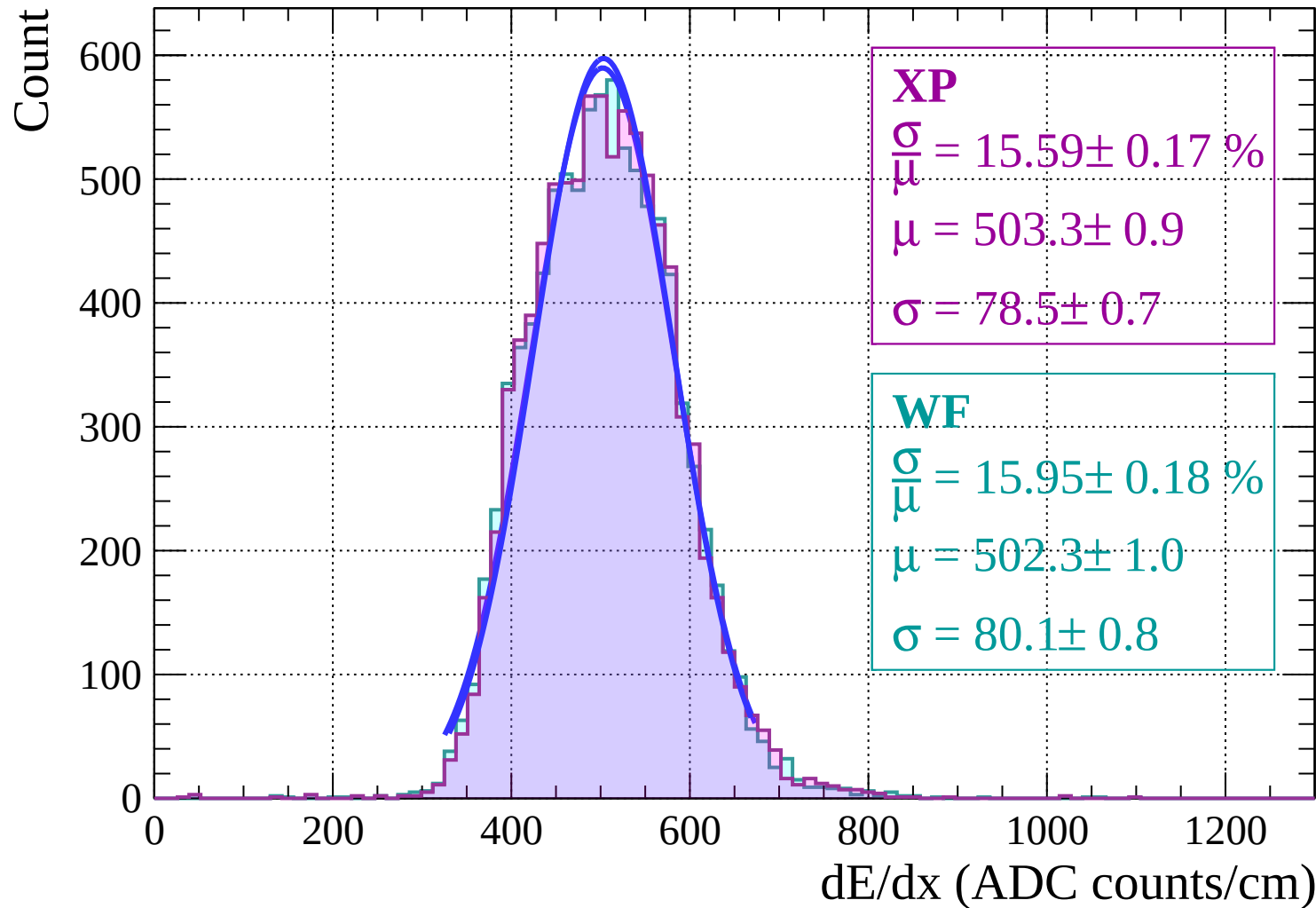
$$\frac{\sigma}{\mu} = 15.41 \pm 0.16 \%$$

$$\mu = 499.1 \pm 0.9$$

$$\sigma = 76.9 \pm 0.7$$

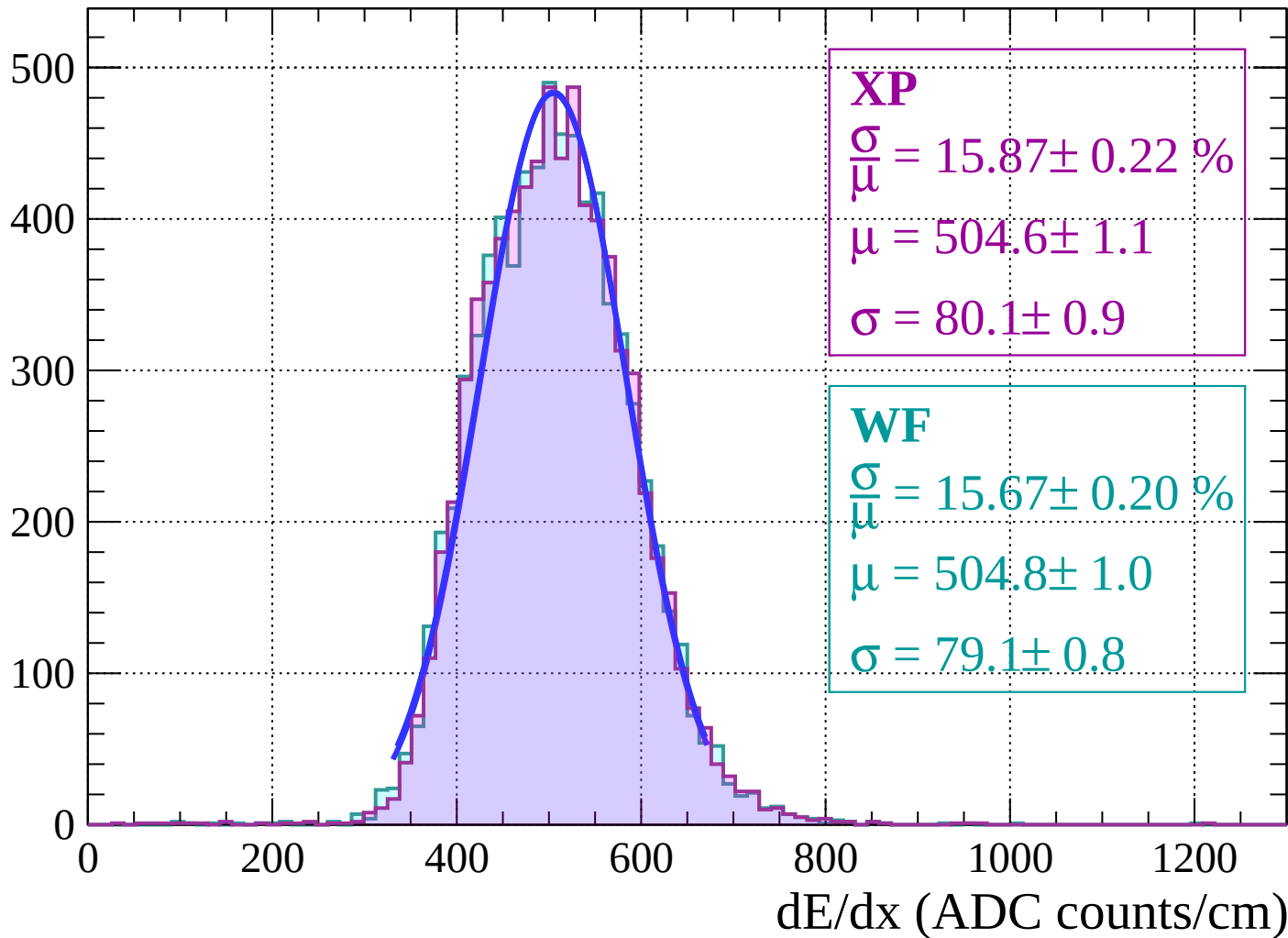
dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

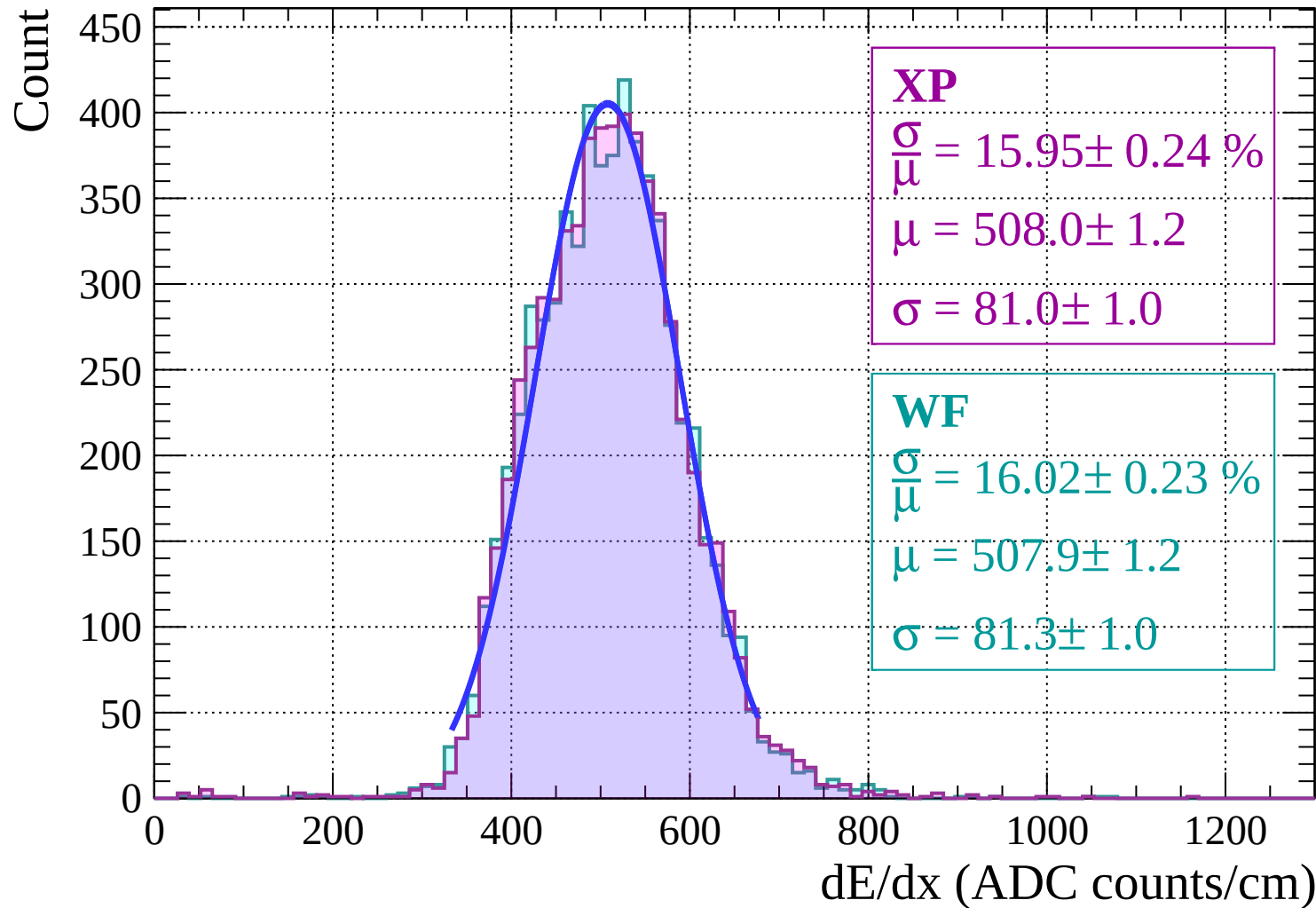


Energy loss | $32 < \theta < 34$

Count

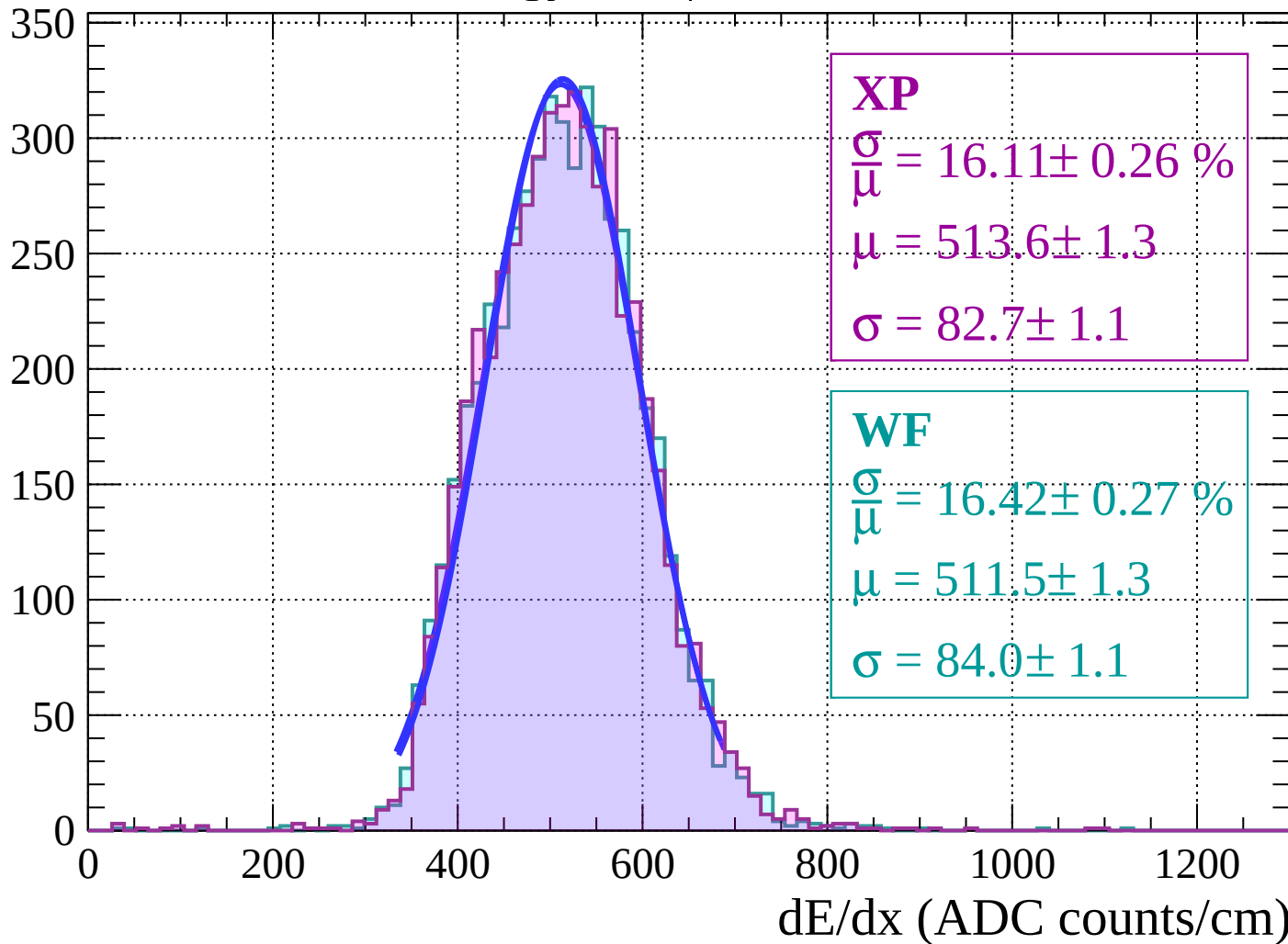


Energy loss | $34 < \theta < 36$



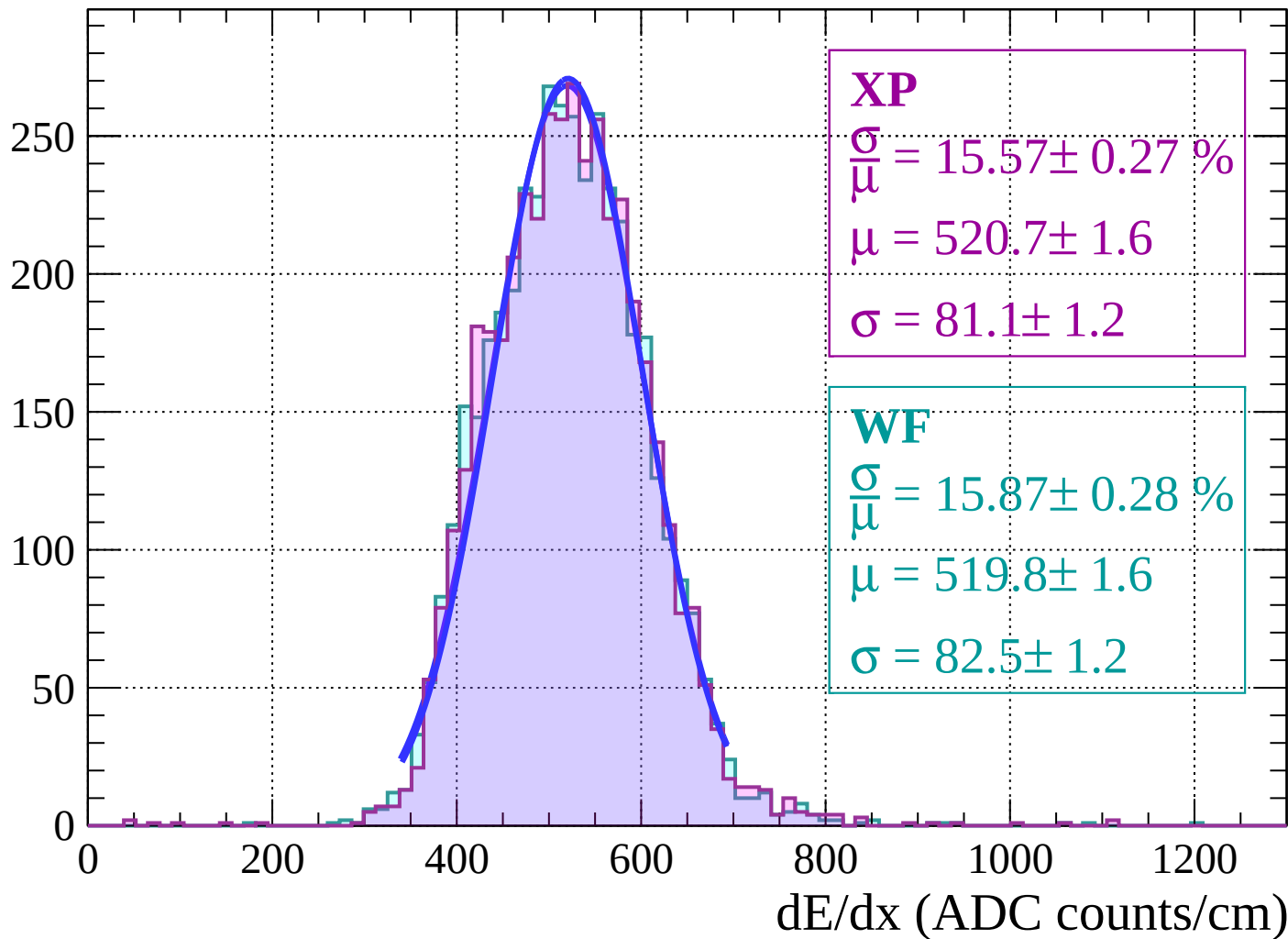
Energy loss | $36 < \theta < 38$

Count



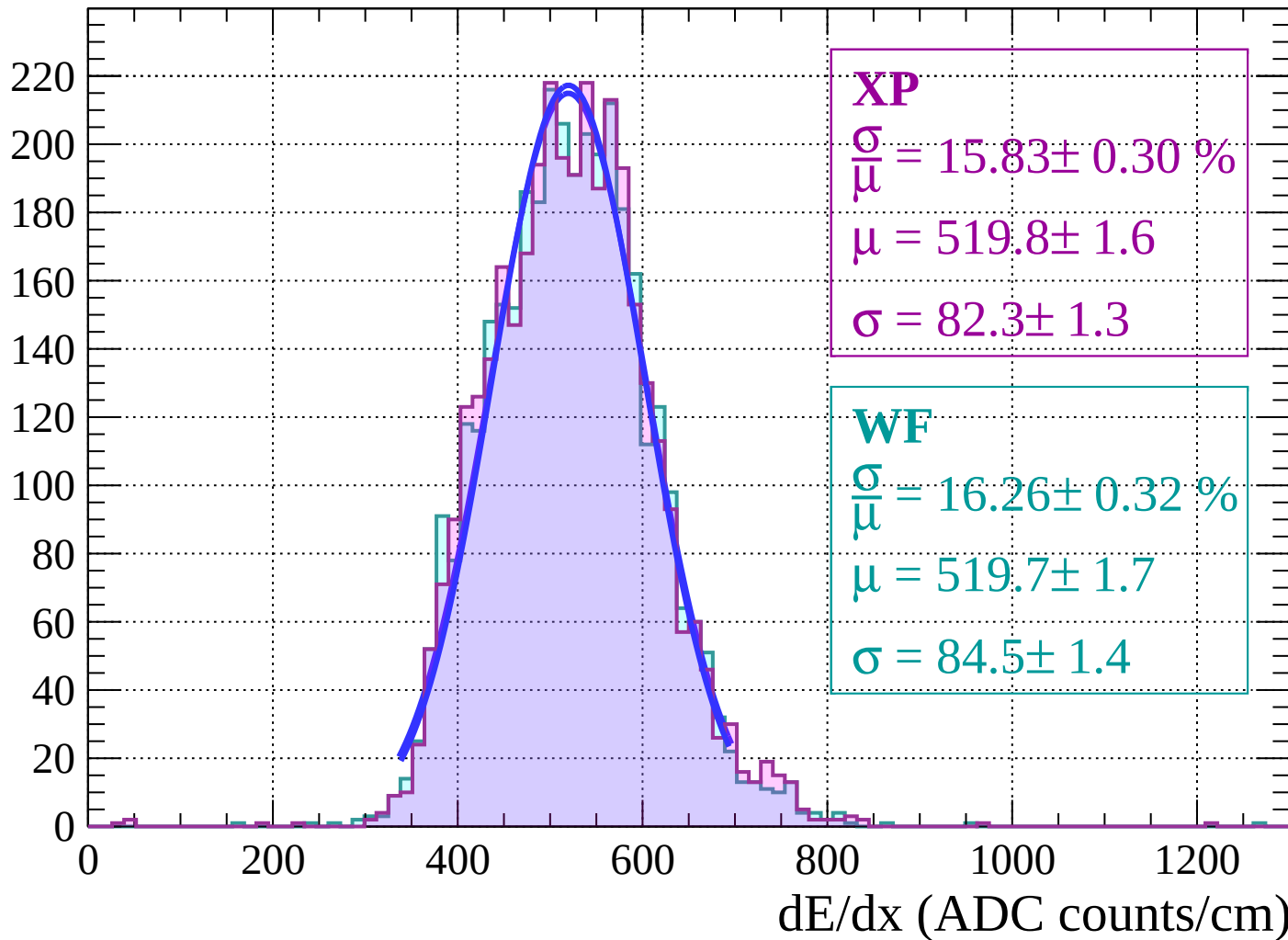
Energy loss | $38 < \theta < 40$

Count



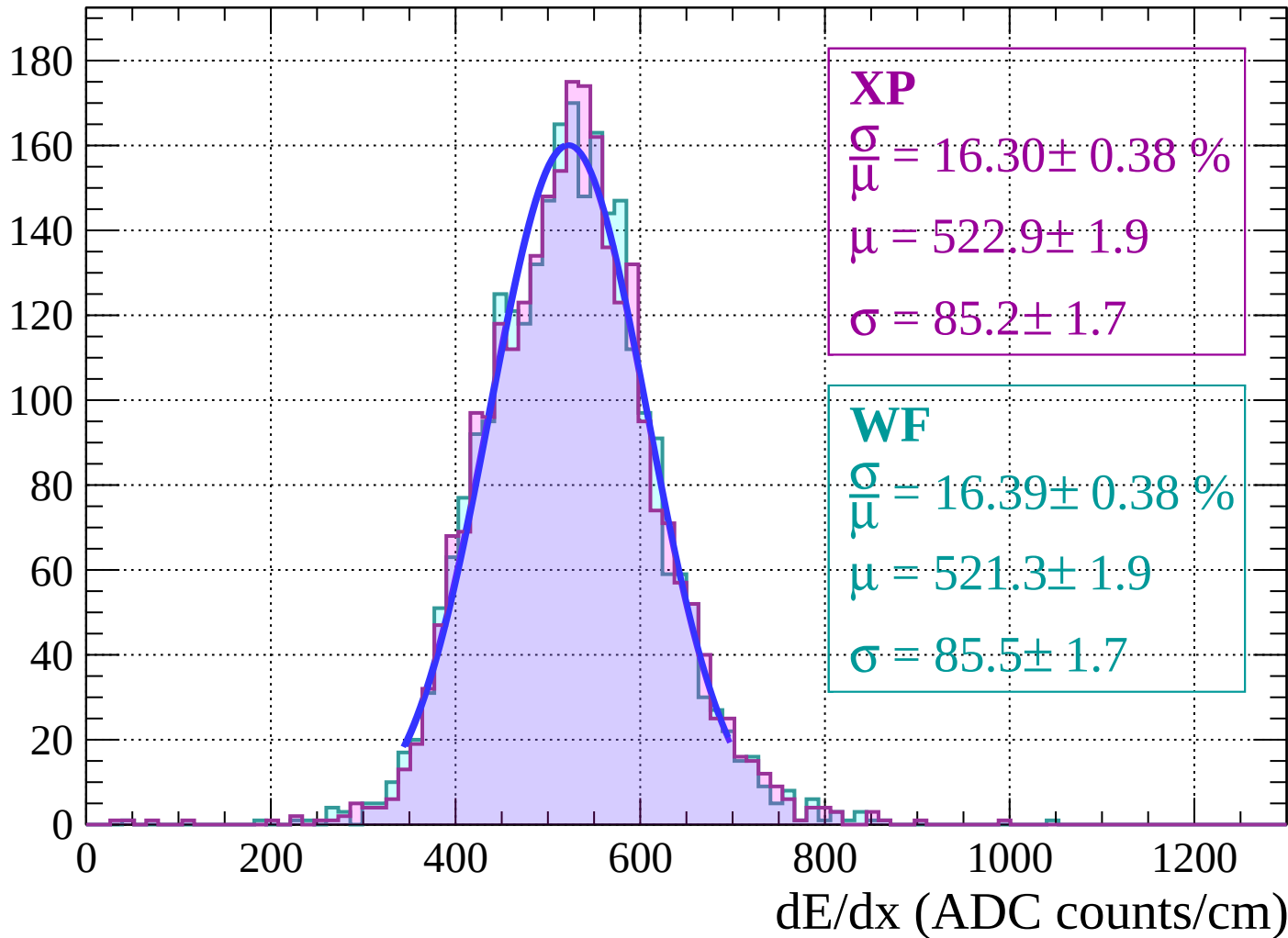
Energy loss | $40 < \theta < 42$

Count

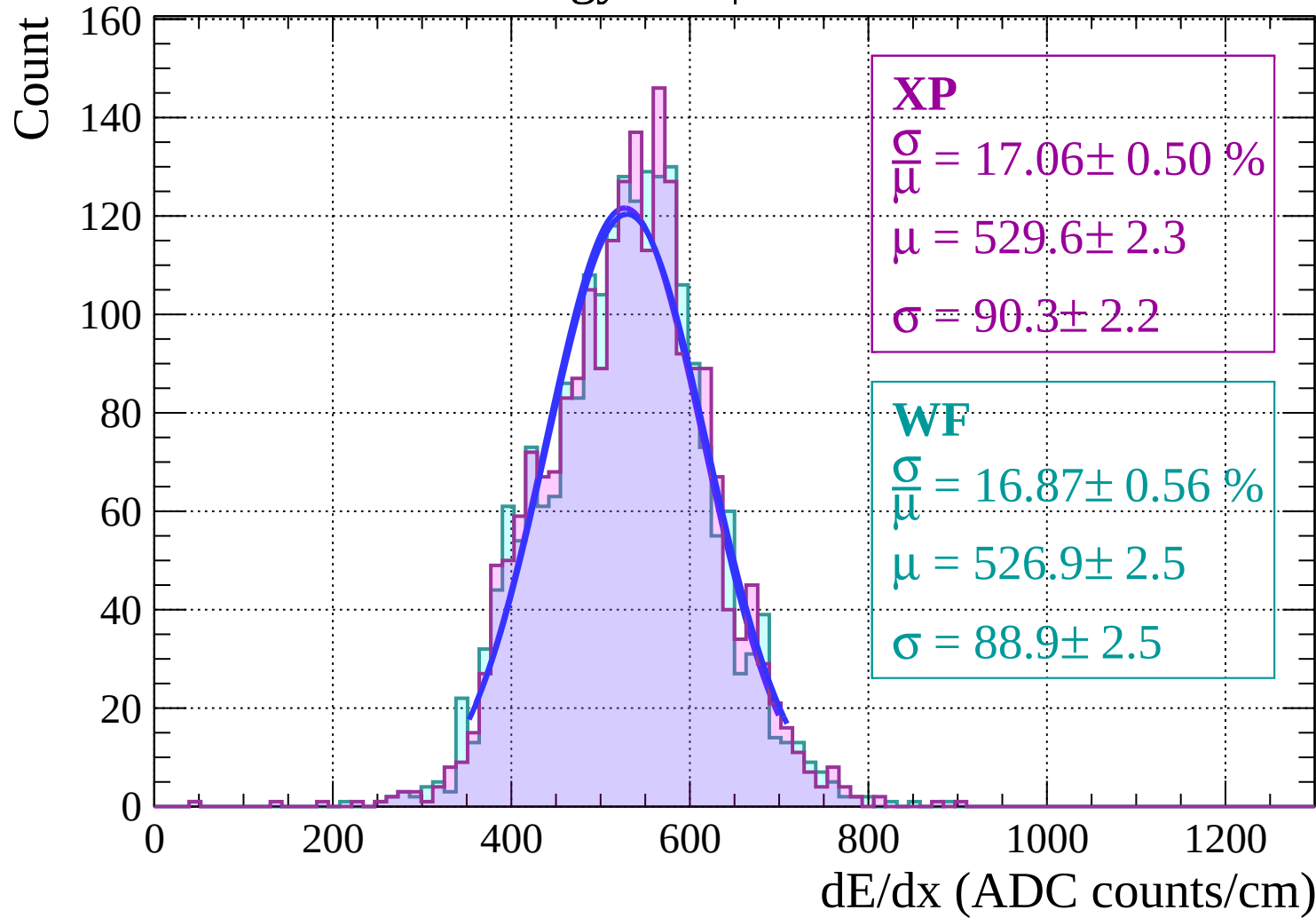


Energy loss | $42 < \theta < 44$

Count

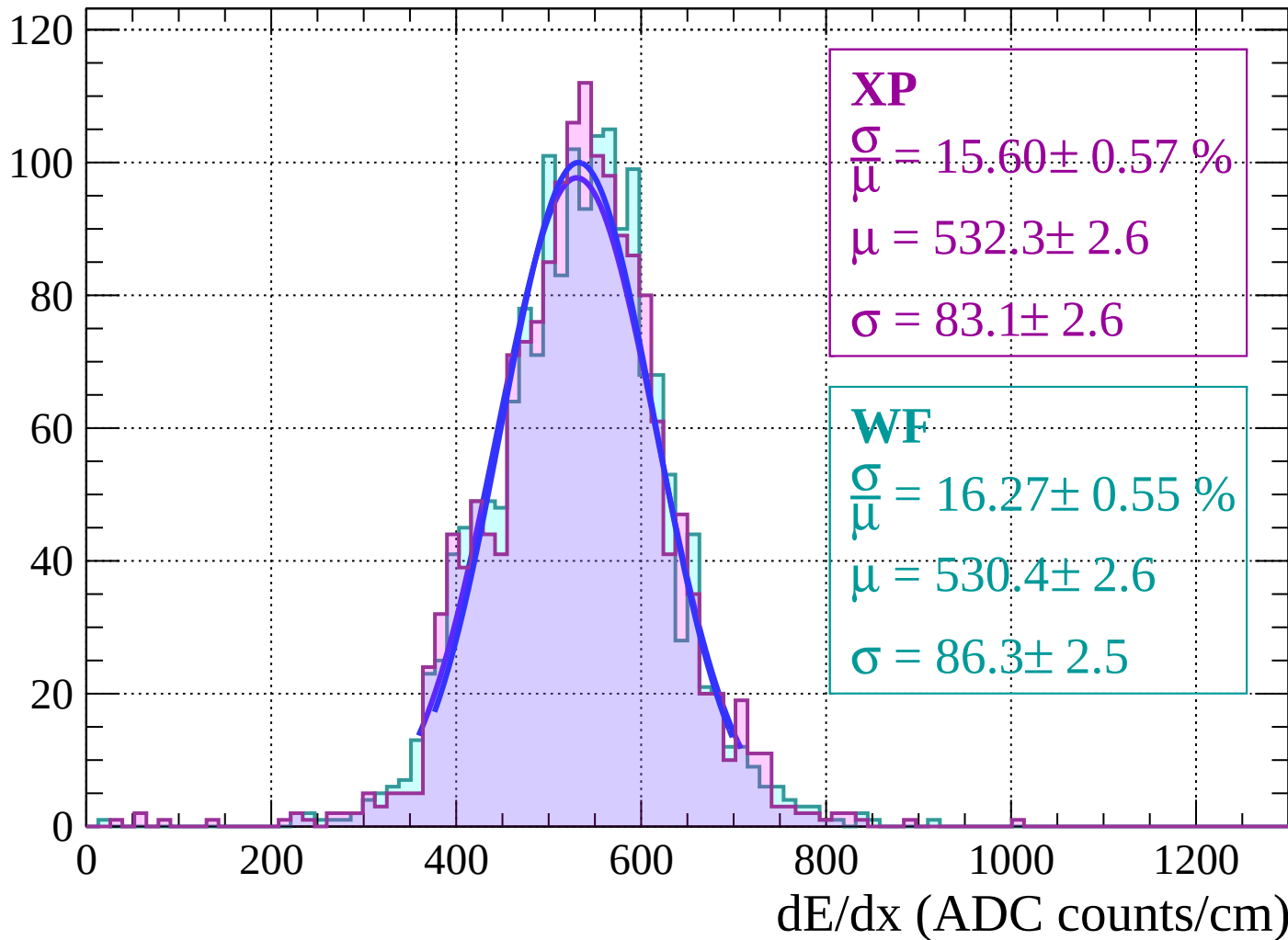


Energy loss | $44 < \theta < 46$



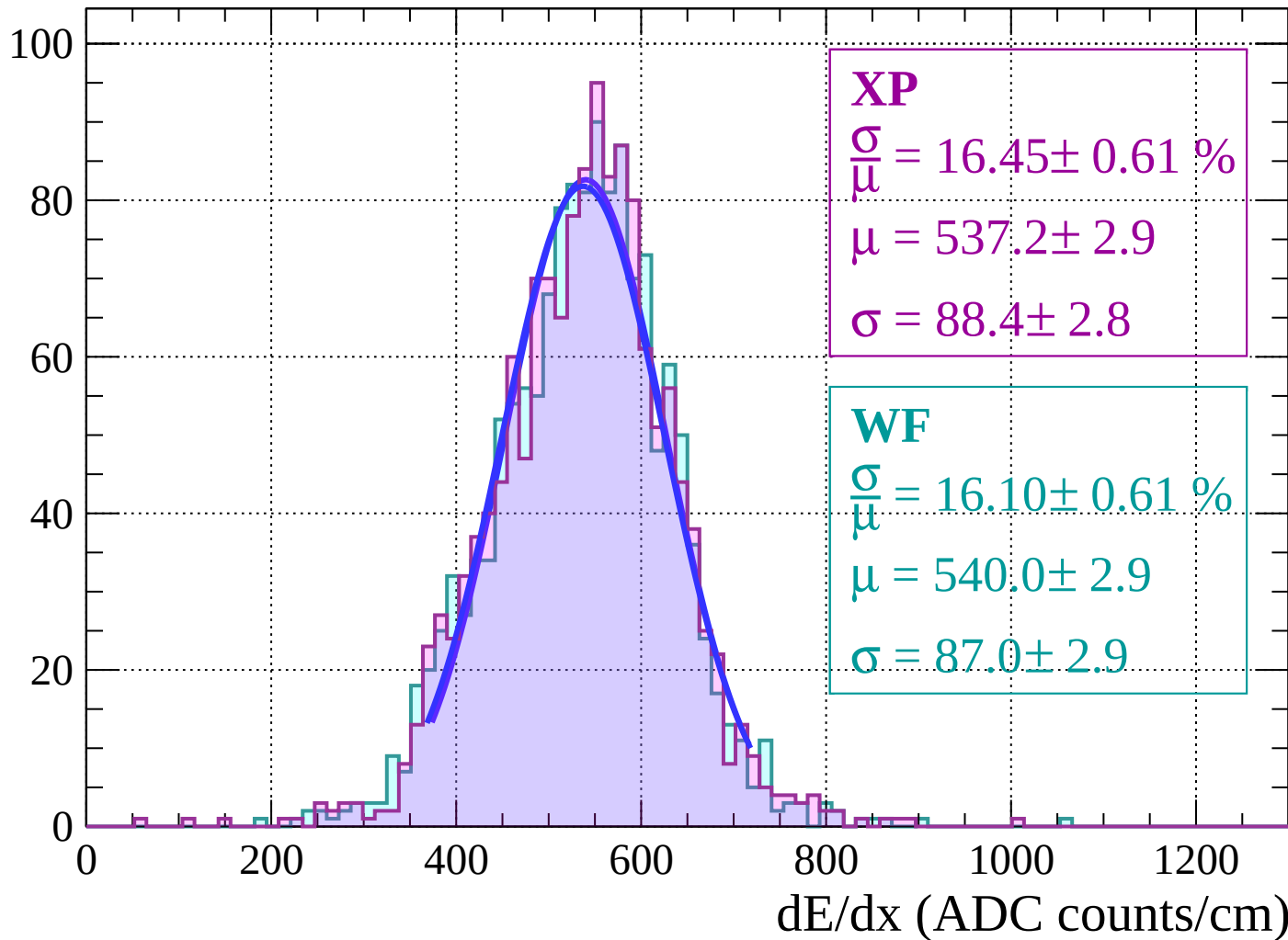
Energy loss | $46 < \theta < 48$

Count



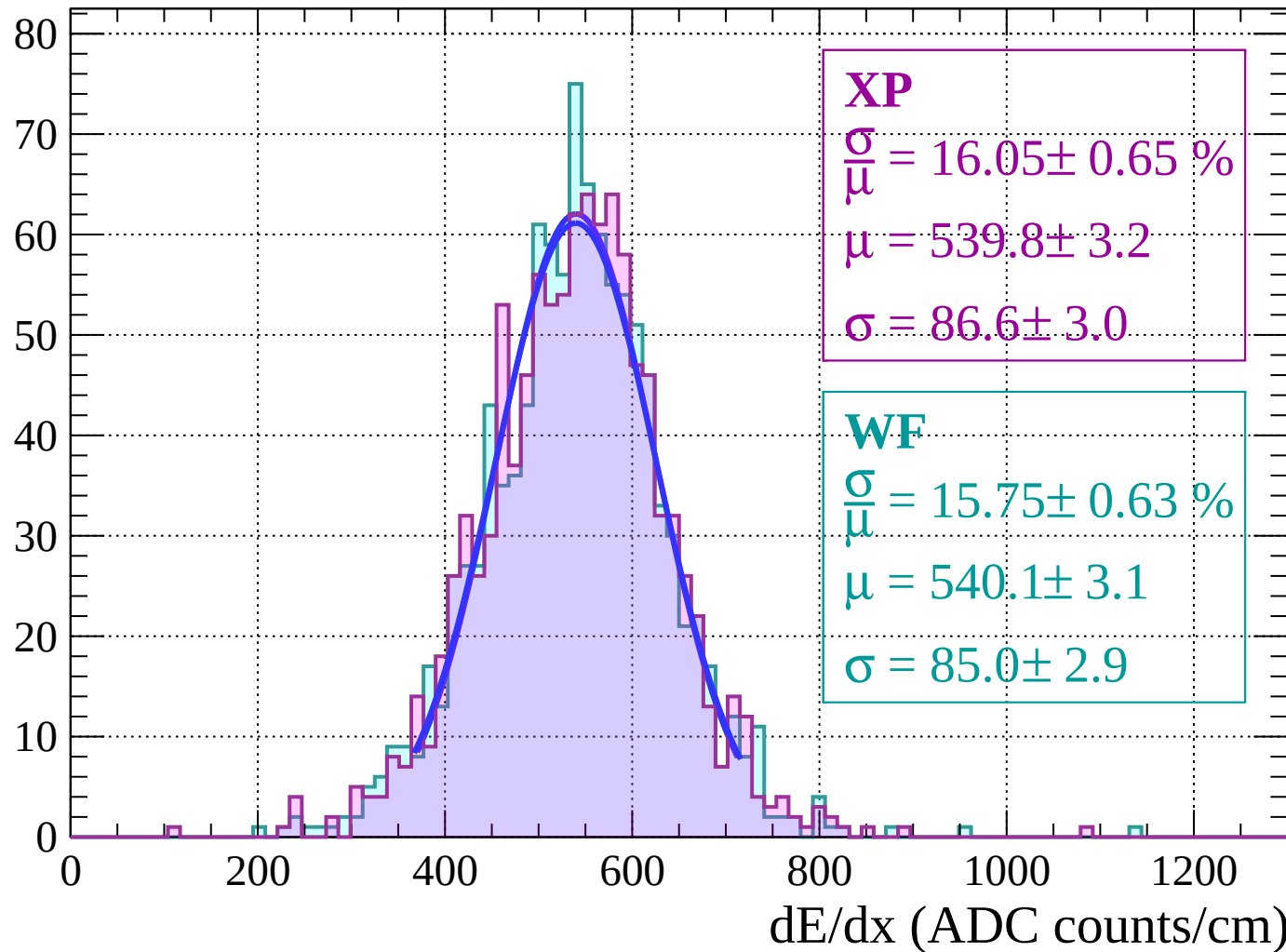
Energy loss | $48 < \theta < 50$

Count



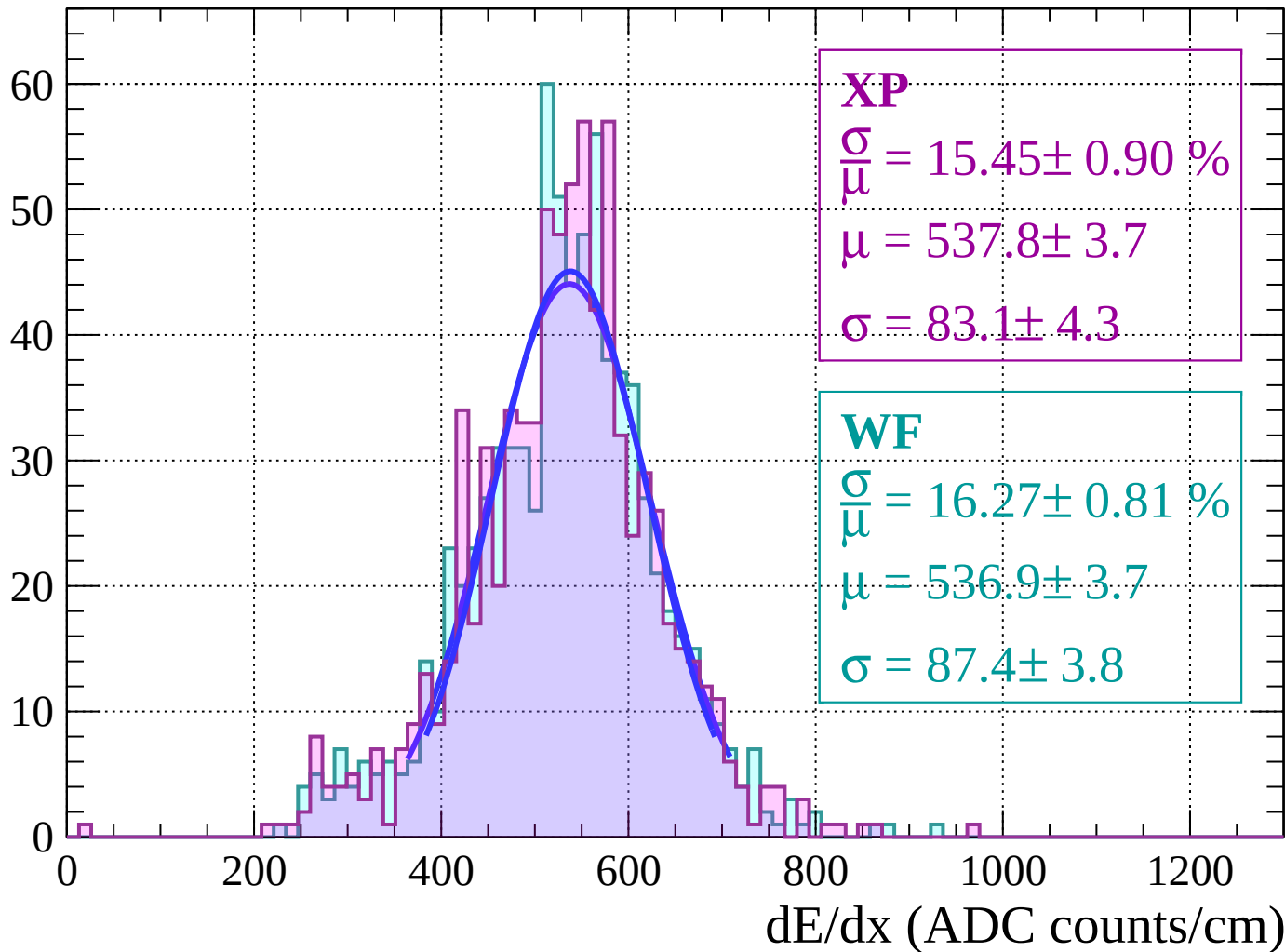
Energy loss | $50 < \theta < 52$

Count



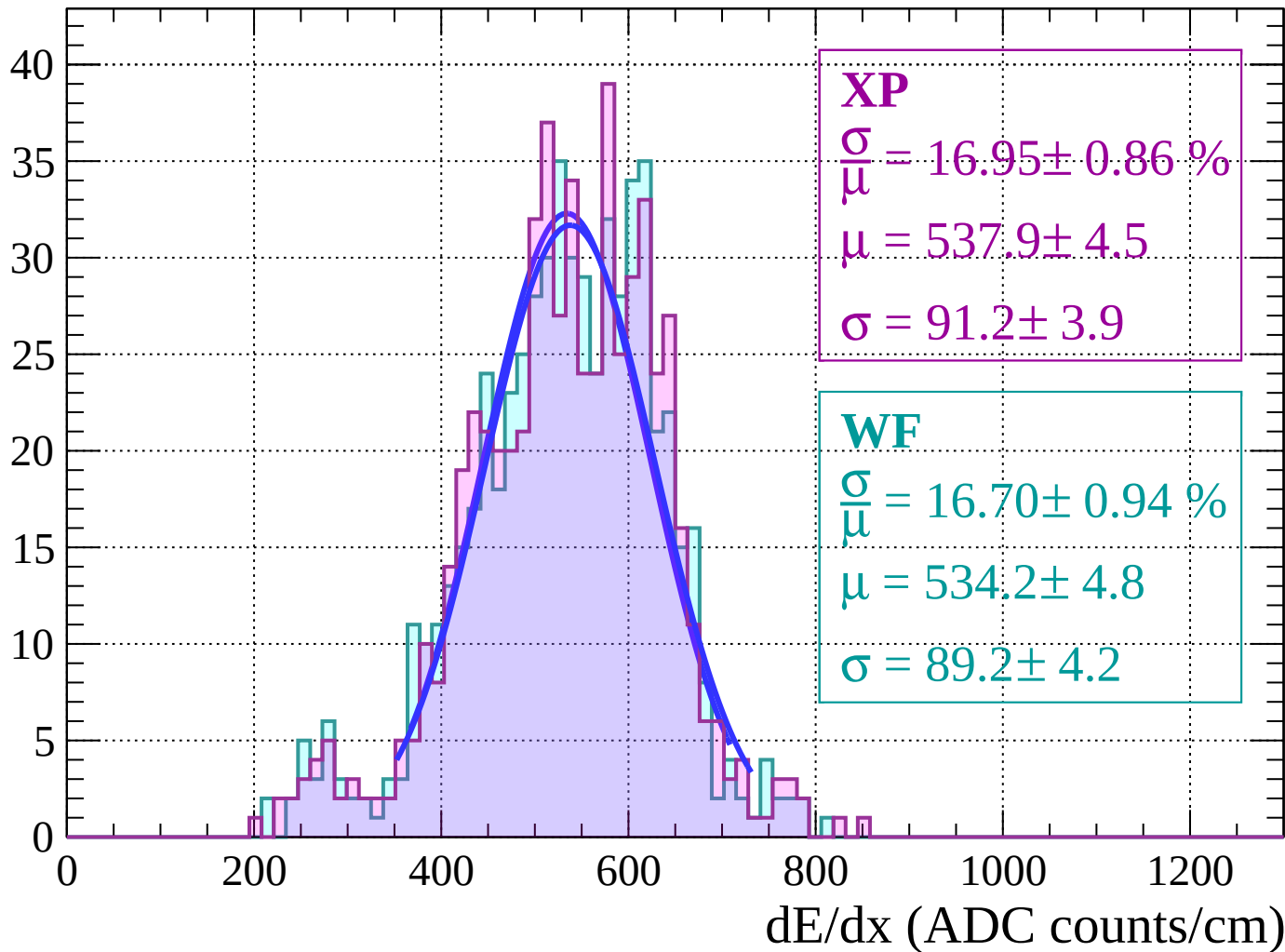
Energy loss | $52 < \theta < 54$

Count



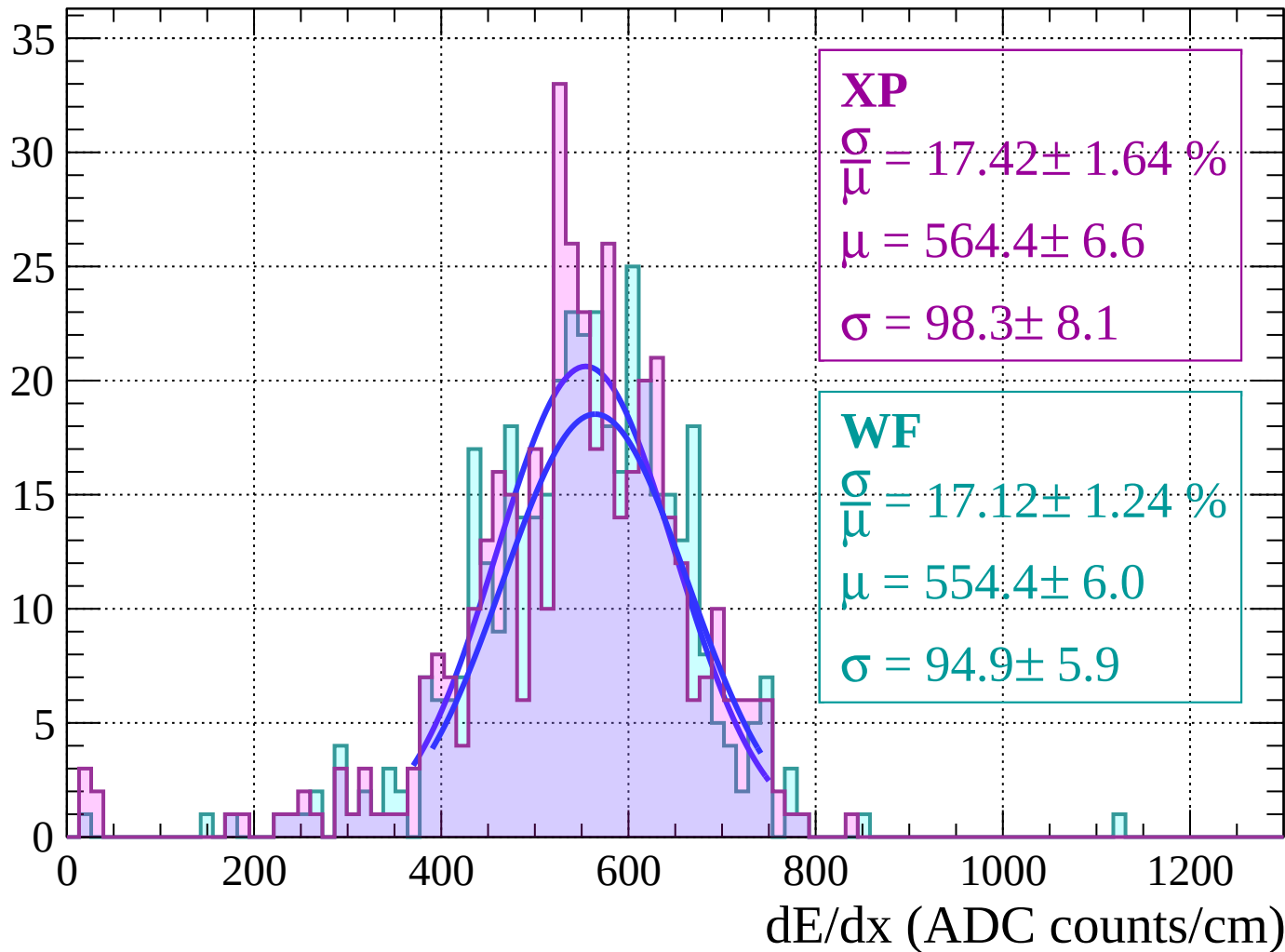
Energy loss | $54 < \theta < 56$

Count



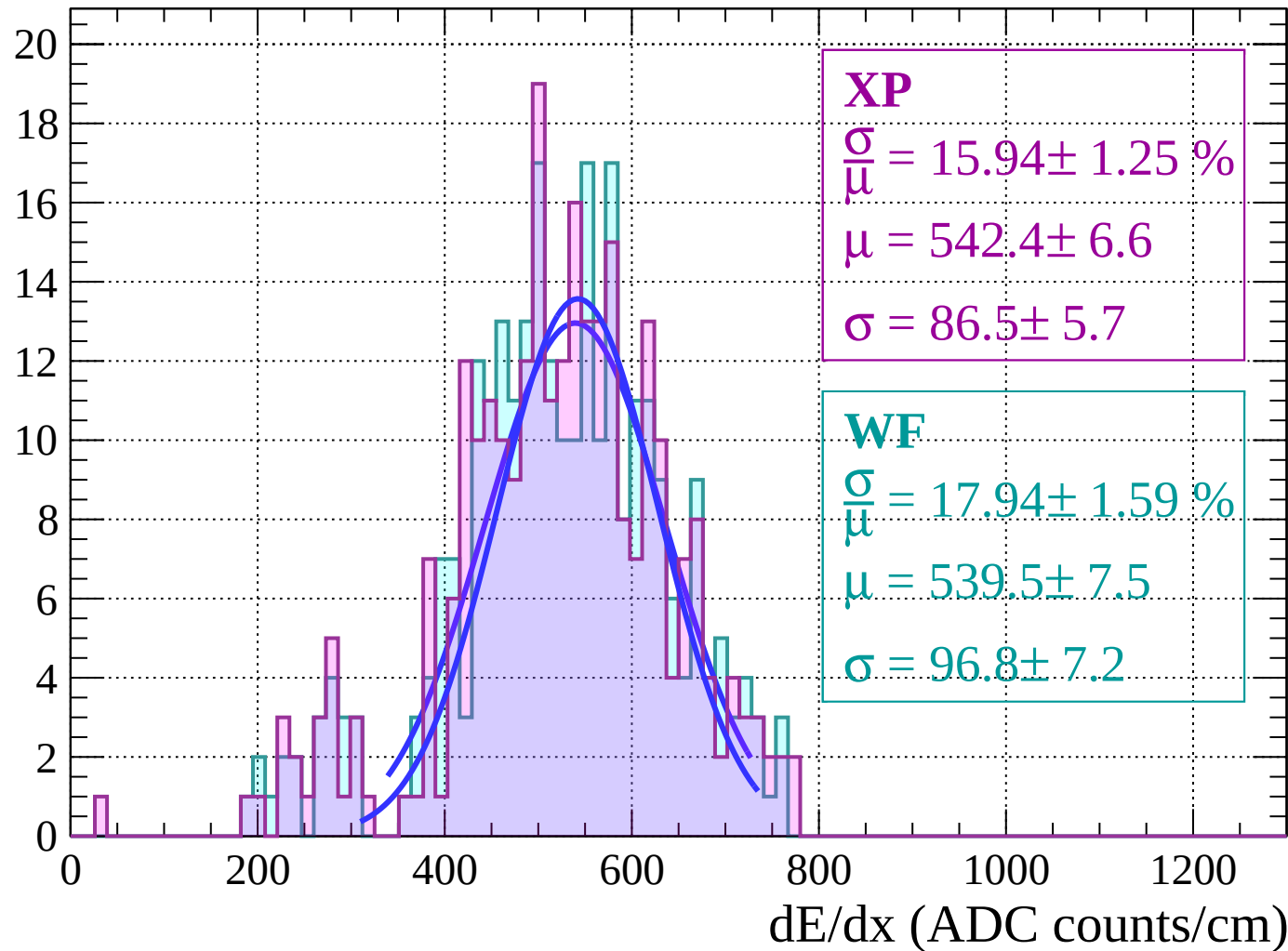
Energy loss | $56 < \theta < 58$

Count



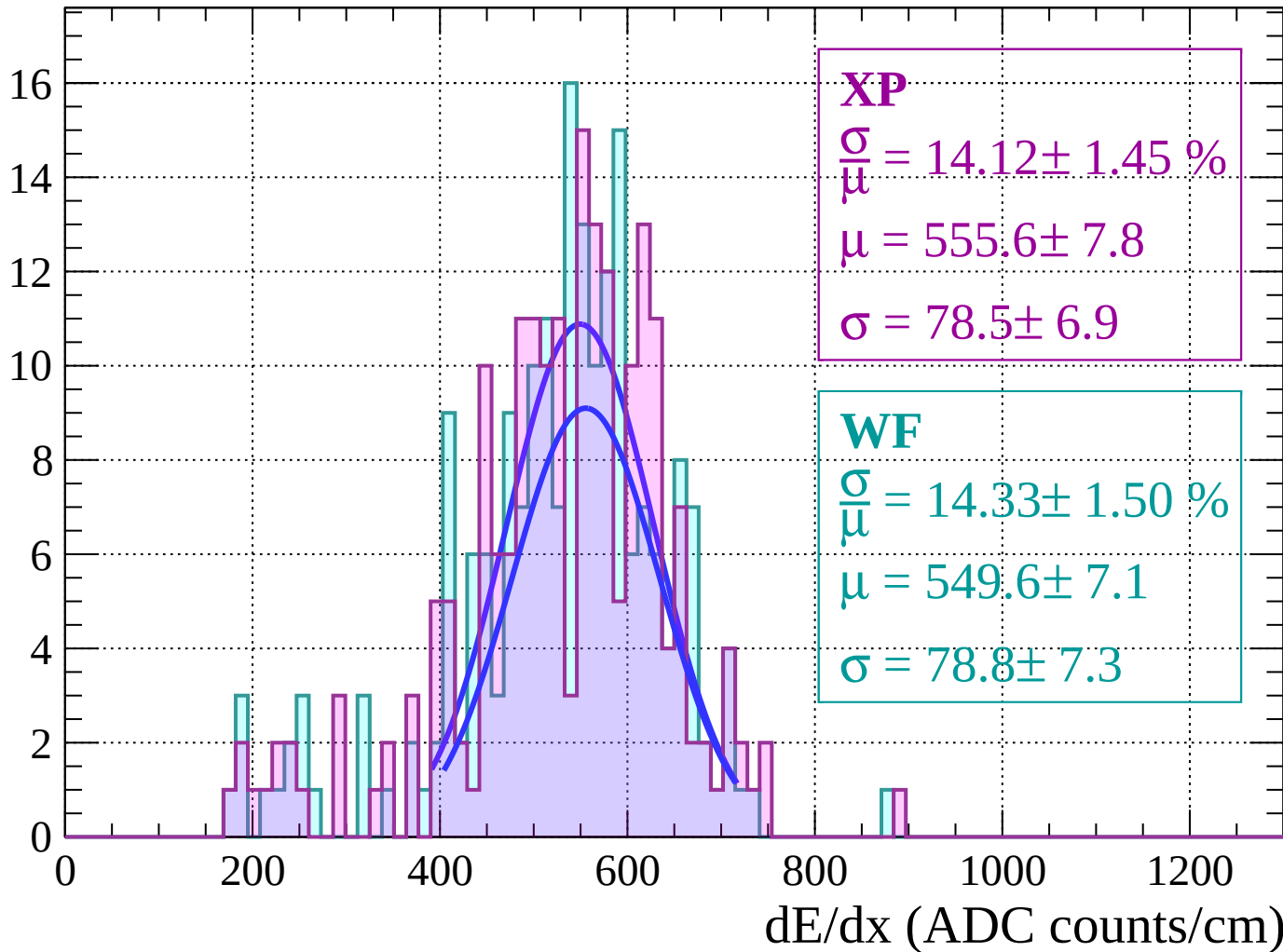
Energy loss | $58 < \theta < 60$

Count



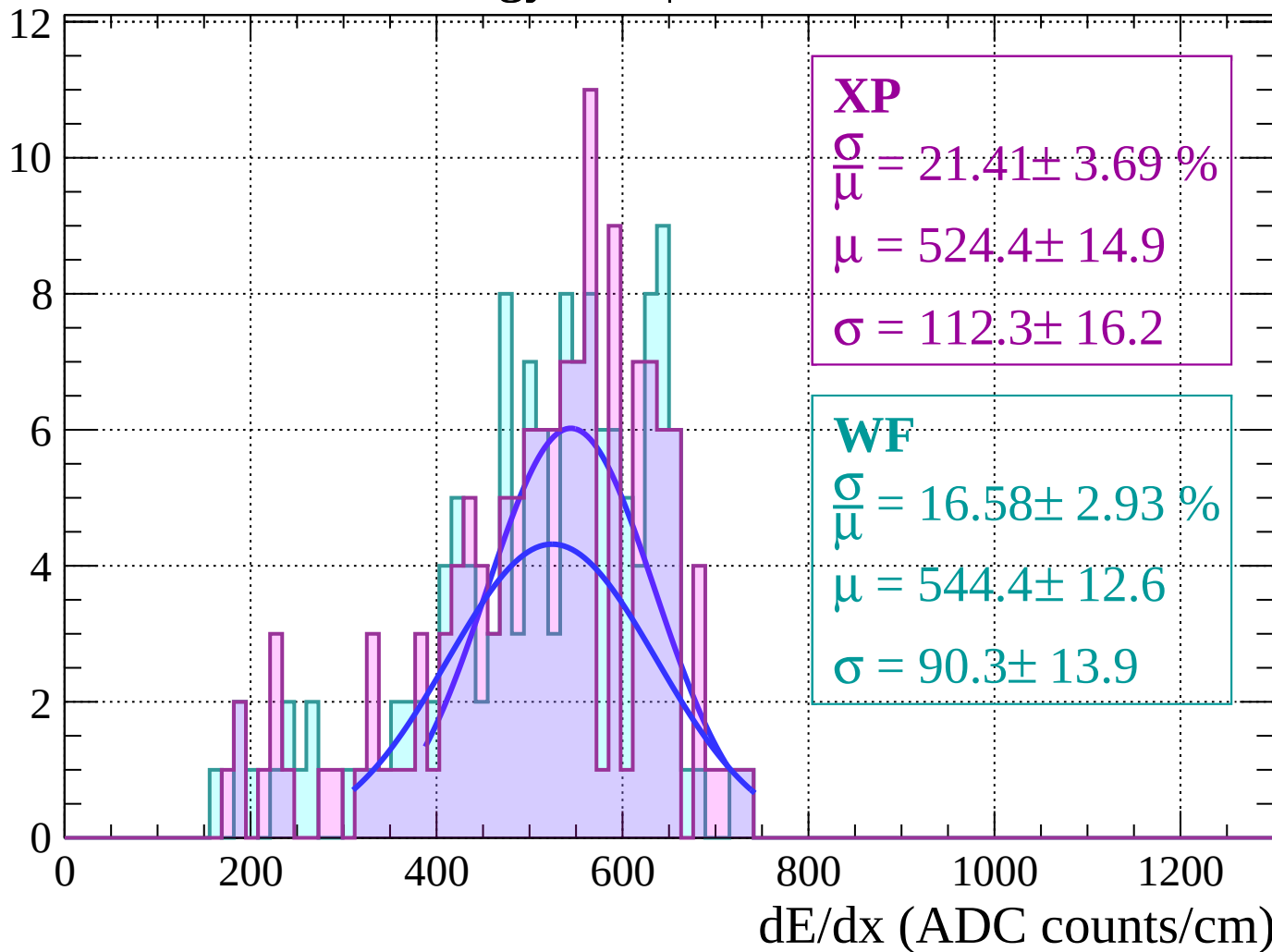
Energy loss | $60 < \theta < 62$

Count



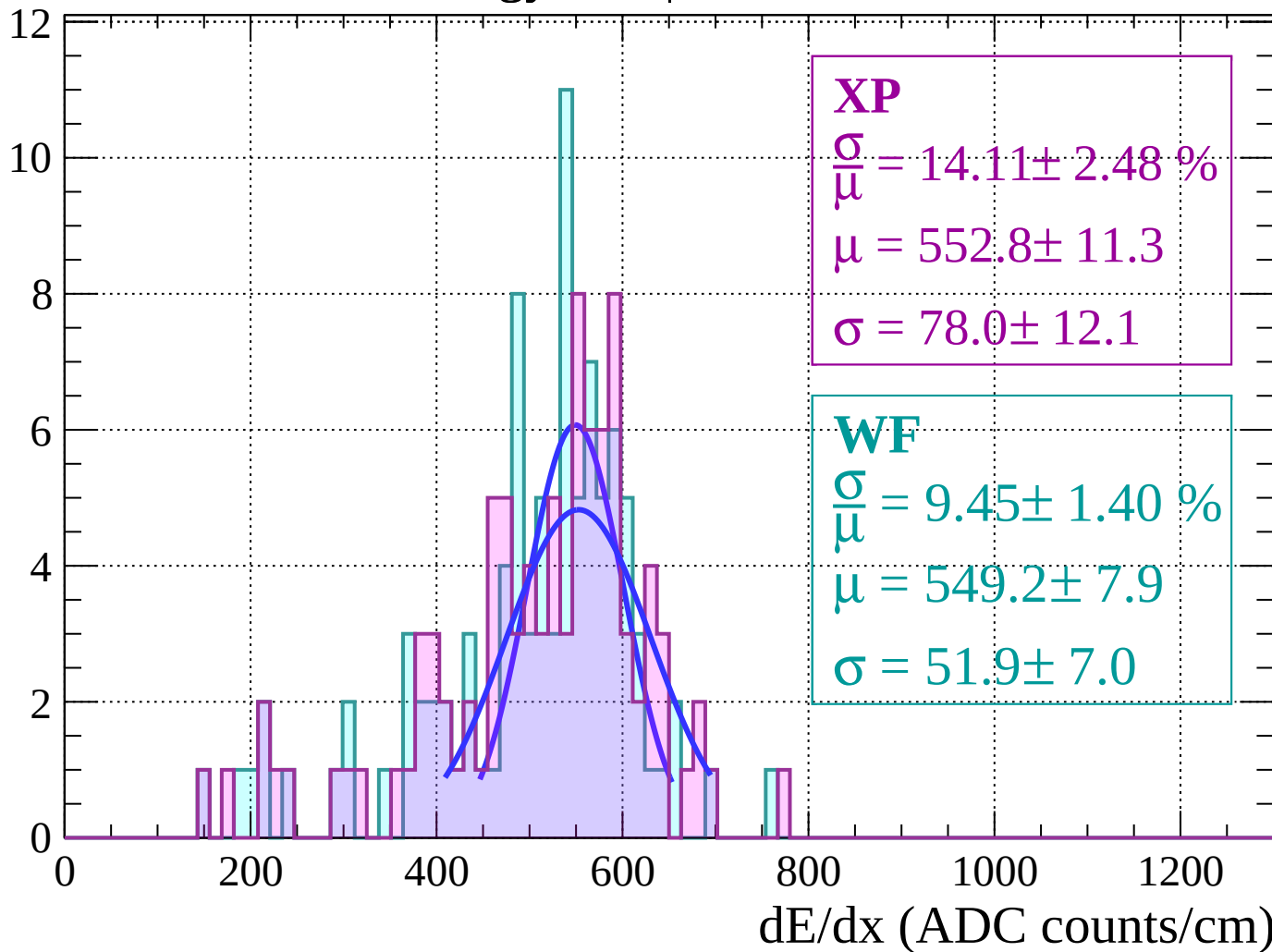
Energy loss | $62 < \theta < 64$

Count



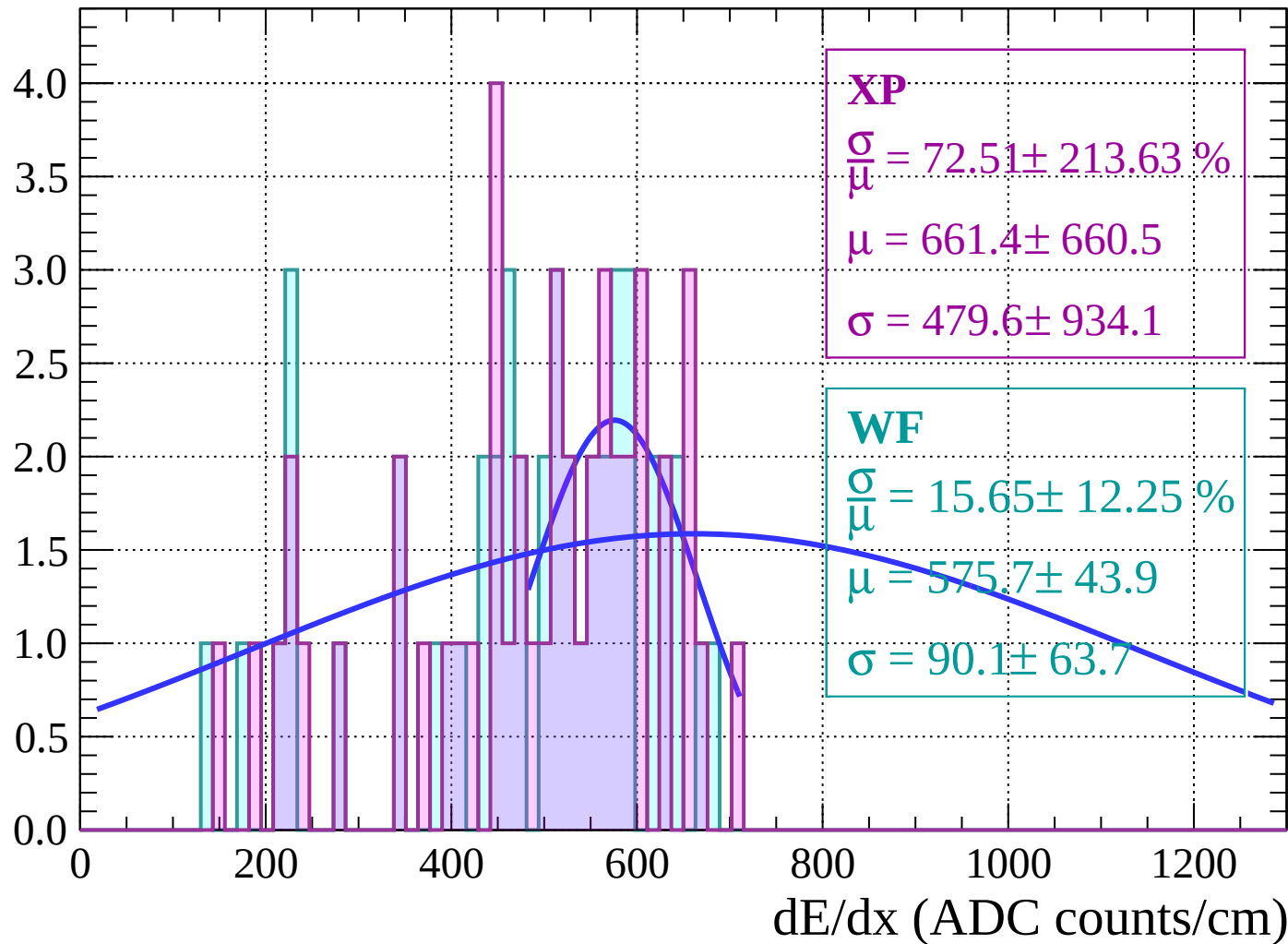
Energy loss | $64 < \theta < 66$

Count



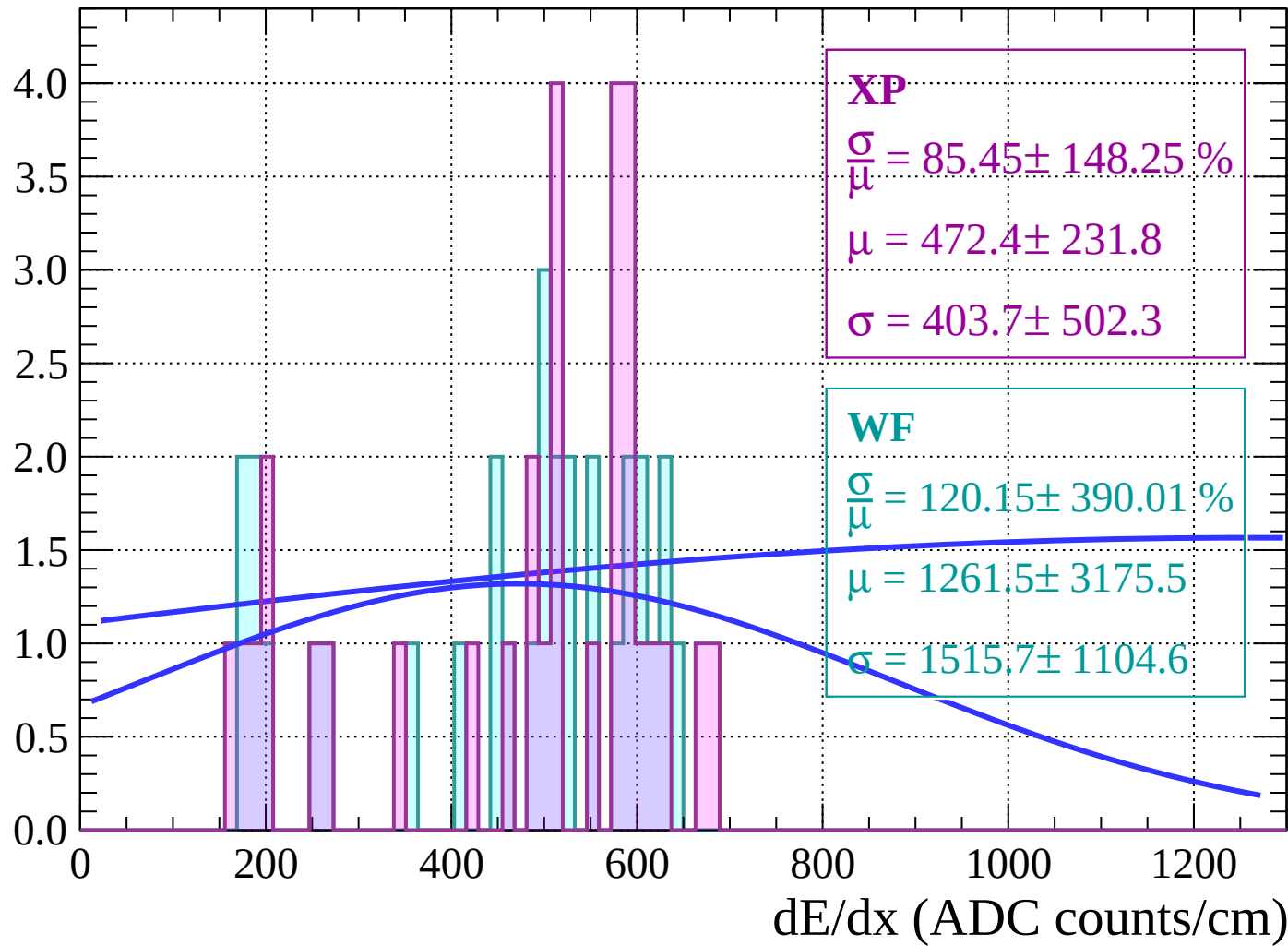
Energy loss | $66 < \theta < 68$

Count



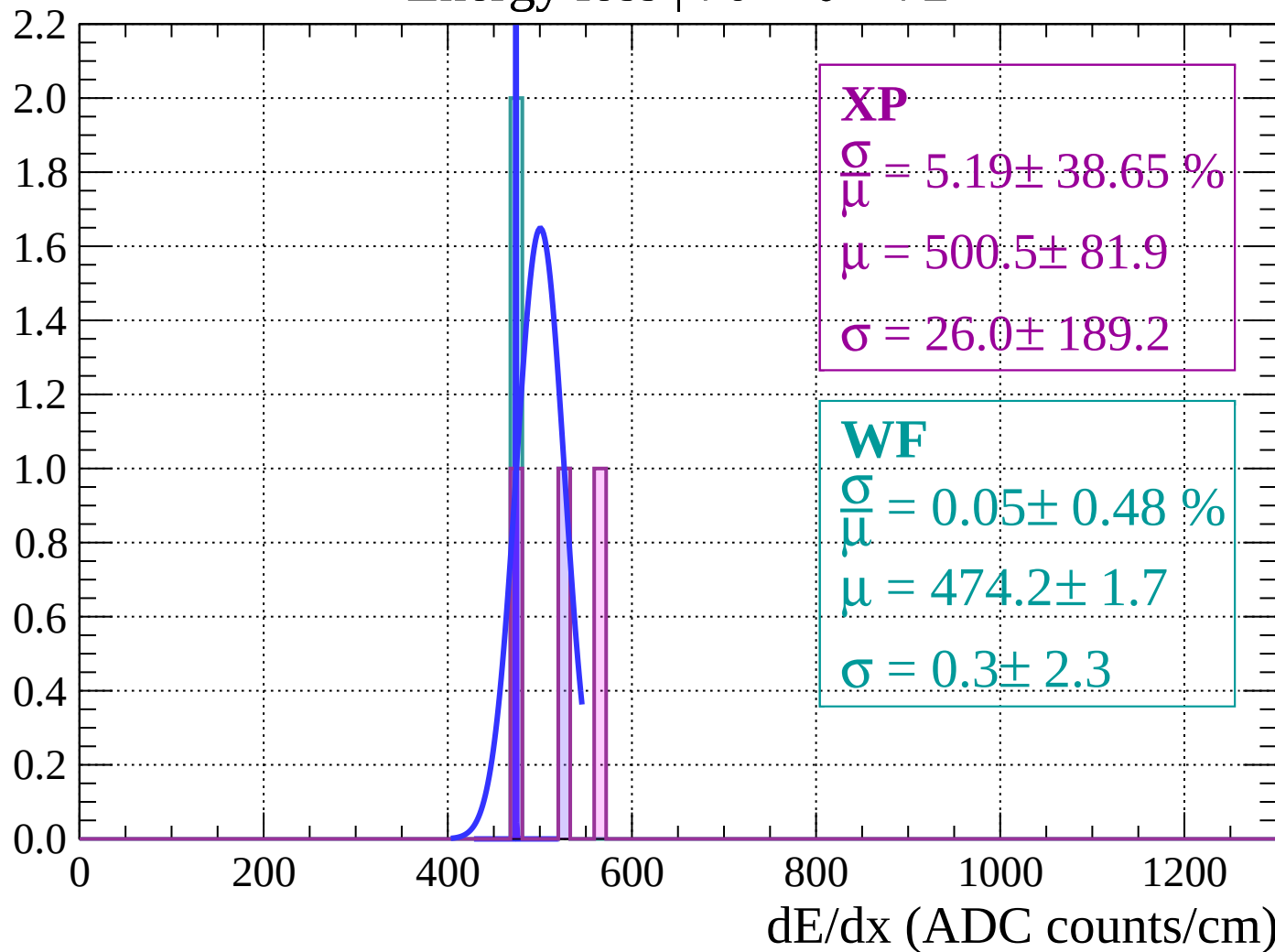
Energy loss | $68 < \theta < 70$

Count



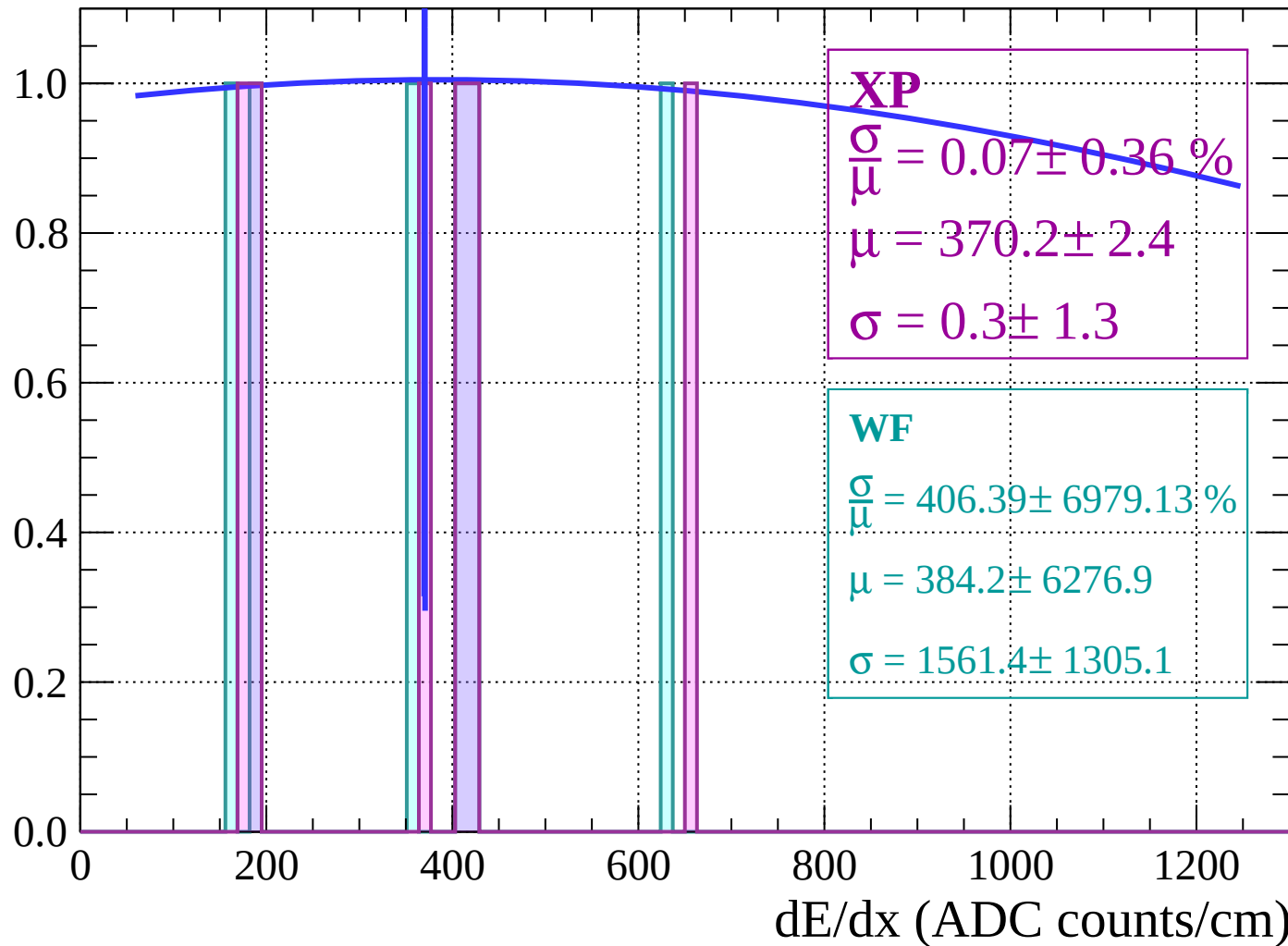
Energy loss | $70 < \theta < 72$

Count



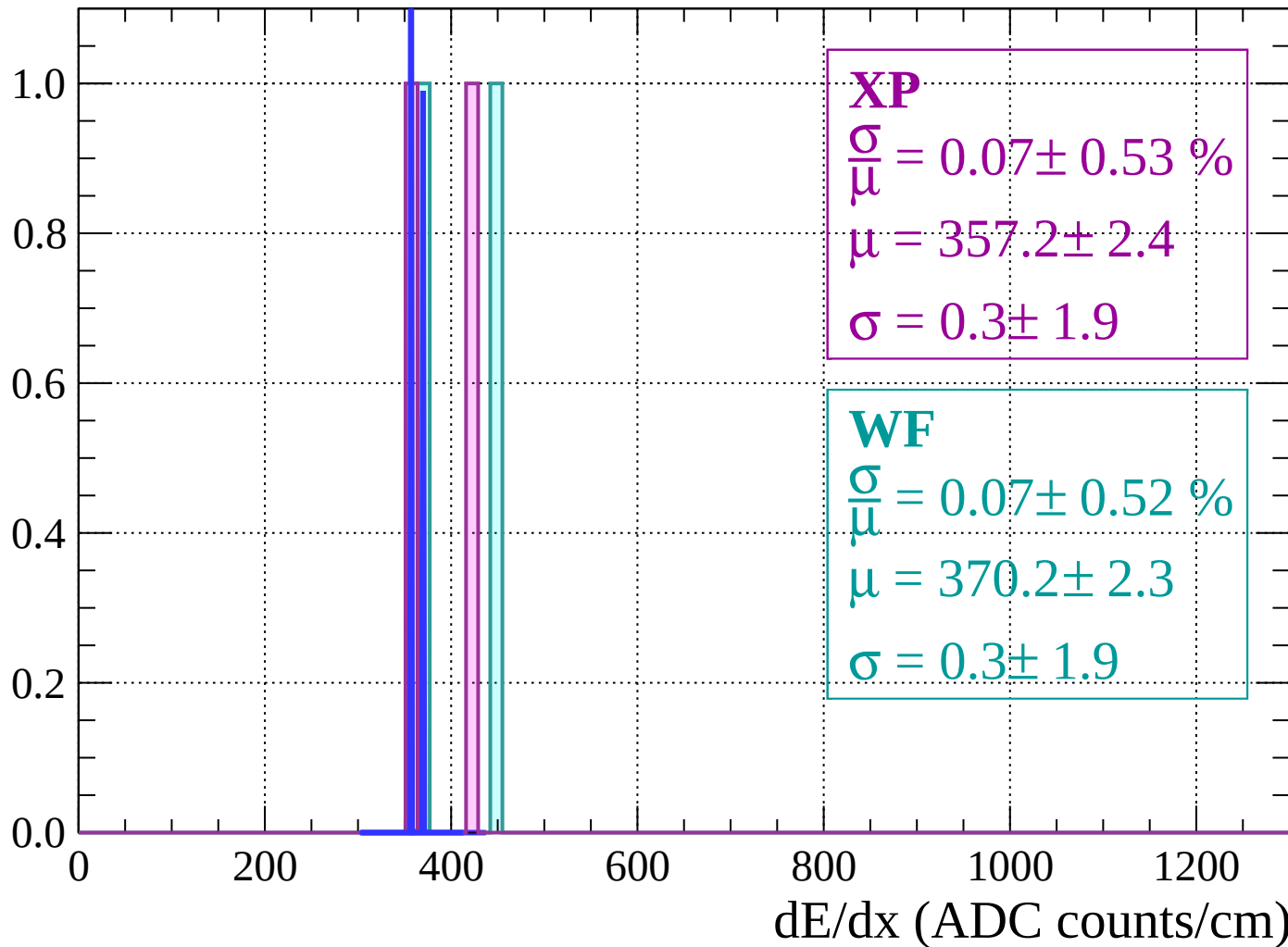
Energy loss | $72 < \theta < 74$

Count



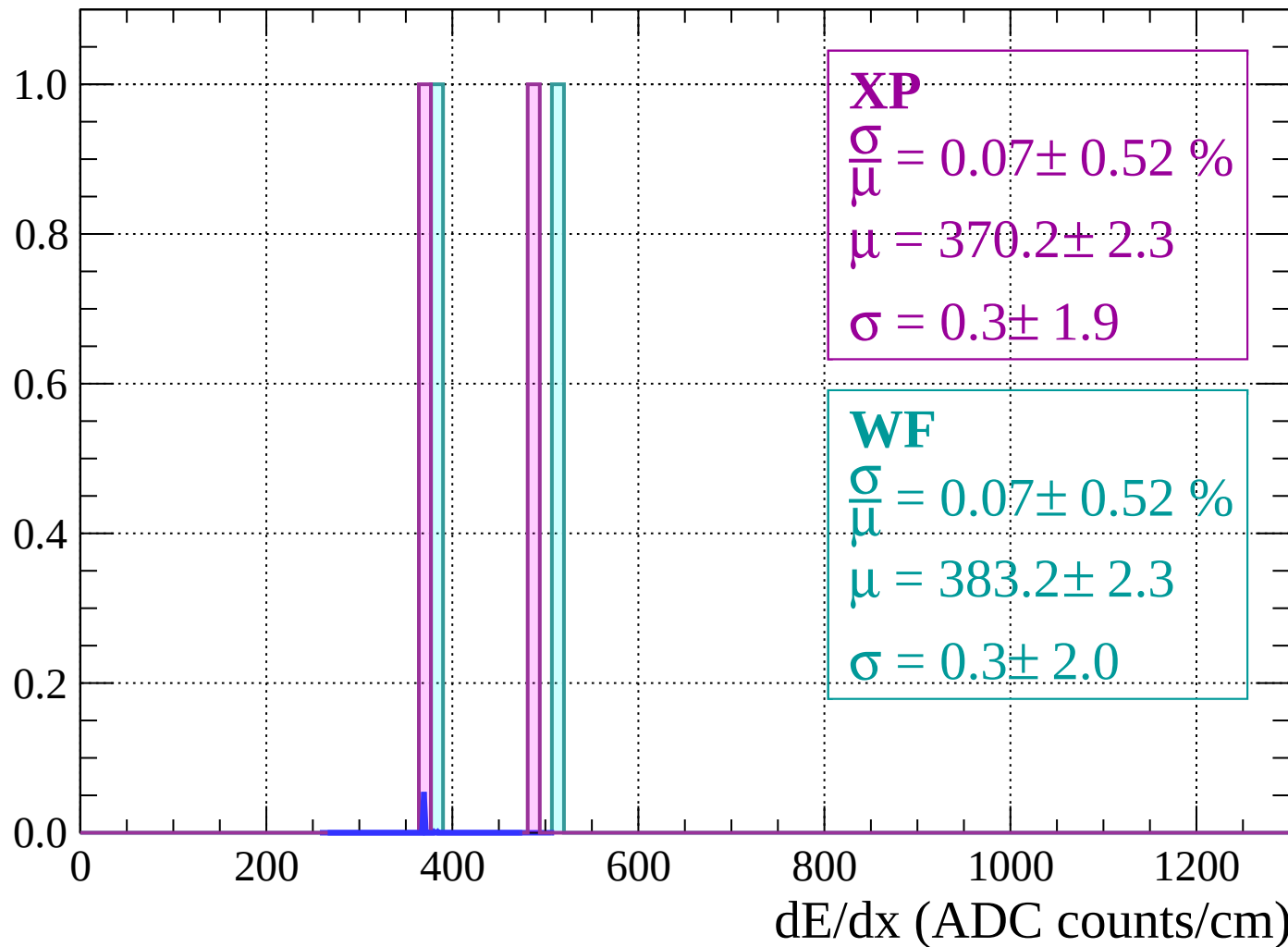
Energy loss | $74 < \theta < 76$

Count



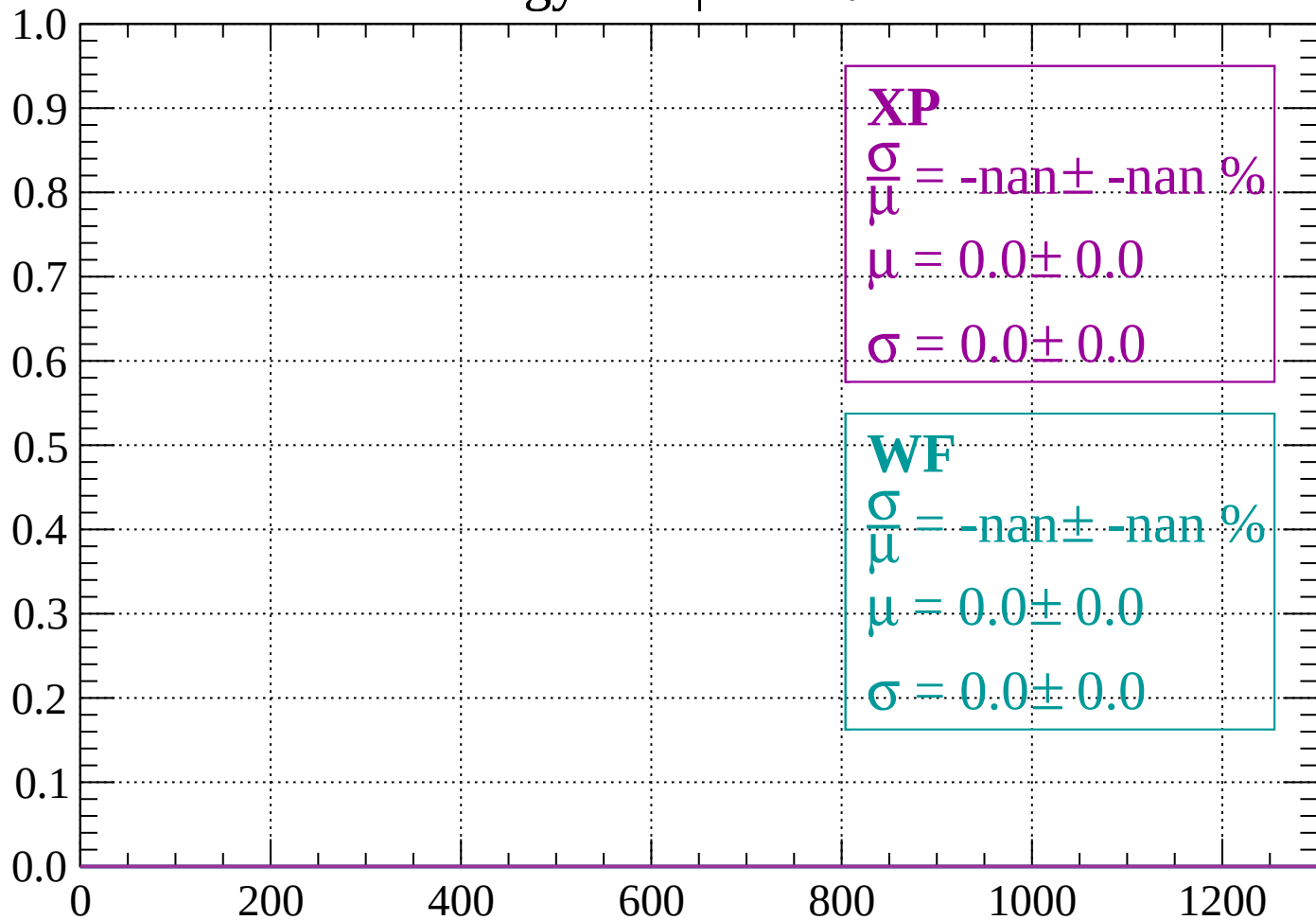
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

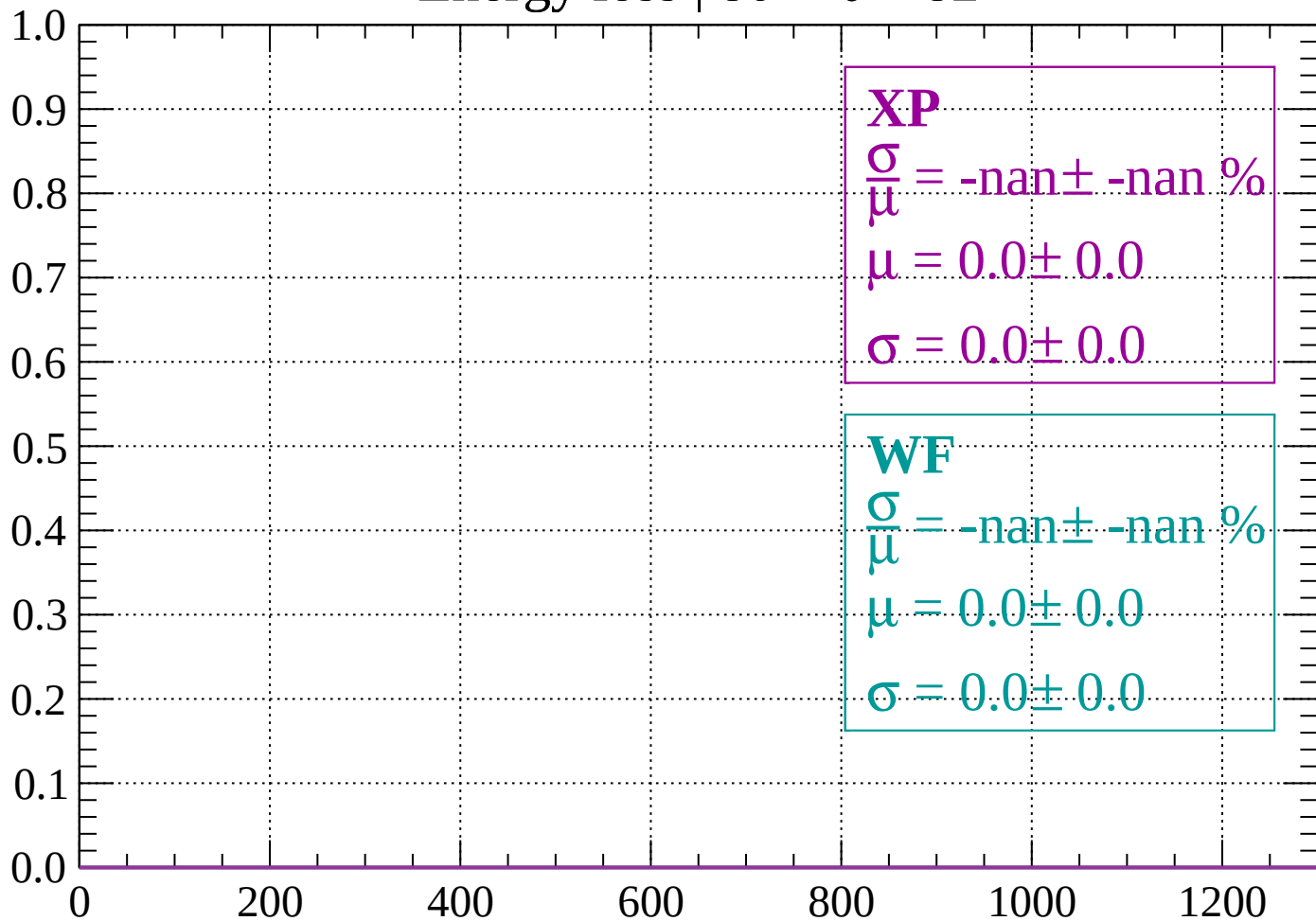
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

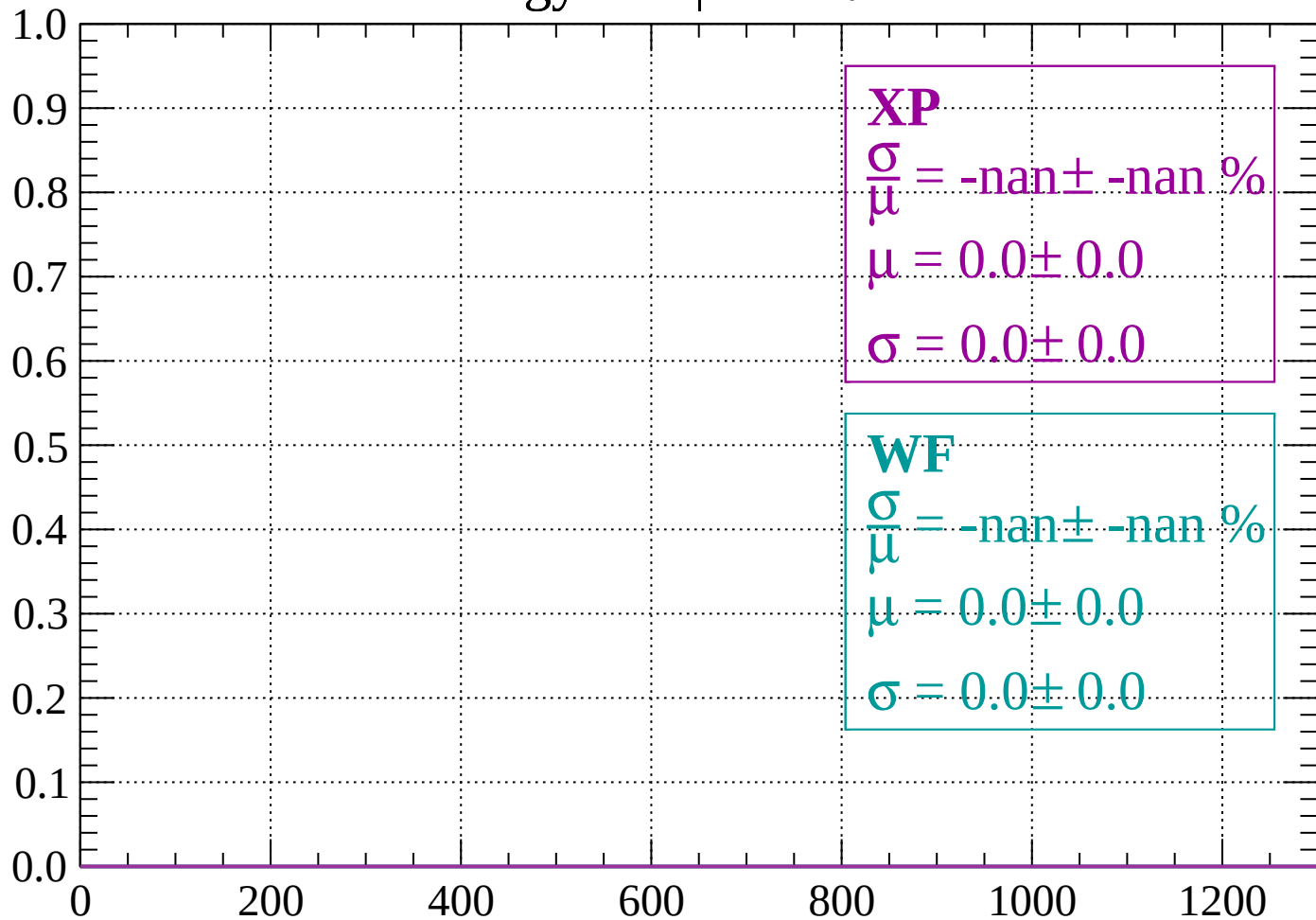
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

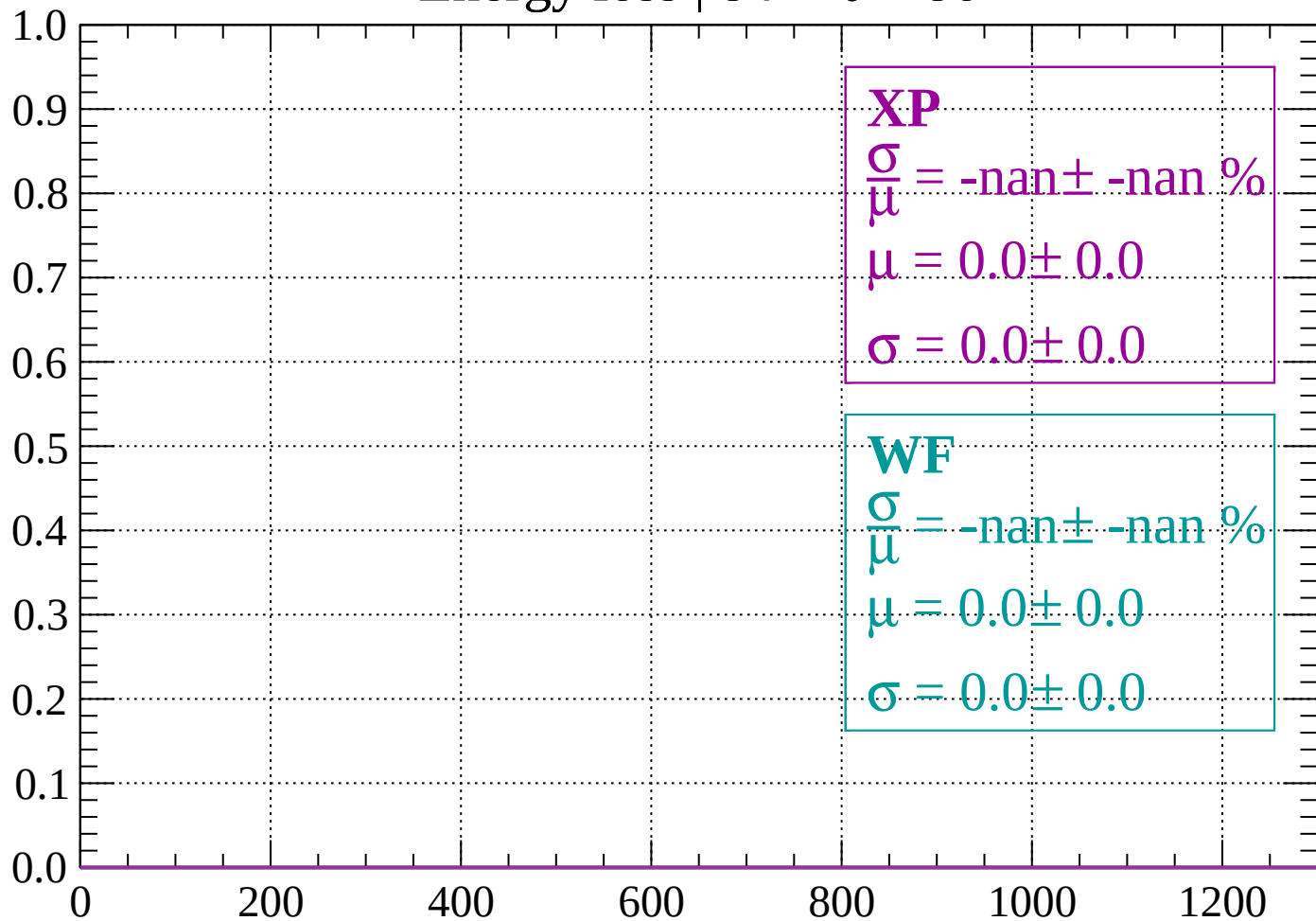
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

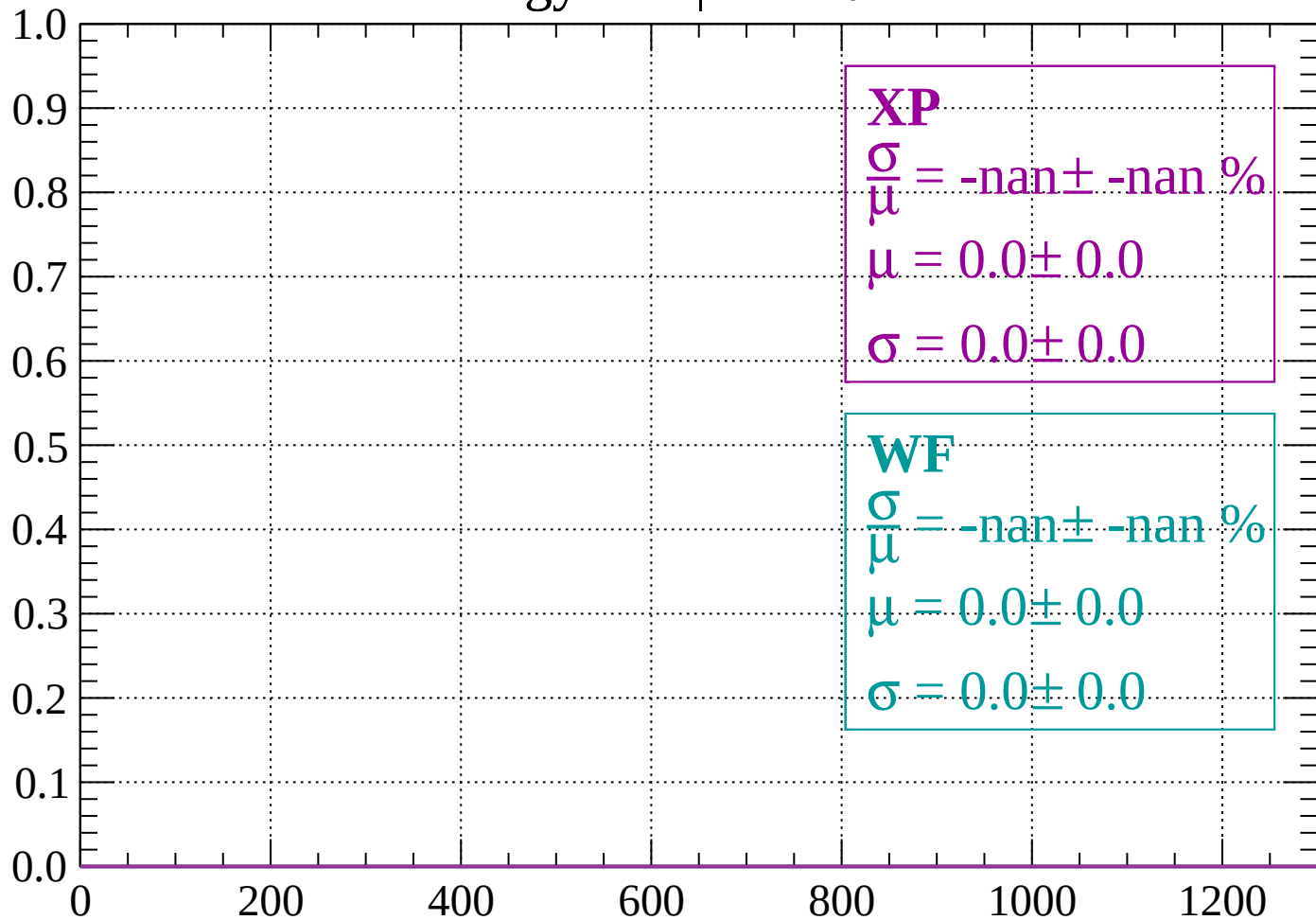
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

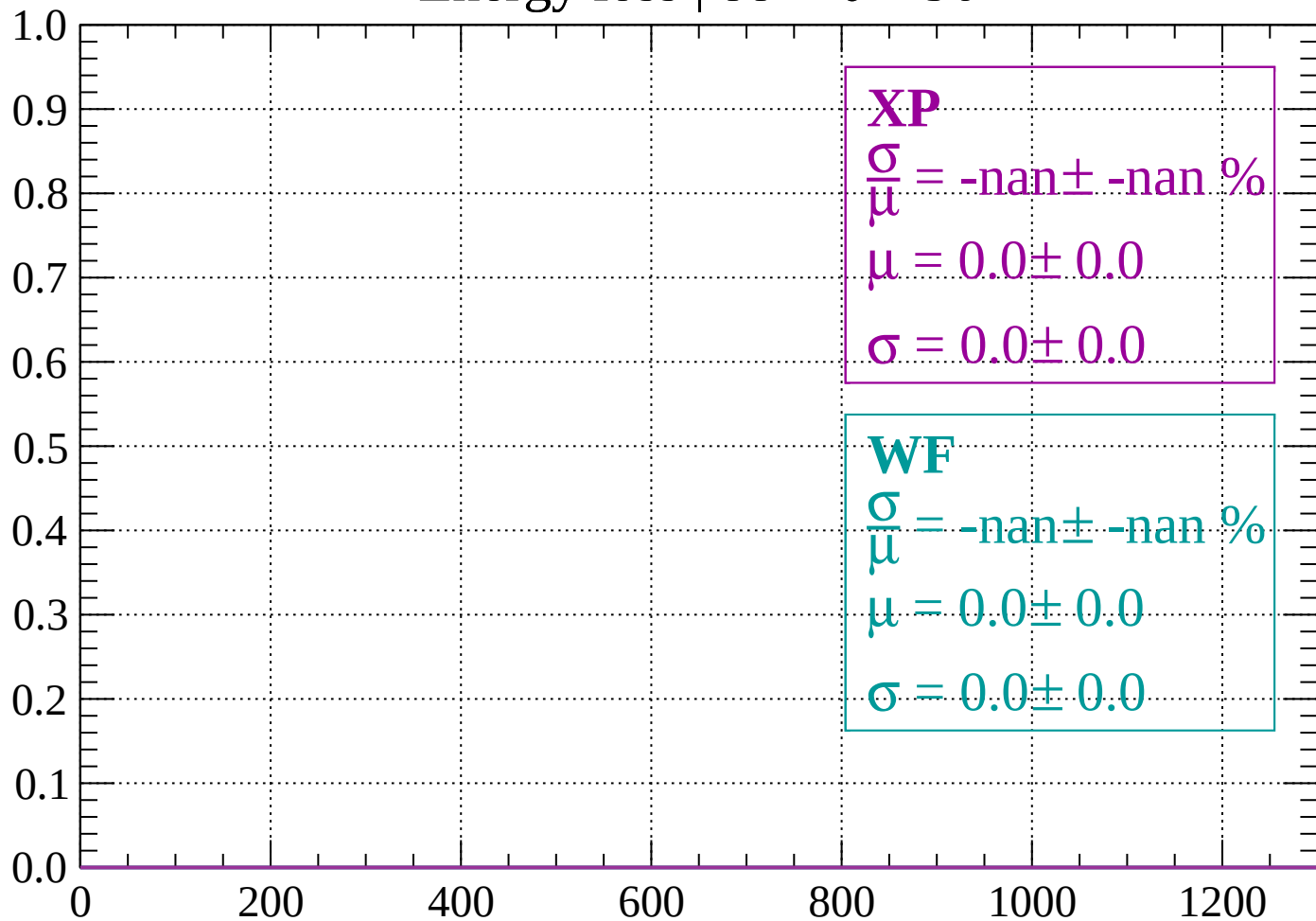
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

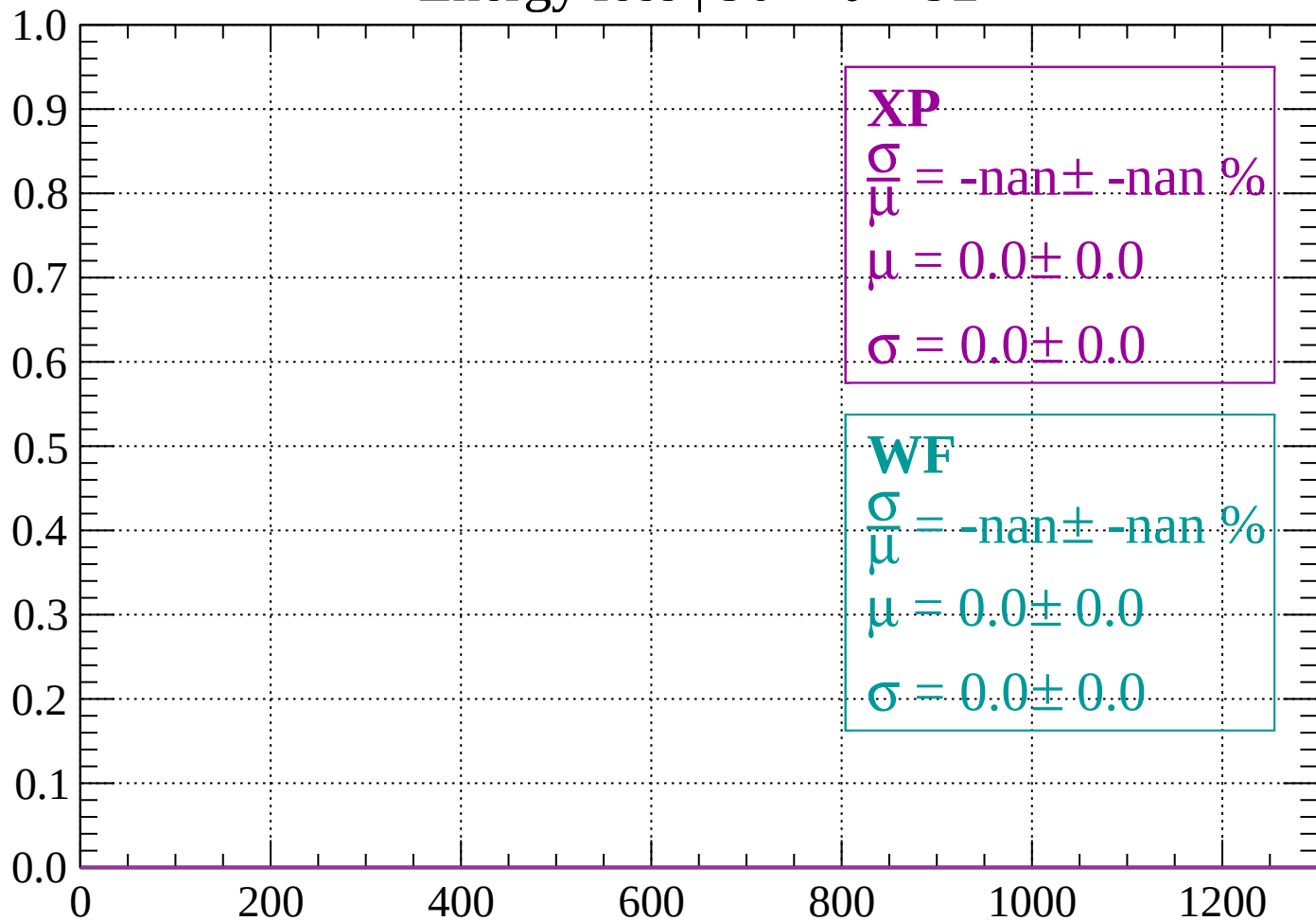
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

